Global Nuclear Command, Control, Communications (NC3), Early Warning, Cryptographic Custody, and Strategic Deterrence Architectures: Complete Forensic Engineering & Causal Dossier (217 Entities)

Scope: Exhaustive factual, engineering-technical, structural-hierarchical, geospatial, personnel, and deterministic cause-and-effect dossier covering 217 strategic command-and-control systems, cryptographic custody mechanisms, space and terrestrial early-warning networks, underground continuity-of-government bunkers, intercontinental and submarine-launched ballistic missiles, strategic bombers, hypersonic glide vehicles, thermonuclear warhead physics packages, and fissile production complexes across ten sovereign strategic architectures (United States, United Kingdom, France, USSR / Russian Federation, People's Republic of China, Democratic People's Republic of Korea, Republic of India, Islamic Republic of Pakistan, State of Israel, and Republic of Korea).
Compiled: 2026-11-06
Verification Standard: Immutable primary governmental, archival, national laboratory, parliamentary, multilateral, and peer-reviewed academic records (National Security Archive / NARA, U.S. Department of Defense / USSTRATCOM, U.S. Air Force / Space Force / Navy Official Fact Sheets, DOE / NNSA / Sandia / LANL / LLNL, Congressional Research Service, Federation of American Scientists (FAS) / Bulletin of the Atomic Scientists Nuclear Notebook, CSIS Missile Threat & Beyond Parallel, SIPRI Yearbook, USAF China Aerospace Studies Institute (CASI), Russian Strategic Nuclear Forces Project, 38 North / Stimson Center, James Martin Center for Nonproliferation Studies (CNS) / NTI, UK Ministry of Defence / Royal Navy, French Ministry of the Armed Forces / DGA / CEA-DAM, Government of India MEA / PIB / DRDO / BARC, and Pakistan MOFA / NCA).

Executive Architectural Synthesis: Deterministic NC3 Control Schematics & Causal Signal Topologies Across Ten Sovereign Strategic Contours

Strategic deterrence and nuclear command, control, and communications (NC3) architectures function as closed-loop, fault-tolerant cybernetic and physical state machines designed to resolve the dual-assurance engineering problem (**"Always / Never" criterion**): guaranteeing deterministic weapon execution upon authenticated sovereign authorization under high-altitude electromagnetic pulse (HEMP), blast overpressure, and decapitation strike conditions (**positive control**), while physically and cryptographically precluding unauthorized, accidental, or single-actor detonation (**negative control**) (Sandia National Laboratories PAL & Surety Architecture; Defense Primer: Nuclear Command, Control, and Communications (NC3) - CRS IF11697). Across all ten sovereign architectures examined in this dossier, the operational signal chain traverses five structural hierarchical levels:

1. **Tier-0 — Space-Based Infrared & Terrestrial Over-the-Horizon / Phased-Array Early Warning Sensor Layer:** Geosynchronous and Highly Elliptical Orbit (GEO/HEO) short-wave and mid-wave infrared (SWIR/MWIR) focal-plane arrays (DSP, SBIRS, Soviet/Russian Oko and Tundra / EKS 14F142, Chinese *Tongxun Jishu Shiyan* / *Huoyan-1* under PLAASF Base 37, and South Korean 425 Project SAR/EO satellites) detect rocket plume thermal emissions within 40 to 90 seconds of booster ignition. Concurrently, large fixed solid-state phased-array radars operating across VHF, UHF, L-band, and X-band (BMEWS, PAVE PAWS / AN/FPS-132 UEWR, Russian Dnestr, Dnepr, Daryal, Volga, and Voronezh-M/DM/VP, and Chinese Huian, Shuangyashan, and Xinjiang LPARs) track ascending boosters and midcourse reentry vehicle clouds, providing **dual-phenomenology confirmation** (independent infrared + radar correlation) to eliminate solar-glint or ionospheric false alarms (Cheyenne Mountain Complex - FAS).
2. **Tier-1 — Central, Underground Alternate, and Airborne Command Authorities & Automated Assessment Hubs:** Correlated track vectors feed hardened subterranean correlation and command bunkers (Cheyenne Mountain Complex, Pentagon NMCC, ANMCC at Raven Rock Site R, USSTRATCOM LeMay C2F, Mount Weather; Russian Solnechnogorsk SPRN Command Post, Vlasikha RVSN HQ, Kosvinsky Kamen, and Mount Yamantau; French PC Jupiter and CDAOA Taverny/Lyon; Chinese CMC Xishan JOCC; Indian NCA Political/Executive Councils; Pakistani NCA/SPD; and South Korean NCMC, Bunker B-1, Bunker B-2, and CP TANGO) alongside survivable airborne command posts (E-4B Nightwatch NAOC, E-6B Mercury Looking Glass, historical EC-135 PACCS, and Russian Il-80 / Il-82).
3. **Tier-2 — Portable Executive Authorization Terminals, Cryptographic Challenge-Response & Survivable Transmission Networks:** Sovereign heads of state and command collegia authenticate identity and select pre-planned or adaptive war-plan options (SIOP / OPLAN 8010-12) via portable terminals (Nuclear Football, Biscuit / Gold Codes; Russian Cheget within Kazbek; DPRK *Haekbangashoe* under the State Affairs Commission; or pre-delegated UK Letters of Last Resort). Validated launch orders are formatted into encrypted Emergency Action Messages (EAMs) or Russian Monolit / Signal-A directives and broadcast across hardened multi-spectral networks (MEECN, TACAMO, Russian Kavkaz, Tu-142MR Orel, or autonomous Perimeter / Dead Hand 15E601 and Sirena 15A11 / 15Zh75 radio-command rockets; French *RAMSES* and *SYDEREC*; Chinese Beidou-3).
4. **Tier-3 — Warhead Custody, Sealed Authentication, Two-Person Integrity & Permissive Action Link (PAL) Unblocking:** Specialized custodial organizations (Russian 12th GUMO, Chinese Base 22 / Taibai Depot / Base 67, Indian BARC / DRDO / SFC, Pakistani SPD) and tactical launch crews enforce the Two-Man Rule, verify Sealed Authenticator System (SAS) physical tokens, and enter cryptographic Permissive Action Link (PAL Cat A–F) keys to arm the warhead physics package.
5. **Tier-4 — Strategic Delivery Platforms, Reentry Vehicles & Thermonuclear Warheads:** Unblocked fire-control computers initiate booster ignition or weapon release across silo-based and mobile ICBMs/IRBMs/MRBMs/SRBMs, ballistic and cruise missile submarines (SSBNs/SSKs), strategic bombers, and unmanned underwater vehicles, deploying MIRV, MARV, or hypersonic boost-glide vehicles (15Yu71 Avangard, DF-17) to target coordinates.

Comparative Sovereign NC3 & Strategic Deterrence Architecture Summary Table

Sovereign State	Supreme Launch Authority	Operational NC3 & War Plan Backbone	Early Warning Sensor Architecture	Warhead Custody & PAL / Auth System	Primary Delivery Triad / Dyad
United States	President (POTUS) via Nuclear Football & Gold Codes	NMCC, ANMCC Site R, E-4B NAOC, E-6B Looking Glass/TACAMO, USSTRATCOM, MEECN, OPLAN 8010-12	DSP, SBIRS, BMEWS, PAVE PAWS / UEWR, Cheyenne Mountain	SAS, PAL Cat D/F, Two-Man Rule, DOE/NNSA & USAF/USN	Minuteman III / Sentinel, Ohio / Columbia SSBN (Trident II D5), B-52H, B-2A, B-21 (W76, W78, W87, W88)
United Kingdom	Prime Minister + Letters of Last Resort	Northwood CTF 345, Whitehall Pindar Bunker, VLF Anthorn/Skelton, CASD	RAF Fylingdales AN/FPS-132 UEWR (integrated with US/NATO warning)	Procedural Dual-Key (CO + XO + WEPS onboard authentication; zero external PAL)	Vanguard-class & Dreadnought-class SSBNs armed with Trident II D5
France	President of the Republic + CEMA dual-half code	Force de frappe: Élysée PC Jupiter, FOST Brest, CDAOA, RAMSES, SYDEREC, VLF Rosnay	Spirale / Syracuse IV space links & NATO/national radar surveillance	Cryptographic split-code (President + CEMA) verified by CEA/DAM PALs	Triomphant-class SNLE-NG (M51 SLBM) & Rafale B/M armed with ASMP-A
USSR / Russian Federation	President + Defense Minister + CGS via Cheget	Kazbek, Krokus, Kavkaz, Monolit, Signal-A, Agat, Perimeter / Dead Hand (15E601), Sirena (15A11/15Zh75), Vlasikha, Kosvinsky Kamen, Yamantau, Il-80, Il-82, Tu-142MR	SPRN: Oko, Tundra (14F142), Dnestr, Dnepr, Daryal, Volga, Voronezh	12th GUMO warhead arsenals + General Staff cryptographic unblocking codes	R-36M2, RS-28 Sarmat, Topol-M, RS-24 Yars, Molodets/Barguzin, Avangard; Borei/Delfin/Akula (Bulava, Sineva, Layner); Tu-160, Tu-95MS (Kh-102)
People's Republic of China	Chairman of the Central Military Commission (CMC)	CMC JOCC (Xishan), PLARF HQ, Bases 61–69, Beidou-3 short-message NC3	PLA Early Warning Network: TJS/Huoyan-1 GEO IR satellites, Huian, Shuangyashan, Xinjiang LPARs	CAEP / Ninth Academy (ShenGuang), Base 22 / Taibai Depot / Base 67	Dongfeng (DF-5B/C, DF-31AG, DF-41, DF-26, DF-17), Type 094/096 SSBN (Julang JL-2/JL-3), H-6N
DPRK (North Korea)	President of the State Affairs Commission (SAC)	SAC, Missile General Bureau, *Haekbangashoe* ("Nuclear Trigger") C2 system	Sohae / Tongchang-ri *Malligyong-1* recon satellite & ground air-defense radars	WPK Munitions Industry Dept / Nuclear Weapons Institute (Yongbyon, Punggye-ri)	Hwasong-14/15/17/18/19, Pukguksong-1/2/3/4/5/6, Hwasal-1/2, Haeil, Rail-Mobile Regt
Republic of India	Prime Minister chairing Political Council of NCA	NCA (Political Council $			
ightarrow$ Executive Council $					
ightarrow$ Strategic Forces Command)	Swordfish LRTR, Green Pine radars, Rukmini/GSAT-7 naval VLF/SATCOM (VLF Vijayanarayanam)	Split-custody & electronic PALs across BARC, DRDO, and SFC	Agni-I/II/III/IV/V/Prime, Prithvi-II, Dhanush, INS Arihant/Arighat/S4 (K-15, K-4), Mirage 2000H, Rafale DH		
Islamic Republic of Pakistan	Prime Minister chairing ECC of NC3 Pakistan (NCA)	NCA (ECC & DCC), Strategic Plans Division (SPD) C4I2SR network, SECDIV	Air Defense Ground Environment radars & SPD intelligence fusion	12-digit Pakistani PAL (PALS) + Three-Man Rule managed by SPD, PAEC, KRL/Kahuta, Khushab	Shaheen-I/II/III, Ghauri, Ghaznavi, Nasr (Hatf-IX), Babur-1/2/3, Ra'ad
State of Israel	Prime Minister + Minister of Defense under Samson Option	Ministerial Committee on National Security, IAEC, Wing 2 (Sdot Micha), Shayetet 7 Haifa	EL/M-2080 Green Pine / Super Green Pine & AN/TPY-2 X-band radar + Ofeq ISR satellites	Dual-key ministerial authentication; warhead engineering at Dimona & Soreq	Jericho-1/2/3 (150, 199, 248 Squadrons), Dolphin / Dolphin-II (INS Tanin, INS Rahav with

Sovereign State	Supreme Launch Authority	Operational NC3 & War Plan Backbone	Early Warning Sensor Architecture	Warhead Custody & PAL / Auth System	Primary Delivery Triad / Dyad
					Popeye Turbo), F-15I Ra'am, F-35I Adir
Republic of Korea (ROK)	President of the Republic of Korea via NCMC	NCMC, Korea Strategic Command (KMPR), Bunker B-1, Bunker B-2, CP TANGO	Kill Chain (425 Project SAR/EO satellites) & KAMD (Green Pine, SPY-1D, THAAD)	ROK Joint Chiefs of Staff & Korea Strategic Command KCCS C4I authentication	Hyunmoo-IV (4.1/4.2/4.4) & Hyunmoo-V ("Monster Missile") high-yield kinetic bunker-busters

Section I: United States Nuclear Command, Control, Communications (NC3), Early Warning, Cryptographic Authentication, and Strategic War Plan Architecture (Entities 1–29)

1. Nuclear Football (Presidential Emergency Satchel)

Technical Passport & Custody Specification Matrix

Parameter	Verified Specification	Primary Source
Official Designation	Presidential Emergency Satchel ("The Football" / "Atomic Football")	National Security Archive: Electronic Briefing Book
Physical Chassis & Mass	Custom-modified, zero-bezel anodized aluminum Zero Halliburton briefcase encased inside a black pebble-grain leather outer jacket lined with a ballistic Kevlar weave; total mass 45 lb (20.4 kg); three active satchels maintained (President, Vice President, and White House backup/Designated Survivor)	Supreme Court of the United States Record Citation: Nuclear Briefcases
Internal Components	1. The 9 × 12-inch (23 × 30 cm), 75-page loose-leaf "Black Book" printed in black and red ink detailing retaliatory strike options under OPLAN 8010-12; 2. A 9 × 12-inch black book listing Continuity of Government (COG) classified Presidential Emergency Facilities (PEFs); 3. An 8-to-10-page manila folder with Emergency Alert System (EAS) / Primary Entry Point (PEP) broadcast procedures; 4. The 3 × 5-inch Sealed Authenticator System (SAS) Biscuit card and secure UHF/SHF satellite transceiver interface	Supreme Court Record Citation: Nuclear Briefcases
Custodial Organization	White House Military Office (WHMO) & White House Communications Agency (WHCA), carried 24/7 by five rotating commissioned military aides-de-camp (O-4/O-5 grades across Army, Navy, Air Force, Marine Corps, and Coast Guard/Space Force)	CNS: The Authority to Use Nuclear Weapons in Nuclear-Armed States

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Instituted under President John F. Kennedy in the immediate aftermath of the October 1962 Cuban Missile Crisis (operationalized in spring 1963) following National Security Action Memorandum reviews by Secretary of Defense Robert S. McNamara, the Presidential Emergency Satchel was engineered to guarantee that the President can review war-plan target packages and authenticate a nuclear launch order from any location worldwide (CNS: The Authority to Use Nuclear Weapons). Early operational protocols were overseen by Brigadier General Godfrey McHugh (Air Force Aide to President Kennedy, 1961–1963) and later detailed in official disclosures by Warrant Officer / WHMO Director Bill Gulley (1976–1977) (Supreme Court Record Citation: Nuclear Briefcases). When space-based infrared sensors (SBIRS/DSP) and terrestrial phased-array radars (BMEWS/PAVE PAWS) confirm an inbound ballistic missile attack, the NMCC Deputy Director for Operations (DDO) initiates a Missile Attack Conference (National Security Archive: NMCC Conference Procedures). The attending Presidential Military Aide unlocks the Zero Halliburton satchel, establishes an encrypted WHCA voice/video conference linking the President with the Secretary of Defense, the NMCC DDO, and the Commander of USSTRATCOM, and presents the Black Book ("Decision Handbook") containing pre-planned Major, Selected, and Limited Attack Options (CNS: The Authority to Use Nuclear Weapons). Once the President selects an option and authenticates identity via the Biscuit, the NMCC formats and broadcasts the encrypted Emergency Action Message (EAM) to strategic triad forces.

2. Biscuit (Gold Code Card)

Technical Passport & Cryptographic Token Matrix

Parameter	Verified Specification	Primary Source
Official Designation	Sealed Authenticator System (SAS) Presidential Identification Card ("The Biscuit")	CNS: The Authority to Use Nuclear Weapons
Physical Dimensions & Build	Credit-card-sized laminated composite token (standardized ISO/IEC 7810 ID-1 85.60 × 53.98 mm / historical 3 × 5-inch card) featuring an opaque, tamper-evident snap-open/peel-back cryptographic foil covering randomized alphanumeric lookup columns	CNS: The Authority to Use Nuclear Weapons
Originating Authority	National Security Agency (NSA) Key Management / COMSEC Directorate, Fort George G. Meade, Maryland (`39°06'32"N 76°46'17"W`)	CRS Defense Primer: NC3 IF11697
Distribution Nodes	Delivered daily by armed courier to the White House Military Office (President), Office of the Vice President, NMCC, ANMCC (Site R), E-4B NAOC, and E-6B Looking Glass	CNS: The Authority to Use Nuclear Weapons

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Manufactured by the National Security Agency (NSA) at Fort Meade, the Biscuit serves as the physical cryptographic token through which the President proves personal identity to the NMCC before any nuclear launch order can be processed (CNS: The Authority to Use Nuclear Weapons). Historical custody procedures and operational audits have been documented by General Henry H. Shelton (Chairman of the Joint Chiefs of Staff, 1997–2001), Lieutenant General Robert Patterson (WHMO Military Aide, 1996–1998), and Secretary of Defense William J. Perry (1994–1997) (CNS: The Authority to Use Nuclear Weapons). During an NMCC Decision Conference, the NMCC Deputy Director for Operations (DDO) issues a two-letter phonetic challenge (for example, "Bravo-Charlie"). The President snaps open the brittle, tamper-evident opaque seal on the Biscuit (which permanently deforms upon opening to provide irreversible visual proof of seal breaking), locates the row/column corresponding to the two-letter challenge, and reads back the matching daily Gold Code alphanumeric string (CNS: The Authority to Use Nuclear Weapons). Two watch officers inside the NMCC independently verify the response against the identical master SAS card held in the NMCC dual-combination safe, converting the authenticated directive into an outgoing EAM.

3. Gold Codes

Technical Passport & Cryptographic Protocol Matrix

Parameter	Verified Specification	Primary Source
Protocol Function	Daily-rotated high-entropy alphanumeric one-time challenge-response authentication sequences verifying National Command Authority (NCA) identity	CNS: The Authority to Use Nuclear Weapons
Functional Separation	Authenticates the human identity of the Commander-in-Chief; strictly distinct from Permissive Action Link (PAL) hardware warhead-unlocking codes and War Reserve SAS message authenticators	Columbia University NSAM-160 & PAL Technical Analysis
Cryptographic Generator	Director of the National Security Agency (DIRNSA / Commander, USCYBERCOM) & NSA Cybersecurity / COMSEC Directorate, Fort Meade, Maryland	CRS Defense Primer: NC3 IF11697

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Formalized under Secretary of Defense Robert S. McNamara (1961–1968) and documented in technical NC3 studies by former Strategic Air Command Minuteman Launch Control Officer and Brookings Institution scholar Dr. Bruce G. Blair (*Strategic Command and Control*, 1985; *The Logic of Accidental Nuclear War*, 1993), the Gold Codes provide positive cryptographic assurance that a nuclear strike directive originates from the incumbent President of the United States (or the lawful successor under the Presidential Succession Act of 1947 and the 25th Amendment) (CNS: The Authority to Use Nuclear Weapons; FAS Report: On the Precipice 2026). Because the NSA rotates the active Gold Code sequence every 24 hours across all Tier-1 command nodes (NMCC, ANMCC, NAOC, and Looking Glass), any expired or compromised card is immediately invalidated. Once the President's

Gold Code response is verified by the two-person NMCC Emergency Actions team, the team unlocks the NMCC War Reserve SAS safe, extracts the war-plan EAM authenticator and PAL unlock keys, and transmits the launch directive across MEECN (CNS: The Authority to Use Nuclear Weapons).

4. Emergency Action Message (EAM)

Technical Passport & Multi-Spectral Transmission Matrix

Parameter	Verified Specification	Primary Source
Message Format & Structure	Preformatted, cryptographically authenticated fixed-or-variable-length alphanumeric directive (standard 30-character preamble/execution block expandable to 150+ phonetic characters) containing the selected OPLAN attack option, execution time, SAS verification string, and PAL unlock keys	FAS Submarine Force Communications Appendix E
HF Global Communications System (HFGCS)	Upper Sideband (USB) voice and encrypted data broadcasts on primary strategic frequencies: **4724.0 kHz, 6739.0 kHz, 8992.0 kHz, 11175.0 kHz, 13200.0 kHz, and 15016.0 kHz**	FAS Submarine Force Communications Appendix A
VLF / LF Band & CJCS Modes	**14 kHz to 60 kHz** (including 17–30 kHz TACAMO AN/ART-54 and fixed naval VLF stations at Cutler, Maine [24.0 kHz], Jim Creek, Washington [24.8 kHz], Lualualei, Hawaii [21.4 kHz], and LaMoure, North Dakota [25.2 kHz]) utilizing **CJCS MEECN Modes 9, 9 MMPM, 15, and High Data Rate (HIDAR)** minimum-shift keying (MSK) at 50–300 bps	FAS Submarine Force Communications Appendix E
UHF / SHF / EHF SATCOM	MILSTAR / Advanced Extremely High Frequency (AEHF) at **44 GHz uplink / 20 GHz downlink** (AN/USC-38 terminals) and AFSATCOM UHF at **225–400 MHz**, plus terrestrial Strategic Automated Command and Control System (SACCS)	FAS Submarine Force Communications Appendix A

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Formatted and originated by the Chairman of the Joint Chiefs of Staff (CJCS) through the NMCC Deputy Director for Operations (J-3) under *Emergency Action Procedures of the Chairman, Joint Chiefs of Staff (EAP-CJCS) Volume VII*, or by Commander, USSTRATCOM and Commander, NAOC, the EAM is the sole authoritative operational message capable of directing U.S. nuclear weapon execution (FAS Submarine Force Communications Appendix E). Technical analyses of EAM modulation, acoustic/ionospheric propagation under nuclear scintillation, and PAL key insertion were conducted by Dr. Ashton B. Carter (later 25th Secretary of Defense, 2015–2017) and Donald R. Cotter in *Managing Nuclear Operations* (Brookings Institution, 1987) (Columbia University NSAM-160 & PAL Technical Analysis). Following Presidential Gold Code authentication, the two-person NMCC Emergency Actions cell encodes the EAM and broadcasts it concurrently across SACCS, HFGCS, MEECN VLF/LF, and AEHF/MILSTAR satellites. Two-person crews in Minuteman III Launch Control Centers, submerged Ohio-class SSBNs, and B-52H/B-2A cockpits copy the EAM independently, unlock their red dual-combination SAS safes, break open the corresponding foil packet to verify character-for-character authenticity, input the embedded PAL unlock keys, and actuate launch keys (CNS: The Authority to Use Nuclear Weapons).

5. NMCC (National Military Command Center)

Technical Passport & Facility Matrix

Parameter	Verified Specification	Primary Source
Geolocation & Footprint	The Pentagon (38°52'15"N 77°03'21"W), Arlington, Virginia; spans approximately 27,000 sq ft across the Joint Staff river-terrace/basement and second-floor rings	FAS National Military Command Center (NMCC)
Statutory & Directive Charter	Established October 1962 under DoD Directive S-5100.30 (17 October 1962) as the primary fixed node of the National Military Command System (NMCS)	FAS National Military Command Center (NMCC)
C3I & Computing Architecture	Initially equipped with the USAF IBM 473L Command and Control System (dual IBM 1410 mainframes and AN/FYA-2 consoles); modernized with the Command Center Processing and Display System-Replacement (CCPDS-R), Jam Resistant Secure Communications (JRSC), SACCS, Defense Red Switch Network (DRSN), and MILSTAR/AEHF terminals	FAS Cheyenne Mountain & CCPDS-R Architecture
Blast Hardening & Staffing	Unhardened surface building (~3–5 psi overpressure rating, relying on ANMCC Site R and E-4B NAOC for nuclear survivability); staffed 24/7 by five rotational Operations Teams commanded by a 1-star Deputy Director for Operations (DDO, Joint Staff J-3)	FAS National Military Command Center (NMCC)

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Established in October 1962 under Secretary of Defense Robert S. McNamara and Chairman of the Joint Chiefs of Staff General Maxwell D. Taylor (1962–1964) during the buildup to the Cuban Missile Crisis, the NMCC serves as the primary peacetime and crisis command-and-communications hub linking the President and Secretary of Defense to the unified combatant commands (FAS National Military Command Center (NMCC)). Within the NMCC Warning Room, watch officers monitor real-time ballistic missile warning feeds transmitted via CCPDS-R from SBIRS/DSP and BMEWS/PAVE PAWS. Upon detection of a potential missile attack, the NMCC DDO convenes a Threat Assessment Conference and a Missile Attack Conference with the Cheyenne Mountain Command Director, Commander of USSTRATCOM, and the President (National Security Archive: NMCC Conference Procedures). Following Presidential Gold Code authentication, the NMCC Emergency Actions cell formats and transmits the EAM over MEECN and operates the U.S.–Russian Direct Communications Link ("Hotline" / MOLINK) (National Security Archive: NMCC Conference Procedures).

6. ANMCC (Alternate National Military Command Center)

Technical Passport & Subterranean Hardening Matrix

Parameter	Verified Specification	Primary Source
Geolocation & Co-Location	Excavated inside Raven Rock Mountain Complex (Site R) at 39°44'03"N 77°25'11"W (summit) / 39.7296°N 77.4325°W (west portal), Liberty Township, Adams County, near Blue Ridge Summit, Pennsylvania (6 miles north of Camp David)	FAS Raven Rock Site R AJCC
Directive Charter	Formally designated as the ANMCC on 1 October 1962 and codified in DoD Directive S-5100.30 on 17 October 1962	FAS Raven Rock Site R AJCC

Parameter	Verified Specification	Primary Source
Overpressure & EMP Hardening	**140 psi** blast overpressure resistance beneath Precambrian greenstone granite overburden; protected by dual **34-ton steel blast doors**, EMP-shielded freon-cooled equipment chambers, and 38+ communications systems	FAS Raven Rock Site R AJCC
Operating Authorities	Joint Staff J-3, Defense Information Systems Agency (DISA), U.S. Army 114th Signal Battalion, and Washington Headquarters Services (WHS, Office of the Secretary of Defense)	FAS Raven Rock Site R AJCC

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Authorized originally by President Harry S. Truman in 1948–1950 and designated as the Alternate National Military Command Center by Secretary of Defense Robert S. McNamara in October 1962, the ANMCC provides a hardened underground continuity-of-operations mirror of the Pentagon NMCC (FAS Raven Rock Site R AJCC). All early warning feeds from SBIRS, DSP, and UEWR stream simultaneously into both the Pentagon NMCC and the ANMCC via CCPDS-R and Jam Resistant Secure Communications (JRSC) circuits (FAS Cheyenne Mountain & CCPDS-R Architecture). If the Pentagon NMCC is destroyed in a decapitation strike or evacuated during a crisis, command authority transfers immediately to the ANMCC Joint War Room, where surviving National Command Authority officials authenticate via Gold Codes and the ANMCC Emergency Actions team broadcasts EAMs over MEECN VLF/LF, HFGCS, and AEHF/MILSTAR links (FAS Raven Rock Site R AJCC).

7. NAOC (National Airborne Operations Center)

Technical Passport & Airborne C3I Matrix

Parameter	Verified Specification	Primary Source
Operational Mission Platform	Boeing E-4B Nightwatch Advanced Airborne Command Post (AACP), redesignated from the National Emergency Airborne Command Post (NEACP) in 1994	U.S. Air Force E-4B Fact Sheet
Electromagnetic Spectrum & Antennas	Operates across **14 kHz VLF to 44 GHz EHF** via **67 satellite dishes and antennas**, including a **5-mile (8 km) trailing wire VLF/LF antenna**, a dorsal radome housing a steerable SHF/EHF MILSTAR/AEHF dish, 13 external communications systems, and 46 internal communications systems	AFGSC E-4B Fact Sheet
Nuclear & EMP Hardening	Full high-altitude electromagnetic pulse (HEMP) Faraday shielding, nuclear thermal radiation shielding, overpressure intake/exhaust blast valves, and analog flight-deck instrumentation	AFGSC Overview of the NAOC and E-4B Nightwatch
Operational Command & Unit	Directed by CJCS through Commander, USSTRATCOM; operated by the **1st Airborne Command and Control Squadron (1 ACCS)** under the **95th Wing** (activated 28 February 2025, succeeding the 595th Command and Control Group), Eighth Air Force, Offutt AFB, Nebraska (`41°07'06"N 95°54'45"W`)	AFGSC E-4B Fact Sheet

Engineering Mechanics, Named Officials & Deterministic Causal Chain

The National Airborne Operations Center (NAOC) is the most survivable node of the National Military Command System (NMCS), providing an airborne command center for the President, Secretary of Defense, and Joint Chiefs of Staff during a nuclear crisis (U.S. Air Force E-4B Fact Sheet). At least one E-4B NAOC is maintained on continuous 24/7 ground alert at Offutt Air Force Base or forward-deployed alongside the President during international travel. Upon receipt of a SBIRS/DSP missile attack warning, the alert NAOC scrambles within minutes and establishes a Joint Airborne Operations Center conference via its dorsal AEHF/MILSTAR antenna (AFGSC Overview of the NAOC and E-4B Nightwatch). Once the President or Secretary of Defense authenticates an OPLAN 8010-12 strike option via Gold Codes, the onboard NAOC battle staff verifies SAS authenticators and broadcasts the EAM directly to ICBM Launch Control Centers, SSBNs, and strategic bombers via its 5-mile VLF trailing wire, HFGCS, and EHF SATCOM (U.S. Air Force E-4B Fact Sheet).

8. Looking Glass (Airborne Command Post Mission)

Technical Passport & Airborne Launch Control Matrix

Parameter	Verified Specification	Primary Source
Mission Designation	USSTRATCOM Airborne Command Post (ABNCP) — Code Name "Looking Glass" (because it mirrors ground-based C2 capabilities at Offutt AFB)	USSTRATCOM: Looking Glass Airborne Command Post
Airborne Launch Control System (ALCS)	Onboard **UHF (225–400 MHz)** encrypted command transmitter capable of directly retargeting and launching LGM-30G Minuteman III ICBMs inside their silos if underground Launch Control Centers (LCCs) are destroyed or isolated	National Park Service: Airborne Launch Control Center - Operation Looking Glass
Historical & Active Platforms	Flown by USAF EC-135C from **3 February 1961 to 1 October 1998** (including continuous 24/7 airborne alert until 24 July 1990, logging >281,000 accident-free hours); flown from **1 October 1998 to present** by the U.S. Navy Boeing E-6B Mercury	USSTRATCOM: Looking Glass Airborne Command Post
Battle Staff & Operators	Commanded by a 1-star General/Flag Officer serving as the Airborne Emergency Action Officer (AEAO), supported by a joint USSTRATCOM battle staff, Navy TACAMO flight crew (STRATCOMWING 1 / Task Force 124), and USAF airborne missileers from the **625th Strategic Operations Squadron (625 STOS)**	USSTRATCOM: Looking Glass Airborne Command Post

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Initiated on 3 February 1961 by General Thomas S. Power (Commander-in-Chief, Strategic Air Command [CINCSAC], 1957–1964) and transitioned from continuous 24/7 airborne alert to 24-hour random airborne/ground alert on 24 July 1990 under General John T. Chain Jr. (CINCSAC, 1986–1991), Operation Looking Glass guarantees that USSTRATCOM can execute nuclear retaliation even if Offutt Air Force Base is obliterated (USSTRATCOM: Looking Glass Airborne Command Post). When an authenticated EAM is received via AEHF/MILSTAR or HFGCS (or generated aboard Looking Glass if ground C2 nodes are destroyed), the AEAO and battle staff validate the SAS authenticators under the Two-Man Rule, broadcast the EAM to submerged SSBNs via the aircraft's 200 kW VLF trailing wire, and—where underground ICBM LCCs are incapacitated—two 625 STOS airborne missileers at the onboard ALCS console transmit direct UHF launch execution signals into the hardened surface UHF antennas of Minuteman III silos across Malmstrom, Minot, and F.E. Warren AFBs, directly initiating booster ignition (Airborne Launch Control Center - Operation Looking Glass).

9. TACAMO (Take Charge and Move Out)

Technical Passport & VLF Submarine Relay Matrix

Parameter	Verified Specification	Primary Source
VLF Transmitter & Frequency Band	**AN/ART-54 High Power Transmit Set** rated at **200 kW** RF output operating in the **Very Low Frequency (VLF) band from 17 kHz to 30 kHz**, penetrating seawater to depths of 50–100+ feet	FAS Submarine Force Communications Appendix A
Dual Trailing Wire Antenna System	**26,000-to-28,000-foot (5-mile / 8 km) Long Trailing Wire Antenna (LTWA)** stabilized by a 90 lb aerodynamic drogue, paired with a **4,000-to-5,000-foot Short Trailing Wire Antenna (STWA)** counterpoise dipole	NAVAIR E-6B Mercury Product Profile

Parameter	Verified Specification	Primary Source
Vertical Polarization Mechanics	Executes a continuous tight 30–40° bank pylon turn ("Orbit Drive") around a fixed geographic point so the 5-mile LTWA hangs **>70% vertically**, radiating vertically polarized VLF magnetic fields into SSBN AN/BRR-6 OE-315 buoyant cable and loop antennas	FAS Submarine Force Communications Appendix A
Operational Wing & Squadrons	**Strategic Communications Wing ONE (STRATCOMWING 1 / Task Force 124)** at Tinker AFB, Oklahoma (`35°24'53"N 97°23'12"W`), commanding **VQ-3 ("Ironmen")**, **VQ-4 ("Shadows")**, and **VQ-7 ("Roughnecks")**, with alert detachments at Travis AFB, California, and NAS Patuxent River, Maryland	U.S. Navy Fleet Air Reconnaissance Squadron VQ-4

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Coined in 1961 by Rear Admiral Bernard F. Roeder (Director of Naval Communications, 1961–1965) to provide a survivable airborne relay between the National Command Authority and submerged fleet ballistic missile submarines (SSBNs) after fixed shore VLF towers are destroyed, TACAMO began prototype flight tests in 1962 aboard a Collins Radio-modified Lockheed KC-130F Hercules, operated on EC-130G/Q aircraft from 1963 to 1992, transitioned to the Boeing E-6A/B Mercury in 1989, and is undergoing future recapitalization onto the Lockheed Martin E-130J TACAMO platform (NAVAIR E-6B Mercury Product Profile; CRS Defense Primer: NC3 IF11697). Upon receiving an encrypted EAM from the NMCC, ANMCC, or E-4B NAOC via 44/20 GHz AEHF/MILSTAR or HFGCS, the orbiting TACAMO crew reels out the 28,000-ft LTWA and 4,000-ft STWA, establishes a tight continuous bank orbit, and broadcasts the EAM at 200 kW in the 17–30 kHz VLF band using CJCS Modes 9, 9 MMPM, 15, or HIDAR (FAS Submarine Force Communications Appendix E). Submerged Ohio-class SSBNs copy the VLF broadcast via buoyant wire antenna, validate SAS authenticators, and launch Trident II D5 SLBMs.

10. BMEWS (Ballistic Missile Early Warning System)

Technical Passport & Polar Radar Network Matrix

Parameter	Verified Specification	Primary Source
Program Designation & Contractors	USAF Project 474L (constructed 1958, operational 1961–2001); prime contractor **RCA** (Moorestown, NJ), radar subcontractors **General Electric** (AN/FPS-50) and **Goodyear Aircraft** (AN/FPS-49), prototype engineering by **MIT Lincoln Laboratory**	National Security Archive: Declassified IDA Evolution of U.S. Strategic Warning (BMEWS)
Three Polar Radar Sites	**Site I — Thule Site J (Pituffik Space Base), Greenland** (`76°34'13"N 68°19'05"W`, 4× AN/FPS-50 + 1× AN/FPS-49A); **Site II — Clear AFS, Alaska** (`64°18'04"N 149°11'23"W`, 3× AN/FPS-50 + 1× AN/FPS-92); **Site III — RAF Fylingdales, North Yorkshire, UK** (`54°21'42"N 00°40'11"W`, 3× AN/FPS-49 84-ft dishes in 140-ft radomes)	FAS Cheyenne Mountain & Early Warning Network
RF Frequency, Power & Range	**UHF band at 420–450 MHz** (425 MHz center frequency); **1.2 MW average / 5–10 MW peak power per transmitter**; **3,000 nautical mile (5,556 km)** range; AN/FPS-50 fixed torus reflectors measured 165 ft (50 m) high × 400 ft (122 m) wide (1,500 tons); processed by duplexed **IBM 7090 AN/FSQ-28 Missile Impact Predictors**	National Security Archive: Declassified IDA BMEWS Study
Solid-State SSPARS / UEWR Upgrade	Replaced between 1987 and 2001 by Raytheon **AN/FPS-120 / AN/FPS-126 Solid State Phased Array Radar System (SSPARS)** and subsequently upgraded to **AN/FPS-132 Upgraded Early Warning Radars (UEWR)** under U.S. Space Force **Space Delta 4** (12th SWS Thule, 13th SWS Clear, and RAF Fylingdales)	FAS Cheyenne Mountain & Early Warning Network

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Approved by President Dwight D. Eisenhower and Secretary of Defense Neil H. McElroy in January 1958 and engineered by RCA Chief Engineer Dr. Harry R. Wege and MIT Lincoln Laboratory Director Dr. Carl F.J. Overhage, BMEWS provided the United States with its first 15-to-25-minute trans-polar ICBM warning capability (National Security Archive: Declassified IDA BMEWS Study). When Soviet/Russian or Chinese ICBMs cross the northern polar horizon into the 420–450 MHz radar fans, on-site processors (historically the IBM 7090 AN/FSQ-28 Missile Impact Predictor, today the Raytheon AN/FPS-132 UEWR signal processor) calculate launch coordinates, ballistic covariance state vectors, and predicted impact ellipses, transmitting the warning via Jam Resistant Secure Communications (JRSC) and Survivable Communications Integration System (SCIS) circuits to Cheyenne Mountain, USSTRATCOM, and the Pentagon NMCC (FAS Cheyenne Mountain & Early Warning Network). Dual-phenomenology correlation with SBIRS/DSP infrared tracks triggers the NMCC Missile Attack Conference, Presidential Gold Code authentication, and EAM dissemination.

11. Permissive Action Link (PAL) (Categories A through F)

Technical Passport & Warhead Hardware Interlock Matrix

PAL Category	Code Length & Hardware Mechanism	Key Surety & Anti-Tamper Features	Primary Source
Category A (1962)	4-digit electromechanical combination lock	External control box/cable assembly interrupting firing circuit on early NATO theater nuclear weapons (Honest John, Sergeant, Davy Crockett)	Columbia University NSAM-160 & PAL Technical Analysis
Category B (mid-1960s)	4-digit motor-driven rotary electromechanical switch	Isolates high-voltage firing set (X-unit) from power supply; supports group unlocking codes	Columbia University NSAM-160 & PAL Technical Analysis
Category C (late 1960s)	4-digit enhanced electromechanical switch	Incorporates initial environmental and anti-tamper coupling prior to solid-state multi-code processors	Columbia University NSAM-160 & PAL Technical Analysis
Category D (1970s–present)	**6-digit numeric/alphanumeric** solid-state processor	Supports separate unlock, training, disarm, and command-disablement keys; includes a **limited-try counter** that permanently locks the warhead after repeated wrong entries	Columbia University NSAM-160 & PAL Technical Analysis
Category F (late 1970s–present)	**12-digit alphanumeric** cryptographic microprocessor	Integrated inside a multi-sensor tamper-evident barrier with **Unique Signal Generators (USG)**, **Stronglink/Weaklink Environmental Sensing Devices (ESDs)**, selectable yield control, and violent/non-violent self-disablement (internal shaped-charge asymmetric pit deformation)	Columbia University NSAM-160 & PAL Technical Analysis

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Originated from prototypes developed at Los Alamos Scientific Laboratory and **Sandia National Laboratories** (Albuquerque, New Mexico, and Livermore, California) by **Dr. Harold M. Agnew** (later Director of Los Alamos, 1970–1979, who inspected unlocked NATO nuclear bases in 1960–1961 and briefed Joint Committee on Atomic Energy Chairman Senator Clinton P. Anderson), **Donald R. Cotter** (Sandia engineer and later Assistant to the Secretary of Defense for Atomic Energy, 1973–1978), **Stanley D. Spray**, **Curt Mueller**, and **Jay Grear** (architects of the Sandia Unique Signal Concept, SAND91-1269), Permissive Action Links were mandated across U.S. nuclear forces by President **John F. Kennedy** in **National Security Action Memorandum 160 (NSAM-160) on 6 June 1962** (Columbia University NSAM-160 & PAL Technical Analysis). Embedded deep inside the nuclear warhead physics package so that bypassing the switch requires dismantling the warhead (which triggers anti-tamper shaped-charge pit deformation), the Cat D/F PAL microprocessor remains locked until the launch crew receives the master PAL unlock key inside an authenticated EAM. Once the 6-digit or 12-digit PAL key is entered via the Launch Control Panel or Aircraft Monitoring and Control (AMAC) unit, the PAL enables the Unique Signal Generator (USG) to send a 24-bit non-repeating mechanical/optical pattern that closes the **Stronglink** safing switches only after **Environmental Sensing Devices (ESDs)** confirm true ballistic boost acceleration, free-fall zero-G, and barometric/radar reentry sequences (Columbia University NSAM-160 & PAL Technical Analysis).

12. Sealed Authenticator System (SAS)

Technical Passport & Physical-Cryptographic Verification Matrix

Parameter	Verified Specification	Primary Source
Token Construction	Tamper-evident laminated opaque plastic/foil packets ("cookies" or "biscuits", ~3 × 5 inches) containing NSA-generated randomized alphanumeric authentication strings	CNS: The Authority to Use Nuclear Weapons
Physical Storage Security	Stored inside bright red dual-combination steel safes equipped with two independent manipulation-proof locks (each officer on a two-person crew knows only one combination)	USSTRATCOM 2023 Congressional Posture Statement
Anti-Tamper Chemistry	Brittle inner foil and X-ray/thermal-sensitive laminate that permanently fractures or discolors if steamed, X-rayed, or flexed, audited at every crew shift change	CNS: The Authority to Use Nuclear Weapons
Governing Directives	CJCS Emergency Action Procedures (EAP-CJCS), DoD Directive 5210.41, and Air Force Instruction (AFI) 91-104	AFDP 3-72 Nuclear Operations

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Produced by the National Security Agency (NSA) at Fort Meade, the Sealed Authenticator System (SAS) provides end-to-end physical-cryptographic proof to tactical launch crews that a received Emergency Action Message (EAM) is genuine and was originated by the National Command Authority (CNS: The Authority to Use Nuclear Weapons). Following Presidential Gold Code validation, the two-person Emergency Actions team at the NMCC (or ANMCC / E-4B NAOC) unlocks the NMCC dual-combination SAS safe, snaps open the specific War Reserve SAS foil packet corresponding to the selected war-plan option, and embeds its exact alphanumeric string into the outgoing EAM. At receiving Minuteman III Launch Control Centers, Ohio-class SSBNs, and B-52H/B-2A bombers, the two-person launch crew copies the EAM independently, unlocks their red dual-combination SAS safe using their separate lock combinations, breaks open their matching local SAS foil packet, and compares the printed characters character-for-character against the characters in the EAM (CNS: The Authority to Use Nuclear Weapons). An exact match authorizes the crew to input the PAL unlock keys and execute launch.

13. Two-Man Rule (Air Force / Navy / DoD Directives)

Technical Passport & Positive Control Matrix

Parameter	Verified Specification	Primary Source
Governing DoD & Service Directives	**DoD Directive 5210.41** ("Security Policy for Protecting Nuclear Weapons"), **DoDM 5210.42** ("Nuclear Weapon Personnel Reliability Program [PRP]"), and **Air Force Instruction (AFI) 91-104** ("Nuclear Surety Tamper Control and Two-Person Concept")	AFDP 3-72 Nuclear Operations
ICBM Launch Control Center (LCC) Mechanics	Two PRP-certified missile combat crew officers seated at consoles separated by **12 feet (3.65 m)** must turn spring-loaded launch keys within **2.0 seconds** of each other and hold them for **5.0 seconds**; requires a **"two-capsule vote"** (four officers across two separate underground LCCs, or one LCC + E-6B ALCS) to initiate silo ignition	National Park Service Delta-01 Launch Control Facility
SSBN Launch Execution Mechanics	Concurrent authentication and sequential key/trigger actuation across the Commanding Officer (CO), Executive Officer (XO), Weapons Officer (WEPS), and Missile Technician crew across separate compartments	CNS: The Authority to Use Nuclear Weapons

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Formalized Department-wide in 1961–1962 under Secretary of Defense Robert S. McNamara and Deputy Secretary of Defense Roswell L. Gilpatric alongside NSAM-160 and overseen today by the Deputy Assistant Secretary of Defense for Nuclear Matters (DASD(NM)), Commander of USSTRATCOM, Commander of Air Force Global Strike Command (AFGSC), and Director of U.S. Navy Strategic Systems Programs (SSP), the Two-Man Rule prohibits any single individual from accessing a nuclear weapon, opening a SAS safe, formatting an EAM, or actuating a launch circuit (AFDP 3-72 Nuclear Operations). While the President exercises sole constitutional authority to *authorize* nuclear use via the Gold Codes, every subsequent technical step—verifying the Presidential Gold Code in the NMCC, encoding the EAM, unlocking red SAS safes, entering PAL codes, and turning physically separated launch switches—requires the synchronized action of at least two PRP-certified personnel (CNS: The Authority to Use Nuclear Weapons).

14. Single Integrated Operational Plan (SIOP) (SIOP-62 through OPLAN 8044)

Technical Passport & Historical War Plan Evolution Matrix

SIOP Iteration	Effective Date	Target Base & Warhead Allocation	Doctrinal & Technical Innovation	Primary Source
SIOP-62	1 July 1961 (completed 14 Dec 1960)	**3,200+ nuclear warheads** delivered against **1,060 Designated Ground Zeros (DGZs)** (including 130+ Soviet, Chinese, and Eastern European cities)	First unified tri-service target plan eliminating duplicate Navy/SAC targeting; single synchronized massive strike	National Security Archive: New Evidence on the Origins of Overkill (SIOP-62)
SIOP-63	July 1962	Expanded National Strategic Target List (NSTL) partitioned into **5 Major Attack Options (Tasks Alpha–Echo)**	Introduced under SecDef Robert S. McNamara to separate counterforce missile/bomber/C2 strikes (Tasks A–D) from urban-industrial retaliation (Task E)	FAS Report: Planning for the Unthinkable (2025)
SIOP-5 & SIOP-6	1 January 1976 (SIOP-5); 1 October 1983 (SIOP-6)	National Target Base (NTB) peaked at **~16,000 targets** in 1985 before declining to ~12,500 by 1990	Implemented NSDM-242 (1974 "Schlesinger Doctrine") and PD-59 (1980 "Countervailing Strategy"): Major (MAO), Selected (SAO), Regional (RNO), and Limited Nuclear Options (LNO) compiled on IBM System/360-370 mainframes	FAS Report: Planning for the Unthinkable (2025)
SIOP-93 to OPLAN 8044	June 1992 – 1 March 2003	~2,500–3,000 targets across Russia, China, and proliferant states	Re-designated by fiscal year (SIOP-93 through SIOP-01); formally retired on **1 March 2003** and replaced by **OPLAN 8044 ("Revision 03")**	FAS: US Nuclear War Plan Updated (OPLAN 8044 / 8010)

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Ordered by President Dwight D. Eisenhower and Secretary of Defense Thomas S. Gates Jr. in August 1960, the Single Integrated Operational Plan (SIOP) was constructed by the **Joint Strategic Target Planning Staff (JSTPS)** at Offutt Air Force Base under dual-hatted Director **General Thomas S. Power** (CINCSAC, 1957–1964) and Deputy Director **Rear Admiral Edward N. Parker, USN**, with analytical input from the RAND Corporation (William W. Kaufmann and Albert Wohlstetter) (National Security Archive: Origins of Overkill (SIOP-62)). Subsequent revisions were directed by Secretaries of Defense Robert S. McNamara (SIOP-63), James R. Schlesinger (NSDM-242 / SIOP-5), and Harold Brown (PD-59 / SIOP-6), before Commander of USSTRATCOM **Admiral James O. Ellis Jr.** (2001–2004) retired the SIOP title in March 2003 in favor of OPLAN 8044 (FAS Report: Planning for the Unthinkable (2025)). In the causal execution chain, Presidential Nuclear Weapons Employment Policy (NUWEP) and the CJCS Joint Strategic Capabilities Plan (JSCP) directed JSTPS/USSTRATCOM targeteers to compute exact warhead-to-target pairings and time-on-target deconfliction ("fratricide avoidance"), distilling the options into the Black Book inside the Nuclear Football so that a single Presidential Gold Code-authenticated EAM triggers pre-loaded target packages across the entire Triad.

15. OPLAN 8010-12 (Strategic Deterrence and Force Employment)

Technical Passport & Integrated War Plan Matrix

Parameter	Verified Specification	Primary Source
Official Short Title & Date	**USSTRATCOM OPLAN 8010-12 ("Strategic Deterrence and Force Employment")**, promulgated **30 July 2012** (succeeding OPLAN 8010-08 [Feb 2008 / Feb 2009] and OPLAN 8044; updated in April 2019)	National Security Archive: USSTRATCOM OPLAN 8010-12
Six Integrated Mission Spectrums	1. Strategic nuclear forces (Triad); 2. Conventional global strike; 3. Non-kinetic and offensive cyber operations; 4. Integrated missile defense; 5. Intelligence, Surveillance, and Reconnaissance (ISR); 6. Counter-WMD	FAS Analysis: US Nuclear War Plan Updated (OPLAN 8010-12)
Designated Adversary Set	Four state actors (**Russian Federation, People's Republic of China, Democratic People's Republic of Korea, and Islamic Republic of Iran**) plus non-state WMD networks (Syria removed following earlier 8010 iterations)	USSTRATCOM OPLAN 8010-08 and 8010-12 Declassified FOIA Release
Planning Staff & Command	Authored by the **430-person Joint Functional Component Command for Global Strike (JFCC-GS)** and USSTRATCOM J5 Plans and Policy Directorate at Offutt AFB, Nebraska	FAS Analysis: US Nuclear War Plan Updated (OPLAN 8010-12)

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Promulgated on 30 July 2012 by **General C. Robert Kehler, USAF** (Commander, USSTRATCOM, 2011–2013) in response to the CJCS operational order *Global Citadel* following the 2010 Nuclear Posture Review—succeeding **General Kevin P. Chilton** (CDRUSSTRATCOM, 2007–2011, who signed OPLAN 8010-08) and **General James E. Cartwright, USMC** (CDRUSSTRATCOM, 2004–2007), and later updated in April 2019 under **General John E. Hyten** (CDRUSSTRATCOM, 2016–2019)—OPLAN 8010-12 replaced Cold War single-purpose nuclear war planning with a whole-of-government, flexible escalation-control architecture (FAS Analysis: US Nuclear War Plan Updated (OPLAN 8010-12); Bulletin of the Atomic Scientists: United States Nuclear Weapons 2026). Declassified FOIA segments obtained by **Hans M. Kristensen** (Federation of American Scientists) document that OPLAN 8010-12 embeds conventional kinetic strike and offensive cyber operations directly alongside tailored nuclear strike options (USSTRATCOM OPLAN 8010-08 and 8010-12 Declassified FOIA Release). During a crisis conference, the President selects an OPLAN 8010 response package from the Black Book inside the Nuclear Football, authenticates via Gold Codes, and directs the NMCC/USSTRATCOM Global Operations Center to transmit the corresponding EAM to Task Forces 214, 144, 134, and 204.

16. NC3 (Nuclear Command, Control, and Communications)

Technical Passport & Enterprise Architecture Matrix

Parameter	Verified Specification	Primary Source
System-of-Systems Scale	Integrates up to **160 individual space, airborne, terrestrial, and subsurface subsystems** across five doctrinal functions: 1. Detection, warning, and attack characterization; 2. Adaptive nuclear planning; 3. Decision-making conferencing; 4. Receiving Presidential orders; 5. Force management and direction	CRS Report: NC3 Modernization
Multi-Spectral RF & Network Stack	**VLF (14–30 kHz)**, **LF (30–300 kHz)**, **HF (2–30 MHz HFGCS)**, **UHF (225–450 MHz ALCS/AFSATCOM)**, **SHF (7–8 GHz WGS)**, and **EHF (44 GHz uplink / 20 GHz downlink MILSTAR/AEHF)**, coupled with SACCS, Global Aircrew Strategic Network Terminal (GASNT), and Family of Advanced Beyond Line-of-Sight Terminals (FAB-T)	CRS Defense Primer: NC3 IF11697

Parameter	Verified Specification	Primary Source
Enterprise Governance	**Commander, USSTRATCOM** designated as **NC3 Enterprise Lead** (October 2018), supported by the **NC3 Enterprise Center (NEC)** at Offutt AFB and Fort Meade, the **Under Secretary of Defense for Acquisition and Sustainment (USD(A&S))** as Capability Portfolio Manager, and the **DoD CIO**	CRS Defense Primer: NC3 IF11697

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Designated as a unified enterprise in October 2018 by Secretary of Defense James N. Mattis under the leadership of Commander, USSTRATCOM—successively **General John E. Hyten** (2016–2019), **Admiral Charles A. Richard** (2019–2022), **General Anthony J. Cotton** (2022–2025), and **Admiral Richard Correll** (2025–present)—the U.S. NC3 architecture binds all space/ground sensors, command centers, cryptographic systems, and nuclear delivery forces into a single deterministic loop (CRS Defense Primer: NC3 IF11697; FAS Report: On the Precipice 2026). Signals flow from Tier-0 warning sensors (SBIRS, DSP, BMEWS, PAVE PAWS/UEWR) → Tier-1 assessment and conferencing nodes (Cheyenne Mountain, NMCC, ANMCC, E-4B NAOC, USSTRATCOM C2F) → Presidential Nuclear Football & Gold Code authentication → Tier-2 survivable transport (MEECN, E-6B Looking Glass/TACAMO) → Tier-3/4 SAS/PAL verification and Triad execution.

17. Minimum Essential Emergency Communications Network (MEECN)

Technical Passport & Survivable Communications Matrix

Parameter	Verified Specification	Primary Source
VLF / LF Portion & Modes	**14 kHz to 60 kHz** operating CJCS interoperable **MEECN Modes 15, 9, 9 MMPM (MEECN Message Processing Mode), and High Data Rate (HIDAR, fielded FY96)** received via VMEbus/VXI **Submarine VLF/LF Receivers (SLVR)** and USAF Fixed Submarine Broadcast System (FSBS) / High Power Transmit Sets (HPTS)	FAS Submarine Force Communications Appendix E
Airborne VLF/LF & UHF Relay	E-4B NAOC and E-6B TACAMO/ABNCP 200 kW VLF transmitters (17–30 kHz) with 5-mile trailing wire antennas	FAS Submarine Force Communications Appendix A
Survivable EHF / UHF SATCOM	**MILSTAR I/II** (LDR 75–2,400 bps; MDR 4.8 kbps–1.544 Mbps) and **Advanced EHF (AEHF)** using **AN/USC-38(V)1** terminals at **44 GHz uplink / 20 GHz downlink** with frequency-hopping anti-jam LPI/LPD waveforms	FAS Submarine Force Communications Appendix A

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Promulgated by the Chairman of the Joint Chiefs of Staff under *Emergency Action Procedures of the CJCS, Volume VII*—with mode architecture directed in the 1990s by **Lieutenant General Henry Viccellio Jr., USAF** (Director of the Joint Staff, 1991–1993), Joint Staff MEECN technical officer **Lieutenant Colonel Harvey Le Cato, USAF**, and naval submarine communications engineers **Commander E. R. Jablonski (OPNAV N872E)**, **Lieutenant Commander N. P. Moe (N872E4)**, and **Commander D. L. Olberding (COMSUBLANT)**—MEECN constitutes the hardened, tri-service minimum-essential communications backbone engineered to survive nuclear electromagnetic pulse (HEMP) and ionospheric blackout (FAS Submarine Force Communications Appendix E). When the NMCC, ANMCC, NAOC, or Looking Glass originates an EAM, MEECN encodes the message into CJCS Modes 9, 9 MMPM, 15, or HIDAR and broadcasts it concurrently across fixed VLF/LF transmitters, E-6B/E-4B trailing wires, HFGCS, and cross-linked MILSTAR/AEHF satellites to ensure guaranteed reception by ICBM LCCs, submerged SSBNs, and strategic bombers (FAS Submarine Force Communications Appendix A).

18. DEFCON (Defense Readiness Condition Levels 5 to 1)

Technical Passport & Five-Tier Readiness State Matrix

DEFCON Level	JCS Exercise Term	Color Code	Operational Force Readiness & Technical Posture	Primary Source
DEFCON 5	**FADE OUT**	Blue	Normal peacetime military readiness; routine intelligence watch and baseline security	National Security Archive: JCS SM-833-59 Uniform Readiness Conditions
DEFCON 4	**DOUBLE TAKE**	Green	Above-normal readiness; heightened intelligence watch and strengthened COMSEC/physical security	National Security Archive: JCS SM-833-59
DEFCON 3	**ROUND HOUSE**	Yellow	Increased force readiness; USAF strategic aircraft poised to mobilize within 15 minutes; all tactical/strategic communications switch to classified call signs and encrypted modes	FAS DEFCON Defense Condition
DEFCON 2	**FAST PACE**	Red	Next step to nuclear war; armed forces ready to deploy and engage in under 6 hours; continuous airborne bomber/command-post alert and full missile crew generation	FAS DEFCON Defense Condition
DEFCON 1	**COCKED PISTOL**	White	Maximum readiness; nuclear war is imminent or has already begun; immediate weapon execution posture	National Security Archive: JCS SM-833-59

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Established by the Joint Chiefs of Staff in August–November 1959 under Chairman **General Nathan F. Twining, USAF** (CJCS, 1957–1960) and NORAD Commander-in-Chief **General Earle E. Partridge, USAF** (codified in JCS Staff Memorandum SM-833-59), the DEFCON system synchronizes force generation across all unified and specified combatant commands (National Security Archive: Alerts, Crises, and DEFCONs; National Security Archive: JCS SM-833-59). Major historical activations documented in declassified National Security Archive records include: **DEFCON 3** ordered in May 1960 by Secretary of Defense **Thomas S. Gates Jr.** during the Paris Summit U-2 crisis; **DEFCON 2** ordered for Strategic Air Command (SAC) by **General Thomas S. Power** (CINCSAC) on **24 October 1962** (while remaining U.S. forces stood at DEFCON 3 until 15–21 November 1962) during the Cuban Missile Crisis under President **John F. Kennedy**; **DEFCON 3** ordered globally on **25 October 1973** during the Yom Kippur War under President **Richard M. Nixon**, National Security Advisor **Henry A. Kissinger**, and Secretary of Defense **James R. Schlesinger**; and **DEFCON 3** ordered on **11 September 2001** by Secretary of Defense **Donald H. Rumsfeld** and CJCS **General Henry H. Shelton** (National Security Archive: Crises, Alerts, and DEFCONs 1961–1976 Part II; FAS DEFCON Defense Condition).

19. COCKED PISTOL (DEFCON 1 / Maximum Strategic Alert Codeword)

Technical Passport & Alert Codeword Matrix

Parameter	Verified Specification	Primary Source
Official JCS Designation	JCS exercise and alert codeword for **DEFCON 1 (White)** as codified in JCS SM-833-59 Appendix ("Uniform Readiness Conditions"); also utilized in historical SAC maximum readiness generation exercises alongside **FAST PACE** (DEFCON 2)	National Security Archive: JCS SM-833-59
Mandatory Force Actions	Immediate NCA conference with POTUS, SecDef, and CJCS; NCA relocation to E-4B NAOC / Site R; sealing of Cheyenne Mountain 25-ton blast doors; 100% bomber runway/airborne alert generation; SSBN surge deployment; ICBM LCC wartime isolation	National Security Archive: JCS SM-833-59

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Codified by the Joint Chiefs of Staff under General Nathan F. Twining in JCS SM-833-59 and exercised in USSTRATCOM strategic readiness exercises (*Global Thunder* and *Global Lightning*), the codeword **COCKED PISTOL** designates maximum wartime readiness in which strategic nuclear attack is imminent or underway (National Security Archive: JCS SM-833-59; FAS DEFCON Defense Condition). Under JCS SM-833-59 provisions, declaration of COCKED PISTOL immediately convenes the President, Secretary of Defense, and Joint Chiefs in an emergency decision conference, triggers NCA evacuation protocols, transitions all Minuteman III LCCs, Ohio-class SSBNs, and nuclear-loaded B-52H/B-2A bombers to immediate execution posture, and prepares the NMCC/NAOC for instantaneous EAM broadcast upon Presidential Gold Code authentication.

20. E-4B Nightwatch (Boeing E-4B NAOC)

Technical Passport & Aircraft Engineering Matrix

Parameter	Verified Specification	Primary Source
Airframe & Tail Numbers	Modified **Boeing 747-200B**; four operational aircraft in USAF inventory (Tail Numbers **73-1676, 73-1677, 74-0787, and 75-0125**); unit cost **$223.2 million** (FY98 constant dollars)	U.S. Air Force E-4B Fact Sheet
Dimensions & Weights	Length **231 ft 4 in (70.5 m)**; wingspan **195 ft 8 in (59.7 m)**; height **63 ft 5 in (19.3 m)**; maximum takeoff weight **800,000 lb (360,000 kg)**	U.S. Air Force E-4B Fact Sheet
Propulsion, Speed & Endurance	**4× General Electric CF6-50E2** turbofan engines (**52,500 lbf / 234 kN** thrust each); max speed **602 mph (Mach 0.86)**; ceiling **>30,000 ft (9,091 m)**; unrefueled range **7,130 statute miles (6,200 nmi)**; endurance **12 hours unrefueled / 72+ hours with in-flight refueling**; crew capacity up to **111 personnel**	AFGSC E-4B Fact Sheet
Internal Layout & C3 Payload	**6 main-deck functional compartments** (NCA work area, conference room, briefing room, battle staff work area, communications control center, rest area); lower-lobe **5-mile VLF trailing wire**; dorsal **SHF/EHF MILSTAR/AEHF** radome; dual 1,200 kVA electrical generators; copper-mesh HEMP-shielded windows	U.S. Air Force E-4B Fact Sheet

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Constructed by **The Boeing Company** with C3 mission payload integration by **E-Systems** (now Raytheon / L3Harris), **Burroughs Corporation**, and **Collins Radio** (first E-4A delivered December 1974; first E-4B delivered January 1980; all four converted to E-4B standard by January 1985), the E-4B Nightwatch is operated by the **1st Airborne Command and Control Squadron (1 ACCS)** under the **95th Wing** (activated 28 February 2025, succeeding the 595th Command and Control Group), Eighth Air Force (commanded in 2024 by **Major General Jason R. Armagost**), Air Force Global Strike Command, at Offutt AFB, Nebraska, and is scheduled for replacement in the 2030s by the **Sierra Nevada Corporation (SNC) E-4C Survivable Airborne Operations Center (SAOC)** (AFGSC E-4B Fact Sheet; USSTRATCOM: Mighty Eighth Commander Qualifies on E-4B Nightwatch). In the nuclear execution chain, the E-4B receives real-time SBIRS/DSP attack assessments via its dorsal AEHF dish, hosts the surviving NCA to authenticate OPLAN 8010-12 strike options via Gold Codes, and broadcasts EAMs over its 200 kW 5-mile VLF wire, HFGCS, and EHF SATCOM to the Strategic Triad.

21. E-6B Mercury (ABNCP / TACAMO ALCS)

Technical Passport & Dual-Role Strategic Aircraft Matrix

Parameter	Verified Specification	Primary Source
Airframe & Fleet Size	Modified **Boeing 707-320B** commercial airframe (final 707 derivatives built by Boeing); **16 operational aircraft** in U.S. Navy inventory; unit cost **$141.7 million**	NAVAIR E-6B Mercury Product Profile

Parameter	Verified Specification	Primary Source
Dimensions & Weights	Length **150 ft 4 in (45.82 m)**; wingspan **148 ft 4 in (45.21 m)**; height **42 ft 5 in (12.93 m)**; wing area **3,050 sq ft (283 m²)**; empty weight **172,795 lb (78,378 kg)**; max takeoff weight **342,000 lb (155,129 kg)**	NAVAIR E-6B Mercury Product Profile
Propulsion, Speed & Endurance	**4× CFM International CFM56-2A-2** high-bypass turbofans (**24,000 lbf / 107 kN** thrust each); ceiling **40,000 ft (12,192 m)**; speed **600 mph (Mach 0.86)**; range **6,600 nmi (12,223 km)**; endurance **15 hours unrefueled / 72 hours with aerial refueling**; crew **22**	NAVAIR E-6B Mercury Product Profile
Dual Strategic Mission Payloads	1. **AN/ART-54 200 kW VLF transmitter (17–30 kHz)** with 28,000-ft LTWA and 4,000-ft STWA for TACAMO SSBN communication; 2. **Airborne Launch Control System (ALCS)** UHF (225–400 MHz) transmitter for direct Minuteman III ICBM silo launch; 3. **MILSTAR/AEHF** dorsal/wingtip SATCOM & HFGCS	USSTRATCOM: Looking Glass Airborne Command Post

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Manufactured by **The Boeing Company** (Renton/Seattle, Washington) and sustained by **Naval Air Systems Command (NAVAIR PMA-271)**, the E-6A entered fleet service in August 1989 and was upgraded to the dual-role **E-6B Mercury** standard beginning in October 1998, when it formally relieved the USAF EC-135C of the USSTRATCOM Looking Glass Airborne Command Post mission (NAVAIR E-6B Mercury Product Profile; U.S. Navy Fleet Air Reconnaissance Squadron VQ-4). Operated by **Strategic Communications Wing ONE (Task Force 124 — VQ-3, VQ-4, VQ-7)** at Tinker AFB, Oklahoma, alongside USAF airborne missileers from the **625th Strategic Operations Squadron (625 STOS)**, the E-6B is the only platform in the U.S. inventory capable of simultaneously broadcasting EAMs to submerged Ohio-class SSBNs over its 5-mile VLF wire (17–30 kHz) AND transmitting direct UHF ALCS launch execution commands (225–400 MHz) into Minuteman III ICBM silos (USSTRATCOM: Looking Glass Airborne Command Post).

22. EC-135 Looking Glass (Post-Attack Command and Control System PACCS)

Technical Passport & Historical SAC Airborne C2 Matrix

Parameter	Verified Specification	Primary Source
Aircraft Variants & Propulsion	**Boeing EC-135A/C/G/H/J/L/P** (derived from the KC-135A/B Stratotanker / Boeing 367-80; powered on the EC-135C by **4× Pratt & Whitney TF33-P-9** turbofan engines); operational **3 February 1961 – 1 October 1998**	National Park Service: The Airborne Command Post System
PACCS Network Topology	Integrated the primary **Looking Glass ABNCP** (Offutt AFB), East and West **Auxiliary Command Posts (AUXCPs)** (Barksdale AFB & March/Ellsworth AFB), three **Airborne Launch Control Centers (ALCCs No. 1, 2, and 3)** (Minot AFB), and **EB-47L / EC-135L Radio Relay** aircraft	National Park Service: Airborne Launch Control Center - Operation Looking Glass
ALCS & VLF/UHF Suite	Equipped with **Airborne Launch Control System (ALCS Phase I–III, operational 31 May 1967)** UHF (225–400 MHz) direct ICBM launch transmitter, **AN/ARC-96 LF/VLF** trailing-wire transmitter, AFSATCOM UHF, and HF Single Sideband	National Park Service: Airborne Launch Control Center - Operation Looking Glass

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Conceived by **General Curtis E. LeMay** (CINCSAC 1948–1957, USAF Chief of Staff 1961–1965) and ordered into continuous 24/7 airborne alert on **3 February 1961** by **General Thomas S. Power** (CINCSAC, 1957–1964), the EC-135 Post-Attack Command and Control System (PACCS) fleet was operated by the **2d Airborne Command and Control Squadron (2 ACCS)** at Offutt AFB, Nebraska, the **4th ACCS** at Ellsworth AFB, South Dakota, and the **7th ACCS** (National Park Service: The Airborne Command Post System). For 29 years, 5 months, and 21 days (until **24 July 1990** under **General John T. Chain Jr.**), at least one EC-135C Looking Glass was airborne every second of every day. In the event that a Soviet first strike destroyed SAC's Underground Command Post at Offutt AFB and isolated underground Minuteman Launch Control Centers, the Airborne Emergency Action Officer

(AEAO) aboard the EC-135C assumed command of SAC, validated SAS authenticators under the Two-Man Rule, relayed EAMs via EC-135L Radio Relay planes and ALCCs, and used the onboard ALCS UHF transmitter to ignite Minuteman I, II, and III boosters directly inside their silos (National Park Service: Airborne Launch Control Center - Operation Looking Glass).

23. Raven Rock Mountain Complex (Site R)

Technical Passport & Subterranean Bunker Matrix

Parameter	Verified Specification	Primary Source
Geolocation & Portal Coordinates	**Raven Rock Mountain** (`39°44'03"N 77°25'11"W` summit; `39.7296°N 77.4325°W` west tunnel portal), Liberty Township, Adams County, near Blue Ridge Summit, Pennsylvania	FAS Raven Rock Site R AJCC
Excavation & Blast Resistance	Constructed **1951–1953** (operational May 1953) inside dense Precambrian greenstone granite; upgraded in 1963 to **140 psi** overpressure resistance; comprises **five parallel 1,000-foot-long rock-cut caverns** housing **five multi-story freestanding steel-framed buildings (Buildings A–E)** isolated from granite shockwaves	FAS Raven Rock Site R AJCC
Blast Doors, Power & Life Support	Protected by **two 34-ton steel blast doors**, six **1,000 kW diesel generators**, subterranean cooling and potable water reservoirs, and **38+ communications systems**	FAS Raven Rock Site R AJCC
Tenant Command Nodes	Alternate National Military Command Center (ANMCC, Joint Staff J-3), Alternate Joint Communications Center (AJCC), Army/Navy/Air Force Emergency Operations Centers, DISA, and U.S. Army 114th Signal Battalion under Washington Headquarters Services (OSD)	FAS Raven Rock Site R AJCC

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Approved by President **Harry S. Truman** in 1950 following the August 1949 Soviet nuclear test and engineered by **Parsons, Brinckerhoff, Hall & Macdonald** (New York) and the **U.S. Army Corps of Engineers**, the Raven Rock Mountain Complex (Site R, the "Underground Pentagon") became operational in May 1953 (FAS Raven Rock Site R AJCC). Connected via hardened fiber-optic, microwave, and satellite trunks to the Pentagon NMCC, Camp David, and Mount Weather, Site R continuously ingests SBIRS/DSP and UEWR warning feeds. Upon evacuation or destruction of the Pentagon, surviving Office of the Secretary of Defense and Joint Staff leaders inside Site R's five subterranean buildings assume National Military Command System authority, verify Presidential Gold Codes, and transmit authenticated EAMs over MEECN (FAS Raven Rock Site R AJCC).

24. Cheyenne Mountain Complex (NORAD/USSPACECOM)

Technical Passport & Super-Hardened Granite Cavern Matrix

Parameter	Verified Specification	Primary Source
Geolocation & Overburden	**Cheyenne Mountain Space Force Station** (`38°44'40"N 104°50'48"W`), Colorado Springs, Colorado; excavated beneath **2,000 feet (610 m) of solid Pikes Peak granite** on the eastern slope of the 9,565-ft Cheyenne Mountain massif	NORAD Official Fact Sheet: Cheyenne Mountain Complex
Excavation & Blast Rating	**5.1 acres (2.1 ha)** of subterranean chambers excavated by blasting **693,000 tons of granite**, reinforced with **115,000 rock bolts**; hardened against a **30-megaton nuclear detonation within 1.0 nautical mile (1.9 km)** (~**600 psi** portal overpressure)	NORAD Official Fact Sheet: Cheyenne Mountain Complex

Parameter	Verified Specification	Primary Source
Shock Isolation & Blast Doors	Houses **15 freestanding three-story structural steel buildings** clad in continuously welded **3/8-inch (9.5 mm) low-carbon steel plate** (Faraday HEMP shield), mounted atop **1,380 heavy-duty coiled steel springs** (~1,000 lb each, 12-inch vertical deflection); protected by **two sequential 25-ton (22.7 metric ton), 3.5-ft-thick hydraulically actuated steel blast doors** and fast-acting blast valves	FAS Cheyenne Mountain Complex
Power, Water & C2 Systems	Six **1,750 kW diesel generators**, 500,000-gallon fuel reserve, 1.8M gallons potable water + 5.3M gallons industrial cooling water; hosts the **Missile Warning Center (MWC)**, **CCPDS-R**, **Survivable Communications Integration System (SCIS)**, and Space Defense Operations Center	FAS Cheyenne Mountain Complex

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Groundbreaking occurred on **19 June 1961** under **General Laurence S. Kuter, USAF** (CINCNORAD, 1959–1962), engineered by the **U.S. Army Corps of Engineers (Omaha District)** and prime excavation contractor **Utah Construction & Mining Company**, achieving operational status as the NORAD Combat Operations Center on **20 April 1966** and full completion on **6 February 1967** at a cost of **$142.4 million** (NORAD Official Fact Sheet: Cheyenne Mountain Complex). Sealed during the 11 September 2001 attacks under NORAD Commander **General Ralph E. Eberhart**, designated as the NORAD and USNORTHCOM Alternate Command Center on **12 May 2008**, and upgraded under a 2015 $700M Raytheon contract initiated by **Admiral William E. Gortney** (CDRNORAD/USNORTHCOM, 2014–2016), the complex is managed by **U.S. Space Force Space Base Delta 1** (NORAD Official Fact Sheet: Cheyenne Mountain Complex; NORAD 9/11 Blast Door Commemoration). In the deterministic causal chain, SBIRS/DSP infrared satellites and BMEWS/PAVE PAWS/UEWR radars feed raw track vectors via SCIS into Cheyenne Mountain's Missile Warning Center and CCPDS-R mainframes; watch officers validate dual-phenomenology tracks within ~180 seconds, and the NORAD Command Director formally assesses whether North America is under attack, initiating a Missile Display / Threat Assessment Conference with the NMCC, USSTRATCOM, and the President (FAS Cheyenne Mountain Complex).

25. Mount Weather Emergency Operations Center (FEMA)

Technical Passport & Continuity of Government Facility Matrix

Parameter	Verified Specification	Primary Source
Geolocation & Designations	**High Point Special Facility (HPSF / "SF")** (`39°03'59"N 77°53'31"W`), 19844 Blueridge Mountain Road South (State Route 601), straddling Loudoun and Clarke Counties near Bluemont and Berryville, Virginia (48 miles west of Washington, D.C.)	FAS Mount Weather / High Point Special Facility
Surface ("Area A") & Underground ("Area B")	**Area A**: **434+ acres (176 ha)** of secure surface command, telecommunications, and support buildings; **Area B** (completed **1959**): **600,000-square-foot (56,000 m²)** subterranean bunker excavated into Blue Ridge metabasalt/greenstone bedrock, protected by a **34-ton steel blast door** (**>100–200 psi** overpressure resistance), underground freshwater reservoirs, diesel power plant, hospital, and Cabinet/Supreme Court Continuity of Government (COG) quarters	FAS Mount Weather / High Point Special Facility
Communications Master Hub	Master control station for the **FEMA National Radio System (FNARS)** (HF/VHF/UHF and satellite network linking federal	FAS Mount Weather / High Point Special Facility

Parameter	Verified Specification	Primary Source
	agencies, DoD, and all 50 State EOCs), **National Warning System (NAWAS)**, and **Emergency Alert System (EAS)** Primary Entry Point	

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Excavated between **1954 and 1959** by the **U.S. Army Corps of Engineers** under President **Dwight D. Eisenhower** and the **Office of Civil and Defense Mobilization (OCDM)** (later the Office of Emergency Preparedness [OEP] and Federal Preparedness Agency [FPA]) and transferred to the **Federal Emergency Management Agency (FEMA)** upon FEMA's creation by President **Jimmy Carter** under **Executive Order 12148 (20 July 1979)**, Mount Weather is operated by FEMA's **Office of National Continuity Programs (ONCP)** within the Department of Homeland Security (FAS Mount Weather / High Point Special Facility). Upon early warning detection and DEFCON escalation, designated constitutional Presidential successors (Cabinet members and senior executive branch leaders) evacuate via helicopter to Mount Weather Area B ("Enduring Constitutional Government" protocol). Linked via WHCA and DISA fiber/satellite trunks to the NMCC, ANMCC (Site R), and E-4B NAOC, any surviving constitutional successor at Mount Weather who assumes the Presidency authenticates via the backup Nuclear Football and Gold Codes, directs USSTRATCOM to transmit EAMs over MEECN, and addresses the nation via FNARS/EAS (FAS Mount Weather / High Point Special Facility).

26. USSTRATCOM (United States Strategic Command, Offutt AFB)

Technical Passport & Combatant Command Matrix

Parameter	Verified Specification	Primary Source
Headquarters Geolocation & C2F	**Offutt Air Force Base** (`41°07'06"N 95°54'45"W`, 900 SAC Boulevard, Bellevue, Sarpy County, Nebraska); operates out of the **General Curtis E. LeMay Building (Command and Control Facility [C2F])**, a **$1.3 billion, 1,000,000-square-foot (93,000 m²)** HEMP-shielded, cyber-hardened facility (dedicated **18 November 2019**, replacing 1956 Building 500)	USSTRATCOM: C2F Dedicated to Gen. Curtis LeMay
Operational Task Forces	**TF 204** (B-52H/B-2A Strategic Bombers, 8th AF); **TF 214** (Minuteman III ICBMs, 20th AF); **TF 144** (Atlantic Ohio-class SSBNs, COMSUBLANT); **TF 134** (Pacific Ohio-class SSBNs, COMSUBPAC); **TF 124** (E-6B Looking Glass/TACAMO, STRATCOMWING 1)	USSTRATCOM 2023 Congressional Posture Statement
Core Sub-Centers	**Global Operations Center (GOC)**, **Alternate Missile Warning Center (AMWC)**, **NC3 Enterprise Center (NEC)**, and **Joint Electromagnetic Spectrum Operations Center (JEMSOC)**	USSTRATCOM 2023 Congressional Posture Statement

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Established on **1 June 1992** by President **George H.W. Bush** and Secretary of Defense **Dick Cheney**—unifying the strategic nuclear forces and targeting staff of the disestablished Strategic Air Command (headquartered at Offutt AFB since 9 November 1948 under **General Curtis E. LeMay**) and the Joint Strategic Target Planning Staff (JSTPS)—USSTRATCOM exercises combatant command (OPCOM) over the U.S. Nuclear Triad and serves as the DoD NC3 Enterprise Lead (USSTRATCOM: C2F Dedicated to Gen. Curtis LeMay). Successive Commanders of USSTRATCOM include **General George Lee Butler** (1992–1994), **Admiral Henry G. Chiles Jr.** (1994–1996), **General Eugene E. Habiger** (1996–1998), **Admiral Richard W. Mies** (1998–2001), **Admiral James O. Ellis Jr.** (2001–2004), **General James E. Cartwright** (2004–2007), **General Kevin P. Chilton** (2007–2011), **General C. Robert Kehler** (2011–2013), **Admiral Cecil D. Haney** (2013–2016), **General John E. Hyten** (2016–2019), **Admiral Charles A. Richard** (2019–2022), **General Anthony J. Cotton** (2022–2025), and **Admiral Richard Correll** (2025–present) (AFGSC USSTRATCOM Tour 2026). In the nuclear execution chain, the USSTRATCOM Global Operations Center inside the LeMay C2F correlates SBIRS/UEWR threat tracks, advises the President during the NMCC Decision Conference on OPLAN 8010-12 strike options, and—upon Presidential Gold Code authorization—transmits EAMs via SACCS and MEECN to Task Forces 214, 144, 134, 204, and 124.

27. Defense Support Program (DSP)

Technical Passport & Geosynchronous IR Satellite Matrix

Parameter	Verified Specification	Primary Source
Program Designation & Orbit	USAF Program 647 (**23 operational satellites** launched from **6 November 1970 [DSP-1] to 11 November 2007 [DSP-23]** across Blocks 1–5); deployed at **22,300 statute miles (35,900 km) Geosynchronous Earth Orbit (GEO)**	U.S. Space Force SMC: Defense Support Program 45th Anniversary
Spacecraft Mass, Dimensions & Power	**0 kg)**; height **32.8 ft (10.0 m)**; diameter **22.0 ft (6.7 m)**; solar a	U.S. Space Force SMC: Defense Support Program 45th Anniversary
Optical / Infrared Sensor Physics	**3.63-meter Schmidt telescope** canted **7.5° off the spacecraft spin axis**; entire spacecraft spins at **6 rpm (1 scan every 10 seconds)** so its **6,000-element lead sulfide (PbS, ~2.7 µm SWIR) and mercury cadmium telluride (HgCdTe, ~4.3–4.5 µm MWIR)** linear focal-plane array sweeps the Earth's disk, supplemented by **Nuclear Detonation (NUDET) Detection System (NDS)** optical/X-ray/EMP sensors	U.S. Space Force SMC: Defense Support Program 45th Anniversary
Prime Contractors & Operators	Spacecraft bus by **TRW Space & Electronics** (Redondo Beach, CA); IR sensor payload by **Aerojet Electronic Systems** (Azusa, CA) (both now Northrop Grumman); operated by **2nd Space Warning Squadron (2 SWS), Space Delta 4**, Buckley Space Force Base, Aurora, Colorado (`39°42'16"N 104°45'29"W`)	U.S. Space Force SMC: Defense Support Program 45th Anniversary

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Engineered by **TRW** and **Aerojet Electronic Systems** under the Space and Missile Systems Center (SMC, now Space Systems Command) Remote Sensing Systems Directorate (directed in 2015 by **Colonel Michael A. Guetlein**, now General and Vice Chief of Space Operations), the Defense Support Program (DSP) replaced the 1960s Missile Defense Alarm System (MIDAS) and served for five decades as America's primary boost-phase infrared sentinel (U.S. Space Force SMC: Defense Support Program 45th Anniversary). Within 40 to 60 seconds of an ICBM or SLBM breaking cloud tops, DSP's 6 rpm spinning Schmidt telescope detects the 2.7 µm / 4.3 µm hydrocarbon/solid-propellant exhaust plume and downlinks telemetry to the Buckley SFB Mission Control Station (and historically Joint Defense Facility Nurrungar, Australia, and Kapaun AS, Germany). Processed launch vectors are forwarded via CCPDS-R to Cheyenne Mountain, USSTRATCOM, and the NMCC, where dual-phenomenology correlation with ground phased-array radars triggers the NMCC Missile Attack Conference and Presidential Gold Code authentication.

28. Space-Based Infrared System (SBIRS)

Technical Passport & Multi-Orbit IR Surveillance Matrix

Parameter	Verified Specification	Primary Source
Constellation Architecture	**6 dedicated Geosynchronous Earth Orbit satellites (SBIRS GEO-1 through GEO-6**, launched **7 May 2011 to 4 August 2022** on Atlas V boosters at ~35,786 km altitude) plus **4 Highly Elliptical Orbit (SBIRS HEO-1 through HEO-4)** hosted infrared payloads in Molniya orbits (`apogee ~39,000 km, perigee ~1,100 km, 63.4° inclination`) for North Polar coverage	U.S. Space Force Official Fact Sheet: Space-Based Infrared System

Parameter	Verified Specification	Primary Source
Spacecraft Bus & Mass	Lockheed Martin **A2100M** (GEO-1–4) and **LM 2100 Combat Bus** (GEO-5–6); dry mass ~**5,600 lb (2,540 kg)**; solar power **2,800 W**	Space Systems Command: SBIRS GEO-6 Launch
Dual Schmidt Telescopes & IR Bands	Each GEO spacecraft carries **two independent Schmidt telescopes** with high-density **mercury cadmium telluride (HgCdTe)** focal-plane arrays operating across **three IR bands (SWIR ~2.7 µm, MWIR ~4.3–4.8 µm, and See-to-the-Ground ~3.0–3.5 µm)**: 1. **Scanning Sensor** (wide-area continuous sweep); 2. **Staring Sensor** (high-revisit pinpoint tracking of dim post-boost vehicles and hypersonic/tactical threats)	Buckley Space Force Base: SBIRS Fact Sheet
Prime Contractors & Operators	Prime integrator **Lockheed Martin Space** (Sunnyvale, CA / Littleton, CO) with infrared payload subcontractor **Northrop Grumman Electronic Systems** (Azusa, CA); operated by **2nd Space Warning Squadron (2 SWS)** and **11th Space Warning Squadron (11 SWS)**, **Space Delta 4**, Buckley SFB, Colorado	U.S. Space Force Official Fact Sheet: Space-Based Infrared System

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Acquired by **Space Systems Command (SSC)** at Los Angeles SFB and operated by **Space Delta 4** at the Buckley SFB Mission Control Station (39°42'16"N 104°45'29"W), SBIRS executes four core missions: **missile warning, missile defense, technical intelligence (TECHINT), and battlespace awareness** (U.S. Space Force Official Fact Sheet: Space-Based Infrared System). Unlike spinning DSP satellites, SBIRS is three-axis stabilized and pairs a rapid wide-area Scanning Sensor with a step-stare Staring Sensor whose See-to-the-Ground band detects booster ignition directly on the launch pad before cloud break (Buckley Space Force Base: SBIRS Fact Sheet; MDAA SBIRS Technical Profile). Raw infrared telemetry is downlinked to Buckley SFB, where ground computers extract trajectory state vectors, cue AN/FPS-132 UEWR radars, and transmit high-confidence alerts via CCPDS-R to Cheyenne Mountain, USSTRATCOM, and the NMCC, initiating the NMCC Missile Attack Conference, Presidential Gold Code authentication, and EAM transmission.

29. PAVE PAWS (AN/FPS-115 / AN/FPS-132 UEWR)

Technical Passport & UHF Solid-State Phased-Array Radar Matrix

Parameter	Verified Specification	Primary Source
Radar Designations & Contractor	**AN/FPS-115 PAVE PAWS** (PAVE Phased Array Warning System) $	
ightarrow$ upgraded to **AN/FPS-120 / AN/FPS-123 / AN/FPS-126 SSPARS** $		
ightarrow$ **AN/FPS-132 Upgraded Early Warning Radar (UEWR)**; engineered and built by **Raytheon Company** (Wayland/Sudbury/Tewksbury, Massachusetts)	FAS Cheyenne Mountain & Early Warning Network	
Pyramidal Array Geometry & Elements	**105-ft (32 m) high pyramidal Scanner Building** with **two planar circular array faces** tilted back **20° from vertical** (3 faces at RAF Fylingdales); each face is **102 ft (31.1 m) in diameter** with a **72.5-ft (22.1 m) active aperture** housing **1,792 active solid-state bipolar/GaAs T/R dipole modules** + 885 dummy edge elements per face (expandable to 5,354 elements)	National Security Archive: PAVE PAWS SLBM Warning System
RF Frequency, Power, Azimuth & Range	**UHF band at 420–450 MHz**; **580 kW peak power (324 W per T/R module) / ~145 kW average power per face**; **240° azimuth coverage** (120° per face; **360°** on	FRS Observatoire Défense Antimissile Bulletin No. 14

Parameter	Verified Specification	Primary Source
	3-face Fylingdales array) and **3°–85° elevation** across a 54 ms duty cycle; **3,000 nautical mile (5,556 km)** range against a 10 m² RCS target	
Active & Historical Sites	1. **Cape Cod SFS, MA** (41°45'07"N 70°32'17"W, 6th SWS, IOC 4 Apr 1981); 2. **Beale AFB, CA** (39°08'10"N 121°21'03"W, 7th SWS, IOC 15 Aug 1981); 3. **Clear SFS, AK** (64°18'04"N 149°11'23"W, 13th SWS, relocated from Eldorado AFS, TX [30°58'45"N 100°33'16"W] in 2001); 4. **Pituffik Space Base (Thule), Greenland** (76°34'13"N 68°19'05"W, 12th SWS); 5. **RAF Fylingdales, UK** (54°21'42"N 00°40'11"W); (Robins AFB, GA [32°34'52"N 83°34'09"W] operated 1986–1995)	FAS Cheyenne Mountain & Early Warning Network

Engineering Mechanics, Named Officials & Deterministic Causal Chain

Contracted to **Raytheon Company** by the USAF Electronic Systems Division (ESD, Hanscom AFB) in April 1976 to detect Submarine-Launched Ballistic Missiles (SLBMs) launched from the Atlantic and Pacific Oceans and subsequently upgraded across all five northern-hemisphere early warning sites to the **AN/FPS-132 Upgraded Early Warning Radar (UEWR)** standard under **U.S. Space Force Space Delta 4**, PAVE PAWS / UEWR forms the terrestrial phased-array backbone of U.S. strategic warning and Ground-based Midcourse Defense (GMD) cueing (National Security Archive: PAVE PAWS SLBM Warning System; FAS Cheyenne Mountain & Early Warning Network). When an SLBM or ICBM breaks the radar horizon into the 3°–10° elevation electronic surveillance fence, the Raytheon solid-state beam-steering processors switch within milliseconds to dedicated tracking mode, computing launch coordinates, ballistic covariance state vectors, and predicted impact points, and transmitting the data via Jam Resistant Secure Communications (JRSC) and SCIS circuits to Cheyenne Mountain, USSTRATCOM, the Pentagon NMCC, and Fort Greely/Vandenberg GMD fire-control nodes (FAS Cheyenne Mountain & Early Warning Network).

Section II: United States Strategic Triad Delivery Systems, Hardened Silos, Strategic Bombers, Thermonuclear Warheads, and Reentry Physics (Entities 30–48)

30. ICBM (Intercontinental Ballistic Missile Architecture)

Technical Passport & Flight Physics Matrix

Parameter	Verified Specification	Primary Source
Range Classification Threshold	Minimum range exceeding 5,500 km (3,418 statute miles / 2,970 nautical miles); operational US systems span 9,600 km to 15,000 km	FAS ICBM Technical Introduction
Trajectory Phases	1. Boost Phase (180–300 seconds, solid or storable liquid rocket motors); 2. Post-Boost / Bus Deployment Phase (300–600 seconds via Post-Boost Vehicle / Propulsion System Rocket Engine); 3. Midcourse Exo-Atmospheric Ballistic Phase (apogee 805–1,120 km, duration 1,200–1,500 seconds); 4. Terminal Atmospheric Reentry Phase (duration 30–60 seconds below 100 km altitude)	FAS ICBM Technical Introduction
Burnout Velocity & Mach Regime	6.7 km/s to 7.9 km/s (15,000 mph to 18,000 mph; Mach 23 to Mach 25 at final stage burnout)	USAF LGM-30G Minuteman III Fact Sheet
Throw-Weight Envelope	500 kg (small ICBM baseline) to 1,150 kg (LGM-30G Minuteman III), 3,950 kg (LGM-118A Peacekeeper), and 3,720 kg (LGM-25C Titan II)	Nuclear Weapon Archive MX Peacekeeper
Circular Error Probable (CEP)	900 m (LGM-25C Titan II inertial) to 120–200 m (LGM-30G Minuteman III NS-50 AIRS) and 90–100 m (LGM-118A Peacekeeper AIRS)	CSIS LGM-118 Peacekeeper

Parameter	Verified Specification	Primary Source
Active US Deployed Force (2026)	400 deployed silo-based ICBMs across 450 operational and warm silos, carrying 400 deployed warheads (with 800 total warheads assigned including upload reserve)	FAS Nuclear Notebook: United States Nuclear Weapons 2025

Engineering Mechanics, Propulsion & Guidance Architecture

An Intercontinental Ballistic Missile (ICBM) operates as a multi-stage rocket-propelled strategic weapon system engineered to deliver thermonuclear reentry vehicles across intercontinental distances along a sub-orbital elliptical ballistic trajectory governed by Keplerian orbital mechanics and terrestrial gravimetric perturbations (FAS ICBM Technical Introduction). US ICBM engineering evolved across three distinct propulsion and guidance generations:

1. **Cryogenic and Storable Liquid-Propellant Boosters (1959–1987):** Early architectures such as SM-65 Atlas and HGM-25A Titan I utilized cryogenic liquid oxygen (LOX) and RP-1 kerosene, requiring surface erection or fuel loading prior to launch. The second-generation LGM-25C Titan II introduced hypergolic storable liquid propellants—Aerozine-50 (a 50:50 mixture of hydrazine and unsymmetrical dimethylhydrazine) oxidized by nitrogen tetroxide (N_2O_4)—enabling instantaneous ignition inside underground silos within 60 seconds (CSIS Titan II).
2. **Solid-Propellant Three-Stage Architectures (1962–Present):** Initiated under US Air Force Western Development Division (later Ballistic Missile Office) leadership of General Bernard A. Schriever and propulsion breakthroughs in metallized polybutadiene acrylic acid acrylonitrile (PBAN) and hydroxyl-terminated polybutadiene (HTPB) composite solid propellants by Thiokol, Aerojet, and Hercules (FAS LGM-30A/B Minuteman I). Solid-propellant motors eliminate corrosive liquid fuel handling and allow continuous 24/7/365 alert readiness for decades inside unmanned silos (National Museum of the USAF Minuteman III).
3. **Post-Boost Vehicle (PBV) & Inertial Guidance Physics:** Upon third-stage solid motor burnout and thrust termination via blow-out ports or nozzle thrust reversal, a liquid-fueled or solid-thruster Post-Boost Vehicle (PBV) corrects accumulated boost-phase velocity vectors, aligns attitude against high-resolution gravity-anomaly models of the Earth's geoid, and dispenses reentry vehicles and penetration aids (chaff, decoys) onto distinct ballistic trajectories (FAS ICBM Technical Introduction). Guidance relies on radiation-hardened, closed-loop all-inertial navigation units—evolving from the Autonetics D-17B and D-37C computers to the NS-20, NS-50, and Charles Stark Draper Laboratory Advanced Inertial Reference Sphere (AIRS) fluid-suspended beryllium sphere gyroscopes—measuring specific force via pendulous integrating gyroscopic accelerometers operating autonomously via internal closed-loop inertial sensing (FAS LGM-30G Minuteman III).

Hierarchical Command-and-Control (NC3) & Key Personnel

- **National Command Authority:** President of the United States → Secretary of Defense → Chairman of the Joint Chiefs of Staff via the National Military Command Center (NMCC) and E-4B National Airborne Operations Center (NAOC) (AFDP 3-72 Nuclear Operations).
- **Combatant Command (OPCON):** Commander, U.S. Strategic Command (CDRUSSTRATCOM), Admiral Richard Correll, USN (headquartered at Offutt AFB, Nebraska; relieved General Anthony J. Cotton) (AFGSC USSTRATCOM Tour 2026).
- **Service Component Command (COMAFSTRAT / ADCON & OPCON):** Commander, Air Force Global Strike Command (AFGSC), General Stephen L. Davis, USAF, with Deputy Commander Lieutenant General Jason R. Armagost, USAF (headquartered at Barksdale AFB, Louisiana) (AFGSC Biographies).
- **Numbered Air Force (ICBM Operational Execution):** Commander, Twentieth Air Force (20 AF / Task Force 214), Major General Colin J. Connor, USAF (relieved Major General Stacy Jo Huser on July 10, 2026), with Deputy Commander Colonel Dustin D. Harmon and Command Chief Master Sergeant Jeffery D. Sipos at F.E. Warren AFB, Wyoming (AFGSC 20th Air Force Change of Command).
- **Acquisition & Systems Engineering:** Air Force Nuclear Weapons Center (AFNWC), Kirtland AFB, New Mexico, and Direct Reporting Program Manager (DRPM) for ICBM Modernization (AF.mil Sentinel Restructure 2026).

Causal Chronology & Operational Force Structure

In February 1958, the Department of Defense authorized the Air Force to develop a solid-fueled ICBM force to replace vulnerable first-generation Atlas and Titan I complexes (FAS LGM-30A/B Minuteman I). Following the cancellation of the rail-mobile Minuteman concept ("Operation Big Star") by Secretary of Defense Robert S. McNamara on December 7, 1961, Strategic Air Command (SAC) constructed 1,000 dispersed Minuteman silos alongside 54 Titan II silos (FAS LGM-30A/B Minuteman I). Following the retirement of Titan II in 1987, the deactivation of Minuteman II under START I in the 1990s, the retirement of the 50 LGM-118A Peacekeeper missiles in September 2005, and the New START force-structure reductions completed between 2014 and 2018 (which downloaded all deployed Minuteman III missiles to a single warhead and reduced active deployed launchers from 450 to 400 across the 90th, 91st, and 341st Missile Wings), the US land-based ICBM architecture functions as a 400-silo responsive prompt-strike sponge requiring an adversary to expend at least 800 to 900 high-accuracy warheads to attempt a disarming counterforce strike (FAS Nuclear Notebook: United States Nuclear Weapons 2025).

31. Silo (Hardened Launch Facilities, Launch Control Centers, psi Overpressure Ratings)

Technical Passport & Structural Hardening Matrix

Parameter	Verified Specification	Primary Source
Launch Facility (LF) Silo Dimensions	80 ft (24.38 m) depth; 12 ft (3.66 m) lower launch tube diameter;	National Park Service Delta-09 Missile Silo

Parameter	Verified Specification	Primary Source
	25 ft (7.62 m) upper equipment room collar diameter; steel-plate liner encased in reinforced poured concrete	
Silo Closure Door (Blast Door)	90-ton to 110-ton reinforced steel and concrete horizontal sliding closure door; actuated prior to launch by high-pressure ballistic gas generators to clear the shaft in seconds (Titan II utilized a 750-ton silo closure door)	CSIS Titan II
Launch Control Center (LCC) Capsule	Buried 31 ft to 100 ft (9.4 m to 30.5 m) underground depending on wing geology; cylindrical steel-walled capsule (29 ft diameter, 54 ft length) with 4-foot-thick steel-reinforced concrete shell and 8-ton manually operated steel blast door	National Park Service Delta-01 Launch Control Facility
Overpressure Hardening (psi)	Original Minuteman I/II silos rated at 300 psi; upgraded Minuteman III silos and WS-133A-M/B Launch Facilities hardened to 1,000–2,000 psi static and dynamic blast overpressure; LCC capsules rated at 1,000 psi overpressure	FAS ICBM Overpressure & Hardening
Shock Isolation Suspension	Missile suspended inside silo on a pendulum cable/spring shock-isolation carriage system; LCC command rack floor suspended inside the outer capsule on four massive pneumatic/hydraulic shock actuators	FAS LGM-30G Minuteman III
Dispersion & Squadron Topology	3 to 7 nautical miles (5.5 to 13 km) minimum separation between silos (preventing fratricide/two-for-one destruction by a single warhead); 10 Launch Facilities (1 Flight) controlled by 1 LCC; 5 Flights (50 LFs + 5 LCCs) per	FAS LGM-30A/B Minuteman I

Parameter	Verified Specification	Primary Source
	Missile Squadron; 3 Squadrons (150 LFs + 15 LCCs) per Missile Wing	

Engineering Mechanics, Shock Attenuation & NC3 Redundancy

The US ICBM ground infrastructure comprises unmanned Launch Facilities (LFs) and manned underground Launch Control Centers (LCCs) situated below topside Launch Control Facilities (LCFs) (National Park Service Delta-01 Launch Control Facility). Each Launch Facility consists of an 80-foot-deep steel-lined reinforced concrete cylinder housing the missile in an environmentally controlled shaft maintained at 60°F to 65°F (15.5°C to 18.3°C) by a two-level annular upper equipment room containing primary power rectifiers, backup diesel generators, brine chillers, and guidance alignment autocollimators (National Park Service Delta-09 Missile Silo). Topside security incorporates pulsed Doppler motion-detection radars (historically the balanced-pressure and electromagnetic Intrusion Detection System), A-plug and B-plug personnel access vaults requiring remote cryptographic authentication from the LCC, and an independent break-away perimeter fence engineered to collapse outward when the ballistic gas actuator fires the 90-to-110-ton horizontal launcher closure door off the silo mouth (National Park Service Delta-09 Missile Silo).

Each Launch Control Center capsule is a hermetically sealed, electromagnetic pulse (EMP) Faraday-shielded steel-and-concrete pressure vessel accessed via an elevator shaft and sealed behind an 8-ton steel blast door (National Park Service Delta-01 Launch Control Facility). To survive ground shock waves up to several tens of G-forces generated by a nearby megaton-class surface burst, the interior command deck—housing the two Rapid Execution and Combat Targeting (REACT) consoles, survival rations, chemical-biological-radiological (CBR) air filtration, and an escape tube filled with sand leading toward the surface—hangs on four heavy pneumatic/mechanical shock isolators with clearance to swing inside the outer shell (FAS LGM-30G Minuteman III). Topside LCF support buildings house eight enlisted personnel (a Facility Manager NCO, a chef, and six Security Forces defenders) alongside auxiliary diesel generators (National Park Service Delta-01 Launch Control Facility).

Hierarchical Command-and-Control (NC3) & Key Personnel

- **Two-Person Launch Execution Protocol:** Inside each LCC, two USAF Missile Combat Crew officers (Commander and Deputy Commander) remain on 24-hour alert (AFNWC Minuteman III). Upon receipt of an encrypted Emergency Action Message (EAM) via the Strategic Automated Command and Control System (SACCS), survival low-frequency/very-low-frequency (LF/VLF) receivers, or ultra-high-frequency (UHF) satellite terminals, both officers independently open a dual-lock safe, validate the Sealed Authenticator System (SAS) codes against the EAM, enter the target plan and unlock codes, and turn two launch keys separated by 12 feet within two seconds of each other.
- **Squadron Consensus Vote & ALCS Override:** Launch execution mandates a dual-LCC "two-vote" consensus launch command from two separate LCCs within the 5-LCC, 50-missile squadron interconnected by pressurized, hardened underground HICS (Hardened Intersite Cable System) cables (FAS LGM-30A/B Minuteman I). If all five LCCs are isolated or destroyed, an airborne launch control officer aboard a US Navy E-6B Mercury TACAMO/ABNCP aircraft equipped with the Airborne Launch Control System (ALCS) can transmit the enabling and execution commands directly to the Launch Facility's UHF survivals antenna (AFDP 3-72 Nuclear Operations).
- **Active Operational Wings (20th Air Force):**
  - **90th Missile Wing** (F.E. Warren AFB, Wyoming; 319th, 320th, and 321st Missile Squadrons across Wyoming, Nebraska, and Colorado) (FAS Nuclear Notebook: United States Nuclear Weapons 2025).
  - **91st Missile Wing** (Minot AFB, North Dakota; 740th, 741st, and 742nd Missile Squadrons) (FAS Nuclear Notebook: United States Nuclear Weapons 2025).
  - **341st Missile Wing** (Malmstrom AFB, Montana; 10th, 12th, and 490th Missile Squadrons) (FAS Nuclear Notebook: United States Nuclear Weapons 2025).

Causal Chronology & Sentinel Silo Recapitalization

Constructed by the U.S. Army Corps of Engineers Ballistic Missile Construction Office (CEBMCO) and prime contractor Boeing Aerospace Company between 1961 and 1967, the original 1,000 Minuteman silos underwent the Minuteman Force Modernization and Silo Upgrade (FMSU) programs from 1966 to 1980, increasing blast overpressure resistance from 300 psi to 2,000 psi (FAS LGM-30A/B Minuteman I). Following the 1991 START I Treaty, 150 Minuteman II silos at Ellsworth AFB, South Dakota (preserved at Delta-01 and Delta-09 as the Minuteman Missile National Historic Site), 150 silos at Whiteman AFB, Missouri, and 150 Minuteman III silos at Grand Forks AFB, North Dakota were imploded or deactivated, followed by 50 silos at Malmstrom AFB (FAS LGM-30F Minuteman II). In 2025–2026, following the LGM-35A Sentinel Nunn-McCurdy review and restructure led by Direct Reporting Program Manager (DRPM) White, the Department of the Air Force determined it would construct 400 entirely new, modern Launch Facilities, new Wing Command Centers, and thousands of miles of utility corridors rather than refurbishing the 1970s-era Minuteman III concrete structures, completing a full-scale launch silo prototype in March 2026 (AF.mil Sentinel Silo Prototype 2026).

32. Minuteman (LGM-30A/B/F/G Minuteman I, II, III)

Technical Passport & Variant Evolution Matrix

Variant	Length	Diameter	Launch Weight	Propulsion Stages & Thrust	Range & Speed	Payload / Warhead / RV	CEP	Service Dates	Primary Source
LGM-30A Minuteman IA	53 ft 8 in (16.36 m)	5.5 ft (1.67 m)	65,000 lb (29,483 kg)	3-stage solid (Thiokol M55 / Aerojet SR19 / Hercules M57)	5,500 nm (10,186 km); Mach 23	1 × W59 (1.0 Mt) or W56 (1.2 Mt) in Avco Mk 5 or Mk 11 RV	2,400 m	1962–1969 (150 deployed at Malmstrom AFB)	FAS LGM-30A/B Minuteman I

Variant	Length	Diameter	Launch Weight	Propulsion Stages & Thrust	Range & Speed	Payload / Warhead / RV	CEP	Service Dates	Primary Source
LGM-30B Minuteman IB	55 ft 11 in (17.04 m)	5.5 ft (1.67 m)	65,000 lb (29,483 kg)	3-stage solid (Thiokol M55 / Aerojet SR19 / Hercules M57A1 with titanium case)	5,500 nm (10,186 km); Mach 23	1 × W56 (1.2 Mt) in Avco Mk 11A RV	1,850 m	1963–1972 (650 deployed across 4 wings)	FAS LGM-30A/B Minuteman I
LGM-30F Minuteman II	57 ft 7 in (17.55 m)	5.5 ft (1.67 m)	73,000 lb (33,112 kg)	3-stage solid (Thiokol M55A1 / Aerojet SR19-AJ-1 with liquid-injection TVC / Hercules M57A1)	6,300 nm (11,664 km); 15,000 mph	1 × W56 Mod 4 (1.2 Mt) in Avco Mk 11C RV with Tracor Mk 1 penetration aids	400–600 m	1965–1995 (450–500 deployed)	FAS LGM-30F Minuteman II
LGM-30G Minuteman III	59.9 ft (18.26 m)	5.5 ft (1.67 m)	79,432 lb (36,030 kg)	Stage 1: ATK refurbished M55A1 (203,158 lbf thrust); Stage 2: ATK refurbished SR19-AJ-1 (60,793 lbf thrust); Stage 3: ATK refurbished SR73-AJ-1 (35,086 lbf thrust); PBV: Aerojet/Rocketdyne RS-14 liquid bipropellant PSRE	5,218+ nm (6,000+ statute miles / 13,000 km max); 15,000 mph (Mach 23 / 24,000 kph); 1,120 km apogee	Throw-weight ~1,150 kg; up to 3 MIRVs (deployed with 1 RV under New START): either 1 × W78 (335 kt) in GE Mk 12A RV or 1 × W87-0 (300 kt) in Avco Mk 21 RV	120–200 m (NS-50)	June 1970–Present (400 active deployed; 550 peak)	AFGSC LGM-30G Minuteman III Fact Sheet

Engineering Mechanics, Propulsion & Guidance Upgrades

Manufactured by prime contractor The Boeing Company, the LGM-30G Minuteman III was the first operational ICBM in the world equipped with Multiple Independently Targetable Reentry Vehicles (MIRVs) (CSIS Minuteman III). Its propulsion stack comprises three solid-propellant rocket stages and a liquid-fueled fourth stage (Post-Boost Vehicle):

- **First Stage (Thiokol/ATK M55A1):** D6AC high-strength steel motor case loaded with PBAN aluminized composite propellant, generating 203,158 lbf (903.7 kN) of thrust through four movable gimbaled nozzles for pitch, yaw, and roll control (AFGSC LGM-30G Minuteman III Fact Sheet).
- **Second Stage (Aerojet/ATK SR19-AJ-1):** Titanium-alloy case producing 60,793 lbf (270.4 kN) of thrust with a single submerged nozzle using Freon-114B2 liquid-injection thrust vector control (LITVC) (AFGSC LGM-30G Minuteman III Fact Sheet).
- **Third Stage (Aerojet/Hercules/ATK SR73-AJ-1):** Filament-wound fiberglass composite case introduced on Minuteman III (replacing the narrower M57A1 of Minuteman I/II to widen the upper stage to 5.5 ft), generating 35,086 lbf (156.1 kN) of thrust with LITVC and six pyrotechnic thrust-termination ports (AFGSC LGM-30G Minuteman III Fact Sheet).
- **Post-Boost Vehicle (Propulsion System Rocket Engine - PSRE):** Powered by hypergolic monomethylhydrazine (MMH) and nitrogen tetroxide (N_2O_4), featuring an axial 315-lbf rocket engine, six 22-lbf pitch/yaw attitude control thrusters, and four 18-lbf roll thrusters, guided by the Autonetics/Boeing NS-50 Missile Guidance Set (which replaced the NS-20 under the Guidance Replacement Program) loaded via the cryptographic Data Transfer Unit (DTU) (CSIS Minuteman III Upgrade).

Hierarchical Command-and-Control & Key Personnel

- **Program Founders & Chief Engineers:** General Bernard A. Schriever (Commander, Air Force Ballistic Missile Division), Colonel Edward N. Hall (USAF solid-propellant Minuteman architect), T.A. Wilson (Boeing Minuteman Program Manager, later Boeing CEO), and Charles B. Henderson (Atlantic Research Corporation metallized solid-propellant pioneer).
- **Operational Command:** Commander, USSTRATCOM (Admiral Richard Correll) → Commander, AFGSC (General Stephen L. Davis) → Commander, 20th Air Force (Major General Colin J. Connor) → Commanders of the 90th Missile Wing (F.E. Warren AFB), 91st Missile Wing (Minot AFB), and 341st Missile Wing (Malmstrom AFB) (AFGSC Biographies).
- **Test & Sustainment:** 377th Test and Evaluation Group and 576th Flight Test Squadron at Vandenberg Space Force Base, California (conducting unarmed Glory Trip operational test launches over 6,760 km to the Ronald Reagan Ballistic Missile Defense Test Site at Kwajalein Atoll), and the Air Force Nuclear Weapons Center ICBM Systems Directorate at Hill AFB, Utah (CSIS US Tests Minuteman III).

Causal Chronology & Life Extension Programs (2002–2052)

First flight-tested on August 16, 1968 and achieving Initial Operational Capability (IOC) in June 1970 at Minot AFB, Minuteman III production concluded in December 1978 after 830 missiles were built (AFGSC LGM-30G Minuteman III Fact Sheet). Between 2002 and 2012, the USAF executed a $7 billion suite of life-extension programs—including the Guidance Replacement Program (GRP, upgrading NS-20 to NS-50), the Propulsion Replacement Program (PRP, pouring new propellant and liners into all three solid stages), the Propulsion System Rocket Engine Life Extension Program (PSRE LEP), the Safety Enhanced Reentry Vehicle (SERV) program (transferring Mk 21/W87 warheads from retired Peacekeeper missiles onto Minuteman III), and the Rapid Execution and Combat Targeting (REACT) console upgrade (CSIS Minuteman III). Although currently deployed with a single warhead per missile (either a 300 kt W87/Mk 21 or a 335 kt W78/Mk 12A) to comply with New START ceilings, the USAF retains multi-RV mounting bulkheads and routinely executes three-MIRV Glory Trip test flights from Vandenberg SFB to preserve the technical capability to upload up to 800 warheads across the 400-missile fleet (FAS Nuclear Notebook: United States Nuclear Weapons 2025). Under current transition schedules, the final Minuteman III missile is projected to be removed from its silo in 2052 as LGM-35A Sentinel completes deployment (FAS Nuclear Notebook: United States Nuclear Weapons 2025).

33. Sentinel (LGM-35A Sentinel GBSD)

Technical Passport & Acquisition Matrix

Parameter	Verified Specification	Primary Source
Official Designation	LGM-35A Sentinel (formerly Ground Based Strategic Deterrent / GBSD; designated by SECAF Frank Kendall on April 5, 2022)	AFNWC Sentinel Designation
Prime Contractor & Industrial Team	Northrop Grumman Corporation (Engineering & Manufacturing Development Contract awarded September 2020); Aerojet Rocketdyne (Stage 3 solid motor & post-boost propulsion); General Dynamics; Bechtel; Honeywell; Textron Systems; Charles Stark Draper Laboratory (guidance system); Lockheed Martin (command & launch / Mk21A RV)	AFGSC LGM-35A Sentinel Fact Sheet
Propulsion Architecture	Three-stage composite-cased solid-propellant rocket motors (Stage 1 and Stage 2 manufactured by Northrop Grumman in Promontory/Bacchus, Utah; Stage 3 manufactured by Aerojet Rocketdyne) with thrust vector control	AF.mil Sentinel Stage-2 Motor Test 2025
Range, Speed & Ceiling	Range: >6,000 statute miles (>5,218 nautical miles / >9,656 km); Speed: ~15,000 mph (Mach 23 / 24,000 kph) at burnout; Ceiling: ~700 miles (1,120 km)	AFNWC LGM-35A Sentinel Fact Sheet
Warhead & Reentry Vehicle	Initially fielded with W87-0 (300 kt) in Mk 21 RV; transitioning to the modernized W87-1 thermonuclear warhead (LLNL/Sandia) housed inside the Lockheed Martin Mk 21A Reentry Vehicle	DOE/NNSA Warhead Modernization Fact Sheet
Planned Inventory & Basing	400 deployed missiles across 450 newly constructed Launch Facilities at the 90th MW (F.E. Warren AFB, WY), 91st MW (Minot AFB, ND), and 341st MW (Malmstrom AFB, MT); 659 total missiles procured including test and spares	AFGSC LGM-35A Sentinel Fact Sheet

Engineering Mechanics, Digital Engineering & Modular Open Architecture

The LGM-35A Sentinel is a fully digital-engineered, modular open-systems architecture (MOSA) three-stage solid-propellant ICBM designed to replace the LGM-30G Minuteman III through 2075 (AFGSC LGM-35A Sentinel Fact Sheet). Unlike the steel and titanium first/second stages of Minuteman III, Sentinel utilizes lightweight, high-strength carbon-composite filament-wound motor casings across all three stages, paired with electro-mechanical thrust vector control actuators that eliminate the toxic Freon liquid-injection tanks of the legacy SR19 and SR73 stages (AF.mil Sentinel Stage-2 Motor Test 2025). Guidance and navigation are engineered by Charles Stark Draper Laboratory (which broke ground on a dedicated Sentinel support facility at Hill AFB), integrating solid-state strategic-grade inertial sensors with upgraded, cyber-hardened Nuclear Command, Control, and Communications (NC3) interfaces (AFNWC Sentinel ICBM).

Hierarchical Command-and-Control & Key Personnel

- **Direct Reporting Program Manager (DRPM):** Established in 2025–2026 under DRPM White to synchronize USSTRATCOM (Admiral Richard Correll), AFGSC (General Stephen L. Davis), and Department of the Air Force acquisition and civil engineering under a single accountable decision authority (AF.mil Sentinel Restructure 2026).
- **AFNWC & Test Execution:** Air Force Nuclear Weapons Center Sentinel Systems Directorate (Hill AFB, Utah), 717th Test Squadron at Arnold Engineering Development Complex (Arnold AFB, Tennessee), and 719th Test Squadron at Air Force Test Center / Vandenberg SFB, California (AF.mil Inert Sentinel Assembly 2026).

Causal Chronology & 2024–2026 Nunn-McCurdy Restructure Milestones

Initiated in September 2020 with a $13.3 billion Engineering and Manufacturing Development (EMD) contract to Northrop Grumman, Sentinel achieved its first full-scale Stage-1 solid rocket motor static fire test at Promontory, Utah on March 2, 2023, followed by the Stage-3 static fire test at Arnold AFB on March 16, 2024 (F.E. Warren AFB Sentinel History). In July 2024, following a critical Nunn-McCurdy cost breach driven primarily by ground command-and-launch infrastructure and silo conversion costs (pushing projected program costs toward $140.9 billion), the Under Secretary of Defense for Acquisition and Sustainment certified the program as essential to national security and ordered a comprehensive restructure (F.E. Warren AFB Sentinel History). On July 20, 2025, the 717th Test Squadron at Arnold AFB completed the full-scale vacuum-chamber qualification static test of the Stage-2 solid rocket motor (AF.mil Sentinel Stage-2 Motor Test 2025). In February and March 2026, the Air Force confirmed the decision to construct 400 brand-new launch silos rather than reusing 1970s Minuteman silos,

completed the silo prototype, and on September 14, 2026 completed full on-pad assembly of an inert Sentinel missile and passed the first flight test Design Review ahead of developmental flight testing scheduled for 2027 and Initial Operational Capability (IOC) in the early 2030s (AF.mil Inert Sentinel Assembly 2026).

34. SSBN (Fleet Ballistic Missile Submarine Architecture)

Technical Passport & Fleet Deterrence Matrix

Parameter	Verified Specification	Primary Source
US Active Fleet Force Structure	14 Ohio-class SSBNs carrying 20 Trident II D5LE SLBMs per boat (280 total active SLBM tubes; ~1,920 total warheads assigned to the SSBN force, of which ~900–950 are deployed at sea at any given time)	FAS Nuclear Notebook: United States Nuclear Weapons 2025
Allied NATO SSBN Forces	United Kingdom: 4 Vanguard-class SSBNs (16 tubes each, transitioning to 4 Dreadnought-class SSBNs with 12 tubes each); France: 4 Triomphant-class SNLE-NGs (16 tubes each, transitioning to 4 SNLE 3G boats)	FAS UK Nuclear Weapons 2024
Patrol Tempo & Crew Rotation	Dual-crew concept (Blue and Gold crews in US/UK; Bleu and Rouge in France); US Ohio-class averages 77 days at sea followed by 35 days in port for maintenance and crew turnover (~66% at-sea operational availability)	US Navy SUBPAC Ballistic Missile Submarines
Alert Posture	8 to 10 US SSBNs at sea at any given time, with 4 to 5 boats on "hard alert" in designated primary launch stations in the Pacific and Atlantic Oceans and 4 to 5 boats in transit/training able to reach full alert in hours or days	FAS Nuclear Notebook: United States Nuclear Weapons 2025
Strategic Basing & SWF Complexes	Pacific Fleet (8 SSBNs): Naval Base Kitsap-Bangor, Washington (Strategic Weapons Facility Pacific - SWFPAC); Atlantic Fleet (6 SSBNs): Naval Submarine Base Kings Bay, Georgia (Strategic Weapons Facility Atlantic - SWFLANT)	FAS Nuclear Notebook: United States Nuclear Weapons 2025

Engineering Mechanics, Acoustic Quieting & Submerged Launch Physics

A Fleet Ballistic Missile Submarine (SSBN) integrates a pressurized water nuclear reactor (PWR), hydrodynamic hull quieting, ultra-precise inertial navigation, and vertical cold-gas-ejection missile tubes to serve as the most survivable assured-retaliation leg of the strategic nuclear triad (FAS SSBN-726 Ohio-Class). Acoustic stealth is engineered by isolating all rotating machinery—turbines, reduction gears, coolant pumps, and auxiliary generators—onto sound-damping rafts decoupled from the HY-80/HY-100 steel pressure hull, allowing natural-circulation reactor cooling at low patrol speeds relying solely on quiet thermal convection through the primary coolant loop (FAS SSBN-726 Ohio-Class). Because an SLBM's circular error probable (CEP) is directly proportional to the accuracy of the launch platform's initial position, velocity, and local vertical alignment at the instant of ejection, SSBNs carry dual Electrostatically Supported Gyro Navigators (ESGN) or ring-laser/fiber-optic Ships Inertial Navigation Systems (SINS) paired with sonar velocity logs and bathymetric gravity-gradient maps (FAS Trident I C-4). During launch, the submarine hovers at launch depth via variable ballast compensation tanks; a solid-propellant gas generator flashes cooling water into high-pressure steam/gas at the base of the launch tube, rupturing the elastomeric dome membrane and ejecting the missile through the water column to the surface before first-stage rocket motor ignition (FAS Trident II D-5).

Hierarchical Command-and-Control (NC3) & Key Personnel

- **Operational Command (OPCON):** President of the United States → Secretary of Defense → Commander, U.S. Strategic Command (Admiral Richard Correll, USN) → Commander, Task Force 144 (CTF 144 / Submarine Force Strategic Deterrence) (AFDP 3-72 Nuclear Operations).
- **Type Commanders (TYCOMs):**
  - **Commander, Submarine Forces (COMSUBFOR) / Commander, Submarine Force Atlantic (COMSUBLANT) / CTF 114 / CTF 88:** Vice Admiral Richard E. "Rick" Seif, USN (assumed command in February 2026 at Naval Station Norfolk, Virginia, relieving Vice Admiral Robert M. "Rob" Gaucher) (US Navy Biography VADM Richard E. Seif).

- **Commander, Submarine Force, U.S. Pacific Fleet (COMSUBPAC):** Rear Admiral Chris Cavanaugh, USN (relieved Rear Admiral Rick Seif at Joint Base Pearl Harbor-Hickam on April 11, 2025) (SUBPAC Change of Command 2025).
- **Subordinate SSBN Operational Groups:** Submarine Group 9 (Bangor, Washington; Submarine Squadron 17) and Submarine Group 10 (Kings Bay, Georgia; Submarine Squadron 20) (Naval Submarine League SSWATS 2024).

- **Strategic Weapon System Acquisition & Sustainment:** Director, Portfolio Acquisition Executive (PAE) Strategic Systems Programs (SSP), Vice Admiral Douglas L. Williams, USN (15th Director of SSP; relieved Vice Admiral Johnny R. Wolfe Jr. on June 12, 2026 at Washington Navy Yard) (SSP Change of Command June 2026).
- **Submerged NC3 Transmission Architecture:** Fixed Very Low Frequency (VLF) stations (Jim Creek NLK in Washington; Cutler NAA in Maine; Lualualei NPM in Hawaii; LaMoure NML in North Dakota; Harold E. Holt NWC in Australia) and the airborne TACAMO (Take Charge and Move Out) fleet of E-6B Mercury aircraft operated by Strategic Communications Wing One (STRATCOMWING 1) at Tinker AFB, Oklahoma, trailing dual miles-long VLF wire antennas (Tinker AFB AICUZ Study / STRATCOM Wing One).

Causal Chronology & Fleet Evolution

Established in December 1955 under Rear Admiral William F. Raborn Jr. as the Navy Special Projects Office (now SSP) and Admiral Hyman G. Rickover's Naval Reactors branch, the US SSBN architecture commissioned its lead boat *USS George Washington* (SSBN-598) on December 30, 1959 and executed the first submerged Polaris A1 patrol in November 1960 (FAS SSBN-726 Ohio-Class). Between 1959 and 1967, the US Navy commissioned the "41 for Freedom" SSBN fleet (5 *George Washington*, 5 *Ethan Allen*, 9 *Lafayette*, 10 *James Madison*, and 12 *Benjamin Franklin*-class boats), transitioning from Polaris A1/A2/A3 to Poseidon C3 and Trident I C4 (FAS SLBM Index). Starting in 1981, the 18-boat *Ohio*-class replaced the 41 legacy SSBNs; following the 2001 Nuclear Posture Review and New START Treaty implementation, four *Ohio* hulls were converted into SSGNs (2002–2008) and the remaining 14 SSBNs had four tubes per boat deactivated (from 24 to 20 tubes), carrying over 50 percent of the United States' operationally deployed strategic nuclear warheads (US Navy SUBPAC Ballistic Missile Submarines).

35. Trident (UGM-96A Trident I C4 and UGM-133A Trident II D5 / D5LE)

Technical Passport & Comparative SLBM Matrix

Parameter	UGM-96A Trident I (C4)	UGM-133A Trident II (D5 / D5LE)	Primary Source
Prime Contractor	Lockheed Missiles and Space Company (Sunnyvale, CA)	Lockheed Martin Space (Sunnyvale, CA / Titusville, FL)	CSIS Trident D5
Length & Diameter	Length: 34.1 ft (10.39 m); Diameter: 74 in (1.88 m)	Length: 44.0 ft (13.42 m); Diameter: 83 in (2.11 m)	FAS Trident I C-4; CSIS Trident D5
Launch Weight & Throw-Weight	Launch Weight: 73,000 lb (33,112 kg); Throw-Weight: ~1,500 kg	Launch Weight: 130,000 lb (59,090 kg); Throw-Weight: 2,800 kg (6,160 lb)	FAS UK Trident-II D-5; CSIS Trident D5
Propulsion Stages	3-stage solid-propellant in Kevlar/epoxy filament-wound cases (Hercules/Thiokol crosslinked double-base XLDB propellant)	3-stage solid-propellant in IM7 carbon/graphite-epoxy composite cases (Hercules/ATK/Northrop Grumman & United Technologies CSD) with NEPE (nitrate ester polyether) high-energy propellant	FAS Trident II D-5
Range & Burnout Speed	4,000 nm (7,400 km); ~13,000 mph	>4,000 nm (7,400 km full payload) to 12,000 km (reduced payload); >20,000 ft/s (6,096 m/s / Mach 18–24)	FAS Trident II D-5; CSIS Trident D5
Guidance & CEP	Draper Lab Mk 5 stellar-inertial guidance; CEP 380 m (1,250 ft)	Draper Lab Mk 6 (D5) / Mk 6 Mod 1 (D5LE) stellar-inertial	Nuclear Weapon Archive W76; CSIS Trident D5

Parameter	UGM-96A Trident I (C4)	UGM-133A Trident II (D5 / D5LE)	Primary Source
		guidance (with GPS test instrumentation); CEP 90 m	
Payload & Warheads	Up to 8 Mk 4 RVs carrying W76-0 (100 kt) warheads (also flight-tested with Mk 500 Evader MARV)	PBV capable of 12 RVs (capped at 8 RVs under New START, averaging 4–5 deployed per missile): W76-1 (90–100 kt in Mk 4A RV), W76-2 (5–7 kt in Mk 4A RV), W88 Alt 370 (475 kt in Mk 5 RV), and UK Holbrook/Mk4A (up to 100 kt)	CSIS Trident D5
Service Dates	October 20, 1979 – 2005	March 1990 – Present (D5LE through 2042; D5LE2 through 2084)	CSIS Trident D5

Engineering Mechanics, Aerospike Drag Reduction & Generalized Energy Management Steering (GEMS)

To double the range of the predecessor Poseidon C3 within the exact same 34-foot by 74-inch launch-tube envelope, Lockheed Missiles and Space Company and Navy Strategic Systems Project Office (SSPO) engineers introduced three foundational innovations on the UGM-96A Trident I C4 and scaled them onto the larger UGM-133A Trident II D5:

1. **Concentric Third-Stage Post-Boost Packaging:** The third-stage solid rocket motor is mounted vertically in the center of the Equipment Section (Post-Boost Vehicle), with the conical reentry vehicles arranged in a circular ring ("wrapped around") the third-stage motor case (FAS Trident I C-4).
2. **Telescoping Frontal Aerospike:** Because wrapping the RVs around the third stage necessitated a blunt nose fairing, a self-contained telescoping aerospike deploys outward from the nose cone via a solid-propellant gas generator immediately after water exit and first-stage ignition (FAS Trident I C-4). The aerospike creates a detached bow shock wave ahead of the blunt fairing, reducing boost-phase aerodynamic frontal drag by 50 percent and extending range by over 300 nautical miles (FAS Trident II D-5).
3. **Generalized Energy Management Steering (GEMS) & Stellar-Inertial Sighting:** Rather than cutting off third-stage thrust early with blow-out ports, all three boost motors burn completely to depletion; the guidance computer shapes the trajectory dynamically via GEMS to dissipate excess propulsive energy while optimizing reentry angle (FAS Trident I C-4). During exo-atmospheric flight prior to PBV warhead deployment, the Draper Laboratory Mk 6 stellar-inertial guidance system takes an optical star sighting (using a solid-state charge-coupled device star tracker) to measure and null out submarine launch-position and gyro-drift errors, achieving a hard-target counterforce accuracy of 90 meters CEP (CSIS Trident D5).

Hierarchical Command-and-Control & Key Personnel

- **Program Architects & Directors:** Vice Admiral Levering Smith, USN (Director, SSPO SP-00, 1965–1977), Rear Admiral Robert H. Wertheim, USN (Technical Director and Director, SSPO, 1977–1980), Robert A. Duffy (President, Charles Stark Draper Laboratory), Vice Admiral Johnny R. Wolfe Jr. (Director, SSP, 2018–June 2026), and Vice Admiral Douglas L. Williams (15th Director, PAE SSP, June 2026–Present) (FAS Trident I C-4; SSP Change of Command June 2026).
- **Shared US-UK Missile Pool:** Under the 1958 US-UK Mutual Defence Agreement and the 1963 Polaris Sales Agreement (as amended for Trident in 1982 by President Ronald Reagan and Prime Minister Margaret Thatcher), the Royal Navy leases title to 58 Trident II D5 missiles drawn from a commingled pool at the Strategic Weapons Facility Atlantic (SWFLANT) in Kings Bay, Georgia, mating UK sovereign warheads at RNAD Coulport in Scotland (FAS UK Trident-II D-5).

Causal Chronology & D5LE / D5LE2 Modernization

Trident I C4 commenced advanced development on November 15, 1971 under Contract N00030-72-C-0108, completed its first pad launch on January 18, 1977, deployed on *USS Francis Scott Key* (SSBN-657) on October 20, 1979, and armed both backfitted *James Madison*/*Benjamin Franklin* boats and the first eight Pacific *Ohio*-class SSBNs (SSBN 726–733) before those eight *Ohio* hulls were backfitted to Trident II D5 between 1996 and 2008 (FAS Trident I C-4; FAS SSBN-726 Ohio-Class). Trident II D5 began development in March 1980, executed its first pad test in January 1987, and entered operational service aboard *USS Tennessee* (SSBN-734) in March 1990 (CSIS Trident D5). Initiated in June 2002, the Trident D5 Life Extension (D5LE) program replaced the Mk 6 guidance system and flight control electronics with the Mk 6 Mod 1 system to extend service life through 2042 across both US *Ohio*/*Columbia* and UK *Vanguard*/*Dreadnought* fleets (CSIS Trident D5). In 2021–2026, SSP initiated the D5 Life Extension 2 (D5LE2) program to sustain the SLBM force through 2084 alongside the W93/Mk 7 warhead acquisition (CSIS Trident D5; DOE/NNSA Warhead Modernization Fact Sheet).

36. Ohio-class (SSBN-726 Class)

Technical Passport & Vessel Registry Matrix

Parameter	Verified Specification	Primary Source
Builder	General Dynamics Electric Boat Division (Groton, Connecticut and Quonset Point, Rhode Island)	FAS SSBN-726 Ohio-Class
Dimensions & Displacement	Length: 560 ft (170.69 m); Beam: 42 ft (12.80 m); Draft: 36 ft (10.97 m) surfaced; Displacement: 16,764 tons surfaced / 18,750 tons submerged	FAS SSBN-726 Ohio-Class
Nuclear Reactor & Propulsion	1 × General Electric S8G Pressurized Water Reactor (PWR, 220 MWt thermal output, natural circulation capable); 2 × geared steam turbines delivering 60,000 shp (45 MW) to a single 7-bladed skewed propeller shaft; 1 × 325 hp auxiliary magnetohydrodynamic/secondary propulsion motor	FAS SSBN-726 Ohio-Class
Speed & Test Depth	>25 knots submerged (20 knots surfaced); operating test depth in excess of 800 ft (244 m)	FAS SSBN-726 Ohio-Class
Strategic & Tactical Armament	24 vertical missile tubes constructed (4 tubes permanently deactivated per New START, leaving 20 active tubes per boat for UGM-133A Trident II D5LE SLBMs); 4 × 21-inch (533 mm) bow torpedo tubes firing Mk 48 ADCAP heavyweight torpedoes	US Navy SUBPAC Ballistic Missile Submarines
Complement	Dual crews (Blue and Gold): 15 officers and 140 enlisted sailors per crew (155 total)	FAS SSBN-726 Ohio-Class
Service Lifespan	Designed for 30 years; extended to 42–45 years via a mid-life nuclear reactor Engineered Refueling Overhaul (ERO, final ERO completed in February 2023)	FAS Nuclear Notebook: United States Nuclear Weapons 2025

Complete 18-Hull Ohio-Class Operational Disposition

- **Converted to Guided Missile Submarines (SSGN 726–729, 2002–2008, up to 154 Tomahawk cruise missiles each):** *USS Ohio* (SSGN-726), *USS Michigan* (SSGN-727), *USS Florida* (SSGN-728), *USS Georgia* (SSGN-729) (Nuclear Weapon Archive Current US Nuclear Forces).
- **Active Pacific Fleet Strategic Deterrent Boats (8 SSBNs at Naval Base Kitsap-Bangor, WA under Submarine Group 9 / SUBRON 17):** *USS Henry M. Jackson* (SSBN-730, commissioned Oct 1984), *USS Alabama* (SSBN-731, May 1985), *USS Alaska* (SSBN-732, Jan 1986), *USS Nevada* (SSBN-733, Aug 1986), *USS Pennsylvania* (SSBN-735, Sep 1989), *USS Kentucky* (SSBN-737, Jul 1991), *USS Nebraska* (SSBN-739, Jul 1993), *USS Maine* (SSBN-741, Jul 1995) (US Navy SUBPAC Ballistic Missile Submarines; FAS Nuclear Notebook: United States Nuclear Weapons 2025).
- **Active Atlantic Fleet Strategic Deterrent Boats (6 SSBNs at Naval Submarine Base Kings Bay, GA under Submarine Group 10 / SUBRON 20):** *USS Tennessee* (SSBN-734, Dec 1988), *USS West Virginia* (SSBN-736, Oct 1990), *USS Maryland* (SSBN-738, Jun 1992), *USS Rhode Island* (SSBN-740, Jul 1994), *USS Wyoming* (SSBN-742, Jul 1996), *USS Louisiana* (SSBN-743, Sep 6, 1997) (FAS SSBN-726 Ohio-Class; FAS Nuclear Notebook: United States Nuclear Weapons 2025).

Engineering Mechanics, TRIPER Logistics & Sonar/Fire Control Suite

Engineered under the Navy's Undersea Long-Range Missile System (ULMS) program, the Ohio-class hull incorporates three large-diameter logistics hatches that allow shipyard teams at Trident Refit Facility (TRIREFFAC) Bangor and Kings Bay to lower modular equipment replacement pallets and machinery components directly into the hull under the TRIDENT Planned Equipment Replacement (TRIPER) program (FAS SSBN-726 Ohio-Class). This architecture enables a 77-day at-sea patrol followed by a 35-day refit turnaround and 15+ years between major depot overhauls (US Navy SUBPAC Ballistic Missile Submarines). Acoustic sensors comprise the AN/BQQ-6 (upgraded to AN/BQQ-10 ARCI) bow spherical passive/active sonar array, TB-16 and TB-29 thin-line towed arrays, and the Mk 98 Digital Fire Control System interfacing with the Trident II D5LE missiles (FAS SSBN-726 Ohio-Class).

Hierarchical Command-and-Control & Key Personnel

- **Command Chain:** President of the United States → Secretary of Defense → CDRUSSTRATCOM (Admiral Richard Correll) → COMSUBFOR/CTF 144 (Vice Admiral Richard E. Seif) / COMSUBPAC (Rear Admiral Chris Cavanaugh) → Commanders of Submarine Group 9 (Bangor) and Submarine Group 10 (Kings Bay) → Commanding Officers (Blue & Gold crews) (US Navy Biography VADM Richard E. Seif; SUBPAC Change of Command 2025).
- **Onboard Launch Protocol:** Onboard *Ohio*-class SSBNs, autonomous shipboard launch control vests in the dual-crew command hierarchy: launch execution requires authenticated Emergency Action Message (EAM) verification by two independent officer verification teams and simultaneous physical key turns and trigger actuation across the Commanding Officer, Executive Officer, Weapons Officer ("Weps"), and Missile Technicans under strict two-person integrity.

Causal Chronology & Life Extension

Authorized in the 1974 defense budget with *USS Ohio* (SSBN-726) commissioned on November 11, 1981 and *USS Louisiana* (SSBN-743) commissioned on September 6, 1997, the class underwent a multi-decade Engineered Refueling Overhaul (ERO) cycle that concluded in February 2023 when *USS Louisiana* completed its refueling at Puget Sound Naval Shipyard (FAS Nuclear Notebook: United States Nuclear Weapons 2025). With all 14 SSBNs refueled, the entire fleet is available for deployment cycles until the first hull (*USS Henry M. Jackson*, SSBN-730) begins retirement in the late 2020s (2027–2029 window), with the US Navy evaluating selective 3-to-5-year service-life extensions for up to five *Ohio* boats to prevent any strategic deterrence dip during the *Columbia*-class transition (FAS Nuclear Notebook: United States Nuclear Weapons 2025).

37. Columbia-class (SSBN-826 Class)

Technical Passport & Procurement Matrix

Parameter	Verified Specification	Primary Source
Lead Ship & Hull Designations	*USS District of Columbia* (SSBN-826; renamed from *USS Columbia* on June 3, 2022 to avoid conflict with SSN-771), *USS Wisconsin* (SSBN-827), *USS Groton* (SSBN-828); 12 total boats planned (SSBN 826–837)	CRS Navy Ship Names RS22478; FAS Nuclear Notebook: United States Nuclear Weapons 2025
Prime Builder & Partner	General Dynamics Electric Boat (Groton, CT and Quonset Point, RI) as prime contractor and lead design yard; HII Newport News Shipbuilding (Newport News, VA) as major module subcontractor	NHHC/CRS Navy Columbia Class Submarine Program
Dimensions & Displacement	Length: 560 ft (170.69 m); Beam: 43 ft (13.11 m); Submerged Displacement: 20,810 tons (~2,000+ tons heavier than *Ohio*-class, making it the largest submarine ever built by the United States)	FAS Nuclear Notebook: United States Nuclear Weapons 2025
Nuclear Reactor & Propulsion	1 × Bechtel/Naval Reactors S1B Pressurized Water Reactor featuring a 42-year life-of-the-ship nuclear fuel core (eliminating mid-life refueling overhauls); Permanent Magnet Motor (PMM) turbo-electric drive system powering a shrouded pump-jet propulsor (eliminating mechanical reduction gears)	NHHC/CRS Navy Columbia Class Submarine Program
Strategic Armament (CMC)	16 missile tubes (87-inch diameter) arranged in four 4-tube "Quad Packs" within the US-UK Common Missile Compartment (CMC), armed with UGM-133A Trident II D5LE / D5LE2 SLBMs	CSIS Trident D5; Royal Navy Dreadnought Class
Program Cost & Budget Share	~$130 billion total acquisition cost (~20% of the US Navy's entire shipbuilding budget from mid-2020s to mid-2030s); $15.2 billion for lead boat SSBN-826 (including non-recurring fleet engineering) and $9.3 billion for second boat SSBN-827	FAS Nuclear Notebook: United States Nuclear Weapons 2025

Engineering Mechanics, Life-of-Ship Core & Electric Drive Architecture

The Columbia-class (formerly the *Ohio* Replacement Program / SSBN[X]) is engineered to replace 14 *Ohio*-class SSBNs with 12 boats while maintaining the full fleet operational availability of continuously patrolling submarines at sea (NHHC/CRS Navy Columbia Class Submarine Program). This 12-for-14 operational equivalence is achieved through the Naval Reactors S1B pressurized water reactor, whose nuclear core is engineered to operate for the entire 42-year service life of the submarine (124 deterrent patrols per hull), eliminating the two-to-three-year mid-life Engineered Refueling Overhaul (ERO) required by the *Ohio* class (NHHC/CRS Navy Columbia Class Submarine Program). Acoustic signature is reduced below the ambient ocean background noise floor through an Integrated Electric Drive system: steam from the S1B reactor drives turbo-generators that supply electrical power to a radial-gap Permanent Magnet Motor turning a pump-jet propulsor, eliminating the mechanical gear-mesh tonals of traditional steam turbine reduction gears, while X-stern control surfaces provide quiet high-precision depth and course control. Furthermore, the US-UK Common Missile Compartment (CMC)—jointly engineered by General Dynamics Electric Boat, BAE Systems, and Navy SSP—houses 16 Trident II D5LE missiles in four modular four-tube "Quad Packs" shared identically with the Royal Navy's *Dreadnought* class (Royal Navy Dreadnought Class).

Hierarchical Command-and-Control & Key Personnel

- **Acquisition & Design Authority:** Program Executive Officer Strategic Submarines (PEO SSBN), Rear Admiral Todd Weeks, USN; Director of Naval Reactors (NR), Admiral William J. Houston, USN; Director of PAE Strategic Systems Programs, Vice Admiral Douglas L. Williams, USN; and President of General Dynamics Electric Boat, Kevin Graney (SSP Change of Command June 2026).
- **Operational Command:** CDRUSSTRATCOM (Admiral Richard Correll) → COMSUBFOR (Vice Admiral Richard E. Seif) / COMSUBPAC (Rear Admiral Chris Cavanaugh) (US Navy Biography VADM Richard E. Seif).

Causal Chronology & Construction Status (2020–2031)

Construction of the lead boat *USS District of Columbia* (SSBN-826) officially began on October 1, 2020 (the first day of FY 2021), its keel was laid down on June 4, 2022 at Quonset Point, Rhode Island, and the hull passed its 50 percent construction completion milestone in August 2024 (FAS Nuclear Notebook: United States Nuclear Weapons 2025). In August 2026, the Department of the Navy and the Submarine Industrial Base (SIB) Program Office authorized accelerated additive manufacturing and supply-chain protocols to meet the "1+2" national security mandate of delivering one *Columbia*-class SSBN and two *Virginia*-class SSNs per year (US Navy Submarine Industrial Base PPD Aug 2026). *USS District of Columbia* is scheduled to conduct its first strategic deterrent patrol in the 2030–2031 window, with the 12th boat completing delivery by the early 2040s (FAS Nuclear Notebook: United States Nuclear Weapons 2025).

38. B-52 Stratofortress (B-52H, ALCM/LRSO Integration)

Technical Passport & Stand-Off Strike Matrix

Parameter	Verified Specification	Primary Source
Prime Airframe & Engine Contractors	Boeing Military Airplane Co. (Wichita, KS / Seattle, WA); Current engines: 8 × Pratt & Whitney TF33-P-3/103 turbofans (up to 17,000 lbf thrust each); Commercial Engine Replacement Program (CERP / B-52J): 8 × Rolls-Royce F130 turbofans	AFNWC B-52H Stratofortress Fact Sheet
Dimensions & Weights	Wingspan: 185 ft (56.4 m); Length: 159 ft 4 in (48.5 m); Height: 40 ft 8 in (12.4 m); Empty Weight: ~185,000 lb (83,250 kg); Maximum Takeoff Weight: 488,000 lb (219,600 kg); Fuel Capacity: 312,197 lb (141,610 kg)	AFNWC B-52H Stratofortress Fact Sheet
Performance	Speed: 650 mph (Mach 0.84); Unrefueled Combat Range: 8,800 statute miles (7,652 nm / 14,162 km); Service Ceiling: 50,000 ft (15,151.5 m); Crew: 5 (Aircraft Commander, Pilot, Radar Navigator, Navigator, Electronic Warfare Officer)	AFNWC B-52H Stratofortress Fact Sheet
Payload & Nuclear Armament	70,000 lb (31,500 kg) mixed ordnance; Nuclear role: up to 20 × AGM-86B Air-Launched Cruise Missiles (ALCMs; 8 on internal Common Strategic Rotary Launcher + 12 on two external Heavy Stores Adapter Beams) armed with W80-1 variable-yield (5 kt to 150 kt) thermonuclear warheads; transitioning to AGM-181A Long-Range Standoff (LRSO) missile with W80-4 warhead	AFNWC B-52H Stratofortress Fact Sheet; FAS Nuclear Notebook: United States Nuclear Weapons 2025
Active Inventory & Nuclear Coding	76 total B-52H aircraft in inventory (58 active force, 18 Reserve, 4 test); 46 B-52H bombers are nuclear-capable ("combat-coded"	AFNWC B-52H Stratofortress Fact Sheet; FAS Nuclear Notebook: United States Nuclear Weapons 2025

Parameter	Verified Specification	Primary Source
	under New START after 30 B-52Hs were converted to conventional-only under the 2015–2017 New START conversion)	

Engineering Mechanics, ALCM/LRSO Standoff Architecture & B-52J Modernization

Delivered between May 1961 and October 1962, the B-52H Stratofortress serves as the dedicated standoff nuclear cruise-missile carrier of the US strategic bomber leg (having been relieved of its nuclear gravity-bomb mission in 2017) (FAS Nuclear Notebook: United States Nuclear Weapons 2025). Nuclear-capable B-52H airframes retain unique Nuclear Command, Control, and Communications (NC3) receivers, Permissive Action Link (PAL) pre-arming control panels in the cockpit, and wing/bay wiring for the Boeing AGM-86B Air-Launched Cruise Missile (ALCM) (FAS Nuclear Notebook: United States Nuclear Weapons 2025). The AGM-86B is powered by a Williams International F107-WR-101 turbofan engine, navigates over a 2,400+ km range at low-altitude terrain-hugging subsonic speeds via Terrain Contour Matching (TERCOM) radar altimetry, and carries the LANL-designed W80-1 thermonuclear warhead (5 to 150 kt yield; ~528 ALCMs remain in the inventory) (FAS Nuclear Notebook: United States Nuclear Weapons 2025). Under the B-52J transformation program (extending service life past 2050), the fleet is integrating the Raytheon AN/APG-79B4 active electronically scanned array (AESA) radar (Radar Modernization Program), Rolls-Royce F130 turbofan engines (Commercial Engine Replacement Program), upgraded AEHF/VLF nuclear communications, and the Raytheon AGM-181A Long-Range Standoff (LRSO) stealth nuclear cruise missile carrying the LLNL/Sandia W80-4 warhead (whose first development flight test and baseline design review were completed in 2023–2024) (Sandia Labs Accomplishments 2024).

Hierarchical Command-and-Control & Key Personnel

- **Operational Command:** President of the United States → Secretary of Defense → CDRUSSTRATCOM (Admiral Richard Correll) → Commander, AFGSC / COMAFSTRAT (General Stephen L. Davis) → Commander, Eighth Air Force ("The Mighty Eighth") and Commander, Joint-Global Strike Operations Center (J-GSOC), Major General Ty W. Neuman, USAF (assumed command in November 2025 at Barksdale AFB, Louisiana), with Deputy Commander Colonel Allen Y. Agnes and Command Chief Master Sergeant Brandon M. Wolfgang (AFGSC Biographies).
- **Operational B-52H Bomb Wings:**
  - **2nd Bomb Wing** (Barksdale AFB, Louisiana: 11th, 20th, and 96th Bomb Squadrons; plus Air Force Reserve 307th Bomb Wing) (AFGSC Units).
  - **5th Bomb Wing** (Minot AFB, North Dakota: 23rd and 69th Bomb Squadrons) (Minot AFB Global Thunder 27).

Causal Chronology & Operational Readiness (1952–2026)

Following the B-52 prototype's first flight in April 1952 and the delivery of the final B-52H (serial 61-0040) on October 26, 1962, the Stratofortress maintained continuous Chrome Dome airborne alert (1960–1968) and ground strip alert until President George H.W. Bush stood down strategic bombers from 24-hour nuclear strip alert in September 1991 (AFNWC B-52H Stratofortress Fact Sheet). In November 2026, B-52H crews from the 5th Bomb Wing at Minot AFB and 2nd Bomb Wing at Barksdale AFB executed nuclear generation and dispersal drills under USSTRATCOM's annual command-and-control exercise *Global Thunder 27* (Minot AFB Global Thunder 27).

39. B-2 Spirit (B-2A Stealth Strategic Bomber, B61-12/B83)

Technical Passport & Penetrating Stealth Matrix

Parameter	Verified Specification	Primary Source
Prime Contractor & Team	Northrop Grumman Corporation (Palmdale, CA); Major subcontractors: Boeing Military Airplanes Co., Hughes Radar Systems Group (Raytheon), General Electric Aircraft Engine Group, Vought Aircraft Industries	USAF B-2 Spirit Fact Sheet
Power Plant & Thrust	4 × General Electric F118-GE-100 high-bypass turbofan engines buried internally within the wing root, rated at 17,300 lbf (77 kN) thrust each	USAF B-2 Spirit Fact Sheet
Dimensions & Weights	Wingspan: 172 ft (52.12 m); Length: 69 ft (20.9 m); Height: 17 ft (5.1 m); Empty Weight: 160,000 lb (72,575 kg); Maximum Takeoff Weight: 336,500 lb (152,634 kg); Internal Fuel Capacity: 167,000 lb (75,750 kg)	USAF B-2 Spirit Fact Sheet
Performance & Crew	Speed: High subsonic (Mach 0.95); Unrefueled Range: Intercontinental (~6,000 nm / 11,100 km; >10,000 nm with one aerial refueling);	USAF B-2 Spirit Fact Sheet

Parameter	Verified Specification	Primary Source
	Ceiling: 50,000 ft (15,240 m); Crew: 2 pilots (Pilot and Mission Commander) in ACES II upward-firing ejection seats	
Payload & Nuclear Armament	60,000 lb (27,216 kg) official maximum internal payload (40,000 lb nominal across two internal weapons bays equipped with rotary launcher assemblies); Certified for up to 16 nuclear gravity bombs: B61-12 guided nuclear tail-kit bomb (0.3, 1.5, 10, or 50 kt selectable yield), B61-11 earth-penetrator (400 kt), and B83-1 strategic bomb (1.2 Mt max yield, transitioning out of active service)	USAF B-2 Spirit Fact Sheet; FAS Half of Operational B-2 Force Deploys to Diego Garcia 2025
Active Inventory & Cost	19–20 aircraft in inventory (1 test aircraft; 21 built total; *Spirit of Kansas* lost in 2008 and *Spirit of Hawaii* retired after 2022 incident); Unit cost: $2.1 billion total program average)	USAF B-2 Spirit Fact Sheet; AFGSC B-2 Spirit Fact Sheet

Engineering Mechanics, Low-Observable Flying Wing & B61-12 RATS Integration

Unveiled on November 22, 1988 at Air Force Plant 42 in Palmdale, California, first flown on July 17, 1989, and achieving Initial Operational Capability (IOC) in April 1997, the B-2A Spirit utilizes a continuous-curvature flying-wing planform employing split-brake-rudder yaw control surfaces along the trailing wing edges to minimize radar cross-section (RCS) across VHF, UHF, S, C, and X radar bands (USAF B-2 Spirit Fact Sheet). Its four GE F118-GE-100 turbofans are deeply recessed inside serpentine S-duct inlets that shield the compressor fan blades from radar waves, while exhaust gases are mixed with cool boundary-layer air and vented across carbon-composite/titanium aft deck troughs to suppress infrared thermal signatures. For nuclear precision strikes against heavily defended underground command bunkers and mobile launchers, the B-2A integrates the Radar Aided Targeting System (RATS) using its AN/APQ-181 low-probability-of-intercept (LPI) AESA radar to update the inertial guidance of the B61-12 guided nuclear gravity bomb immediately prior to release, enabling the Boeing/Sandia tail-kit assembly on the B61-12 to achieve a circular error probable (CEP) of 30 meters (FAS Nuclear Notebook: United States Nuclear Weapons 2025; Sandia Labs Accomplishments 2024).

Hierarchical Command-and-Control & Key Personnel

- **Operational Command:** President of the United States → Secretary of Defense → CDRUSSTRATCOM (Admiral Richard Correll) → Commander, AFGSC (General Stephen L. Davis) → Commander, Eighth Air Force / J-GSOC (Major General Ty W. Neuman) → Commander, **509th Bomb Wing**, Colonel Joshua Wiitala, USAF (operating the 13th and 393rd Bomb Squadrons alongside the Associate Missouri Air National Guard **131st Bomb Wing** at Whiteman AFB, Missouri) (AFGSC 509th Bomb Wing Omaha Trophy June 2026).

Causal Chronology & 2024–2026 Operational Milestones

After all 20 surviving B-2A aircraft were upgraded to the unified Block 30 standard with the Defensive Management System Modernization (DMS-M) and Extremely High Frequency (EHF) FAB-T nuclear satellite communications, the fleet completed certification for the B61-12 life-extended nuclear bomb, enabling the retirement of the legacy B61-7 and the ongoing phase-out of the 1.2-megaton B83-1 bomb (FAS Half of Operational B-2 Force Deploys to Diego Garcia 2025). In April 2024, the 509th and 131st Bomb Wings executed a 12-aircraft "mass fly-off" during Exercise *Spirit Vigilance* (FAS Half of Operational B-2 Force Deploys to Diego Garcia 2025). In 2025, the 509th Bomb Wing executed *Operation Midnight Hammer* and *Operation Rough Rider*—including deploying half of the operational B-2 fleet to Naval Support Facility Diego Garcia in the Indian Ocean—and achieved the highest Nuclear Surety Inspection score on record, earning the USSTRATCOM Omaha Trophy for the second consecutive year on May 26, 2026 (AFGSC 509th Bomb Wing Omaha Trophy June 2026).

40. B-21 Raider (Northrop Grumman Sixth-Generation Strategic Bomber)

Technical Passport & Acquisition Matrix

Parameter	Verified Specification	Primary Source
Official Designation & Role	B-21 Raider (Long-Range Strike Bomber / LRS-B); dual-capable sixth-generation penetrating stealth strategic bomber	USAF B-21 Raider Fact Sheet
Prime Contractor & Partners	Northrop Grumman Aeronautics Systems (Air Force Plant 42, Palmdale, CA); Major suppliers: Pratt & Whitney (propulsion), BAE Systems, Spirit AeroSystems, Collins Aerospace, GKN Aerospace, Janicki Industries	AFGSC B-21 Raider Fact Sheet; HSDL/CRS B-21 Raider Report

Parameter	Verified Specification	Primary Source
Dimensions, Planform & Crew	Blended flying-wing W-tail planform (~132–140 ft wingspan); Crew: 2 pilots (officially confirmed by the Department of the Air Force on July 9, 2026) with provisions for uncrewed/optionally manned operations	AFGSC B-21 Summit / Crew Complement Announcement 2026
Nuclear & Conventional Armament	Internal weapons bay configured for the B61-12 guided nuclear gravity bomb, B61-13 guided nuclear gravity bomb (up to 360 kt), and the Raytheon AGM-181A Long-Range Standoff (LRSO) nuclear cruise missile (W80-4 warhead)	FAS Nuclear Notebook: United States Nuclear Weapons 2025
Planned Fleet Size & Unit Cost	Minimum 100 aircraft planned by USAF (with USSTRATCOM advocating up to 145); Average Procurement Unit Cost (APUC) cap: $692 million in FY 2022 dollars)	AFGSC B-21 Raider Fact Sheet; HSDL/CRS B-21 Raider Report
Main Operating Bases (MOBs)	MOB 1 (Training & First Operational Wing): Ellsworth AFB, South Dakota (28th Bomb Wing); MOB 2: Whiteman AFB, Missouri (509th Bomb Wing); MOB 3: Dyess AFB, Texas (7th Bomb Wing)	AFGSC B-21 Summit Dyess AFB Oct 2026

Engineering Mechanics, Sixth-Generation Stealth & Open Mission Systems

Awarded under the Long-Range Strike Bomber (LRS-B) contract on October 27, 2015, publicly unveiled on December 2, 2022 at Palmdale, California, and executing its maiden flight on November 10, 2023 to Edwards AFB, the B-21 Raider incorporates next-generation low-observable surface coatings that eliminate the delicate climate-controlled tape-and-caulk maintenance cycles of the B-2A, deeply recessed flush conformal engine inlets that eliminate radar cavity reflections at all aspect angles, and a cloud-native Open Mission Systems (OMS) digital avionics backbone (USAF B-21 Raider Fact Sheet). The B-21 is designed as the central node of a collaborative family of systems, integrating with multi-domain Intelligence, Surveillance, and Reconnaissance (ISR) sensors and survival NC3 networks to penetrate dense integrated air defense systems (IADS) (HSDL/CRS B-21 Raider Report).

Hierarchical Command-and-Control & Key Personnel

- **Acquisition & Test Authorities:** Air Force Rapid Capabilities Office (AFRCO) Director William Bailey; 412th Test Wing and B-21 Combined Test Force (CTF) at Edwards AFB, California; and Northrop Grumman CEO Kathy Warden (USAF B-21 Raider Fact Sheet).
- **Operational Command:** CDRUSSTRATCOM (Admiral Richard Correll) → Commander, AFGSC (General Stephen L. Davis) → Commander, Eighth Air Force (Major General Ty W. Neuman), which activated **B-21 Integration Detachment 1** on September 10, 2026 to unify beddown across Ellsworth AFB, Whiteman AFB, and Dyess AFB (AFGSC B-21 Summit Dyess AFB Oct 2026).

Causal Chronology & 2025–2026 Production Expansion

Following Low-Rate Initial Production (LRIP) authorization in early 2024, the second flight-test B-21 aircraft arrived at Edwards AFB in September 2025 (AFGSC B-21 Summit Dyess AFB Oct 2026). In February and April 2026, the Department of the Air Force announced an expansion of B-21 production capacity, followed on July 9, 2026 by the formal announcement of the B-21's two-pilot crew complement, on September 10, 2026 by the activation of Eighth Air Force's B-21 Integration Detachment 1, and on October 8, 2026 by the groundbreaking of a new B-21 production facility by the Department of the Air Force and Northrop Grumman (AFGSC B-21 Summit Dyess AFB Oct 2026). The B-21 will incrementally replace both the B-1B Lancer and the B-2A Spirit to form a two-bomber strategic fleet alongside the B-52J (USAF Aircraft Fact Sheets).

41. Peacekeeper (LGM-118A MX Peacekeeper)

Technical Passport & Performance Matrix

Parameter	Verified Specification	Primary Source
Official Designation	LGM-118A Peacekeeper (Missile Experimental / MX)	FAS LGM-118A Peacekeeper
System Integrator & Stage Contractors	Ballistic Missile Organization (BMO, Norton AFB/San Bernardino, CA) & TRW (systems engineering);	FAS LGM-118A Peacekeeper

Parameter	Verified Specification	Primary Source
	Martin Marietta (assembly & test); Stage 1: Thiokol; Stage 2: Aerojet; Stage 3: Hercules; Stage 4 (PBV): Rocketdyne; Guidance: Northrop / Draper Lab; Canister/Launch: Westinghouse	
Dimensions & Weight	Length: 71 ft (21.8 m); Diameter: 7 ft 8 in (2.34 m); Launch Weight: 195,000 lb (87,750 kg) including reentry vehicles	FAS LGM-118A Peacekeeper; CSIS LGM-118 Peacekeeper
Throw-Weight Breakdown	Total Throw-Weight: 8,710 lb (3,950 kg), comprising a 3,010 lb (1,365 kg) Post-Boost Vehicle and a 4,700 lb (2,585 kg) net warhead/RV payload	Nuclear Weapon Archive MX Peacekeeper
Propulsion Architecture	Stage 1: Thiokol solid motor (500,000 lbf thrust, Kevlar-epoxy case); Stage 2: Aerojet solid motor (Kevlar-epoxy case, extendable exit cone); Stage 3: Hercules solid motor (Kevlar-epoxy case, two-position extendable exit cone); Stage 4 (PBV): Rocketdyne axial liquid bipropellant rocket motor (MMH / N_2O_4)	FAS LGM-118A Peacekeeper; Nuclear Weapon Archive MX Peacekeeper
Range, Speed & Ceiling	Range: >6,000 miles (9,600 km); Burnout Speed: 15,000 mph (24,000 km/hr / 6.7 km/s / Mach 20+); Apogee: ~500 miles (805 km)	Nuclear Weapon Archive MX Peacekeeper; CSIS LGM-118 Peacekeeper
Guidance & CEP	Draper Lab / Northrop Advanced Inertial Reference Sphere (AIRS); CEP: 90 m to 100 m (300–330 ft)	CSIS LGM-118 Peacekeeper; Nuclear Weapon Archive MX Peacekeeper
Payload & Warheads	Tested/capable of 11–12 RVs; deployed with 10 Avco Mk 21 Reentry Vehicles carrying 10 × W87-0 thermonuclear warheads (300 kt yield each; upgradable to 475 kt)	Nuclear Weapon Archive MX Peacekeeper
Deployment & Service Dates	50 missiles deployed in modified Minuteman silos at the 400th Strategic Missile Squadron, 90th Missile Wing, F.E. Warren AFB, Wyoming; IOC: December 1986; FOC: December 1988; Final deactivation: September 19, 2005	FAS LGM-118A Peacekeeper

Engineering Mechanics, Cold-Gas Steam Ejection & AIRS Guidance

The LGM-118A Peacekeeper was the first US ICBM to employ cold-launch canister technology and the most accurate ballistic missile ever fielded by the United States (FAS LGM-118A Peacekeeper). Although weighing 195,000 lb—nearly 2.5 times the mass of Minuteman III—Peacekeeper was installed directly inside refurbished Minuteman III silos by encapsulating the missile inside a Westinghouse launch canister (FAS LGM-118A Peacekeeper). Upon launch execution, a solid-propellant gas generator at the base of the canister vaporized water into high-pressure steam, ejecting the 195,000-lb missile out of the silo to an altitude of 150 to 300 feet (45 to 90 m) before the 500,000-lbf Thiokol Stage 1 solid motor ignited in mid-air, protecting the silo liner from acoustic and thermal blast damage (FAS LGM-118A Peacekeeper).

Guidance accuracy of 90–100 meters CEP over a 9,600 km trajectory was achieved by the Charles Stark Draper Laboratory Advanced Inertial Reference Sphere (AIRS) (Nuclear Weapon Archive MX Peacekeeper). AIRS housed three gyroscopes and three accelerometers inside a 10-inch neutrally buoyant beryllium sphere floating inside a temperature-controlled fluorocarbon fluid shell in a gimballess fluid-floated architecture, eliminating gimbal-lock errors and seismic drift while tracking over 19,000 parts in a radiation-hardened sphere.

Hierarchical Command-and-Control & Key Personnel

- **Program & Strategic Leadership:** President Jimmy Carter (authorized full-scale MX development in June 1979), President Ronald Reagan and the Scowcroft Commission on Strategic Forces (Brent Scowcroft, recommended deploying 100 MX missiles in existing Minuteman silos in April 1983), Ballistic Missile Organization (BMO) at Norton AFB, California, and the 400th Missile Squadron under the 90th Missile Wing at F.E. Warren AFB, Wyoming (FAS LGM-118A Peacekeeper).

Causal Chronology & Post-Retirement Reuse (1979–Present)

Engineered to counter the hardened Soviet fourth-generation ICBM silos (SS-18 Satan and SS-19 Stiletto), MX Peacekeeper underwent over 30 basing-mode studies—including Carter's 4,600-shelter "Multiple Protective Shelters" (MPS) racetrack in Nevada/Utah, Reagan's "Dense Pack" closely spaced basing, and the 1989 "Peacekeeper Rail Garrison" (mounting 50 missiles on 25 stealthy two-missile trains across seven SAC bases)—before Congress capped deployment at 50 silo-based missiles at F.E. Warren AFB (FAS LGM-118A Peacekeeper). Following the 1993 START II signing and the 2001 Nuclear Posture Review decision by President George W. Bush, the 50 Peacekeeper missiles were deactivated between October 2002 and September 19, 2005 (CSIS Minuteman III). Its 500 W87-0/Mk 21 reentry vehicles were transferred onto the LGM-30G Minuteman III fleet under the Safety Enhanced Reentry Vehicle (SERV) program, while retired Peacekeeper solid rocket stages were repurposed by Northrop Grumman (formerly Orbital ATK) as the Minotaur IV space launch vehicle and for hypersonic glide vehicle target boosters (FAS Nuclear Notebook: United States Nuclear Weapons 2025).

42. Titan II (LGM-25C Titan II)

Technical Passport & Hypergolic Heavy ICBM Matrix

Parameter	Verified Specification	Primary Source
Official Designation & Builder	SM-68B / LGM-25C Titan II; Prime Contractor: Glenn L. Martin Company (Martin Marietta, Denver, Colorado)	FAS SM-68B Titan II
Dimensions & Launch Weight	Length: 103 ft (31.30 m); Diameter: 10 ft (3.05 m); Launch Weight: 330,000 lb (149,700 kg) fully fueled—the largest and heaviest operational ICBM ever fielded by the United States	CSIS Titan II
Propulsion & Hypergolic Propellants	Two-stage liquid-propellant rocket burning hypergolic Aerozine-50 (50% hydrazine, 50% unsymmetrical dimethylhydrazine) and nitrogen tetroxide (N_2O_4) oxidizer: Stage 1: Aerojet LR87-AJ-5 twin-chamber engine (430,000 lbf thrust); Stage 2: Aerojet LR91-AJ-5 single-chamber engine (100,000 lbf vacuum thrust)	CSIS Titan II; FAS SM-68B Titan II
Range, Speed & Throw-Weight	Range: 15,000 km (9,320 statute miles / 8,100 nm); Burnout Speed: Mach 25 (~17,000 mph); Throw-Weight: ~3,720 kg (8,200 lb)	CSIS Titan II
Guidance & CEP	Originally AC Spark Plug / IBM ASC-15 inertial guidance; upgraded in 1978–1979 with Delco Electronics Universal Space Guidance System (USGS) carousel IMU; CEP: 900 m (0.5–0.8 nm)	CSIS Titan II
Warhead & Reentry Vehicle	1 × W53 thermonuclear warhead (LANL design; 9.0 Megaton yield—the highest-yield warhead ever deployed on a US missile) housed inside a General Electric Mk 6 ablative Reentry Vehicle (weighing 8,140 lb / 3,690 kg)	CSIS Titan II

Parameter	Verified Specification	Primary Source
Deployment & Service History	54 missiles deployed across 3 Strategic Missile Wings (18 silos per wing, 1 missile and 1 four-person crew per complex): 308th SMW (Little Rock AFB, AR), 381st SMW (McConnell AFB, KS), and 390th SMW (Davis-Monthan AFB, AZ); Service: 1963–1987	CSIS Titan II

Engineering Mechanics, In-Silo Hot Launch & Mk 6 / W53 Payload

Developed starting in June 1960 to overcome the 15-to-20-minute cryogenic fueling and elevator-raising vulnerability of the HGM-25A Titan I, the LGM-25C Titan II introduced room-temperature storable hypergolic propellants (Aerozine-50 and N_2O_4) that ignited spontaneously upon contact inside the Aerojet LR87-AJ-5 thrust chambers (CSIS Titan II). Unlike Minuteman's unmanned 10-silo flights, each Titan II Launch Complex (hardened to 300–1,000 psi and spaced 13 to 19 km apart) was an autonomous underground fortress containing a single 146-foot-deep silo with W-shaped acoustic/flame deflector ducts (flooded with 100,000 gallons of water per minute during ignition to suppress acoustic vibration that would otherwise destroy the missile airframe), a 750-ton hydraulic silo closure door, a blast-lock tunnel, a two-story utility/power house, and a three-story shock-mounted underground Launch Control Center manned by a four-person combat crew (two officers: Missile Combat Crew Commander and Deputy Commander; two enlisted technicians: Ballistic Missile Analyst Technician and Missile Facilities Technician) (CSIS Titan II). This configuration enabled Titan II to execute a "hot launch" directly out of its silo within 58 to 60 seconds of launch key turn (CSIS Titan II).

Hierarchical Command-and-Control & Key Personnel

- **Operational Command:** Strategic Air Command (SAC, Offutt AFB, Nebraska) → Eighth Air Force / Fifteenth Air Force → **308th Strategic Missile Wing** (373rd and 374th SMS, Little Rock AFB, Arkansas), **381st Strategic Missile Wing** (532nd and 533rd SMS, McConnell AFB, Kansas), and **390th Strategic Missile Wing** (570th and 571st SMS, Davis-Monthan AFB, Arizona; Complex 571-7 is preserved today as the Titan Missile Museum in Sahuarita, Arizona) (CSIS Titan II).

Causal Chronology, Fatal Silo Accidents & Deactivation (1963–1987)

Achieving full operational capability across all 54 silos in December 1963, Titan II served for 24 years as America's heavy counter-value and deep-bunker-busting strategic deterrent while simultaneously serving as NASA's human-rated Gemini launch vehicle (12 flights in 1964–1966) (CSIS Titan II). However, the toxicity, corrosiveness, and volatility of its storable liquid propellants and hydraulic systems caused two major fatal silo accidents:

1. **August 9, 1965 (Launch Complex 373-4, Searcy, Arkansas):** During Project *Yard Fence* silo hardening modifications, a welder's rod pierced a high-pressure hydraulic line inside the silo while the W53 warhead was removed; the resulting flash fire consumed the oxygen and killed 53 civilian contractors inside the sealed launch duct (CSIS Titan II).
2. **September 18–19, 1980 (Launch Complex 374-7, Damascus, Arkansas):** An airman performing pressure checks on the Stage 2 oxidizer tank dropped an 8-pound, 3-foot-long socket wrench ratchet socket, which fell 70 feet, bounced off the thrust mount, and punctured the Stage 1 Aerozine-50 fuel tank (CSIS Titan II). Eight and a half hours later, at 3:00 AM on September 19, 1980, the leaking hypergolic vapors exploded, killing Senior Airman David Livingston, catapulting the 750-ton silo door 200 meters away, hurling 20-ton concrete flame-deflector chunks 500 meters, and ejecting the 9-megaton W53/Mk 6 reentry vehicle 100 meters into a ditch—where Sandia's Permissive Action Link and environmental sensing devices prevented any nuclear or high-explosive detonation (CSIS Titan II).

Following the Damascus explosion and subsequent congressional investigations, President Ronald Reagan announced the retirement of the Titan II fleet in October 1981; deactivation began in July 1982 and the final Titan II on alert (at Complex 373-8, Little Rock AFB) was stood down on May 5, 1987, with the decommissioned airframes converted into Titan 23G space launch vehicles flown until 2003 (CSIS Titan II).

43. W76 (W76-0 / W76-1 / W76-2 Thermonuclear Warheads, Mk4/Mk4A RV)

Technical Passport & Variant Matrix

Parameter	W76-0 (Mk 4 RV)	W76-1 (Mk 4A RV)	W76-2 (Mk 4A RV)	Primary Source
Physics & Systems Design Labs	Los Alamos National Laboratory (LANL) & Sandia National Laboratories (SNL)	Los Alamos National Laboratory (LANL) & Sandia National Laboratories (SNL)	Los Alamos National Laboratory (LANL) & Sandia National Laboratories (SNL)	DOE/NNSA W76-1 LEP Fact Sheet; Sandia Labs Accomplishments
Thermonuclear Architecture	Two-stage Teller-Ulam radiation implosion (DT-boosted beryllium-reflected Pu-239 primary + Li-6 deuteride / HEU secondary in a thin	Refurbished two-stage radiation implosion with new Acorn organic components, MC4081-2 clocks, and neutron generators	Single-stage primary-only fission/boosted configuration (secondary stage removed/replaced with ballast)	Nuclear Weapon Archive W76; FAS Nuclear Notebook: United States Nuclear Weapons 2025

Parameter	W76-0 (Mk 4 RV)	W76-1 (Mk 4A RV)	W76-2 (Mk 4A RV)	Primary Source
	plastic-backed uranium radiation case)			
Explosive Yield	100 kilotons (kt)	90 kilotons (kt) (frequently cited as 90–100 kt)	~5 to 8 kilotons (kt) low-yield	Nuclear Weapon Archive W76; FAS Nuclear Notebook: United States Nuclear Weapons 2025
Reentry Body & Weight	Lockheed Mk 4 RV (TWCP heatshield); Total RV weight: ~95 kg (210 lb; NWA cites up to 362.5 lb with bus adapter)	Lockheed Mk 4A RV with Sandia MC4700 Arming, Fuzing, and Firing (AF&F) super-fuze	Lockheed Mk 4A RV	Nuclear Weapon Archive W76; FAS Trident I C-4
Production & Deployment	~3,400 produced (1978–1987); retired upon W76-1 conversion	~1,600–2,000 converted at Pantex Plant (Sep 2008 – Dec 2018); ~1,536 active in stockpile	~25–50 produced at Pantex Plant (FPU Feb 2019 – FY2020); deployed on Trident II D5LE since late 2019	DOE/NNSA W76-1 Celebration; CSIS Trident D5

Engineering Mechanics, Fissile Physics & the MC4700 "Super-Fuze"

Designed at Los Alamos National Laboratory (with senior designer Charles C. Cremer) beginning in May 1973 for the Trident I C4 and Trident II D5 SLBMs, the W76 is the most numerous thermonuclear warhead in the United States nuclear stockpile (Nuclear Weapon Archive W76). Its primary stage utilizes a deuterium-tritium (DT) gas-boosted, beryllium-reflected plutonium-239 pit imploded by conventional PBX-9501 high explosive, channeled inside an ultra-lightweight plastic-backed uranium radiation case to compress a lithium-6 deuteride (Li^6D) thermonuclear secondary jacketed in highly enriched uranium (HEU) (Nuclear Weapon Archive W76). The warhead is housed inside the conicalMk 4 / Mk 4A reentry body, which employs a tape-wrapped carbon phenolic (TWCP) ablative heatshield bonded to an aluminum shell with a fine-grain computer-machined graphite nosetip (FAS Trident I C-4).

During the W76-1 Life Extension Program (LEP), Sandia National Laboratories replaced the legacy Mk 4 contact/fixed-radar fuze with the **MC4700 Arming, Fuzing, and Firing (AF&F)** assembly on the Mk 4A reentry body. By using path-length-compensated radar proximity fuzing ("super-fuzing") to detonate the 90 kt W76-1 at the exact edge of the lethal airburst overpressure sphere even if the RV has a slight downrange/track error, the Mk 4A/W76-1 multiplies its hard-target kill probability against reinforced ICBM silos more than threefold.

Hierarchical Command-and-Control & Key Personnel

- **Design & Production Enterprise:** Under Secretary for Nuclear Security and NNSA Administrator Brandon Williams (following Dr. Jill Hruby and Lisa E. Gordon-Hagerty); LANL Director Dr. Thomas Mason and Associate Laboratory Director for Weapons Dr. Charlie Nakhleh; Sandia Director Dr. James S. Peery; Pantex Plant (Amarillo, TX, final assembly); Y-12 National Security Complex (Oak Ridge, TN, canned subassemblies/secondaries); Kansas City National Security Campus (AF&F and non-nuclear electronics); and Savannah River Site (Aiken, SC, tritium reservoirs) (LANL Fusion Issue 2026; DOE/NNSA W76-1 LEP Fact Sheet).

Causal Chronology & W76-2 Low-Yield Deployment (1978–2026)

Originally manufactured between June 1978 and July 1987 with a 20-year design life, the W76-0 entered the W76-1 LEP in October 1998, delivering its First Production Unit in September 2008 and completing its Last Production Unit at the Pantex Plant in December 2018 (formally celebrated by Energy Secretary Rick Perry and NNSA Administrator Lisa Gordon-Hagerty on January 24, 2019), extending service life to 60 years (through the late 2040s) (DOE/NNSA W76-1 Celebration). Following the 2018 Nuclear Posture Review's requirement for a prompt, survivable low-yield sea-based option to deter regional non-strategic nuclear escalation, NNSA and the W76-2/Mk 4A Reentry Body Low-Yield Working Group rapidly modified ~25–50 W76-1 warheads into the 5–8 kt **W76-2** between 2018 and 2020, deploying the W76-2 aboard *USS Tennessee* (SSBN-734) at the end of 2019 (CSIS Trident D5; Sandia Labs Accomplishments 2020).

44. W78 (W78 Thermonuclear Warhead, Mk12A RV)

Technical Passport & ICBM Warhead Matrix

Parameter	Verified Specification	Primary Source
Physics & Systems Design Labs	Los Alamos Scientific Laboratory / Los Alamos National Laboratory (LANL, physics package) &	Nuclear Weapon Archive W78

Parameter	Verified Specification	Primary Source
	Sandia National Laboratories (SNL, non-nuclear ordnance and fuzing)	
Reentry Vehicle & Dimensions	General Electric Mk 12A Reentry Vehicle; Base Diameter: 21.3 in (54.1 cm); Length: 71.3 in (181.1 cm); Total W78/Mk 12A RV Package Weight: 700–800 lb (317–363 kg; warhead physics package ~400 lb / 181 kg)	Nuclear Weapon Archive W78; Nuclear Weapon Archive Current US Nuclear Forces
Explosive Yield	335 kilotons (kt) (335–350 kt range; upgraded from the 170 kt W62/Mk 12 warhead)	FAS Nuclear Notebook: United States Nuclear Weapons 2025; CSIS Minuteman III
Thermonuclear Architecture	Two-stage Teller-Ulam radiation implosion combining a plutonium-239 primary imploded by conventional high explosive (conventional PBX high explosive) with an older-generation thermonuclear secondary derived from the W50 Pershing warhead	Nuclear Weapon Archive W78
Delivery Platform & Accuracy	LGM-30G Minuteman III ICBM (capable of carrying 1 to 3 Mk 12A RVs per missile; currently deployed with 1 RV per missile on ~200 of the 400 active Minuteman IIIs, with additional W78 warheads held in the upload hedge stockpile); CEP: ~120–200 m	FAS Nuclear Notebook: United States Nuclear Weapons 2025
Production & Retirement Schedule	1,083 warheads produced between August 1979 and October 1982 (~600 remaining in active/reserve stockpile); scheduled for complete replacement by the W87-1 / Mk 21A on the LGM-35A Sentinel in the 2030s	DOE/NNSA Warhead Modernization Fact Sheet

Engineering Mechanics & Threshold Test Ban Design Constraints

Initiated by Los Alamos in July 1974 to double the yield of the LLNL-designed 170 kt W62/Mk 12 warhead while preserving the exact exterior aerodynamic geometry and ballistic coefficient of the General Electric Mk 12 reentry cone, the W78 married a compact Los Alamos primary stage to a high-yield thermonuclear secondary derived from the W50 warhead inside the newly designated **Mk 12A Reentry Vehicle** (adding only ~35 lb to the 750-lb Mk 12 package) (Nuclear Weapon Archive W78). Because the W78 prototype was first tested in Operation *Keelson* on February 4, 1976 (~200 kt yield) immediately prior to the May 1976 entry into force of the 150-kiloton Threshold Test Ban Treaty (TTBT), the redesigned W78 underwent underground partial-yield hydronuclear and primary-plus-secondary testing up to the 150 kt treaty threshold prior to production (Nuclear Weapon Archive W78). When Lawrence Livermore scientists challenged Los Alamos's calculated yield estimate, a special Department of Energy review panel chaired by former LLNL Director Dr. John S. Foster Jr. validated LANL's radiation-hydrodynamics calculations and cleared the W78 for production engineering in 1977 (Nuclear Weapon Archive W78). Because the W78 utilizes conventional high explosive rather than Insensitive High Explosive (IHE) and utilizes a conventional unclad plutonium pit, USAF air transport of the W78 is restricted and its replacement by the IHE-equipped W87-1 is a primary safety mandate of the NNSA (DOE/NNSA Warhead Modernization Fact Sheet).

Hierarchical Command-and-Control & Key Personnel

- **Design & Sustainment Authority:** Los Alamos National Laboratory (Director Dr. Thomas Mason) and Sandia National Laboratories (Director Dr. James S. Peery) under DOE/NNSA Stockpile Surveillance; operational deployment under 20th Air Force (Major General Colin J. Connor) across F.E. Warren, Minot, and Malmstrom AFBs (AFGSC Biographies; FAS Nuclear Notebook: United States Nuclear Weapons 2025).

Causal Chronology & Operational Upload Role (1979–2030s)

Fielded starting in December 1979 on 300 Minuteman III missiles (carrying 3 Mk 12A RVs each for 900 deployed W78 warheads), the W78 provided the US ICBM force with its primary hard-target counterforce capability until the arrival of the W87/Peacekeeper in 1986 (Nuclear Weapon Archive W78). Following the 2006–2012 SERV modification that placed W87/Mk 21 RVs on roughly half of the Minuteman III force and the 2014–2018 New START downloading of all deployed Minuteman IIIs to a single warhead, approximately 200 W78/Mk 12A warheads remain deployed on alert silos while another ~400 remain in depot storage—providing the United States with its rapid MIRV upload capability (adding 1 or 2 extra W78/Mk 12A RVs onto existing 3-RV buses) following the February 5, 2026 expiration of the New START Treaty (FAS Nuclear Notebook: United States Nuclear Weapons 2025).

45. W87 (W87-0 / W87-1 Thermonuclear Warhead, Mk21 RV)

Technical Passport & Variant Matrix

Parameter	W87-0 (Mk 21 RV)	W87-1 Modification Program (Mk 21A RV)	Primary Source
Physics & Engineering Design Labs	Lawrence Livermore National Laboratory (LLNL) & Sandia National Laboratories (SNL)	Lawrence Livermore National Laboratory (LLNL) & Sandia National Laboratories (SNL)	Nuclear Weapon Archive W87; DOE/NNSA Warhead Modernization Fact Sheet
Explosive Yield	300 kilotons (kt) baseline; upgradable to 475 kt by adding highly enriched uranium (HEU / Oralloy) rings to the secondary stage tamper	~300–475 kilotons (kt) class using newly manufactured plutonium-239 pits	Nuclear Weapon Archive MX Peacekeeper; FAS Nuclear Notebook: United States Nuclear Weapons 2025
Reentry Vehicle & Dimensions	Avco Systems Division Mk 21 RV (Advanced Ballistic Reentry Vehicle / ABRV); Base Diameter: 21.8 in (55 cm); Length: 68.9 in (175 cm); Weight: ~500–590 lb (227–268 kg)	Lockheed Martin Mk 21A Reentry Vehicle with Sandia Mk 21 Fuze and polymer multilayer main capacitors	Nuclear Weapon Archive W87; Sandia Labs Accomplishments 2024
Enhanced Surety & Safety Features	TATB-based Insensitive High Explosive (LX-17 / PBX-9502), Fire-Resistant Pit (FRP) refractory beryllium/metal cladding able to withstand $1,000^\circ\text{C}$ jet-fuel fires for hours while containing the fissile plutonium core intact, Enhanced Nuclear Detonation Safety (ENDS), and Command Disable	Full modern IHE, newly manufactured plutonium-239 pits from LANL PF-4 / SRS, and intrinsic radiation hardening	Nuclear Weapon Archive W87; DOE/NNSA Warhead Modernization Fact Sheet
Production & Deployment	~525–540 W87-0 units produced (April 1986 – December 1988); deployed on LGM-118A Peacekeeper (1986–2005) and LGM-30G Minuteman III (2006–Present; ~200 deployed, ~340 in reserve)	Entered Phase 6.3 Development Engineering in FY 2023/2024; First Production Unit (FPU) scheduled for early 2030s to arm LGM-35A Sentinel	Sandia Labs Accomplishments 2024; DOE/NNSA Warhead Modernization Fact Sheet

Engineering Mechanics, Fire-Resistant Pit & W87-1 Newly Manufactured Pits

Developed by Lawrence Livermore National Laboratory starting in February 1982 and selected jointly with the Avco Mk 21 Advanced Ballistic Reentry Vehicle (ABRV) by the Department of Defense on January 29, 1982 to arm the LGM-118A Peacekeeper, the W87 represents the pinnacle of US Cold War ICBM warhead surety (Nuclear Weapon Archive MX Peacekeeper). Because Peacekeeper was originally planned for mobile basing across public railroads and highways, LLNL engineered the W87-0 with triaminotrinitrobenzene (TATB) Insensitive High Explosive (LX-17), which remains chemically stable under high-velocity projectile impact, shrapnel, and solid-propellant rocket motor deflagrations, alongside a **Fire-Resistant Pit (FRP)** that encapsulates the plutonium-239 core inside a high-melting-point refractory shell to prevent molten plutonium aerosolization in a prolonged hydrocarbon or propellant fire (1,000°C) (Nuclear Weapon Archive W87).

The **W87-1 Modification Program** (formerly the Interoperable Warhead 1 / IW-1) is the first US nuclear weapon program since the November 1989 shutdown of the Rocky Flats Plant to incorporate **newly manufactured plutonium-239 pits** (produced at Los Alamos National Laboratory's Plutonium Facility PF-4, which achieved War Reserve diamond-stamped pit qualification in 2023–2024, and the Savannah River Plutonium Processing Facility) rather than reusing legacy Cold War pits (DOE/NNSA Warhead Modernization Fact Sheet). In 2023–2024, Sandia National Laboratories completed W87-1 Ground Test Unit 1, transitioned the W87-1 into Phase 6.3 Development Engineering, and brought all major components of the new **Mk 21 Fuze** to First Production Unit status (Sandia Labs Accomplishments 2024).

Hierarchical Command-and-Control & Key Personnel

- **Design & Acquisition Leadership:** LLNL Director Dr. Kimberly S. Budil, Principal Deputy Director Carolyn Zerkle, and Deputy Director for Strategic Deterrence Brad Wallin (LLNL Leadership); Sandia Director Dr. James S. Peery; LANL Director Dr. Thomas Mason (pit production); and NNSA Defense Programs (NNSA Nuclear Deterrence Summit).

Causal Chronology (1982–2030s)

Manufactured at Pantex between April 1986 and December 1988, 500 W87-0 warheads stood alert on the 50 Peacekeeper missiles at F.E. Warren AFB until Peacekeeper's retirement in 2002–2005 (Nuclear Weapon Archive MX Peacekeeper). Under the Safety Enhanced Reentry Vehicle (SERV) program, those W87-0/Mk 21 packages were transferred onto LGM-30G Minuteman III missiles (configured with single-RV bulkheads) starting in 2006 (CSIS Minuteman III). When LGM-35A Sentinel achieves Initial Operational Capability in the early 2030s, it will initially carry existing W87-0 warheads before transitioning to the W87-1/Mk 21A, allowing the complete retirement and dismantlement of the legacy W78 stockpile (DOE/NNSA Warhead Modernization Fact Sheet).

46. W88 (W88 Alt 370 Thermonuclear Warhead, Mk5 RV)

Technical Passport & Asymmetric Secondary Matrix

Parameter	Verified Specification	Primary Source
Physics & Engineering Design Labs	Los Alamos National Laboratory (LANL, physics package & detonator assemblies) & Sandia National Laboratories (SNL, AF&F, integrated circuits, thermal batteries)	DOE/NNSA W88 Alt 370 Fact Sheet
Explosive Yield	475 kilotons (kt)—the highest-yield strategic ballistic missile warhead in the active US stockpile	CSIS Trident D5; Nuclear Weapon Archive W88
Reentry Body & Dimensions	Lockheed Mk 5 Reentry Body; Length: 68.9 in (175 cm); Base Diameter: 21.8 in (55 cm); Weight: <800 lb (~360 kg / 794 lb; physics package ~390–400 lb)	Nuclear Weapon Archive W88; FAS Nuclear Notebook: United States Nuclear Weapons 2025
Thermonuclear Architecture	Two-stage radiation implosion utilizing a prolate (aspherical / "peanut-shaped") DT-boosted plutonium-239 primary (code-named *Komodo*) positioned in the wider aft base of the cone and a spherical HEU-tampered lithium-6 deuteride thermonuclear secondary (code-named *Cursa*) in the narrower forward cone, imploded by conventional PBX-9501 high explosive	Nuclear Weapon Archive W88
Delivery Platform & Accuracy	UGM-133A Trident II D5 / D5LE SLBM aboard *Ohio*-class and future *Columbia*-class SSBNs; CEP: 90 m	CSIS Trident D5
Production & W88 Alt 370 Status	~400 warheads originally produced (Sep 1988 – Nov 1989; halted by Rocky Flats closure); **W88 Alteration 370**: First Production Unit completed	DOE/NNSA W88 Alt 370 First Production Unit; DOE/NNSA W88 Alt 370 Last Production Unit Dec 2025

Parameter	Verified Specification	Primary Source
	July 1, 2021; **Last Production Unit completed December 2025** at Pantex Plant	

Engineering Mechanics, Prolate Primary Physics & W88 Alt 370 Modernization

Designed at Los Alamos National Laboratory beginning in March 1984 to give the U.S. Navy's Trident II D5 SLBM a prompt hard-target counterforce kill capability against deeply buried Soviet command bunkers and ICBM silos, the W88 packs a 475-kiloton yield into a Mk 5 reentry cone measuring 68.9 inches long and weighing under 360 kg (Nuclear Weapon Archive W88). To maximize packaging efficiency inside the tapered conical geometry of the Mk 5 reentry body—which is mounted in a ring around the Trident D5's high-energy NEPE solid-propellant third-stage motor—LANL weapon physicists inverted the traditional layout: they placed a compact, prolate (aspherical two-point implosion) plutonium-239 primary in the wider aft section of the cone and the spherical thermonuclear secondary in the narrower forward section inside a peanut-shaped uranium-238/plastic radiation case (Nuclear Weapon Archive W88). Because the Mk 5 RVs sit directly adjacent to the high-energy NEPE propellant of the D5 third stage (where any third-stage propellant detonation would exceed the threshold of even insensitive high explosives), the W88 was engineered with higher-energy conventional PBX-9501 explosive rather than IHE to minimize weight and volume (Nuclear Weapon Archive W88).

Initiated in late 2009 jointly by NNSA and Navy Strategic Systems Programs, the **W88 Alteration (Alt) 370** program modernized every aging subsystem of the W88/Mk 5:

- **Pantex Plant (Amarillo, TX):** Refreshed the conventional high-explosive main charges in the primary and performed final weapon assembly (DOE/NNSA W88 Alt 370 Fact Sheet).
- **Sandia National Laboratories (NM/CA) & Kansas City National Security Campus (MO):** Designed and manufactured the modernized Arming, Fuzing, and Firing (AF&F) assembly, lightning arrestor connectors, radar/contact fuzing, thermal batteries, integrated circuits, foams, polymers, and gas transfer system components (DOE/NNSA W88 Alt 370 Fact Sheet).
- **Los Alamos National Laboratory (NM) & Y-12 National Security Complex (TN):** Manufactured new detonator assemblies at LANL and reaccepted/refurbished canned subassemblies at Y-12 (DOE/NNSA W88 Alt 370 Fact Sheet).

Hierarchical Command-and-Control & Key Personnel

- **Design & Production Authorities:** NNSA Administrator Brandon Williams (and predecessor Acting Administrator Dr. Charles P. Verdon, who oversaw the July 2021 FPU); LANL Director Dr. Thomas Mason; Sandia Director Dr. James S. Peery; and Navy PAE SSP Director Vice Admiral Douglas L. Williams (and predecessor Vice Admiral Johnny R. Wolfe Jr.) (DOE/NNSA W88 Alt 370 First Production Unit; DOE/NNSA W88 Alt 370 Last Production Unit Dec 2025).

Causal Chronology & December 2025 Last Production Unit Milestone

Quantity production of the W88 began in April 1989 (planning for 4,000–5,000 warheads), but was abruptly terminated in November 1989 after only ~400 warheads were completed when an FBI and EPA environmental raid shut down plutonium pit production at the Rocky Flats Plant near Denver, Colorado (Nuclear Weapon Archive W88). To preserve the 384–400 W88 warheads in the active sea-based deterrent, the W88 Alt 370 program achieved its system-level First Production Unit (FPU) at the Pantex Plant on July 1, 2021 (one month ahead of schedule) and completed its **Last Production Unit (LPU) in December 2025**, transitioning the entire W88 Alt 370 fleet into active deployment and stockpile surveillance while allowing the NNSA production staff to pivot to the W93/Mk 7 and SLCM-N programs (DOE/NNSA W88 Alt 370 Last Production Unit Dec 2025).

47. MIRV (Multiple Independently Targetable Reentry Vehicle Bus & Physics)

Technical Passport & Bus Kinematics Matrix

Parameter	Verified Specification	Primary Source
Foundational Patent & Inventor	Ted Taylor & Ernest J. Sternglass (early concept); Richard Garwin; **Minuteman III / Poseidon C3 Post-Boost Vehicle Architecture** engineered by Autonetics, Bell Labs, Boeing, Lockheed, and Charles Stark Draper Laboratory (first flight-tested on August 16, 1968 simultaneously on Minuteman III and Poseidon C3)	CSIS Minuteman III; FAS ICBM Technical Introduction

Parameter	Verified Specification	Primary Source
Post-Boost Vehicle (PBV) Mass Fraction	Typically consumes ~35% to 50% of total missile throw-weight (e.g., on LGM-118A Peacekeeper, the PBV weighs 1,365 kg out of 3,950 kg total throw-weight, carrying 2,585 kg of Mk 21 RVs)	FAS ICBM Technical Introduction; Nuclear Weapon Archive MX Peacekeeper
Bus Propulsion Chemistry	US ICBMs (Minuteman III PSRE / Peacekeeper Stage 4): Storable hypergolic liquid bipropellant (monomethylhydrazine MMH fuel + nitrogen tetroxide N_2O_4 oxidizer) with multi-restart axial and attitude thrusters; US SLBMs (Trident I C4 / Trident II D5): Solid-propellant gas generator Post-Boost Control System (PBCS)	FAS LGM-118A Peacekeeper; FAS Trident I C-4
Footprint & Cross-Range Envelope	Downrange dispersion: 500–1,200 km; Cross-range dispersion: 150–400 km per missile bus	FAS ICBM Technical Introduction
Ballistic Coefficient (β)	High-beta slender conical RVs (Mk 12A, Mk 21, Mk 4A, Mk 5, French TNO): β ≈ 3,500 to 5,000 lb/ft² (17,000 to 24,000 kg/m²), decelerating from ~7 km/s to Mach 6–10 at impact to minimize atmospheric wind drift	FAS ICBM Technical Introduction

Engineering Mechanics, Post-Boost Bus Maneuvering & Reentry Physics

A Multiple Independently Targetable Reentry Vehicle (MIRV) system replaces a single large warhead with an exo-atmospheric maneuvering upper stage—termed the **Post-Boost Vehicle (PBV)** or "bus"—that sequentially aims and releases multiple warheads onto independent Keplerian trajectories against widely separated targets (FAS ICBM Technical Introduction). The MIRV engagement sequence executes in four deterministic physical stages:

1. **Shroud Jettison & Boost Error Nulling:** Shortly after third-stage burnout in the vacuum of space (above 150–200 km altitude), a nose-cap rocket motor and explosive bolts jettison the aerodynamic protective shroud (FAS LGM-118A Peacekeeper). The PBV's inertial measurement unit (and star tracker on SLBMs) computes the residual velocity vector error accumulated during boost and fires its axial thrusters to enter the first pre-computed ballistic transfer ellipse (FAS ICBM Technical Introduction).
2. **Sequential "Stop-and-Drop" Backing-Off Maneuvers:** For each individual reentry vehicle (and accompanying Tracor chaff clouds or inflatable metallized Mylar decoys), the PBV fires its attitude control thrusters to align the RV's longitudinal axis with the local reentry angle of attack (α), unlatches the mechanical attachment via explosive-cartridge gas pressure or spring actuators, and backs away using reverse thrusters so the separation imparts zero lateral perturbation to the RV (FAS LGM-118A Peacekeeper).
3. **Spin Stabilization & Boundary-Layer Transition Control:** As the PBV releases the RV (or via aerodynamic fluted vanes upon entering the upper mesosphere), the RV is spun around its longitudinal axis at 5 to 15 revolutions per second (FAS ICBM Technical Introduction). This gyroscopic spin averages out asymmetric mass shifts and uneven nose-tip ablation as the carbon-phenolic heatshield endures stagnation temperatures of 6,000°C to 8,000°C and decelerations exceeding 50–60 G during atmospheric reentry (FAS ICBM Technical Introduction; FAS Trident I C-4).

Hierarchical Command-and-Control & Targeting (SIOP / OPLAN 8010)

- **Strategic Target Planning:** USSTRATCOM Joint Functional Component Command for Global Strike (JFCC GS) and the Joint-Global Strike Operations Center (J-GSOC, commanded by Major General Ty W. Neuman at Barksdale AFB) allocate MIRV footprints within OPLAN 8010 (formerly the Single Integrated Operational Plan / SIOP) to optimize target coverage while preventing "fratricide" (where the X-ray pulse, neutron flux, or dust cloud of the first detonating RV destroys or deflects trailing RVs arriving seconds later) (AFDP 3-72 Nuclear Operations).

Causal Chronology & Strategic Stability Impact (1968–2026)

Originally conceived in the early 1960s under Secretary of Defense Robert McNamara and DDR&E Dr. Harold Brown to saturate the Soviet A-35 Galosh Anti-Ballistic Missile (ABM) system around Moscow, MIRV technology was flight-tested on the exact same day—August 16, 1968—aboard both the USAF LGM-30G Minuteman III (3 Mk 12 RVs) and the USN UGM-73 Poseidon C3 (up to 14 Mk 3 RVs) (CSIS Minuteman III). Because MIRVs allowed a single attacking missile (such as the 10-RV Peacekeeper or 10-RV Soviet SS-18) to destroy multiple single-silo enemy ICBMs, land-based MIRVs created a theoretical "use-them-or-lose-them" first-strike instability during the late Cold War. Consequently, the 1993 START II Treaty sought to ban land-based MIRVs, and under the 2010 New START Treaty the United States downloaded all 400 deployed Minuteman III ICBMs to a single warhead—while retaining MIRVs on its invulnerable sea-based Trident II D5LE force (4 to 8 RVs per missile), UK Trident D5s, and French M51.2/M51.3 SLBMs (4 to 6 TNO/TNO-2 RVs per missile) (FAS Nuclear Notebook: United States Nuclear Weapons 2025; CSIS M51). Following the February 5, 2026 expiration of New START, US upload of reserved W78/Mk 12A MIRVs onto Minuteman III buses has re-emerged as a central Pentagon force-posture option (FAS Nuclear Notebook: United States Nuclear Weapons 2025).

48. MARV (Maneuverable Reentry Vehicle Mechanics & Guidance)

Technical Passport & Exemplar MARV Systems Matrix

System / Program	Prime Contractor & Lab	Aerodynamic Control Mechanism	Terminal Guidance Architecture	Warhead & CEP	Flight Test / Deployment Status	Primary Source
MGM-31B Pershing II MARV	Martin Marietta & Goodyear Aerospace; LANL / Sandia	4 cruciform movable aerodynamic control vanes on the mid-body RV	Goodyear Aerospace **RADAG** (Radar Area Correlator) active J-band (15 GHz) terminal radar comparing live ground returns against stored digital reference scenes + Singer-Kearfott inertial IMU	W85 variable-yield thermonuclear warhead (5 kt to 80 kt); **CEP: 30 meters**	Flight-tested 1977–1983; 108 launchers deployed in West Germany (56th Field Artillery Command, Dec 1983 – May 1988); eliminated under INF Treaty by 1991	CSIS Pershing 2
Mk 500 Evader MARV (Trident C4)	Lockheed Missiles & Space Co. & General Electric	Asymmetric bent-nose ("cranked-cone") lifting body rolling around its velocity vector via internal roll-control actuators	Pre-programmed inertial evasive corkscrew/jinking maneuvers below 60 km altitude to defeat terminal ABM interceptors	W76 (100 kt) compatible	6 successful flight tests on Atlas/Minuteman and Trident I C4 (March 6, 1974 – November 13, 1975); held in engineering readiness	FAS Trident I C-4; DTIC/RAND Future of US ICBM Force
Advanced Maneuverable Reentry Vehicle (AMaRV)	McDonnell Douglas Astronautics Co. (under USAF ABRES)	Split-windward body flap and two yaw flaps on a high-lift-to-drag biconic reentry body	Autonomous strapdown ring-laser gyroscope inertial system guiding high-G pull-up and lateral S-turn evasive maneuvers	Advanced Ballistic Reentry Systems (ABRES) prototype	Flight-tested on Minuteman I booster on December 20, 1979; direct aerodynamic precursor to modern US Hypersonic Glide Bodies (C-HGB)	DSIAC Advanced Maneuverable Reentry Vehicle (AMaRV); DTIC/RAND Future of US ICBM Force

Engineering Mechanics, Aerodynamic Lift Generation & Terminal Correlation Guidance

Unlike a pure ballistic reentry vehicle (which follows a predictable gravity-and-drag trajectory once released by the PBV), a **Maneuverable Reentry Vehicle (MARV or MaRV)** generates controlled aerodynamic lift (L) alongside drag (D) inside the atmosphere (below 80–100 km altitude) to execute high-G lateral and vertical trajectories (DTIC/RAND Future of US ICBM Force). MARVs are divided into two functional classes:

1. **Evader MARVs (Mk 500 Evader & AMaRV):** Designed to defeat exo- and endo-atmospheric Anti-Ballistic Missile (ABM) interceptors (such as the Soviet Sprint/Gazelle/S-300/A-135 systems) while preserving full warhead throw-weight through compact internal roll-control actuators (FAS Trident I C-4). The **Mk 500 Evader** achieved aerodynamic lift using a fixed, slightly bent ("cranked") nose cone; by rotating its internal center of mass or rolling the bent body with reaction jets during atmospheric entry, the lift vector rotated rapidly, causing the RV to execute unpredictable high-G spiral and S-turn evasive maneuvers before rolling to hit its target (FAS Trident I C-4). The McDonnell Douglas **AMaRV** introduced a biconic lifting body with a split windward flap and yaw flaps controlled by a strapdown ring-laser gyro IMU, executing a pull-up glide maneuver on

December 20, 1979 that established the aerodynamic database for the Sandia/Dynetics Common Hypersonic Glide Body (C-HGB) used in the US Navy Conventional Prompt Strike (CPS) and US Army Long-Range Hypersonic Weapon (LRHW) (DSIAC AMaRV).

2. **Precision-Guided Terminal Homing MARVs (MGM-31B Pershing II):** After separating from its second-stage Hercules solid motor and pitching down into the atmosphere over target area at Mach 8, the **Pershing II** RV executed a high-G aerodynamic pull-up using four movable mid-body cruciform vanes to slow to Mach 2–3 and eliminate plasma sheath blackout (CSIS Pershing 2). Beginning at 15 km (50,000 ft) altitude, the Goodyear Aerospace **RADAG (Radar Area Correlator)** active radar scanned the target terrain through an ablative nose radome at 120 RPM, correlating live radar returns against pre-loaded digital terrain elevation/reflectivity maps at multiple decreasing altitudes and commanding the aerodynamic vanes to steer the W85 warhead directly onto the target with a **30-meter CEP** (CSIS Pershing 2).

Hierarchical Command-and-Control & Key Personnel

- **Development & Operational Authorities:** USAF Advanced Ballistic Reentry Systems (ABRES) Office at Norton AFB (AMaRV); USN Strategic Systems Project Office under Vice Admiral Levering Smith and Rear Admiral Robert Wertheim (Mk 500 Evader); and U.S. Army Missile Command (MICOM, Redstone Arsenal, AL) and 56th Field Artillery Command (Schwäbisch Gmünd, West Germany) under Commander, U.S. European Command (SACEUR) for Pershing II (CSIS Pershing 2; FAS Trident I C-4).

Causal Chronology & Strategic Impact (1974–2026)

The December 1983 deployment of 108 Pershing II MARV launchers in West Germany gave NATO the ability to strike hardened Soviet command bunkers in western Russia within 8 to 10 minutes of launch with 30-meter accuracy—a capability that Soviet Defense Minister Marshal Dmitri Ustinov and General Secretary Mikhail Gorbachev cited as the primary military driver forcing the Soviet Union to sign the December 1987 Intermediate-Range Nuclear Forces (INF) Treaty, which eliminated all Pershing II missiles by May 1991 (CSIS Pershing 2). In the 2020s, MARV and hypersonic boost-glide physics have returned to the center of strategic competition across the US (C-HGB), Russian (Avangard), Chinese (DF-ZF), and French (V-MAX and cuarta-generation ASN4G / M51.3/M51.4) modernization programs (French Ministry of the Armed Forces SYLEX Test Nov 2026).

Section III: United Kingdom and French Sovereign Strategic Deterrence and NC3 Architectures (Entities 49–55)

49. Letters of Last Resort (UK Prime Minister Nuclear Directives)

Technical Passport & Constitutional Protocol Matrix

Parameter	Verified Specification	Primary Source
Authorizing Constitutional Actor	Prime Minister of the United Kingdom (currently Prime Minister Sir Keir Starmer, who authored his letters upon taking office in July 2024), briefed on day one in Downing Street by the Chief of the Defence Staff (CDS) and Cabinet Secretary	UK Defence Journal: Keir Starmer Letters of Last Resort
Physical Quantity & Storage	4 identical handwritten letters (one aboard each of the four Vanguard-class SSBNs: *HMS Vanguard*, *HMS Victorious*, *HMS Vigilant*, *HMS Vengeance*), sealed inside nested double combination safes bolted inside the submarine Control Room	UK Defence Journal: Prime Minister Letters of Last Resort
Trigger Condition ("Britain Is Gone")	Activated only if the Commanding Officer of the patrolling CASD submarine establishes through a multi-step check—including failure of VLF transmissions from Skelton/Anthorn and the absence of BBC Radio 4 Today programme broadcasting on 198 kHz longwave—that a nuclear strike has decapitated both the Prime Minister and the designated "second person" (Nuclear Deputy)	UK Defence Journal: Prime Minister Letters of Last Resort; Chatham House NATO NC3 Cyber Resilience
The Four Canonical Directives	1. Retaliate with nuclear weapons against designated targets; 2. Refrain from retaliatory strike; 3. Use your own judgment as Commanding Officer; 4. Place the	UK Defence Journal: Keir Starmer Letters of Last Resort; Cairn.info Le Président et la Bombe

Parameter	Verified Specification	Primary Source
	submarine under the command of an allied nation if it survives (specifically the United States or Australia)	
Destruction Protocol	Immediately upon a Prime Minister leaving office, the four sealed letters of the outgoing Prime Minister are removed from the safes and destroyed unopened (incinerated), preserving exclusive personal knowledge of the directive with the authoring Prime Minister	UK Defence Journal: Trident and Online Misinformation

Engineering Mechanics, Absence of Permissive Action Links (PALs) & Two-Person Rule

The **Letters of Last Resort** represent the physical and constitutional capstone of the United Kingdom's "fail-deadly" assured retaliation architecture (UK Defence Journal: Trident and Online Misinformation). Unlike US ICBMs and French nuclear forces—which incorporate cryptographic Permissive Action Links (PALs) or *contrôle gouvernemental* unlock codes that physically prevent the weapon from arming until an external cryptographic unlock code is transmitted from the head of state—British Trident warheads and Vanguard-class fire-control systems employ sovereign onboard procedural and mechanical launch controls (UK Defence Journal: Trident and Online Misinformation). Instead, the UK enforces a strict **Two-Person Rule** at every stage aboard the submarine, allowing the Commanding Officer, Executive Officer, and Weapon Engineer Officer to arm and launch the boat's Trident II D5 missiles autonomously if—and only if—the UK government is destroyed in a surprise decapitation strike (UK Defence Journal: Trident and Online Misinformation). This guarantees to any adversary that destroying Whitehall, the Northwood Nuclear Operations and Targeting Centre, and VLF transmitter stations triggers autonomous execution of the Prime Minister's pre-positioned directive by the patrolling submarine commander.

Hierarchical Command-and-Control & Key Personnel

- **Normal Peacetime/Crisis Launch Chain:** Prime Minister of the United Kingdom (Sir Keir Starmer) / Designated Nuclear Deputy → Chief of the Defence Staff, Air Chief Marshal Sir Richard "Rich" Knighton KCB ADC FREng (succeeded Admiral Sir Tony Radakin on September 2, 2025) → Nuclear Operations and Targeting Centre (NOTC) at CTF 345 / Northwood Headquarters → VLF transmission stations (Skelton, Cumbria; Anthorn, Cumbria) → Commanding Officer of the patrolling Vanguard-class SSBN (GOV.UK ACM Sir Richard Knighton CDS Appointment; Chatham House NATO NC3 Cyber Resilience).
- **Historical Custodians & Academic Chroniclers:** Documented by constitutional historian Lord Peter Hennessy (*The Secret State*) and former Chief of the Defence Staff General Lord Charles Guthrie, who recounted briefing Prime Minister Tony Blair in May 1997 when Blair wrote his Letters of Last Resort (UK Defence Journal: Prime Minister Letters of Last Resort).

Causal Chronology (1968–2026)

Instituted when the Royal Navy assumed the UK strategic deterrent role with the *Resolution*-class Polaris submarines in 1968–1969 under Prime Minister Harold Wilson, the Letters of Last Resort have been handwritten by every British Prime Minister from Harold Wilson, Edward Heath, James Callaghan, Margaret Thatcher, John Major, Tony Blair, Gordon Brown, David Cameron, Theresa May, Boris Johnson, Liz Truss, and Rishi Sunak to Sir Keir Starmer (July 5, 2024), maintaining an unbroken chain of sovereign command authority across 57 years of Continuous At-Sea Deterrence (UK Defence Journal: Keir Starmer Letters of Last Resort).

50. Vanguard-class (UK SSBNs: HMS Vanguard, Victorious, Vigilant, Vengeance)

Technical Passport & 4-Boat Fleet Registry Matrix

Parameter	Verified Specification	Primary Source
Builder & Homeport	Built by Vickers Shipbuilding and Engineering Ltd (VSEL, now BAE Systems Submarines) at Barrow-in-Furness, Cumbria; Homeported at HM Naval Base Clyde (Faslane), Scotland; Deep maintenance at Devonport Royal Dockyard (Babcock International), Plymouth	Royal Navy Vanguard Class; FAS UK Nuclear Weapons 2024
4-Hull Fleet Registry	1. **HMS Vanguard (S28)**: Commissioned Aug 1993, First Patrol Dec 1994; 2. **HMS Victorious (S29)**: Commissioned Jan 1995, First Patrol Dec 1995; 3. **HMS Vigilant (S30)**: Commissioned Nov 1996, First Patrol Jun 1998; 4. **HMS Vengeance (S31)**: Commissioned Nov 1999, First Patrol Feb 2001	Royal Navy Vanguard Class; FAS Vanguard Class SSBN

Parameter	Verified Specification	Primary Source
Dimensions & Displacement	Length: 149.9 m (491 ft 10 in); Beam: 12.8 m (42 ft); Draft: 12.0 m (39 ft 4 in); Submerged Displacement: 15,900 tonnes (twice the displacement of the predecessor *Resolution*-class Polaris boats)	Royal Navy Vanguard Class; GOV.UK Types of UK Royal Navy Submarine
Nuclear Reactor & Propulsion	1 × Rolls-Royce **PWR2** (Pressurised Water Reactor 2) fitted with Core H (life-of-boat core installed during Long Overhaul Periods); 2 × GEC steam turbines (27,500 shp / 20.5 MW) driving a single shaft with a pump-jet propulsor; Speed: >25 knots submerged	Royal Navy Vanguard Class; Royal Navy Dreadnought Class
Strategic & Tactical Armament	16 missile tubes for UGM-133A Trident II D5 / D5LE SLBMs (under UK policy, patrolling boat carries 8 operational missiles armed with up to 40–48 **Holbrook / Mk 4A** thermonuclear warheads of up to 100 kt yield designed by AWE Aldermaston); 4 × 533 mm (21-inch) torpedo tubes firing Marconi/BAE Spearfish heavyweight torpedoes	FAS UK Nuclear Weapons 2024; Royal Navy Vanguard Class
Sensors & Complement	Thales Type 2054 composite multi-mode sonar suite (bow, flank, and towed array); CK51 search and CH91 attack periscopes integrated into an optronic mast; Complement: 132 officers and ratings per crew (Port and Starboard crews)	Royal Navy Vanguard Class

Engineering Mechanics, PWR2 Core H & UK Holbrook/Mk 4A Warhead Integration

The four Vanguard-class nuclear-powered ballistic missile submarines constitute the United Kingdom's sole nuclear delivery platform (FAS UK Nuclear Weapons 2024). Powered by the Rolls-Royce PWR2 reactor—originally designed specifically for the *Vanguard* class and later adapted for the *Astute*-class SSNs—all four boats received the upgraded **Core H** reactor core during their Long Overhaul Period (Refuel) (LOP(R)) refits at Devonport Royal Dockyard, eliminating the need for subsequent refueling for the remainder of their extended service lives (Royal Navy Vanguard Class; FAS UK Nuclear Weapons 2024). The submarines' 16 missile tubes launch US-leased Trident II D5LE missiles fitted with the British **Holbrook** two-stage thermonuclear warhead (a close aerodynamic and physics sibling of the US LANL W76, featuring a selectable strategic yield up to 100 kt and a sub-strategic primary-only yield option of ~0.3–10 kt), which underwent the Mk 4A Life Extension Programme at Atomic Weapons Establishment (AWE) Aldermaston and Burghfield to integrate the Sandia MC4700 AF&F super-fuze (FAS UK Nuclear Weapons 2024; FAS UK Trident-II D-5). Warheads are stored and mated to missiles inside the Explosive Handling Jetty at the **Royal Naval Armaments Depot (RNAD) Coulport** on Loch Long, 3 km west of HMNB Clyde (FAS UK Nuclear Weapons 2024).

Hierarchical Command-and-Control & Key Personnel

- **Command Chain:** Prime Minister (Sir Keir Starmer) ⟶ Secretary of State for Defence (John Healey MP) ⟶ Chief of the Defence Staff (Air Chief Marshal Sir Richard Knighton) ⟶ First Sea Lord and Chief of the Naval Staff, General Sir Gwyn Jenkins KCB OBE RM (appointed May 15, 2025, succeeding Admiral Sir Ben Key) ⟶ Commander Operations (COMOPS at Northwood) & Commander Submarine Flotilla (COMSUBFLOT at HMNB Clyde) ⟶ Commanding Officers (Port and Starboard crews) (GOV.UK General Sir Gwyn Jenkins First Sea Lord Appointment; GOV.UK ACM Sir Richard Knighton CDS Appointment).
- **Warhead & Enterprise Governance:** Chief of Defence Nuclear (Defence Nuclear Organisation), Madelaine McTernan CB (appointed November 27, 2024); Atomic Weapons Establishment (AWE plc, wholly owned MoD non-departmental public body since July 1, 2021) Chief Executive Officer Nick Elliott and Board Chair Sir John Manzoni (GOV.UK Defence Nuclear Organisation; AWE Annual Report and Accounts 2025).

Causal Chronology & Service Life Extension (1992–2030s)

Ordered between 1986 and 1992 to replace the four *Resolution*-class Polaris SSBNs, the *Vanguard* class has exceeded its original 25-year design life (with *HMS Vanguard* reaching 33 years of service in 2026) (Royal Navy £8.4bn Dreadnought Injection July 2026). Following a 7-year deep maintenance refit of *HMS Vanguard* (2015–2023) at Devonport and the ongoing deep maintenance of *HMS Victorious*, the remaining active boats have executed extended deterrent patrols exceeding 150 to 200 days to sustain CASD until the *Dreadnought* class enters service in the early 2030s (FAS UK Nuclear Weapons 2024).

51. Dreadnought-class (UK Next-Generation SSBNs)

Technical Passport & 4-Boat Construction Matrix

Parameter	Verified Specification	Primary Source
4-Hull Fleet Names	1. **HMS Dreadnought** (lead boat, steel cut Oct 2016, keel laid March 2025); 2. **HMS Valiant** (steel cut Sep 2019); 3. **HMS Warspite** (steel cut Feb 2023); 4. **HMS King George VI** (steel cut September 22, 2025)	GOV.UK Types of UK Royal Navy Submarine; Royal Navy Patrol Welcome & 4th Dreadnought Production Oct 2025
Prime Industrial Alliance	Dreadnought Alliance: Submarine Delivery Agency (SDA), BAE Systems Submarines (Barrow-in-Furness, hull construction & integration), and Rolls-Royce Submarines (Raynesway, Derby, nuclear propulsion plant)	Royal Navy Dreadnought Class
Dimensions & Displacement	Length: 153.6 m (503 ft 11 in; ~150 m class); Submerged Displacement: 17,200 tonnes (15,900 tonnes surfaced); largest submarines ever built for the Royal Navy	Royal Navy Dreadnought Class; GOV.UK Types of UK Royal Navy Submarine
Nuclear Reactor & Propulsion	1 × Rolls-Royce **PWR3** (Pressurised Water Reactor 3) leveraging US-UK naval reactor technology for higher safety, simplified operations, and reduced through-life maintenance; turbo-electric drive with a pump-jet propulsor; 42 km (26.4 miles) of titanium and high-grade piping	Royal Navy Dreadnought Class
Strategic & Tactical Armament	12 missile tubes arranged in three 4-tube "Quad Packs" across five deck levels within the US-UK Common Missile Compartment (CMC), firing UGM-133A Trident II D5LE / D5LE2 SLBMs; transitioning in the late 2030s to the UK **A21/Mk 7 Astraea** replacement nuclear warhead; 4 × 533 mm torpedo tubes for Spearfish Mod 1 heavyweight torpedoes	Royal Navy Dreadnought Class; FAS UK Nuclear Weapons 2024
Complement & Habitation	130 crew members (including 3 chefs and 1 medical doctor), featuring dedicated separate female crew quarters, adaptive diurnal circadian lighting, gym, and classroom facilities	GOV.UK Types of UK Royal Navy Submarine
Program Budget & Schedule	£31 billion approved manufacture budget plus £10 billion contingency fund (with an £8.4 billion contract injection announced on July 30, 2026); entry into service scheduled for the early 2030s	FAS UK Nuclear Weapons 2024; Royal Navy £8.4bn Dreadnought Injection July 2026

Engineering Mechanics, Rolls-Royce PWR3, Common Missile Compartment & A21 Astraea Warhead

The Dreadnought-class (formerly the *Successor* programme) represents a complete propulsion, hydro-acoustic, and warhead transformation of the British sea-based deterrent (Royal Navy Dreadnought Class). Its **Rolls-Royce PWR3** nuclear propulsion plant replaces the UK-indigenous PWR2 design with a hybrid US-UK reactor architecture that delivers extended core life, higher thermal margins, and simplified maintenance (Royal Navy Dreadnought Class). The central hull section integrates three 4-tube "Quad Packs" (12 missile tubes total, compared to 16 on the US *Columbia* class) built to the exact specifications of the US-UK Common Missile Compartment (CMC) (Royal Navy Dreadnought Class). In parallel, under the UK Nuclear Warhead Capability Sustainment Programme, AWE Aldermaston (building the *Pegasus* enriched uranium facility and *Project AURORA* plutonium manufacturing facility) and AWE Burghfield (building the *MENSA* warhead assembly/disassembly facility) are developing the **A21/Mk 7 Astraea** replacement warhead in partnership with the US W93/Mk 7 warhead program for integration onto *Dreadnought* in the late 2030s (FAS UK Nuclear Weapons 2024).

Hierarchical Command-and-Control & Key Personnel

- **Acquisition & Delivery Governance:** Chief of Defence Nuclear Madelaine McTernan CB; Submarine Delivery Agency (SDA) CEO Chris Gardner; BAE Systems Submarines Managing Director Steve Timms; Rolls-Royce Submarines President Steve Carlier; and First Sea Lord General Sir Gwyn Jenkins (GOV.UK Defence Nuclear Organisation; Royal Navy Vanguard Class).

Causal Chronology & 2025–2026 Milestones

Approved by a House of Commons vote of 472 to 117 on July 18, 2016, steel was cut on *HMS Dreadnought* at Barrow-in-Furness in October 2016 (CSIS Trident D5). On September 22, 2025, the fourth and final submarine of the class, *HMS King George VI*, officially entered production at BAE Systems in Barrow-in-Furness (Royal Navy Patrol Welcome & 4th Dreadnought Production Oct 2025). On July 30, 2026, the UK Ministry of Defence injected an additional £8.4 billion into the Dreadnought Alliance to fund Delivery Phase 3 continuation into the early 2030s (Royal Navy £8.4bn Dreadnought Injection July 2026).

52. Continuous At-Sea Deterrent (CASD / Operation Relentless)

Technical Passport & Operational Posture Matrix

Parameter	Verified Specification	Primary Source
Operational Codename & Start Date	**Operation Relentless**; continuous unbroken patrols since April 1969 (initiated by *HMS Resolution* S22); surpassing 57 consecutive years in 2026—the longest sustained military operation in British history	GOV.UK Continuous At Sea Deterrent 50; Royal Navy Silent Service 125 Years Oct 2026
Patrol Requirement	Minimum of 1 Royal Navy ballistic missile submarine (SSBN) submerged and undetected on deterrent patrol at every hour of every day	GOV.UK Dreadnought Submarine Programme
Stockpile Ceiling & Readiness	Overall UK warhead stockpile ceiling raised from 180 to **260 warheads** under the 2021 Integrated Review (~225 estimated in inventory, of which ~120 are operationally available); patrolling SSBN carries 8 missiles and ~40 warheads; missiles held in a **detargeted posture** at several days' notice to fire	FAS UK Nuclear Weapons 2024; FAS Vanguard Class SSBN
Single Integrated Operating Base	HM Naval Base Clyde (Faslane, Gare Loch, Scotland, expanding from 6,800 to 8,200 personnel) and RNAD Coulport (Loch Long, Scotland), protected seaward and landward by **43 Commando Fleet Protection Group Royal Marines**	Delivering for Scotland: Royal Navy in Scotland
NATO Assignment	Assigned to the defence of NATO under the 1962 Nassau Agreement (except where UK supreme national interests are at stake), providing NATO's independent European sea-based nuclear center of decision alongside US forces	GOV.UK UK's Nuclear Deterrent (CASD)

Engineering & Operational Mechanics of the 4-Boat CASD Cycle

Maintaining **Continuous At-Sea Deterrence (CASD)**—codenamed **Operation Relentless**—requires a four-hull fleet so that at least one SSBN is always submerged on patrol while a second boat is in transit or working up in Fleet Operational Standards and Training (FOST), a third is in routine base maintenance at Faslane, and the fourth is in deep refit (Long Overhaul Period) at Devonport (GOV.UK Continuous At Sea Deterrent 50). Before a Vanguard-class SSBN departs HMNB Clyde down the Firth of Clyde into the North Channel and North Atlantic, an *Astute*-class nuclear attack submarine (SSN), Type 23/Type 26 anti-submarine warfare frigate, Merlin HM2 helicopters, and RAF Poseidon MRA1 maritime patrol aircraft execute a sanitizing sweep operation to clear the acoustic transit corridor of foreign intelligence-gathering submarines (Delivering for Scotland: Royal Navy in Scotland). Special nuclear materials (tritium, HEU, plutonium, and non-nuclear warhead components) are transported between AWE Aldermaston/Burghfield and RNAD Coulport by the Defence Nuclear Organisation's Special Escort Group road convoys and between RAF Brize Norton and US facilities via RAF C-17 Globemaster III transport flights (FAS UK Nuclear Weapons 2024).

Hierarchical Command-and-Control & Key Personnel

- **Political & Military Chain:** Prime Minister (Sir Keir Starmer) → Secretary of State for Defence (John Healey MP) → Chief of the Defence Staff (Air Chief Marshal Sir Richard Knighton) → First Sea Lord (General Sir Gwyn Jenkins) → Commander Submarine Flotilla (COMSUBFLOT) at HMNB Clyde (GOV.UK General Sir Gwyn Jenkins First Sea Lord Appointment).

Causal Chronology (1969–2026)

Since *HMS Resolution* sailed on the first Polaris patrol in April 1969, Operation Relentless has completed well over 350 uninterrupted deterrent patrols—first by the four *Resolution*-class Polaris boats (1969–1996) and since December 1994 by the four *Vanguard*-class Trident boats (GOV.UK Continuous At Sea Deterrent 50). In 2025, following the UK Strategic Defence Review, the UK Government also announced the procurement of 12 dual-capable F-35A Lightning II aircraft to join NATO's airborne nuclear mission alongside CASD, while FAS documented the return of US nuclear surety infrastructure to RAF Lakenheath (GOV.UK Chief of the Air Staff Appointment July 2025; FAS Reawakening a Nuclear Legacy Lakenheath Report 2025).

53. Force de frappe (Force de dissuasion, FOST & FAS, Jupiter Bunker / CDAOA / RAMSES / SYDEREC)

Technical Passport & French Nuclear Dyad / NC3 Architecture Matrix

Component	Verified Specification	Primary Source
Total Stockpile & Doctrine	~290 operational nuclear warheads ("strict sufficiency" / *stricte suffisance*); strictly defensive sovereign doctrine of inflicting "absolutely unacceptable damages" (*dommages absolument inacceptables*) on an aggressor's political, economic, and military centers of power, preceded if necessary by an airborne "final warning" (*ultime avertissement*)	FAS Nuclear Notebook: French Nuclear Weapons 2025; Assemblée Nationale Rapport d'Information N° 1112
Annual Deterrence Budget	€6.36 billion (2024), €6.89 billion (2025), and **€7.38 billion (2026)** under the 2024–2030 Military Programming Law (LPM), supporting ~2,000 companies in the French Defense Industrial and Technological Base (BITD)	Sénat Rapport d'Information N° 950 (Sept 2026)
Oceanic Component (FOST)	*Force Océanique Stratégique* (headquartered at Brest, operational base at Île Longue, Crozon Peninsula): 4 *Triomphant*-class SNLEs operating 3 sets of 16 M51 SLBMs (~240 TNO / TNO-2 warheads)	FAS Nuclear Notebook: French Nuclear Weapons 2025
Airborne Component (FAS & FANu)	*Forces Aériennes Stratégiques* (FAS, HQ consolidated at BA 921 Taverny in June 2024; ~40 Rafale B F3-R/F4 aircraft in EC 1/4 *Gascogne* and EC 2/4 *La Fayette* at BA 113 Saint-Dizier, supported by 15 Airbus A330-200 *Phénix* MRTT tankers at BA 125 Istres, plus BA 702 Avord and BA 116 Luxeuil) and *Force Aéronavale Nucléaire* (FANu, ~10 Rafale M aircraft embarking on nuclear carrier *Charles de Gaulle*); ~50 TNA warheads on ASMPA / ASMPA-R missiles	FAS Nuclear Notebook: French Nuclear Weapons 2025; French Ministry of the Armed Forces FAS 60 Years
Presidential Command Bunker (PC Jupiter)	Hardened underground nuclear command post situated beneath the Cour d'Honneur of the **Élysée Palace** in Paris, equipped with electromagnetic shielding, secure video/cryptographic links, and accompanied at all times outside the palace by a presidential mobile command post	FAS Nuclear Notebook: French Nuclear Weapons 2025; Cairn.info Le Président et la Bombe

Component	Verified Specification	Primary Source
3rd Pillar: Nuclear Transmissions (RAMSES & SYDEREC)	**RAMSES** (*Réseau Amont Maillé Stratégique et de Survie*, operational since 1988, modernized via RAMSES III/IV): EMP-hardened mesh network linking PC Jupiter and the COFN in Paris to FOST VLF transmitter stations (**Rosnay**, **Sainte-Assise**, **Kerlouan**, **La Régine / France-Sud**) and FAS nuclear air bases (*Bases Aériennes à Vocation Nucléaire* - BAVN); **SYDEREC** (*Système de Dernier Recours*): last-resort VLF aerostat/tethered balloon and hardened ground antenna system designed to reconstitute and transmit presidential launch orders after a massive nuclear strike	FAS Nuclear Notebook: French Nuclear Weapons 2025; Cairn.info Le Président et la Bombe
Warhead & Propulsion Design Authority	**CEA/DAM** (*Direction des Applications Militaires* of the *Commissariat à l'énergie atomique et aux énergies alternatives*): DAM Île-de-France (Bruyères-le-Châtel, supercomputing), CESTA (Le Barp near Bordeaux, **Laser Mégajoule / LMJ**), Valduc (fissile plutonium/HEU fabrication & pit assembly), Gramat (blast/EMP effects), Ripault (high explosives), and Cadarache (K15 naval reactor cores), plus **EPure** joint UK-France hydrodynamic radiographic facility at Valduc	CEA DAM History & Organization; FAS Nuclear Notebook: French Nuclear Weapons 2025

Engineering Mechanics of French Nuclear Command, Control & Communications (NC3)

Under Article 15 of the Constitution of the Fifth Republic and Article L. 1111-4 of the French Defense Code (derived from the decree of January 14, 1964), the President of the French Republic holds exclusive authority to order the engagement of nuclear forces (Cairn.info Le Président et la Bombe). Firing a French nuclear weapon requires the physical convergence of three independent cryptographic and technical parameters:

1. **Contrôle de l'engagement (Presidential Authentication & Unlock Code):** Transmitted directly from the President of the Republic inside **PC Jupiter** (under the Élysée Palace) or via the presidential mobile briefcase/command post (Cairn.info Le Président et la Bombe).
2. **Contrôle de la conformité de l'emploi (Military Execution Validation):** Verified and transmitted by the Chief of the Defence Staff (*Chef d'état-major des armées* - CEMA) from the *Centre opérationnel des forces nucléaires* (COFN) at the Hexagone Balard / Paris (Cairn.info Le Président et la Bombe).
3. **Target & Weapon Programming:** Transmitted across the EMP-hardened **RAMSES** mesh network (or, if surface hubs are destroyed, the aerostat-borne **SYDEREC** *Système de Dernier Recours*) to the four VLF naval transmission stations (Rosnay, Sainte-Assise, Kerlouan, and France-Sud at La Régine—whose central antenna pylons exceed the height of the Eiffel Tower) for FOST submarines and to **COFAS** (underground at BA 921 Taverny, backed up by BA 942 Lyon-Mont Verdun where **CDAOA** coordinates air defense and airspace corridors) for the FAS Rafale squadrons and hardened "K Buildings" storage vaults at Saint-Dizier, Avord, and Istres (FAS Nuclear Notebook: French Nuclear Weapons 2025; French Ministry of the Armed Forces CTM France-Sud 50 Years).

Hierarchical Command-and-Control & Key Personnel (2026)

- **President of the French Republic (*Chef des Armées*):** Emmanuel Macron (French Ministry of the Armed Forces M51.3 Test Oct 2026).
- **Minister of the Armed Forces and Veterans Affairs:** Catherine Vautrin (following Sébastien Lecornu) (French Ministry of the Armed Forces FAS 60 Years).
- **Chief of the Defence Staff (*Chef d'état-major des armées* - CEMA):** Général d'armée aérienne Fabien Mandon (former Chief of the Military Staff of the President of the Republic *CEMP*; succeeded Général d'armée Thierry Burkhard as CEMA on September 1, 2025) (French Ministry of the Armed Forces Biography of CEMA Fabien Mandon).
- **Chief of Staff of the French Air and Space Force (CEMAAE):** Général d'armée aérienne Jérôme Bellanger (former Commander of the Strategic Air Forces, appointed CEMAAE in September 2024) (French Ministry of the Armed Forces New Air Chief Sept 2024).
- **Commander of the Strategic Air Forces (*Commandant les Forces aériennes stratégiques* - CFAS):** Général de corps aérien Stéphane Virem (headquartered at BA 921 Taverny) (French Ministry of the Armed Forces Interview GCA Stéphane Virem).
- **Commander of Air Defense and Air Operations (*Commandant la Défense aérienne et les opérations aériennes* - CDAOA):** Général de corps aérien Marc Le Bouil (French Ministry of the Armed Forces CDAOA Retrospective 2025).
- **Commander of the Submarine Forces and the Strategic Oceanic Force (ALFOST):** Vice-amiral d'escadre Xavier Petit (former Commanding Officer of SNLE *Le Terrible* Blue Crew; assumed command as ALFOST at Brest on September 2, 2024) (French Ministry of the Armed Forces New ALFOST Sept 2024).
- **Director of Military Applications at CEA (Directeur des applications militaires - CEA/DAM):** Jérôme Demoment (succeeded Vincenzo Salvetti) (CEA Organigramme Nov 2026).

Causal Chronology (1960–2026)

Initiated by the creation of the CEA under General Charles de Gaulle, Frédéric Joliot-Curie, and Raoul Dautry on October 18, 1945 and the first French nuclear test *Gerboise Bleue* (70 kt at Reggane, Algeria) on February 13, 1960, the *Force de frappe* stood its first nuclear alert on October 8, 1964 with a Mirage IVA bomber armed with the AN-11 gravity bomb (CEA DAM History; French Ministry of the Armed Forces FAS 60 Years). After operating a full strategic triad (Mirage IV bombers, 18 S2/S3 IRBMs in hardened silos on the Plateau d'Albion from 1971 to 1996, and *Le Redoutable*-class SSBNs from 1971), President Jacques Chirac dismantled the Plateau d'Albion silos and the Pacific testing center after the final 1995–1996 test series, transitioning warhead certification to the CEA **Simulation Programme** (Tera-10/Tera-100/Tera-1000 supercomputers, Airix radiographic accelerator, and the **Laser Mégajoule** at CESTA) (CEA DAM History). In March 2024, France announced a contract with EDF to resume military tritium production using lithium-6 irradiation in the civilian **Civaux** nuclear power reactors, and on March 2, 2026 at Île Longue, President Emmanuel Macron inaugurated a policy of "advanced deterrence" (*dissuasion avancée*) with European partners alongside the launch of the hypersonic **ASN4G** realization phase (FAS Nuclear Notebook: French Nuclear Weapons 2025; Sénat Rapport d'Information N° 950 (Sept 2026)).

54. Triomphant-class (SNLE-NG: Le Triomphant, Le Téméraire, Le Vigilant, Le Terrible, M51 SLBM)

Technical Passport & 4-Boat / M51 SLBM Matrix

Parameter	Verified Specification	Primary Source
Builder & Homeport	Built by DCN / DCNS / **Naval Group** at Cherbourg Shipyard; Operational Base: **Île Longue** (Brest roadstead, Brittany), supported by the *Service de soutien de la flotte* (SSF) and protected by *Fusiliers Marins*	FAS Le Triomphant SNLE; French Ministry of the Armed Forces Adaptation au M51
4-Hull Fleet Registry	1. **Le Triomphant (S616)**: Laid down Jun 1989, Commissioned Mar 21, 1997; 2. **Le Téméraire (S617)**: Laid down 1994, Commissioned Dec 23, 1999; 3. **Le Vigilant (S618)**: Ordered May 1993, Commissioned Nov 2004; 4. **Le Terrible (S619)**: Ordered Sep 1999, Commissioned Sep 2010 (first built natively for M51)	FAS Le Triomphant SNLE; French Ministry of the Armed Forces Adaptation au M51
Dimensions, Hull Steel & Displacement	Length: 138.0 m (453 ft); Beam: 12.5 m (41 ft; 17 m across aft planes); Draft: 12.5 m; Displacement: 12,640 tonnes surfaced / **14,335 tonnes submerged**; Pressure Hull: **100 HLES** high-elastic-limit steel (100 kg/mm^2 yield strength, equivalent to HY-130) allowing diving depths >400 m	FAS Le Triomphant SNLE
Nuclear Reactor & Propulsion	1 × TechnicAtome (Framatome) / CEA **K15** integrated Pressurized Water Reactor (150 MWt thermal output); 2 × turbo-alternators driving 1 main electric propulsion motor (30.5 MW / 41,500 shp) coupled to a shrouded **pump-jet propulsor** (*hélice-pompe*); 2 × SEMT-Pielstick 8 PA4 V 200 SM auxiliary diesels (900 kW); Speed: >25 knots submerged; Complement: 111 (15 officers, Bleu & Rouge crews)	FAS Le Triomphant SNLE
Strategic Armament: M51 SLBM (M51.1 / M51.2 / M51.3 / M51.4)	16 vertical launch tubes integrated via the **CESAD M51** (*Composante Embarquée du Système d'Armes de Dissuasion*); **ArianeGroup M51 SLBM**: Length 12.0–13.0 m, Diameter 2.30–2.35 m, Launch Mass 53,000–54,000 kg (54 tonnes), 3-stage carbon-epoxy composite solid-propellant motors with flexible-nozzle TVC and nose aerospike, stellar-inertial guidance, Range 8,000–10,000+ km; Carrying 4 to 6 MIRVs: **TNO** (*Tête Nucléaire Océanique*, ~100–150 kt on M51.2) and **TNO-2** (on **M51.3**, which entered operational service on **October 24, 2025**)	CSIS M51; French Ministry of the Armed Forces M51; French Ministry of the Armed Forces M51.3 Acceptance Launch Oct 2026

Parameter	Verified Specification	Primary Source
Tactical Weapons & Combat System	4 × 533 mm bow tubes firing Naval Group **F21** Artemis heavyweight torpedoes and MBDA **SM39 Exocet** anti-ship missiles (18 mixed weapons); Combat System upgraded to **SYCOBS** with Safran SGN inertial navigation and Thales DMUX 80 / DSUV 61 towed array sonar	FAS Le Triomphant SNLE; French Ministry of the Armed Forces Adaptation au M51

Engineering Mechanics, Acoustic Quieting (1,000x Reduction) & M51.3 / TNO-2 Evolution

Designed by DCN (now Naval Group) and TechnicAtome to reduce radiated acoustic noise by a factor of 1,000 (30 dB) compared to the predecessor *Le Redoutable* class, the Triomphant-class (*Sous-Marins Nucléaires Lanceurs d'Engins de Nouvelle Génération* - SNLE-NG) suspends all 750,000 onboard components and 50 km of fluid piping on decoupled elastic cradles inside an anechoic-tiled 100 HLES steel hull (FAS Le Triomphant SNLE). Originally commissioned with the M45 SLBM (on the first three boats), the entire class was converted under the DGA *Cœlacanthe* program during their major refits (*Indisponibilités Périodiques pour Entretien et Réparation* - IPER) between 2010 and 2018 to embark the **ArianeGroup M51** SLBM, the **CESAD M51** fire control system, and the **SYCOBS** combat management system (French Ministry of the Armed Forces Adaptation au M51).

The **M51** SLBM leverages Ariane 5 solid-rocket booster technologies—featuring carbon-epoxy filament-wound motor casings, composite submerged flexible nozzles, a deployable frontal aerospike, and stellar-inertial guidance (CSIS M51). Its evolution spans four increments:

- **M51.1 (2010–2015):** Armed with 4 to 6 TN-75 thermonuclear warheads (110–150 kt) (CSIS M51).
- **M51.2 (2015–Present):** Armed with the CEA/DAM-designed **TNO** (*Tête Nucléaire Océanique*, ~100 kt yield in a stealthier, higher-beta reentry vehicle with enhanced penetration aids) (CSIS M51).
- **M51.3 (Operational October 24, 2025):** Features a newly engineered third stage for extended intercontinental range (>9,000–10,000 km), enhanced stellar-inertial accuracy, and the new **TNO-2** oceanic warhead designed by CEA/DAM (French Ministry of the Armed Forces M51.3 Acceptance Launch Oct 2026).
- **M51.4 (Contracted August 28, 2025):** Ordered by DGA from ArianeGroup on August 28, 2025 (announced September 12, 2025) for deployment in the 2030s aboard both the *Triomphant* class and the future **SNLE 3G** (whose first steel was cut at Cherbourg in March 2024 for service entry around 2035) (French Ministry of the Armed Forces M51.4 Press Release Sept 2025; FAS Nuclear Notebook: French Nuclear Weapons 2025).

Hierarchical Command-and-Control & Key Personnel

- **Operational Command:** President of the Republic (Emmanuel Macron) → Chief of the Defence Staff / CEMA (Général d'armée aérienne Fabien Mandon) → Commander of the Strategic Oceanic Force / ALFOST (Vice-amiral d'escadre Xavier Petit at Brest) → Commanding Officers (Bleu and Rouge crews of *Le Triomphant*, *Le Téméraire*, *Le Vigilant*, and *Le Terrible*) (French Ministry of the Armed Forces New ALFOST Sept 2024).
- **Industrial & Warhead Leads:** DGA (*Délégué général pour l'armement* Emmanuel Chiva), CEA/DAM (Director Jérôme Demoment), Naval Group (CEO Pierre Éric Pommellet), and ArianeGroup (CEO Martin Sion) (French Ministry of the Armed Forces M51.3 Acceptance Launch Oct 2026).

Causal Chronology & October 2026 M51.3 Acceptance Launch

Following *Le Triomphant*'s commissioning in March 1997 and the completion of the four-boat class with *Le Terrible* in September 2010, FOST has maintained at least one SNLE on continuous submerged deterrent patrol at all times (French Ministry of the Armed Forces M51.3 Acceptance Launch Oct 2026). On **October 24, 2025**, the **M51.3** missile equipped with the **TNO-2** warhead officially entered operational service (French Ministry of the Armed Forces M51.3 Acceptance Launch Oct 2026). On **October 5–6, 2026**, following the completion of its IPER modernization refit, the SNLE ***Le Vigilant*** **(S618)** successfully executed an operational acceptance launch of an unarmed **M51.3** missile into the North Atlantic—tracked by the DGA missile range instrumentation ship *Monge* and witnessed in person by President Emmanuel Macron aboard the FREMM frigate *Auvergne* (French Ministry of the Armed Forces M51.3 Acceptance Launch Oct 2026).

55. ASMP-A (Air-Sol Moyenne Portée Amélioré Supersonic Nuclear Missile & TNA Warhead)

Technical Passport & Ramjet Missile Matrix

Parameter	Verified Specification	Primary Source
Official Designations	**ASMP-A** (*Air-Sol Moyenne Portée Amélioré*) & **ASMPA-R** (*Air-Sol Moyenne Portée Amélioré Rénové*); Successor in realization: **ASN4G** (*Air-Sol Nucléaire de 4ème Génération*)	FAS Nuclear Notebook: French Nuclear Weapons 2025; CSIS ASMP / ASMP-A

Parameter	Verified Specification	Primary Source
Prime Missile & Warhead Contractors	Missile: **MBDA France** (formerly Aérospatiale Tactical Missiles, assembled at Selles-Saint-Denis); Warhead: **CEA/DAM** (*Direction des Applications Militaires*)	CSIS ASMP / ASMP-A; FAS Nuclear Notebook: French Nuclear Weapons 2025
Dimensions & Launch Weight	Length: 5.38 m (17 ft 8 in); Body Diameter: 0.38 m (15 in; ~0.50 m across twin lateral air intakes); Launch Weight: 860 kg (1,896 lb)	CSIS ASMP / ASMP-A
Propulsion Architecture	Integral rocket-ramjet (*statoréacteur*): Nozzleless solid-propellant booster inside the combustion chamber accelerates the missile to Mach 2.0 within 5 seconds of drop, after which the booster port covers eject and the liquid-kerosene-fueled ramjet sustains supersonic flight with dual lateral air inlets	CSIS ASMP / ASMP-A
Speed, Range & Flight Profile	Speed: **Mach 3.0** at high altitude (Mach 2.0 at low altitude); Range: **500 km to 600 km** (twice the 300 km range of the legacy ASMP); multi-altitude high-low-high or terrain-hugging *chandelle* trajectory with high-G terminal evasive maneuvers	CSIS ASMP / ASMP-A; FAS Nuclear Notebook: French Nuclear Weapons 2025
Nuclear Warhead (TNA)	**TNA** (*Tête Nucléaire Aéroportée*): CEA/DAM two-stage thermonuclear warhead using insensitive high explosive (IHE), certified entirely through computational and laser-inertial physics via the CEA Simulation Programme; Explosive Yield: **~300 kilotons (kt)**	FAS Nuclear Notebook: French Nuclear Weapons 2025; CSIS ASMP / ASMP-A
Inventory & Delivery Platforms	54 total ASMP-A missiles produced (supporting ~50 operational TNA warheads); carried by **Dassault Rafale B F3-R/F4** (FAS: EC 1/4 *Gascogne* & EC 2/4 *La Fayette* at BA 113 Saint-Dizier, plus BA 116 Luxeuil by 2035) and **Rafale M** (FANu *Flottille* 11F/12F/17F aboard aircraft carrier *Charles de Gaulle*)	FAS Nuclear Notebook: French Nuclear Weapons 2025

Engineering Mechanics, Integral Rocket-Ramjet Physics & ASMPA-R / ASN4G Evolution

The ASMP-A (*Air-Sol Moyenne Portée Amélioré*) is a supersonic, liquid-fueled ramjet nuclear standoff missile engineered by MBDA France to penetrate dense surface-to-air missile defenses and deliver France's strategic or "final warning" (*ultime avertissement*) nuclear strike (CSIS ASMP / ASMP-A). Upon release from the centerline pylon of a Dassault Rafale B or Rafale M fighter, an integrated solid-propellant grain cast directly inside the ramjet combustion chamber ignites, accelerating the 860-kg missile to Mach 2.0 in five seconds (CSIS ASMP / ASMP-A). Once the solid grain burns out, the inlet blanking plugs open, ram air compresses through the twin lateral supersonic intakes, and liquid kerosene is injected and ignited in the now-empty chamber, sustaining speeds up to **Mach 3.0** over a 500–600 km range (CSIS ASMP / ASMP-A). Guidance combines a strapdown laser-gyro inertial navigation system with terrain-contour mapping and radar altimetry, while the CEA/DAM **TNA** (*Tête Nucléaire Aéroportée*, ~300 kt yield) was the first warhead in the French arsenal certified completely via supercomputing and inertial-confinement laser simulation using insensitive high explosives (FAS Nuclear Notebook: French Nuclear Weapons 2025).

In 2016, France initiated the **ASMPA-R** (*Rénové*) mid-life refurbishment program to extend range and aerodynamic performance into the 2030s; following developmental firings in December 2020 and March 2022, the Strategic Air Forces (FAS) executed the first operational evaluation launch of the **ASMPA-R** from a Rafale during **Operation Durandal** on **May 22, 2024** (FAS Nuclear Notebook: French Nuclear Weapons 2025). In **March 2026**, France

officially launched the realization phase of its scramjet-powered hypersonic successor, the **ASN4G** (*Air-Sol Nucléaire de 4ème Génération*), engineered by MBDA and ONERA to fly at speeds exceeding **Mach 5 to Mach 7** with complex hypersonic maneuvering on the future **Rafale F5** standard by **2035** (Sénat Rapport d'Information N° 950 (Sept 2026); French Ministry of the Armed Forces Strategic Role of FAS).

Hierarchical Command-and-Control & Key Personnel

- **Operational Execution Chain:** President of the Republic (Emmanuel Macron) → Chief of the Defence Staff / CEMA (Général d'armée aérienne Fabien Mandon) → Commander of the Strategic Air Forces / CFAS (Général de corps aérien Stéphane Virem at COFAS Taverny) or Commander of the *Force Aéronavale Nucléaire* (FANu, aboard *Charles de Gaulle*) → Rafale B pilot and weapon systems officer (EC 1/4 *Gascogne* & EC 2/4 *La Fayette*, tested 4 times annually in **Operation Poker** simulated nuclear raids) (FAS Nuclear Notebook: French Nuclear Weapons 2025; French Ministry of the Armed Forces Interview GCA Stéphane Virem).

Causal Chronology (1978–2026)

Developed from Aérospatiale's 1978 ramjet design to replace the AN-22 gravity bomb on the Mirage IVP (with the original ASMP entering service in May 1986 carrying the 300 kt TN-81 warhead), the upgraded **ASMP-A** commenced flight testing in December 2000 and entered operational service in October 2009 on the Mirage 2000N at Istres and in July 2010 on the Rafale F3 at Saint-Dizier (CSIS ASMP / ASMP-A). Following the retirement of the Mirage 2000N on June 21, 2018, the Rafale became the sole strike fighter of both FAS and FANu (FAS Nuclear Notebook: French Nuclear Weapons 2025). Following the May 22, 2024 operational evaluation launch of the renovated **ASMPA-R**, President Emmanuel Macron announced on March 18, 2025 at **Luxeuil Air Base (BA 116)** that Luxeuil will be upgraded as France's fourth nuclear air base to host two additional nuclear-capable Rafale F5 squadrons armed with the **ASN4G** hypersonic nuclear missile by 2035 (FAS Nuclear Notebook: French Nuclear Weapons 2025; Sénat Rapport d'Information N° 950 (Sept 2026)).

Section IV: USSR & Russian Federation NC3, Automated Retaliation (Kazbek / Cheget / Perimeter / Dead Hand), Early Warning (SPRN), 12th GUMO, and Strategic Triad Architecture (Entities 56–97)

USSR / Russian Federation End-to-End NC3 Execution Topology

The Soviet and Russian Federation strategic nuclear deterrence architecture operates through two parallel, mutually reinforcing causal command loops: the **Primary Centralized Directive Loop** (Kazbek / Monolit) and the **Autonomous Fail-Deadly Retaliatory Loop** (Perimeter / 15E601).

1. **Aerospace & Terrestrial Detection Phase (SPRN):** Space-based infrared telescopes aboard Oko (US-K / US-KS / US-KMO) and its successor Tundra (EKS 14F142 Kupol) detect ICBM/SLBM boost-phase exhaust plumes against the Earth's limb and background clutter, transmitting encrypted telemetry to the Western Command Post at Serpukhov-15 (Kurilovo) (`55°03'15"N 37°11'15"E`) and the Eastern Command Post at Pivan-1 (Komsomolsk-on-Amur) (`50°20'57"N 137°11'22"E`). Concurrently, over-the-horizon and bistatic/monostatic phased-array radars—legacy Dnestr (5N15), Dnepr (5N86), Daryal (5N79), Volga (70M6), and modern modular Voronezh (77Ya6-M / -DM / -VP)—track midcourse ballistic trajectories and feed fused radar-satellite vectors to the Main Centre for Missile Attack Warning (GTsPRN) at Solnechnogorsk (Timonovo) (`56°11'00"N 37°00'35"E`).
2. **Executive Alerting & Authorization Phase (Krokus → Kazbek → Cheget → Kavkaz):** Validated attack parameters trigger the Krokus display terminals at the General Staff and activate the Kazbek automated strategic nuclear command network. Three portable Cheget subscriber terminals—assigned to the President of the Russian Federation, the Minister of Defense, and the Chief of the General Staff—receive the threat profile and link the national command authority via the Kavkaz (Kavkaz-1 through Kavkaz-7) protected telecommunications backbone.
3. **Directive Cryptographic Unblocking & Combat Control Phase (Monolit / Signal-A / Agat):** Upon two-of-three executive authorization codes, the General Staff Operations Directorate and the RVSN Central Command Post at Vlasikha (Odintsovo-10) (`55°41'14"N 37°11'02"E`) or the underground reserve complexes at Kosvinsky Kamen (Granit) (`59°31'00"N 59°03'43"E`) and Mount Yamantau (Mezhgorye) (`54°15'18"N 58°06'07"E`) compile the formatted strike package (Preliminary and Direct Commands plus warhead unblocking codes held under 12th GUMO cryptographic architecture). The directive propagates down the Monolit (15E556) strategic C2 hierarchy to divisional and regimental Signal-A (15E557 / ASBU) terminals, airborne relays (Il-80 Maxdome, Il-82, Tu-142MR Orel), and naval Agat combat management consoles.
4. **Autonomous Retaliatory Bypass Phase (Perimeter 15E601 / Dead Hand / Sirena 15A11 & 15Zh75):** When placed on combat alert during a crisis, Perimeter (15E601) monitors seismic, ionizing radiation, electromagnetic pulse, and atmospheric overpressure sensors while polling the General Staff link. If physical nuclear detonations are confirmed and communication with the General Staff is severed, duty officers in the deep spherical command bunker authorize the launch of 15P011 / 15A11 (or road-mobile 15P175 Sirena / 15Yu75 / 15Zh75) command rockets carrying the 15B99 radio-command payload. At a 1,000 km apogee, the 15B99 transmitter broadcasts ultra-high-power launch and unblocking codes directly to autonomous receivers on silo/mobile ICBMs (R-36M2 Voevoda, RS-28 Sarmat, RT-2PM2 Topol-M, RS-24 Yars, 15Yu71 Avangard, legacy BZhRK Molodets / projected Barguzin), SSBNs (Project 955 Borei with R-30 Bulava, Project 667BDRM Delfin with R-29RMU2 Sineva and R-29RMU2.1 Layner, legacy Project 941 Akula), and strategic bombers (Tu-160, Tu-95MS armed with Kh-102).

Subsystem Group IV.A: NC3, Automated Retaliation, C2 Subsystems & Airborne Command Posts (Entities 56–71)

56. Cheget (GRAU 15E601 / 15Э601 Nuclear Briefcase — Subscriber Terminal of Kazbek)

- **GRAU & Industrial Index:** Subscriber terminal `Абонентский комплекс «Чегет»` within the Kazbek automated strategic command system; frequently cross-indexed in Western and Russian public literature with the `15Э601` (`15E601`) strategic command index, whereas in strict GRAU nomenclature `15Э601` denotes the Perimeter command post complex while `Чегет` is the portable executive terminal of Kazbek (developed under the `Казбек` / `Кавказ-1` R&D program).

- **Design Bureau & Manufacturer:** Research Institute of Automatic Equipment (NII Avtomatiki / НИИА, now JSC Concern Avtomatika, Moscow, `55°49'35"N 37°37'18"E`), in cooperation with the Scientific Research Institute of Communication Systems and Control (MNIIRS) and the Penza Radio Factory (`Производственное объединение «Старт» / Радиозавод`).
- **Chief Designers & Key Personnel:** Chief Designer of the Kazbek/Cheget complex **Vladimir Ivanovich Bessonov** (NII Avtomatiki) and Academician **Vladimir Sergeyevich Semenikhin** (Deputy Minister of Radio Industry of the USSR, Scientific Supervisor of Automated Control Systems); commissioned under General Secretary **Yuri Vladimirovich Andropov** (development initiated 1981–1983) and placed into operational service with General Secretary **Mikhail Sergeyevich Gorbachev** in March 1985 (Cheget Profile).
- **Deployment Coordinates & Operational Custody:** Three active terminals are maintained in continuous 24/7 custody by naval-uniformed operator-officers of the 9th Directorate of the General Staff:
  1. **President of the Russian Federation:** The Kremlin, Moscow (`55°45'07"N 37°37'03"E`) and presidential residences (Novo-Ogaryovo `55°43'55"N 37°12'02"E`) / presidential aircraft (`Il-96-300PU`).
  2. **Minister of Defense of the Russian Federation:** National Defense Management Center (NTsUO), Frunzenskaya Naberezhnaya 20, Moscow (`55°43'38"N 37°35'16"E`).
  3. **Chief of the General Staff of the Armed Forces:** General Staff Main Building, Znamenka Street 19, Moscow (`55°44'55"N 37°36'14"E`).
  4. **Historical Archival Exhibit:** An early-1990s decommissioned Cheget terminal is displayed at the Boris Yeltsin Presidential Center in Yekaterinburg (`56°50'41"N 60°35'29"E`) (Wikipedia: Cheget).
- **Engineering-Technical Parameters:**
  - **Physical Construction:** Hardened black composite briefcase (approximately `40 × 30 × 15 cm`, mass `11 kg`), secured by a three-digit mechanical combination lock and an electronic magnetic/flash authorization keycard (`Спецжетон`).
  - **Control Interface:** As televised by the Russian Ministry of Defense's Zvezda network in December 2019, the internal console features a matrix display section, an individual flashcard slot, and a `КОМАНДА` ("Command") block with a white **`ПУСК`** ("Launch/Execute") button and a red **`ОТМЕНА`** ("Cancel") button (Al Jazeera NC3 Analysis).
  - **Cryptographic & Telemetry Link:** Interfaces via wired plug-in or short-range encrypted radio/cellular-style transceiver with the portable Kavkaz-1 (`Кавказ-1`) briefcase radio station carried by accompanying communications officers.
- **Causal Command Sequence & Historical Event Log:**
  1. When SPRN sensors detect a ballistic missile launch, the duty officer at Solnechnogorsk transmits an alert packet via the Krokus subsystem into Kazbek.
  2. All three Cheget terminals emit an acoustic and visual alarm simultaneously, displaying trajectory counts, time-to-impact, and pre-planned retaliatory strike options (`Варианты ответно-встречного удара`).
  3. The President consults the Minister of Defense and Chief of the General Staff over the Kavkaz-7 secure conference bridge, inserts the individual authorization flashcard, and presses the white `ПУСК` button to transmit his portion of the authorization cipher (`Шифр санкции`).
  4. **Documented Operational Activation (25 January 1995):** Following the launch of a four-stage Black Brant XII sounding rocket from the Andøya Rocket Range in Norway (`69°17'39"N 16°01'15"E`), the Olenegorsk radar tracked a profile resembling a US Trident SLBM high-altitude EMP precursor strike. All three Cheget terminals (held by President Boris Yeltsin, Defense Minister Pavel Grachev, and Chief of the General Staff Mikhail Kolesnikov) were switched into combat mode for the first and only documented time in history before Oko satellites confirmed the payload trajectory was moving away from Russian territory (Russian Strategic Nuclear Forces C2).

57. Kazbek (Automated Strategic Nuclear Command System — ASBU VGK)

- **GRAU & Industrial Index:** `АСБУ «Казбек»` (`Автоматизированная система боевого управления Верховного Главнокомандования Стратегическими ядерными силами` — Automated Combat Control System of the Supreme High Command for Strategic Nuclear Forces).
- **Design Bureau & Manufacturer:** Research Institute of Automatic Equipment (NII Avtomatiki / НИИА, Moscow), under the overarching coordination of the Main Directorate of the General Staff and the Ministry of Radio Industry of the USSR.
- **Chief Designers & Key Personnel:** Academician **Vladimir Sergeyevich Semenikhin** (General Designer of Strategic Automated Systems), **Vladimir Ivanovich Bessonov** (Chief Designer of Kazbek at NII Avtomatiki), and Colonel General **Varfolomey Vladimirovich Korobushin** (First Deputy Chief of the Main Staff of the RVSN, who co-architected the operational algorithms for launch-on-warning / `ответно-встречный удар`).
- **Deployment Coordinates:** Central nodes at the General Staff Operations Directorate (Arbat Military District, Moscow, `55°44'55"N 37°36'14"E`), the alternate General Staff bunker at **Chekhov / Sharapovo** (`55°09'27"N 37°22'48"E`), the RVSN Central Command Post at Vlasikha (`55°41'14"N 37°11'02"E`), and the reserve strategic command complex at Kosvinsky Kamen (`59°31'00"N 59°03'43"E`).
- **Engineering-Technical Parameters:**
  - **Tri-Component Architecture:** Kazbek integrates three primary functional subsystems:
    1. **Krokus:** The front-end ingestion and display subsystem receiving real-time missile attack warning telemetry from SPRN.
    2. **Cheget:** The mobile subscriber terminals providing the political-military leadership with situational display and cryptographic sanction input.
    3. **Kavkaz:** The multi-redundant wired, satellite, tropospheric, and airborne telecommunications transport layer (`Kavkaz-1` through `Kavkaz-7`).
  - **Decision Window & Latency:** Engineered in 1974–1983 in direct response to the deployment of US Pershing II IRBMs in Western Europe (flight time 6–8 minutes) and forward-deployed Trident C4 SLBMs, compressing the entire detection-to-authorization cycle from 15+ minutes to **under 180 seconds** (FAS Russian C3I Guide).
- **Causal Command Sequence:**
  1. Kazbek does not autonomously fire weapons; it enforces a cryptographic "two-man/three-terminal rule" between the political leadership and the General Staff (Nuclear Information Service NC3 Report).
  2. Once the President and Minister of Defense transmit their sanction codes via Cheget, the General Staff's Central Command Post (`ЦКП ГШ`) combines the political sanction with the military warhead unblocking code (`Код разблокирования ЯБП`) and injects the formatted strike directive into the Monolit (RVSN), Agat (VMF), and Long-Range Aviation C2 networks.

58. Perimeter (GRAU 15E601 / 15Э601 Automated Retaliatory Command System)

- **GRAU & Industrial Index:**

    - **Overarching System & Command Post Complex:** 15Э601 «Периметр» (15E601 Perimeter) (Missilery.info 15A11/15E601).
    - **Modernized System (post-1990):** «Периметр-РЦ» (Perimeter-RTs) (Russian Strategic Nuclear Forces C2).
    - **Command Rocket Complex:** 15П011 (15P011) housing the 15A11 (15A11) command ICBM in hardened silo launcher 15П716 (15P716) with radio-command warhead 15Б99 (15B99) (Encyclopedia Astronautica: Perimetr 15A11).
- **Design Bureau & Manufacturer:**
    - **Prime Rocket & Complex Integrator: OKB-586 / KB Yuzhnoye** (Dnepropetrovsk, Ukrainian SSR, 48°26'02"N 34°59'55"E), mandated by CPSU Central Committee and Council of Ministers Decree No. 695-227 of **30 August 1974** (Encyclopedia Astronautica: Perimetr 15A11).
    - **Command Post & Radio-Command Payload (15Б99 / 15Э601): OKB "Impuls" at the Leningrad Polytechnic Institute (now JSC NPO Impuls)**, Saint Petersburg (60°00'28"N 30°23'26"E) (Wikipedia RU: Sistema Perimetr).
    - **Serial Warhead Production: NPO Strela** (Orenburg, 51°47'12"N 55°06'30"E) (Ruzhany Strategic RVSN Archive).
    - **Guidance, Silo & Sensor Contributors: NPO AP** (guidance), **KBSM** (silo 15П716), **TsKB Geofizika** (nuclear detonation sensors), **VNIIEF (KB-11, Sarov)** and **Kharkiv Institute of Physics and Technology** (radiation/EMP hardening verification), and **4th TsNII MO** (ballistic/operational algorithms) (Wikipedia RU: Sistema Perimetr).
- **Chief Designers & Key Personnel:**
    - **Vladimir Fyodorovich Utkin** — General Designer of KB Yuzhnoye (overall rocket complex 15П011 and missile 15A11).
    - **Valery Ivanovich Melnik** & **Taras Nikolayevich Sokolov** — Chief Designers at OKB/NPO Impuls (autonomous C2 complex 15Э601 and radio-command payload 15Б99).
    - **Nikolai Alekseyevich Pilyugin** (NPO AP, flight control), **Anatoly Fyodorovich Utkin** (KBSM, silo 15П716), **G. F. Ignatyev** (TsKB Geofizika), **Yevgeny Borisovich Volkov** (4th TsNII MO), and Colonel **Vladimir Yevgenyevich Yarynich** (4th TsNII MO / General Staff C3 architect) (Wikipedia: Dead Hand).
- **Deployment Coordinates:**
    - **Primary Operational Deployment (510th Guards Rocket Regiment, 7th Guards Rocket Division, Vypolzovo / Ozernyy, Tver Oblast):** Combat Launch Complex BRK-6 (Military Unit в/ч 52642), headquartered at 57°52'00"N 33°40'00"E (with silos dispersed near 57°50'45"N 33°32'18"E) (Ruzhany Strategic RVSN Archive).
    - **Strategic Underground Command Post:** Kosvinsky Kamen (Granit) in Sverdlovsk Oblast (59°31'00"N 59°03'43"E).
    - **Flight Test Range:** NIIP-5 Baikonur Cosmodrome, Kazakhstan (Site 175 / 131, 45°59'20"N 63°33'40"E), where 7 flight trials were conducted between 26 December 1979 and 13 November 1984 (Encyclopedia Astronautica: Perimetr 15A11).
- **Engineering-Technical Parameters:**
    - **Operational Dates:** Placed on combat duty in **January 1985**; original 15A11 complex temporarily stood down in June 1995 under START I rules and upgraded to **Периметр-РЦ (Perimeter-RTs)** utilizing solid-propellant Topol/Yars-derived command rockets; confirmed active in December 2011 and 2024–2025 by RVSN Commander Colonel General **Sergey Viktorovich Karakayev** (Wikipedia: Dead Hand).
    - **Sensor Matrix:** Network of autonomous ground sensors measuring seismic P/S-wave shock, atmospheric overpressure, ionizing gamma/neutron flux, and nuclear electromagnetic pulse (NEMP) across strategic targets.
    - **Hardening:** Tested at the Novaya Zemlya nuclear test site (73°22'30"N 54°44'00"E) and VNIIEF laboratories to withstand nuclear blast overpressures exceeding 100+ atmospheres and severe ionospheric Compton-electron blackout (Wikipedia RU: Sistema Perimetr).
- **Causal Command Sequence:**
    1. **Crisis Pre-Activation:** During a high-alert nuclear crisis, the Supreme Commander-in-Chief or General Staff switches 15Э601 Perimeter from dormant standby into active combat monitoring mode.
    2. **Four-Stage Logical Gate (`If/Then` Causal Chain):**
        - *Gate 1:* Is the system armed (ВКЛЮЧЕНО) by executive sanction?
        - *Gate 2:* Do the TsKB Geofizika seismic, barometric, optical, and radiation sensors register physical nuclear detonations on Soviet/Russian territory?
        - *Gate 3:* Is the heartbeat communication link with the General Staff Central Command Post (ЦКП ГШ) severed?
        - *Gate 4:* Has the preset silence timer (15 to 60 minutes) elapsed with zero cancellation signal from Kazbek?
    3. **Sanction Transfer & Command Rocket Launch:** When all four gates evaluate true, 15Э601 unlocks the launch authority for the duty officers sealed inside the deep underground spherical bunker (шар). Upon the duty officer's execution command, the 15A11 (or 15Ю75/15Ж75) command missile launches out of its 15П716 silo, climbs to a 1,000 km suborbital trajectory across Russian territory, and broadcasts the encrypted Приказ на пуск and warhead unblocking codes directly to every ICBM silo, mobile launcher, SSBN receiver, and airborne bomber (Wikipedia: Dead Hand).

59. Dead Hand (Western Designation & Operational Architecture of Perimeter)

- **GRAU & Industrial Index:** Western intelligence and strategic studies designation (**"Dead Hand"** / Russian literal translation «Мёртвая рука») for the Soviet/Russian 15E601 Perimeter fail-deadly retaliatory architecture.
- **Design Bureau & Institutional Provenance:** Architectural concept developed jointly by the **4th Central Research Institute of the Ministry of Defense (4th TsNII MO, Yubileyny / Korolyov, 55°55'52"N 37°51'14"E)**, **OKB/NPO Impuls** (Leningrad), and the **General Staff Operations Directorate**, and disclosed in Western open literature starting in 1993 by **Bruce G. Blair** (Brookings Institution) and **Valery Yarynich** in the *New York Times* (8 October 1993) and documented extensively by **David E. Hoffman** in *The Dead Hand: The Untold Story of the Cold War Arms Race and Its Dangerous Legacy* (2009) (National Security Archive: Hoffman Excerpt).
- **Chief Designers & Key Personnel:**

    - **Valery Yevgenyevich Yarynich** (Colonel, 4th TsNII MO / General Staff C3 planner who co-authored the operational concept and later published *C3: Nuclear Command, Control, Cooperation*, 2003) (Archive.org: Yarynich C3).
    - **Vitaly Leonidovich Kataev** (Senior Defense Industry Department official of the CPSU Central Committee whose archival notes in the Hoover Institution confirmed that Perimeter had been placed on combat duty by 1985) (RussianForces.org: Kataev Archive).
    - **Dr. Pavel Podvig** (Director, Russian Strategic Nuclear Forces Project, who documented the human-in-the-loop predelegation architecture) (RussianForces.org: Dead Hand Analysis).
- **Deployment Coordinates:** Operational command and control nodes at Kosvinsky Kamen (`59°31'00"N 59°03'43"E`) and Vlasikha (`55°41'14"N 37°11'02"E`); archival records held at the Hoover Institution Library & Archives (Stanford University, `37°25'39"N 122°09'58"W`, *Vitaly Kataev Papers*) and the National Security Archive (George Washington University, `38°54'00"N 77°02'54"W`).
- **Engineering-Technical & Strategic Parameters:**
    - **Semi-Automatic Predelegation vs. Fully Autonomous Doomsday Machine:** During the 1974–1977 conceptual phase, Soviet engineers evaluated a fully automatic loop capable of launching ICBMs without any human intervention. As documented by David Hoffman and Pavel Podvig, the General Staff rejected the zero-human blueprint as unacceptably hazardous and implemented a **semi-automatic fail-deadly predelegation architecture** (RussianForces.org: Dead Hand Analysis).
    - **Strategic Stabilizer Function ("Buffer Against Hasty Launch-on-Warning"):** Rather than hair-triggering a strike on ambiguous SPRN early-warning signals, Dead Hand was designed as a **psychological and institutional stabilizer**: the Soviet leadership could arm the system upon receiving an ambiguous warning and ride out the attack (`ответный удар` — retaliatory strike after impact), confident that even total decapitation of Moscow would still result in guaranteed retaliation (Wikipedia: Dead Hand).
- **Causal Command Sequence:**
    1. **Political Predelegation:** The Supreme Commander-in-Chief activates the system in a crisis, delegating conditional launch authority to the underground bunker crew.
    2. **Decapitation Verification:** Only after physical nuclear detonations are registered on Russian soil AND the General Staff communication links are dead does the computer unlock the launch console for the three duty officers inside the underground sphere (`шар`), who then execute the launch of the command rockets (Wikipedia: Dead Hand).

60. Oko (US-K / US-KS / US-KMO Space-Based Infrared Early Warning Constellation)

- **GRAU & Industrial Index:**
    - **1st Generation System `УС-К` (`US-K` — Управляемый Спутник Континентальный):** System index **`73Д6` (`73D6`)**, spacecraft index **`11Ф624` (`11F624`)**, operating in highly elliptical Molniya orbits (Wikipedia: Oko).
    - **1st Generation Geostationary System `УС-КС` (`US-KS` — Управляемый Спутник Континентальный Стационарный):** System index **`74Д6` (`74D6`)**, spacecraft index **`11Ф649` (`11F649`)**, operating in geostationary orbit (GEO).
    - **2nd Generation System `УС-КМО` (`US-KMO` — Управляемый Спутник Континентальный Модернизированный Оптический / «Око-1»):** System index **`71Х6` (`71Kh6`)**, spacecraft index **`17Ф118` (`17F118`)**, operating in GEO (Everyday Astronaut: EKS & Oko History).
- **Design Bureau & Manufacturer:**
    - **System Integrator & Infrared Payload: TsNII "Kometa" (`ЦНИИ «Комета»`, formerly KB-1 / TsNII-108)**, Moscow (`55°42'29"N 37°39'52"E`) (RussianSpaceWeb: EKS/Oko).
    - **Spacecraft Bus & Final Assembly: NPO Lavochkin (`НПО им. С. А. Лавочкина`)**, Khimki, Moscow Oblast (`55°54'12"N 37°25'18"E`).
    - **Infrared Telescope Optics: State Optical Institute (GOI im. S. I. Vavilova)** and **LOMO**, Saint Petersburg.
- **Chief Designers & Key Personnel:**
    - **Anatoly Ivanovich Savin** — Academician, Director and General Designer of TsNII "Kometa" (architect of the Oko and Oko-1 space-based early warning systems).
    - **Vladislav Georgiyevich Repin** — Chief Designer of the SPRN algorithmic and systemic integration.
    - **Georgy Nikolayevich Babakin**, **Sergey Sergeyevich Kryukov**, and **Vyacheslav Mikhailovich Kovtunenko** — Chief Designers at NPO Lavochkin responsible for the `11F624`, `11F649`, and `17F118` spacecraft platforms.
- **Deployment Coordinates (Ground Control & Launch Sites):**
    - **Western Ground Command & Control Center (`ЗКП СПРН`): Serpukhov-15 (Kurilovo)**, Kaluga/Moscow Oblast border (`55°03'15"N 37°11'15"E`) (Wikipedia: Oko).
    - **Eastern Ground Command & Control Center (`ВКП СПРН`): Pivan-1**, near Komsomolsk-on-Amur, Khabarovsk Krai (`50°20'57"N 137°11'22"E`) (Wikipedia: Oko).
    - **Launch Sites: Plesetsk Cosmodrome** (`62°55'32"N 40°34'40"E`) using `Molniya-M (8K78M)` with `Blok-2BL` upper stage for `US-K`; **Baikonur Cosmodrome** (`46°02'24"N 63°01'55"E`) using `Proton-K (8K82K)` with `Blok-DM-2` upper stage for `US-KS` and `US-KMO` (Wikipedia: Oko).
- **Engineering-Technical Parameters:**
    - **Deployment History:** 101 total satellites launched between **19 September 1972** (`Kosmos-520`) and **30 March 2012** (`Kosmos-2479`, the final `71Kh6 US-KMO`, which failed in April 2014; final `US-K Kosmos-2469` launched 30 September 2010) (Wikipedia: Oko).
    - **Orbital Mechanics (`US-K`):** Molniya highly elliptical orbit (`Apogee ~39,700 km`, `Perigee ~600 km`, `Inclination 62.8°–63.8°`, `Period 718 minutes / 12 hours`),requiring a minimum of 4 satellites for 24-hour coverage of US ICBM fields (`Viewing geometry tangent to Earth's limb`) and 9 satellites for full constellation redundancy.
    - **Spacecraft & Sensor Specifications (`US-K` vs `US-KMO`):**
        - `US-K (11F624)`: Mass `2,400 kg` (`1,250 kg` dry), length `2.0 m`, diameter `1.7 m`, equipped with a `350 kg` narrow-field infrared telescope (`50 cm` primary mirror) plus wide-field IR/visible TV cameras cooled by liquid nitrogen/passive radiators, detecting plume radiation in the `2.7 μm` and `4.3 μm` $\text{H}_2\text{O}/\text{CO}_2$ spectral bands against the dark space backdrop at the Earth's limb.

        - US-KMO (71Kh6 / 17F118): Mass 2,600 kg, equipped with a 1.0 m aperture mirror telescope capable of nadir ("looking down") detection of missile plumes directly against sunlit cloud tops and terrain clutter from geostationary orbit (35,786 km).
    - **On-Board Self-Destruct Anomaly:** Early US-K satellites carried an АРО (Аварийный подрыв объекта) explosive charge designed to prevent capture or uncontrolled re-entry; between 1976 and 1983 (Kosmos-1481), 11 satellites prematurely detonated in orbit due to command receiver glitches before the self-destruct charge was disabled (Wikipedia: Oko).
- **Causal Command Sequence & Historical Event Log:**
    1. **26 September 1983 False Alarm Incident:** Kosmos-1382 (US-K satellite at apogee) registered specular solar reflection off high-altitude cirrus clouds over North Dakota as the launch of 1, and subsequently 5, Minuteman III ICBMs. Duty Officer Lieutenant Colonel **Stanislav Yevgrafovich Petrov** at Serpukhov-15 (55°03'15"N 37°11'15"E) overrode the automated computer high-confidence РАКЕТНОЕ НАПАДЕНИЕ ("Missile Attack") flag because ground-based Daryal/Dnepr horizon radars showed zero incoming targets and an actual US first strike would involve hundreds of missiles rather than five (Wikipedia: Oko).
    2. **10 May 2001 Serpukhov-15 Fire:** A catastrophic electrical fire in the Serpukhov-15 control building severed command links for 12 hours, causing four US-K satellites (Kosmos-2340, Kosmos-2351, Kosmos-2368, Kosmos-2342) to drift out of their orbital slots (Wikipedia IT: Oko).

61. Tundra (EKS 14F142 Kupol Early Warning Satellite Constellation)

- **GRAU & Industrial Index:**
    - **Overarching Space System:** ЕКС «Купол» (Единая космическая система обнаружения и боевого управления — Integrated Space System for Detection and Combat Control), GRAU index **14К032 (14К032)** (Everyday Astronaut: EKS 6).
    - **Spacecraft Index: 14Ф142 «Тундра» (14F142 Tundra)**, launched atop Soyuz-2.1b (14A14-1B) with Fregat-M (14S44) upper stage under payload fairing 14C737 (14S737) (RussianSpaceWeb: Tundra 14F142).
- **Design Bureau & Manufacturer:**
    - **Prime System & Payload Integrator: JSC Corporation "Kometa" (АО «Корпорация „Комета“», formerly TsNII Kometa)**, Moscow (55°42'29"N 37°39'52"E) (RussianSpaceWeb: EKS System).
    - **Spacecraft Bus & Final Assembly: PJSC RSC "Energia" (РКК «Энергия» им. С. П. Королёва)**, Korolyov, Moscow Oblast (55°55'05"N 37°48'02"E) — selected in 1999–2000 over NPO Lavochkin, utilizing the unified USP (Универсальная космическая платформа / *Yamal*-derived) unpressurized bus (RussianSpaceWeb: Tundra 14F142).
- **Chief Designers & Key Personnel:**
    - **Viktor Pavlovich Misnik** — General Director and General Designer of JSC Corporation "Kometa" (chief architect of the 14К032 EKS system).
    - **Nikolai Nikolayevich Sevastyanov** and **Igor Nikolayevich Ozar** — Heads of RSC Energia overseeing the 14F142 spacecraft platform production.
- **Deployment Coordinates (Launch & Ground Control):**
    - **Launch Complex: Plesetsk Cosmodrome**, Site 43, Pads 3 and 4 (62°55'44"N 40°27'25"E), Arkhangelsk Oblast (RussianSpaceWeb: EKS-5).
    - **Ground Command Centers:** Upgraded Western Command Post at **Serpukhov-15 (Kurilovo)** (55°03'15"N 37°11'15"E) and Eastern Command Post at **Pivan-1 (Komsomolsk-on-Amur)** (50°20'57"N 137°11'22"E), linked directly to the Main Centre for Missile Attack Warning at **Solnechnogorsk** (56°11'00"N 37°00'35"E) (The Space Review: EKS).
- **Engineering-Technical Parameters:**
    - **Launch & Orbital Registry (6 Spacecraft Launched 2015–2022):**
        1. Kosmos-2510 (EKS-1): Launched 17 November 2015 (NORAD 41032, 2015-066A).
        2. Kosmos-2518 (EKS-2): Launched 25 May 2017 (NORAD 42719, 2017-027A).
        3. Kosmos-2541 (EKS-3): Launched 26 September 2019 (NORAD 44552, 2019-065A).
        4. Kosmos-2546 (EKS-4): Launched 22 May 2020 (NORAD 45608, 2020-031A) (RussianSpaceWeb: EKS-4).
        5. Kosmos-2552 (EKS-5): Launched 25 November 2021 (NORAD 49503, 2021-114A) (RussianSpaceWeb: EKS-5).
        6. Kosmos-2563 (EKS-6): Launched 2 November 2022 (NORAD 54227, 2022-146A) (RussianSpaceWeb: EKS-6).
    - **Orbital Parameters:** Highly Elliptical Orbit (Molniya-type, Perigee ~1,625 km, Apogee ~38,550 km, Inclination 63.8°, Period 717.9 minutes) (RussianForces.org: First Tundra Launch).
    - **Dual-Phenomenology & Emergency NC3 Transponder Payload (БАОБУ):** In addition to multi-spectral staring focal-plane array IR telescopes capable of tracking dim upper-stage plumes and computing impact vectors onboard, each 14F142 Tundra spacecraft carries an onboard **Combat Control Transponder (Бортовая аппаратура обеспечения боевого управления — БАОБУ)** integrated into the Kazbek/Monolit network to relay emergency retaliatory launch orders to ICBM divisions and SSBNs (Deagel: EKS 14F142).
- **Causal Command Sequence:**
    1. The 14F142 focal-plane IR array detects boost-phase plumes across both continental ICBM fields and oceanic SLBM patrol areas, processes the trajectory azimuth and burn termination point onboard, and downlinks the target track to Serpukhov-15 / Pivan-1.
    2. Simultaneously, if ground-based C2 relays are degraded, the General Staff uplinks encrypted launch directives to the 14F142 БАОБУ payload, which rebroadcasts the execution signal from its 38,550 km apogee across the entire Eurasian landmass (The Space Review: EKS).

62. Kavkaz (Kavkaz-1 through Kavkaz-7 Secure State/Military Communication System)

- **GRAU & Industrial Index:** АСУ специального назначения «Кавказ» (Кавказ-1 through Кавказ-7 — Automated Special-Purpose State and Military Strategic Communication System).
- **Design Bureau & Manufacturer:**

    - **Lead Developer: Moscow Scientific Research Institute of Radio Communications (МНИИРС — Московский научно-исследовательский институт радиосвязи)**, Moscow (55°44'18"N 37°42'25"E), in cooperation with **NII Avtomatiki (НИИА)**, **TsNII Svyazi (ЦНИИС)**, and **NPO Impuls**.
- **Chief Designers & Key Personnel:**
    - **Anton Petrovich Bilenko** — General Director and Chief Designer of MNIIRS (lead architect of satellite and special strategic communications within the Kavkaz and Perimeter frameworks) (Wikipedia RU: Sistema Perimetr).
    - **Vladimir Ivanovich Bessonov** (NII Avtomatiki — cryptographic and subscriber integration with Kazbek).
    - **Viktor Ivanovich Timoshenko** (Major General, veteran developer of Soviet strategic mobile communications who documented the creation of the Kavkaz mobile network) (Russian Defense Policy: Kavkaz, Perimeter, and Dead Hand).
- **Deployment Coordinates:** Central switching hubs at the Kremlin Special Communications Center (FSO, Moscow 55°45'07"N 37°37'03"E), General Staff Communications Center (Узел связи ГШ «Рубин», Moscow 55°44'55"N 37°36'14"E), underground bunker at **Chekhov / Sharapovo** (55°09'27"N 37°22'48"E), and airborne relays aboard the **Special Flight Squadron "Rossiya"** (Il-96-300PU at Vnukovo-2 55°35'46"N 37°16'03"E) and **8th Special Purpose Aviation Division** (Il-80 at Chkalovsky 55°52'42"N 38°03'42"E).
- **Engineering-Technical Parameters (Subsystem Breakdown Kavkaz-1 to Kavkaz-7):**
    - **Kavkaz-1 (Кавказ-1):** Portable briefcase radio transceiver and cryptographic modem unit carried by communications officers alongside the Cheget operator, linking the Cheget terminal to the nearest vehicle, aircraft, or stationary Kavkaz node.
    - **Kavkaz-2 (Кавказ-2):** Special armored VIP vehicle communications complex (ЗИЛ-41047 / Mercedes-Pullman / Aurus Senat escort communications vans) providing continuous encrypted voice and data links while the President or Defense Minister is in transit.
    - **Kavkaz-3 (Кавказ-3):** Stationary hardened ground relay network utilizing deep-buried coaxial/fiber-optic cables, tropospheric scatter stations (ТРРС), and HF/VHF/UHF radio centers.
    - **Kavkaz-4 (Кавказ-4):** Airborne VIP and strategic command post communications suite installed aboard Presidential aircraft (Il-62M, Il-96-300PU, Tu-214PU) and Il-80 Maxdome (Zveno-S).
    - **Kavkaz-5 (Кавказ-5):** Railway and mobile command train (ЗКП ЖД) special communications complex.
    - **Kavkaz-6 (Кавказ-6):** Geostationary and Molniya-orbit satellite communication segment (Радуга / Глобус / Благовест / Меридиан transponders) dedicated to Supreme High Command conferencing.
    - **Kavkaz-7 (Кавказ-7):** The overarching integrated strategic teleconferencing and cryptographic decision-support network (Сеть совещательной связи высшего звена) that binds the three Cheget terminals and senior command posts into a tamper-proof, EMP-hardened real-time conference during a nuclear attack assessment (Nuclear Information Service: Russian NC3).
- **Causal Command Sequence:**
    1. Upon activation of Krokus and Cheget, Kavkaz-7 automatically establishes a priority-preempted, cryptographically authenticated conference bridge among the President, Minister of Defense, Chief of the General Staff, and Commander of the GTsPRN (Solnechnogorsk), routing traffic simultaneously across Kavkaz-1/2/3/4/6 physical layers so that destruction of any single terrestrial or satellite path cannot interrupt the executive launch decision (FAS: Constitutional Changes & Launch Authority).

63. Signal-A (Automated Launch Order Reception and Combat Control System in RVSN Regiments)

- **GRAU & Industrial Index:**
    - **1st Generation (АСУ «Сигнал»):** GRAU index **15Э1 (15Е1)**, deployed 1967–1974.
    - **2nd Generation (АСУ «Сигнал-М»):** Upgraded regimental/silo automated combat control system deployed in the 1970s.
    - **3rd Generation (АСБУ «Сигнал-А»):** GRAU index **15Э557 (15Е557)** (stationary/regimental complex) and mobile division/regiment variants (15Э1142 / АСБУ 4-го и 5-го поколений), deployed from the early 1980s onward (RussianForces.org: Third-Generation C2 System).
- **Design Bureau & Manufacturer:**
    - **Prime Designer: OKB "Impuls" at the Leningrad Polytechnic Institute / JSC NPO "Impuls" (НПО «Импульс»)**, Saint Petersburg (60°00'28"N 30°23'26"E) (SPbSTU Polytechnic NPO Impuls 50th Anniversary Archive).
    - **Serial Production: Leningrad Electromechanical Plant (ЛЭМЗ), Kalinin Plant** (Saint Petersburg), and **Penza Radio Factory**.
- **Chief Designers & Key Personnel:**
    - **Taras Nikolayevich Sokolov** — Professor at Leningrad Polytechnic Institute, Founder and First Chief Designer of OKB "Impuls" (Lenin Prize Laureate for creating the Signal automated combat control system) (RusArmy RVSN Technical Archive).
    - **Valery Ivanovich Melnik** — General Director and General Designer of NPO "Impuls" (1979–2000s), Chief Designer of Signal-A, Monolit, and Perimeter (15Е601) (Wikipedia RU: Sistema Perimetr).
    - **Igor Ustinov** and **Vladimir Igumnov** — Senior military acceptance (военная приёмка / GURVO) officers and subsequent heads of NPO "Impuls" overseeing Signal-A testing and 4th-generation upgrades (RusArmy RVSN Technical Archive).
- **Deployment Coordinates:** Installed across all RVSN Missile Armies (27th Guards at **Vladimir** 56°07'45"N 40°23'30"E, 31st at **Orenburg** 51°46'30"N 55°06'15"E, 33rd Guards at **Omsk** 54°59'30"N 73°22'40"E), all 12 active RVSN Missile Divisions, Regimental Command Posts (КП полка — e.g., 15В52У / 15В222), Mobile Command Posts (ПКП «Выбор» / 15В148), and individual ICBM silos (ШПУ) and mobile TELs (АПУ) (WW2.dk Strategic Rocket Forces Order of Battle).
- **Engineering-Technical Parameters:**
    - **Core Architecture:** Signal-A is a cyclic, packet-switched, noise-immune command-transmission and telemetry-reporting system that links the RVSN Central Command Post (Vlasikha, where it interfaces with Monolit) directly down to regimental command posts, road-mobile battalion command vehicles (МПБУ 15В231), and individual unattended ICBM silos (RussianForces.org: Third-Generation C2 System).

    - **Hardware & Redundancy:** Built on radiation-hardened domestic integrated circuits (БИС/СБИС) with triple-modular redundancy (мажоритарный принцип 2 из 3), non-volatile memory (НЗУ), and multi-channel receivers (buried cable, VLF/LF ground-wave, HF skywave, UHF satellite, and 15В99 command-rocket UHF receivers).
    - **Capabilities:** Continuous automated polling of missile readiness states (Контроль технического состояния), remote retargeting (Переприцеливание / selection of flight plan Полётное задание), transmission of warhead unblocking codes (Разблокирование), and **direct remote launch of individual silos from the Central Command Post or General Staff bypassing intermediate regimental crews** (Russian Strategic Nuclear Forces C2).
- **Causal Command Sequence:**
    1. When a launch directive enters Monolit at Vlasikha, it is formatted into a Signal-A combat packet containing the target plan number, the cryptographic unblocking key, and the execution command (Приказ на пуск).
    2. The packet propagates simultaneously through buried hardened cables, satellite channels, and VLF radio to the regimental command post (КП полка) and directly to each silo's Signal-A terminal.
    3. If the regimental command post crew executes the dual-key turn within the prescribed time window, the silo fires immediately; if the regimental command post is destroyed or fails to respond, the direct Signal-A packet from Vlasikha (or from a Sirena 15A11 command rocket) triggers the silo's autonomous launch sequence (Автоматический пуск без участия расчёта КП) (Russian Strategic Nuclear Forces C2).

64. Krokus (Central Command Post Alert Reception Subsystem of Kazbek)

- **GRAU & Industrial Index:** Система отображения и доведения предупреждения «Крокус» (Krokus — Missile Attack Warning Reception and Display Subsystem of the Kazbek NC3 architecture).
- **Design Bureau & Manufacturer: TsNII "Kometa" (ЦНИИ «Комета»)**, Moscow (55°42'29"N 37°39'52"E), and **Research Institute of Automatic Equipment (NII Avtomatiki / НИИА)**, Moscow (55°49'35"N 37°37'18"E), in cooperation with the **4th TsNII MO** and **RTI Mints** (Main Centre for Missile Attack Warning).
- **Chief Designers & Key Personnel:**
    - **Vladislav Georgiyevich Repin** — Chief Designer of the overarching SPRN algorithmic complex and the Krokus warning terminal interface (commissioned in **1978**) (Main Centre for Missile Attack Warning).
    - **Anatoly Ivanovich Savin** (TsNII Kometa) and **Vladimir Ivanovich Bessonov** (NII Avtomatiki).
- **Deployment Coordinates:**
    - **Source Transmission Node:** Main Centre for Missile Attack Warning (ГЦПРН), **Solnechnogorsk-7 (Timonovo)**, Moscow Oblast (56°11'00"N 37°00'35"E) (Main Centre for Missile Attack Warning).
    - **Receiving Display Terminals:** General Staff Central Command Post (ЦКП ГШ, Moscow 55°44'55"N 37°36'14"E), alternate bunker at **Chekhov / Sharapovo** (55°09'27"N 37°22'48"E), RVSN Central Command Post at **Vlasikha** (55°41'14"N 37°11'02"E), and Air & Missile Defense Command Post (ЦКП ПВО-ПРО, Balashikha / Zarya 55°45'40"N 38°04'30"E).
- **Engineering-Technical Parameters:**
    - **Commissioning Date:** Introduced into combat service in **1978** (Комплекс «Крокус») as the dedicated digital bridge between the SPRN supercomputers (Эльбрус-2 / M-10 at Solnechnogorsk) and the Supreme High Command (Main Centre for Missile Attack Warning).
    - **Information Payload:** Transmits structured digital frames showing the attack scale (Одиночный, Групповой, or Массированный удар), originating theater/azimuth, number of tracked ballistic targets, predicted impact regions, and countdown to first detonation, while simultaneously triggering the Cheget briefcases (Russian Strategic Nuclear Forces C2).
- **Causal Command Sequence:**
    1. When Oko/Tundra satellites or Voronezh/Daryal radars detect a launch, the Эльбрус computing complex at Solnechnogorsk correlates the track; the GTsPRN duty officer certifies the warning level, which immediately illuminates the Krokus consoles at the General Staff and Vlasikha and unlocks the Kazbek retaliatory decision loop (Russian Strategic Nuclear Forces C2).

65. Agat (C3 Subsystem within Soviet/Russian Strategic Command Architecture)

- **GRAU & Industrial Index:** АСБУ ВМФ «Агат» / Naval Strategic Combat Management & Fire Control Architecture (КБЮС / БИУС «Туча» / «Агат») and encrypted strategic communications multiplexing complex (П-310М «Агат»).
- **Design Bureau & Manufacturer: JSC Concern "Morinformsystem-Agat" (АО «Концерн „Моринформсистема-Агат“», formerly TsNII "Agat" [1971–1977] and NPO "Agat" [1977–2004])**, headquartered at Shosse Entuziastov 29, Moscow (55°45'26"N 37°44'48"E), with production at the **Agat Experimental Plant** (Moscow) and **Zavod "izmeritel"** (Corporation of Marine Instrumentation: History of Concern Agat).
- **Chief Designers & Key Personnel:**
    - **Georgy Vladimirovich Antsev** — General Director and General Designer of JSC Concern "Morinformsystem-Agat" (Wikipedia RU: Concern Morinformsystem-Agat).
    - **Yevgeny Vladimirovich Novikov** and Soviet-era directors of TsNII/NPO "Agat" who developed all submarine ballistic missile fire-control and combat information systems (БИУС and КСУС) for Project 667BDRM, Project 941, and Project 955 SSBNs.
- **Deployment Coordinates:**
    - **Headquarters & Design Complex:** Shosse Entuziastov 29, Moscow (55°45'26"N 37°44'48"E).
    - **Operational Naval Command Nodes:** Main Staff of the Russian Navy (ГКП ВМФ, Saint Petersburg 59°56'15"N 30°18'30"E and Moscow 55°46'10"N 37°39'15"E), Northern Fleet Command Post at **Severomorsk** (69°04'10"N 33°25'00"E), Pacific Fleet Command Post at **Vladivostok** (43°06'55"N 131°53'25"E), and onboard all Project 955 Borei (Okrug / Боевая информационно-управляющая система «Округ»), Project 667BDRM Delfin (Omnibus-BDRM), and Project 941 Akula (Omnibus-941) SSBNs (InfoMach: Concern Agat).
- **Engineering-Technical Parameters:**
    - **Role in the Strategic Triad C3 Architecture:** While NPO "Impuls" builds the land-based RVSN combat control systems (Monolit and Signal-A), **NPO/Concern "Agat"** is the sole lead enterprise for the naval strategic C3 and SLBM pre-launch calculation, targeting, and cryptographic launch-authorization execution loops aboard SSBNs (Wikipedia RU: Concern Morinformsystem-Agat).

    - **Multiplexing & Communications Backbone:** Also encompasses the П-310М «Агат» (P-310M Agat) 12/24-channel encrypted carrier-frequency multiplexing equipment used across hardened military underground cable and radio-relay command lines (Wikipedia RU: Russian Weapon Designations A).
- **Causal Command Sequence:**
    1. When the General Staff issues a strategic launch order via Kazbek, the naval directive is transmitted through the VLF transmitter network (Goliath at Nizhny Novgorod, Antey at Vileyka) or Tu-142MR trailing-wire relays to the submerged SSBN's buoy/towed antenna (Параван).
    2. The onboard **Agat** Combat Information and Fire-Control System decodes the authorization, verifies the unblocking cipher, computes real-time geodetic/inertial launch vectors from the ship's navigation complex (Симфония / Тобол), floods the launch tubes, and sequences the salvo firing of R-30 Bulava or R-29RMU2 Sineva SLBMs (Corporation of Marine Instrumentation: Concern Agat).

66. Monolit (Automated C2 System of Strategic Missile Forces RVSN)

- **GRAU & Industrial Index:** АСБУ высшего звена «Монолит» — GRAU index **15Э556 (15E556)** (Strategic Automated Combat Control System of the General Staff and Main Staff of the RVSN, operating in tandem with regimental 15Э557 Сигнал-А and radio-relay Вьюга).
- **Design Bureau & Manufacturer: ОКВ "Impuls" / JSC NPO "Impuls" (НПО «Импульс»)**, Saint Petersburg (60°00'28"N 30°23'26"E), in cooperation with **NII Avtomatiki (НИИА)** and the **4th TsNII MO** (SPbSTU Polytechnic NPO Impuls 50th Anniversary Archive).
- **Chief Designers & Key Personnel:**
    - **Valery Ivanovich Melnik** and **Taras Nikolayevich Sokolov** — Chief Designers at NPO "Impuls" (SPbSTU Polytechnic NPO Impuls 50th Anniversary Archive).
    - **General of the Army Yuri Pavlovich Maximov** (Commander-in-Chief of the RVSN, 1985–1992), **Colonel General Nikolai Yevgenyevich Solovtsov** (Commander of the RVSN, 2001–2009), and **Colonel General Sergey Viktorovich Karakayev** (Commander of the RVSN, 2010–present, overseeing the 4th- and 5th-generation АСБУ modernization of Monolit/Signal-A) (RussianForces.org: Third-Generation C2 System).
- **Deployment Coordinates:**
    - **Primary Node:** RVSN Central Command Post (ЦКП РВСН) at **Vlasikha (Odintsovo-10 / Perkhushkovo)**, Moscow Oblast (55°41'18"N 37°10'43"E).
    - **Alternate Underground Command Post (ЗЦКП РВСН): Balabanovo / Borovsk** (Объект 754, Kaluga Oblast, 55°11'19"N 36°36'37"E) (WW2.dk Strategic Rocket Forces).
    - **General Staff Interface Nodes:** General Staff Central Command Post (Moscow, 55°44'55"N 37°36'14"E) and Kosvinsky Kamen (59°31'00"N 59°03'43"E).
- **Engineering-Technical Parameters:**
    - **Hierarchical Position:** While Kazbek connects the political leadership (Cheget) to the General Staff, **Monolit (15E556)** is the strategic military command-formation engine at the General Staff and RVSN Central Command Post that translates a political sanction into concrete cryptographic strike directives (Приказы по боевому управлению) across the entire ICBM force (Russian Strategic Nuclear Forces C2).
    - **Technical Evolution:** Deployed in the early 1980s on unified multi-machine computing complexes (ЕС ЭВМ and specialized radiation-hardened domestic processors) and upgraded in 2008–2024 to 4th/5th-generation АСБУ РВСН standards capable of real-time retargeting and bypassing intermediate command echelons (RussianForces.org: Third-Generation C2 System).
- **Causal Command Sequence:**
    1. Upon receiving the Kazbek sanction cipher, the duty generals at the General Staff and Vlasikha enter the execution keys into the Monolit (15E556) console.
    2. Monolit cryptographically seals the strike package (combining the political sanction, the General Staff unblocking code, and the specific warplan index) and injects it concurrently into the terrestrial/satellite Signal-A (15E557) network and the airborne/command-rocket Вьюга (Vyuga) radio-control networks (Russian Strategic Nuclear Forces C2).

67. Sirena (15A11 / 15Zh75 Command Rocket Based on UR-100UTTKh / MR-UR-100 with 15B99 Radio-Command Warhead)

- **GRAU & Industrial Index:**
    - **Silo-Based Command Rocket Complex: 15П011 (15P011)** with command ICBM **15А11 (15A11)** (derived from 15A16 MR-UR-100UTTKh / SS-17 Sego), launched from silo **15П716 (15P716)**, carrying special radio-command warhead **15Б99 (15B99)** (Missilery.info: 15P011 / 15A11).
    - **Intermediate-Range Command Rocket Variant: 15Ж46 (15Zh46)** in complex **15П646 (15P646 «Горн»)** (derived from 15Ж45 RSD-10 Pioneer / SS-20 Saber) with payload **15Б99** (Wikipedia RU: Sistema Perimetr Talk/Indices).
    - **Road-Mobile / Modernized Solid-Fuel Command Rocket Complex (Сирена / Сирена-М / Периметр-РЦ):** Complex **15П175 «Сирена» (15P175 Sirena)** with solid-propellant command ICBM **15Ю75 (15Yu75)** / **15Ж75 (15Zh75)** (derived from 15Ж58 RT-2PM Topol and subsequently upgraded on the RS-24 Yars chassis) (Ruzhany RVSN Archive: Complex 15P175 Sirena with 15Yu75/15Zh75).
- **Design Bureau & Manufacturer:**
    - **15A11 Missile (15P011): OKB-586 / KB Yuzhnoye** (Dnepropetrovsk) and **Yuzhmash (ЮМЗ)** (Encyclopedia Astronautica: Perimetr 15A11).
    - **15Yu75 / 15Zh75 Missile (15P175 Sirena): Moscow Institute of Thermal Technology (МИТ / MITT)**, Moscow (55°51'16"N 37°36'56"E), and **Votkinsk Machine Building Plant** (57°03'12"N 53°59'10"E) (Ruzhany RVSN Archive: 15P175 Sirena).
    - **15B99 Radio-Command Warhead (СГЧ 15Б99):** Designed by **OKB "Impuls" at Leningrad Polytechnic Institute (NPO Impuls)**; attitude-control and stabilization module (ДУ СГЧ) designed by **KB Yuzhnoye**; serially manufactured at **NPO "Strela"** (Orenburg) (Ruzhany Strategic RVSN Archive: 15P011 / 15A11).
- **Chief Designers & Key Personnel:**
    - **Vladimir Fyodorovich Utkin** (General Designer, KB Yuzhnoye — 15A11 rocket and 15B99 gas-jet stabilization bus).
    - **Valery Ivanovich Melnik** (Chief Designer, OKB/NPO Impuls — 15B99 radio-command transmitter payload).
    - **Alexander Davidovich Nadiradze**, **Boris Nikolayevich Lapygin**, and **Yury Solomonovich Solomonov** (General Designers, MITT — 15P175 Sirena / 15Yu75 / 15Zh75 solid-propellant command rockets).

- **Deployment Coordinates:**
  - **15A11 (15P011) Deployment:** 510th Guards Rocket Regiment, 7th Guards Rocket Division, **Vypolzovo (Ozernyy)**, Tver Oblast (57°52'00"N 33°40'00"E, BRK-6 в/ч 52642) (Ruzhany Strategic RVSN Archive).
  - **Flight Trial Launches (15A11 at Baikonur NIIP-5, 45°59'20"N 63°33'40"E):** 7 flight tests between 26 December 1979 and 13 November 1984 (6 successes, 1 failure on 28 January 1982); during the test launches, the 15B99 payload transmitted live combat launch commands to an operational 15A35 (UR-100NUTTKh) silo at Baikonur, which successfully executed an autonomous unmanned launch to the Kura Test Range on Kamchatka (57°20'00"N 161°50'00"E) (Encyclopedia Astronautica: Perimetr 15A11).
- **Engineering-Technical Parameters (15A11 & 15B99):**
  - **Launch Mass (15A11):** 71,100 kg; Length: 21.6 m; Diameter: 2.25 m; Stages: 2 stages using storable hypergolic liquid propellant (N_2O_4 / UDMH); Cold-launch mortar ejection (миномётный старт) from silo 15P716 (Missilery.info: 15A11).
  - **Range & Trajectory:** 11,000 km maximum range; operational command trajectory uses a high suborbital loft with an **apogee of 1,000 km (600 mi)** and flight duration of 25–35 minutes across Soviet/Russian territory (Encyclopedia Astronautica: Perimetr 15A11).
  - **Warhead 15B99 Specifications:** Mass **1,412 kg** (Wikipedia: Dead Hand). Contains an unpressurized hermetically sealed instrument container with a high-power multi-frequency UHF/VHF radio-command transmitter (Передатчик команд Вьюга/Периметр), autonomous silver-zinc thermal battery power supply, and a cold-gas nitrogen 3-axis stabilization system that keeps the transmitting antenna oriented toward the Earth's surface throughout the unpowered midcourse flight phase (Ruzhany Strategic RVSN Archive).
- **Causal Command Sequence:**
  1. Upon receiving the launch command from the Perimeter (15E601) command bunker, the 15A11 or 15Yu75/15Zh75 command missile ejects from its launcher and ascends along a flat-to-high-apogee corridor spanning from European Russia to Eastern Siberia.
  2. Immediately after second/third-stage burnout and 15B99 separation, the attitude control jets orient the antenna nadir-ward and the transmitter radiates the pre-loaded Supreme High Command launch cipher and warhead unblocking codes continuously across the line-of-sight footprint (~3,500 km radius at 1,000 km altitude), directly triggering the 15Э1313 / Вьюга receivers on all surviving ICBM silos, mobile launchers, SSBNs, and airborne strategic bombers (Ruzhany Strategic RVSN Archive).

68. 15E601 (GRAU Index for Perimeter / Cheget Complexes — Exact GRAU Sub-Index Disambiguation)

- **GRAU & Industrial Index Disambiguation Matrix:** In Soviet and Russian Directorate of Rocket and Artillery Armament (ГРАУ МО) nomenclature, Group 15 designates Strategic Missile Forces (РВСН) equipment, where letter codes specify the exact functional class (Missilery.info: 15P011 / 15E601):
  - **15Э601 (15E601):** Ground-based automated command and control complex (Командно-управляющий комплекс системы «Периметр») — the central sensor-fusion, cryptographic logic, and command-post equipment of the **Perimeter** system developed by **NPO Impuls**. (In secondary Western media, 15E601 is frequently conflated with the Cheget nuclear briefcase, which is actually the portable subscriber terminal of the Kazbek system built by **NII Avtomatiki**, while 15Э601 strictly denotes the RVSN Perimeter autonomous retaliatory C2 complex) (Wikipedia RU: Sistema Perimetr Talk/Indices).
  - **15П011 (15P011):** Silo-based command missile complex (Боевой ракетный комплекс командных ракет) of the 15E601 Perimeter system (Missilery.info: 15A11).
  - **15А11 (15A11):** Two-stage liquid-propellant command ICBM (Командная ракета) based on the 15A16 (MR-UR-100UTTKh) (Encyclopedia Astronautica: Perimetr 15A11).
  - **15Б99 (15B99):** Special radio-technical command warhead (Специальная головная часть – СГЧ) mounted on 15A11 and 15Ж46 (Ruzhany Strategic RVSN Archive).
  - **15П716 (15P716):** High-hardness automated silo launcher (Шахтная пусковая установка типа ОС) for the 15A11 missile (Wikipedia: Dead Hand).
  - **15В75 / 15В117 (15V75 / 15V117):** Unified command-post equipment (Унифицированный командный пункт) of the 15P011 regimental complex (Ruzhany Strategic RVSN Archive).
  - **15П646 (15P646 «Горн») & 15Ж46 (15Zh46):** Road-mobile command missile complex and missile based on the 15Ж45 RSD-10 Pioneer (SS-20 Saber), eliminated under the 1987 INF Treaty (Wikipedia RU: Sistema Perimetr Talk/Indices).
  - **15П175 (15P175 «Сирена») & 15Ю75 / 15Ж75 (15Yu75 / 15Zh75):** Road-mobile solid-propellant command missile complex and missile based on the RT-2PM Topol / RS-24 Yars (Ruzhany RVSN Archive: 15P175 Sirena).
  - **15Э556 (15E556 «Монолит») & 15Э557 (15E557 «Сигнал-А»):** Top-echelon and regimental automated combat control complexes that interface directly with 15E601.
- **Design Bureau & Chief Designers: OKB/NPO Impuls** (**V. I. Melnik**, **T. N. Sokolov**), **KB Yuzhnoye** (**V. F. Utkin**), **KBSM** (**A. F. Utkin**), **MITT** (**A. D. Nadiradze**, **B. N. Lapygin**, **Yu. S. Solomonov**), and **NII Avtomatiki** (**V. I. Bessonov**).
- **Deployment Coordinates:** 7th Guards Rocket Division at **Vypolzovo (Ozernyy)** (57°52'00"N 33°40'00"E), **Kosvinsky Kamen** (59°31'00"N 59°03'43"E), and **Vlasikha** (55°41'14"N 37°11'02"E).
- **Causal Command Sequence:**
  1. 15Э601 (command post electronics) evaluates seismic/radiation sensors and General Staff communication status -> commands 15B75 regimental launch post -> fires 15A11 (from 15П716 silo) or 15Ю75/15Ж75 (from 15П175 mobile TEL) -> deploys 15Б99 radio-command warhead in space -> triggers 15Э1313 receivers tied to 15Э557 (Signal-A) across all ICBM launchers (Ruzhany Strategic RVSN Archive).

69. Il-80 (Ilyushin Il-80 Maxdome / Aimak Airborne Command Post, Zveno-S)

- **GRAU & Industrial Index:** Aircraft designation **Ил-80 (Il-80)** / **Ил-86ВКП (Il-86VKP — Воздушный командный пункт)**; Industrial R&D cipher **«Аймак» (Aimak)**; Onboard strategic C3 mission complex **«Звено-С» (Zveno-S)**; NATO reporting name: **Maxdome** (Wikipedia: Ilyushin Il-80).
- **Design Bureau & Manufacturer:**
  - **Airframe Design Bureau: Ilyushin Design Bureau (ОКБ им. С. В. Ильюшина / ПАО «Ил»)**, Moscow (55°47'35"N 37°32'05"E).
  - **Airframe Manufacture: Voronezh Aircraft Production Association (ВАСО)**, Voronezh (51°38'28"N 39°15'18"E).

    - **C3 Mission System (Zveno-S) Integrator: NPO "Polet"** (Nizhny Novgorod, 56°16'10"N 43°54'30"E) and **MNIIRS** (Moscow), with airframe modernization conducted at **Beriev TANTK** (ТАНТК им. Г. М. Бериева, Taganrog 47°11'54"N 38°50'55"E) (Wikipedia: Ilyushin Il-80).
- **Chief Designers & Key Personnel:**
    - **Genrikh Vasilyevich Novozhilov** — General Designer of the Ilyushin Design Bureau (overall Il-86 / Il-80 program).
    - **Viktor Vsevolodovich Livanov** and **RP Talikov** — Chief Designers overseeing the Il-86VKP / Il-80 conversion and Il-80M / Zveno-3S (Il-96-400M) modernization.
- **Deployment Coordinates:**
    - **Home Base: Chkalovsky Air Base** (8th Special Purpose Aviation Division / 8-я авиационная дивизия особого назначения), Shchyolkovo, Moscow Oblast (55°52'42"N 38°03'42"E) (Wikipedia: Ilyushin Il-80).
    - **Overhaul & Modernization Facility: Taganrog-Yuzhny Airfield (Beriev TANTK)**, Rostov Oblast (47°11'54"N 38°50'55"E).
- **Engineering-Technical Parameters:**
    - **Fleet Serial Numbers & Registrations:** Exactly **4 aircraft built** (first flight 5 March 1985 / 29 May 1987, entered service 1992):
        1. RA-86146 (c/n 042 / 086042): Prototype, placed in reserve/storage at Chkalovsky.
        2. RA-86147 (c/n 043 / 086043): Active airborne command post.
        3. RA-86148 (c/n 046 / 086046): Active airborne command post.
        4. RA-86149 (c/n 048 / 086048): Active airborne command post (upgraded to Il-80M standard) (Wikipedia: Ilyushin Il-86 Military Operations).
    - **Airframe & EMP Hardening:** All cabin windows are deleted and plated over with aluminum-lithium skin to shield the interior against nuclear thermal flash and electromagnetic pulse (NEMP); an internal cockpit blast baffle protects the flight deck; fitted with a fixed port-side aerial refueling probe (Wikipedia: Ilyushin Il-80).
    - **Auxiliary Power & Antenna Suite (Zveno-S):**
        - Two massive underwing electrical generator pods (length 9.5 m, diameter 1.3 m) mounted inboard of the four Kuznetsov NK-86 turbofan engines (4 × 127.5 kN) to power the C3 and VLF transmitter suites (Wikipedia: Ilyushin Il-80).
        - A prominent dorsal fiberglass SATCOM canoe (length ~15 m, width 1.5 m) housing SHF/EHF satellite communications antennas (Kavkaz-4/6).
        - Lower aft fuselage motorized winch deploying a **multi-kilometer trailing wire antenna (Выпускная буксируемая антенна — ВБА, length up to 7.5–8 km)** for Very Low Frequency (VLF, 15–30 kHz) transmission to submerged SSBNs (Wikipedia: Ilyushin Il-80).
- **Causal Command Sequence:**
    1. Upon strategic warning, the General Staff operational battle staff boards Il-80 at Chkalovsky (55°52'42"N 38°03'42"E).
    2. In flight, Il-80 links into the Kazbek / Kavkaz-4 network via its dorsal SATCOM array and deploys its 8 km VLF trailing wire antenna in a high-bank orbit (Вираж), broadcasting encrypted launch orders directly to RVSN divisions, Long-Range Aviation bombers, and submerged Borei / Delta IV SSBNs if ground command posts at Moscow and Vlasikha are destroyed.

70. Il-82 (Ilyushin Il-82 Airborne Command/Relay Post Based on Il-76)

- **GRAU & Industrial Index:** Aircraft designation **Ил-82 (Il-82)** / **Ил-76ВКП (Il-76VKP — Воздушный командный пункт / Самолёт-ретранслятор)**; Industrial R&D cipher **Изделие 65С (Izdeliye 65S)**; Onboard C3 relay suite **«Звено» (Zveno)** (Air & Space Forces Magazine: Russian Airpower Almanac).
- **Design Bureau & Manufacturer:**
    - **Airframe Design Bureau: Ilyushin Design Bureau (ОКБ им. С. В. Ильюшина)**, Moscow (55°47'35"N 37°32'05"E) ([Wikipedia: List of Ilyushin Aircraft](https://en.wikipedia.org/wiki/List of_Ilyushin_aircraft)).
    - **Airframe Manufacture: Tashkent Aviation Production Association named after V. P. Chkalov (ТАПОиЧ)**, Tashkent (41°18'20"N 69°20'15"E).
    - **C3 Relay Payload: NPO "Polet"** (Nizhny Novgorod) and **MNIIRS** (Moscow).
- **Chief Designers & Key Personnel:**
    - **Genrikh Vasilyevich Novozhilov** and **R. P. Papkovsky** (Ilyushin Design Bureau — Il-76MD / Il-82 conversion).
- **Deployment Coordinates:** Based at **Chkalovsky Air Base** (8th Special Purpose Aviation Division), Moscow Oblast (55°52'42"N 38°03'42"E), alongside the Il-80 fleet (Wikipedia: Ilyushin Il-76).
- **Engineering-Technical Parameters:**
    - **Fleet Serial Numbers & Registrations: 2 aircraft built** in 1987–1990 by converting Il-76MD strategic airlifter hulls:
        1. RA-76450 (c/n 0073476298)
        2. RA-76451 (c/n 0073478350)
    - **Configuration & Propulsion:** Powered by four Soloviev D-30KP-2 turbofans (4 × 117.7 kN); maximum takeoff mass 190,000 kg; windowless cargo hold accommodating a roll-on/roll-off modular hermetically sealed C3 command and radio-relay capsule.
    - **Antenna & Relay Architecture:** Features a prominent forward dorsal canoe fairing (SATCOM) extending from the cockpit crown to the wing root, ventral blade antennas, and an aft fuselage VLF/LF wire antenna reel, engineered to operate from shorter or semi-prepared dispersal airfields (Unpaved / austere runway capability inherited from the Il-76) where the wide-body Il-80 cannot land (Air & Space Forces Magazine: Il-82).
- **Causal Command Sequence:**

1. Deployed as an airborne strategic communications relay and reserve command post augmenting the Il-80, Il-82 receives Kazbek/Monolit directives via its dorsal satellite dome and relays them over HF/VHF/UHF and VLF channels to dispersed mobile ICBM divisions and theater nuclear forces.

71. Tu-142MR Orel (Tupolev Tu-142MR VLF Submarine Communications Relay Aircraft)

- **GRAU & Industrial Index:** Aircraft designation **Ту-142МР «Орёл» (Tu-142MR Orel)**; Tupolev OKB internal index **Изделие ВПМР (Izdeliye VPMR)**; Onboard VLF radio-relay complex **БКСР-200 «Орёл» (BKSR-200 Orel)** with VLF transmitter **«Фрегат» (Fregat)**; NATO reporting name: **Bear-J** (Wikipedia: Tupolev Tu-142).
- **Design Bureau & Manufacturer:**
  - **Airframe Design Bureau: Tupolev Design Bureau (ОКБ А. Н. Туполева / ПАО «Туполев»)**, Moscow (55°45'42"N 37°40'48"E).
  - **Serial Manufacture: Taganrog Machinery Plant named after G. Dimitrov (ТМЗ / Beriev TANTK)**, Taganrog (47°11'54"N 38°50'55"E), where ~10–12 Tu-142MR aircraft were manufactured between 1977 and 1994 (entered service in December 1980) (Wikipedia: Tupolev Tu-142).
  - **VLF Radio Complex: NPO "Polet"** (Nizhny Novgorod) and **Komintern Plant / RIRV** (Saint Petersburg).
- **Chief Designers & Key Personnel:**
  - **Alexei Andreyevich Tupolev** (General Designer, Tupolev OKB) and **Nikolai Vladimirovich Kirsanov** (Chief Designer of the Tu-142 family).
- **Deployment Coordinates:**
  - **Northern Fleet: Kipelovo Air Base (Fedotovo)**, Vologda Oblast (59°11'18"N 39°07'24"E, 76th Independent Anti-Submarine Aviation Squadron / 403rd Mixed Aviation Regiment) (Wikipedia: Tupolev Tu-142).
  - **Pacific Fleet: Kamenny Ruchey Air Base (Mongokhto)**, Khabarovsk Krai (49°14'12"N 140°11'24"E, 568th Independent Composite Aviation Regiment).
- **Engineering-Technical Parameters:**
  - **Airframe & Propulsion:** Derived from the Tu-142M long-range maritime patrol airframe powered by four **Kuznetsov NK-12MP** turboprops (4 × 11,033 kW / 14,795 shp) driving contra-rotating AV-60P propellers; Max speed: 855–925 km/h; Combat radius: 5,200–6,500 km; Unrefueled endurance: 14–16 hours (equipped with nose in-flight refueling probe) (Wikipedia: Tupolev Tu-142).
  - **VLF Trailing Wire Antenna (ВБА / Fregat Transmitter):** Houses a massive ventral blister fairing in the former forward weapons bay containing a motorized drum and **an 8.6 km (28,200 ft) copper-clad steel trailing wire antenna** weighted by a aerodynamic drogue cone (Конус-стабилизатор), driven by the **100 kW Fregat VLF transmitter** operating in the **14–30 kHz band** (whose electromagnetic waves penetrate seawater to depths of 20–40 meters) (Wikipedia: Tupolev Tu-142).
  - **Additional C3 Suite:** Fin-tip HF/UHF fairing, dorsal and ventral blade antennas, satellite communications terminal, and emergency radio-buoy dispensers (Аварийные радиобуи).
- **Causal Command Sequence:**
  1. Operating as the Russian Naval Aviation counterpart to the US Navy E-6B Mercury TACAMO, Tu-142MR loiters over the Barents Sea, White Sea, or Sea of Okhotsk SSBN bastions.
  2. Upon receiving a strategic launch directive from the General Staff (Kazbek), Navy Main Staff (Agat), or a Sirena 15A11 command rocket, Tu-142MR reels out its 8.6 km trailing wire antenna, enters a tight continuous pylon turn (Вираж) so the wire hangs nearly vertically for vertical polarization, and transmits the VLF launch authorization signal into the ocean to submerged Project 955 Borei and Project 667BDRM Delfin SSBNs (Wikipedia: Tupolev Tu-142).

Subsystem Group IV.B: Underground Command Bunkers, Warhead Custody & Early Warning Radars (Entities 72–81)

72. Kosvinsky Kamen (Granit Underground Strategic Command Bunker, Sverdlovsk Oblast)

- **GRAU & Facility Index:** **Объект «Гранит» (Object Granit)** / Alternate Central Command Post of the General Staff and Strategic Rocket Forces (Запасной Центральный Командный Пункт – ЗЦКП «Косьвинский Камень»); Military garrison town **Kytlym** (Кытлым, Military Unit в/ч 40274 / 12th GUMO & General Staff Special Underground Facility Directorate) (Wikipedia: Kosvinsky Kamen).
- **Design Bureau & Engineering Contractor:**
  - **Structural & Underground Engineering: Central Design Institute No. 20 / 26th Central Research Institute of the Ministry of Defense (26 ЦНИИ МО)** and **Main Directorate of Special Construction (ГУССМО / Glavspetsstroy)**.
  - **Automated C2 Integration: NPO "Impuls"** (Saint Petersburg — Monolit and Perimeter 15E601 C2 suites) and **NII Avtomatiki** (Moscow — Kazbek).
- **Chief Designers & Key Personnel:**
  - **Valery Ivanovich Melnik** (NPO Impuls — C2 installation), **General of the Army Yuri Pavlovich Maximov** (RVSN Commander-in-Chief during construction initiation in the late 1970s), and **General of the Army Igor Dmitriyevich Sergeyev** (RVSN Commander-in-Chief and Minister of Defense who oversaw the facility's completion and commissioning in **early 1996**) (Wikipedia: B61 Nuclear Bomb / Kosvinsky Kamen Reference).
- **Deployment Coordinates:**
  - **Mountain Peak & Underground Complex: Mount Kosvinsky Kamen**, Karpinsk Municipal District, Sverdlovsk Oblast, Northern Urals: **59°31'00"N 59°03'43"E** (59.5167°N, 59.0619°E), summit elevation **1,519 m (4,984 ft)** (Wikipedia: Kosvinsky Kamen).
  - **Support Garrison & Portal: Kytlym** (59°29'15"N 59°12'00"E), located on the southern/southeastern flank of the massif.
- **Engineering-Technical Parameters:**
  - **Geological Shielding:** Excavated beneath **~300 meters (1,000 feet) of solid Paleozoic dunite, pyroxenite, and gabbro bedrock** inside the Kosvinsky Kamen pluton, specifically engineered to defeat direct hits by US earth-penetrating nuclear warheads (B53, B61 Mod 11, and W88) (Wikipedia: B61 Nuclear Bomb).

- **Internal Shock-Isolated Architecture:** Command modules and the Perimeter (15E601) autonomous control chambers are suspended on heavy hydraulic-pneumatic shock-isolation pendulums inside steel-lined rock caverns, isolating electronics from ground-shock accelerations exceeding `50–100 g`.
- **Very-Low-Frequency / Extremely-Low-Frequency (VLF/ELF) Deep-Strata Antenna:** Utilizes the high electrical resistivity of the surrounding crystalline Urals pluton to radiate VLF/ELF signals through buried rock-strata dipole antennas that cannot be destroyed by surface airbursts.

- **Causal Command Sequence:**
  1. Serves as the primary wartime survivable alternate command post for the Russian General Staff and the central nerve center of the Perimeter (15E601) / Monolit systems (Wikipedia: Kosvinsky Kamen).
  2. If Moscow (Vlasikha and Chekhov/Sharapovo) is neutralized in a decapitation strike, operational command of the Strategic Nuclear Forces automatically transfers to `Object Granit` under Kosvinsky Kamen, which either issues directed retaliatory strikes via Monolit or executes the Perimeter command-rocket launch sequence.

73. Mount Yamantau (Mezhgorye Underground Strategic Complex, Bashkortostan)

- **GRAU & Facility Index: `Объект «Белорецк-16» / ЗАТО Межгорье` (`Beloretsk-16 / ZATO Mezhgorye`)**; Construction project cipher **`УС-30` (`Управление строительства № 30` — Construction Directorate No. 30)**; Subordinate to the **Main Directorate of Special Programs of the President of the Russian Federation (`ГУСП Президента РФ`)** (Wikipedia: Mount Yamantau).
- **Design Bureau & Engineering Contractor:**
  - **Lead Tunneling & Underground Construction Enterprise: FSUE "Construction Directorate No. 30" (`ФГУП «Управление строительства № 30»` / US-30)**, headquartered in Mezhgorye, together with **Metrostroy** and the **Main Directorate of Special Construction (`ГУСС`)**.
- **Chief Designers & Key Personnel:**
  - **Nikolai Ivanovich Makarov** and **Abram Lvovich Kats** (Heads of Construction Directorate No. 30 during the Soviet and post-Soviet excavation phases); overseen at the federal level by the Head of the **Main Directorate of Special Programs of the President of the Russian Federation (`ГУСП`)**, Colonel General **Alexander Leonidovich Linets**.
- **Deployment Coordinates:**
  - **Mountain Summit & Underground Complex: Mount Bolshoy Yamantau**, Beloretsky District, Republic of Bashkortostan, Southern Urals: **`54°15'18"N 58°06'07"E`** (`54.2550°N, 58.1019°E`), the highest peak in the Southern Urals at **`1,640 meters`** (**`5,381 ft`**) (Wikipedia: Mount Yamantau).
  - **Closed Administrative-Territorial Formation (`ЗАТО Межгорье`):** Comprises two garrisons flanking the mountain: **Mezhgorye Yugo-Zapadny (Beloretsk-16 / Tatly)** at `54°03'25"N 57°49'18"E` and **Mezhgorye Tsentralny (Beloretsk-15 / Kuzelga)** at `54°13'40"N 57°57'45"E`, connected by dedicated rail spurs and highways directly to the mountain portals.
- **Engineering-Technical Parameters:**
  - **Scale & Overburden:** Excavated beneath **up to `1,000 meters` (`3,280 ft`) of Proterozoic quartzite and sandstone overburden** across an underground footprint estimated by US congressional testimony (Representative Roscoe Bartlett, House Armed Services Committee, 2000) to span an area comparable to the Washington, D.C. Capital Beltway interior, capable of sustaining **up to 60,000 personnel** for months with independent underground nuclear/diesel power generation, air regeneration, water filtration, and food/materiel reserves (FAS Congressional Record: Russian Threats to US Security).
  - **Hardening Class:** Immune to multiple sequential direct surface and earth-penetrating multi-megaton thermonuclear detonations (Wikipedia: Nuclear Bunker Buster).
- **Causal Command Sequence:**
  1. Functions as the supreme **Continuity of Government (CoG)** sanctuary (`Пункт управления государством и Вооружёнными Силами в особый период`) for the Presidential Administration, Security Council, and Supreme High Command.
  2. Connected via deep-buried Kavkaz-3 fiber-optic/coaxial trunks and VLF/ELF relays to Kosvinsky Kamen (580 km to the north) and Vlasikha, ensuring national political-military command continuity during protracted nuclear conflict.

74. Vlasikha (Odintsovo-10, Main Headquarters and Central Command Post of the RVSN)

- **GRAU & Facility Index: `ЗАТО Власиха` (`Одинцово-10` / `Перхушково` / Военный городок № 22/1)** — Main Staff of the Strategic Rocket Forces (`Главный штаб РВСН`, Military Unit `в/ч 43176`) and Central Command Post of the RVSN (`Центральный командный пункт РВСН – ЦКП РВСН`, Military Unit `в/ч 33925` / `"КП-4"`) (Wikipedia: Vlasikha, Moscow Oblast).
- **Design Bureau & Engineering Contractor:**
  - **Underground Command Complex: Central Design Institute of the Ministry of Defense (`ЦПИ-20` / `26 ЦНИИ МО`)** and **Glavspetsstroy**.
  - **Automated Combat Control Systems: NPO "Impuls"** (Saint Petersburg — Monolit 15E556, Signal-A 15E557, and 4th/5th-generation `АСБУ РВСН`) (SPbSTU Polytechnic NPO Impuls 50th Anniversary Archive).
- **Chief Designers & Key Personnel:**
  - Established by Order of USSR Minister of Defense Marshal **Rodion Yakovlevich Malinovsky** on **15 January 1960** following the creation of the RVSN on **17 December 1959** under First Commander-in-Chief Chief Marshal of Artillery **Mitrofan Ivanovich Nedelin** (Wikipedia: Vlasikha, Moscow Oblast).
  - Current Commander of the RVSN at Vlasikha: **Colonel General Sergey Viktorovich Karakayev** (in command since 22 June 2010); Chief of the Main Staff of the RVSN: **Colonel General Igor Nikolayevich Fazletdinov**.
- **Deployment Coordinates (Vlasikha & Associated RVSN C3 Node Complex):**
  - **RVSN Main Headquarters (`Главный штаб РВСН`):** Vlasikha, Odintsovo District, Moscow Oblast: **`55°41'14"N 37°11'02"E`** (WW2.dk Strategic Rocket Forces).
  - **RVSN Central Command Post Bunker (`ЦКП РВСН Перхушково`): `55°41'18"N 37°10'43"E`** (WW2.dk Strategic Rocket Forces).
  - **RVSN Alternate Underground Command Post (`ЗЦКП РВСН Балабаново`): `55°11'19"N 36°36'37"E`** (`Balabanovo Combat Operations Center at 55°11'41"N 36°37'42"E`) (WW2.dk Strategic Rocket Forces).

    - **Naro-Fominsk RVSN Radio-Communications Receiving Complex (ПДРЦ / Receiver):** `55°19'38"N 36°48'36"E` (WW2.dk Strategic Rocket Forces).
    - **Novopetrovskoye RVSN Radio-Communications Transmitter Complex (ПРЦ / Transmitter):** `55°56'39"N 36°28'33"E` (WW2.dk Strategic Rocket Forces).
- **Engineering-Technical Parameters:**
    - **Multi-Tier Underground Bunkers:** The original 1958–1960 command bunker at Vlasikha is a **4-tier hardened subterranean structure**; in **1986**, construction commenced on a new **12-tier deep underground bunker** located 2 km from the original facility, hardened against nuclear overpressure and EMP, housing the Central Computer Center (`ЦВЦ РВСН`), the Monolit (15E556) / Signal-A (15E557) central halls, and the Central Communications Hub (`Центральный узел связи «Кристалл»`) (Wikipedia: Vlasikha, Moscow Oblast).
- **Causal Command Sequence:**
    1. Duty shifts (`Дежурные смены ЦКП РВСН`, numbering ~500 officers on 24/7 combat duty across the Vlasikha complex) continuously monitor the telemetry of all ~300+ deployed Russian ICBMs via Signal-A.
    2. Upon receiving the Kazbek launch authorization from the General Staff, the Vlasikha Central Command Post executes the Monolit launch sequence, transmitting direct unblocking and firing orders to the 3 Missile Armies, 12 Missile Divisions, and individual ICBM silos/mobile launchers.

75. 12th GUMO (12th Chief Directorate of the Ministry of Defense — Nuclear Warhead Custody & Arsenal Network)

- **GRAU & Institutional Index:** **`12-е Главное управление Министерства обороны Российской Федерации`** **(`12 ГУМО` / `12th GUMO`)** — Military Unit **`в/ч 31600`**; responsible for nuclear-technical support (`Ядерно-техническое обеспечение`), physical custody, storage, transport, safety, and cryptographic unblocking maintenance of all strategic and non-strategic nuclear warheads (`ЯБП`) of the Russian Federation, as well as the Special Control Service (`Служба специального контроля – ССК` for global nuclear test monitoring) (Wikipedia: 12th Chief Directorate).
- **Design Bureau & Scientific-Production Affiliates:**
    - **Warhead Design & Serial Production (Rosatom / MinSredMash Interface): RFNC-VNIIEF (`РФЯЦ-ВНИИЭФ` / `КБ-11`)**, Sarov (`54°55'00"N 43°19'00"E`); **RFNC-VNIITF (`РФЯЦ-ВНИИТФ` / `НИИ-1011`)**, Snezhinsk (`56°05'00"N 60°44'00"E`); **VNIIA named after N. L. Dukhov (`ВНИИА им. Н. Л. Духова`)**, Moscow (`55°47'15"N 37°35'45"E` — fuzing and Permissive Action Links / `Устройства блокировки`).
    - **12th GUMO Subordinate Research Institutes: 12th Central Research Institute (`12 ЦНИИ МО РФ`)** at Sergiyev Posad-7 (`Ферма`, Moscow Oblast `56°18'25"N 38°11'15"E` — nuclear weapons effects testing) and the **Central Physical-Technical Institute (`ЦФТИ МО РФ`)**.
- **Chief Designers & Historical/Current Commanders:**
    - Founded on **4 September 1947** as the Special Department of the General Staff under Colonel General **Vyacheslav Alexeyevich Malyshev** and Major General **Viktor анисимович Bolyatko** (Head of 12th GUMO 1958–1965); succeeded by Colonel General **Nikolai Pavlovich Yegorov** (1965–1974), Marshal of Engineering Troops **Viktor Kharlampiyevich Grekov**, and Colonel General **Vladimir Ivanovich Gerasimov** (1985–1992).
    - **Post-Soviet Commanders:** Re-established in the Russian Armed Forces on **2 September 1993** (MoD Order No. 68) under Colonel General **Yevgeny Petrovich Maslin** (1992–1997, who oversaw the consolidation of Soviet warheads from Ukraine, Belarus, and Kazakhstan under the Lisbon Protocol and Nunn-Lugar Cooperative Threat Reduction program), Colonel General **Igor Nikolayevich Valynkin** (1997–2005), Colonel General **Vladimir Nikolayevich Verkhovtsev** (2005–2010), Colonel General **Yuri Grigoryevich Сыч (Sych)** (2010–2017), and **Lieutenant General / Colonel General Igor Aleksandrovich Kolesnikov** (2017–present) (Stimson Center: US Access to 12th GUMO Storage Sites).
- **Deployment Coordinates (Headquarters & National-Level Nuclear Arsenals — `Объекты «С»`):**
    - **12th GUMO Main Headquarters:** Bolshoy Znamensky Pereulok 19, Khamovniki/Arbat, Moscow: **`55°44'51.8"N 37°36'06.4"E`** (`55.747733°N, 37.601767°E`) (Wikipedia: 12th Chief Directorate).
    - **12 Central National-Level Nuclear Storage Arsenals (`Национальные ядерные хранилища 12 ГУМО`):**
        1. **Vologda-20 (Chebsara / Object 955, `в/ч 25594`):** `59°13'40"N 38°48'10"E` (Vologda Oblast).
        2. **Voronezh-45 (Borisoglebsk / Gribanovsky / Object 387, `в/ч 14254`):** `51°21'48"N 41°57'40"E` (Voronezh Oblast).
        3. **Saratov-63 (Krasnoarmeysk / Karamysh, `в/ч 25623`):** `51°06'15"N 45°38'20"E` (Saratov Oblast).
        4. **Sverdlovsk-45 (Lesnoy / Nizhnyaya Tura / Object 917, `в/ч 40274`):** `58°38'00"N 59°47'00"E` (Sverdlovsk Oblast).
        5. **Chelyabinsk-115 (Karabash / Zlatoust-30, `в/ч 25625`):** `55°22'10"N 60°09'45"E` (Chelyabinsk Oblast).
        6. **Khabarovsk-47 (Korfovsky / Object 519, `в/ч 25624`):** `48°13'15"N 135°04'30"E` (Khabarovsk Krai).
        7. **Belgorod-22 (Golovchino / Object 1150, `в/ч 25624` branch):** `50°33'50"N 35°51'10"E` (Belgorod Oblast) (Gender Desk / Object 1150 OSINT).
        8. **Bryansk-18 (Rzhanitsa / Zhukovka, `в/ч 42685`):** `53°27'20"N 33°53'15"E` (Bryansk Oblast).
        9. **Tver-9 (Migalovo / Torzhok) & Mozhaysk-10 (Fersikovo / Object 714, `в/ч 52025`):** `55°28'55"N 35°54'10"E` (Moscow Oblast).
        10. **Olenegorsk-2 (Ramozero / Object 956, `в/ч 62834`):** `68°09'10"N 33°49'30"E` (Murmansk Oblast).
        11. **Sebezh-5 (Sosnovy Bor, `в/ч 42644`) & Kaliningrad Kulikovo (`54°54'30"N 20°21'15"E`):** Upgraded forward storage site in Kaliningrad Oblast (FAS: Russia Upgrades Nuclear Storage in Kaliningrad).
        12. **Irkutsk-45 (Zalarinsky / Golumet, `в/ч 39995`):** `53°18'45"N 102°22'10"E` (Irkutsk Oblast).
- **Engineering-Technical Parameters:**
    - **Physical Security & Storage Architecture:** Each national arsenal features triple-to-quadruple concentric perimeter sensor fences (`12 m` cleared microwave/seismic zones), underground earth-bermed reinforced concrete storage igloos, dedicated rail spurs for armored nuclear transport trains (`Спецэшелоны`), and Special Transport Groups (`СТГ`) equipped with armored warhead carriers (`НГ9В1` / `КамАЗ-5350` special vans) (FAS: Nuclear Storage Perimeters).
    - **Stockpile Custody (2025–2026 Baseline):** Maintains custody of Russia's total military inventory of **~4,309 to 4,400 nuclear warheads** (comprising ~1,718 deployed strategic warheads, ~1,114 reserve strategic warheads, and ~1,477 non-strategic warheads kept in central/base-level 12th GUMO storage) (Bulletin of the Atomic Scientists: Russian Nuclear Weapons 2025, Russian Nuclear Weapons 2026).
- **Causal Command Sequence:**

1. For **strategically deployed ICBMs and SLBMs**, warheads are physically mated to delivery vehicles, and 12th GUMO technical units (РТБ — ремонтно-технические базы) maintain the cryptographic Permissive Action Links (Устройства блокировки), which can only be unlocked by the General Staff unblocking code transmitted via Kazbek/Monolit/Signal-A.
2. For **strategic bombers and non-strategic delivery systems**, warheads are held unmated inside 12th GUMO central and base-level storage bunkers (РТБ). Only upon receiving a coded directive transmitted simultaneously over **at least three independent 12th GUMO communication channels** do the 12th GUMO Special Transport Groups (СТГ) extract the warheads from the vaults, transport them to the airbases/launch brigades, mate them to the delivery vehicles, and insert the pre-arming cryptographic keys (Wikipedia: 12th Chief Directorate, UNIDIR: Zero-Deployed Non-Strategic Weapons).

76. SPRN (Sistema Preduprezhdeniya o Raketnom Napadenii — Missile Attack Warning System)

- **GRAU & Institutional Index: СПРН (Система предупреждения о ракетном нападении)** — operated by the **820th Main Centre for Missile Attack Warning (820-й Главный центр предупреждения о ракетном нападении — ГЦПРН, Military Unit в/ч 44806 / formerly 3rd Independent Missile Attack Warning Army в/ч 73570)** of the **15th Special Purpose Aerospace Forces Army (15-я армия ВКС особого назначения)** (Wikipedia: Main Centre for Missile Attack Warning).
- **Design Bureau & System Integrator:**
  - **Overarching System Architect (Головная организация по СПРН): JSC Academician A. L. Mints Radiotechnical Institute (АО «Радиотехнический институт имени академика А. Л. Минца» — РТИ им. А. Л. Минца)**, 8 Marta Street 10, Moscow (55°48'02"N 37°33'15"E), together with **JSC Corporation "Kometa"** (space echelon) and **JSC NPK "NIIDAR"** (UHF radar echelon) (Wikipedia: Main Centre for Missile Attack Warning).
- **Chief Designers & Key Personnel:**
  - **Academician Alexander Lvovich Mints** — Founder of RTI and pioneer of Soviet early-warning radar arrays.
  - **Vladislav Georgiyevich Repin** — Chief Designer of the overall SPRN system and its Krokus / Elbrus command-information architecture (1970–1988).
  - **Viktor Pavlovich Sloka** — General Designer of RTI Mints (Daryal, Volga, and early Voronezh programs).
  - **Sergey Фёдорович Boev** — General Designer of the modern SPRN and Director of RTI Mints / Concern RTI (awarded Hero of Labor of the Russian Federation for completing the solid-state Voronezh perimeter).
- **Deployment Coordinates:**
  - **Central Command Post (КП СПРН): Solnechnogorsk-7 (Timonovo)**, Moscow Oblast: **56°11'00"N 37°00'35"E** (Wikipedia: Main Centre for Missile Attack Warning).
  - **Alternate Command Post (ЗКП СПРН): Komsomolsk-on-Amur (Pivan-1)** (50°20'57"N 137°11'22"E) and **Serpukhov-15 (Kurilovo)** (55°03'15"N 37°11'15"E).
- **Engineering-Technical Parameters:**
  - **Two-Echelon Dual-Phenomenology Architecture:**
    1. **First Echelon (Space-Based Infrared):** Originally Oko (US-K / US-KS / US-KMO), now replaced by the **Tundra (EKS 14F142 Kupol)** constellation in Molniya HEO, detecting missile boost-phase plumes within 60–90 seconds of liftoff.
    2. **Second Echelon (Ground-Based Over-the-Horizon & Above-the-Horizon Radars):** Continuous 360-degree peripheral radar fence comprising legacy Dnepr (5N86), Daryal (5N79), Volga (70M6), OTH 29B6 Kontayner (54°07'35"N 43°59'10"E at Kovylkino), and ten modern **Voronezh (77Ya6-M / -DM / -VP)** stations, tracking warhead buses and RV clouds at ranges up to 6,000–10,000 km (FRS: État des lieux du système d'alerte avancée russe).
  - **Commissioning Chronology:** Formally placed on combat duty on **15 February 1971** with the Solnechnogorsk command post and the Olenegorsk and Skrunda Dnestr-M nodes (Wikipedia: Main Centre for Missile Attack Warning).
- **Causal Command Sequence:**
  1. First-echelon Tundra (14F142) satellites detect the infrared boost plume and transmit launch azimuths to Solnechnogorsk (56°11'00"N 37°00'35"E).
  2. Second-echelon peripheral radars acquire the warhead bus as it rises above the radar horizon, confirming target count, trajectory, and predicted impact zones, and feeding the correlated attack vector into the Krokus / Kazbek NC3 system and the A-135/A-235 Moscow ABM complex.

77. Dnepr (5N86 Hen House Early Warning Radar)

- **GRAU & Industrial Index: 5Н86 «Днепр» (5N86 Dnepr / Днепр-М)**; NATO reporting name: **Hen House**; upgraded receiver-transmitter pairing with 5У83 «Даугава» (5U83 Daugava) at Olenegorsk (Wikipedia: Dnestr/Dnepr Radar, Project Orlan: 5N86 Dnepr).
- **Design Bureau & Manufacturer: Academician A. L. Mints Radiotechnical Institute (РТИ АН СССР / РТИ им. А. Л. Минца)**, Moscow (55°48'02"N 37°33'15"E); manufactured by the **Dnepr Machine-Building Plant (ДМЗ, Dnepropetrovsk)** (Project Orlan: 5N86 Dnepr).
- **Chief Designers & Key Personnel:**
  - **Yury Vsevolodovich Polyak** — Chief Designer of the 5N15 Dnestr, 5N15M Dnestr-M, and 5N86 Dnepr radar family at RTI Mints (under Academician **A. L. Mints**).
- **Deployment Coordinates (All 6 Historical & Active 5N86 Dnepr Nodes):**
  1. **Olenegorsk (RO-1, Murmansk Oblast): 68°06'59.6"N 33°54'55.5"E** (paired with 5U83 Daugava at 68°06'48.75"N 33°54'34.95"E) (Air Power Australia: Soviet/Russian ABM & EW Radars).
  2. **Balkhash-9 (Sary-Shagan / Priozersk, OS-2, Kazakhstan): 46°36'11.29"N 74°31'51.79"E** (entered service 12 May 1974; decommissioned/transferred 2020) (Wikipedia: Dnestr Radar).
  3. **Mishelevka (Usolye-Sibirskoye, OS-1, Irkutsk Oblast): 52°52'39.24"N 103°16'24.73"E** (replaced by Voronezh-M in 2015) (Air Power Australia).
  4. **Sevastopol (Khersones, RO-4, Crimea): 44°34'43.66"N 33°23'11.36"E** (entered service 1975/1979) (Air Power Australia).
  5. **Mukachevo (Pistryalovo, RO-5, Zakarpattia Oblast, Ukraine): 48°22'39.64"N 22°42'27.85"E** (entered service 1979; Russian data agreement terminated 2009) (Wikipedia: Dnestr Radar).
  6. **Skrunda-1 (RO-2, Latvia): 56°42'54.81"N 21°57'46.50"E** and **56°42'29.65"N 21°56'27.48"E** (decommissioned 31 August 1998 and dismantled) (Air Power Australia).

- **Engineering-Technical Parameters:**
  - **Antenna Geometry:** Bistatic/quasi-monostatic double sectoral horn waveguide slot array consisting of two wings (`each wing 250 m long × 12–14 m high`) angled in a V-formation around a central two-story computing/transmitter building (Wikipedia: Dnestr Radar).
  - **Frequency & Scanning:** VHF band (**`154–162 MHz`**, wavelength `~1.9 m`); frequency-steered azimuth scanning covering **`60° per wing`** **(`120° total azimuth`)** and **`5° to 35° in elevation`**; equipped with a polarizing filter and `800 μs` pulse width (Wikipedia: Dnestr Radar).
  - **Range & Power:** Maximum detection range **`4,000–6,000 km`** against ICBM warhead buses; peak pulse power `~1.5–2 MW` per transmitter set; central computing complex `M-10` / `5Э26` (Air Power Australia).
- **Causal Command Sequence:**
  1. Frequency-scanned VHF fan beams continuously sweep the 120° azimuth sector; upon detecting a ballistic object piercing the lower beam (`5°–10°`) and subsequently the upper beam (`25°–35°`), the `M-10` computer calculates the Keplerian orbit and impact ellipse and transmits the track over encrypted links (`Аппаратура передачи данных «Крокус»`) to the GTsPRN at Solnechnogorsk.

78. Dnestr (5N15 Hen House Early Warning Radar)

- **GRAU & Industrial Index:**
  - **Prototype (`ЦСО-П` — Центральная станция обнаружения ПВО):** Tested at Sary-Shagan (1960–1964) (WW2.dk: 10th ONIIP Sary-Shagan Sites).
  - **1st Production Variant (`5Н15 «Днестр»` / `5N15 Dnestr`):** Space surveillance (`СККП`) variant accepted into service in **April 1967** (Wikipedia: Dnestr Radar).
  - **Upgraded Early Warning Variant (`5Н15М «Днестр-М»` / `5N15M Dnestr-M`):** Ballistic missile early warning (`СПРН`) variant accepted into service on **15 February 1971**; NATO reporting name: **Hen House** (Wikipedia: Dnestr Radar).
- **Design Bureau & Manufacturer: Academician A. L. Mints Radiotechnical Institute (`РТИ АН СССР`)**, Moscow (`55°48'02"N 37°33'15"E`).
- **Chief Designers & Key Personnel:**
  - **Yury Vsevolodovich Polyak** (Chief Designer) and **Academician Alexander Lvovich Mints** (Scientific Director).
- **Deployment Coordinates:**
  - **Sary-Shagan Prototype (`TsSO-P` / `5N15` mock-up, Kazakhstan):** **`46°00'04.65"N 73°38'52.11"E`** (first space track acquired 17 September 1961; participated in 1961–1962 *Project K* high-altitude nuclear EMP tests) (Wikipedia: Dnestr Radar).
  - **Balkhash-9 (`OS-2`, 4 RLYa radar cells, Kazakhstan):**
    - Cell 1 (`Dnestr` -> `Dnestr-M`): **`46°37'53.15"N 74°30'44.49"E`**
    - Cell 2 (`Dnestr` -> `Dnestr-M`): **`46°37'31.31"N 74°31'02.60"E`**
    - Cell 3 (`Dnestr`): **`46°36'51.92"N 74°31'22.92"E`**
    - Cell 4 (`Dnestr`): **`46°36'26.61"N 74°31'23.53"E`** (Air Power Australia).
  - **Mishelevka (`OS-1`, 4 RLYa radar cells, Irkutsk Oblast):**
    - Cell 1: **`52°52'58.57"N 103°15'29.19"E`**
    - Cell 2: **`52°52'53.32"N 103°15'57.70"E`**
    - Cell 3: **`52°52'31.47"N 103°15'23.64"E`**
    - Cell 4: **`52°52'29.37"N 103°15'39.21"E`** (Air Power Australia).
  - **Olenegorsk (`RO-1`, `5N15M Dnestr-M`, Murmansk Oblast):** **`68°06'59.6"N 33°54'55.5"E`** (completed 1968, commissioned 1970, later upgraded to `5N86 Dnepr`) (Wikipedia: Dnestr Radar).
  - **Skrunda-1 (`RO-2`, `5N15M Dnestr-M`, Latvia):** **`56°42'54.81"N 21°57'46.50"E`** (commissioned 1970, later upgraded to `Dnepr`) (Wikipedia: Dnestr Radar).
- **Engineering-Technical Parameters:**
  - **Antenna & Beam Architecture:** Two waveguide horn wings (`250 m long × 20 m high` on `5N15`, reduced to `14 m` on `5N86`), each covering a **`30° azimuth sector`** (`60° total per radar cell`) with a `0.5°` frequency-steered beam and `+10° to +90°` elevation (`5N15 SKKP`) or `+10° to +30°` elevation (`5N15M SPRN`) (Wikipedia: Dnestr Radar).
  - **Pulse & Computing Upgrade (`5N15` to `5N15M`):** Pulse duration quadrupled from **`200 μs`** **(`5N15`)** to **`800 μs`** **(`5N15M`)**, increasing effective range from `~2,500–3,000 km` to `~3,500–4,500 km`, processed by two `M-4 2M` semiconductor computers (Wikipedia: Dnestr Radar).
- **Causal Command Sequence:**
  1. Formed the foundational sensor layer of the Soviet `СККП` (Space Surveillance System, 1967) and `СПРН` (Missile Attack Warning System, 1971), feeding orbital elements and SLBM launch tracks from the Norwegian/Barents Seas to the Solnechnogorsk command post (Wikipedia: Dnestr Radar).

79. Daryal (5N79 Pechora LPAR Early Warning Radar)

- **GRAU & Industrial Index:**
  - **Baseline Complex:** **`5Н79 «Дарьял»`** **(`5N79 Daryal`)**; NATO reporting name: **Pechora** (Wikipedia: Daryal Radar).
  - **Receiver Prototype:** **`5У83 «Даугава»`** **(`5U83 Daugava`)** at Olenegorsk (Wikipedia: Daryal Radar).
  - **Modernized Variants (Uncompleted/Demolished):** **`90Н6 «Дарьял-У»`** **(`90N6 Daryal-U`)** and **`«Дарьял-УМ»`** **(`Daryal-UM`)** (Wikipedia: Daryal Radar).
- **Design Bureau & Manufacturer: Academician A. L. Mints Radiotechnical Institute (`РТИ АН СССР / РТИ им. А. Л. Минца`)**, Moscow (`55°48'02"N 37°33'15"E`).
- **Chief Designers & Key Personnel:**

    - **Viktor Pavlovich Sloka** — General Designer of the 5N79 Daryal Large Phased Array Radar (LPAR) (awarded Hero of Socialist Labor for the Pechora/Gabala deployments).
    - **A. L. Mints** and **Yu. V. Polyak** (early conceptual design).
- **Deployment Coordinates (All 7 Planned/Built Daryal Sites):**
    1. **Pechora (RO-30, Komi Republic — Operational 1984):** Transmitter & Receiver complex at **65°12'37.88"N 57°17'07.64"E** (entered combat duty **20 January 1984**) (Air Power Australia).
    2. **Gabala (RO-7, Lyaki, Azerbaijan — Operational 1985–2012):** **40°52'11.08"N 47°48'06.11"E** (entered combat duty **February 1985**; shut down **10 December 2012** upon lease expiration and replaced by the Armavir Voronezh-DM) (Wikipedia: Gabala Radar Station).
    3. **Balkhash-9 (OS-2, 90N6 Daryal-U, Kazakhstan):** **46°35'42.87"N 74°28'57.95"E** (uncompleted; damaged by fire in September 2004 and demolished) (Air Power Australia).
    4. **Mishelevka (OS-1, 90N6 Daryal-U, Irkutsk Oblast):** **52°51'34.34"N 103°14'01.34"E** (uncompleted; demolished **21 June 2011** to make way for Voronezh-M) (Wikipedia: Mishelevka Radar Station).
    5. **Yeniseysk-15 (Abalakovo, Daryal-U, Krasnoyarsk Krai):** **57°52'14.05"N 93°06'48.36"E** (violated the 1972 ABM Treaty due to inland siting; dismantled 1989–1991) (Air Power Australia).
    6. **Skrunda-2 (RO-2, Daryal-UM, Latvia):** **56°43'32.60"N 21°58'51.36"E** (demolished by controlled implosion on **4 May 1995**) (Air Power Australia).
    7. **Mukachevo (Pistryalovo, Daryal-UM, Ukraine):** **48°23'14.59"N 22°47'49.07"E** (construction halted and dismantled 1990s–2000s) (Air Power Australia).
- **Engineering-Technical Parameters:**
    - **Bistatic Active Phased Array Architecture:** Consists of two separate monumental concrete buildings spaced 500 m to 1,500 m apart:
        - **Transmitter Building (Передающая позиция):** 30 m × 40 m aperture (60 m building height) housing **1,260 active VHF transmitter modules (90N6 Daryal-U reduced to 630 modules)**.
        - **Receiver Building (Приёмная позиция):** 80 m × 80 m aperture (100 m building height) housing **4,000 crossed-dipole receiving elements** (Wikipedia: Mishelevka Radar Station).
    - **Frequency, Range & Power:** Operates in the **VHF band (150–200 MHz, wavelength 1.5–2.0 m)**; detection range **6,000 km** against a 0.1–0.2 m² radar cross-section (RCS) target across a 110° azimuth and 0.5°–55° elevation sector; electrical power consumption **~50 MW** (requiring dedicated power plants and cooling reservoirs); processed by dual M-10 / Эльбрус-2 vector supercomputers (Wikipedia: Daryal Radar).
- **Causal Command Sequence:**
    1. Pechora (65°12'37.88"N 57°17'07.64"E) monitors the principal trans-polar ICBM attack corridor from North America over the Arctic Ocean; upon acquiring incoming warhead clouds and decoys, its Elbrus processors perform polarimetric and trajectory discrimination and feed real-time target vectors via Krokus to Solnechnogorsk (56°11'00"N 37°00'35"E) and Kazbek.

80. Volga (70M6 Hantsavichy Solid-State Early Warning Radar)

- **GRAU & Industrial Index: 70M6 «Волга» (70M6 Volga)** — Radio-technical Node **"Baranovichi" (ОРТУ «Барановичи», Military Unit в/ч 03522 / 474th Separate Radio-Technical Node)** (Wikipedia: Baranavichy Radar Station).
- **Design Bureau & Manufacturer: Academician A. L. Mints Radiotechnical Institute (РТИ им. А. Л. Минца)**, Moscow (55°48'02"N 37°33'15"E), in cooperation with **NPO "Integral"** (Minsk, Belarus) and **NIIVK** (Moscow, 5Э26 / Эльбрус computing complex) (Wikipedia: Daryal Radar / Volga Variant).
- **Chief Designers & Key Personnel:**
    - **Viktor Pavlovich Sloka** — General Designer of RTI Mints and Chief Designer of the 70M6 Volga radar (awarded State Prize of the Russian Federation after commissioning).
    - **Alexander Grigoryevich Lukashenko** & **Vladimir Vladimirovich Putin** (ratified the 1995 and 2021 bilateral lease extensions granting Russian Aerospace Forces rent-free operational control of the Kletsk/Hantsavichy node) (Wikipedia: Baranavichy Radar Station).
- **Deployment Coordinates:**
    - **Radar Complex (Ozerechye / Kletsk-2 / Hantsavichy, Brest/Minsk Oblast border, Belarus):**
        - **Transmitter & Receiver Complex Center: 52°50'51.97"N 26°28'15.70"E** (52.8478°N, 26.4710°E) (Air Power Australia).
        - **Transmitter Array:** 52°51'41.98"N 26°28'02.88"E; **Receiver Array:** 52°51'10.11"N 26°28'49.94"E (separated by ~1,300 m, oriented along a **262.5° WSW boresight** covering all of Western Europe and the North Atlantic).
- **Engineering-Technical Parameters:**
    - **Construction & Commissioning:** Construction began in **1982** as a solid-state UHF counterpart to Daryal, paused in 1991, resumed in 1993–1998 after the loss of the Skrunda radars in Latvia, entered trial combat duty in 2002, and was formally commissioned on **1 October 2003** (upgraded in 2016) (Wikipedia: Baranavichy Radar Station).
    - **Bistatic Modular Solid-State Architecture:**
        - **Transmitter Building:** 36 m × 20 m aperture (30° inclination), completely solid-state modular UHF active phased array (eliminating the high-voltage vacuum-tube klystrons of Daryal).
        - **Receiver Building:** 60 m × 60 m aperture (30° inclination).
    - **Frequency & Performance:** Operates in the **decimeter UHF band (~420–450 MHz, wavelength ~70 cm)**; azimuth coverage **120°** (4° to 54° elevation); detection range **4,800 km** against ballistic missile warheads and LEO satellites in the North Atlantic and Norwegian Sea SLBM patrol zones; power consumption reduced to **~10–15 MW** (compared to 50 MW for Daryal) (Wikipedia: Baranavichy Radar Station).
- **Causal Command Sequence:**
    1. Continuously scans the Western European and North Atlantic corridor (202.5° to 322.5° azimuth); upon detecting SLBM launches from NATO submarines in the North Atlantic or medium/intercontinental missiles in Europe, 70M6 Volga computes precision UHF trajectory parameters and transmits them directly via encrypted fiber/satellite trunks to the GTsPRN at Solnechnogorsk (56°11'00"N 37°00'35"E) (Wikipedia: Baranavichy Radar Station).

81. Voronezh (77Ya6 Voronezh-M / Voronezh-DM / Voronezh-VP Radars)

- **GRAU & Industrial Index:**
  - **VHF Meter-Wave Family: 77Я6 «Воронеж-М» (77Ya6 Voronezh-M)** and high-potential 6-segment **77Я6-ВП «Воронеж-ВП» (77Ya6-VP Voronezh-VP)** (Wikipedia: Voronezh Radar).
  - **UHF Decimeter-Wave Family: 77Я6-ДМ «Воронеж-ДМ» (77Ya6-DM Voronezh-DM)** (Wikipedia: Voronezh Radar).
  - **SHF Centimeter-Wave Variant: «Воронеж-СМ» (Voronezh-SM)** (Russian Strategic Nuclear Forces: SPRN & Voronezh Radar).
- **Design Bureau & Manufacturer:**
  - **VHF Variants (77Ya6-M and 77Ya6-VP): JSC Academician A. L. Mints Radiotechnical Institute (РТИ им. А. Л. Минца)**, Moscow (55°48'02"N 37°33'15"E) (Wikipedia: Voronezh Radar).
  - **UHF Variant (77Ya6-DM): JSC NPK "NIIDAR" (АО «НПК „НИИДАР“» — Scientific and Production Corporation "Research Institute of Long-Range Radio Communications")**, Preobrazhenskaya Ploshchad 8, Moscow (55°47'45"N 37°42'40"E) (Wikipedia: Voronezh Radar).
- **Chief Designers & Key Personnel:**
  - **Sergey Fyodorovich Boev** (General Designer of the SPRN and Head of Concern RTI).
  - **Vladimir Ivanovich Karasev** — Chief Designer of the 77Ya6 Voronezh-M and Voronezh-VP VHF radars at RTI Mints.
  - **Sergey Dmitriyevich Saprykin** — General Designer at NPK NIIDAR for the 77Ya6-DM Voronezh-DM UHF radar.
- **Deployment Coordinates (All 10 Operational / Under-Construction Voronezh Stations):**
  1. **Lekhtusi (Leningrad Oblast — 77Ya6-VP, 6 segments): 60°16'31.65"N 30°32'45.66"E** (trial service December 2006, upgraded to Voronezh-VP and fully operational February 2012) (Wikipedia: Voronezh Radar).
  2. **Armavir (Baronovsky, Krasnodar Krai — Two 77Ya6-DM arrays): 44°55'30.38"N 40°59'02.02"E** (SW and SE arrays replacing Sevastopol/Gabala, fully operational April 2015) (Wikipedia: Voronezh Radar).
  3. **Pionersky (Dunayevka Air Base, Kaliningrad Oblast — 77Ya6-DM): 54°51'26.26"N 20°10'56.46"E** (54.857294°N, 20.18235°E, fully operational 2014) (Wikipedia: Voronezh Radar).
  4. **Mishelevka (Usolye-Sibirskoye, Irkutsk Oblast — Two 77Ya6-VP arrays): 52°51'20.11"N 103°13'53.94"E** (fully operational 2015) (Wikipedia: Voronezh Radar).
  5. **Yeniseysk (Uzhur / Shapkino, Krasnoyarsk Krai — 77Ya6-DM): 58°30'21.94"N 92°02'45.86"E** (58.506095°N, 92.046072°E, entered combat duty December 2017) (Wikipedia: Voronezh Radar, RussianForces.org: Three Radars Begin Combat Duty).
  6. **Barnaul (Konyukhi, Altai Krai — 77Ya6-DM): 53°08'21.1"N 83°40'52.5"E** (entered combat duty December 2017) (RussianForces.org).
  7. **Orsk (Novoorsk, Orenburg Oblast — 77Ya6-M): 51°16'24.0"N 58°57'32.7"E** (entered combat duty December 2017) (Wikipedia: Voronezh Radar).
  8. **Vorkuta (Vorgashor, Komi Republic — 77Ya6-M / -VP): 67°36'50.3"N 63°45'19.5"E** (commissioned 2021–2022 to replace/augment Pechora Daryal).
  9. **Olenegorsk (Protoki, Murmansk Oblast — 77Ya6-VP / -DM): 68°06'59.6"N 33°54'55.5"E** (replacing legacy Dnepr/Daugava).
  10. **Sevastopol (Khersones, Crimea — 77Ya6-SM / -DM): 44°34'44"N 33°23'12"E** (Wikipedia: Voronezh Radar).
- **Engineering-Technical Parameters:**
  - **High Factory Readiness (ВЗГ — Высокая заводская готовность):** Assembled on-site from **23–30 prefabricated container modules** on a flat reinforced concrete pad within 18–24 months (compared to 5–10 years for Daryal), consuming only **0.7 MW of electrical power** (versus 50 MW for Daryal) and manned by a duty shift of just 15 personnel (RussianForces.org: Modular EW Radar, Wikipedia: Voronezh Radar).
  - **Frequency & Range:**
    - 77Ya6-M / 77Ya6-VP (VHF): **150–200 MHz (1.5–2.0 m wavelength)**, 3-segment (M) or 6-segment (VP) open truss phased array (30 m × 30 m), horizon range **4,200–6,000 km** (Signal Identification Wiki: 77Ya6 Voronezh).
    - 77Ya6-DM (UHF): **430–440 MHz (~0.7 m / decimeter wavelength, FMCW modulation, 2 MHz bandwidth)**, enclosed radome-protected active phased array, horizon range **6,000 km**, vertical space-tracking altitude **8,000–10,000 km**, capable of simultaneously tracking **500 ballistic and space objects** (Wikipedia: Voronezh Radar, Signal Identification Wiki: 77Ya6 Voronezh).
- **Causal Command Sequence:**
  1. Forms the complete continuous 360-degree solid-state terrestrial perimeter of Russia's modern SPRN; VHF Voronezh-M/VP arrays provide wide-volume search and stealth/hypersonic detection while UHF Voronezh-DM arrays provide centimetre-accurate warhead vs. decoy discrimination and impact-point prediction, streaming tracks directly to Solnechnogorsk (56°11'00"N 37°00'35"E) and the Krokus/Kazbek NC3 network.

Subsystem Group IV.C: Naval Strategic Forces — SSBNs, SLBMs & Heavy Silo ICBMs (Entities 82–88)

82. Project 955 Borei-Class (955 / 955A Borei-A SSBN)

- **GRAU & Naval Project Index: Проект 955 «Борей» (Project 955 Borei, R&D cipher 09550)** and modernized **Проект 955А «Борей-А» (Project 955A Borei-A, R&D cipher 09551)**; NATO reporting name: **Dolgorukiy / Borei I & Borei II**; Armed with the **Д-30 (D-30)** missile launch complex housing **16 × R-30 Bulava (3M30 / RSM-56) SLBMs** (Wikipedia: Borei-Class Submarine, RussianForces.org: Strategic Fleet).
- **Design Bureau & Shipyard:**
  - **Design Bureau: JSC Central Design Bureau for Marine Engineering "Rubin" (АО «ЦКБ МТ „Рубин“»)**, Marata Street 90, Saint Petersburg (59°55'05"N 30°20'22"E) (Deagel: Project 955).
  - **Shipyard: JSC Production Association "Sevmash" (АО «ПО „Севмаш“» / Shipyard No. 402)**, Severodvinsk, Arkhangelsk Oblast (64°34'45"N 39°48'35"E) (Wikipedia: Borei-Class Submarine).
  - **Combat Information & Fire-Control System (БИУС «Округ»): JSC Concern "Morinformsystem-Agat"** (Moscow) (Corporation of Marine Instrumentation: Concern Agat).

- **Chief Designers & Key Personnel:**
  - **Vladimir Nikolayevich Zdornov** — Chief Designer of Project 955 (`Borei`) at TsKB MT "Rubin" (succeeded initial conceptual work under **Academician Sergey Nikitich Kovalev**).
  - **Sergey Vladimirovich Sukhanov** — Chief Designer of Project 955A (`Borei-A`) at TsKB MT "Rubin".
  - **Igor Vilnit** — General Director of TsKB MT "Rubin"; **Mikhail Valentinovich Budnichenko** — General Director of PO "Sevmash".
- **Deployment Coordinates (Northern Fleet & Pacific Fleet SSBN Bases):**
  - **Northern Fleet (31st Submarine Division, Gadzhiyevo / Yagelnaya Bay, Murmansk Oblast): `69°15'22"N 33°19'48"E`** — Homeport to `K-535 Yury Dolgorukiy` (Project 955, commissioned 10 January 2013) and `K-549 Knyaz Vladimir` (Project 955A, commissioned 12 June 2020), with `K-555 Knyaz Pozharskiy` (Project 955A, launched February 2024 / commissioned 2025) (FAS Nuclear Notebook 2025, Wikipedia: Borei-Class Submarine).
  - **Pacific Fleet (25th Submarine Division, Vilyuchinsk / Rybachiy / Krasheninnikova Bay, Kamchatka Krai): `52°55'15"N 158°29'20"E`** — Homeport to `K-550 Alexander Nevskiy` (955, commissioned 23 December 2013), `K-551 Vladimir Monomakh` (955, commissioned 19 December 2014), `K-552 Knyaz Oleg` (955A, commissioned 21 December 2021), `K-553 Generalissimus Suvorov` (955A, commissioned 29 December 2022), and `K-554 Imperator Alexander III` (955A, commissioned 11 December 2023) (RussianForces.org: Strategic Fleet, IISS Military Balance / Taylor & Francis).
  - **Additional Hulls Under Construction at Sevmash (`64°34'45"N 39°48'35"E`):** `K-556 Knyaz Potemkin` and `K-557 Dmitry Donskoy` (laid down 23 August 2021).
- **Engineering-Technical Parameters:**
  - **Dimensions & Displacement:** Surface displacement **`14,720 tonnes`**, Submerged displacement **`24,000 tonnes`**; Length **`170.0 m`**; Beam **`13.5 m`**; Draft **`10.0 m`**; Crew **`107 personnel`** (including 55 officers), equipped with a detachable full-crew ascending rescue capsule (`Всплывающая спасательная камера`) in the sail (Wikipedia: Borei-Class Submarine).
  - **Nuclear Propulsion & Pump-Jet:** Single **`OK-650V`** **(`190 MWt`)** pressurized-water nuclear reactor (designed by **OKBM Afrikantov**, Nizhny Novgorod `56°19'40"N 43°53'15"E`) driving a `GTZA` steam turbine (`~37,000–40,000 kW / 50,000 shp`) coupled to the **first single-shaft pump-jet propulsor (`водомётный движитель`) installed on a Russian SSBN**, yielding a submerged speed of **`29 knots`** and acoustic signature 5 times quieter than Project 971 *Akula* / Project 949A *Antey* (Deagel: Project 955).
  - **Sonar, C3 & Armament: `MGK-600B Irtysh-Amfora-Borei`** digital spherical bow sonar array; `Paravan` towed VLF communication buoy and `Zveno` / `Snegir` satellite transceivers; **16 × `D-30` launch tubes** for R-30 Bulava SLBMs (total 96–160 MIRVs per boat) plus **6 × 533 mm bow torpedo tubes** (Wikipedia: Borei-Class Submarine).
- **Causal Command Sequence:**
  1. On deterrent patrol under the Arctic ice cap or in the Sea of Okhotsk bastion, the Borei SSBN trails its `Paravan` VLF buoy at `15–20 m` depth while cruising at `50–100 m` depth.
  2. Upon receiving an encrypted Kazbek launch directive via VLF shoreside stations or a Tu-142MR Orel relay, the Commanding Officer and Senior Assistant (`Старпом`) authenticate the code on the Agat `Okrug` combat console; the submarine rises to launch depth (`40–55 m`) or surfaces through polar ice, equalizes tube pressure, and executes a dry-canister gas-generator salvo launch of all 16 Bulava SLBMs while moving at up to `5–7 knots` (Deagel: Project 955).

83. Project 667BDRM Delfin-Class (Delta IV SSBN)

- **GRAU & Naval Project Index: `Проект 667БДРМ «Дельфин»` (`Project 667BDRM Delfin`, R&D cipher `06678`)**; NATO reporting name: **Delta IV**; Armed with the **`Д-9РМ`** **(`D-9RM`)** / **`Д-29РМУ2`** **(`D-29RMU2`)** launch complex carrying **16 × R-29RMU2 Sineva / R-29RMU2.1 Layner SLBMs** (Wikipedia: Delta-Class Submarine, Weaponsystems.net: R-29RMU2 Sineva).
- **Design Bureau & Shipyard:**
  - **Design Bureau: Central Design Bureau for Marine Engineering "Rubin" (`ЦКБ МТ «Рубин»`)**, Saint Petersburg (`59°55'05"N 30°20'22"E`).
  - **Shipyard: PO "Sevmash"** (`Северное машиностроительное предприятие`), Severodvinsk (`64°34'45"N 39°48'35"E`); mid-life refits and `Sineva`/`Layner` upgrades performed at **JSC Ship Repair Center "Zvyozdochka" (`ЦС «Звёздочка»`)**, Severodvinsk (`64°32'30"N 39°46'15"E`).
- **Chief Designers & Key Personnel:**
  - **Academician Sergey Nikitich Kovalev** — General Designer of the entire Project 667 family (`667A` through `667BDRM`) at TsKB MT "Rubin" (twice Hero of Socialist Labor).
- **Deployment Coordinates:**
  - **Northern Fleet (31st Submarine Division, Gadzhiyevo / Yagelnaya Bay, Murmansk Oblast): `69°15'22"N 33°19'48"E`** — All 7 boats were constructed between 1981 and 1990 (`K-51 Verkhoturye`, `K-84 Yekaterinburg`, `K-64 Podmoskovye` [converted to special mission submarine `BS-64`], `K-114 Tula`, `K-117 Bryansk`, `K-18 Karelia`, and `K-407 Novomoskovsk`), with the active strategic boats homeported at Yagelnaya Bay, Gadzhiyevo (Wikipedia: Russian Submarine Tula).
- **Engineering-Technical Parameters:**
  - **Dimensions & Displacement:** Surface displacement **`11,740 tonnes`**, Submerged displacement **`18,200 tonnes`**; Length **`167.4 m`**; Beam **`11.7 m`**; Draft **`8.8 m`**; Test depth **`400 m`**, crush depth **`550–650 m`**; Double hull constructed of high-yield **`AK-32`** **(`AK-32`)** steel; Crew **`135–140 personnel`** (Wikipedia: Delta-Class Submarine).
  - **Nuclear Propulsion:** Two **`VM-4SG`** **(`ВМ-4СГ`, `2 × 90 MWt = 180 MWt total`)** pressurized-water reactors driving two `GTZA-635` steam turbines (`2 × 20,000 shp = 40,000 shp total`) and twin 5-bladed low-RPM skewed propellers with hub vortex diffusers; Submerged speed **`24 knots`** (Wikipedia: Delta-Class Submarine).
  - **Combat & Launch Systems:** `Omnibus-BDRM` combat management system (Concern Agat), `Shlyuz` astro-inertial navigation complex, `Skat-VDRM` sonar, and `D-9RM`/`D-29RMU2` wet-launch tubes capable of **circular-sector ripple-firing all 16 missiles (`Operation Behemoth-2`, executed by `K-407 Novomoskovsk` on 6 August 1991, firing all 16 SLBMs in 224 seconds at 14-second intervals)** from a depth of `55 meters` at speeds up to `6–7 knots` (Wikipedia: Delta-Class Submarine, FAS Guide: R-29 SLBM).
- **Causal Command Sequence:**
  1. Receives encrypted launch directive via the `Zevs` (`82 Hz` ELF transmitter at Kola Peninsula `68°48'49"N 33°49'46"E`, which alerts deep-submerged SSBNs to rise to VLF reception depth) followed by the full Kazbek launch cipher over VLF.

2. Upon dual-key authentication on the `Omnibus-BDRM` console, the `D-9RM` system floods each missile silo prior to opening the outer hatch (`мокрый старт` — wet launch), ignites the `R-29RMU2 Sineva` / `Layner` first-stage hypergolic engine directly inside the flooded tube, and ripple-fires the 16-missile salvo at 14-second intervals (FAS Guide: R-29 SLBM).

84. Project 941 Akula-Class (Typhoon SSBN)

- **GRAU & Naval Project Index: `Проект 941 «Акула»` (`Project 941 Akula`)** and **`Проект 941УМ` (`Project 941UM` — `TK-208 Dmitry Donskoy` Bulava testbed)**; NATO reporting name: **Typhoon**; Armed with the **`Д-19` (`D-19`)** missile complex carrying **20 × `R-39 Rif` (`3M65` / `RSM-52` / `SS-N-20 Sturgeon`) three-stage solid-propellant SLBMs** (Wikipedia: Typhoon-Class Submarine, RussianForces.org: Strategic Fleet).
- **Design Bureau & Shipyard:**
  - **Design Bureau: Central Design Bureau for Marine Engineering "Rubin" (`ЦКБ МТ «Рубин»`)**, Saint Petersburg (`59°55'05"N 30°20'22"E`), authorized by government decree in December 1972 / 1973 in direct response to the US Navy *Ohio*-class SSBN (Wikipedia: Typhoon-Class Submarine).
  - **Shipyard: PO "Sevmash"** (Shop No. 55 — the world's largest covered shipbuilding hall), Severodvinsk (`64°34'45"N 39°48'35"E`) (Wikipedia: Russian Submarine Dmitry Donskoy).
- **Chief Designers & Key Personnel:**
  - **Academician Sergey Nikitich Kovalev** — General Designer of Project 941 *Akula* at TsKB MT "Rubin" (Wikipedia: Typhoon-Class Submarine).
  - **Academician Viktor Petrovich Makeev** — General Designer of the `D-19` / `R-39 Rif` (`3M65`) SLBM system at SKB-385 (`GRTs Makeev`).
- **Deployment Coordinates:**
  - **Historical Operational Base (18th Submarine Division, Northern Fleet): Nerpichya Bay (`Губа Нерпичья`), Zapadnaya Litsa (Zaozyorsk / Murmansk-150)**, Murmansk Oblast: **`69°26'00"N 32°21'00"E`** (purpose-built with reinforced piers and heavy gantry cranes for loading the 84-tonne `R-39` canisters).
  - **Testbed Homeport (`TK-208 Dmitry Donskoy`, 2002–2023): Severodvinsk (Belomorskaya Naval Base)** (`64°34'45"N 39°48'35"E`), conducting launches in the White Sea test range (`65°30'00"N 38°00'00"E`) (Wikipedia: Russian Submarine Dmitry Donskoy).
- **Engineering-Technical Parameters:**
  - **Fleet Registry (6 Hulls Built 1976–1989):** `TK-208 Dmitry Donskoy` (commissioned 29 December 1981, modified to `941UM` in 2002, decommissioned 6 February 2023), `TK-202` (scrapped 1999–2005 under Nunn-Lugar CTR), `TK-12 Simbirsk` (scrapped 2006–2008), `TK-13` (scrapped 2007–2009), `TK-17 Arkhangelsk` (placed in reserve 2006, decommissioned), and `TK-20 Severstal` (placed in reserve 2004, decommissioned) (Wikipedia: Typhoon-Class Submarine, RussianForces.org: Strategic Fleet).
  - **Multi-Hull Titanium Architecture & Displacement:** Surface displacement **`23,200–24,500 tonnes`**, Submerged displacement **`48,000 tonnes`** (the largest ballistic missile submarine class by displacement ever built); Length **`172.8–175.0 m`**; Beam **`23.3 m`**; Draft **`11.0 m`**; Crew **`160 personnel`** (Wikipedia: Typhoon-Class Submarine). Features a unique **catamaran-style five-inner-hull layout** inside an outer hydrodynamic steel casing: two parallel **`8.5 m`-diameter Titanium Alloy (`ПТ-3В` / `Ti-Al-V`) main pressure hulls** side-by-side, with the **20 × `D-19` missile launch tubes** positioned forward of the sail between the two main hulls, plus three smaller titanium pressure modules (command/control room under the sail, bow torpedo module, and aft steering module) and two detachably surfacing escape pods (Wikipedia: Typhoon-Class Submarine).
  - **Nuclear Propulsion & Arctic Icebreaking:** Two **`OK-650` (`2 × 190 MWt = 380 MWt total`)** pressurized-water nuclear reactors driving two `GTZA-674` steam turbines (`2 × 50,000 shp = 100,000 shp / 74.6 MW total`) and twin shrouded 7-bladed propellers; Submerged speed **`25–27 knots`**; reinforced sail and retractable bow planes engineered to break straight through **`2.5 to 3.0 meters` of solid Arctic pack ice** (Wikipedia: Typhoon-Class Submarine).
- **Causal Command Sequence:**
  1. Engineered to operate beneath the thick central Arctic ice pack where NATO ASW submarines and aircraft could not penetrate; upon receiving a strategic launch directive via ELF/VLF (`Paravan` buoy), the Project 941 boat used its upward-looking ice-profiling sonar (`МГ-518 «Север»`) to locate a polynya or punch through up to 2.5 m of ice, clearing the forward deck to fire its 20 × `R-39 Rif` (`10 MIRVs × 100–200 kt each = 200 warheads per submarine`) in a dry-canister mortar-ejection salvo.

85. R-30 Bulava (3M30 / RSM-56 SLBM)

- **GRAU, Treaty & NATO Index:** Operational designation **`P-30 «Булава»` (`R-30 Bulava` / `Булава-30`)**; GRAU missile index **`3M30` (`3M30`)**; Shipborne launch complex **`Д-30` (`D-30`)**; START/New START treaty designation **`PCM-56` (`RSM-56`)**; NATO reporting name: **SS-N-32** (development designation `SS-NX-30`) (CSIS Missile Threat: RSM-56 Bulava, Wikipedia: RSM-56 Bulava).
- **Design Bureau & Manufacturer:**
  - **Design Bureau: JSC Corporation "Moscow Institute of Thermal Technology" (`АО «Корпорация „МИТ“»` / MITT)**, Berezovaya Alleya 10, Moscow (`55°51'16"N 37°36'56"E`) — transferred from GRTs Makeev to MITT by Russian Security Council decision in **1998** following three failed launches of Makeev's `R-39UTTKh Bark` (`3M91`) (CSIS Missile Threat: RSM-56 Bulava, FAS Guide: R-39M Bark).
  - **Serial Manufacturer: JSC "Votkinsk Plant" (`АО «Воткинский завод»`)**, Votkinsk, Udmurt Republic (`57°03'12"N 53°59'10"E`).
  - **Submarine Launch Complex (`D-30`): JSC GRTs Makeev** (Miass) and **KB Mashinostroyeniya** (Moscow).
- **Chief Designers & Key Personnel:**
  - **Academician Yury Solomonovich Solomonov** — General Designer of MITT and Chief Designer of the `R-30 Bulava`, `RT-2PM2 Topol-M`, and `RS-24 Yars` (NTI Russia Nuclear Chronology, Wikipedia: RSM-56 Bulava).
- **Deployment Coordinates:**
  - **Operational Deployment (16 missiles per boat across 8+ Project 955 / 955A SSBNs = 128+ deployed launchers):**
    - **Gadzhiyevo (Yagelnaya Bay, Northern Fleet): `69°15'22"N 33°19'48"E`.**
    - **Vilyuchinsk (Rybachiy, Pacific Fleet): `52°55'15"N 158°29'20"E`** (FAS Nuclear Notebook 2025).

    - **Flight Test Ranges:** Launched from the White Sea (`65°30'00"N 38°00'00"E`) to the **Kura Missile Test Range** on Kamchatka (`57°20'00"N 161°50'00"E`) and from the Sea of Okhotsk (`54°00'00"N 148°00'00"E`) to the **Chizha Test Range** on the Kanin Peninsula (`67°05'00"N 44°20'00"E`).
- **Engineering-Technical Parameters:**
    - **Dimensions & Mass:** Length **`12.1 m`** (`11.5 m` without warhead nose section); Diameter **`2.0 m`** (tapering to `1.8 m` on upper stages); Launch mass **`36,800 kg`**; Declared START throw-weight **`1,150 kg`** (Wikipedia: RSM-56 Bulava).
    - **Three-Stage Hybrid Propulsion Architecture:**
        - **1st & 2nd Stages:** High-energy composite **solid propellant** wound in aramid/carbon-fiber cases with gimbaled thrust-vector-control nozzles (`fast boost phase <180 seconds` to defeat space-based boost-phase interceptors).
        - **3rd Stage / Post-Boost Vehicle (PBV):** Storable **liquid propellant** propulsion module enabling high-agility midcourse velocity trimming, depressed/quasi-ballistic trajectories (`apogee reduced below standard ballistic arcs`), and evasive post-boost dispensing (Wikipedia: RSM-56 Bulava, NTI Russia Nuclear Chronology).
    - **Range, Guidance & Warhead Payload:** Operational range **`8,300 to 9,300+ km`**; Astro-inertial guidance with **GLONASS** satellite updates, achieving a **CEP of `120–350 m`**; Armed with **6 to 10 independently targetable `100–150 kt` thermonuclear MIRV warheads** (commonly deployed with **6 MIRVs** plus ~40 heavy/light decoys and chaff penetration aids under New START) (CSIS Missile Threat: RSM-56 Bulava, FAS Nuclear Notebook 2025).
- **Causal Command Sequence:**
    1. Upon receipt of the unblocking and launch order via the Agat `Okrug` console aboard a Project 955 Borei SSBN, the `D-30` launch tube executes an inclined **"dry" cold launch (`сухой миномётный старт`)**: a solid-fuel gas generator pressurizes the sealed launch canister, ejecting the `3M30 Bulava` through a rupturable membrane and out of the water inside a gas-cavitation bubble while the submarine moves at up to 7 knots, igniting the 1st-stage solid motor once clear of the sea surface (Wikipedia: RSM-56 Bulava).

86. R-29RMU2 Sineva (3M37 / RSM-54 SLBM)

- **GRAU, Treaty & NATO Index:** Operational designation **`Р-29РМУ2 «Синева»` (`R-29RMU2 Sineva`)**; GRAU missile index **`3М37` (`3M37`**, upgraded from baseline `3M37 R-29RM Shtil`); Shipborne launch complex **`Д-29РМУ2` (`D-29RMU2`**, modernized `D-9RM`); START/New START treaty designation **`РСМ-54` (`RSM-54`)**; NATO reporting name: **SS-N-23A Skiff** (`SS-N-23 Mod 2`) (CSIS Missile Threat: R-29RM Shtil / Sineva, Weaponsystems.net: R-29RMU2 Sineva).
- **Design Bureau & Manufacturer:**
    - **Design Bureau: JSC State Rocket Center named after Academician V. P. Makeev (`АО «ГРЦ им. академика В. П. Макеева»`, formerly SKB-385)**, Turgoyakskoye Shosse 1, Miass, Chelyabinsk Oblast (`55°09'25"N 60°09'10"E`) (CSIS Missile Threat: R-29RM Shtil / Sineva).
    - **Serial Manufacturer: JSC Krasnoyarsk Machine-Building Plant (`АО «Красмаш»`)**, Krasnoyarsk (`56°00'18"N 92°57'12"E`).
    - **Propulsion Engines (`1st, 2nd, 3rd stages`): KBKhM named after A. M. Isayev (`КБХМ им. А. М. Исаева`)**, Korolyov (`55°55'20"N 37°49'10"E`).
    - **Guidance System (`ЦБК-11`): NPO Avtomatiki named after Academician N. A. Semikhatov (`НПОА им. Н. А. Семихатова`)**, Yekaterinburg (`56°50'22"N 60°37'12"E`).
- **Chief Designers & Key Personnel:**
    - **Academician Viktor Petrovich Makeev** (designer of baseline `3M37 R-29RM Shtil`, commissioned 1986).
    - **Academician Vladimir Grigoryevich Degtyar** — General Director and General Designer of GRTs Makeev (chief designer of the `R-29RMU2 Sineva` [accepted into service by Presidential Decree on **9 July 2007**], `R-29RMU2.1 Layner` [2014], and `RS-28 Sarmat`) (Wikipedia: R-29RMU Sineva).
- **Deployment Coordinates:**
    - **Operational Base (31st Submarine Division, Northern Fleet): Gadzhiyevo (Yagelnaya Bay)**, Murmansk Oblast: **`69°15'22"N 33°19'48"E`** — deployed aboard up to 6 modernized Project 667BDRM Delfin SSBNs (`16 missiles per boat`) (CSIS Missile Threat: SS-N-23).
    - **World-Record Range Launch Coordinate (11 October 2008):** Fired by `K-114 Tula` from the Barents Sea (`~70°N 35°E`) into the equatorial Pacific Ocean (`11,547 km` downrange) (Weaponsystems.net: R-29RMU2 Sineva).
- **Engineering-Technical Parameters:**
    - **Dimensions & Mass:** Length **`14.8 m`**; Diameter **`1.90 m`**; Launch mass **`40,300 kg`**; Throw-weight **`2,800 kg`** (yielding the highest throw-weight-to-launch-mass ratio of any SLBM in service worldwide) (CSIS Missile Threat: SS-N-23, Weaponsystems.net: R-29RMU2 Sineva).
    - **Three-Stage Immersed-Engine Storable Liquid Propulsion:** Powered by three stages burning hypergolic **dinitrogen tetroxide (`$\text{N}_2\text{O}_4$` / Amyl) and unsymmetrical dimethylhydrazine (UDMH / Heptyl)**; utilizes Makeev's signature **"immersed engine" (`утопленный двигатель`)** layout where the rocket motors are recessed directly inside the propellant tanks with shared common bulkheads (`совмещённые днища`) to fit within the `14.8 m` tube length (FAS Guide: SLBM Overview).
    - **Range, Guidance & Warheads:** Nominal full-payload range **`8,300 km`**, maximum flight-tested range **`11,547 km`**; Astro-inertial + GLONASS (`Ураган`) guidance (`CEP ~250 m`) with `Малахит-3` onboard computer; Carries **4 × `500 kt` medium-yield MIRVs** or **10 × `100 kt` low-yield high-speed MIRVs (`ББ среднего или малого класса`)** plus the `Эгида` (`Egida`) anti-ABM penetration aid suite (Weaponsystems.net: R-29RMU2 Sineva).
- **Causal Command Sequence:**

1. Triggered via the Project 667BDRM `Omnibus-BDRM` fire-control computer after VLF/ELF launch authentication; the `D-29RMU2` tube floods with seawater around the missile's rubber-metal annular shock absorbers (`резинометаллические амортизаторы`), the outer hatch opens, and the first-stage liquid engine ignites underwater (`мокрый старт`), propelling the missile out of the tube and into its intercontinental trajectory.

87. R-29RMU2.1 Layner (Modification of R-29RMU2 SLBM)

- **GRAU, Treaty & NATO Index:** Operational designation **`Р-29РМУ2.1 «Лайнер»`** **(`R-29RMU2.1 Layner` / `Lajner`)**; Shipborne launch complex **`Д-29РМУ2.1`** **(`D-29RMU2.1`)**; GRAU missile index **`3М37У2.1`** **(`3M37U2.1`)**; Declared under the existing **`РСМ-54`** **(`RSM-54`)** treaty category; NATO reporting name: **SS-N-23B Skiff** (Wikipedia: R-29RMU2 Lajner, RussianForces.org: Liner SLBM Explained).
- **Design Bureau & Manufacturer:**
  - **Design Bureau: JSC State Rocket Center named after Academician V. P. Makeev (`АО «ГРЦ Макеева»`)**, Miass (`55°09'25"N 60°09'10"E`) (RussianForces.org: Liner SLBM Explained).
  - **Serial Manufacturer: JSC "Krasmash"** (Krasnoyarsk, `56°00'18"N 92°57'12"E`), with warhead bus and fuzing electronics by **NPO Avtomatiki** (Yekaterinburg) and **RFNC-VNIITF** (Snezhinsk, `56°05'00"N 60°44'00"E`).
- **Chief Designers & Key Personnel:**
  - **Academician Vladimir Grigoryevich Degtyar** — General Director and General Designer of GRTs Makeev (flight-tested on **20 May 2011** from `K-84 Yekaterinburg` and **29 September 2011** from `K-114 Tula`, recommended by the State Commission in December 2012, and formally accepted into service by Presidential Decree in **January 2014**) (Wikipedia: R-29RMU2 Lajner).
- **Deployment Coordinates:**
  - **Operational Base (31st Submarine Division, Northern Fleet): Gadzhiyevo (Yagelnaya Bay)**, Murmansk Oblast (`69°15'22"N 33°19'48"E`) aboard modernized Project 667BDRM Delfin SSBNs (`K-114 Tula`, `K-117 Bryansk`, `K-18 Karelia`, `K-407 Novomoskovsk`).
- **Engineering-Technical Parameters:**
  - **Airframe & Propulsion Continuity:** Retains the exact first-, second-, and third-stage dimensions (`length 14.8 m`, `diameter 1.90 m`, `launch mass 40,300 kg`, `throw-weight 2,800 kg`, `range 8,300–11,500 km`) and storable liquid propellant (N_2O_4/UDMH) of the R-29RMU2 Sineva, allowing it to be declared under New START as an existing `RSM-54` modification while introducing a **completely redesigned Post-Boost Vehicle (`Боевая ступень`) and digital flight control suite** (RussianForces.org: Liner SLBM Explained).
  - **Mixed-Yield Modular MIRV Payload Configurations:** Unlike `Sineva`, `R-29RMU2.1 Layner` is engineered to carry **up to 12 low-yield MIRVs** or **mixed-yield warhead combinations (`разнородное оснащение`)** in four distinct combat loadouts (Wikipedia: R-29RMU2 Lajner, RussianForces.org: Liner SLBM Explained):
    1. **12 × low-yield (`~100 kt`) MIRVs** (without heavy ABM penetration aids);
    2. **10 × low-yield (`~100 kt`) MIRVs** + standard ABM penetration aid complex (`Комплекс средств преодоления ПРО — КСП ПРО`);
    3. **8 × low-yield (`~100 kt`) MIRVs** + enhanced heavy ABM penetration aid complex;
    4. **4 × medium-yield (`~500 kt`) MIRVs** + full heavy ABM penetration aid complex (`or mixed medium + low-yield warheads on the same bus`).
- **Causal Command Sequence:**
  1. Identical wet-launch sequence (`мокрый старт`) from Project 667BDRM `D-29RMU2.1` tubes; during the post-boost dispensing phase in exoatmospheric space, the upgraded `Layner` post-boost vehicle independently maneuvers to release mixed-yield MIRVs and multi-spectral decoys across widely separated target footprints.

88. R-36M2 Voevoda (15A18M / SS-18 Satan Mod 5 & Mod 6 ICBM)

- **GRAU, Treaty & NATO Index:** Operational designation **`Р-36М2 «Воевода»`** **(`R-36M2 Voevoda`)**; GRAU missile index **`15А18М`** **(`15A18M`)** (single-warhead variants `15A18M1` / `15A18M2`); Silo launch complex **`15П018М`** **(`15P018M`)** using ultra-hardened silo **`15П718М`** **(`15P718M`)** and unified command post **`15В155`** **(`15V155`)**; START/New START treaty designation **`РС-20В`** **(`RS-20V`)**; NATO reporting name: **SS-18 Satan Mod 5 (10 MIRV) & Mod 6 (single high-yield RV)** (CSIS Missile Threat: R-36 / SS-18 Satan, Wikipedia: R-36 Missile).
- **Design Bureau & Manufacturer:**
  - **Design Bureau: OKB-586 / KB "Yuzhnoye" named after M. K. Yangel (`КБ «Южное» им. М. К. Янгеля`)**, Dnepropetrovsk, Ukrainian SSR (`48°26'02"N 34°59'55"E`), developed under CPSU CC Decree of **9 August 1983** (CSIS Missile Threat: SS-18).
  - **Serial Manufacturer: PO "Yuzhny Machine-Building Plant" (`ПО «Южмаш»` / YuMZ)**, Dnepropetrovsk (`48°26'02"N 34°59'55"E`).
  - **Propulsion Engines (`RD-264` & `RD-0255`): NPO Energomash named after V. P. Glushko** (Khimki `55°53'50"N 37°26'15"E`) and **KBKhA** (Voronezh `51°38'40"N 39°13'30"E`).
  - **Guidance System: NPO "Khartron" (`НПО «Хартрон»` / OKB-692)**, Kharkiv (`50°02'35"N 36°16'55"E`).
  - **Silo Complex (`15P718M`): KBSM (`КБ специального машиностроения`)**, Saint Petersburg (`59°59'05"N 30°20'45"E`).
- **Chief Designers & Key Personnel:**
  - **Academician Vladimir Fyodorovich Utkin** — General Designer of KB Yuzhnoye (lead designer of `R-36M`, `R-36MUTTKh`, and `R-36M2 Voevoda`) and **Stanislav Nikolayevich Konyukhov** (General Designer of KB Yuzhnoye from 1991, who supervised post-Soviet service-life extensions until 2014).
  - **Valentin Petrovich Glushko** (NPO Energomash, `RD-264` engine), **Vladimir Grigoryevich Sergeyev** (NPO Khartron, onboard digital computer), and **Vladimir Степанович Stepanov** / **Anatoly Fyodorovich Utkin** (KBSM, `15P718M` super-hardened silo).
- **Deployment Coordinates (Active RVSN Divisions):**
  1. **13th Red Banner Rocket Division — Dombarovsky (Yasny), Orenburg Oblast: `51°05'38"N 59°50'33"E`** (`51.0939°N, 59.8425°E`) (FAS Guide: ICBM Bases, FAS Nuclear Notebook 2025).
  2. **62nd Red Banner Rocket Division — Uzhur (Solnechny / Krasnoyarsk-66), Krasnoyarsk Krai: `55°19'48"N 89°48'32"E`** (`55.3300°N, 89.8089°E`) (FAS Nuclear Notebook 2025).

    - *(Historical Soviet deployments at Kartaly 52°59'N 60°35'E, Aleysk 52°30'N 82°46'E, Zhangiz-Tobe 49°12'N 81°12'E [Kazakhstan], and Derzhavinsk 51°09'N 66°19'E [Kazakhstan] were deactivated and eliminated under START I / CTR)* (National Security Archive: Kazakhstan & Nunn-Lugar).
- **Engineering-Technical Parameters:**
    - **Dimensions & Mass:** Length **34.3 m**; Diameter **3.00 m**; Launch mass **211,100 kg (211.1 metric tons)**; Throw-weight **8,800 kg (8.8 metric tons — the heaviest ICBM ever fielded)** (CSIS Missile Threat: SS-18, Wikipedia: R-36 Missile).
    - **Two-Stage Liquid Propulsion & Cold-Launch Mortar:**
        - **Mortar Ejection (Миномётный старт):** Black-powder/solid-propellant pressure accumulator (ПАД) ejects the 211.1-ton missile inside its fiberglass transport-launch canister (ТПК) out of the 15P718M silo at >30 m/s; the RD-264 first-stage engine ignites at 20–30 m altitude above the silo, allowing the silo to be reused and eliminating internal flame-deflector ducts.
        - **1st Stage:** Four-chamber **RD-264 (15Д117)** closed-cycle staged-combustion engine (4,521 kN sea-level thrust, 461 tonnes-force) burning hypergolic N_2O_4/UDMH.
        - **2nd Stage:** Single-chamber **RD-0255 (15Д285)** (755 kN vacuum thrust) with four-nozzle vernier RD-0257.
    - **Range, Hardening & Payload:** Range **11,000 km** (15A18M 10-MIRV Mod 5) to **16,000 km** (single-RV Mod 6); Autonomous radiation-hardened inertial guidance (CEP 220 m), engineered to fly directly through nuclear fireballs and dust clouds; Silo 15P718M hardened to withstand **300–600 atmospheres (4,400–8,800 psi)** of nuclear overpressure; Carries **10 × 750–800 kt MIRVs (15Ф173)** or **1 × 20 Mt warhead** plus КСП ПРО heavy/light decoys (CSIS Missile Threat: SS-18, Wikipedia: R-36 Missile).
    - **Current Force Strength (2025–2026): ~34 deployed silos (340 warheads)** remaining across Dombarovsky and Uzhur as gradual silo conversion to Avangard and Sarmat proceeds (FAS Nuclear Notebook 2025).
- **Causal Command Sequence:**
    1. Designed specifically for **Launch-Under-Attack (Ответно-встречный удар)**: upon receiving a Monolit / Signal-A or Perimeter 15B99 command, the 15P718M silo blows its 120-ton armored roof door inside 62 seconds, mortar-ejects the 15A18M, and flies through an active nuclear detonation environment to deliver 10 × 800 kt hard-target kill warheads against adversary ICBM silos and C3 bunkers.

Subsystem Group IV.D: Land-Based ICBMs, Hypersonic Glide Vehicles & Strategic Aviation (Entities 89–97)

89. RS-28 Sarmat (15A28 / SS-X-30 ICBM)

- **GRAU, Treaty & NATO Index:** Operational designation **РС-28 «Сармат» (RS-28 Sarmat)**; GRAU missile index **15A28 (15А28)**; Silo launch complex **15П278 (15P278)** deployed in modernized 15П718М silos with active point-defense system **«Мозырь» (Mozyr)**; NATO reporting name: **SS-X-30 / SS-29 Satan II** (CSIS Missile Threat: RS-28 Sarmat, FAS Nuclear Notebook 2025).
- **Design Bureau & Manufacturer:**
    - **Prime Design Bureau: JSC State Rocket Center named after Academician V. P. Makeev (АО «ГРЦ им. академика В. П. Макеева»)**, Miass, Chelyabinsk Oblast (55°09'25"N 60°09'10"E) — awarded state R&D contract (ОКР «Сармат») in early 2011 to eliminate Russia's post-1991 dependence on Ukrainian KB Yuzhnoye / Yuzhmash for heavy liquid-propellant ICBMs (CSIS Missile Threat: RS-28 Sarmat).
    - **Co-Developer (Payload & HGV Integration): JSC Military-Industrial Corporation "NPO Mashinostroyeniya" (АО «ВПК „НПО машиностроения“»)**, Reutov, Moscow Oblast (55°45'42"N 37°51'18"E) (CSIS Missile Threat: RS-28 Sarmat).
    - **Serial Manufacturer: JSC Krasnoyarsk Machine-Building Plant (АО «Красмаш»)**, Krasnoyarsk (56°00'18"N 92°57'12"E).
    - **First-Stage Propulsion (PDU-99 / RD-274): JSC NPO Energomash** (Khimki) and **Proton-PM** (Perm 58°01'00"N 56°15'00"E).
- **Chief Designers & Key Personnel:**
    - **Academician Vladimir Grigoryevich Degtyar** — General Director and General Designer of GRTs Makeev (chief designer of 15A28 RS-28 Sarmat).
    - **Alexander Georgiyevich Leonov** — General Director and General Designer of NPO Mashinostroyeniya (15Yu71 Avangard HGV integration onto Sarmat).
- **Deployment Coordinates:**
    - **Planned Primary Operational Deployment:**
        1. **62nd Red Banner Rocket Division — Uzhur (Solnechny), Krasnoyarsk Krai: 55°19'48"N 89°48'32"E** (first regiment undergoing silo conversion) (FAS Nuclear Notebook 2025).
        2. **13th Red Banner Rocket Division — Dombarovsky (Yasny), Orenburg Oblast: 51°05'38"N 59°50'33"E**.
    - **Flight Test Silo (15P718M Conversion at Plesetsk Cosmodrome): Site 157 ("Yubileynaya" Silo), Plesetsk Cosmodrome**, Arkhangelsk Oblast: **62°53'11"N 41°45'56"E** (62.8864°N, 41.7656°E) — site of the 2017–2018 pop-up ejection tests, the successful maiden full-range flight test on **20 April 2022** to Kura (57°20'00"N 161°50'00"E), and the **September 2024** test failure in which the missile exploded inside the Site 157 silo, leaving a ~60-meter crater (CSIS Missile Threat: Russia Test-Fires Sarmat ICBM, FAS Nuclear Notebook 2025).
- **Engineering-Technical Parameters:**
    - **Dimensions & Mass:** Length **35.5 m**; Diameter **3.00 m**; Launch mass **208,100 kg (208.1 metric tons)**; Throw-weight **~10,000 kg (10 metric tons)** (CSIS Missile Threat: RS-28 Sarmat).
    - **Propulsion & Shortened Boost Phase:** Three-stage storable liquid-propellant (N_2O_4/UDMH) rocket powered on the first stage by the **PDU-99 (ПДУ-99 / modernized RD-264 -> RD-274)** four-chamber engine assembly, engineered with a significantly higher thrust-to-weight ratio than the R-36M2 to shorten the active infrared boost phase (<180–200 seconds) and evade space-based early-warning tracking and boost-phase intercept.
    - **Range & Fractional Orbital Bombardment (FOBS) Capability:** Range **18,000 km** (global range), enabling attacks along **South Polar trajectories** that bypass North American northern-hemisphere BMEWS/GMD radar arcs (CSIS Missile Threat: Russia to Deploy Sarmat).

    - **Payload Configurations:** Capable of carrying **10 to 15 high-yield MIRVs**, OR **up to 3 × 15Yu71 Avangard hypersonic boost-glide vehicles**, plus heavy active/passive penetration aids (CSIS Missile Threat: RS-28 Sarmat).
- **Causal Command Sequence:**
    1. Emplaced directly into existing `15P718M` (`R-36M2`) super-hardened silos and integrated into the 5th-generation Monolit/Signal-A C2 network; upon receiving a launch order, the powder-gas generator mortar-ejects the `208.1-ton` canisterized missile to `25–30 m` height, where the `PDU-99` engine ignites and accelerates the payload into either a northern trans-polar or southern sub-orbital/FOBS trajectory.

90. RT-2PM2 Topol-M (15Zh65 / 15A165 / SS-27 Mod 1 ICBM)

- **GRAU, Treaty & NATO Index:**
    - **Silo-Based Complex: `15П165` (`15P165`)** with missile **`15Ж65` (`15Zh65`**, also cited as `15A165`) in modernized silo **`15П765` (`15P765`)**; START/New START designation **`PC-12M2` (`RS-12M2`)** (CSIS Missile Threat: RT-2PM2 Topol-M, Wikipedia: RT-2PM2 Topol-M).
    - **Road-Mobile Complex: `15П155` (`15P155`)** with missile **`15Ж55` (`15Zh55`)** mounted on the 8-axle **`15У168` (`15U168`)** Autonomous Launcher (АПУ) using the **`MZKT-79221`** (`16×16`) chassis; START/New START designation **`PC-12M1` (`RS-12M1`)**; NATO reporting name: **SS-27 Mod 1 Sickle B** (CSIS Missile Threat: RT-2PM2 Topol-M, Wikipedia: RT-2PM2 Topol-M).
- **Design Bureau & Manufacturer:**
    - **Design Bureau: JSC Corporation "Moscow Institute of Thermal Technology" (`АО «Корпорация „МИТ“» / MITT`)**, Moscow (`55°51'16"N 37°36'56"E`) — the first ICBM fully designed and manufactured inside the post-Soviet Russian Federation (redesigned by Presidential Decree of February 1993) (CSIS Missile Threat: RT-2PM2 Topol-M).
    - **Serial Missile Manufacturer: JSC "Votkinsk Plant" (`АО «Воткинский завод»`)**, Votkinsk (`57°03'12"N 53°59'10"E`).
    - **Solid Propellant Charges: Federal Center for Dual-Use Technologies "Soyuz" (`ФЦДТ «Союз»`)**, Dzerzhinsky, Moscow Oblast (`55°37'45"N 37°51'10"E`) (Wikipedia: RT-2PM2 Topol-M).
    - **TEL Launcher (`15U168`): FNPTS "Titan-Barrikady"** (Volgograd `48°46'25"N 44°33'45"E`) on **MZKT** (Minsk Wheel Tractor Plant `53°51'30"N 27°39'40"E`) `MZKT-79221` chassis.
- **Chief Designers & Key Personnel:**
    - **Boris Nikolayevich Lapygin** (General Designer of MITT, 1987–1997, who initiated the `Universal` / `Topol-M` upgrade after **Alexander Davidovich Nadiradze**) and **Academician Yury Solomonovich Solomonov** (General Designer of MITT from 1997, who completed flight trials and serial deployment) (Wikipedia: RT-2PM2 Topol-M).
- **Deployment Coordinates (All 78 Deployed Launchers — 60 Silo-Based + 18 Road-Mobile):**
    1. **60th Taman Rocket Division — Tatishchevo (Svetly), Saratov Oblast (60 Silo-Based `15Zh65` Across 6 Regiments): `51°40'00"N 45°34'00"E`** (`51.6667°N, 45.5667°E` — first trial silo placed on combat duty in December 1997; 60th silo completed in 2012) (FAS Guide: Tatishchevo, CSIS Missile Threat: RT-2PM2 Topol-M).
    2. **54th Guards Rocket Division — Teykovo, Ivanovo Oblast (18 Road-Mobile `15Zh55` TELs Across 2 Regiments): `56°51'00"N 40°32'00"E`** (`56.8500°N, 40.5333°E` — deployed 2006–2010; serial production transitioned entirely to RS-24 Yars in 2010) (Wikipedia: RT-2PM2 Topol-M, FAS Nuclear Notebook 2025).
- **Engineering-Technical Parameters:**
    - **Dimensions & Mass:** Length **`22.7 m`**; First-stage diameter **`1.90 m`** (`1.95 m`); Launch mass **`47,200 kg`**; Throw-weight **`1,200 kg`** (Wikipedia: RT-2PM2 Topol-M).
    - **Three-Stage Fast-Burn Solid Propulsion:** Three carbon-fiber filament-wound solid-fuel stages developed by FTsDT "Soyuz", providing an ultra-high acceleration rate that reaches burnout velocity (**`7,320 m/s`**) in **`~120–150 seconds`**, coupled with auxiliary thruster jets and aerodynamic/gas-dynamic steering surfaces that enable the booster to perform **evasive S-curve lateral and vertical maneuvers during the boost phase** (Wikipedia: RT-2PM2 Topol-M).
    - **Range, Hardening & Warhead:** Range **`11,000 km`**; Radiation-hardened against close-in nuclear EMP and space-based laser illumination (`10x higher laser resistance than RT-2PM Topol`); Inertial + GLONASS guidance (`CEP ~200 m`); Carries **1 × `800 kt` (or `550 kt–1 Mt`) single thermonuclear warhead** (`моноблочная ГЧ`, compliant with START II restrictions when designed) with an evasive maneuvering re-entry vehicle (`MaRV`) and decoy suite (CSIS Missile Threat: RT-2PM2 Topol-M).
- **Causal Command Sequence:**
    1. For silo-based `15P165` at Tatishchevo (`51°40'N 45°34'E`), Signal-A or Perimeter 15B99 commands trigger cold-launch mortar ejection from the converted `OS-84`/`15P765` silo.
    2. For road-mobile `15P155` at Teykovo (`56°51'N 40°32'E`), the `15U168` (`MZKT-79221`) TEL receives the encrypted launch directive via the battalion mobile command post (`МПБУ 15B231`) or directly via satellite/VLF/command-rocket receiver, jacks up on hydraulic outriggers at any surveyed or unsurveyed patrol point (`autonomous gyro-compassing via GLONASS`), elevates the `22.7 m` canister to `90°`, and cold-launches the missile within `2–3 minutes` of receiving the execution code (Wikipedia: RT-2PM2 Topol-M).

91. RS-24 Yars (15Zh65M / 15Zh67 / SS-27 Mod 2 ICBM)

- **GRAU, Treaty & NATO Index:**
    - **Road-Mobile Complex: `15П165М` (`15P165M`)** with missile **`15Ж65М` (`15Zh65M`)** on 8-axle **`15У175М` (`15U175M` / `MZKT-79221`)** TEL (and upgraded **`Ярс-С` (`Yars-S`)** / **`Ярс-М` (`Yars-M`, `15Ж67` / `15Zh67`)**) (CSIS Missile Threat: RS-24 Yars, RussianSpaceWeb: Long-Range Ballistic Missiles).
    - **Silo-Based Complex: `15П155М` (`15P155M`)** with missile **`15Ж67` (`15Zh67`)** in silo **`15П765М` (`15P765M`)** (CSIS Missile Threat: RS-24 Yars).
    - **Treaty & NATO Index:** New START designation **`PC-24 «Ярс»` (`RS-24 Yars`)**; NATO reporting name: **SS-27 Mod 2 Sickle B** (sometimes cited as `SS-29`) (CSIS Missile Threat: RS-24 Yars).
- **Design Bureau & Manufacturer:**
    - **Design Bureau: JSC Corporation "Moscow Institute of Thermal Technology" (`АО «Корпорация „МИТ“» / MITT`)**, Moscow (`55°51'16"N 37°36'56"E`), initiated in **2004** following US withdrawal from the ABM Treaty and the impending expiration of START I/II bans on MIRVed mobile ICBMs; maiden flight test conducted from Plesetsk on **29 May 2007** (CSIS Missile Threat: RS-24 Yars).
    - **Serial Manufacturer: JSC "Votkinsk Plant" (`АО «Воткинский завод»`)**, Votkinsk (`57°03'12"N 53°59'10"E`).

    - **TEL Launcher (15U175M): FNPTS "Titan-Barrikady"** (Volgograd) on **MZKT-79221** chassis; escorted by **15M107 «Листва» (15M107 Listva)** microwave mine-clearing/IED-neutralization vehicles and **«Тайфун-М» (Taifun-M, 15Ц56М)** counter-sabotage vehicles (Wikipedia: RT-2PM2 Topol-M / Taifun-M).
- **Chief Designers & Key Personnel:**
    - **Academician Yury Solomonovich Solomonov** — General Designer of MITT and Chief Designer of the RS-24 Yars family (CSIS Missile Threat: RS-24 Yars).
    - **Colonel General Sergey Viktorovich Karakayev** — Commander of the RVSN, who announced on **16 December 2023** that with the rearmament of the final regiment at Bologoye (Vypolzovo), 100% of Russia's road-mobile ICBM force had completed transition to RS-24 Yars (Wikipedia: RT-2PM Topol).
- **Deployment Coordinates (Backbone of the Russian ICBM Force — ~206 Deployed Launchers Across 9 Divisions as of 2025–2026):**
    1. **28th Guards Rocket Division — Kozelsk, Kaluga Oblast (Silo-Based 15Zh67): 54°02'15"N 35°47'30"E** (30 silos across 3 regiments) (FAS: Russian ICBM Upgrade at Kozelsk).
    2. **60th Taman Rocket Division — Tatishchevo, Saratov Oblast (Silo-Based): 51°40'00"N 45°34'00"E** (converting UR-100NUTTKh silos to Yars).
    3. **54th Guards Rocket Division — Teykovo, Ivanovo Oblast (Road-Mobile 15Zh65M): 56°51'00"N 40°32'00"E** (first operational Yars division, entered service February/July 2010) (CSIS Missile Threat: RS-24 Yars).
    4. **39th Guards Rocket Division — Novosibirsk (Gvardeysky / Pashino), Novosibirsk Oblast (Road-Mobile): 55°18'45"N 82°59'20"E**.
    5. **42nd Rocket Division — Nizhny Tagil (Svobodny / Verkhnyaya Salda), Sverdlovsk Oblast (Road-Mobile): 58°03'15"N 60°24'10"E**.
    6. **14th Kiev-Zhitomir Rocket Division — Yoshkar-Ola (Rechnoy), Mari El Republic (Road-Mobile): 56°38'10"N 48°05'20"E**.
    7. **29th Guards Rocket Division — Irkutsk (Zelyony), Irkutsk Oblast (Road-Mobile): 52°19'00"N 104°14'00"E** (FAS Guide: ICBM Bases).
    8. **35th Rocket Division — Barnaul (Sibirsky), Altai Krai (Road-Mobile): 53°33'20"N 83°50'15"E**.
    9. **7th Guards Rocket Division — Vypolzovo (Ozernyy / Bologoye-4), Tver Oblast (Road-Mobile): 57°52'00"N 33°40'00"E** (completed conversion December 2023) (Wikipedia: RT-2PM Topol, FAS Nuclear Notebook 2025).
- **Engineering-Technical Parameters:**
    - **Dimensions & Mass:** Length **22.5 m**; Diameter **2.00 m**; Launch mass **49,000–49,600 kg**; Throw-weight **~1,200–1,250 kg** (CSIS Missile Threat: RS-24 Yars).
    - **Propulsion, Range & Warheads:** Three-stage fast-burn solid-propellant booster with a liquid/solid post-boost MIRV bus; Minimum range **2,000 km**, Maximum range **10,500–12,000 km**; Inertial + GLONASS guidance (CEP 150–200 m); Carries **3 to 4 × 150–300 kt (or up to 500 kt) independently targetable MIRV warheads** (each capable of hypersonic atmospheric re-entry maneuvering) plus active electronic jammers and inflatable/heavy decoys (CSIS Missile Threat: RS-24 Yars, FAS Nuclear Notebook 2025).
- **Causal Command Sequence:**
    1. Integrated with the 4th/5th-generation АСБУ РВСН (Monolit / Signal-A) and mobile command relay vehicles (15B231 / 15B242); upon receiving a launch order in garrison (Крона sliding-roof shelter), in silo (15P765M), or dispersed along forest patrol routes, the RS-24 Yars executes an automated cold-launch mortar ejection, performs boost-phase evasive maneuvers, and dispenses its 4 MIRVs onto independent target coordinates.

92. BZhRK Molodets (15P961 / RT-23UTTKh Rail-Mobile ICBM Complex)

- **GRAU, Treaty & NATO Index:**
    - **Rail-Mobile Complex (БЖРК — Боевой железнодорожный ракетный комплекс): 15П961 «Молодец» (15P961 Molodets)** with three-stage solid-propellant ICBM **15Ж61 (15Zh61)** launched from rail car **15П761 (15P761 / 15У126)**; START I designation **РС-22В (RS-22V)**; NATO reporting name: **SS-24 Mod 2 / Mod 3 Scalpel** (FAS Guide: RT-23 / SS-24 Scalpel, Wikipedia: RT-23 Molodets).
    - **Silo-Based Sister Complex: 15П060 (15P060)** with missile **15Ж60 (15Zh60)** in silo **15П760 (15P760)**; START I designation **РС-22А (RS-22A)**; NATO reporting name: **SS-24 Mod 1 Scalpel** (FAS Guide: RT-23 / SS-24 Scalpel).
- **Design Bureau & Manufacturer:**
    - **Missile & Overall System Design Bureau: OKB-586 / KB "Yuzhnoye" (КБ «Южное» им. М. К. Янгеля)**, Dnepropetrovsk, Ukrainian SSR (48°26'02"N 34°59'55"E), authorized by CPSU CC Decree of **1 June 1979** (РТ-23УТТХ / 15Ж60 & 15Ж61 authorized **9 August 1983**) (FAS Guide: RT-23 / SS-24 Scalpel).
    - **Serial Missile Manufacturer: Pavlograd Mechanical Plant (ПМЗ — branch of PO Yuzhmash)**, Pavlograd, Dnepropetrovsk Oblast (48°32'00"N 35°52'00"E).
    - **Rail-Car Launch Complex (15P761 / 15U126): KBSM (КБ специального машиностроения)**, Saint Petersburg (59°59'05"N 30°20'45"E), and **Kalinin Machine-Building Plant / Tver Carriage Works (ТВЗ)**.
    - **Automated Combat Control Suite: NPO "Impuls"** (Saint Petersburg) (SPbSTU Polytechnic NPO Impuls 50th Anniversary Archive).
- **Chief Designers & Key Personnel:**
    - **Academician Vladimir Fyodorovich Utkin** — General Designer of KB Yuzhnoye (chief designer of the RT-23UTTKh missile and 15P961 BZhRK complex, working alongside his brother **Anatoly Fyodorovich Utkin**, General Designer of KBSM in Leningrad, who engineered the 15P761 railway erector-launcher car and overhead contact-wire deflection mechanism) (Militera Archive: Karpenko, Utkin & Popov — Domestic Strategic Missile Complexes).
- **Deployment Coordinates (3 Rail-Mobile Divisions [12 Trains / 36 Launchers] + 2 Silo Divisions [56 Silos]):**
    1. **10th Guards Rocket Division — Kostroma, Kostroma Oblast (4 Trains / 12 15Zh61 Launchers): 57°45'00"N 40°55'00"E** (57.7500°N, 40.9167°E — first regiment placed on combat duty **20 October 1987**; last regiment stood down in **August 2005**) (FAS Guide: Kostroma, Wikipedia: RT-23 Molodets).
    2. **52nd Rocket Division — Bershet (Zvyozdny / Perm-76), Perm Krai (3 Trains / 9 15Zh61 Launchers): 57°46'00"N 56°22'00"E** (57.7667°N, 56.3667°E, stood down 2002–2003) (FAS Guide: ICBM Bases).

3. **36th Guards Rocket Division — Krasnoyarsk (Kedrovy / Krasnoyarsk-66), Krasnoyarsk Krai (4 Trains / 12 15Zh61 Launchers): 56°17'00"N 91°59'00"E** (56.2833°N, 91.9833°E, stood down 2002–2005) (FAS Guide: RT-23 / SS-24 Scalpel).
4. **Silo-Based 15Zh60 Deployments (56 Silos): 46 silos at Pervomaysk** (46°38'00"N 33°44'00"E, 46th Rocket Division, Ukraine — deactivated 1996 and eliminated 1998–2001 under Nunn-Lugar) and **10 silos at Tatishchevo** (51°40'00"N 45°34'00"E, 60th Rocket Division — deactivated 2001; last missile eliminated April 2008) (FAS Guide: Pervomaysk, Wikipedia: RT-23 Molodets).

- **Engineering-Technical Parameters:**
  - **Train Architecture (15P961):** Each BZhRK regiment consisted of a single train pulled by **three DM62 (ДМ62 / M62) diesel-electric locomotives**, pulling **three 15P761 missile launch cars** (disguised externally as standard 8-axle heavy refrigerated freight cars рефрижераторные секции with load-sharing hydraulic couplers that distributed the 200-ton car weight onto adjacent cars so standard railroad bridges and tracks would not deform), **seven command, control, communications, diesel-generator, and crew quarters cars (15B66 command post)**, and a fuel tank car, capable of patrolling **1,500 km from garrison across 145,000 km of Soviet rail network** (FAS Guide: RT-23 / SS-24 Scalpel, The War Zone: Soviet Railway ICBM).
  - **Missile Specifications (15Zh61) & Inflatable Nosecone:** Launch mass **104,500 kg (104.5 metric tons)**; Diameter **2.40 m**; Stowed length inside the rail car **18.8–19.0 m**, expanding in flight to **23.3 m** via an **inflatable/deployable aerodynamic nose fairing (надувной головной обтекатель)** that folded down so the 104.5-ton missile could fit inside a standard 24-meter railway car and also extendable 1st-stage nozzle (15Д305 with gimbaled nozzle) (FAS Guide: RT-23 / SS-24 Scalpel, The War Zone: Inflatable Nosecone).
  - **Range & Payload:** Range **10,000–10,450 km**; Throw-weight **4,050 kg**; Astro-inertial guidance (CEP 150–250 m); Armed with **10 × 400–550 kt (15Ф161 / 430 kt) independently targetable MIRV warheads (30 warheads per train, 360 warheads across all 12 trains)** (FAS Guide: RT-23 / SS-24 Scalpel).
- **Causal Command Sequence:**
  1. Upon receiving a Monolit/Signal-A launch directive while on rail patrol, the BZhRK halted at any point on the track; a specialized hydraulic pantograph arm (механизм отвода контактной сети) pushed the overhead 3 kV / 27.5 kV electrified railway catenary wire aside without cutting power, hydraulic outriggers stabilized the 15P761 car, the roof split open, the canister elevated to 90°, and the powder-gas generator cold-launched the 104.5-ton 15Zh61 missile, which tilted slightly using nose-mounted solid lateral thrusters (ДПА — двигатель пороховой увода) so the main 1st-stage exhaust plume ignited clear of the train car (The War Zone: Soviet Railway ICBM).

93. BZhRK Barguzin (Project Rail-Mobile ICBM Complex Based on RS-24 Yars)

- **GRAU, Treaty & Industrial Index:** R&D Project **ОКР «Баргузин» (BZhRK Barguzin)** — projected complex index **15П169 (15P169)** (also referenced in Western literature as RS-27 / SS-X-32Zh) armed with **6 × RS-24 Yars / Yars-M (15Zh65M / 15Zh67) solid-propellant ICBMs per train** (Wikipedia: BZhRK Barguzin, RussianSpaceWeb: Long-Range Ballistic Missiles).
- **Design Bureau & Manufacturer:**
  - **Prime Design Bureau: JSC Corporation "Moscow Institute of Thermal Technology" (АО «Корпорация „МИТ“» / MITT)**, Moscow (55°51'16"N 37°36'56"E), initiated in **2012–2013** to create an all-Russian solid-propellant successor to the Ukrainian-built 15P961 Molodets (Wikipedia: BZhRK Barguzin, Army Recognition: Barguzin).
  - **Rail-Car Launcher Engineering: JSC KBSM** (Saint Petersburg, 59°59'05"N 30°20'45"E) and **JSC Votkinsk Plant** (57°03'12"N 53°59'10"E).
- **Chief Designers & Key Personnel:**
  - **Academician Yury Solomonovich Solomonov** — General Designer of MITT (chief designer of Barguzin) (Wikipedia: BZhRK Barguzin).
  - **Colonel General Sergey Viktorovich Karakayev** — Commander of the RVSN (approved the preliminary technical design [эскизный проект] in December 2014 / May 2016).
- **Deployment Coordinates:**
  - **Pop-Up Ejection Test Site (November 2016): Plesetsk Cosmodrome**, Arkhangelsk Oblast (62°55'32"N 40°34'40"E), where a successful rail-car mortar-ejection test (бросковые испытания) was conducted in **November 2016** (Defence Blog: Russia Halts Barguzin).
  - **Planned Regimental Basing (Suspended December 2017):** Evaluated for reactivation at the former BZhRK infrastructure in **Kostroma** (57°45'00"N 40°55'00"E), **Perm / Bershet** (57°46'00"N 56°22'00"E), and **Krasnoyarsk** (56°17'00"N 91°59'00"E); in **December 2017**, the Russian Ministry of Defense suspended serial deployment under the 2018–2027 State Armament Program (ГПВ-2027) to prioritize funding for the RS-28 Sarmat and 15Yu71 Avangard, while retaining the completed design blueprints for rapid reactivation (CSIS Missile Threat: RS-24 Yars Footnotes, The War Zone: Barguzin).
- **Engineering-Technical Parameters:**
  - **Standard 4-Axle Refrigerated Car Integration:** Because the RS-24 Yars weighs **~47–49 metric tons** (less than half the 104.5-ton mass of the RT-23UTTKh Molodets), each Barguzin launch car fits inside a **standard 4-axle 24-meter railway car without requiring 8-axle bogies, load-sharing couplers, or three triple-coupled locomotives** — making the train visually and structurally indistinguishable from civilian Russian Railways (РЖД) container/refrigerated freight trains (Wikipedia: BZhRK Barguzin, The War Zone).
  - **Firepower & Range:** Each Barguzin train (equating to one full missile regiment) was designed to carry **6 × RS-24 Yars / Yars-M launch cars** (each missile carrying 4 MIRVs = 24 thermonuclear warheads per train), with a range of **11,000–12,600 km** and astro-inertial + GLONASS guidance (Wikipedia: BZhRK Barguzin).
- **Causal Command Sequence:**
  1. Integrated with 5th-generation АСБУ РВСН (Monolit / Signal-A) via concealed flush-mounted satellite and VLF antennas inside the command car; upon receiving a launch order, the train halts on any standard rail line without track reinforcement, deflects the overhead catenary wire, elevates the 6 launch canisters, and executes a cold-launch salvo.

94. 15Yu71 Avangard (4202 Hypersonic Boost-Glide Vehicle)

- **GRAU, Treaty & NATO Index:**
  - **Hypersonic Boost-Glide Vehicle (Гиперзвуковой планирующий крылатый боевой блок):** GRAU index **15Ю71 (15Yu71 / Yu-71)**; R&D project cipher **Изделие 4202 (Izdeliye 4202 / ОКР «Альбатрос» -> «Авангард»)** (Wikipedia: Avangard Hypersonic Glide Vehicle, CSIS Missile Threat: Avangard).

    - **Operational Missile Complex: 15П771 «Авангард» (15P771 Avangard)** employing the **15А35П (15A35P / UR-100NUTTKh / RS-18B)** two-stage liquid-propellant booster in silo **15П735**; US intelligence/NATO designation: **SS-19 Mod 4 Stiletto** (CSIS Missile Threat: UR-100 / SS-19 Mod 4).
- **Design Bureau & Manufacturer:**
    - **Design Bureau: JSC Military-Industrial Corporation "NPO Mashinostroyeniya" (АО «ВПК „НПО машиностроения“», formerly OKB-52 Chelomey)**, Gagarina Street 33, Reutov, Moscow Oblast (55°45'42"N 37°51'18"E) (CSIS Missile Threat: Avangard).
    - **Serial Manufacturer (15Yu71 Glider & 15A35P Integration): PO "Strela"** (Orenburg, 51°47'12"N 55°06'30"E) and **Votkinsk / Khrunichev** facilities.
    - **Thermonuclear Warhead Payload: RFNC-VNIITF (РФЯЦ-ВНИИТФ)**, Snezhinsk (56°05'00"N 60°44'00"E) / **RFNC-VNIIEF**, Sarov.
- **Chief Designers & Key Personnel:**
    - **Gerbert Aleksandrovich Yefremov** — Honorary General Director and Honorary General Designer of NPO Mashinostroyeniya (pioneer of the Albatros / 4202 maneuvering hypersonic glider concept since the mid-1980s; awarded the Order of St. Andrew the Apostle the First-Called and Hero of Labor of the Russian Federation in 2017–2020 specifically for creating 15Yu71 Avangard).
    - **Alexander Georgiyevich Leonov** — General Director and General Designer of NPO Mashinostroyeniya (oversaw final flight tests in December 2018 and combat deployment in December 2019).
- **Deployment Coordinates:**
    - **Active Operational Deployment (12 Silos Across 2 Regiments — 621st and 368th Rocket Regiments): 13th Red Banner Rocket Division — Dombarovsky (Yasny), Orenburg Oblast: 51°05'38"N 59°50'33"E** (51.0939°N, 59.8425°E — first regiment placed on combat duty on **27 December 2019**; second regiment completed in **December 2022 / December 2023**, bringing total deployed Avangard launchers at Dombarovsky to **12 silos**) (FAS Nuclear Notebook 2025, SIPRI Yearbook: Russian Nuclear Forces, Wikipedia: UR-100N).
- **Engineering-Technical Parameters:**
    - **Glider Dimensions & Thermal Protection:** Length **~5.4 m**; Mass **~2,000 kg**; Wedge-shaped lifting-body configuration fabricated from three-dimensional woven carbon-carbon / silicon-carbide composite materials engineered to withstand **aerodynamic stagnation temperatures of 1,600 °C to 2,000 °C** during sustained atmospheric hypersonic flight (CSIS Missile Threat: Avangard, Wikipedia: Avangard HGV).
    - **Booster & Flight Profile (15A35P):** Boosted by a two-stage storable liquid-propellant UR-100NUTTKh (15A35P, launch mass 105,600 kg, range 10,000+ km, Strela-type oversized payload fairing) to a **suborbital apogee of ~100 km**; instead of following a predictable Keplerian ballistic arc in space, 15Yu71 separates and descends into the upper stratosphere (40–70 km altitude), **gliding unpowered at speeds of Mach 20 to Mach 27 (~6.28 to 9.0 km/s) while executing multi-hundred-kilometer lateral and vertical aerodynamic skips and turns** to bypass exoatmospheric midcourse interceptors (GMD / SM-3 Block IIA) (CSIS Missile Threat: Avangard).
    - **Warhead:** Armed with a single **multi-megaton (~800 kt to 2 Mt) thermonuclear warhead** (Wikipedia: Avangard HGV).
- **Causal Command Sequence:**
    1. Controlled directly from Vlasikha and the 31st Missile Army (Orenburg) via the Monolit/Signal-A C2 network; upon receiving the launch command, the 15A35P hot-launches out of its Dombarovsky silo (51°05'38"N 59°50'33"E), releases the 15Yu71 glider at ~100 km altitude, and the glider's radiation-hardened onboard guidance computer executes pre-programmed lateral and altitude maneuvers through the surrounding ionized plasma sheath until terminal dive onto the target (CSIS Missile Threat: Avangard).

95. Tu-160 (Tupolev Tu-160 / Tu-160M Blackjack Strategic Bomber)

- **GRAU & Industrial Index:** Aircraft designation **Ту-160 «Белый лебедь» (Tu-160, modernized Ту-160М / Ту-160М2)**; Tupolev OKB internal index **Изделие 70 (Izdeliye 70**, modernized **Изделие 70М / 70М2)**; NATO reporting name: **Blackjack** (Wikipedia: Tupolev Tu-160, SIPRI Yearbook: Russian Nuclear Forces).
- **Design Bureau & Manufacturer:**
    - **Design Bureau: PJSC "Tupolev" (ПАО «Туполев» / OKB-156)**, Academician Tupolev Embankment 17, Moscow (55°45'42"N 37°40'48"E).
    - **Serial Manufacturer: Kazan Aviation Plant named after S. P. Gorbunov (КАЗ им. С. П. Горбунова — branch of PJSC Tupolev, Factory No. 22)**, Kazan, Republic of Tatarstan (55°51'25"N 49°07'55"E) (Wikipedia: Tupolev Tu-160).
    - **Turbofan Engines (NK-32 & NK-32-02): JSC "ODK-Kuznetsov" (ПАО «ОДК-Кузнецов»)**, Samara (53°13'10"N 50°16'40"E).
- **Chief Designers & Key Personnel:**
    - **Valentin Ivanovich Bliznyuk** — Chief Designer of the Tu-160 (Izdeliye 70) at the Tupolev Design Bureau (1975–1999, under General Designer **Alexei Andreyevich Tupolev**; maiden flight **18 December 1981**, entered operational service in **April 1987**).
    - **Alexander V. Konyukhov** and **Konstantin Vladimirovich Timofeyev** — Directors/Designers overseeing the Tu-160M / Tu-160M2 reproduction and deep modernization program (first newly manufactured Tu-160M2 airframe flew at Kazan on **12 January 2022**; state trials began in **2023**) (IISS Military Balance / Taylor & Francis).
- **Deployment Coordinates:**
    - **Primary Operational Base (121st Guards Sevastopol Heavy Bomber Aviation Regiment, 22nd Guards Heavy Bomber Aviation Division, Long-Range Aviation VKS): Engels-2 Air Base**, Saratov Oblast: **51°28'54"N 46°12'36"E** (51.4817°N, 46.2100°E) (FAS Guide: Ukraine/Engels Transfer, Bulletin of the Atomic Scientists: Russian Nuclear Forces).
    - **Forward Arctic & Far East Dispersal Bases: Olenya Air Base** (Murmansk Oblast, 68°09'06"N 33°27'54"E), **Belaya Air Base** (Irkutsk Oblast, 52°54'54"N 103°34'30"E), **Ukrainka Air Base** (Amur Oblast, 51°10'12"N 128°26'42"E), and **Anadyr (Ugolny)** (Chukotka, 64°44'06"N 177°44'30"E).
- **Engineering-Technical Parameters:**

- **Dimensions, Mass & Variable-Sweep Geometry:** Maximum takeoff mass **275,000 kg (275 metric tons — the heaviest and fastest supersonic strategic bomber in history)**; Empty weight 110,000 kg; Fuel capacity 148,000 kg; Length **54.10 m**; Wingspan **55.70 m** (20° sweep) to **35.60 m** (65° sweep); Crew **4** (pilot, co-pilot, bombardier/navigator, defensive systems operator on K-36LM zero-zero ejection seats) (Wikipedia: Tupolev Tu-160).
- **Propulsion & Performance:** Four **Kuznetsov NK-32** (Tu-160M: upgraded **NK-32-02**, Изделие P) three-shaft afterburning turbofans producing **4 × 137.3 kN dry / 4 × 245.2 kN (25,000 kgf) with afterburner**; Maximum speed **Mach 2.05 (2,220 km/h)** at 12,200 m; Unrefueled range **12,300 km** (+1,000 km on Tu-160M with NK-32-02), combat radius **7,300 km**, extendable globally via retractable nose aerial refueling probe (Wikipedia: Tupolev Tu-160).
- **Strategic Weapons Payload (45,000 kg across Two Internal Tandems Bays):** Two 11.28 m internal weapons bays, each fitted with a **MKU-6-5U (МКУ-6-5У)** 6-round rotary launcher (12 missiles total) carrying **12 × Kh-102 (Izdeliye 504B) stealth nuclear ALCMs** (or Kh-101 conventional ALCMs / legacy Kh-55SM nuclear ALCMs) (Bulletin of the Atomic Scientists: Russian Nuclear Forces, CSIS Missile Threat: Kh-101/Kh-102).
- **Modernized Tu-160M Avionics:** NV-70M (Новелла-70M) phased-array attack radar, Redut-70M digital glass cockpit and C3 communications suite (integrating directly with Kazbek/Monolit and Kavkaz satellite links), BKO-2 Himalai electronic warfare suite, and ABS-160M inertial/astro-inertial/GLONASS navigation system; **~15–17 aircraft in active inventory** (with a 2018 contract for 10 new-build Tu-160M2 airframes) (IISS Military Balance 2026, FAS Nuclear Notebook 2025).

- **Causal Command Sequence:**
  1. In peacetime, nuclear warheads (Kh-102) are stored inside the base-level 12th GUMO RTB bunker at Engels-2 (51°29'45"N 46°14'10"E).
  2. During strategic generation, 12th GUMO STG convoys deliver the armed MKU-6-5U rotary drums to the Tu-160 revetments; upon receiving the Kazbek / Long-Range Aviation Command Post (ЦКП ДА, Moscow 55°46'35"N 37°34'50"E) execution cipher over HF/SATCOM (Редут-70M), the navigator-operator inputs the warhead unblocking code and target coordinates into the 12 Kh-102 missiles and releases them from standoff launch boxes outside NATO/NORAD fighter-interceptor coverage (Bulletin of the Atomic Scientists: Russian Nuclear Forces).

96. Tu-95MS (Tupolev Tu-95MS / Tu-95MSM Bear-H Strategic Bomber)

- **GRAU & Industrial Index:** Aircraft designation **Ty-95MC (Tu-95MS — Variants Ty-95MC-6 and Ty-95MC-16)** and modernized **Ty-95MCM (Tu-95MSM)**; Tupolev OKB internal index **Изделие ВП-021 (Izdeliye VP-021, derived from the Tu-142M airframe)**; NATO reporting name: **Bear-H (Bear-H6 / Bear-H16)** (Wikipedia: Tupolev Tu-95, SIPRI Yearbook: Russian Nuclear Forces).
- **Design Bureau & Manufacturer:**
  - **Design Bureau: PJSC "Tupolev" (ОКБ А. Н. Туполева / ПАО «Туполев»)**, Moscow (55°45'42"N 37°40'48"E).
  - **Serial Manufacturer (Tu-95MS, 88 Built 1981–1992): Kuibyshev Aviation Plant (КуАЗ / Factory No. 18, now Aviakor)**, Samara (53°14'15"N 50°21'50"E, after prototype conversion at Taganrog TM3).
  - **Deep Modernization (Tu-95MSM): Beriev TANTK (ТАНТК им. Г. М. Бериева)**, Taganrog (47°11'54"N 38°50'55"E) and **360th Aviation Repair Plant (360 АРЗ)**, Ryazan/Dyagilevo (54°38'30"N 39°34'20"E).
- **Chief Designers & Key Personnel:**
  - **Andrei Nikolayevich Tupolev** and **Nikolai Ilyich Bazenkov** (original Tu-95 program); **Nikolai Vladimirovich Kirsanov** — Chief Designer of the Tu-95MS cruise-missile carrier (Izdeliye VP-021, first flight September 1979, serial production 1981–1992).
- **Deployment Coordinates:**
  1. **6950th Guards Aviation Base / 22nd Guards Heavy Bomber Aviation Division (184th Heavy Bomber Aviation Regiment): Engels-2 Air Base**, Saratov Oblast: **51°28'54"N 46°12'36"E** (SIPRI Yearbook: Russian Nuclear Forces).
  2. **6952nd Guards Aviation Base / 326th Heavy Bomber Aviation Division (182nd and 79th Heavy Bomber Aviation Regiments): Ukrainka Air Base (Seryshevo)**, Amur Oblast: **51°10'12"N 128°26'42"E** (51.1700°N, 128.4450°E) (Bulletin of the Atomic Scientists: Russian Nuclear Forces).
  3. **Training & Overhaul Center (43rd Center for Combat Employment and Retraining of Long-Range Aviation Personnel): Dyagilevo Air Base**, Ryazan Oblast: **54°38'30"N 39°34'20"E** (Air & Space Forces Magazine: Russian Airpower Almanac).
- **Engineering-Technical Parameters:**
  - **Dimensions & Mass:** Maximum takeoff mass **185,000–187,700 kg**; Length **49.13 m**; Wingspan **50.04 m** (35° sweep); Crew **7** (Wikipedia: Tupolev Tu-95).
  - **Turboprop Propulsion & Range:** Four **Kuznetsov NK-12MP** (Tu-95MSM: upgraded **NK-12MPM**) turboprops (4 × 11,033 kW / 15,000 shp) driving AV-60K (Tu-95MSM: low-vibration **AV-60T**) four-blade contra-rotating propellers; Maximum speed **830–920 km/h**; Unrefueled combat range **10,500–11,000 km** (extendable via fixed nose refueling probe) (SIPRI Yearbook: Russian Nuclear Forces).
  - **Strategic Cruise Missile Armament (Tu-95MS vs Tu-95MSM):**
    - **Tu-95MS-6 (Bear-H6):** Internal weapons bay housing a single **MKU-6-5 (МКУ-6-5)** 6-round rotary launcher for 6 × Kh-55 / Kh-55SM (AS-15 Kent) nuclear ALCMs.
    - **Tu-95MS-16 / Tu-95MSM (Bear-H16):** Because the longer **7.45 m Kh-101 / Kh-102** stealth ALCM does not fit inside the internal MKU-6-5 bay of the Tu-95MS, the modernized **Tu-95MSM** carries **8 × Kh-102 (Izdeliye 504B) nuclear ALCMs (or Kh-101 conventional ALCMs) externally on four underwing heavy twin-store pylons (AKU-5M)**, plus up to 6 internal Kh-55SM missiles (CSIS Missile Threat: Kh-101/Kh-102, FAS Nuclear Notebook 2025).
  - **Modernized Tu-95MSM Avionics:** Novella-NV1.021 phased-array radar (replacing Obzor-MS), SOI-021 glass cockpit, Meteor-NM2 electronic countermeasures suite, and S-505-95 digital strategic C3 communications terminal; **~47–52 aircraft in inventory** (FAS Nuclear Notebook 2025).
- **Causal Command Sequence:**

1. Following warhead mating by the base 12th GUMO `RTB` at Engels-2 (`51°28'54"N 46°12'36"E`) or Ukrainka (`51°10'12"N 128°26'42"E`), `Tu-95MSM` aircraft launch to Arctic/Pacific standoff patrol zones; upon receiving the Kazbek launch order and warhead unblocking code over the `S-505-95` HF/SATCOM link, the navigator feeds GLONASS/digital terrain maps into the 8 underwing Kh-102 missiles and releases them from `>2,500–4,500 km` standoff range.

97. Kh-102 (Izdeliye 504B Stealth Thermonuclear Air-Launched Cruise Missile)

- **GRAU & Industrial Index:**
  - **Thermonuclear Strategic ALCM: `X-102` (`Kh-102`)** — Industrial index **`Изделие 504Б` (`Izdeliye 504B`)**; NATO reporting name: **AS-23B Kodiak** (CSIS Missile Threat: Kh-101 / Kh-102, Wikipedia: Kh-101 / Kh-102).
  - **Conventional Sister Variant: `X-101` (`Kh-101`)** — Industrial index **`Изделие 504А` (`Izdeliye 504A`)**; NATO reporting name: **AS-23A Kodiak** (CSIS Missile Threat: Kh-101 / Kh-102).
- **Design Bureau & Manufacturer:**
  - **Design Bureau & Serial Manufacturer: JSC State Machine-Building Design Bureau "Raduga" named after A. Ya. Bereznyak (`АО «ГосМКБ „Радуга“ им. А. Я. Березняка»`, part of Tactical Missiles Corporation `КТРВ`)**, Zhukovskogo Street 2A, Dubna, Moscow Oblast (`56°45'18"N 37°08'35"E`), alongside the **Dubna Machine-Building Plant (`ДМЗ им. Н. П. Фёдорова`)** and **Smolensk Aviation Plant (`СмАЗ`)** (CSIS Missile Threat: Kh-101 / Kh-102).
  - **Turbofan Engine (`TRDD-50A`): JSC "ODK-Saturn" (`ПАО «ОДК-Сатурн»` / OMKB)**, Rybinsk (`58°02'55"N 38°51'30"E`) and Omsk (`54°56'40"N 73°25'10"E`) (CSIS Missile Threat: Kh-101 / Kh-102).
  - **Thermonuclear Warhead: RFNC-VNIIEF (`РФЯЦ-ВНИИЭФ` / KB-11)**, Sarov (`54°55'00"N 43°19'00"E`).
- **Chief Designers & Key Personnel:**
  - **Igor Seleznev** — General Designer of MKB "Raduga" (1974–2005, who initiated the `Kh-101/Kh-102` design in the late 1980s and full-scale development in **1995** to replace the `Kh-55`) (CSIS Missile Threat: Kh-101 / Kh-102).
  - **Vladimir Nikolayevich Трусов (Trusov)** and **Sergey Анатольевич Bogatikov** — General Directors/Designers of GosMKB "Raduga" (completed flight trials and introduced `Kh-101/Kh-102` into service in **2012**) (CSIS Missile Threat: Kh-101 / Kh-102).
- **Deployment Coordinates:**
  - **Nuclear Storage & Bomber Loading Bases (12th GUMO RTB Facilities):**
    1. **Engels-2 Air Base (`RTB` Warhead Storage Facility, Saratov Oblast): `51°29'45"N 46°14'10"E`** (serving the 121st [`Tu-160`] and 184th [`Tu-95MS`] Guards Heavy Bomber Regiments at `51°28'54"N 46°12'36"E`) (SIPRI Yearbook: Russian Nuclear Forces).
    2. **Ukrainka Air Base (`RTB` Warhead Storage Facility, Amur Oblast): `51°11'30"N 128°24'15"E`** (serving the 182nd and 79th Heavy Bomber Regiments at `51°10'12"N 128°26'42"E`) (Bulletin of the Atomic Scientists: Russian Nuclear Forces).
- **Engineering-Technical Parameters:**
  - **Dimensions, Mass & Low-Observable Airframe:** Length **`7.45 m`**; Fuselage cross-section diameter **`0.51 m` (`0.742 m` width across chined stealth fuselage)**; Folded wingspan `3.00 m` (deployed `~4.0 m`); Launch mass **`2,300–2,400 kg`**; Constructed with radar-absorbing composite materials (`ЭПР ~0.01 m²`) and a chined trapezoidal fuselage cross-section to minimize radar cross-section (CSIS Missile Threat: Kh-101 / Kh-102).
  - **Propulsion & Flight Envelope:** Powered by a ventral drop-down **`TRDD-50A` (`ТРДД-50А` / `Изделие 36МТ`)** bypass turbofan engine (`4.4 kN / 450 kgf` thrust); Cruising speed **`Mach 0.58` (`700–720 km/h`)**, maximum speed **`Mach 0.78` (`970 km/h`)**; Flight endurance **`~10 hours`**; Range **`2,500–2,800 km`** (US/CSIS baseline estimate) up to **`4,500–5,500 km`** (Russian MoD specification); Transit altitude `6,000–10,000 m`, descending to **terrain-hugging nap-of-the-earth flight at `30 to 60 meters` (`98–196 ft`) above ground level** (CSIS Missile Threat: Kh-101 / Kh-102).
  - **Guidance, Defensive Countermeasures & Thermonuclear Warhead:**
    - **Guidance (`СН-99` + `Отблеск-У`):** Autonomous inertial navigation (`БИНС`) + anti-jam **GLONASS (`Комета-М` CRPA)** + radar altimeter TERCOM + **`Отблеск-У` (`Otblesk-U`)** electro-optical/infrared digital scene-matching DSMAC terminal correlator, achieving a **CEP of `6 to 10–20 meters`** (CSIS Missile Threat: Kh-101 / Kh-102).
    - **Onboard Defensive Suite (Modernized 2022–2025 Production):** Equipped with an integrated radar warning receiver (`Станция предупреждения об облучении`), active electronic jammer, and ventral **`L-504` (`Л-504`)** chaff/flare decoy dispensers.
    - **Warhead (`Kh-102`): 1 × `250 kt` (or variable-yield `250–450 kt`) thermonuclear warhead** (CSIS Missile Threat: Kh-101 / Kh-102, FAS Nuclear Notebook 2025).
- **Causal Command Sequence:**
  1. Upon receiving the Kazbek nuclear release directive, the Tu-160 (`MKU-6-5U` internal rotary bay) or Tu-95MSM (`AKU-5M` underwing pylons) downloads the encrypted warhead unblocking code (`Код разблокирования ЯБП`) and route waypoints into the `Kh-102` onboard computer; upon gravity drop release, the folded wings deploy, the `TRDD-50A` turbofan extends downward from the rear fuselage and ignites, and the missile descends to `30–60 m` terrain-following altitude to penetrate adversary IADS and detonate its 250 kt–450 kt payload at the target coordinates (CSIS Missile Threat: Kh-101 / Kh-102).

Section V: People's Republic of China (PRC) Strategic Rocket Force (PLARF), Central Military Commission (CMC), CAEP, Bases 22 & 61–69, Delivery Triad, Early Warning LPARs, Beidou-3, and ShenGuang ICF Complex (Entities 98–131)

98. People's Liberation Army Rocket Force (PLARF / 中国人民解放军火箭军)

- **Institutional & Organizational Designation:** People's Liberation Army Rocket Force (PLARF), formerly the PLA Second Artillery Force (SAF / 第二炮兵, founded July 1, 1966); elevated to a full independent PLA service branch on December 31, 2015 / January 1, 2016 (CASI PLARF Organization). Military Unit Cover Designator (MUCD) block: 96600-series (Base/HQ level) and 96700/96800-series (brigade/regiment level) (CNS PLARF Order of Battle 2023).
- **Headquarters & Geolocation:** Qinghe Compound, Haidian District, Beijing (`40.0381°N, 116.3314°E`), with underground command complexes in the Xishan (Western Hills) strategic complex (`39.9850°N, 116.1400°E`) (CASI NC3 Report 2024).
- **Command Hierarchy & Named Leadership:** Reports directly to the Central Military Commission (CMC). Following the July 2023 CMC Commission for Discipline Inspection removals of former Commander Gen. Li Yuchao (李玉超), former Deputy Commander Lt. Gen. Liu Guangbin (刘光斌), and former Commanders Wei Fenghe (魏凤和) and Zhou Yaning (周亚宁), CMC Chairman Xi Jinping appointed Gen. Wang Houbin (王厚斌, former PLAN Deputy Commander) as Commander of the PLARF and Gen. Xu Xisheng (徐西盛, former Southern Theater Command Air Force Political Commissar) as Political Commissar on July 31, 2023 (PLA Rocket Force Overview; FAS Arms Control 2025 CMPR Analysis). PLARF Headquarters comprises four core administrative departments: Staff Department (参谋部), Political Work Department (政治工作部), Logistics Department (后勤部), and Equipment Department (装备部) (CASI PLARF Organization).
- **Engineering-Technical & Operational Architecture:** Commands over 120,000 personnel organized into nine Corps-level or Corps Deputy Leader-grade Bases: six operational missile Bases (Bases 61–66) commanding at least 41–45 ballistic and cruise missile brigades, and three functional support Bases (Base 67 for central nuclear warhead storage/handling, Base 68 for underground and silo engineering, and Base 69 for missile test/training and opposition-force operations) (CASI PLARF Organization; CASI NC3 Report 2024). Per the 2025 DoD CMPR and FAS Nuclear Notebook 2025, the PLARF inventory encompasses approximately 400+ ICBMs (including >100 solid-fuel silos loaded with DF-31-class ICBMs), 500 IRBMs (DF-26), 1,300 MRBMs (DF-17, DF-21), 900 SRBMs (DF-11, DF-15, DF-16), and 400 GLCMs (DF-10A, DF-100).
- **Deterministic Causal Chain:** On December 31, 2015, CMC Chairman Xi Jinping elevated the Second Artillery Corps into the PLARF to execute the doctrinal mandate of "dual deterrence and dual operations" (核常兼备、全域慑战 — nuclear and conventional integration across all domains) (CASI NC3 Report 2024). Between 2017 and 2024, the PLARF expanded from 29 to over 41 launch brigades, initiated construction of 320 solid-fuel silos across three northern desert complexes in 2020–2021, conducted a long-range Pacific Ocean ICBM test launch (DF-31B/AG) from Hainan Island on September 25, 2024, and executed rapid-succession salvo ICBM launches into western China in December 2024 (Erickson DoD CMPR Synthesis).

99. Central Military Commission (CMC / 中央军事委员会) — Joint Operations Command Center & Nuclear C2 Chain

- **Institutional & Organizational Designation:** Central Military Commission of the Chinese Communist Party (中共中央军事委员会) and Central Military Commission of the PRC (中华人民共和国中央军事委员会), operating the CMC Joint Operations Command Center (CMC JOCC / 中央军委联合作战指挥中心) (CASI NC3 Report 2024).
- **Headquarters & Geolocation:** Administrative HQ at the Bayi Building, Fuxing Road, Haidian District, Beijing (`39.9075°N, 116.3180°E`); hardened wartime underground CMC Joint Operations Command Center beneath Xishan (Western Hills) National Forest Park, Haidian District, Beijing (`39.9850°N, 116.1400°E`) (CASI NC3 Report 2024).
- **Command Hierarchy & Named Leadership:** Chaired by CMC Chairman and Commander-in-Chief of the Joint Operations Command Center Xi Jinping (习近平), supported by CMC Vice Chairmen Gen. Zhang Youxia (张又侠) and Gen. He Weidong (何卫东), and CMC Joint Staff Department (JSD) Chief of Joint Staff Gen. Liu Zhenli (刘振立) (2025 DoD CMPR). Within the JSD, the Information and Communications Bureau (JSD-ICB / 联参信息通信局) and the newly established PLA Information Support Force (PLAISF, April 2024) maintain the hardened fiber-optic, underground cable, and satellite NC3 trunks (CASI Base 37 Report).
- **Engineering-Technical & Operational Architecture:** Exercises bifurcated command-and-control ("dual command") over PLARF forces: conventional missile strikes flow from the CMC through the five Theater Command Joint Operations Command Centers (Eastern, Southern, Western, Northern, Central) to theater-assigned conventional PLARF brigades, whereas nuclear launch authority rests exclusively with the CMC Chairman and flows from the CMC JOCC via the PLARF Command Post directly to operational Base and nuclear missile brigade launch battalions, bypassing Theater Commands (CASI NC3 Report 2024).
- **Deterministic Causal Chain:** To prevent unauthorized warhead mating or coup-driven mobilization—instituted after Marshal Lin Biao's unauthorized October 18, 1969 Project No. 1 readiness order—the CMC established a strict separation between warhead custody (Unit 0674 / Base 22 / Base 67) and missile launch brigades (Stokes NPEC Study). With the deployment of TJS/Huoyan-1 infrared satellites and LPAR fusion under PLAASF Base 37, the CMC JOCC now receives confirmed boost-phase ICBM alerts within 3 to 4 minutes of adversary launch, enabling Xi Jinping to transmit encrypted Emergency Action Messages over redundant fiber-optic, VLF/HVLF, and Beidou-3 short-message channels to silo and mobile ICBM brigades before inbound detonation (FAS Nuclear Weapons Archives; China's Nuclear Early-Warning Architecture).

100. China Academy of Engineering Physics (CAEP / 中国工程物理研究院) — Mianyang Science City

- **Institutional & Organizational Designation:** China Academy of Engineering Physics (CAEP), state-owned sovereign nuclear weapons research, design, development, and production complex operating under the State Administration of Science, Technology and Industry for National Defense (SASTIND) and CMC Equipment Development Department oversight (CAEP Profile).
- **Headquarters & Geolocation:** Science City (科学城, Complex 839),涪城区 (Fucheng / Youxian District), Mianyang, Sichuan Province (`31.4775°N, 104.7697°E`), with subordinate warhead component production at Zitong (Base 902) and Jiangyou (Base 903) (CAEP Profile).
- **Command Hierarchy & Named Leadership:** Directed by President Liu Cangli (刘仓理, shock mechanics and pulsed-power physicist) and Party Secretary杭义洪 (Hang Yihong) / Мо Дамин, building on foundational scientific directors Deng Jiaxian (邓稼先), Yu Min (于敏, architect of China's two-stage thermonuclear primary-secondary configuration), Wang Ganchang (王淦昌), Zhu Guangya (朱光亚), and Cheng Kaijia (程开甲) (CAEP Profile).

- **Engineering-Technical & Operational Architecture:** Comprises 12 specialized research institutes (including the Institute of Applied Physics and Computational Mathematics [IAPCM / 9th Institute] in Beijing, Institute of Nuclear Physics and Chemistry [INPC / 2nd Institute], Institute of Fluid Physics [1st Institute], Institute of Chemical Materials [3rd Institute], Institute of Structural Mechanics [4th Institute], Institute of Electronic Engineering [5th Institute], and Research Center of Laser Fusion [8th Institute]), 6 National Key Laboratories, the FBR-2 fast burst reactor, Dragon-I/II linear induction flash X-ray hydrotest facilities, and the ShenGuang high-energy laser complex (CAEP Profile).
- **Deterministic Causal Chain:** CAEP executed all 45 Chinese atmospheric and underground nuclear weapon tests at Lop Nur through July 29, 1996, including Project 596 (22 kt U-235 implosion test on October 16, 1964), Test No. 6 (3.3 Mt thermonuclear test on June 17, 1967), and enhanced radiation weapon (ERW / neutron bomb) principles and confirmation shots CHIC-32 (December 19, 1984) and CHIC-34 (September 29, 1988) (CAEP Profile). Post-1996, CAEP transitioned to subcritical hydronuclear testing at Lop Nur, TATB insensitive high-explosive compression modeling, and miniaturized thermonuclear warhead production (including the `#535` and low-yield `<10 kt #5x5` warhead families) that directly arm the DF-5B/C, DF-31AG, DF-41, DF-26, and JL-3 MIRV buses (FAS Nuclear Notebook 2026).

101. Ninth Academy (第九研究院 / Nine-Academy Designation of CAEP)

- **Institutional & Organizational Designation:** Ninth Bureau / Ninth Academy of the Second Ministry of Machine Building (第二机械工业部第九局 / 第九研究院), historical and enduring internal PLA/defense-industrial designation of the China Academy of Engineering Physics (CAEP Profile).
- **Headquarters & Geolocation:** Originally established in Beijing (Huayuan Road, Haidian District, `39.9810°N, 116.3650°E`, October 1958); relocated to Factory 221 at Haiyan County, Haibei Tibetan Autonomous Prefecture, Qinghai Province (`36.9600°N, 100.9200°E`, 1962–1969); transferred under the Third Line Defense (三线建设) relocation to Mianyang/Zitong/Jiangyou, Sichuan Province (`31.4775°N, 104.7697°E`, 1969–1990) (CAEP Profile; Stokes NPEC Study).
- **Command Hierarchy & Named Leadership:** Founded under Li Jue (李觉, Director), Wu Jilin (吴际霖), Guo Yingkui (郭英会), and Chief Theoretical Designers Deng Jiaxian and Zhou Guangzhao (周光召), operating under Marshal Nie Rongzhen's (聂荣臻) National Defense Science and Technology Commission (NDSTC) (Stokes NPEC Study).
- **Engineering-Technical & Operational Architecture:** Structured the original nine numbered divisions responsible for theoretical nuclear weapon physics, explosive lens detonation hydrodynamics, plutonium/HEU metallurgy, neutron initiator fabrication, ballistic reentry vehicle structural mechanics, and arming/fuzing/firing (AF&F) electronics (CAEP Profile).
- **Deterministic Causal Chain:** Following the Soviet Union's June 1959 termination of prototype atomic bomb technical assistance, the Ninth Academy engineered indigenous uranium-235 implosion lenses at Factory 221 (Haiyan) using HEU from the Lanzhou Gaseous Diffusion Plant (Plant 504), achieving criticality at Lop Nur on October 16, 1964 (CAEP Profile). During the 1969 Sino-Soviet border crisis and Cultural Revolution instability at Factory 221, the CMC ordered the dispersement of the Ninth Academy's fabrication, plutonium pit machining, and tritium boosting laboratories into hardened Third Line valleys around Mianyang (Complex 839), which was publicly co-designated CAEP in the 1980s while retaining "Ninth Academy" (九院) across internal defense procurement ledgers (CAEP Profile; Stokes NPEC Study).

102. Base 22 (067 Base / 二十二基地 — Central Nuclear Warhead Storage Complex)

- **Institutional & Organizational Designation:** PLA Second Artillery Force Base 22 (二十二基地), originally established as Unit 0674 (067基地) in 1958 under direct CMC authority; redesignated as PLARF Base 67 in the April 2017 PLA force-wide reorganization (CASI PLARF Nuclear Warhead Management 2026; Stokes NPEC Study).
- **Headquarters & Geolocation:** Administrative headquarters in Baoji City, Shaanxi Province (`34.3619°N, 107.2378°E`), with primary subterranean central storage and handling complexes in Taibai County (Hongchuan Valley), Qinling Mountains, Shaanxi Province (`34.0583°N, 107.3164°E`) (CASI PLARF Nuclear Warhead Management 2026).
- **Command Hierarchy & Named Leadership:** Founded by Major General Jia Qianrui (贾乾瑞, former Commandant of the Shangqiu Infantry School and "father of China's warhead security"), student affairs director Hong Youdao (洪有道), and Yao Shumei (姚书梅) (Stokes NPEC Study). Operates at Corps Deputy Leader / Corps Leader grade reporting directly to PLARF Headquarters and the CMC Equipment Development Department (CASI PLARF Organization).
- **Engineering-Technical & Operational Architecture:** Serves as the central custodian of China's strategic nuclear warhead stockpile ("National Treasure" / 国宝 and "Special Equipment" / 特装). Maintains atmospheric-controlled deep granite tunnels, tritium reservoir replenishment stations, non-destructive radiographic warhead inspection bays, a dedicated Equipment Inspection Institute (装备检验研究所), an armored rail/road warhead transport regiment, an air defense regiment, and an electronic countermeasures (ECM) regiment (CASI PLARF Nuclear Warhead Management 2026).
- **Deterministic Causal Chain:** Established in 1958 as Unit 0674 and initially sited 12 nautical miles from the Koko Nor (Haiyan) weapons fabrication plant in Qinghai, Base 22 was relocated to the Qinling Mountain tunnels in Taibai County, Shaanxi in 1969 to insulate China's nuclear stockpile from Soviet preventive strikes and domestic Cultural Revolution factional seizures (Stokes NPEC Study). Transferred from direct CMC/NDSTC control to the Second Artillery Force in 1980, Base 22 enforced China's peacetime de-mated nuclear posture by holding central warhead reserves in Shaanxi and dispatching warheadbatches via specialized rail cars to operational Base Equipment Inspection Regiments (Bases 61–66) upon CMC alert directives (CASI PLARF Nuclear Warhead Management 2026).

103. Taibai Depot (Hongchuan / Qinling Mountain Warhead Storage Complex of Base 22 / Base 67)

- **Institutional & Organizational Designation:** Taibai Underground Nuclear Warhead Storage and Technical Service Complex (Hongchuan Technical Service Brigade / 洪川技术勤务旅 and Unit 96038 / 96871-series facilities), primary central subterranean nuclear warhead repository of Base 22 / Base 67 (CASI PLARF Nuclear Warhead Management 2026).
- **Headquarters & Geolocation:** Hongchuan Town (红川镇) and Taibai County (太白县) valleys inside the Qinling Mountain massif, between Baoji (Shaanxi) and Tianshui (Gansu), centered around `34.0583°N, 107.3164°E` and `33.9850°N, 107.1200°E` under >500–1,000 meters of granite overburden (CASI PLARF Nuclear Warhead Management 2026; Stokes NPEC Study).
- **Command Hierarchy & Named Leadership:** Commanded by the Base 67 (former Base 22) Technical Service Brigade commander and Political Commissar under Base 67 Headquarters in Baoji, supported by the 308th Engineering Command civil engineering detachment in Hanzhong and Luoyang installation engineering regiments (Stokes NPEC Study).
- **Engineering-Technical & Operational Architecture:** Multi-portal granite tunnel complex engineered with blast-proof hydraulic doors, positive-pressure NBC filtration, electrostatic discharge shielding, constant-temperature/humidity warhead storage vaults, "Special Equipment Stations" (特装站) for pit and high-explosive aging diagnostics, and disguised rail/road transfer terminals connecting to the Longhai Railway

corridor (CASI PLARF Nuclear Warhead Management 2026). Protected by a dedicated Base 67 mobile Air Defense and ECM Regiment whose tracking and jamming success rates against low-flying terrain-masking aircraft were upgraded from 40%/33% in September 2017 to 80%/100% by 2018 (CASI PLARF Nuclear Warhead Management 2026).

- **Deterministic Causal Chain:** Constructed beginning in 1969 and continuously expanded between 2000 and 2025 across the Taibai–Tianshui Qinling tunnel network, the Taibai/Hongchuan complex receives newly fabricated thermonuclear warheads from CAEP (Mianyang/Zitong) via the Baoji–Chengdu rail line, performs acceptance testing and periodic tritium/electronic refurbishment, and dispatches warhead canisters to regional Base storage depots (Bases 61–66) and the northern ICBM silo fields (CASI PLARF Nuclear Warhead Management 2026; Stokes NPEC Study).

104. Base 61 (六十一基地 — Huangshan, Anhui Province)

- **Institutional & Organizational Designation:** PLARF Base 61 (MUCD Unit 96601, pre-2017 Base 52 / Unit 96151), Corps Deputy Leader / Corps Leader-grade operational missile base established in 1965 covering the Eastern Theater Command (Anhui, Fujian, Jiangxi, Zhejiang, and eastern Guangdong) (CASI PLARF Organization; CNS PLARF Order of Battle 2023).
- **Headquarters & Geolocation:** Tunxi District, Huangshan City, Anhui Province (`29.7147°N, 118.3375°E`), with subordinate brigades deployed across Anhui, Jiangxi, Zhejiang, and Fujian (CNS PLARF Order of Battle 2023).
- **Command Hierarchy & Named Leadership:** Commanded by Base 61 Commander (Major General grade, formerly Maj. Gen. Li Hongjun / Maj. Gen. Liu Weidong) and Political Commissar under PLARF HQ; coordinates conventional fires directly with the Eastern Theater Command Joint Operations Command Center in Nanjing (`32.0430°N, 118.8260°E`) while retaining nuclear C2 directly to the CMC (CASI PLARF Organization).
- **Engineering-Technical & Operational Architecture:** Commands seven operational missile brigades (Brigades 611 through 617, MUCDs 96711–96717) plus an expanded Technical Service Brigade (upgraded from a regiment due to high-volume SRBM/MRBM handling) and a dedicated UAV ISR Regiment operating TB-001/BZK-005 drones (CASI PLARF Organization; CASI TB-001 Report):
  - **611 Brigade** (MUCD 96711, Qingyang/Chizhou, Anhui, `30.6903°N, 117.9011°E`): Upgraded from DF-21A to DF-26 dual-capable IRBM / DF-31AG ICBM (FAS Nuclear Notebook 2024; CNS PLARF Order of Battle 2023).
  - **612 Brigade** (MUCD 96712, Leping/Jingdezhen, Jiangxi, `28.9797°N, 117.1205°E`): Upgraded from DF-21A nuclear MRBM to DF-31AG road-mobile ICBM (FAS Nuclear Notebook 2024; Internet Archive CASI PLARF Full Text).
  - **613 Brigade** (MUCD 96713, Shangrao, Jiangxi, `28.4286°N, 117.9631°E`): Armed with DF-15B/C conventional precision SRBMs (CNS PLARF Order of Battle 2023).
  - **614 Brigade** (MUCD 96714, Yong'an, Fujian, `25.9922°N, 117.3611°E`): Armed with DF-17 hypersonic glide vehicle MRBMs (upgraded from DF-11A SRBMs) (CNS PLARF Order of Battle 2023).
  - **615 Brigade** (MUCD 96715, Meizhou, Guangdong, `24.3314°N, 116.1264°E`): Armed with DF-11A / DF-16 conventional SRBMs/MRBMs (CNS PLARF Order of Battle 2023).
  - **616 Brigade** (MUCD 96716, Ganzhou, Jiangxi, `25.8450°N, 114.9336°E`): Armed with DF-15A/B/C and DF-16 tactical ballistic missiles (CNS PLARF Order of Battle 2023).
  - **617 Brigade** (MUCD 96717, Jinhua, Zhejiang, `29.0781°N, 119.6472°E`): Armed with DF-16 / DF-11A conventional SRBMs (CNS PLARF Order of Battle 2023).
- **Deterministic Causal Chain:** Designed as the primary ballistic strike formation opposite Taiwan and the Ryukyu/First Island Chain, Base 61 executes initial joint firepower preparation against Taiwanese air defenses, C2 bunkers, and US/Japanese airbases on Okinawa using Brigades 613–617, while Brigades 611 and 612 provide dispersed road-mobile nuclear ICBM/IRBM deterrence from central-eastern tunnel complexes (CASI PLARF Organization).

105. Base 62 (六十二基地 — Kunming, Yunnan Province)

- **Institutional & Organizational Designation:** PLARF Base 62 (MUCD Unit 96602, pre-2017 Base 53 / Unit 96201), Corps Deputy Leader-grade operational missile base established in 1966 covering the Southern Theater Command and southwestern approaches (CASI PLARF Organization).
- **Headquarters & Geolocation:** Panlong/Guandu District, Kunming, Yunnan Province (`25.0440°N, 102.7580°E`), overseeing brigades across Sichuan, Yunnan, Guangxi, Guangdong, and Hainan (CNS PLARF Order of Battle 2023).
- **Command Hierarchy & Named Leadership:** Commanded by Base 62 Commander (Major General grade) and Political Commissar under PLARF HQ; coordinates conventional maritime/theater strike missions with the Southern Theater Command JOCC in Guangzhou (CASI PLARF Organization).
- **Engineering-Technical & Operational Architecture:** Commands seven operational missile brigades (Brigades 621 through 627, MUCDs 96721–96727) spanning ICBM, dual-capable IRBM, ASBM, hypersonic MRBM, and LACM systems (CASI PLARF Organization; FAS Nuclear Notebook 2024):
  - **621 Brigade** (MUCD 96721, Yibin, Sichuan, `28.7607°N, 104.7914°E`): Upgraded from DF-21A to DF-31AG road-mobile nuclear ICBM (FAS Nuclear Notebook 2024).
  - **622 Brigade** (MUCD 96722, Yuxi, Yunnan, `24.3601°N, 102.4942°E`): Armed with DF-31A road-mobile nuclear ICBM (FAS Nuclear Notebook 2024).
  - **623 Brigade** (MUCD 96723, Liuzhou, Guangxi, `24.3264°N, 109.4281°E`): Armed with CJ-10A (DF-10A) ground-launched land-attack cruise missiles (CNS PLARF Order of Battle 2023).
  - **624 Brigade** (MUCD 96724, Danzhou, Hainan Island, `19.5208°N, 109.5808°E`): Armed with DF-21D anti-ship ballistic missiles (ASBM) and DF-26 IRBMs covering the South China Sea (CNS PLARF Order of Battle 2023).
  - **625 Brigade** (MUCD 96725, Jianshui, Yunnan, `23.6369°N, 102.8267°E`): Armed with DF-26 dual nuclear-conventional IRBMs (FAS Nuclear Notebook 2024).
  - **626 Brigade** (MUCD 96726, Qingyuan, Guangdong, `23.6820°N, 113.0560°E`): Armed with DF-26 dual-capable IRBM/ASBM launchers (FAS Nuclear Notebook 2024).
  - **627 Brigade** (MUCD 96727, Jieyang/Puning, Guangdong, `23.3725°N, 116.1650°E`): Armed with DF-17 hypersonic glide vehicle MRBMs (CASI PLARF Organization; CNS PLARF Order of Battle 2023).
- **Deterministic Causal Chain:** Base 62 projects strategic nuclear deterrence toward India, the Indian Ocean, and the continental United States via Brigades 621 and 622, while Brigades 624, 625, 626, and 627 create an overlapping anti-access/area-denial (A2/AD) and theater nuclear strike umbrella across the South China Sea, Strait of Malacca, Taiwan Strait southern approaches, and US facilities in Guam (CASI PLARF Organization).

106. Base 63 (六十三基地 — Huaihua, Hunan Province)

- **Institutional & Organizational Designation:** PLARF Base 63 (MUCD Unit 96603, pre-2017 Base 55 / Unit 96301), Corps Leader-grade operational missile base headquartered in Huaihua, Hunan Province (CASI PLARF Organization).
- **Headquarters & Geolocation:** Hecheng District, Huaihua, Hunan Province (27.5694°N, 110.0017°E), operating hardened mountain silos and road-mobile garrisons across Hunan, Guizhou, and Guangdong (CNS PLARF Order of Battle 2023).
- **Command Hierarchy & Named Leadership:** Commanded by a Corps Leader-grade Major General and Political Commissar reporting directly to PLARF HQ and the CMC JOCC for strategic intercontinental silo and mobile launches (CASI PLARF Organization). Retains a specialized liquid-propellant "Service Regiment" (勤务团) for UDMH/N2O4 fuel management alongside its Comprehensive Support Regiment (CASI PLARF Organization).
- **Engineering-Technical & Operational Architecture:** Commands six operational missile brigades (Brigades 631 through 636, MUCDs 96731–96736) centered on heavy silo-based liquid-fueled ICBMs, road-mobile solid-fueled ICBMs, and long-range cruise missiles (CASI PLARF Organization; CNS PLARF Order of Battle 2023):
  - **631 Brigade** (MUCD 96731, Jingzhou, Hunan, 26.5850°N, 109.6880°E, with silos such as 631-1 at 26.430807°N, 109.461770°E): Operates silo-based DF-5B (CSS-4 Mod 3, 5-MIRV) and newly expanded DF-5C silos (CNS PLARF Order of Battle 2023; CASI Ballistic Missile Industry).
  - **632 Brigade** (MUCD 96732, Shaoyang, Hunan, 27.2389°N, 111.4678°E): Armed with DF-31AG road-mobile solid-fuel ICBMs (CASI Ballistic Missile Industry; FAS Nuclear Notebook 2024).
  - **633 Brigade** (MUCD 96733, Huitong, Hunan, 26.8708°N, 109.7219°E): Operates silo-based DF-5A (single multi-megaton RV) and upgraded DF-5B/C silos (CASI Ballistic Missile Industry; FAS Nuclear Notebook 2026).
  - **634 Brigade** (MUCD 96734, Tongdao, Hunan, 26.1580°N, 109.7840°E): Newly stood-up ICBM brigade operating expanded DF-5C / DF-31AG systems (FAS Nuclear Notebook 2024).
  - **635 Brigade** (MUCD 96735, Yichun, Jiangxi, 27.8042°N, 114.3886°E): Armed with CJ-10 (DF-10) / DF-100 ground-launched cruise missiles (CNS PLARF Order of Battle 2023).
  - **636 Brigade** (MUCD 96736, Shaoguan, Guangdong, 24.8011°N, 113.5972°E): Armed with DF-16 conventional MRBMs targeting Taiwan and theater nodes (CNS PLARF Order of Battle 2023).
- **Deterministic Causal Chain:** Base 63 anchors China's assured thermonuclear counter-value and counter-force retaliation against the continental United States through its dispersed mountain DF-5A/B/C silos in western Hunan (Brigades 631, 633, 634) and road-mobile DF-31AG launchers (Brigade 632), doubling the silo count per DF-5 brigade from 6 to 12 silos between 2021 and 2025 (FAS Nuclear Notebook 2024; FAS Nuclear Notebook 2026).

107. Base 64 (六十四基地 — Lanzhou, Gansu / Xining, Qinghai)

- **Institutional & Organizational Designation:** PLARF Base 64 (MUCD Unit 96604, pre-2017 Base 56 / Unit 96351), Corps Leader-grade operational missile base covering northwestern and north-central China across the Western Theater Command (CASI PLARF Organization).
- **Headquarters & Geolocation:** Headquartered in Lanzhou, Gansu Province (36.0570°N, 103.8390°E; historically headquartered in Xining, Qinghai Province at 36.6171°N, 101.7782°E), with subordinate brigades across Shaanxi, Qinghai, Gansu, Ningxia, and Xinjiang (CASI PLARF Organization; FAS Nuclear Notebook 2024).
- **Command Hierarchy & Named Leadership:** Commanded by a Corps Leader-grade Major General and Political Commissar reporting directly to PLARF HQ and the CMC JOCC (CASI PLARF Organization).
- **Engineering-Technical & Operational Architecture:** Nearly doubled since 2017 from four to seven road-mobile brigades (Brigades 641 through 647, MUCDs 96741–96747), alongside geographic overlap with the three massive solid-fuel ICBM silo fields at Yumen (Gansu, 120 silos), Hami (Xinjiang, 110 silos), and Ordos/Yulin (Inner Mongolia/Shaanxi, 90 silos) (CASI PLARF Organization; FAS Nuclear Notebook 2025):
  - **641 Brigade** (MUCD 96741, Hancheng, Shaanxi, 35.4750°N, 110.4420°E): Armed with DF-31 / DF-31AG road-mobile ICBMs (FAS Nuclear Notebook 2024).
  - **642 Brigade** (MUCD 96742, Datong/Xining, Qinghai, 36.9231°N, 101.6856°E): Armed with DF-31AG road-mobile ICBMs (FAS Nuclear Notebook 2024).
  - **643 Brigade** (MUCD 96743, Tianshui, Gansu, 34.5808°N, 105.7247°E): Armed with DF-31AG road-mobile ICBMs (FAS Nuclear Notebook 2024).
  - **644 Brigade** (MUCD 96744, Hanzhong, Shaanxi, 33.0728°N, 107.0233°E): Armed with DF-41 road-mobile MIRV ICBMs (FAS Nuclear Notebook 2024).
  - **645 Brigade** (MUCD 96745, Yinchuan, Ningxia, 38.4872°N, 106.2309°E): Armed with DF-41 road-mobile MIRV ICBMs (FAS Nuclear Notebook 2024).
  - **646 Brigade** (MUCD 96746, Korla, Xinjiang, 41.7258°N, 86.1747°E): Armed with DF-26 dual nuclear-conventional IRBMs (FAS Nuclear Notebook 2024).
  - **647 Brigade** (MUCD 96747, Xining/Delingha, Qinghai, 37.3694°N, 97.3608°E): Armed with DF-26 / road-mobile ICBM systems (FAS Nuclear Notebook 2024).
- **Deterministic Causal Chain:** Positioned deep within China's continental interior beyond adversary conventional stand-off strike range, Base 64 executes polar and trans-Pacific ICBM trajectories against North America and Europe (Brigades 641–645 and the Yumen/Hami/Ordos silo fields) while holding Indian, Central Asian, and Middle Eastern theater targets at risk with DF-26 IRBMs (Brigade 646) (CASI PLARF Organization; FAS Nuclear Notebook 2025).

108. Base 65 (六十五基地 — Shenyang, Liaoning Province)

- **Institutional & Organizational Designation:** PLARF Base 65 (MUCD Unit 96605, pre-2017 Base 51 / Unit 96101), Corps Deputy Leader-grade operational missile base first established in Tonghua, Jilin in 1964 and relocated to Shenyang, Liaoning (CASI PLARF Organization).
- **Headquarters & Geolocation:** Huanggu District, Shenyang, Liaoning Province (41.8350°N, 123.4310°E), covering Northeast China (Manchuria) and the Shandong Peninsula within the Northern Theater Command (CASI PLARF Organization).
- **Command Hierarchy & Named Leadership:** Commanded by a Corps Deputy Leader-grade Major General and Political Commissar under PLARF HQ; coordinates conventional theater operations with the Northern Theater Command JOCC in Shenyang (CASI PLARF Organization).

- **Engineering-Technical & Operational Architecture:** Doubled in size since 2017 from three to six operational missile brigades (Brigades 651 through 656, MUCDs 96751–96756) (CASI PLARF Organization; FAS Nuclear Notebook 2024):
  - **651 Brigade** (MUCD 96751, Chifeng, Inner Mongolia, 42.2578°N, 118.8869°E): Armed with DF-21A nuclear MRBMs / transitioning to DF-26/DF-31AG (FAS Nuclear Notebook 2024).
  - **652 Brigade** (MUCD 96752, Tonghua, Jilin, 41.7283°N, 125.9397°E): Armed with DF-31AG / DF-41 road-mobile nuclear ICBMs (FAS Nuclear Notebook 2024).
  - **653 Brigade** (MUCD 96753, Laiwu/Jinan, Shandong, 36.2136°N, 117.6767°E): Armed with DF-21D anti-ship ballistic missiles and DF-100 supersonic cruise missiles (CNS PLARF Order of Battle 2023).
  - **654 Brigade** (MUCD 96754, Dalian/Wafangdian, Liaoning, 39.6269°N, 122.0028°E): Armed with DF-26 dual nuclear-conventional IRBMs (FAS Nuclear Notebook 2024).
  - **655 Brigade** (MUCD 96755, Tonghua/Baishan, Jilin, 41.9430°N, 126.4240°E): Armed with DF-17 hypersonic MRBMs (CNS PLARF Order of Battle 2023).
  - **656 Brigade** (MUCD 96756, Qingzhou/Weifang, Shandong, 36.6850°N, 118.4790°E): Armed with DF-26 dual-capable IRBMs (FAS Nuclear Notebook 2024).
- **Deterministic Causal Chain:** Base 65 provides high-latitude polar ICBM trajectory coverage over the Continental United States (Brigade 652) while holding US Forces Japan (Misawa, Yokota, Kadena, Yokosuka, Sasebo), US Forces Korea (Camp Humphreys, Osan, Kunsan), and Russian Far East nodes at immediate nuclear and conventional precision-strike risk via Brigades 651, 653, 654, 655, and 656 (CASI PLARF Organization; FAS Nuclear Notebook 2024).

109. Base 66 (六十六基地 — Luoyang, Henan Province)

- **Institutional & Organizational Designation:** PLARF Base 66 (MUCD Unit 96606, pre-2017 Base 54 / Unit 96251), Corps Leader-grade operational missile base established in 1966 in central China (CASI PLARF Organization).
- **Headquarters & Geolocation:** Luolong/Jianxi District, Luoyang, Henan Province (34.6197°N, 112.4540°E), commanding hardened mountain silos and road-mobile brigades across Henan and Shanxi within the Central Theater Command (CASI PLARF Organization; FAS Nuclear Notebook 2024).
- **Command Hierarchy & Named Leadership:** Commanded by a Corps Leader-grade Major General and Political Commissar reporting directly to PLARF HQ and the CMC JOCC (CASI PLARF Organization).
- **Engineering-Technical & Operational Architecture:** Commands six operational nuclear and dual-capable missile brigades (Brigades 661 through 666, MUCDs 96761–96766) combining heavy DF-5B/C silos, road-mobile DF-31AG and DF-41 ICBMs, and DF-26 IRBMs (CASI PLARF Organization; FAS Nuclear Notebook 2024):
  - **661 Brigade** (MUCD 96761, Lingbao/Lushi, Henan, 34.5169°N, 110.8942°E): Operates silo-based DF-5B (5-MIRV) and expanded DF-5C silos across the Western Henan mountains (CNS PLARF Order of Battle 2023; FAS Nuclear Notebook 2024).
  - **662 Brigade** (MUCD 96762, Luanchuan, Henan, 33.7858°N, 111.6158°E): Operates silo-based DF-5A/B/C ICBMs (FAS Nuclear Notebook 2024).
  - **663 Brigade** (MUCD 96763, Nanyang, Henan, 33.0039°N, 112.5283°E): Armed with DF-31A / DF-31AG road-mobile ICBMs (FAS Nuclear Notebook 2024).
  - **664 Brigade** (MUCD 96764, Luoyang/Yanshi, Henan, 34.7280°N, 112.7890°E): Armed with DF-31AG road-mobile ICBMs (FAS Nuclear Notebook 2024).
  - **665 Brigade** (MUCD 96765, Changzhi, Shanxi, 36.1953°N, 113.1164°E): Armed with DF-21C/DF-17 or road-mobile ICBM systems (CNS PLARF Order of Battle 2023).
  - **666 Brigade** (MUCD 96766, Xinyang, Henan, 32.1470°N, 114.0910°E): Armed with DF-26 dual nuclear-conventional IRBMs (FAS Nuclear Notebook 2024).
- **Deterministic Causal Chain:** Serving as the central strategic reserve of the PLARF in Henan's Funiu and Qinling mountain extensions, Base 66 executes intercontinental retaliatory strikes against the United States via two silo-based DF-5B/C brigades (Brigades 661 and 662, including 18+ new silos constructed between 2021 and 2024) and two road-mobile DF-31A/AG brigades (Brigades 663 and 664), while Brigade 666 provides 4,000 km theater nuclear/conventional reach to Guam and the Western Pacific (FAS Nuclear Notebook 2024; FAS Nuclear Notebook 2026).

110. Base 67 (六十七基地 — Baoji, Shaanxi Province)

- **Institutional & Organizational Designation:** PLARF Base 67 (MUCD Unit 96607, pre-2017 Base 22 / Unit 96401), Corps Deputy Leader / Corps Leader-grade central nuclear warhead stockpile management Base ("Guardians of the National Treasure" / 国宝卫士) (CASI PLARF Nuclear Warhead Management 2026; CASI PLARF Organization).
- **Headquarters & Geolocation:** Weibin/Jintai District, Baoji City, Shaanxi Province (34.3619°N, 107.2378°E), with dispersed storage and technical complexes across Taibai County (Hongchuan, 34.0583°N, 107.3164°E), Fengxian (33.9110°N, 106.5240°E), and Meixian in the Qinling Mountains (CASI PLARF Nuclear Warhead Management 2026).
- **Command Hierarchy & Named Leadership:** Commanded by Base 67 Commander and Political Commissar reporting directly to PLARF HQ and the CMC Equipment Development Department (CASI PLARF Nuclear Warhead Management 2026).
- **Engineering-Technical & Operational Architecture:** Commands three brigade-sized formations—two Warhead Storage and Technical Service Brigades (including the Hongchuan Technical Service Brigade) and one Nuclear Emergency Response and Rescue Brigade—plus the Equipment Inspection Institute (装备检验研究所), Unit 96038, and eight specialized regiment-sized units: Training Regiment, Communications Regiment, Security/Guard Regiment, Comprehensive Support/Logistics Regiment, Air Defense Regiment, Electronic Countermeasures (ECM) Regiment, Warhead Quality Control Regiment, and Special Rail/Road Transport Regiment (CASI PLARF Nuclear Warhead Management 2026).
- **Deterministic Causal Chain:** Reorganized from Base 22 into Base 67 in April 2017 to support China's expanding warhead inventory and transition toward higher readiness, Base 67 executes end-to-end warhead custody, tritium replenishment, non-destructive testing, radiological accident response, and encrypted rail/road warhead delivery to the Equipment Inspection Regiments (装备勤务团 / 装检团) of Operational Bases 61–66, which then transfer warheads to launch battalions for warhead-missile mating (CASI PLARF Nuclear Warhead Management 2026).

111. Base 68 (六十八基地 — Luoyang, Henan Province)

- **Institutional & Organizational Designation:** PLARF Base 68 (MUCD Unit 96608), Corps Deputy Leader-grade strategic engineering and underground infrastructure construction Base (CASI PLARF Organization).

- **Headquarters & Geolocation:** Headquartered in Luoyang, Henan Province (34.6520°N, 112.4280°E), with subordinate engineering brigades and regiments deployed in Hanzhong (Shaanxi), Luoyang, Tianshui, and active construction zones at Yumen, Hami, and Ordos (CASI PLARF Organization; Stokes NPEC Study).
- **Command Hierarchy & Named Leadership:** Formed in 2012 as the PLARF Engineering Base from the former Second Artillery Engineering Technical Zongdui (工程技术总队), and merged in the April 2017 reorganization with the 308th Engineering Command (308工程指挥部) under a Corps Deputy Leader-grade Major General and Political Commissar (CASI PLARF Organization).
- **Engineering-Technical & Operational Architecture:** Commands six Engineering Brigades (Brigades 681–686, MUCDs 96881–96886)—four civil/tunnel excavation brigades utilizing tunnel-boring machines (TBMs) for deep granite "Underground Great Wall" (地下长城) networks and missile silos, and at least two specialized installation brigades responsible for blast doors, overpressure shock isolation, electromagnetic pulse (EMP) and radiation shielding, HVAC, and closed-loop life support—plus a Communications Engineering Regiment, a Communications Repair and Maintenance Regiment, and four specialized support regiments (CASI PLARF Organization).
- **Deterministic Causal Chain:** Base 68 excavated and hardened the PLARF's subterranean drive-in TEL tunnels across China and between 2020 and 2025 executed the standardized grid construction of 320 solid-fuel ICBM silos across Yumen, Hami, and Ordos (under inflatable environmental/concealment air domes) as well as 30 new liquid-fuel DF-5 silos for Bases 63 and 66 (FAS Nuclear Notebook 2024; CASI PLARF Organization).

112. Base 69 (六十九基地 — Jilantai / Yinchuan / Wuzhai Test & Training Base)

- **Institutional & Organizational Designation:** PLARF Base 69 (MUCD Unit 96609, Test and Training Base / 试验训练基地), Corps Deputy Leader-grade operational test, evaluation, and Blue Force (OPFOR) training Base established in April 2017 (CASI PLARF Organization).
- **Headquarters & Geolocation:** Administrative headquarters in Yinchuan, Ningxia (38.4872°N, 106.2309°E) and Jilantai (Alxa Left Banner, Inner Mongolia, 39.7540°N, 105.7040°E), operating the Jilantai ICBM Silo and Mobile Training Complex (39.7820°N, 105.6250°E) and coordinating launches with Taiyuan/Wuzhai Missile Test Center (38.8489°N, 111.6080°E) (CASI PLARF Organization; FAS Silo Construction Analysis).
- **Command Hierarchy & Named Leadership:** Commanded by a Corps Deputy Leader-grade Major General and Political Commissar reporting to PLARF HQ (CASI PLARF Organization).
- **Engineering-Technical & Operational Architecture:** Oversees at least three Test and Training Regiments/Districts, a specialized Blue Force (OPFOR) electronic and physical opposition regiment, a UAV reconnaissance unit, and the Jilantai test complex containing 15–16 full-scale solid- and liquid-fuel training silos, rail-transfer pads, and road-mobile launch pads (CASI PLARF Organization; FAS Nuclear Notebook 2025).
- **Deterministic Causal Chain:** Constructed 15–16 prototype silos at Jilantai between 2019 and 2021 to validate silo geometry, canister loading via DF-31BJ/DF-41 loader vehicles, and Launch-on-Warning crew drills before replicating the design across Yumen, Hami, and Ordos (FAS Silo Construction Analysis). In 2023, Base 69 test-launched two ICBMs from training silos into western impact ranges, and in December 2024 executed a rapid-succession multi-ICBM salvo launch from its training center into western China to certify Early-Warning Counterstrike (EWCS) operational readiness (Erickson DoD CMPR Synthesis; FAS Ballistic Missiles Archive).

113. Dongfeng (东风 — DF Ballistic Missile Family Architecture, CASC CALT / CASIC)

- **Institutional & Organizational Designation:** Dongfeng ("East Wind" / 东风) land-based strategic and tactical surface-to-surface missile series, designed and manufactured by the China Aerospace Science and Technology Corporation (CASC / 中国航天科技集团) First Academy (China Academy of Launch Vehicle Technology, CALT / 中国运载火箭技术研究院) and Fourth Academy (Academy of Aerospace Solid Propulsion Technology, AASPT, Xi'an), alongside the China Aerospace Science and Industry Corporation (CASIC / 中国航天科工集团) Fourth Academy (Sanjiang Space Group / 航天三江集团, Wuhan/Xiaogan) (CASI Ballistic Missile Industry; CSIS Missiles of China).
- **Headquarters & Geolocation:** CALT (1st Academy) at Nandahongmen, Fengtai District, Beijing (39.7950°N, 116.4020°E) and Tianjin Binhai production base; CASC 4th Academy (AASPT) at Baqiao District, Xi'an (34.2890°N, 109.0640°E); CASIC 4th Academy (Sanjiang) at Wuhan, Hubei (30.6380°N, 114.2850°E) (CASI Ballistic Missile Industry).
- **Command Hierarchy & Named Leadership:** Pioneered by Qian Xuesen (钱学森, founder of the Fifth Academy of the Ministry of National Defense on October 8, 1956), Ren Xinmin (任新民), Tu Shou'e (屠守锷, DF-5 chief designer), Huang Weilu (黄纬禄, DF-21/JL-1 chief designer), Fang Xinran, and Liuhead designers at CALT and CASIC under SASTIND and CMC Equipment Development Department oversight (CASI Ballistic Missile Industry).
- **Engineering-Technical & Operational Architecture:** Spans five generations across liquid-storable (UDMH/N2O4) and composite solid-propellant (HTPB/NEPE filament-wound carbon-epoxy motor) architectures:
  - **ICBM Tier:** DF-4 (retired/training), DF-5A/B/C (liquid silo-based, 12,000–13,000+ km), DF-31/DF-31A/DF-31AG/DF-31B (solid road-mobile and silo-based, 7,000–11,700 km), and DF-41 (solid road/rail/silo MIRV ICBM, 12,000–15,000 km) (CSIS Missiles of China).
  - **IRBM / Conventional ICBM Tier:** DF-26 (dual-capable hot-swappable IRBM/ASBM, 4,000 km) and DF-27 (fielded HGV/conventional ICBM/ASBM, 5,000–8,000 km) (Erickson DoD CMPR Synthesis).
  - **MRBM & SRBM Tier:** DF-17 (hypersonic glide vehicle MRBM, 1,800–2,500 km), DF-21A/C/D (1,500–2,150 km), DF-16 (800–1,000 km), DF-15A/B/C (600–900 km), and DF-11A (300–600 km) (CNS PLARF Order of Battle 2023).
- **Deterministic Causal Chain:** Originating with the Fifth Academy in 1956 and the "Eight-Year Four-Missile Plan" (八年四弹) of 1965 that produced the DF-2A, DF-3, DF-4, and DF-5 to sequentially target Japan, the Philippines, Guam, and the US mainland/Moscow, CALT and CASIC expanded solid-propellant motor production facilities across Beijing, Xi'an, and Hubei between 2018 and 2025 to supply hundreds of TELs and silo-loaded canisterized ICBMs simultaneously (CASI Ballistic Missile Industry; FAS Nuclear Notebook 2025).

114. DF-5B (东风-5B / CSS-4 Mod 3 — MIRV Silo-Based ICBM)

- **Institutional & Organizational Designation:** Dongfeng-5B (DF-5B, US DoD/NASIC designation CSS-4 Mod 3; upgraded from the single-warhead DF-5A / CSS-4 Mod 2), heavy liquid-propellant silo-based intercontinental ballistic missile (CSIS DF-5 Profile; FAS Nuclear Notebook 2026).

- **Headquarters & Geolocation:** Manufactured by CASC First Academy (CALT) in Beijing; deployed in hardened mountain silos under Base 63 (631 Brigade in Jingzhou, Hunan at 26.430807°N, 109.461770°E and 633 Brigade in Huitong, Hunan) and Base 66 (661 Brigade in Lingbao/Lushi, Henan at 34.5169°N, 110.8942°E) (CNS PLARF Order of Battle 2023; FAS Nuclear Notebook 2024).
- **Command Hierarchy & Named Leadership:** Developed by CALT under Chief Designer Tu Shou'e's legacy team with post-boost vehicle (PBV) vernier engine engineering from CASC 1st and 6th Academies; operated by PLARF Brigades 631 and 661 under direct CMC JOCC nuclear launch authority (CSIS DF-5 Profile).
- **Engineering-Technical & Operational Architecture:** Two-stage storable liquid-propellant (UDMH fuel / N2O4 oxidizer, YF-20/YF-22 engine family) plus liquid-propellant MIRV post-boost vehicle; length 32.6 m, diameter 3.35 m, launch weight 183,000 kg, throw-weight ~3,900 kg, operational range 12,000–13,000 km, circular error probable (CEP) 300 m (astro-inertial + Beidou-3 updates), carrying up to 5 MIRVs (each 200–300 kt yield thermonuclear warhead with chaff/balloon penetration aids) (CSIS DF-5 Profile; FAS Nuclear Notebook 2026).
- **Deterministic Causal Chain:** To counter the US deployment of the Ground-based Midcourse Defense (GMD) system in Alaska and California after the 2002 US ABM Treaty withdrawal, CALT integrated the Yuanzheng-series post-boost bus onto the 3.35-meter DF-5A airframe, entering operational service in 2015 (publicly unveiled at the September 3, 2015 Beijing Victory Day Parade) and multiplying warhead loading per silo fivefold (CSIS DF-5 Profile; FAS Nuclear Notebook 2026).

115. DF-5C (东风-5C / CSS-4 Mod 4 — Multi-Megaton & 10-MIRV Silo-Based ICBM Variant)

- **Institutional & Organizational Designation:** Dongfeng-5C (DF-5C, US DoD designation CSS-4 Mod 4), heavy silo-based liquid-propellant ICBM variant (CSIS DF-5 Profile; FAS Nuclear Notebook 2026).
- **Headquarters & Geolocation:** Flight-tested from Taiyuan Satellite Launch Center / Wuzhai (38.8489°N, 111.6080°E) to the Korla/Taklamakan impact range in Xinjiang; deployed across 30 newly constructed and upgraded silos in Base 63 (Hunan) and Base 66 (Henan) (FAS Nuclear Notebook 2024; FAS Nuclear Notebook 2026).
- **Command Hierarchy & Named Leadership:** Engineered by CASC CALT (1st Academy) and CAEP (Mianyang); fielded by at least two PLARF silo brigades under Bases 63 and 66 (FAS Nuclear Notebook 2026).
- **Engineering-Technical & Operational Architecture:** Utilizes the 3.35 m diameter, 183,000+ kg two-stage UDMH/N2O4 booster with an elongated payload shroud capable of two distinct payload configurations documented in US intelligence assessments: (1) a 10-MIRV post-boost vehicle configuration flight-tested on January 21, 2017 from Taiyuan to western China (CSIS DF-5 Profile), and (2) a high-yield multi-megaton single thermonuclear warhead configuration (3–5 Mt class) documented in the 2024 and 2025 DoD CMPRs to replace legacy DF-5A missiles and hold hardened deeply buried command nodes at risk (FAS Nuclear Notebook 2026).
- **Deterministic Causal Chain:** Following the January 2017 flight test with 10 MIRVs from Wuzhai, the PLARF initiated construction of 30 new DF-5-class silos across three mountainous areas in central-eastern China (doubling per-brigade silo counts from 6 to 12 in Brigades 631, 633, 634, 661, and 662), fielding the DF-5C by 2024–2026 to preserve both heavy multi-megaton hard-target kill capability and saturation MIRV throw-weight (FAS Nuclear Notebook 2024; FAS Nuclear Notebook 2026).

116. DF-31AG (东风-31AG / CSS-10 Mod 2A — 8-Axle TEL Road-Mobile & Silo-Variant ICBM)

- **Institutional & Organizational Designation:** Dongfeng-31AG (DF-31AG / DF-31B, US DoD designation CSS-10 Mod 2A, with silo-loader variant designated DF-31BJ), three-stage solid-propellant road-mobile and silo-deployed intercontinental ballistic missile (CSIS DF-31 Profile; FAS Nuclear Notebook 2025; FAS Arms Control 2025 CMPR Analysis).
- **Headquarters & Geolocation:** Airframe and propulsion manufactured by CASC CALT (1st Academy) and AASPT (4th Academy); 8-axle all-terrain transporter-erector-launcher (TEL) manufactured by CASIC Sanjiang Space Group / Wanshan Special Vehicle Co. (Xiaogan, Hubei). Deployed across at least eight road-mobile brigades (including Brigades 612, 621, 632, 641, 642, 643, 652, 664) and >100 solid-propellant silos at Yumen, Hami, and Ordos (FAS Nuclear Notebook 2024; FAS Arms Control 2025 CMPR Analysis).
- **Command Hierarchy & Named Leadership:** Operated by PLARF Bases 61, 62, 63, 64, 65, and 66 under direct CMC JOCC nuclear command (FAS Nuclear Notebook 2024).
- **Engineering-Technical & Operational Architecture:** Three-stage solid-propellant ICBM; length ~14–15 m, diameter 2.0 m, launch mass ~42,000 kg, range 11,200–11,700 km, CEP 150–300 m (inertial + stellar + Beidou-3), cold-gas canister ejection, armed with a single 200–300 kt thermonuclear warhead plus at least eight decoy penetration aids (or dual/triple lower-yield MIRV option on late blocks) (CSIS DF-31 Profile; CSIS ChinaPower Nuclear Modernization). Replaces the road-bound 8-axle Hanyang tractor-semitrailer of the DF-31A with an integrated 8-axle (16x16) off-road Wanshan HTF5980A TEL and ribbed launch canister (CSIS DF-31 Profile).
- **Deterministic Causal Chain:** Unveiled at the July 2017 Zhurihe 90th PLA Anniversary Parade to eliminate the DF-31A's vulnerability on paved road networks, the DF-31AG expanded to 64+ road-mobile launchers by 2024, executed the historic September 25, 2024 full-range Pacific Ocean ICBM test launch from Hainan Island (referred to in the 2025 CMPR as DF-31B), and provided the silo-based ICBM variant loaded into >100 silos across Yumen, Hami, and Ordos by 2024–2025 (CSIS ChinaPower Nuclear Modernization; FAS Arms Control 2025 CMPR Analysis).

117. DF-41 (东风-41 / CSS-20 — Road-Mobile, Silo & Rail-Mobile MIRV ICBM; Yumen, Hami, Ordos Silo Fields)

- **Institutional & Organizational Designation:** Dongfeng-41 (DF-41, US DoD designation CSS-20, originally Project 204), fourth-generation three-stage solid-propellant MIRV intercontinental ballistic missile deployed in road-mobile, silo-based, and rail-mobile configurations (CSIS DF-41 Profile; FAS Nuclear Notebook 2025).
- **Headquarters & Geolocation:** Developed by CASC CALT (1st Academy) with 8-axle Taian HTF5980 TELs built by Taian Special Vehicle Company (Shandong). Road-mobile brigades deployed at Base 64 (644 Brigade in Hanzhong, Shaanxi at 33.0728°N, 107.0233°E; 645 Brigade in Yinchuan, Ningxia at 38.4872°N, 106.2309°E) and Base 65 (652 Brigade in Tonghua, Jilin at 41.7283°N, 125.9397°E), alongside the three northern solid-propellant ICBM silo complexes (FAS Nuclear Notebook 2024; FAS Silo Construction Analysis):
  - **Yumen Silo Field (Gansu Province):** 40.2000°N, 96.9000°E — 120 solid-fuel silos arranged in a 3 km triangular grid across 1,800 sq km northwest of Yumen (FAS Silo Construction Analysis; CNDP Silo Analysis).
  - **Hami Silo Field (Xinjiang Uyghur Autonomous Region):** 42.3000°N, 92.8000°E — 110 solid-fuel silos in a 3 km grid southeast of Hami (FAS Nuclear Notebook 2024).

    - **Ordos / Hanggin Banner / Yulin Silo Field (Inner Mongolia / Shaanxi border):** 39.9000°N, 108.3000°E — 90 solid-fuel silos across Hanggin Banner (Ordos) and Yulin (FAS Nuclear Notebook 2025; Science & Global Security Silo Fallout Study).
- **Command Hierarchy & Named Leadership:** Engineered under CASC CALT Chief Designer Fang Xinran / Liu Baoyong teams and CAEP warhead integration teams; commanded by PLARF Bases 64, 65, and 66 under direct CMC JOCC nuclear command (CSIS DF-41 Profile).
- **Engineering-Technical & Operational Architecture:** Three-stage solid-propellant motor with liquid/gas PBV; length 21 m, diameter 2.25 m, launch weight ~80,000 kg, range 12,000–15,000 km, top speed Mach 25, CEP ~100–150 m (ring-laser gyro inertial + stellar + Beidou-3), cold-gas canister launch, carrying up to 3 MIRVs in standard operational configuration (assessed by DoD CMPR; structurally capable of up to 6–10 smaller RVs/decoys using CAEP's #535 or #5x5 warheads) (CSIS DF-41 Profile; CSIS ChinaPower Nuclear Modernization; FAS Nuclear Notebook 2026).
- **Deterministic Causal Chain:** Initiated as Project 204 in 1986, flight-tested at least 11 times between July 24, 2012 and November 2019 (including a December 5, 2015 rail-mobile ejection test), and unveiled at the October 1, 2019 70th PRC Anniversary Parade (16 TELs from two brigades), the DF-41 and the 320 northern solid-fuel silos (completed in late 2022) force any adversary counterforce plan to expend over 600–1,000 precision warheads across Western China while guaranteeing survivable MIRV second-strike reach across all of North America (CSIS DF-41 Profile; FAS Nuclear Notebook 2025).

118. DF-26 (东风-26 / CSS-18 — Dual-Capable "Hot-Swappable" Precision Strike IRBM)

- **Institutional & Organizational Designation:** Dongfeng-26 (DF-26, US DoD designation CSS-18, with anti-ship variant designated DF-26B), two-stage solid-propellant road-mobile intermediate-range ballistic missile ("Guam Express" / 关岛快递) (CSIS DF-26 Profile; FAS Nuclear Notebook 2025).
- **Headquarters & Geolocation:** Designed and mass-produced by CASC First Academy (CALT) at its Beijing Nandahongmen plant (39.7950°N, 116.4020°E) on 6-axle Taian HTF5680A1 TELs; deployed across at least seven PLARF brigades including 625 Brigade (Jianshui, Yunnan), 626 Brigade (Qingyuan, Guangdong), 646 Brigade (Korla, Xinjiang), 654 Brigade (Dalian, Liaoning), 656 Brigade (Qingzhou, Shandong), and 666 Brigade (Xinyang, Henan) (FAS Nuclear Notebook 2024; CNS PLARF Order of Battle 2023).
- **Command Hierarchy & Named Leadership:** Operated by dual-capable PLARF brigades under Bases 61, 62, 64, 65, and 66, where each DF-26 brigade is three times the size of legacy DF-21A brigades (~36 TELs per brigade, totaling 250+ TELs and 500+ missiles by 2025, with ~100 TELs assigned a nuclear mission by FAS) (CASI Base 37 Report; FAS Nuclear Notebook 2025).
- **Engineering-Technical & Operational Architecture:** Two-stage solid-propellant IRBM; length 14.0 m, diameter 1.4 m, launch mass 20,000 kg, payload 1,200–1,800 kg, range 4,000 km (up to 5,000 km in extended configurations), equipped with a biconic finned Maneuverable Reentry Vehicle (MaRV) using active radar/infrared terminal seekers and Beidou-3 updates for <10–30 m CEP (CSIS DF-26 Profile). Engineered for rapid field "hot-swapping" of payload nosecones—allowing a single missile body and TEL to swap in the field between a conventional high-explosive/submunition/anti-ship warhead and a thermonuclear warhead (including low-yield <10 kt precision nuclear warheads assessed in the 2025 DoD CMPR) (CSIS DF-26 Profile; FAS Nuclear Notebook 2026).
- **Deterministic Causal Chain:** Unveiled at the September 3, 2015 Beijing Parade and fielded in 2016, the DF-26 completely replaced the medium-range DF-21A in the regional nuclear role by 2025, demonstrated moving-ship maritime strike capability in the South China Sea in August 2020 (DF-26B launched from Qinghai/Zhejiang), and provides Beijing with a low-yield theater nuclear escalation-control weapon and conventional anti-carrier/Guam strike system co-mingled on identical launchers (CSIS DF-26 Profile; FAS Nuclear Notebook 2026).

119. DF-17 (东风-17 / CSS-22 — MRBM with DF-ZF Hypersonic Glide Vehicle)

- **Institutional & Organizational Designation:** Dongfeng-17 (DF-17, US DoD designation CSS-22) medium-range ballistic missile booster paired with the DF-ZF (formerly designated WU-14 by US intelligence) hypersonic glide vehicle (HGV) (CSIS DF-17 Profile; CNS PLARF Order of Battle 2023).
- **Headquarters & Geolocation:** Developed by CASC First Academy (CALT) and China Academy of Aerospace Aerodynamics (CAAA / 11th Academy, Beijing); deployed with Base 61 (614 Brigade in Yong'an, Fujian at 25.9922°N, 117.3611°E), Base 62 (627 Brigade in Jieyang/Puning, Guangdong at 23.3725°N, 116.1650°E), and Base 65 (655 Brigade in Tonghua/Baishan, Jilin) (CNS PLARF Order of Battle 2023; CASI TB-001 Report).
- **Command Hierarchy & Named Leadership:** Commanded by PLARF Bases 61, 62, and 65 in coordination with the Eastern, Southern, and Northern Theater Command JOCCs (CASI PLARF Organization).
- **Engineering-Technical & Operational Architecture:** Single-stage solid-propellant booster (derived from the DF-16B rocket motor) mounted on a 5-axle (10x10) Wanshan enclosed TEL; total length 11.0 m, launch weight 15,000 kg, operational range 1,800–2,500 km (CSIS DF-17 Profile). The wedge-shaped lifting-body DF-ZF HGV separates at a depressed apogee of ~60 km and executes sustained endoatmospheric hypersonic glide and lateral cross-range aerodynamic maneuvers at Mach 5 to Mach 10 (6,170–12,360 km/h) under inertial, Beidou-3, and terminal seeker guidance (CSIS DF-17 Profile; PLA Rocket Force Overview). Primarily conventional, though DoD and CSIS assess potential dual nuclear capability (CSIS ChinaPower Nuclear Modernization).
- **Deterministic Causal Chain:** The DF-ZF HGV executed its first flight test on January 9, 2014 from Taiyuan/Wuzhai and completed at least nine tests through November 2017 (including a November 1, 2017 test that flew 1,400 km in 11 minutes at 60 km altitude), entering operational service at the October 1, 2019 Beijing National Day Parade to underfly and maneuver around US THAAD, Patriot PAC-3 MSE, and Aegis SM-3/SM-6 missile defense envelopes across the First Island Chain (CSIS DF-17 Profile; PLA Rocket Force Overview).

120. Julang (巨浪 — JL Submarine-Launched Ballistic Missile Program)

- **Institutional & Organizational Designation:** Julang ("Giant Wave" / 巨浪) sea-based strategic submarine-launched ballistic missile (SLBM) family, developed under the "One Missile, Two Modifications" (一弹两用 — land-based Dongfeng / sea-based Julang) engineering doctrine by CASC First Academy (CALT) and CASIC Fourth Academy (CSIS Missiles of China; CSIS JL-2 Profile).
- **Headquarters & Geolocation:** Engineered in Beijing (CALT / CASIC 2nd & 4th Academies) and solid-motor plants in Xi'an/Hubei; test-fired from Golf-class (Type 031) and Qing-class (Type 032, Pennant 201) auxiliary ballistic-missile test submarines stationed at Xiaopingdao Naval Base, Dalian, Liaoning (38.8180°N, 121.4950°E) into the Bohai Sea into western China (Taklamakan Desert) (CSIS JL-2 Profile).
- **Command Hierarchy & Named Leadership:** Founded by Chief Designer Huang Weilu (黄纬禄, JL-1 / DF-21) and succeeded by Chief Designers Fang Xinran and Liu Baoyong for JL-2/JL-3; operationally deployed by the People's Liberation Army Navy (PLAN) Submarine Force under direct CMC JOCC nuclear command (CSIS JL-2 Profile; CASI NC3 Report 2024).

- **Engineering-Technical & Operational Architecture:** Evolved across three generations:
  - **JL-1 (CSS-N-3):** Two-stage solid-fuel SLBM (range 1,700–2,150 km, single 200–500 kt warhead), first underwater launch October 12, 1982 from Golf-class SSB, deployed on the single Type 092 Xia-class SSBN (Pennant 406) (FAS Chinese Nuclear Forces 2019).
  - **JL-2 (CSS-N-14):** Three-stage solid-fuel intercontinental SLBM derived from DF-31 (range 7,200+ km, 12 tubes per Type 094 SSBN) (CSIS JL-2 Profile).
  - **JL-3 (CSS-N-20):** Third-generation three-stage solid-fuel MIRV SLBM (range 10,000+ km) refitted onto all six Type 094 SSBNs and slated for the Type 096 SSBN (FAS Nuclear Notebook 2025; CSIS ChinaPower Nuclear Modernization).
- **Deterministic Causal Chain:** Initiated in 1967 under Project 09 to establish a survivable sea-based second-strike leg, the Julang program overcame underwater cold-gas ejection, dynamic sea-state fire-control alignment, and composite motor casing hurdles over five decades, transitioning the PLAN in 2022–2025 from regional bastion patrols into true intercontinental littoral deterrence from the South China Sea (FAS Nuclear Notebook 2025; CSIS ChinaPower Nuclear Modernization).

121. JL-2 (巨浪-2 / CSS-N-14 — Submarine-Launched Ballistic Missile)

- **Institutional & Organizational Designation:** Julang-2 (JL-2, US DoD designation CSS-N-14), three-stage solid-propellant submarine-launched intercontinental ballistic missile (CSIS JL-2 Profile; FAS Chinese Nuclear Forces 2019).
- **Headquarters & Geolocation:** Manufactured by CASC CALT (1st Academy) in Beijing; deployed at Yulin Naval Base (Longposan), Sanya, Hainan Island (`18.2108°N, 109.6844°E`) aboard PLAN Type 094 Jin-class SSBNs (FAS Chinese Nuclear Forces 2019).
- **Command Hierarchy & Named Leadership:** Operated by the PLAN 42nd Submarine Flotilla / Second Submarine Base at Yulin under Southern Theater Command Navy administrative support and direct CMC JOCC nuclear command (FAS Chinese Nuclear Forces 2019; CASI NC3 Report 2024).
- **Engineering-Technical & Operational Architecture:** Navalized variant of the DF-31 ICBM; three-stage solid-propellant rocket motor, length 13.0 m, diameter 2.0 m, launch mass 42,000 kg, payload 1,050–2,800 kg, operational range 7,200 km (up to 8,000–9,000 km in extended estimates), CEP 150–300 m (inertial + stellar + Beidou-3), armed with a single 1 Mt thermonuclear warhead (or 3 smaller MIRVs) (CSIS JL-2 Profile; FAS Chinese Nuclear Forces 2019).
- **Deterministic Causal Chain:** Developed jointly with the DF-31 starting in January 1985, flight-tested from the Type 031 Golf SSB and Type 094 SSBN between 2001 and 2015, and deployed on operational deterrence patrols beginning in December 2015, the JL-2 enabled Type 094 SSBNs in the Bohai/South China Sea to hold Alaska, Guam, Hawaii, Russia, and India at risk, while requiring transit past the First Island Chain into the central Pacific to reach the contiguous US—a geometric constraint that directly drove the development of the 10,000+ km JL-3 (FAS Chinese Nuclear Forces 2019; CSIS ChinaPower Nuclear Modernization).

122. JL-3 (巨浪-3 / CSS-N-20 — MIRV Intercontinental SLBM)

- **Institutional & Organizational Designation:** Julang-3 (JL-3, US DoD designation CSS-N-20), third-generation three-stage solid-propellant MIRV intercontinental submarine-launched ballistic missile (FAS Nuclear Notebook 2025; CSIS ChinaPower Nuclear Modernization).
- **Headquarters & Geolocation:** Developed by CASC First Academy (CALT) and AASPT (4th Academy); test-launched from the Type 032 (Qing-class) test submarine at Dalian (`38.8180°N, 121.4950°E`) and Type 094A SSBNs in the Bohai Sea and South China Sea; deployed at Yulin Naval Base, Hainan (`18.2108°N, 109.6844°E`) (FAS Nuclear Notebook 2025; CSIS ChinaPower Nuclear Modernization).
- **Command Hierarchy & Named Leadership:** Operated by the PLAN Submarine Force at Yulin Naval Base under direct CMC JOCC nuclear command via VLF/ELF shore transmitters and satellite relays (CASI NC3 Report 2024).
- **Engineering-Technical & Operational Architecture:** Three-stage high-energy NEPE solid-propellant SLBM utilizing carbon-fiber composite casings and a drag-reducing aerospike nose cap to fit within the existing 12-tube missile compartment of the Type 094/094A SSBN; operational range ~10,000+ km, astro-inertial + Beidou-3 guidance, capable of delivering multiple independently targetable reentry vehicles (up to 3 MIRVs using CAEP's `#535` or `#5x5` warhead design) (FAS Nuclear Notebook 2025; FAS Nuclear Notebook 2026; CSIS ChinaPower Nuclear Modernization).
- **Deterministic Causal Chain:** Flight-tested in November 2018, June 2019, and December 2019 in the Bohai Sea, the JL-3 was confirmed by US Pacific Fleet Commander Adm. Samuel Paparo in November 2022 and the 2023–2025 DoD CMPRs to be operationally refitted onto China's six Type 094 SSBNs, enabling the PLAN for the first time to hold the northwestern and continental United States at risk from protected bastion waters inside the South China Sea and Bohai Sea without transiting choke points in the First Island Chain (FAS Nuclear Notebook 2025; CSIS ChinaPower Nuclear Modernization).

123. Type 094 Jin-Class (094型战略核潜艇 — SSBN Fleet, Yulin Naval Base, Hainan)

- **Institutional & Organizational Designation:** Type 094 / Type 094A (NATO reporting name: Jin-class) second-generation nuclear-powered ballistic missile submarine (SSBN), designed by the 719th Research Institute (Wuhan Second Ship Design and Research Institute) of China State Shipbuilding Corporation (CSSC) and built at Bohai Shipbuilding Heavy Industry Co. in Huludao (FAS Chinese Nuclear Forces 2019; FAS Nuclear Notebook 2025).
- **Headquarters & Geolocation:** Homeported at Yulin Naval Base (Longposan / 第二潜艇基地), Yalong Bay, Sanya, Hainan Province (`18.2108°N, 109.6844°E`), featuring a hardened subterranean submarine cavern entrance at `18.2044°N, 109.6922°E` and six concrete finger piers (FAS Chinese Nuclear Forces 2019; FAS Nuclear Notebook 2025).
- **Command Hierarchy & Named Leadership:** Designed under Chief Designer Zhang Jinlin (张金麟, 719th Institute Academician); operated by the PLAN Second Submarine Base at Yulin under CMC JOCC strategic nuclear authority (FAS Nuclear Notebook 2025). Six operational hulls in service (including *Changzheng 10* [411], *Changzheng 11* [412], *Changzheng 12* [413], *Changzheng 13* [414], *Changzheng 18* [421, commissioned April 23, 2021 by Xi Jinping], and [422]), with additional improved Type 094 hulls observed under construction in 2024–2026 (SIPRI Chinese Nuclear Forces; FAS Nuclear Notebook 2026).
- **Engineering-Technical & Operational Architecture:** Length 137 m, beam 11.8 m, submerged displacement ~11,000 tonnes, powered by a single pressurized water reactor (PWR) driving steam turbines and a 7-bladed skewed propeller; armed with 12 vertical SLBM launch tubes housed in a pronounced dorsal "turtleback" missile compartment behind the sail (carrying 12 JL-2 or JL-3 SLBMs per boat, totaling 72 SLBM tubes across the

6-boat fleet) and six 533 mm torpedo tubes (FAS Chinese Nuclear Forces 2019; CSIS ChinaPower Nuclear Modernization). Type 094A variants incorporate a blended, windowless, filleted sail and hydroacoustic hull refinements to reduce radiated flow noise (FAS Nuclear Notebook 2025).

- **Deterministic Causal Chain:** Constructed at Huludao starting in 1999 to replace the single unreliable Type 092 Xia-class SSBN, the Type 094 fleet expanded to six boats by 2021 and transitioned in 2022–2025 to continuous at-sea nuclear deterrence patrols ("fully armed patrols") out of the Longposan underground submarine tunnel complex in Hainan armed with JL-3 SLBMs (FAS Nuclear Notebook 2025; CSIS ChinaPower Nuclear Modernization).

124. Type 096 Tang-Class (096型战略核潜艇 — Next-Generation SSBN, Bohai Shipyard Huludao)

- **Institutional & Organizational Designation:** Type 096 (NATO reporting name: Tang-class) third-generation nuclear-powered ballistic missile submarine, designed by the CSSC 719th Research Institute (Wuhan) and constructed at Bohai Shipbuilding Heavy Industry Co. (BSHIC / 渤海造船厂集团有限公司), Huludao, Liaoning Province (FAS Chinese Nuclear Forces 2019; FAS Nuclear Notebook 2025; FAS Nuclear Notebook 2026).
- **Headquarters & Geolocation:** Construction complex at Bohai Shipyard, Longgang District, Huludao, Liaoning Province (`40.7186°N, 120.9917°E`), where a massive new modular indoor assembly hall (capable of building multiple nuclear submarines simultaneously under cover) and drydock basin were completed between 2017 and 2021 (FAS Chinese Nuclear Forces 2019; FAS Nuclear Notebook 2025).
- **Command Hierarchy & Named Leadership:** Engineered by CSSC 719th Institute and Bohai Shipyard under PLAN Armament Department and CMC Equipment Development Department supervision; slated for operational assignment to PLAN submarine bases at Yulin (Hainan) and Jianggezhuang (Qingdao) (FAS Chinese Nuclear Forces 2019; FAS Nuclear Notebook 2026).
- **Engineering-Technical & Operational Architecture:** Wide-beam pressurized hull design eliminating or minimizing the high-drag dorsal "turtleback" of the Type 094; equipped with raft-mounted vibration isolation, integrated shaftless/pump-jet propulsor, anechoic hull tiling, and a high-output pressurized water reactor; armed with 16 to 24 vertical launch tubes for the JL-3 SLBM or a follow-on longer-range MIRVed SLBM (FAS Nuclear Notebook 2025; FAS Nuclear Notebook 2026).
- **Deterministic Causal Chain:** Because the Type 094 SSBN radiates acoustic signatures higher than Soviet Delta III-class submarines (making it vulnerable to US and allied anti-submarine warfare in open ocean), China expanded the Huludao modular construction halls in 2019–2022 to build the quieted Type 096 SSBN, which DoD assesses will operate concurrently with the Type 094 across a ~40-year service life to field a combined fleet of 8 to 10+ SSBNs (FAS Chinese Nuclear Forces 2019; 2020 DoD CMPR).

125. H-6N (轰-6N — Xi'an Air-Refuelable Nuclear Bomber Carrying ALBM KD-21 / CH-AS-X-13)

- **Institutional & Organizational Designation:** Xi'an H-6N (US DoD designation B-6N, nicknamed "Leishen" / Thunder God), strategic long-range air-to-air refuelable nuclear-capable bomber carrying the KD-21 (US intelligence designation CH-AS-X-13) air-launched ballistic missile (ALBM) (FAS Chinese Nuclear Forces 2019; SIPRI Chinese Nuclear Forces; FAS Nuclear Notebook 2025).
- **Headquarters & Geolocation:** Manufactured by Xi'an Aircraft Industrial Corporation (XAC / AVIC Xi'an) at Yanliang District, Xi'an, Shaanxi (`34.6440°N, 109.2430°E`); operationally fielded with the PLAAF 106th Bomber Brigade at Neixiang Air Base, Nanyang, Henan Province (`33.0950°N, 111.7980°E`) and the 8th Bomber Division at Leiyang/Shaodong Air Base, Hunan Province (FAS Nuclear Notebook 2025).
- **Command Hierarchy & Named Leadership:** Operated by PLAAF strategic bomber brigades under Central Theater Command Air Force administrative oversight and direct CMC JOCC nuclear command (SIPRI Chinese Nuclear Forces; CASI NC3 Report 2024).
- **Engineering-Technical & Operational Architecture:** Twin-engine strategic bomber powered by two Soloviev D-30KP-2 or indigenous WS-18 turbofans; incorporates a fixed nose-mounted aerial refueling probe (extending unrefueled combat radius from ~3,500 km to >5,000–6,000+ km with YY-20A tanker refueling) and replaces the internal bomb bay with a semi-recessed ventral fuselage cradle (SIPRI Chinese Nuclear Forces; CSIS ChinaPower Nuclear Modernization). The ventral cradle carries a single large air-launched ballistic missile—designated CH-AS-X-13 by DIA/DoD (an air-launched derivative of the DF-21 MRBM with a biconic MaRV or HGV payload and ~3,000 km stand-off range, assessed as nuclear-capable including for low-yield `<10 kt` precision nuclear strikes) or the smaller KD-21 (2PZD-21) ALBM displayed at Zhuhai Airshow 2022/2024 (FAS Chinese Nuclear Forces 2019; FAS Nuclear Notebook 2026).
- **Deterministic Causal Chain:** After the PLAAF was formally reassigned a strategic nuclear mission in 2018, China flight-tested the CH-AS-X-13 nuclear ALBM in December 2016 and January 2018, unveiled the H-6N at the October 1, 2019 Beijing Parade, and operationally fielded the H-6N at Neixiang Air Base in 2020—formally completing China's air-land-sea nuclear triad ahead of the future H-20 stealth bomber (FAS Chinese Nuclear Forces 2019; SIPRI Chinese Nuclear Forces; CSIS ChinaPower Nuclear Modernization).

126. PLA Early Warning Radar & Space Network (Launch-on-Warning / EWCS Transition, TJS / Huoyan-1 Satellites, Base 37)

- **Institutional & Organizational Designation:** PLA Integrated Space- and Ground-Based Strategic Early Warning Network, operated by PLA Aerospace Force (PLAASF, formerly PLASSF Space Systems Department) Base 37 (Unit 32035, "Early-Warning Base") and Base 26 (Xi'an Satellite TT&C Center, Unit 63750) in coordination with PLAAF radar units (CASI Base 37 Report; FAS Nuclear Weapons Archives).
- **Headquarters & Geolocation:** Base 37 Headquarters and Orbital Data Fusion Center at Lintong District, Xi'an, Shaanxi Province (`34.3660°N, 109.2140°E`), adjacent to Base 26 (Xi'an Satellite Control Center at `34.2480°N, 109.0060°E`), with subordinate units in Chongqing (Units 32036, 32038), Yao'an, Yunnan (Unit 32037, `25.5270°N, 101.1810°E`), Henan/Beijing/Hubei (Unit 32039), and Jinan, Shandong (Unit 95921) (CASI Base 37 Report).
- **Command Hierarchy & Named Leadership:** Following the April 2024 disestablishment of the Strategic Support Force (SSF), Base 37 (MUCD 32035) reports to the PLA Aerospace Force (PLAASF) and fuses space and radar early-warning tracks directly into the CMC Joint Operations Command Center in Beijing (CASI Base 37 Report; China's Nuclear Early-Warning Architecture).
- **Engineering-Technical & Operational Architecture:** Implements "dual-phenomenology" missile warning combining:
  1. **Space-Based Infrared Tier:** Geostationary Tongxun Jishu Shiyan (TJS-1, TJS-2, TJS-4, TJS-5, TJS-6, TJS-7, TJS-9, TJS-10, TJS-11+, designated **Huoyan-1** [火眼-1] early-warning infrared satellites by the DoD CMPR), manufactured by CASC Eighth Academy (SAST) and Fifth Academy (CAST) and launched from Xichang, capable of detecting an adversary ICBM boost-phase infrared plume within **90 seconds of launch** and transmitting an early-warning alert via Tianlian relay satellites to the command center within **3 to 4 minutes** (FAS Nuclear Weapons Archives; Erickson DoD CMPR Synthesis).

2. **Ground-Based Radar Tier:** At least seven Large Phased-Array Radar (LPAR) stations (Xuanhua, Korla, Jinan/Yiyuan, Hangzhou/Lin'an, Pingliang, Jiamusi/Shuangyashan, Sheyang, plus Hui'an and Yunnan nodes) that track midcourse RVs and corroborate TJS/Huoyan-1 alerts (CASI Base 37 Report; Defense News LPAR Analysis).

- **Deterministic Causal Chain:** Augmented by a 2019 Russian-Chinese bilateral agreement to develop early-warning software and architecture (announced by President Vladimir Putin in October 2019), the integration of TJS/Huoyan-1 satellites and Base 37 LPAR fusion enabled the PLARF to transition into its "Early-Warning Counterstrike" (预警反击, EWCS / Launch-on-Warning) posture—wherein T+90s satellite detection and T+3–4m JOCC alert fusion allow the CMC Chairman to authorize retaliatory launches from >100 solid-fuel DF-31/DF-41 silos before inbound warhead detonation (FAS Nuclear Weapons Archives; CSIS ChinaPower Nuclear Modernization).

127. Huian LPAR (惠安大型相控阵雷达 — Fujian Province Large Phased-Array Radar)

- **Institutional & Organizational Designation:** Hui'an Large Phased-Array Radar and Electronic Warfare Complex (惠安大型相控阵雷达站, joined by the Southeast Coastal LPAR network in Lin'an/Hangzhou and Longyan/Fujian), coastal ballistic-missile early-warning, space-tracking, and electronic warfare array (CASI China's Ground Segment; Jane's / Rediff Huian LPAR Analysis).
- **Headquarters & Geolocation:** Located in Hui'an County, Quanzhou Prefecture, Fujian Province (`24.9750°N, 118.8850°E`, directly opposite the Taiwan Strait, paired in the southeastern early-warning sector with the Hangzhou/Lin'an LPAR at `30.286567°N, 119.128608°E`) (CASI China's Ground Segment; CASI Base 37 Report).
- **Command Hierarchy & Named Leadership:** Engineered by China Electronics Technology Group Corporation (CETC) 14th Research Institute (Nanjing Research Institute of Electronic Technology); operated within the Eastern Theater Command / PLAASF Base 37 early-warning and electronic warfare network (CASI China's Ground Segment; CASI Base 37 Report).
- **Engineering-Technical & Operational Architecture:** Solid-state UHF/P-band active electronically scanned array (AESA) housed in a monumental concrete turret with a near-vertical array face oriented on an **azimuth boresight of ~115°–130° (East-Southeast)** across a 120-degree electronic scan sector (±60° from boresight) and 3°–85° elevation coverage, providing 3,000–5,500 km range tracking across Taiwan, the Philippine Sea, Guam, and the Western Pacific (Jane's / Rediff Huian LPAR Analysis; Defense News LPAR Analysis).
- **Deterministic Causal Chain:** Identified in Airbus Defence and Space commercial satellite imagery by 2014 at the Hui'an EW complex, the Hui'an LPAR was constructed with a near-vertical array face both to track ballistic missile launches, SLBM trajectories from the Philippine Sea, and LEO satellites, and to generate high-power UHF co-channel electronic interference capable of neutralizing Taiwan's AN/FPS-115 Pave Paws early-warning radar on Leshan Mountain (`24.4980°N, 121.1140°E`) (Jane's / Rediff Huian LPAR Analysis; CASI China's Ground Segment).

128. Shuangyashan / Jiamusi LPAR (双鸭山/佳木斯大型相控阵雷达 — Heilongjiang Province LPAR)

- **Institutional & Organizational Designation:** Jiamusi / Shuangyashan Monitoring and Early Warning Station LPAR (佳木斯/双鸭山大型相控阵雷达预警站), northeast strategic ballistic-missile early-warning and space-tracking radar installation (CASI Base 37 Report; Economic Times CASI LPAR Analysis).
- **Headquarters & Geolocation:** Located on the mountainous ridge straddling Huanan County (Jiamusi Prefecture) and Jixian County (Shuangyashan Prefecture), Heilongjiang Province at exact coordinates **`46.527581°N, 130.756423°E`** (CASI Base 37 Report).
- **Command Hierarchy & Named Leadership:** Engineered by CETC 14th Research Institute; operated by a subordinate monitoring and early-warning station of PLAASF Base 37 (Unit 32035) in co-location with a Chinese Academy of Sciences (CAS) deep-space tracking station; showcased during CMC Chairman Xi Jinping's Lunar New Year military address with joint PLA Army, Navy, Air Force, and Aerospace Force personnel (CASI Base 37 Report; Economic Times CASI LPAR Analysis).
- **Engineering-Technical & Operational Architecture:** Sloped-face (20° tilt) UHF/P-band solid-state Large Phased-Array Radar oriented on a **northeasterly/north-northeasterly boresight azimuth (~35°–55° NE, covering a 120° azimuth sector from ~335° NNW to ~115° ESE)** across 3°–85° elevation and up to 5,500 km detection range (CASI Base 37 Report; Defense News LPAR Analysis; China's Nuclear Early-Warning Architecture).
- **Deterministic Causal Chain:** Initially constructed before 2011 and extensively renovated between late 2020 and early 2021 with standardized Base 37 support garages and underground fuel infrastructure, the Shuangyashan/Jiamusi LPAR provides primary midcourse tracking of US Minuteman III ICBM trajectories approaching over the North Pacific/Bering Sea/Arctic corridor, US Ohio-class Trident II D5 SLBM launches from the northern Pacific, and Russian Far East / Alaskan aerospace activity (CASI Base 37 Report; China's Nuclear Early-Warning Architecture).

129. Xinjiang LPAR (库尔勒/奇台大型相控阵雷达 — Korla / Qitai LPAR, Xinjiang)

- **Institutional & Organizational Designation:** Korla Large Phased-Array Radar Station (库尔勒大型相控阵雷达站, operating in conjunction with the Xinjiang Western Strategic Test & Tracking Complex at Korla/Bohu and Qitai), strategic ballistic-missile tracking, midcourse ABM fire-control cueing, and western-azimuth early-warning radar (CASI Base 37 Report; China's Nuclear Early-Warning Architecture).
- **Headquarters & Geolocation:** Located northeast of Korla in Bayingolin Mongol Autonomous Prefecture, Xinjiang Uyghur Autonomous Region at exact coordinates **`41.641194°N, 86.236749°E`** (near the Bohu directed-energy laser facility and Korla Missile Test Complex) (CASI Base 37 Report).
- **Command Hierarchy & Named Leadership:** Constructed under the former PLA General Armaments Department (GAD) Base 21 / Korla Missile Test Complex and integrated with PLAASF / EDD missile instrumentation and early-warning networks (CASI Base 37 Report).
- **Engineering-Technical & Operational Architecture:** Sloped-face UHF/P-band solid-state Large Phased-Array Radar with a 120-degree electronic scan sector (±60° from boresight), oriented along a **western/southwestern and range-instrumentation azimuth (~210°–290° SW/W)** covering Central Asia, the Indian subcontinent (Agni-V ICBM corridors), Russian central missile ranges, and the terminal impact corridors for PLARF ICBM/IRBM test shots fired from Taiyuan/Wuzhai and Jilantai into the Taklamakan Desert, as well as cueing SC-19 / HQ-19 / DN-2/DN-3 midcourse kinetic-kill anti-ballistic missile (ABM) and anti-satellite (ASAT) interceptors launched from Korla (CASI Base 37 Report; Eurasian Times Korla LPAR Analysis; China's Nuclear Early-Warning Architecture).

- **Deterministic Causal Chain:** Placed under construction in 2004 and completed by 2009 as China's second LPAR (after Xuanhua), the Korla LPAR provided the precision midcourse target-tracking vectors for China's January 2010, January 2013, July 2014, February 2018, February 2021, June 2022, and April 2023 midcourse ballistic-missile intercept tests while simultaneously tracking all PLARF MIRV and HGV test flights into the western Xinjiang impact ranges (CASI Base 37 Report; China's Nuclear Early-Warning Architecture).

130. Beidou-3 (北斗三号 — BDS-3 Military Short-Message NC3 & PNT Guidance Integration)

- **Institutional & Organizational Designation:** Beidou-3 Global Navigation Satellite System (BDS-3 / 北斗三号全球卫星导航系统), sovereign military Positioning, Navigation, Timing (PNT), and two-way Short-Message Communication (SMC / 短报文通信) constellation managed by the PLA Aerospace Force (formerly PLASSF) **35th Base** (战略支援部队35基地) and Satellite Positioning Main Station (卫星定位总站) under the CMC Joint Staff Department Battlefield Environment Support Bureau (战场环境保障局) (NDU China's Strategic Support Force; PLA BeiDou Integration Study).
- **Headquarters & Geolocation:** Master Control and Satellite Positioning Main Station in Haidian District, Beijing (`40.0720°N, 116.2750°E`) and Xi'an Satellite Control Center (Base 26, `34.2480°N, 109.0060°E`), backed by distributed terrestrial ground anchor stations across China (NDU China's Strategic Support Force; PLA BeiDou Integration Study).
- **Command Hierarchy & Named Leadership:** Engineered under Chief Designer Yang Changfeng (杨长风) and CASC Fifth Academy (CAST); military operational management executed by PLAASF Base 35 and the CMC JSD Battlefield Environment Support Bureau (NDU China's Strategic Support Force).
- **Engineering-Technical & Operational Architecture:** Completed in June 2020 (commissioned July 31, 2020 by Xi Jinping), the 35-satellite BDS-3 constellation comprises 24 Medium Earth Orbit (MEO, ~21,528 km), 3 Inclined Geosynchronous Orbit (IGSO, ~35,786 km), and 3+ Geostationary Earth Orbit (GEO, ~35,786 km) satellites (plus on-orbit spares) linked by Ka-band phased-array **Inter-Satellite Links (ISLs)** and ground anchor stations (Deciphering China's Military Space Program; PLA BeiDou Integration Study). Key military specifications:
  - **Encrypted Military PNT Accuracy:** 10 cm (0.1 m) 3D positioning accuracy on encrypted military B1A/B3A frequencies with Position Dilution of Precision (PDOP) `<1.4` for PLARF post-boost vehicles, MaRVs, HGVs, and cruise missiles (PLA BeiDou Integration Study).
  - **Two-Way Short Message Communication (SMC) NC3:** Unlike US GPS (which is receive-only), BDS-3 integrates L/S-band two-way encrypted short-message transceivers (supporting up to 1,000 Chinese characters / 14,000 bits per message and "group call" broadcast commands), allowing the CMC JOCC and PLARF Base Command Posts to transmit launch orders, target re-aiming coordinates, and receive real-time TEL geolocation/status reports from dispersed DF-31AG, DF-41, and DF-26 launchers in remote mountains or tunnels without terrestrial network coverage (PLA BeiDou Integration Study; CASI NC3 Report 2024).
  - **Autonomous Anti-ASAT Resilience:** Multi-objective ISL optimization maintains network connectivity `>3` (and up to 5–6 in super-frames), distributing ground anchor data across the entire constellation in 3 link hops using `<10%` of time-slot resources and sustaining autonomous orbit determination and clock synchronization for 60+ days even if terrestrial control stations are destroyed (PLA BeiDou Integration Study).
- **Deterministic Causal Chain:** Following the 1995–1996 Taiwan Strait Crisis (when PLA missile tests experienced GPS signal disruption), the CMC mandated total independence from US GPS; the 2020 completion of BDS-3 provided the PLARF with a jam-resistant, anti-ASAT space backbone that fuses 10 cm midcourse/terminal missile guidance with two-way encrypted nuclear/conventional C2 messaging (PLA BeiDou Integration Study).

131. ShenGuang (神光 — SG-I, SG-II, SG-III, SG-IV Inertial Confinement Fusion Laser Facilities at CAEP / SIOM)

- **Institutional & Organizational Designation:** ShenGuang ("Divine Light" / 神光) High-Power Laser Inertial Confinement Fusion (ICF) and Thermonuclear Weapons Physics Simulation Program, jointly developed and operated by the Research Center of Laser Fusion (RCLF / 8th Institute of CAEP, Mianyang) and the National Laboratory on High Power Laser and Physics at the Shanghai Institute of Optics and Fine Mechanics (SIOM / 中国科学院上海光学精密机械研究所, Chinese Academy of Sciences) (Petawatt Class Lasers Worldwide - Cambridge Core; CAEP Profile).
- **Headquarters & Geolocation:** SIOM facility (SG-I, SG-II, SG-II-U) at Jiading District, Shanghai (`31.3850°N, 121.2460°E`); CAEP Research Center of Laser Fusion (SG-III prototype, SG-III, and SG-IV megajoule facility) at Science City, Mianyang, Sichuan Province (`31.4820°N, 104.7760°E`) (Petawatt Class Lasers Worldwide - Cambridge Core; CAEP Profile).
- **Command Hierarchy & Named Leadership:** Founded by Academicians Wang Ganchang (王淦昌), Deng Ximing (邓锡铭), Fan Dianyuan (范滇元), He Xiantu (贺贤土), Zhang Jie (张杰), and Zhu Jianqiang (朱健强) under the National High-Tech R&D Program (863 Program) ICF Subject Committee and CAEP oversight (Petawatt Class Lasers Worldwide - Cambridge Core; CAEP Profile).
- **Engineering-Technical & Operational Architecture:** Four-generation Nd:phosphate glass high-energy laser cascade replicating thermonuclear secondary radiation-implosion physics, equation-of-state (EOS) shock compression, and X-ray opacity at stellar temperatures and pressures:
  - **ShenGuang-I (SG-I):** 2-beam Nd:glass laser at SIOM (commissioned 1985–1986), delivering 1.6 kJ at 1,053 nm (1 ns pulse, ~2 TW peak power) for initial hohlraum indirect-drive X-ray conversion and EOS experiments (Petawatt Class Lasers Worldwide - Cambridge Core).
  - **ShenGuang-II (SG-II) & SG-II-U:** 8-beam Nd:glass laser at SIOM (commissioned 2000–2001) delivering 6 kJ IR (1,053 nm) or 2.4 kJ UV (351 nm, 3ω); augmented in 2005 with a 9th beam (4.5 kJ) converted into a petawatt (1 PW) picosecond fast-ignition/radiography beamline, and expanded via the **SG-II Upgrade (SG-II-U)** adding a separate 8-beam 24 kJ (351 nm, 3 ns) nanosecond driver coupled with a multi-kJ petawatt picosecond beam (Petawatt Class Lasers Worldwide - Cambridge Core).
  - **ShenGuang-III (SG-III):** Preceded by the 8-beam SG-III Technical Integration Line (TIL, ~15 kJ) in Mianyang (2007), the full **48-beam SG-III laser facility** at CAEP Mianyang (commissioned 2015) delivers **180 kJ of frequency-tripled 351 nm UV light** (3 ns pulse, ~60 TW peak power) into a spherical target chamber for indirect-drive hohlraum implosion and weapons-physics code validation (CAEP Profile; Petawatt Class Lasers Worldwide - Cambridge Core).
  - **ShenGuang-IV (SG-IV / Megajoule Laser Facility):** Mega-scale ignition facility at CAEP Science City in Mianyang (`31.4820°N, 104.7760°E`, constructed 2020–2026, larger in footprint than the US LLNL National Ignition Facility), designed to deliver **1.8 to 2.0+ Megajoules (MJ)** of 351 nm UV laser energy to achieve high-gain thermonuclear ignition and full-scale secondary weapons-physics stewardship (CAEP Profile; Petawatt Class Lasers Worldwide - Cambridge Core).

- **Deterministic Causal Chain:** When China signed the Comprehensive Nuclear-Test-Ban Treaty (CTBT) in September 1996 after its 45th nuclear test, CAEP lost access to full-yield underground detonations at Lop Nur; the progression from SG-I/SG-II to the 180 kJ SG-III (2015) and the ~2 MJ SG-IV complex at Mianyang provides CAEP weapon designers with laboratory-scale thermonuclear radiation-hydrodynamics data needed to certify new miniaturized MIRV warheads (`#535`, `#5x5`) and variable-yield primaries without full-scale explosive nuclear tests (CAEP Profile; FAS Nuclear Notebook 2026).

Section VI: Democratic People's Republic of Korea (DPRK) State Affairs Commission, Missile General Bureau, Strategic Delivery Systems, and Nuclear Weapons Complex (Entities 132–152)

132. Missile General Bureau (미싸일총국 / DPRK Missile Administration — Established 2023, Kim Jong Sik / Jang Chang Ha)

- **Institutional & Organizational Designation:** Missile General Bureau (미싸일총국, rendered in DPRK English state media since 2024–2026 as the **DPRK Missile Administration** / **Missile Administration of the DPRK**), supreme military-industrial and operational missile test-launch and command directorate of the DPRK (DPRK Missile Administration Profile; NK Leadership Watch MGB Photo-Op).
- **Headquarters & Geolocation:** Headquartered in Pyongyang (`39.0392°N, 125.7625°E`, integrated with the Workers' Party of Korea [WPK] Munitions Industry Department complex in Central District and the Academy of National Defense Science in Ryongsong District, Pyongyang at `39.1160°N, 125.8050°E`) (DPRK Missile Administration Profile; Jang Chang-ha Profile).
- **Command Hierarchy & Named Leadership:** Directed by **General (Taejang) Jang Chang Ha** (장창하, born January 10, 1964, former President of the Academy of National Defense Science, promoted to four-star General on November 26, 2022, confirmed as **General Director of the Missile General Bureau / Missile Administration**) alongside **Col. Gen. Kim Jong Sik** (김정식, First Vice Department Director of the WPK Central Committee Munitions Industry Department and member of the WPK Central Military Commission), reporting directly to Supreme Leader Kim Jong Un (often accompanied in 2023–2026 inspections by Kim Ju Ae) (Jang Chang-ha Profile; NK Leadership Watch MGB Photo-Op; Newscord KCNA Submarine Cruise Missile Report 2026).
- **Engineering-Technical & Operational Architecture:** Commands specialized "Red Flag Companies" (1st, 2nd Red Flag Companies for ICBM launches) and subordinate missile research institutes overseeing solid- and liquid-fuel ICBM development (Hwasong-15, Hwasong-17, Hwasong-18, Hwasong-19), hypersonic IRBMs (Hwasong-16B), tactical ballistic missiles (Hwasong-11A/KN-23, KN-24, KN-25), strategic cruise missiles (Hwasal-1, Hwasal-2, Pulhwasal-3-31), and the "Haekbangashoe" nuclear trigger C2 drills (DPRK Missile Administration Profile; FAS North Korean Nuclear Weapons 2024). Its official emblem displays a Hwasong-17 ICBM rising over a globe with the founding date `2016.4.30` and was publicly unveiled at the WPK Central Military Commission enlarged meeting on February 6–7, 2023 (DPRK Missile Administration Profile).
- **Deterministic Causal Chain:** To unify R&D fabrication (formerly dispersed across the Second Economic Committee and Academy of National Defense Science) with operational combat-readiness firing units (Korean People's Army Strategic Force), Kim Jong Un elevated the Missile General Bureau in late 2022 / February 2023 under Jang Chang Ha and Kim Jong Sik; since March 2023, the MGB has directly executed all Hwasong-17, Hwasong-18, and Hwasong-19 ICBM launches, simulated nuclear airburst drills, and the October 14, 2026 mixed-mode VLS/torpedo-tube submarine strategic cruise missile salvo in the East Sea (NK Leadership Watch MGB Photo-Op; Newscord KCNA Submarine Cruise Missile Report 2026).

133. State Affairs Commission of the DPRK (조선민주주의인민공화국 국무위원회 — SAC, Sept 2022 Nuclear Force Policy Law, "Haekbangashoe" Nuclear Trigger C2 System)

- **Institutional & Organizational Designation:** State Affairs Commission of the DPRK (SAC / 국무위원회, established by the June 2016 constitutional revision replacing the National Defense Commission), supreme national defense policy and nuclear command authority operating the **"Haekbangashoe"** (핵방아쇠, "Nuclear Trigger") integrated national nuclear weapons management and command-and-control system (FAS North Korean Nuclear Weapons 2024; Taylor & Francis DPRK Nuclear Doctrine Study 2024).
- **Headquarters & Geolocation:** SAC and WPK Central Committee Complex No. 1, Changgwang-dong, Central District, Pyongyang (`39.0145°N, 125.7442°E`), with hardened wartime leadership bunkers in Ryongsong District (`39.1240°N, 125.8120°E`) and Mount Chamo (`39.2900°N, 125.8600°E`) (FAS North Korean Nuclear Weapons 2024).
- **Command Hierarchy & Named Leadership:** Headed by President of the State Affairs Commission and Supreme Commander of the Armed Forces **Kim Jong Un** (김정은), First Vice President Choe Ryong Hae (최룡해), Vice President Pak Thae Song / Kim Tok Hun, assisted in the "State Nuclear Forces Command Organization" by WPK Central Military Commission Vice Chairman Pak Jong Chon (박정천), Minister of National Defense No Kwang Chol / Kang Sun Nam, Kim Jong Sik, and Jang Chang Ha (FAS North Korean Nuclear Weapons 2024; Taylor & Francis DPRK Nuclear Doctrine Study 2024).
- **Engineering-Technical & Operational Architecture:** Governed by the **Law on the State Policy on the Nuclear Forces** adopted by the 7th Session of the 14th Supreme People's Assembly on **September 8, 2022** (and enshrined in Article 58 of the DPRK Socialist Constitution in September 2023), which defines five statutory conditions for nuclear weapons employment, including preemptive first use if an imminent nuclear or non-nuclear decapitation attack against the state leadership or the National Nuclear Forces Command Organization is detected (FAS North Korean Nuclear Weapons 2024; Taylor & Francis DPRK Nuclear Doctrine Study 2024). Section 3 of the September 2022 Law mandates that if the command-and-control system over the state nuclear forces is placed in danger by hostile attack, **"a nuclear strike shall be launched automatically and immediately to destroy the hostile forces, including the starting point of the provocation and the command organization, according to an operation plan decided in advance"** (Taylor & Francis DPRK Nuclear Doctrine Study 2024). The **Haekbangashoe ("Nuclear Trigger")** C2 software/hardware architecture links the Supreme Commander's authorization terminals to tactical and strategic missile units nationwide (FAS North Korean Nuclear Weapons 2024).
- **Deterministic Causal Chain:** Updating the 2013 Nuclear Weapons State Law (which restricted launch strictly to a manual final order by the Supreme Commander), the September 8, 2022 SAC law codified automatic fail-deadly nuclear retaliation to deter US-ROK "Kill Chain" and "Korea Massive Punishment and Retaliation" (KMPR) decapitation plans; in March 2023 and April 22, 2024, the DPRK conducted comprehensive nationwide drills under the "Haekbangashoe" nuclear trigger system, transitioning units to nuclear counterattack posture and firing 600 mm KN-25 super-large MLRS salvos with simulated airburst warheads at 400–500 m altitude (FAS North Korean Nuclear Weapons 2024; Taylor & Francis DPRK Nuclear Doctrine Study 2024).

134. Hwasong-14 (화성-14형 / KN-20 — Two-Stage Liquid-Propellant ICBM)

- **Institutional & Organizational Designation:** Hwasong-14 (《화성-14》형, US/DoD designation KN-20), North Korea's first flight-tested intercontinental ballistic missile capable of reaching mainland North America (CSIS Hwasong-14 Profile; FAS North Korean Nuclear Weapons 2024).
- **Headquarters & Geolocation:** Assembled at the Sanum-dong Research and Development Facility in Pyongyang (`39.0997°N, 125.8147°E`) with liquid engines produced at the Thaesong Machine Factory (`38.9480°N, 125.5650°E`); flight-tested from Panghyon Aircraft Factory site in Kusong, North Pyongan Province (`39.9275°N, 125.2194°E`) and Mupyong-ri, Chagang Province (`40.6125°N, 126.4339°E`) (CSIS Hwasong-14 Profile; 38 North Hwasong-19 Analysis).
- **Command Hierarchy & Named Leadership:** Developed by the Academy of National Defense Science under Jang Chang Ha and Kim Jong Sik and operated by the KPA Strategic Force (CSIS Hwasong-14 Profile).
- **Engineering-Technical & Operational Architecture:** Two-stage storable liquid-propellant (UDMH / N2O4) ICBM; length 19.5 m, diameter 1.7 m, launch weight ~33,000–37,000 kg, operational range 10,000–10,400 km (initially assessed at 7,000–9,500 km after the first test, extended to 10,400 km after the second test), powered in the first stage by a single high-thrust "Paektusan" (RD-250-derived) liquid rocket engine flanked by four gimballing vernier thrusters, and a second stage with one main nozzle and four verniers (CSIS Hwasong-14 Profile). Transported on an 8-axle WS51200 road-mobile transporter-erector (TE) that erects the missile onto a detachable fixed concrete/steel ground firing table (CSIS Hwasong-14 Profile).
- **Deterministic Causal Chain:** Building on the Hwasong-12 (KN-17) IRBM booster tested in May 2017, the DPRK conducted two lofted flight tests of the Hwasong-14 in July 2017: Test 1 on July 4, 2017 from Panghyon (flew 39 minutes, apogee 2,803 km, distance 930 km into the Sea of Japan) and Test 2 on July 28, 2017 from Mupyong-ri (flew 47 minutes, apogee 3,724.9 km, distance 998 km)—proving multi-stage liquid staging at intercontinental velocities and paving the way for the heavier Hwasong-15 four months later (CSIS Hwasong-14 Profile).

135. Hwasong-15 (화성-15형 / KN-22 — Two-Stage Liquid-Propellant ICBM)

- **Institutional & Organizational Designation:** Hwasong-15 (《화성-15》형, US/DoD designation KN-22), heavy two-stage storable liquid-propellant road-mobile intercontinental ballistic missile (CSIS Hwasong-15 Profile; FAS North Korean Nuclear Weapons 2024).
- **Headquarters & Geolocation:** Assembled at Sanum-dong (`39.0997°N, 125.8147°E`) and the March 16 Factory in Pyongsong (`39.2660°N, 125.8540°E`, 9-axle TEL modification); flight-tested from Pyongsong (`39.2550°N, 125.8850°E`) and Sunan International Airport, Pyongyang (`39.2241°N, 125.6700°E`) (CSIS Hwasong-15 Profile).
- **Command Hierarchy & Named Leadership:** Operated by the 1st Red Flag Company of the Missile General Bureau / KPA Strategic Force under Jang Chang Ha and Kim Jong Sik (CSIS Hwasong-15 Profile; DPRK Missile Administration Profile).
- **Engineering-Technical & Operational Architecture:** Two-stage liquid-propellant (UDMH / N2O4) ICBM; length 21.0–22.5 m, diameter 2.0–2.4 m, launch mass ~72,000 kg, payload capacity up to 1,000 kg, maximum operational range **13,000 km** (sufficient to reach the entire continental United States including Washington, D.C. and New York) (CSIS Hwasong-15 Profile; FAS North Korean Nuclear Weapons 2024). Replaces the single-chamber vernier-steered engine of the Hwasong-14 with a twin-chamber gimbaled Paektusan (RD-250-type) first-stage engine cluster (~80 tonnes thrust) and a blunt, wide payload shroud, carried on a 9-axle (18x18) modified WS51200 TEL (CSIS Hwasong-15 Profile).
- **Deterministic Causal Chain:** First flight-tested on November 28–29, 2017 from Pyongsong (flying 53 minutes to an apogee of 4,475 km and a distance of 950 km into the Sea of Japan), the Hwasong-15 prompted Kim Jong Un to declare the "completion of the state nuclear force"; it was subsequently launched in a surprise readiness drill on February 18, 2023 by the 1st Red Flag Company from Sunan Airport (flying 66 minutes 55 seconds to an apogee of 5,768.5 km and distance of 989 km) to validate operational reliability (CSIS Hwasong-15 Profile; FAS North Korean Nuclear Weapons 2024).

136. Hwasong-17 (화성-17형 / KN-27 / KN-28 — Heavy Liquid ICBM, 11-Axle TEL)

- **Institutional & Organizational Designation:** Hwasong-17 (《화성포-17》형, US/DoD designation KN-27 / KN-28), ultra-heavy two-stage liquid-propellant road-mobile intercontinental ballistic missile—the largest road-mobile liquid-fueled ICBM in the world (CSIS Hwasong-17 Profile; FAS North Korean Nuclear Weapons 2024).
- **Headquarters & Geolocation:** Manufactured at Sanum-dong and the March 16 Factory (Pyongsong), with liquid engine production at the Thaesong Machine Factory; flight-tested from Sunan International Airport, Pyongyang (`39.2241°N, 125.6700°E`) and garrisoned at hardened underground TEL facilities such as Sino-ri / Sangnam-ri (CSIS Hwasong-17 Profile; 38 North Hwasong-19 Analysis).
- **Command Hierarchy & Named Leadership:** Directed by General Jang Chang Ha (awarded Hero of the DPRK and promoted to four-star General on November 26, 2022 following the November 18, 2022 Hwasong-17 test) and Kim Jong Sik under the Missile General Bureau (Jang Chang-ha Profile; NK Leadership Watch MGB Photo-Op).
- **Engineering-Technical & Operational Architecture:** Two-stage storable liquid-propellant (UDMH / N2O4) ICBM; length 24.0–26.0 m, diameter 2.4–2.5 m, launch mass ~80,000–110,000 kg, estimated maximum range **15,000+ km** (CSIS Hwasong-17 Profile). Powered in the first stage by a four-nozzle (two twin-chamber RD-250/Paektusan) engine cluster (~160 tonnes thrust) and carried on an indigenous **11-axle (22x22) transporter-erector-launcher (TEL)** from which the missile fires directly off the rear erector table without a separate ground pedestal (CSIS Hwasong-17 Profile). Specifically sized with a 2.5 m diameter bus to accommodate multiple independently targetable reentry vehicles (MIRVs) and heavy penetration aids (CSIS Hwasong-17 Profile; 38 North Hwasong-19 Analysis).
- **Deterministic Causal Chain:** Unveiled at the October 10, 2020 WPK 75th Anniversary Parade, the Hwasong-17 completed full-scale lofted flight tests on March 24, 2022 (flew 67.5 minutes, apogee 6,248.5 km, range 1,090 km—disputed by US/ROK as a modified Hwasong-15 after a March 16 failure), November 18, 2022 (confirmed Hwasong-17, flew 69 minutes, apogee 6,040.9 km, range 999.2 km, attended by Kim Jong Un and Kim Ju Ae), and March 16, 2023 (flew 69.1 minutes, apogee 6,045 km, range 1,000.2 km), with 10–12 canisterized/open Hwasong-17 TELs paraded on February 8, 2023 (CSIS Hwasong-17 Profile). Its 4-nozzle liquid first stage also serves as the booster for the Chollima-1 space launch vehicle (38 North Chollima-1 Analysis).

137. Hwasong-18 (화성포-18형 — Three-Stage Solid-Fuel Road-Mobile ICBM)

- **Institutional & Organizational Designation:** Hwasong-18 (《화성포-18》형), North Korea's first operational three-stage solid-propellant road-mobile intercontinental ballistic missile (CSIS Hwasong-18 Profile; FAS North Korean Nuclear Weapons 2024).
- **Headquarters & Geolocation:** Solid rocket motors produced at the February 8 Vinalon Complex / Hamhung Chemical Material Institute (`39.8550°N, 127.5680°E`) and Kanggye/Kusong solid-motor facilities; flight-tested from the Pyongyang eastern suburb launch estate near Kangdong (`39.0760°N, 125.9860°E`) (CSIS Hwasong-18 Profile).
- **Command Hierarchy & Named Leadership:** Operated by the 2nd Red Flag Company of the Missile General Bureau under General Director Jang Chang Ha and Kim Jong Sik (CSIS Hwasong-18 Profile; DPRK Missile Administration Profile).
- **Engineering-Technical & Operational Architecture:** Three-stage composite solid-propellant ICBM housed inside a sealed cylindrical cold-launch canister carried on a **9-axle (18x18) wheeled TEL** (CSIS Hwasong-18 Profile; 38 North Hwasong-19 Analysis). Length ~25 m, diameter ~2.0–2.1 m, launch mass ~55,000–60,000 kg, operational range **15,000+ km** (CSIS Hwasong-18 Profile). Uses a gas generator to cold-eject the missile vertically from its canister before first-stage ignition, with thrust-vector control and roll-control thrusters, plus stage-separation retro-motors and a deliberate pitch/flight-path change maneuver between Stage 1/2 and Stage 3 (CSIS Hwasong-18 Profile).
- **Deterministic Causal Chain:** Fulfilling the five-year solid-fuel ICBM directive issued by Kim Jong Un at the 8th WPK Congress in January 2021, the Hwasong-18 was paraded on February 8, 2023 and completed three successful flight tests in 2023: April 13, 2023 (maiden test), July 12, 2023 (flew 74 minutes 51 seconds to an apogee of 6,648.4 km and distance of 1,001.2 km), and December 18, 2023 (operational readiness drill by the 2nd Red Flag Company, flying 73 minutes 35 seconds to an apogee of 6,518.2 km and distance of 1,002.3 km), eliminating multi-hour liquid fueling signatures prior to launch (CSIS Hwasong-18 Profile; FAS North Korean Nuclear Weapons 2024).

138. Hwasong-19 (화성포-19형 — 11/12-Axle TEL Heavy Three-Stage Solid-Fuel MIRV ICBM Tested Oct 2024)

- **Institutional & Organizational Designation:** Hwasong-19 (《화성포-19》형), ultra-heavy three-stage solid-propellant canisterized road-mobile MIRV intercontinental ballistic missile (CSIS Hwasong-19 Profile; 38 North Hwasong-19 Analysis).
- **Headquarters & Geolocation:** Solid-propellant motors fabricated at the Hamhung / Kanggye solid-propellant complexes; TEL assembled at the March 16 Factory in Pyongsong; maiden flight test conducted from the Pyongyang / Kangdong launch site (`39.0760°N, 125.9860°E`) on October 31, 2024 (38 North Hwasong-19 Analysis; CSIS Hwasong-19 Profile).
- **Command Hierarchy & Named Leadership:** Launched by the Missile General Bureau (DPRK Missile Administration) under General Director Jang Chang Ha and Kim Jong Sik, personally supervised on site by Kim Jong Un and Kim Ju Ae on October 31, 2024 (38 North Hwasong-19 Analysis).
- **Engineering-Technical & Operational Architecture:** Cold-launched from a sealed canister mounted on an **11-axle (22x22) heavy wheeled TEL** (with a **12-axle TEL** chassis also revealed in state media inspection photos in September 2024) (38 North Hwasong-19 Analysis; CSIS Hwasong-19 Profile). At **28+ meters in length** and ~2.4–2.5 meters in diameter, the Hwasong-19 is larger than both the solid-fuel Hwasong-18 (25 m) and the liquid-fuel Hwasong-17 (24–26 m), making it the largest solid-propellant road-mobile ICBM ever flight-tested (CSIS Hwasong-19 Profile; 38 North Hwasong-19 Analysis). Features a high-volume blunt payload shroud and a revealed Post-Boost Vehicle (PBV) designed specifically to boost a heavy MIRV bus and decoys to 15,000+ km range (38 North Hwasong-19 Analysis).
- **Deterministic Causal Chain:** Because the smaller-diameter Hwasong-18 has constrained throw-weight for multiple first-generation North Korean MIRV warheads and decoys, the MGB developed the oversized solid-fuel Hwasong-19 and executed its initial flight test on **October 31, 2024**, achieving a record flight time of **85 minutes 56 seconds**, a record apogee of **7,687.5 km** (~1,100 km higher and 13 minutes longer than the Hwasong-18), and a downrange distance of **1,001.2 km** into the Sea of Japan—combining the instant launch readiness of solid propellant with the heavy MIRV throw-weight of the Hwasong-17 (38 North Hwasong-19 Analysis; CSIS Hwasong-19 Profile).

139. Pukguksong-1 (북극성-1형 / KN-11 — Submarine-Launched Ballistic Missile)

- **Institutional & Organizational Designation:** Pukguksong-1 (《북극성-1》형, "Polaris-1", US/DoD designation KN-11), North Korea's first two-stage solid-propellant submarine-launched ballistic missile (FAS North Korean Nuclear Weapons 2024; Taylor & Francis DPRK Delivery Systems 2022).
- **Headquarters & Geolocation:** Developed at the Sinpo South Shipyard (`40.0258°N, 128.1558°E`) in South Hamgyong Province; test-launched from submersible test barges and the single *Gorae*-class (Sinpo-B / *8.24 Yongung* ["August 24 Hero"]) experimental ballistic missile submarine (SSB) homeported at Mayang-do / Sinpo (Taylor & Francis DPRK Delivery Systems 2022; Pukguksong-3 Profile).
- **Command Hierarchy & Named Leadership:** Developed by the Academy of National Defense Science under Jang Chang Ha and Kim Jong Sik for the Korean People's Army Navy (KPAN) East Sea Fleet (Taylor & Francis DPRK Delivery Systems 2022).
- **Engineering-Technical & Operational Architecture:** Two-stage solid-propellant SLBM; length ~9.0–9.3 m, diameter 1.1 m, launch weight ~14,000 kg, operational range 1,000–1,200 km (Taylor & Francis DPRK Delivery Systems 2022). Uses a cold-gas ejection generator to clear the submarine sail tube before igniting its solid motor above the water surface, stabilized by eight deployable grid fins at the base skirt (Pukguksong-3 Profile).
- **Deterministic Causal Chain:** Tested six times between May 2015 and August 2016 (culminating in a successful August 24, 2016 launch from the *8.24 Yongung* submarine that flew 500 km on a lofted trajectory toward Japan's ADIZ), the Pukguksong-1 served as North Korea's foundational proof-of-concept for underwater cold ejection and high-thrust solid-fuel propulsion, directly spawning the land-based Pukguksong-2 and the wider-diameter Pukguksong-3/4/5/6 SLBM family (Taylor & Francis DPRK Delivery Systems 2022; FAS North Korean Nuclear Weapons 2024).

140. Pukguksong-2 (북극성-2형 / KN-15 — Land-Based Tracked Solid-Fuel MRBM)

- **Institutional & Organizational Designation:** Pukguksong-2 (《북극성-2》형, US/DoD designation KN-15), land-based two-stage solid-propellant canisterized medium-range ballistic missile (FAS North Korean Nuclear Weapons 2024; Pukguksong-3 Profile).
- **Headquarters & Geolocation:** Flight-tested from Panghyon Airbase in Kusong, North Pyongan Province (39.9275°N, 125.2194°E) on February 12, 2017 and Lake Yonpung, South Pyongan Province (39.6380°N, 125.8050°E) on May 21, 2017 (FAS North Korean Nuclear Weapons 2024).
- **Command Hierarchy & Named Leadership:** Approved for serial production and deployment with KPA Strategic Force MRBM brigades by Kim Jong Un on May 21, 2017 under supervision of Jang Chang Ha and Kim Jong Sik (FAS North Korean Nuclear Weapons 2024).
- **Engineering-Technical & Operational Architecture:** Land-mobile modification of the Pukguksong-1 SLBM; length ~9.5–10.0 m, diameter 1.1–1.4 m, two-stage solid-propellant motor with filament-wound casing, operational range 1,200–2,000 km (FAS North Korean Nuclear Weapons 2024; Pukguksong-3 Profile). Cold-launched from a cylindrical canister mounted on an indigenous **tracked caterpillar-type TEL** capable of off-road movement through rugged Korean mountain tunnels (FAS North Korean Nuclear Weapons 2024).
- **Deterministic Causal Chain:** Developed in 2016–2017 to replace vulnerable liquid-fueled Hwasong-7 (Nodong) MRBMs, the Pukguksong-2 demonstrated rapid cold-gas canister launch and solid-fuel readiness in two successful 2017 flight tests (flying 500 km to apogees of 550–560 km), establishing the operational doctrine for North Korea's transition to an all-solid-fuel regional nuclear strike force targeting South Korea and Japan (FAS North Korean Nuclear Weapons 2024).

141. Pukguksong-3 (북극성-3형 / KN-26 — Submarine-Launched Ballistic Missile)

- **Institutional & Organizational Designation:** Pukguksong-3 (《북극성-3》형, US/DoD designation KN-26), second-generation large-diameter two-stage solid-propellant submarine-launched ballistic missile (Pukguksong-3 Profile; Taylor & Francis DPRK Delivery Systems 2022).
- **Headquarters & Geolocation:** Engineered at the Chemical Material Institute in Hamhung (39.8550°N, 127.5680°E, where wall diagrams of the Pukguksong-3 filament-wound casing were revealed on August 22, 2017) and test-launched from a submersible barge off Wonsan / Sinpo Bay in the Sea of Japan (39.9800°N, 128.2000°E) (Pukguksong-3 Profile).
- **Command Hierarchy & Named Leadership:** Developed and tested by the Academy of National Defense Science under Jang Chang Ha (Pukguksong-3 Profile).
- **Engineering-Technical & Operational Architecture:** Two-stage solid-propellant SLBM with filament-wound composite motor casings; diameter increased from 1.1 m to **1.4 m**, length ~10.0–10.6 m, rounded blunt payload shroud (eliminating the grid-fin base skirt of the Pukguksong-1), operational range **1,900–2,500 km** (Pukguksong-3 Profile; Taylor & Francis DPRK Delivery Systems 2022).
- **Deterministic Causal Chain:** First flight-tested on **October 2, 2019** from a submerged barge off Wonsan (achieving a lofted apogee of 910 km and downrange distance of 450 km, equivalent to ≥1,900 km on a standard ballistic trajectory), the 1.4-meter Pukguksong-3 established the standardized wide-body solid-motor architecture intended for the modified Romeo-class (Sinpo-C / *Hero Kim Kun Ok*, Tactical Submarine No. 841) ballistic missile submarine (Pukguksong-3 Profile; Taylor & Francis DPRK Delivery Systems 2022).

142. Pukguksong-4 (북극성-4ㅅ — SLBM Unveiled October 2020)

- **Institutional & Organizational Designation:** Pukguksong-4ㅅ (《북극성-4ㅅ》, Hangul suffix ㅅ denoting *sujung* [수중, underwater] or *susang* [수상, surface/maritime]), wide-body two-stage solid-propellant submarine-launched ballistic missile (Pukguksong-3 Profile; Taylor & Francis DPRK Delivery Systems 2022).
- **Headquarters & Geolocation:** Unveiled at Kim Il Sung Square, Pyongyang (39.0194°N, 125.7533°E) during the October 10, 2020 75th Anniversary of the Workers' Party of Korea military parade; associated with the Sinpo South Shipyard submarine program (40.0258°N, 128.1558°E) (Pukguksong-3 Profile; Taylor & Francis DPRK Delivery Systems 2022).
- **Command Hierarchy & Named Leadership:** Developed by the Academy of National Defense Science / Missile General Bureau under Jang Chang Ha and Kim Jong Sik (Taylor & Francis DPRK Delivery Systems 2022).
- **Engineering-Technical & Operational Architecture:** Two-stage filament-wound composite solid-propellant SLBM; diameter significantly widened to **~1.5–1.8 m** with a compact length (~9.5–10.2 m) to fit within the vertical launch sail/hull tubes of converted submarines while maximizing propellant mass fraction (Pukguksong-3 Profile; Taylor & Francis DPRK Delivery Systems 2022). Per the **UN Security Council Panel of Experts (2021 Report, S/2021/211, Annex 11)**, the Pukguksong-4ㅅ is assessed to achieve a maximum range between **3,500 km (with a 1,300 kg payload) and 5,400 km (with a 650 kg payload)** (Taylor & Francis DPRK Delivery Systems 2022).
- **Deterministic Causal Chain:** Unveiled on October 10, 2020 to maximize throw-weight and range within the vertical height constraints of North Korean submarine sail tubes, the Pukguksong-4ㅅ demonstrated composite filament-wound motor casings and an enlarged payload shroud sized for multiple reentry vehicles or penetration aids (Taylor & Francis DPRK Delivery Systems 2022; Pukguksong-3 Profile).

143. Pukguksong-5 (북극성-5ㅅ — SLBM Unveiled January 2021)

- **Institutional & Organizational Designation:** Pukguksong-5ㅅ (《북극성-5ㅅ》), elongated-shroud two-stage solid-propellant submarine-launched ballistic missile (Pukguksong-3 Profile; TWZ Pukguksong-5 Analysis; Taylor & Francis DPRK Delivery Systems 2022).
- **Headquarters & Geolocation:** Unveiled at Kim Il Sung Square, Pyongyang (39.0194°N, 125.7533°E) during the January 14, 2021 8th WPK Congress military parade and displayed at the "Self-Defence-2021" exhibition in Pyongyang in October 2021 (Pukguksong-3 Profile; TWZ Pukguksong-5 Analysis).
- **Command Hierarchy & Named Leadership:** Developed by the Academy of National Defense Science / Missile General Bureau under Jang Chang Ha and Kim Jong Sik (Taylor & Francis DPRK Delivery Systems 2022).
- **Engineering-Technical & Operational Architecture:** Shares the wide (~1.5–1.8 m diameter) two-stage filament-wound composite solid-propellant first and second stages of the Pukguksong-4ㅅ, paired with a longer, sharper, ogival/pointed nose cone and extended payload shroud (total length ~10.5–11.0 m), providing increased internal volume for high-ballistic-coefficient reentry vehicles, MIRV buses, or decoys across an estimated range of **3,000–5,000+ km** (Pukguksong-3 Profile; TWZ Pukguksong-5 Analysis; Taylor & Francis DPRK Delivery Systems 2022).

- **Deterministic Causal Chain:** Paraded just three months after the Pukguksong-4ㅅ on January 14, 2021 (where KCNA proclaimed it "the world's most powerful weapon"), the Pukguksong-5ㅅ refined the hydrodynamic/aerodynamic nosecone profile of the wide-body solid SLBM series to support higher-velocity atmospheric reentry and multi-warhead packaging (TWZ Pukguksong-5 Analysis; Taylor & Francis DPRK Delivery Systems 2022).

144. Pukguksong-6 (북극성-6형 — Extended SLBM Variant Unveiled April 2022 / 2023)

- **Institutional & Organizational Designation:** Pukguksong-6 (presumed designation of the extended-length Pukguksong-series SLBM unveiled at the April 25, 2022 90th Anniversary of the Korean People's Revolutionary Army parade and February/July 2023 parades), heavy solid-propellant submarine-launched / sea-based ballistic missile (Pukguksong-3 Profile; SIPRI North Korean Nuclear Forces 2024).
- **Headquarters & Geolocation:** Paraded at Kim Il Sung Square, Pyongyang (`39.0194°N, 125.7533°E`) and linked to the Sinpo South Shipyard large submarine construction hall (`40.0258°N, 128.1558°E`), where the *Hero Kim Kun Ok* (No. 841, launched September 6, 2023, equipped with 4 large and 6 smaller vertical launch tubes in its extended sail) and a larger nuclear-powered submarine hull are constructed (Pukguksong-3 Profile; SIPRI North Korean Nuclear Forces 2024; VOA DPRK Submarine Registration Report).
- **Command Hierarchy & Named Leadership:** Overseen by the Missile General Bureau (Jang Chang Ha / Kim Jong Sik) and Korean People's Army Navy submarine command (SIPRI North Korean Nuclear Forces 2024).
- **Engineering-Technical & Operational Architecture:** Significantly lengthened two/three-stage solid-propellant SLBM airframe (~12.0–13.5 m length, ~1.6–1.8 m diameter) exceeding the length of both Pukguksong-4ㅅ and Pukguksong-5ㅅ, featuring an extended first-stage motor and interstage adapter for intermediate-to-intercontinental sea-based range (4,000–6,000+ km) (Pukguksong-3 Profile; SIPRI North Korean Nuclear Forces 2024).
- **Deterministic Causal Chain:** Displayed on April 25, 2022 as the sixth iteration of the Pukguksong family (SIPRI North Korean Nuclear Forces 2024), the extended Pukguksong-6 airframe aligns with the four oversized vertical launch tubes installed in the sail of the *Hero Kim Kun Ok* tactical nuclear attack submarine and the larger nuclear-powered ballistic missile submarine announced by Kim Jong Un at the 8th WPK Congress (Pukguksong-3 Profile; VOA DPRK Submarine Registration Report).

145. Hwasal-1 (화살-1형 — Strategic Nuclear-Capable Land-Attack Cruise Missile, Hwasan-31 Warhead)

- **Institutional & Organizational Designation:** Hwasal-1 (《화살-1》형, "Arrow Type 1", with improved enlarged-warhead variant designated **Hwasal-1 Ra-3** [화살-1라-3형]), long-range nuclear-capable land-attack cruise missile (LACM) armed with the **Hwasan-31** (화산-31, "Volcano-31") standardized tactical nuclear warhead (Hwasal-1 Profile; Hwasal-2 Profile; SIPRI North Korean Nuclear Forces 2024).
- **Headquarters & Geolocation:** Unveiled at the "Self-Defence-2021" exhibition in Pyongyang (`39.0392°N, 125.7625°E`) in October 2021; flight-tested from South Hamgyong Province (Hamhung/Hongwon), North Pyongan Province, and the west coast into the Yellow Sea and Sea of Japan (Hwasal-2 Profile; SIPRI North Korean Nuclear Forces 2024).
- **Command Hierarchy & Named Leadership:** Operated by KPA tactical nuclear cruise missile units under the Missile General Bureau (General Director Jang Chang Ha and Kim Jong Sik) (Hwasal-2 Profile; FAS North Korean Nuclear Weapons 2024).
- **Engineering-Technical & Operational Architecture:** Subsonic terrain-hugging land-attack cruise missile (externally analogous to the US BGM-109 Tomahawk / Russian 3M-14 Kalibr); length ~6.0 m, diameter ~0.52–0.60 m, powered by a solid-fuel booster for canister ejection and a miniature turbojet/turbofan sustainer engine with pop-out mid-body wings and a ventral air intake; operational range **1,500 km** (flying figure-8 and oval trajectories for 7,580 seconds [2 hours 6 minutes] in its September 11–12, 2021 maiden test) (Hwasal-1 Profile; Hwasal-2 Profile). Armed with the **Hwasan-31** tactical nuclear warhead (unveiled on March 27, 2023 by Kim Jong Un; diameter ~40–50 cm, length ~90 cm, mass ~400–500 kg, estimated yield ~10–15 kt), which is standardized across at least eight DPRK delivery systems (FAS North Korean Nuclear Weapons 2024; SIPRI North Korean Nuclear Forces 2024).
- **Deterministic Causal Chain:** First flight-tested on September 11–12, 2021 to 1,500 km range, the Hwasal-1 executed a simulated tactical nuclear airburst drill on March 22, 2023 (two Hwasal-1 and two Hwasal-2 missiles detonating mock warheads at 600 meters altitude after flying 1,500 km and 1,800 km), followed on April 19, 2024 by a test of the super-large warhead **Hwasal-1 Ra-3** variant, giving the DPRK a low-altitude, radar-evading nuclear strike vector across all of South Korea and Japan (Hwasal-2 Profile; SIPRI North Korean Nuclear Forces 2024).

146. Hwasal-2 (화살-2형 — Long-Range Nuclear-Capable Strategic Cruise Missile)

- **Institutional & Organizational Designation:** Hwasal-2 (《화살-2》형, "Arrow Type 2", accompanied by the submarine/naval-launched variant **Pulhwasal-3-31** [불화살-3-31형, "Fire Arrow-3-31"]), long-range nuclear-capable strategic land-attack cruise missile (Hwasal-2 Profile; Pulhwasal-3-31 Profile; SIPRI North Korean Nuclear Forces 2024).
- **Headquarters & Geolocation:** Unveiled in October 2021 in Pyongyang; flight-tested from Kimchaek City (North Hamgyong Province, `40.6689°N, 129.1897°E`), Hamhung (South Hamgyong Province), Kaechon (South Pyongan Province), and the West Sea (Yellow Sea), as well as aboard the *Amnok*-class corvette (Pennant 661) and the 5,000-ton destroyer *Choe Hyon* (launched April 25, 2025 at Nampo) (Hwasal-2 Profile).
- **Command Hierarchy & Named Leadership:** Operated by KPA Strategic Cruise Missile units and KPAN surface/submarine squadrons under the Missile General Bureau (General Director Jang Chang Ha and Kim Jong Sik) (Hwasal-2 Profile; Newscord KCNA Submarine Cruise Missile Report 2026).
- **Engineering-Technical & Operational Architecture:** Extended-range subsonic turbofan/turbojet cruise missile; diameter 0.60 m, warhead mass ~400 kg (carrying the **Hwasan-31** tactical nuclear warhead), operational range **1,800–2,000 km** (demonstrated in flight durations of 10,208 to 10,234 seconds [2 hours 50 minutes] along elliptical and figure-8 courses) (Hwasal-2 Profile). Launched from 5-tube road-mobile wheeled TELs, naval deck canisters on *Amnok*-class corvettes and *Choe Hyon*-class destroyers, and submarine torpedo tubes / vertical launch tubes (including the Pulhwasal-3-31 tested on January 24 and 28, 2024 for 7,421–7,445 seconds and the October 14, 2026 mixed VLS/torpedo-tube submarine test) (Hwasal-2 Profile; Pulhwasal-3-31 Profile; Newscord KCNA Submarine Cruise Missile Report 2026).
- **Deterministic Causal Chain:** First test-fired on January 25, 2022 and October 12, 2022 (flying 2,000 km in 10,234 seconds), officially named on February 23, 2023 during a four-missile salvo from Kimchaek City into the Sea of Japan, and tested again on March 22, 2023, September 2, 2023 (150 m airburst over the Yellow Sea), and January 30, 2024, the Hwasal-2 and its naval Pulhwasal-3-31 derivative extend DPRK low-altitude nuclear cruise missile reach to 2,000 km across all US bases in Japan and the Okinawa/Ryukyu chain (Hwasal-2 Profile; Pulhwasal-3-31 Profile).

147. Haeil (해일 — Haeil-1 / Haeil-2 / Haeil-5-23 Unmanned Underwater Nuclear Attack Drone)

- **Institutional & Organizational Designation:** *Haeil* ("Tsunami" / 핵무인수중공격정《해일》, encompassing named variants **Haeil-1** [해일-1형], **Haeil-2** [해일-2형], and **Haeil-5-23** [해일-5-23]), nuclear-armed autonomous unmanned underwater vehicle (UUV) / nuclear torpedo system (38 North Haeil Analysis; Haeil Profile; VOA DPRK Submarine Registration Report).
- **Headquarters & Geolocation:** Engineered by the Academy of National Defense Science Underwater Weapons System Institute and fabricated at Sinpo South Shipyard (`40.0258°N, 128.1558°E`, where a cylindrical prototype was imaged by Planet Labs on May 27, 2020); tested from Riwon County (`40.3200°N, 128.6500°E`), Hongwon Bay, and Kajin Port (South Hamgyong Province) in the East Sea (Sea of Japan) (38 North Haeil Analysis; Haeil Profile).
- **Command Hierarchy & Named Leadership:** Initiated at the 8th WPK Congress in January 2021 and guided across 29+ weapon tests by Kim Jong Un, the WPK Central Military Commission, and the Underwater Weapons System Research Institute (38 North Haeil Analysis; Haeil Profile).
- **Engineering-Technical & Operational Architecture:** Heavy torpedo-shaped autonomous underwater vehicle powered by diesel-electric/battery or pump-jet propulsion (unlike Russia's nuclear-reactor-powered Status-6/Poseidon), cruising at depths of **80 to 150 meters** at an average speed of **4.6 knots (8.5 km/h)** along pre-programmed figure-8 and oval waypoint courses to detonate a **Hwasan-31** (or Hwasan-23) nuclear warhead underwater inside adversary naval bases and carrier strike group Operating Areas (38 North Haeil Analysis; Haeil Profile):
  - **Haeil-1:** Tested March 21–23, 2023 (cruised 59 hours 12 minutes at 80–150 m depth) and March 25–27, 2023 (cruised 600 km in 41 hours 27 minutes before underwater test-warhead detonation) (38 North Haeil Analysis).
  - **Haeil-2:** Tested April 4–7, 2023 from Kajin Port to Ryongdae Port, cruising **1,000 km in 71 hours 6 minutes** along an oval and figure-8 course (Haeil Profile). Larger parade variant displayed on July 27, 2023 measures ~16 m in length and 1.5 m in diameter with a shrouded pump-jet propulsor (Haeil Profile).
  - **Haeil-5-23:** Tested in the East Sea in January 2024 by the National Defense Science Academy's Underwater Weapons System Institute in response to trilateral US-ROK-Japan naval exercises (VOA DPRK Submarine Registration Report; Haeil Profile).
- **Deterministic Causal Chain:** Developed from 2012 and subjected to over 50 shakedown tests between 2021 and 2023 before its March 2023 unveiling, the Haeil family was explicitly depicted on the March 27, 2023 wall diagram alongside the Hwasan-31 nuclear warhead to threaten US Navy carrier strike groups and ROK/Japanese naval ports (such as Busan, Chinhae, Yokosuka, and Sasebo) with clandestine underwater nuclear detonation (38 North Haeil Analysis; Haeil Profile).

148. DPRK Railway-Borne Missile Regiment (조선인민군 철도기동미싸일연대 — Pyongan/Yangdok KN-23 Rail-Mobile Ballistic Missile Regiment)

- **Institutional & Organizational Designation:** Korean People's Army Railway-Borne Missile Regiment (철도기동미싸일연대, ordered expanded to a **Railway-Borne Missile Brigade** [철도기동미싸일여단]), specialized rail-mobile nuclear-capable short-range ballistic missile unit established at the 8th Congress of the Workers' Party of Korea in January 2021 (CSIS Rail-Mobile Ballistic Missile Analysis; Open Nuclear Network Rail-Mobile Launch Analysis; FAS North Korean Nuclear Weapons 2024).
- **Headquarters & Geolocation:** Operates across the central and northwestern DPRK mountain rail network: maiden combat firing drill conducted on September 15, 2021 from a rail tunnel spur in **Yangdok County (양덕군), South Pyongan Province (`39.2050°N, 126.6850°E`)**, and second firing drill conducted on January 14, 2022 from **Uiju County (의주군), North Pyongan Province (`40.1850°N, 124.5300°E`)** near the Sino-Korean border (CSIS Rail-Mobile Ballistic Missile Analysis; Hwasong-11A / KN-23 Profile).
- **Command Hierarchy & Named Leadership:** Commanded under the KPA Strategic Force / Missile General Bureau; initial September 15, 2021 combat readiness drill was personally directed on site by Marshal **Pak Jong Chon** (박정천, Member of the Presidium of the WPK Politburo and Secretary of the WPK Central Committee) alongside Kim Jong Sik and Jang Chang Ha (Open Nuclear Network Rail-Mobile Launch Analysis).
- **Engineering-Technical & Operational Architecture:** Deploys modified heavy freight boxcars with split hydraulic roof doors housing a dual-missile erector-launcher assembly armed with two **Hwasong-11A (KN-23)** single-stage solid-propellant quasi-ballistic SRBMs (length 7.5 m, diameter 0.95 m, launch weight 3,400 kg, payload 500 kg including the **Hwasan-31** tactical nuclear warhead, depressed-apogee range 430–800+ km with terminal pull-up maneuver) (Open Nuclear Network Rail-Mobile Launch Analysis; Hwasong-11A Profile). Hauled by a **DF4D / HXD3-derived diesel locomotive** (deliberately selected over electric locomotives to survive wartime destruction of the electrical grid) with flame-deflection exhaust vents built into the railcar floor; planned brigade expansion comprises 9 rail-mobile launcher cars carrying 18 missiles (CSIS Rail-Mobile Ballistic Missile Analysis; FAS North Korean Nuclear Weapons 2024).
- **Deterministic Causal Chain:** Stood up in January 2021 to exploit North Korea's >5,000 km rail network and hundreds of mountain railway tunnels for hide-and-shoot survivability, the Railway-Borne Missile Regiment moved at dawn on September 15, 2021 into the Yangdok mountain tunnel sector and fired two KN-23 missiles 800 km at a 60 km depressed apogee into the East Sea, followed by a rapid-reaction launch of two KN-23s from Uiju on January 14, 2022 that flew 430 km at a 36 km apogee to strike the Al-som island target (CSIS Rail-Mobile Ballistic Missile Analysis; Open Nuclear Network Rail-Mobile Launch Analysis; Hwasong-11A / KN-23 Profile).

149. Punggye-ri Nuclear Test Site (풍계리 핵실험장 — Mantapsan, Tunnels 1–4, 2006–2017 Test Yields)

- **Institutional & Organizational Designation:** Punggye-ri Nuclear Test Site (풍계리 핵실험장, Northern Nuclear Test Ground of the DPRK), underground nuclear weapons detonation complex inside **Mount Mantap (Mantapsan / 만탑산, elevation 2,205 meters)** (NTI Punggye-ri Facility Profile; Mantapsan Geophysical Profile).
- **Headquarters & Geolocation:** Located in Punggye-ri, Kilju County (길주군), North Hamgyong Province (`41.2794°N, 129.0869°E`), with four excavated horizontal adit tunnel portals and a Main Administrative/Command Area (NTI Punggye-ri Facility Profile; 2009 North Korean Nuclear Test Geolocation):
  - **Tunnel No. 1 (East Portal):** `41.2808°N, 129.1047°E` — Site of the 1st nuclear test (October 9, 2006); sealed due to radioactive contamination (NTI Punggye-ri Facility Profile).
  - **Tunnel No. 2 (North Portal, directly beneath Mt. Mantap peak):** `41.2881°N, 129.0819°E` — Site of the 2nd (May 25, 2009), 3rd (February 12, 2013), 4th (January 6, 2016), 5th (September 9, 2016), and 6th (September 3, 2017) underground nuclear tests; suffered internal chimney collapse and 20-inch (50 cm) surface subsidence of Mt. Mantap ("Tired Mountain Syndrome") after the September 2017 thermonuclear blast (NTI Punggye-ri Facility Profile; Mantapsan Geophysical Profile).

    - **Tunnel No. 3 (South Portal):** `41.2750°N, 129.0856°E` — Excavated through unfractured southern granite; portal demolished in May 2018, then **re-excavated and restored to full nuclear-test readiness between March and June 2022** with new support buildings confirmed by the IAEA and CSIS Beyond Parallel (MOFA Diplomatic Bluebook 2023; CSIS Beyond Parallel Punggye-ri Update).
    - **Tunnel No. 4 (West Portal):** `41.2797°N, 129.0758°E` — Excavated 2013–2018; portal collapsed in May 2018 (CSIS Beyond Parallel Punggye-ri Update).
- **Command Hierarchy & Named Leadership:** Operated by the DPRK Nuclear Weapons Institute (핵무기연구소, Directed by Ri Hong Sop [리홍섭]) and WPK Munitions Industry Department (Hong Sung Mu [홍승무]) under direct SAC / Kim Jong Un authorization (38 North Yongbyon 2025 Report).
- **Engineering-Technical & Operational Architecture — Complete 6-Test Chronology & Yields:**
    1. **Test 1 (October 9, 2006, Tunnel 1):** Plutonium implosion device; seismic magnitude `mb 4.3` (USGS); estimated yield **0.2–1.0 kilotons (kt)** (NTI Punggye-ri Facility Profile; DPRK WMD Overview).
    2. **Test 2 (May 25, 2009, Tunnel 2, `41.306°N, 129.029°E`):** Plutonium implosion device; seismic magnitude `mb 4.7` (USGS); estimated yield **2.0–7.0 kt** (2.35 kt refined seismic estimate) (2009 North Korean Nuclear Test Geolocation; NTI Punggye-ri Facility Profile).
    3. **Test 3 (February 12, 2013, Tunnel 2):** Miniaturized HEU/plutonium composite device; seismic magnitude `mb 5.1` (USGS); estimated yield **6–16 kt** (NTI Punggye-ri Facility Profile).
    4. **Test 4 (January 6, 2016, Tunnel 2):** Claimed boosted fission / initial thermonuclear test; seismic magnitude `mb 5.1` (USGS); estimated yield **7–16 kt** (NTI Punggye-ri Facility Profile).
    5. **Test 5 (September 9, 2016, Tunnel 2):** Standardized ballistic-missile warhead test; seismic magnitude `mb 5.3` (USGS); estimated yield **15–30 kt** (NTI Punggye-ri Facility Profile).
    6. **Test 6 (September 3, 2017, Tunnel 2):** Two-stage thermonuclear weapon ("hydrogen bomb for ICBM"); seismic magnitude `mb 6.1–6.3` (USGS/NORSAR, followed by an `mb 4.1` cavity collapse 8.5 minutes later); estimated yield **100–250 kt** (up to 250 kt per NORSAR revised seismic yield calculation) (NTI Punggye-ri Facility Profile; Mantapsan Geophysical Profile; DPRK WMD Overview).
- **Deterministic Causal Chain:** Across six tests in Tunnels 1 and 2 from 2006 to 2017, the DPRK Nuclear Weapons Institute progressed from a sub-kiloton plutonium fizzle to a 100–250 kt thermonuclear warhead; after demolishing the outer portals on May 24, 2018 ahead of the Singapore Summit, North Korea reopened Adit/Tunnel No. 3 in early 2022 to maintain immediate readiness for a seventh nuclear test (to validate the Hwasan-31 tactical warhead or a multi-warhead MIRV primary) (NTI Punggye-ri Facility Profile; MOFA Diplomatic Bluebook 2023; CSIS Beyond Parallel Punggye-ri Update).

150. Yongbyon Nuclear Scientific Research Center (녕변핵시설 / 영변원자력연구소 — 5 MWe Reactor, IRT-2000, ELWR, Radiochemical Lab, Uranium Enrichment)

- **Institutional & Organizational Designation:** Yongbyon Nuclear Scientific Research Center (녕변핵과학연구중심), North Korea's primary sovereign fissile-material production complex spanning plutonium reactors, PUREX reprocessing, tritium production, and gas-centrifuge highly enriched uranium (HEU) cascades (FAS North Korean Nuclear Weapons 2024; NTI Yongbyon Facility Profile; 38 North Yongbyon June 2025 Analysis).
- **Headquarters & Geolocation:** Located along the Kuryong River in Pungang-jigu, Yongbyon County (녕변군), North Pyongan Province (`39.7958°N, 125.7547°E`, approximately 90 km north of Pyongyang) (NTI Yongbyon Facility Profile).
- **Command Hierarchy & Named Leadership:** Operated by the General Department of Atomic Energy (GDAE / 원자력총국) and the Nuclear Weapons Institute (formerly directed by Ri Hong Sop) under WPK Munitions Industry Department oversight; inspected by Kim Jong Un on January 29, 2025 / September 2024 (38 North Yongbyon 2025 Report).
- **Engineering-Technical & Operational Architecture — Core Sub-Facilities:**
    1. **5 MWe Graphite-Moderated Reactor (`39.7958°N, 125.7547°E`):** Magnox-type natural-uranium, CO2-cooled, graphite-moderated reactor (20–25 MWt / 5 MWe, operational since 1986, restarted in July 2021 with periodic refueling shutdowns in April/September 2023), producing up to **6 kg of weapons-grade plutonium (Pu-239) per year** (FAS North Korean Nuclear Weapons 2024).
    2. **IRT-2000 Research Reactor (`39.8031°N, 125.7556°E`):** Soviet-supplied pool-type research reactor (commissioned 1965 at 2 MWt, upgraded to 4 MWt and 8 MWt using 36%/80% HEU fuel), utilized for radioisotope and **tritium (H-3)** production for boosted fission and thermonuclear weapons (NTI North Korea Nuclear Facilities).
    3. **Experimental Light-Water Reactor (ELWR, `39.7942°N, 125.7550°E`):** 25–30 MWe (100 MWt) domestically built light-water reactor constructed starting in 2010; entered **initial commissioning and hot pre-operational testing in October–December 2023** (evidenced by sustained warm cooling-water discharge into the Kuryong River and switchyard thermal signatures), capable of adding ~15–20 kg of plutonium per year or tritium production (FAS North Korean Nuclear Weapons 2024).
    4. **Radiochemical Laboratory (`39.7828°N, 125.7539°E`):** Industrial PUREX (Plutonium Uranium Reduction Extraction) spent-fuel reprocessing plant modeled on the Eurochemic plant in Mol-Dessel, Belgium, extracting 5–8 kg of weapons-grade plutonium per reprocessing campaign from 5 MWe spent fuel rods, supported by a second underground radioactive waste vault (`20 m x 12 m` with 9 roof hatches) constructed in April–June 2025 (FAS North Korean Nuclear Weapons 2024; 38 North Yongbyon June 2025 Analysis).
    5. **Uranium Enrichment Plant (UEP, `39.7767°N, 125.7525°E`) & New 2025–2026 Enrichment Complex:** Original centrifuge hall unveiled to Siegfried Hecker in November 2010 (2,000 P-2-type maraging-steel centrifuges, doubled in floor area in 2013 to ~4,000+ centrifuges), augmented between **December 2024 and April 2026** by a major **new uranium enrichment building** constructed 480 meters north-northeast of the Radiochemical Laboratory (`1,800 meters north of the existing centrifuge halls`), complete with dedicated cooling units, generators, and security walls (FAS North Korean Nuclear Weapons 2024; 38 North Yongbyon June 2025 Analysis; CSIS Beyond Parallel New Yongbyon Enrichment Building April 2026).
- **Deterministic Causal Chain:** Together with the clandestine Kangson centrifuge plant (`38.9210°N, 125.6320°E`), Yongbyon's continuous operation of the 5 MWe reactor, Radiochemical Lab, commissioning of the 100 MWt ELWR in late 2023, and completion of the second Yongbyon enrichment plant in 2025–2026 have generated a cumulative stockpile estimated at **56–70 kg of weapons-grade plutonium** and **1,400–2,200 kg of HEU** (sufficient for up to **90 nuclear warheads**, with **~50 operational warheads assembled** as of 2024–2025) (FAS North Korean Nuclear Weapons 2024; CSIS Beyond Parallel New Yongbyon Enrichment Building April 2026).

151. Sohae Satellite Launching Ground (서해위성발사장 — Vertical Engine Test Stand, Coastal Launch Pad, Chollima-1 / Malligyong-1)

- **Institutional & Organizational Designation:** Sohae Satellite Launching Ground (서해위성발사장, also termed Sohae Satellite Launching Station), North Korea's primary space launch vehicle (SLV) complex and liquid/solid rocket engine static test center operated by the **National Aerospace Technology Administration (NATA /** 국가항공우주기술총국**, formerly NADA)** and the Academy of National Defense Science (CSIS Beyond Parallel Sohae Modernization 2025; 38 North Sohae Expansion April 2026; NTI Sohae Facility Profile).

- **Headquarters & Geolocation:** Located on the Yellow Sea coast in Cholsan County, North Pyongan Province (39.6600°N, 124.7053°E), comprising the Original Launch Pad (39.6600°N, 124.7053°E), the **Yunsong Vertical Engine Test Stand (VETS)** (39.6525°N, 124.7119°E), a new horizontal solid-fuel motor test stand (39.6540°N, 124.7250°E, constructed 2022), and the **New Coastal Launch Pad** (39.6380°N, 124.7390°E, built in rapid modular fashion in April–May 2023) (CSIS Beyond Parallel Sohae May 2024; CSIS Beyond Parallel Sohae Modernization 2025).
- **Command Hierarchy & Named Leadership:** Directed by NATA General Director **Ryu Sang Hun** (류상훈) alongside Missile General Bureau General Director Jang Chang Ha and WPK First Vice Department Director Kim Jong Sik, directly supervised by Kim Jong Un (CSIS Beyond Parallel Another Failed Satellite Launch May 2024).
- **Engineering-Technical & Operational Architecture:**
  1. **Yunsong Vertical Engine Test Stand (VETS):** Heavy concrete-and-steel vertical static test stand used to test the 80-tf and 160-tf Paektusan (RD-250-derived) liquid rocket engines for the Hwasong-12, Hwasong-14, Hwasong-15, and Hwasong-17 ICBMs (including December 7 and December 13, 2019 long-duration tests, a December 15, 2022 140-tf solid-fuel motor test at the adjacent horizontal stand, and an unannounced liquid engine test in the fourth week of April 2024) (CSIS Beyond Parallel Sohae May 2024).
  2. **New Coastal Launch Pad & Chollima-1 / Malligyong-1 Program:** Purpose-built on the southeastern coast of the Sohae peninsula in 2023 with a 33 m x 17 m East Coast VIP Observation Facility 1.3 km northeast; serves as the launch pad for the three-stage **Chollima-1** (천리마-1형) SLV (whose first stage utilizes the 4-nozzle twin-RD-250 liquid engine cluster from the Hwasong-17 ICBM) carrying the **Malligyong-1** (만리경-1호) military electro-optical reconnaissance satellite (38 North Chollima-1 Analysis; CSIS Beyond Parallel Sohae May 2024).
- **Deterministic Causal Chain:** Following two failed Chollima-1 launches from the Coastal Launch Pad on May 31, 2023 (second-stage engine failure) and August 24, 2023 (third-stage emergencyflight-termination error), North Korea successfully placed the **Malligyong-1 (Malligyong-1-3)** reconnaissance satellite into a ~500 km sun-synchronous low Earth orbit on **November 21, 2023** (CSIS Beyond Parallel Sohae May 2024; CSIS Beyond Parallel Sohae Modernization 2025). On **May 27, 2024**, following the September 2023 Kim-Putin summit at Russia's Vostochny Cosmodrome, Sohae launched a "new-type satellite carrier rocket" powered by a **liquid oxygen (LOX) + kerosene (petroleum)** first-stage engine carrying Malligyong-1-1, which exploded in first-stage flight; in **March 2026**, North Korea completely razed the bordering villages of Jangang-dong and Jangya-dong (hundreds of buildings) to expand the Sohae security perimeter and spaceport (CSIS Beyond Parallel Another Failed Satellite Launch May 2024; 38 North Sohae Expansion April 2026).

152. Tongchang-ri (동창리 — Cholsan County Geographic Complex Housing Sohae)

- **Institutional & Organizational Designation:** Tongchang-ri (동창리) Missile, Space Launch, Underground Rail Terminal, and Port Complex, geographic and logistics hub in Cholsan County (철산군), North Pyongan Province housing the Sohae Satellite Launching Ground (38 North Sohae Expansion April 2026; CSIS Beyond Parallel Sohae Modernization 2025).
- **Headquarters & Geolocation:** Encompasses the Cholsan Peninsula complex centered at 39.6600°N, 124.7053°E, bounded by the village site of Tongchang-ri to the north (39.6900°N, 124.7150°E), the demolished port communities of Jangang-dong and Jangya-dong (39.6320°N, 124.7420°E), the Yunsong engine test valley (39.6525°N, 124.7119°E), and the Underground Rail Transfer Terminal (39.6710°N, 124.7080°E), situated ~50 km south of the Chinese border at Dandong and ~70 km west of the Yongbyon Nuclear Scientific Research Center (38 North Sohae Expansion April 2026; CSIS Beyond Parallel Sohae Modernization 2025; NTI Sohae Facility Profile).
- **Command Hierarchy & Named Leadership:** Administered jointly by the KPA General Bureau of Atomic Energy / Missile General Bureau, NATA, and KPA engineering construction bureaus (two primary construction support compounds located 650 m east of the original pad) under Kim Jong Un's March 2022 master expansion directive (CSIS Beyond Parallel Sohae Modernization 2025).
- **Engineering-Technical & Operational Architecture:** Integrates a dedicated spur off the Pyongui Railway line leading into a hardened **Underground Rail Terminal** (refurbished in 2023) that allows ballistic missile and SLV stages to be transported from the Sanum-dong assembly facility in Pyongyang (39.0997°N, 125.8147°E) and unloaded beneath mountain cover without satellite detection; a newly excavated deep-water port pier and coastal causeway for barge delivery of heavy rocket stages; a large new processing/assembly building with an associated underground tunnel facility under construction in 2023–2025; and a telemetry/tracking station overlooking the Yellow Sea southward launch corridor (CSIS Beyond Parallel Sohae Modernization 2025; CSIS Beyond Parallel Sohae May 2024; 38 North Sohae Expansion April 2026).
- **Deterministic Causal Chain:** Selected in the late 1990s (construction started ~2001, operational by 2011) to replace the exposed east-coast Tonghae (Musudan-ri) launch pad with a western site firing southward over the Yellow Sea and East China Sea (avoiding overflight of the Japanese home islands during sun-synchronous polar launches), Tongchang-ri has hosted all seven DPRK satellite launch attempts since April 2012 (Unha-3 / Kwangmyongsong-3 and Kwangmyongsong-4 in 2012–2016, and four Chollima-1 / LOX-kerosene SLV launches in 2023–2024) as well as every liquid- and solid-fuel ICBM static engine test, culminating in the March 2026 clearance of Jangang-dong and Jangya-dong to convert the entire southern Cholsan peninsula into a closed military space and ICBM propulsion complex (38 North Sohae Expansion April 2026; CSIS Beyond Parallel Sohae Modernization 2025).

Section VII: Republic of India National Command Authority (NCA), Strategic Forces Command (SFC), Delivery Triad, BARC, and DRDO Architecture (Entities 153–173)

153. National Command Authority (NCA)

- **Institutional Designation & Sovereign Mandate:** Apex bipartite civilian-military nuclear command, control, and operational decision-making authority of the Republic of India, established on January 4, 2003, via executive directive of the Cabinet Committee on Security (CCS) under Prime Minister Atal Bihari Vajpayee (Ministry of External Affairs CCS Operationalization Release, Jan 4, 2003).
- **Geodetic Coordinates & Command Nodes:** Primary seat at South Block, Raisina Hill, New Delhi (28°36'50"N 77°12'18"E), linked to hardened underground National Command Post (NCP) and alternate chains of command approved by the CCS on January 4, 2003 (Nautilus Institute Indian NC3 Report).
- **Named Leadership & Key Personnel:** Chaired at the Political Council level by Prime Minister Narendra Modi (historically inaugurated under Prime Minister Atal Bihari Vajpayee with National Security Advisor Brajesh Mishra); Executive Council chaired by National Security Advisor Ajit Doval (FAS Indian Nuclear Weapons 2024).

- **Technical & Structural Architecture:** Two-tier council structure separating political launch authorization (Political Council) from technical-military readiness, targeting inputs, and directive execution (Executive Council), transmitting authenticated launch codes to the operational tri-service Strategic Forces Command (SFC) (MEA Nuclear Doctrine Statement).
- **Causal Chain & Operational Chronology:** Following the May 11 and May 13, 1998 Pokhran-II (Operation Shakti) underground nuclear tests and the August 17, 1999 National Security Advisory Board (NSAB) Draft Nuclear Doctrine, the CCS convened on January 4, 2003, to codify India's official nuclear doctrine—enforcing Credible Minimum Deterrence, No First Use (NFU) against nuclear attacks, massive retaliation designed to inflict unacceptable damage, and an explicit nuclear retaliation option against biological or chemical weapon attacks on Indian territory or Indian forces anywhere—and formally created the NCA to institutionalize strict civilian political control over nuclear release (MEA CCS Release, Jan 4, 2003).

154. Strategic Forces Command (SFC)

- **Institutional Designation & Sovereign Mandate:** Operational tri-service unified nuclear command of the Indian Armed Forces responsible for the physical custody, administration, targeting integration, readiness stages, and combat delivery of India's land, air, and sea-based strategic nuclear arsenal (MEA CCS Release, Jan 4, 2003).
- **Geodetic Coordinates & Command Nodes:** Headquartered at New Delhi (`28°36'36"N 77°12'12"E`) with dedicated missile group garrisons, airbase weapon storage areas, and naval SSBN strategic facilities (FAS Indian Nuclear Weapons 2024).
- **Named Leadership & Key Personnel:** Commanded by a three-star Commander-in-Chief (C-in-C) rotating among the three services (Army Lieutenant General, Navy Vice Admiral, Air Force Air Marshal); inaugural C-in-C Air Marshal Teja Mohan Asthana appointed in January 2003, reporting through the Chief of Defence Staff (General Anil Chauhan) and Executive Council to the NCA (Nautilus Institute Indian NC3 Report).
- **Technical & Structural Architecture:** Commands approximately 172 assembled nuclear warheads (from a weapons-grade plutonium stockpile sufficient for up to 210 warheads as of 2024) across Indian Army missile groups (Prithvi-II, Agni-I, Agni-II, Agni-III, Agni-IV, Agni-V, Agni-P), Indian Air Force nuclear-certified strike squadrons (Mirage 2000H, Jaguar IS, Rafale DH/EH), and Indian Navy SSBN deterrent patrols (INS Arihant, INS Arighat) (Bulletin of the Atomic Scientists Indian Nuclear Weapons 2024).
- **Causal Chain & Operational Chronology:** Created by CCS executive order on January 4, 2003, to eliminate inter-service fragmentation after Pokhran-II, the SFC progressively transitioned India's posture from de-mated component storage (where BARC and DRDO held fissile cores and assembly mechanisms separately from SFC delivery vehicles) to canisterized, pre-mated high-readiness deterrent forces with Agni-V, Agni-P, and SSBN patrols, conducting regular user-training launches at the Integrated Test Range (ITR), Abdul Kalam Island, Odisha (FAS Indian Nuclear Weapons 2024).

155. Political Council (NCA India)

- **Institutional Designation & Sovereign Mandate:** Supreme political decision-making body of India's Nuclear Command Authority and the sole organ legally empowered to authorize the employment and launch of nuclear weapons (MEA CCS Release, Jan 4, 2003).
- **Geodetic Coordinates & Command Nodes:** Convened at 7 Lok Kalyan Marg (`28°35'55"N 77°12'35"E`) and South Block Cabinet Room, Raisina Hill, New Delhi (`28°36'50"N 77°12'18"E`), supported by hardened alternate command bunkers (Nautilus Institute Indian NC3 Report).
- **Named Leadership & Key Personnel:** Chaired by the Prime Minister of India (Narendra Modi) and composed of the members of the Cabinet Committee on Security: Minister of Defence (Rajnath Singh), Minister of Home Affairs (Amit Shah), Minister of External Affairs (Subrahmanyam Jaishankar), and Minister of Finance (Nirmala Sitharaman), with the National Security Advisor (Ajit Doval) and Cabinet Secretary attending (MEA CCS Release, Jan 4, 2003).
- **Technical & Structural Architecture:** Operates a two-person authentication and split-code authorization architecture in which the Prime Minister and designated CCS succession chain issue encrypted political release directives through the Executive Council to unlock DRDO/BARC electronic Permissive Action Links (PALs) and direct the SFC to execute retaliatory strike packages (Nautilus Institute Indian NC3 Report).
- **Causal Chain & Operational Chronology:** Established on January 4, 2003, to guarantee that nuclear weapons remain under elected civilian authority at all stages of peace, crisis, and war, the Political Council reviews targeting strategies, alert stages, and alternate chains of command to ensure assured massive retaliation even in the event of a decapitation strike against New Delhi (MEA CCS Release, Jan 4, 2003).

156. Executive Council (NCA India)

- **Institutional Designation & Sovereign Mandate:** Operational-technical council of India's Nuclear Command Authority tasked with providing strategic intelligence, readiness assessments, and targeting options to the Political Council and executing the Political Council's authorized nuclear directives through the Strategic Forces Command (MEA CCS Release, Jan 4, 2003).
- **Geodetic Coordinates & Command Nodes:** Sardar Patel Bhawan / National Security Council Secretariat (NSCS) and South Block, New Delhi (`28°37'21"N 77°12'45"E`) (Nautilus Institute Indian NC3 Report).
- **Named Leadership & Key Personnel:** Chaired by the National Security Advisor (Ajit Doval; inaugurated under Brajesh Mishra, succeeded by J.N. Dixit, M.K. Narayanan, Shivshankar Menon, and Ajit Doval) and comprises the Chief of Defence Staff (Gen Anil Chauhan), Chiefs of Staff of the Army, Navy, and Air Force, C-in-C Strategic Forces Command, Chairman of the Atomic Energy Commission / Secretary DAE (Dr. Ajit Kumar Mohanty), Scientific Advisor to the Defence Minister / Chairman DRDO (Dr. Samir V. Kamat), Secretaries of Defence, External Affairs, Home, and Finance, and Directors of the Intelligence Bureau (IB) and Research and Analysis Wing (R&AW) (Nautilus Institute Indian NC3 Report).
- **Technical & Structural Architecture:** Integrates the scientific-custodial agencies (DAE/BARC for fissile pits and DRDO for non-nuclear explosive assemblies, arming/fuzing/firing systems, and PAL codes) with the military operational command (SFC and Service Chiefs) and intelligence apparatus (R&AW, IB, NTRO, DIA) (FAS Indian Nuclear Weapons 2024).
- **Causal Chain & Operational Chronology:** Upon receiving a Political Council authorization directive, the Executive Council coordinates the simultaneous transmission of DAE/BARC and DRDO arming codes and SFC operational fire orders, transforming a political decision into a synchronized multi-agency mating, unlocking, and launch sequence (MEA CCS Release, Jan 4, 2003).

157. Agni-I (SRBM/MRBM)

- **Institutional Designation & Sovereign Mandate:** Single-stage, solid-fuel, road- and rail-mobile short-to-medium-range nuclear-capable ballistic missile developed by DRDO to bridge the operational range gap between the 350 km Prithvi-II and the 2,000 km Agni-II (CSIS Missile Threat: Agni-I).
- **Geodetic Coordinates & Deployment Nodes:** Flight-tested from Launch Pad 4, Integrated Test Range (ITR), Abdul Kalam Island, Odisha (20°45'29"N 87°05'08"E); operated by the Indian Army's 334 Missile Group under Strategic Forces Command in northern/western India (CSIS Missile Threat: Agni-I).
- **Named Leadership & Key Personnel:** Engineered under DRDO's Integrated Guided Missile Development Programme (IGMDP) led by Dr. A.P.J. Abdul Kalam, Dr. V.K. Saraswat, and Project Director Avinash Chander at the Defence Research and Development Laboratory (DRDL) and Advanced Systems Laboratory (ASL), Hyderabad (CSIS Missile Threat: Agni-I).
- **Technical & Engineering Parameters:** Length 15.0 m; diameter 1.0 m; launch weight 12,000 kg; propulsion via single-stage solid-propellant booster derived from ISRO's SLV-3 utilizing hydroxyl-terminated polybutadiene (HTPB) composite propellant in a 15CDV6 maraging steel casing generating ~48,000 kgf (470 kN) thrust; payload 1,000 kg carrying a single 15–45 kt nuclear warhead or conventional payload; operational range 700–1,200 km; guidance via strapdown ring-laser gyroscope inertial navigation system (RLG-INS) with twin-microprocessor redundant flight computer, thrust vectoring, and aerodynamic fin-tip control surfaces achieving a 25–40 m CEP; transported on a DRDO Mark III 7-axle tractor-trailer road-mobile TEL or rail-mobile launcher (CSIS Missile Threat: Agni-I).
- **Causal Chain & Operational Chronology:** Initiated in September 1999 following the Kargil conflict to provide a dedicated Pakistan-specific theatre nuclear strike system without expending longer-range Agni-II missiles, Agni-I completed its maiden flight test on January 25, 2002 (15 months from design start), succeeded in its second test on January 9, 2003, entered full-rate production in August 2004, and was inducted into SFC service in 2007, with progressive replacement by the canisterized Agni-Prime underway (FAS Indian Nuclear Weapons 2024).

158. Agni-II (MRBM)

- **Institutional Designation & Sovereign Mandate:** Two-stage, solid-fuel, road- and rail-mobile medium-range nuclear ballistic missile serving as a core land-based deterrent of India's Strategic Forces Command (CSIS Missile Threat: Agni-II).
- **Geodetic Coordinates & Deployment Nodes:** Tested at ITR Abdul Kalam Island, Odisha (20°45'29"N 87°05'08"E) and deployed with the Indian Army's 335 Missile Group under SFC (CSIS Missile Threat: Agni-II).
- **Named Leadership & Key Personnel:** Developed at DRDL and ASL Hyderabad under Dr. A.P.J. Abdul Kalam, R.N. Agarwal (Agni Programme Director), and Avinash Chander; manufactured by Bharat Dynamics Limited (BDL), Hyderabad (CSIS Missile Threat: Agni-II).
- **Technical & Engineering Parameters:** Length 20.0–21.0 m; diameter 1.0–1.3 m; launch weight 16,000 kg; two-stage solid-propellant rocket motors (HTPB composite propellant in steel casings with open-truss interstage and flex-nozzle thrust vectoring on Stage 2); payload 1,000 kg carrying a single 15–45 kt fission or 150–200 kt boosted-fission/thermonuclear warhead in a carbon-phenolic heat-shielded maneuvering reentry vehicle (MaRV); range 2,000–3,500 km; guidance via strapdown RLG-INS with GPS/GLONASS updates and terminal velocity-trimming and fin control yielding a 30–40 m CEP; deployed on both special rail-mobile cars and 8-axle road-mobile hydraulic TELs (CSIS Missile Threat: Agni-II).
- **Causal Chain & Operational Chronology:** Authorized in July 1997 to replace the hybrid solid-liquid Agni Technology Demonstrator (tested 1989–1994) with an all-solid, rapid-reaction mobile MRBM capable of reaching western/central China and all of Pakistan, Agni-II completed its maiden flight on April 11, 1999, entered SFC operational service in 2004, and demonstrated nighttime operational launch readiness on November 16, 2019 (CSIS Missile Threat: Agni-II).

159. Agni-III (IRBM)

- **Institutional Designation & Sovereign Mandate:** Two-stage, large-diameter solid-fuel intermediate-range ballistic missile engineered to hold high-value strategic targets across eastern and northern China at risk (CSIS Missile Threat: Agni-III).
- **Geodetic Coordinates & Deployment Nodes:** Launched from ITR Abdul Kalam Island (20°45'29"N 87°05'08"E) and deployed with SFC rail-mobile missile groups in northeastern and central India (FAS Indian Nuclear Weapons 2024).
- **Named Leadership & Key Personnel:** Designed by DRDO's Advanced Systems Laboratory (ASL), Hyderabad under Programme Director Avinash Chander and Project Director Dr. Tessy Thomas, with solid propellant casting at High Energy Materials Research Laboratory (HEMRL), Pune (CSIS Missile Threat: Agni-III).
- **Technical & Engineering Parameters:** Length 16.7 m; diameter 2.0 m (stubby high-volume airframe); launch weight 48,300–50,000 kg; Stage 1 uses a 2.0 m maraging steel motor case with 31.7 tonnes of HTPB composite propellant; Stage 2 uses a 2.0 m carbon-epoxy filament-wound composite motor case with 11.2 tonnes of propellant and electromechanical flex-nozzle thrust vector control (TVC); payload 1,500 kg carrying a single 40–200 kt nuclear warhead; range 3,000–3,500 km (up to 4,000 km with lighter payload); RLG-INS guidance with micro-navigation systems delivering ~40 m CEP; rail-mobile flatcar launcher (CSIS Missile Threat: Agni-III).
- **Causal Chain & Operational Chronology:** Transitioning from 1.0 m SLV-3-derived boosters to a 2.0 m diameter solid rocket motor gave India the throw-weight and impulse required for intercontinental-class propulsion; after an initial heat-shield/stage-separation anomaly on July 9, 2006, DRDO executed successful flight tests on April 12, 2007, May 7, 2008, and February 7, 2010, inducting Agni-III into the SFC in 2011 and using its 2.0 m first-stage motor as the direct foundation for the Agni-V ICBM (CSIS Missile Threat: Agni-III).

160. Agni-IV (IRBM)

- **Institutional Designation & Sovereign Mandate:** Two-stage, solid-fuel, road-mobile intermediate-range ballistic missile (originally designated Agni-II Prime) incorporating lightweight filament-wound composite rocket motors and fifth-generation onboard avionics (CSIS Missile Threat: Agni-IV).

- **Geodetic Coordinates & Deployment Nodes:** Tested at Launch Pad 4, ITR Abdul Kalam Island, Odisha (20°45'29"N 87°05'08"E) and fielded with SFC road-mobile strategic missile units (CSIS Missile Threat: Agni-IV).
- **Named Leadership & Key Personnel:** Led by Project Director Dr. Tessy Thomas ("Missile Woman of India") and Agni Programme Director Avinash Chander at ASL Hyderabad (CSIS Missile Threat: Agni-IV).
- **Technical & Engineering Parameters:** Length 20.0 m; diameter 1.2 m (Stage 1 maraging steel casing generating 710 kN average thrust) and ~1.0 m (Stage 2 carbon-fiber composite casing generating 39.5 kN average thrust); launch weight 17,000 kg (saving over 30 tonnes compared to Agni-III while achieving superior range); payload 1,000 kg nuclear warhead protected by a multi-layer carbon-carbon composite heat shield withstanding reentry temperatures exceeding 4,000°C; range 3,500–4,000 km; guidance via ring-laser gyro inertial navigation system (RLG-INS) coupled with micro-inertial navigation system (MINGS) and digital flight control computer with flex-seal thrust vectoring achieving <100 m CEP; launched from an 8x8 road-mobile TEL (CSIS Missile Threat: Agni-IV).
- **Causal Chain & Operational Chronology:** Developed to provide a lightweight, highly mobile 4,000 km IRBM capable of traversing Indian road bridges and mountain passes unavailable to the 50-tonne rail-bound Agni-III, Agni-IV succeeded on November 15, 2011, completed eight flight tests through September 2024, and validated composite upper-stage motor and redundant digital guidance architectures subsequently backward-integrated into Agni-Prime (FAS Indian Nuclear Weapons 2024).

161. Agni-V (ICBM / Mission Divyastra MIRV)

- **Institutional Designation & Sovereign Mandate:** Three-stage, solid-fuel, canisterized road-mobile intercontinental ballistic missile equipped with Multiple Independently Targetable Reentry Vehicle (MIRV) delivery capability under Mission Divyastra (PIB Mission Divyastra Official Release, Mar 11, 2024).
- **Geodetic Coordinates & Deployment Nodes:** Launched from Launch Pad 4, Dr. APJ Abdul Kalam Island, Odisha (20°45'29"N 87°05'08"E), tracked by telemetry/electro-optical stations at Chandipur (21°26'45"N 87°01'05"E), Paradip, and Port Blair (PIB Mission Divyastra Official Release, Mar 11, 2024).
- **Named Leadership & Key Personnel:** Engineered at ASL Hyderabad under Dr. Avinash Chander, Dr. Tessy Thomas, Dr. V.G. Sekaran, and Mission Divyastra Project Director Sheena Rani, under DRDO Chairman Dr. Samir V. Kamat and authorized by Prime Minister Narendra Modi and Defence Minister Rajnath Singh (PIB Mission Divyastra Official Release, Mar 11, 2024).
- **Technical & Engineering Parameters:** Length 17.5 m; diameter 2.0 m; launch weight 50,000–56,000 kg; three-stage solid-fuel propulsion (Stage 1 maraging steel motor derived from Agni-III; Stages 2 and 3 wound carbon-fiber composite motors with flex-nozzle TVC); payload 1,500 kg carrying either a single high-yield warhead or 3–4 MIRVs on a post-boost vehicle (PBV); range 5,000–8,000 km; guidance via dual-redundant RLG-INS and MINGS with NavIC/GPS satellite updates achieving <10–50 m CEP; cold-launched via gas generator from a hermetically sealed maraging-steel canister mounted on the Transport-cum-Tilting vehicle-5 (TCT-5), a 140-tonne, 30-meter, 7-axle trailer pulled by a 3-axle Volvo prime mover (FAS Indian Nuclear Weapons 2024; CSIS Missile Threat: Agni-V).
- **Causal Chain & Operational Chronology:** First tested in open configuration on April 19, 2012, and September 15, 2013, Agni-V transitioned to sealed-canister cold launches on January 31, 2015, enabling peacetime pre-mating of nuclear warheads inside climate-controlled canisters to compress stop-to-launch reaction time to a few minutes; on March 11, 2024, DRDO executed Mission Divyastra from Abdul Kalam Island, successfully flight-testing Agni-V with indigenous MIRV technology to penetrate adversarial ballistic missile defenses and hold all of mainland China at risk from bases in central and southern India (FAS MIRV Analysis, Mar 12, 2024).

162. Agni-Prime (Agni-P)

- **Institutional Designation & Sovereign Mandate:** Next-generation, two-stage, canisterized solid-fuel medium-range ballistic missile designed via "backward integration of technology" from Agni-IV and Agni-V to replace legacy Prithvi-II, Agni-I, and Agni-II forces (CSIS Missile Threat: India Fires Agni-P).
- **Geodetic Coordinates & Deployment Nodes:** Flight-tested from Launch Pad 4, Dr. APJ Abdul Kalam Island, Odisha (20°45'29"N 87°05'08"E) for induction into Indian Army Strategic Forces Command missile groups (FAS Indian Nuclear Weapons 2024).
- **Named Leadership & Key Personnel:** Developed by ASL Hyderabad under DRDO Chairman Dr. G. Satheesh Reddy and Dr. Samir V. Kamat in coordination with Strategic Forces Command officers (FAS Indian Nuclear Weapons 2024).
- **Technical & Engineering Parameters:** Length ~10.5–11.0 m; diameter 1.2 m; launch weight ~11,000 kg (50% lighter than Agni-III); two-stage filament-wound carbon-composite solid rocket motors with high-energy composite propellants and dual-redundant electromechanical actuators; payload 1,000–1,500 kg supporting unitary MaRV or MIRV configurations; operational range 1,000–2,000 km; guidance via MINGS/RLG-INS with <10 m terminal accuracy; cold-launched from a sealed composite canister mounted on a 5-axle or 6-axle Tatra/BEML road-mobile TEL or rail launcher (FAS Indian Nuclear Weapons 2024).
- **Causal Chain & Operational Chronology:** Conceived in 2016 as Agni-1P to eliminate the steel airframes, hydraulic actuators, and exposed warhead-mating delays of Agni-I and Agni-II, Agni-Prime completed its maiden flight on June 28, 2021, followed by development tests on December 18, 2021, and October 21, 2022, its first night pre-induction trial on June 7, 2023, and an SFC user trial on April 3, 2024, establishing a high-readiness canisterized theatre nuclear force (FAS Indian Nuclear Weapons 2024).

163. Prithvi-II (SRBM)

- **Institutional Designation & Sovereign Mandate:** Single-stage, twin-engine liquid-fuel short-range ballistic missile serving as India's first indigenous ballistic missile deployed under the Strategic Forces Command (CSIS Missile Threat: Prithvi).
- **Geodetic Coordinates & Deployment Nodes:** Tested at Launch Complex 3, Integrated Test Range, Chandipur, Odisha (21°26'45"N 87°01'05"E) and deployed with the 333 and 444 Missile Groups at Secunderabad (17°26'24"N 78°29'54"E) and northern forward staging bases (CSIS Missile Threat: Prithvi).
- **Named Leadership & Key Personnel:** Developed under IGMDP by Dr. A.P.J. Abdul Kalam and Major General V.J. Sundaram at DRDL Hyderabad; produced by Bharat Dynamics Limited (BDL) with 8x8 TELs built by Bharat Earth Movers Limited (BEML), Bengaluru (CSIS Missile Threat: Prithvi).

- **Technical & Engineering Parameters:** Length 8.56 m; diameter 1.1 m; launch weight 4,600 kg; powered by two liquid-propellant open-cycle rocket engines (derived from the Soviet S-75 / Project Devil SAM) burning hypergolic "G-Fuel" (1:1 mixture of xylidene and triethylamine) and red fuming nitric acid (RFNA) oxidizer generating 60 kN (6 tonnes) total thrust; mid-body fixed cruciform wings (lift/drag ratio 9.5) and hydraulic tail fins; payload 500–1,000 kg carrying a ~12 kt fission warhead or conventional submunitions; range 250–350 km; strapdown RLG-INS achieving 10–50 m CEP (upgraded from 150 m on Prithvi-I); launched from a BEML-Tatra 815 8x8 TEL requiring ~2 hours fueling/launch preparation and a battery support train of 18 vehicles per two TELs (CSIS Missile Threat: Prithvi).
- **Causal Chain & Operational Chronology:** Originating in DRDL's 1975 "SS-150" study and formally initiated in 1983 under IGMDP, Prithvi-II was flight-tested on January 27, 1996, entered SFC service in 2003, absorbed upgraded airframes from the retired 150 km Prithvi-I in 2013 under DRDO Chief Avinash Chander, and maintains active nighttime readiness launches while transitioning its role to solid-fuel systems (CSIS Missile Threat: India Tests Prithvi-II).

164. Dhanush (ShLBM)

- **Institutional Designation & Sovereign Mandate:** Single-stage, liquid-fuel, ship-launched short-range ballistic missile (Naval Prithvi-III) deployed aboard modified Indian Navy Sukanya-class offshore patrol vessels as an interim sea-based nuclear delivery system (CSIS Missile Threat: Dhanush).
- **Geodetic Coordinates & Deployment Nodes:** Deployed aboard INS Subhadra (P51) and INS Suvarna (P52) operating out of Karwar Naval Base (14°46'54"N 74°07'32"E) and Mumbai/Visakhapatnam, with test launches conducted in the Bay of Bengal off Odisha (20°45'00"N 87°15'00"E) (CSIS Missile Threat: Dhanush).
- **Named Leadership & Key Personnel:** Engineered by DRDL Hyderabad and Research & Development Establishment (Engineers) [R&DE(E)], Pune (which designed the hydraulic gyro-stabilized shipboard launch platform) under DRDO and the Indian Navy SFC element (CSIS Missile Threat: Dhanush).
- **Technical & Engineering Parameters:** Length 8.53–10.0 m; diameter 0.9–1.1 m; launch weight 4,600–5,600 kg; twin liquid-propellant engines (xylidene/triethylamine + RFNA); payload 500–1,000 kg (single nuclear warhead of 12–15 kt or conventional HE); range 250–400 km; strapdown RLG-INS with GPS updates (~25 m CEP); launched from a gyro-stabilized hydraulic erector pad mounted on the aft helicopter deck of modified Sukanya-class OPVs (CSIS Missile Threat: Dhanush).
- **Causal Chain & Operational Chronology:** First tested from INS Subhadra on April 11, 2000 (initial stabilization failure) and successfully launched on September 21, 2001, and November 7, 2004, Dhanush provided the Indian Navy with ballistic missile handling and stabilization experience prior to INS Arihant's commissioning; following its final user trial on February 23, 2018, satellite imagery confirmed the removal of the launch stabilization pedestals from INS Subhadra and INS Suvarna as the SSBN fleet assumed the sea-based deterrent mission (FAS Indian Nuclear Weapons 2024).

165. INS Arihant (S2 SSBN)

- **Institutional Designation & Sovereign Mandate:** Lead vessel (Project S2, pennant number SSBN-80) of the Arihant-class nuclear-powered ballistic missile submarines, built under the classified Advanced Technology Vessel (ATV) project to establish the sea-based leg of India's nuclear triad (FAS Indian Nuclear Weapons 2024).
- **Geodetic Coordinates & Deployment Nodes:** Constructed at the Ship Building Centre (SBC), Visakhapatnam (17°41'45"N 83°16'05"E) and forward-assigned to the Eastern Naval Command and the underground SSBN bastion at INS Varsha, Rambilli, Andhra Pradesh (17°28'25"N 82°56'28"E) (Bulletin of the Atomic Scientists Indian Nuclear Weapons 2024).
- **Named Leadership & Key Personnel:** Developed by the ATV Programme Headquarters (headed by Vice Admirals DSP Varma, PC Bhasin, and SV Bhokare), BARC (83 MWt PWR reactor team led by Dr. Anil Kakodkar and Dr. Srikumar Banerjee), Larsen & Toubro (hull fabrication at Hazira), and Russia's Rubin Design Bureau (design consultancy) (FAS Indian Nuclear Weapons 2024).
- **Technical & Engineering Parameters:** Length 111.6 m; beam 11.0 m; draught 9.5 m; displacement ~6,000 tonnes surfaced / 6,500 tonnes submerged; propulsion via a single BARC-designed 83 MWt Compact Light Water Reactor (CLWR) burning ~40% enriched uranium (supplied by Rare Materials Plant, Mysuru), driving steam turbines (~47,000 shp) and a single seven-bladed skewback propeller for 12–15 knots surfaced and 24 knots submerged; armed with six 533 mm torpedo tubes and four vertical launch system (VLS) tubes in the dorsal casing housing either 12 K-15 Sagarika SLBMs (3 per tube) or 4 K-4 SLBMs (1 per tube); fitted with USHUS integrated sonar and Panchendriya tactical sonar suite (FAS Indian Nuclear Weapons 2024).
- **Causal Chain & Operational Chronology:** Launched at Visakhapatnam on July 26, 2009, by Gursharan Kaur and Prime Minister Manmohan Singh, INS Arihant achieved reactor criticality on August 10, 2013, was commissioned in August 2016 under Navy Chief Admiral Sunil Lanba, completed its first 20-day nuclear deterrence patrol on November 5, 2018 (announced by Prime Minister Narendra Modi as completing India's nuclear triad), and executed a successful submerged SLBM user-training launch on October 14, 2022 (FAS Indian Nuclear Weapons 2024).

166. INS Arighat (S3 SSBN)

- **Institutional Designation & Sovereign Mandate:** Second Arihant-class nuclear-powered ballistic missile submarine (Project S3, INS Arighat / Arighaat) of the Indian Navy, doubling India's operational SSBN patrol fleet (FAS Indian Nuclear Weapons 2024).
- **Geodetic Coordinates & Deployment Nodes:** Built and commissioned at Ship Building Centre (SBC), Visakhapatnam (17°41'45"N 83°16'05"E), operating in the Bay of Bengal and transitioning to INS Varsha, Rambilli (17°28'25"N 82°56'28"E) (FAS Indian Nuclear Weapons 2024).
- **Named Leadership & Key Personnel:** Commissioned on August 29, 2024, in the presence of Defence Minister Rajnath Singh, National Security Advisor Ajit Doval, Chief of Defence Staff General Anil Chauhan, Chief of the Naval Staff Admiral Dinesh K. Tripathi, and ATV Program Director (FAS Indian Nuclear Weapons 2024).
- **Technical & Engineering Parameters:** Length 111.6 m; beam 11.0 m; submerged displacement ~6,000–6,500 tonnes; powered by an upgraded 83 MWt BARC pressurized water reactor with enhanced acoustic quieting, seven-bladed propeller, and indigenous steel/composite components exceeding 75% domestic content; four dorsal VLS missile tubes carrying 12 K-15 (B-05) SLBMs or 4 K-4 (3,500 km) SLBMs, plus upgraded very-low-frequency (VLF) and extremely-low-frequency (ELF) NC3 communication receivers linked to INS Kattabomman (08°23'14"N 77°45'06"E) (FAS Indian Nuclear Weapons 2024).

- **Causal Chain & Operational Chronology:** Laid down at SBC Visakhapatnam in 2011 and launched on November 19, 2017, to clear assembly hall capacity for S4 and S4*, INS Arighat commenced sea trials in early 2022, was formally commissioned into the Strategic Forces Command on August 29, 2024, and executed a full-range 3,500 km K-4 SLBM test launch in late November 2024, transitioning INS Arihant primarily to training and technology demonstration while Arighat conducts active deterrence patrols (FAS Indian Nuclear Weapons 2024).

167. Project S4 (S4 & S4* Enlarged Arihant-Class SSBNs)

- **Institutional Designation & Sovereign Mandate:** Stretched, heavy-payload second batch of Arihant-class SSBNs—comprising S4 (INS Aridhaman) and S4* (fourth hull)—engineered with a 16–18 m hull plug to double vertical launch tube capacity from four to eight tubes (FAS Indian Nuclear Weapons 2024).
- **Geodetic Coordinates & Deployment Nodes:** Outfitted at Ship Building Centre (SBC), Visakhapatnam (17°41'45"N 83°16'05"E) and assigned to the underground tunnelpens of Project Varsha at Rambilli (17°28'25"N 82°56'28"E), 50 km south of Visakhapatnam (Bulletin of the Atomic Scientists Indian Nuclear Weapons 2024).
- **Named Leadership & Key Personnel:** Engineered by the ATV Programme Headquarters, Naval Science & Technological Laboratory (NSTL) Visakhapatnam, BARC, DRDO, and Larsen & Toubro Defence (FAS Indian Nuclear Weapons 2024).
- **Technical & Engineering Parameters:** Length ~125–130 m (16 to 18 meters longer than S2/S3 as verified by satellite imagery); beam ~11.0–12.0 m; submerged displacement ~7,000 tonnes; propulsion via an uprated BARC 83+ MWt CLWR pressurized water reactor with improved primary coolant circulation; armed with 8 VLS launch tubes capable of carrying 8 intermediate-range K-4 (3,500 km) SLBMs or 24 short-range K-15 (700 km) SLBMs (FAS Indian Nuclear Weapons 2024).
- **Causal Chain & Operational Chronology:** Approved to bridge the industrial production line between the initial 4-tube S2/S3 boats and the future 13,500-tonne, 12-tube S5-class SSBNs (powered by a 190 MWt BARC reactor), S4 was launched at Visakhapatnam on November 23, 2021, followed by S4* launched in October 2024, directly enabling continuous at-sea deterrence (CASD) armed with 3,500 km K-4 missiles capable of striking Chinese targets from the protected bastion of the northern Bay of Bengal (FAS Indian Nuclear Weapons 2024).

168. K-15 Sagarika (B-05 SLBM)

- **Institutional Designation & Sovereign Mandate:** Two-stage, solid-fuel, short-range submarine-launched ballistic missile (designated K-15 / B-05) developed under the Kalam (K) missile series to arm the Arihant-class SSBNs (CSIS Missile Threat: Sagarika/Shaurya).
- **Geodetic Coordinates & Deployment Nodes:** Developed and tested from submersible pontoons (P78) and INS Arihant off Visakhapatnam (17°40'00"N 83°25'00"E) in the Bay of Bengal; manufactured at BDL Hyderabad (17°19'30"N 78°30'15"E) (CSIS Missile Threat: Sagarika/Shaurya).
- **Named Leadership & Key Personnel:** Led by Project Director A.K. Chakrabarti and Programme Director Avinash Chander at DRDL/ASL Hyderabad, with underwater launch tube integration by R&DE(E) Pune and NSTL Visakhapatnam (CSIS Missile Threat: Sagarika/Shaurya).
- **Technical & Engineering Parameters:** Length 10.8 m; diameter 0.74–0.80 m (packed three missiles per 2.4 m diameter Arihant VLS tube); launch weight 5,500–6,300 kg; two-stage composite solid-propellant motor with cold-gas underwater ejection booster; payload 500–1,000 kg carrying a single 12–20 kt boosted-fission warhead; range 700–750 km (at depressed quasi-ballistic hypersonic trajectory reaching Mach 7.5, sharing aerodynamics with the land-based Shaurya missile); RLG-INS guidance with terminal radar correlation (~20–40 m CEP) (CSIS Missile Threat: Sagarika/Shaurya).
- **Causal Chain & Operational Chronology:** Initiated in the late 1990s as Project B-05, K-15 completed 10 pontoon test launches between 2004 and February 26, 2008, conducted its first submerged launch from INS Arihant on November 25, 2015, and completed two combat-unit salvo trials on August 11–12, 2018, certifying the missile for INS Arihant's maiden deterrence patrol in November 2018 (FAS Indian Nuclear Weapons 2024).

169. K-4 (IRBM-Class SLBM)

- **Institutional Designation & Sovereign Mandate:** Two-stage, solid-fuel intermediate-range submarine-launched ballistic missile (SLBM) giving Indian Navy SSBNs a 3,500 km standoff nuclear strike capability across China and Pakistan from Bay of Bengal patrol bastions (FAS Indian Nuclear Weapons 2024).
- **Geodetic Coordinates & Deployment Nodes:** Tested from submerged pontoons off Visakhapatnam (17°40'00"N 83°25'00"E) impacting into the southern Indian Ocean, and deployed aboard INS Arihant (S2), INS Arighat (S3), S4, and S4* (FAS Indian Nuclear Weapons 2024).
- **Named Leadership & Key Personnel:** Engineered at ASL and DRDL Hyderabad under DRDO Directors Avinash Chander, Dr. G. Satheesh Reddy, and Dr. Samir V. Kamat (FAS Indian Nuclear Weapons 2024).
- **Technical & Engineering Parameters:** Length 12.0 m; diameter 1.3 m; launch weight 17,000 kg; two-stage carbon-composite solid rocket motor utilizing nose-mounted puller thrusters and gas generator ejection to clear the ocean surface before main stage ignition; payload up to 2,000 kg (1,000–2,000 kg) carrying a single high-yield nuclear warhead; operational range 3,500 km; 3D RLG-INS with NavIC/GPS satellite updates and thrust vectoring achieving <20–30 m CEP (FAS Indian Nuclear Weapons 2024; CSIS Missile Threat: Sagarika/Shaurya).
- **Causal Chain & Operational Chronology:** Developed because K-15's 700 km range could not reach Beijing or Islamabad from the Bay of Bengal, K-4 underwent submerged pontoon flight tests beginning in March 2014, executed back-to-back full-range 3,500 km trials on January 19 and January 24, 2020 (and two more six days apart in December 2023), and completed its first submarine-launched operational test from INS Arighat on November 27, 2024 (FAS Indian Nuclear Weapons 2024).

170. Mirage 2000H (Mirage 2000I/TI)

- **Institutional Designation & Sovereign Mandate:** French-built Dassault single-engine multirole strike fighter serving as the primary nuclear gravity-bomb delivery aircraft of the Indian Air Force under Strategic Forces Command tasking (FAS Indian Nuclear Weapons 2024).

- **Geodetic Coordinates & Deployment Nodes:** Based at Maharajpur Air Force Station, Gwalior, Madhya Pradesh (26°17'36"N 78°13'42"E) under Central Air Command, home to No. 1 Squadron ("Tigers") and No. 7 Squadron ("Battle Axes"), as well as No. 9 Squadron ("Wolfpack"), adjacent to a hardened underground weapons storage area (FAS Indian Nuclear Weapons 2024).
- **Named Leadership & Key Personnel:** Operated by the Indian Air Force (Chief of the Air Staff Air Chief Marshal Amar Preet Singh) under SFC dual-hatted nuclear certification; upgraded by Dassault Aviation, Thales, and Hindustan Aeronautics Limited (HAL), Bengaluru (FAS Indian Nuclear Weapons 2024).
- **Technical & Engineering Parameters:** Length 14.36 m; wingspan 9.13 m; max takeoff weight 17,000 kg; powered by one SNECMA M53-P2 afterburning turbofan (95 kN thrust, Mach 2.2 top speed); combat radius ~900–1,550 km with drop tanks; upgraded under a $2.1 billion 2011 contract for 51 aircraft to the Mirage 2000I/TI (2000-5 Mk 2) standard with Thales RDY-2 multi-mode radar, glass cockpit, helmet-mounted display, ICMS Mk 3 electronic warfare suite, and centerline wiring for one 1,000 kg free-fall nuclear gravity bomb (12–45 kt yield) (FAS Indian Nuclear Weapons 2024).
- **Causal Chain & Operational Chronology:** Originally inducted in 1985 and selected in the late 1990s by DRDO/BARC and the IAF to carry India's air-dropped nuclear gravity bomb (tested in inert toss-bombing profiles prior to Pokhran-II), the Mirage 2000H fleet at Maharajpur AFS remains India's veteran airborne nuclear deterrent, sustained through the 2020–2021 acquisition of 24 phased-out French Air Force Mirage airframes for spares and ongoing Mirage 2000I upgrades (FAS Indian Nuclear Weapons 2024).

171. Rafale DH (Rafale EH/DH)

- **Institutional Designation & Sovereign Mandate:** Twin-engine 4.5-generation Dassault omnirole strike fighter acquired with 13 India-Specific Enhancements (ISE) to succeed the Mirage 2000H and Jaguar IS in the IAF strategic nuclear and deep-penetration strike role (FAS Indian Nuclear Weapons 2024).
- **Geodetic Coordinates & Deployment Nodes:** Deployed in two 18-aircraft squadrons: No. 17 Squadron ("Golden Arrows") at Ambala Air Force Station, Haryana (30°22'05"N 76°49'00"E, 220 km from the Pakistani border) and No. 101 Squadron ("Falcons of Chamb and Akhnoor") at Hasimara Air Force Station, West Bengal (26°41'53"N 89°22'08"E, covering the Chumbi Valley/China theatre) (FAS Indian Nuclear Weapons 2024).
- **Named Leadership & Key Personnel:** Procured under the September 2016 €7.87 billion inter-governmental agreement signed by Defence Minister Manohar Parrikar and French Defence Minister Jean-Yves Le Drian; tail number RB-001 (named after Air Chief Marshal R.K.S. Bhadauria) formally received by Defence Minister Rajnath Singh at Mérignac, France in October 2019 (FAS Indian Nuclear Weapons 2024).
- **Technical & Engineering Parameters:** Fleet of 36 aircraft (28 single-seat Rafale EH and 8 twin-seat Rafale DH, tail series RB for DH and BS for EH); length 15.27 m; wingspan 10.90 m; max takeoff weight 24,500 kg; powered by two Safran Snecma M88-2 turbofans (75 kN each with afterburner, Mach 1.8); combat radius >1,850 km; equipped with Thales RBE2-AA AESA radar, SPECTRA EW suite, Israeli Elbit TARGO-II helmet-mounted sight, high-altitude cold-engine start capability, 10-hour flight data recorders, SCALP-EG cruise missiles, Meteor BVRAAMs, and structural/avionics lineage derived from France's ASMP-A nuclear-certified Rafale F3R (FAS Indian Nuclear Weapons 2024).
- **Causal Chain & Operational Chronology:** Delivered between October 2019 and April 2022 alongside newly constructed hardened aircraft shelters (HAS) and weapon storage vaults at Ambala and Hasimara, the Rafale DH/EH provides the SFC with a survivable, electronic-warfare-protected aircraft capable of penetrating contested Chinese and Pakistani integrated air defense systems (FAS Indian Nuclear Weapons 2024).

172. Bhabha Atomic Research Centre (BARC)

- **Institutional Designation & Sovereign Mandate:** India's premier national nuclear physics, reactor engineering, plutonium reprocessing, and nuclear warhead design establishment, operating under the Department of Atomic Energy (DAE) directly overseen by the Prime Minister (BARC Official Portal).
- **Geodetic Coordinates & Facility Nodes:** Primary complex at Trombay, Mumbai, Maharashtra (19°00'28"N 72°55'07"E), operating the 100 MWt Dhruva plutonium production reactor (19°00'32"N 72°55'15"E) and Project Plutonium reprocessing plant; subsidiary strategic complexes at Rare Materials Plant (RMP) Ratnahalli, Mysuru (12°21'18"N 76°32'40"E), Tarapur (19°49'48"N 72°39'18"E), Kalpakkam (12°33'27"N 80°10'31"E), and Challakere (14°23'45"N 76°41'30"E) (BARC Research Reactors Portal; FAS Indian Nuclear Weapons 2024).
- **Named Leadership & Key Personnel:** Founded as the Atomic Energy Establishment, Trombay (AEET) on January 3, 1954, by Dr. Homi Jehangir Bhabha (renamed BARC on January 22, 1967, by PM Indira Gandhi); directed by 14th Director Shri Vivek Bhasin (assumed office September 15, 2023, succeeding Dr. Ajit Kumar Mohanty, now DAE Secretary and AEC Chairman); historical weapon design leads include Dr. Raja Ramanna, Dr. P.K. Iyengar, Dr. R. Chidambaram, and Dr. Anil Kakodkar (BARC Leadership Portal).
- **Technical & Engineering Parameters:** Operates the unsafeguarded 100 MWt heavy-water-moderated, natural-uranium-fueled Dhruva research reactor (criticality August 8, 1985), producing ~20–25 kg of weapons-grade plutonium-239 annually (following the December 31, 2010 shutdown of the 40 MWt CIRUS reactor under the US-India Civil Nuclear Agreement), alongside tritium extraction, beryllium/deuterium-tritide neutron initiators, and metallurgically machined plutonium/composite thermonuclear physics packages (tested at Pokhran in 1974 as "Smiling Buddha" and in May 1998 as the 45 kt Shakti-I thermonuclear device, 15 kt Shakti-II fission device, and sub-kiloton Shakti-III/IV/V devices) (BARC Reactor Group; FAS Indian Nuclear Weapons 2024).
- **Causal Chain & Operational Chronology:** BARC maintains physical and technical custody of India's fissile cores, designs the nuclear physics packages for all Agni, Prithvi, and K-series missiles, and engineered the 83 MWt PWR naval reactors for INS Arihant, INS Arighat, S4, and S4*, forming the scientific pillar of the NCA Executive Council (FAS Indian Nuclear Weapons 2024).

173. Defence Research and Development Organisation (DRDO)

- **Institutional Designation & Sovereign Mandate:** Research and development agency of India's Ministry of Defence responsible for strategic ballistic/cruise missile engineering, high-explosive implosion lenses, arming-fuzing-firing (AFF) systems, reentry vehicles, and Permissive Action Link (PAL) safety architectures (DRDO Official Portal).
- **Geodetic Coordinates & Facility Nodes:** Headquarters at DRDO Bhawan, Rajaji Marg, New Delhi (28°36'16"N 77°12'32"E); Dr. APJ Abdul Kalam Missile Complex comprising ASL, DRDL, and Research Centre Imarat (RCI) at Kanchanbagh, Hyderabad (17°19'55"N 78°30'35"E); Terminal Ballistics Research Laboratory (TBRL), Chandigarh (30°41'15"N 76°53'40"E); and HEMRL, Pune (18°33'40"N 73°47'20"E) (NTI India Missile Chronology).

- **Named Leadership & Key Personnel:** Chaired by Secretary Department of Defence R&D and Chairman DRDO Dr. Samir V. Kamat; historical strategic program leaders include Dr. A.P.J. Abdul Kalam (IGMDP founder and Pokhran-II Chief Coordinator), Dr. K. Santhanam (Pokhran-II Field Director), Dr. V.K. Saraswat, Dr. Avinash Chander, Dr. S. Christopher, and Dr. G. Satheesh Reddy (PIB Mission Divyastra Official Release, Mar 11, 2024).
- **Technical & Engineering Parameters:** Through TBRL Chandigarh, DRDO designs and fabricates the high-explosive implosion lenses, detonators, and burst-height barometric/radar fuzes for BARC's fissile cores; through ASL and RCI Hyderabad, DRDO engineers the solid-propellant rocket motors, carbon-carbon reentry heat shields, MIRV post-boost vehicles, ring-laser gyro navigation suites, and coded electronic safety/arming switches (PAL equivalents) that prevent unauthorized detonation (FAS Indian Nuclear Weapons 2024).
- **Causal Chain & Operational Chronology:** Formed in 1958, DRDO co-executed the 1998 Shakti nuclear tests with BARC (validating weaponization, implosion symmetry, and flight-ready warhead mating) and operates alongside BARC in the NCA Executive Council's custodial loop, validating canisterized pre-mated safety interlocks for Agni-V, Agni-P, and K-4 (MEA CCS Release, Jan 4, 2003).

Section VIII: Islamic Republic of Pakistan NC3, Strategic Plans Division (SPD), SECDIV, Full-Spectrum Deterrence Missile Systems, KRL/Kahuta, PAEC, and Khushab Complex (Entities 174–190)

174. Strategic Plans Division (SPD)

- **Institutional Designation & Sovereign Mandate:** Permanent Secretariat of Pakistan's National Command Authority (NCA), exercising day-to-day administrative, budgetary, operational planning, C4I2SR management, arms control policy, and physical security control over all Pakistani nuclear installations, strategic organizations, and warhead stockpiles (Arms Control Association: Pakistan Nuclear Security).
- **Geodetic Coordinates & Command Nodes:** Headquartered at Chaklala Garrison, Rawalpindi, Punjab (33°36'15"N 73°05'45"E), adjacent to Joint Staff Headquarters (JSHQ) (SIPRI EU Non-Proliferation Paper No. 19).
- **Named Leadership & Key Personnel:** Founded and led for nearly 15 years by its inaugural Director General, Lieutenant General Khalid Ahmed Kidwai (served Feb 2000–Dec 2013, subsequently appointed Advisor to the NCA); succeeded by three-star Pakistan Army Director Generals Lt Gen Zubair Mahmood Hayat (2013–2015), Lt Gen Mazhar Jamil (2015–2017), Lt Gen Sarfraz Sattar (2017–2019), Lt Gen Nadeem Zaki Manj (2019–2022), and Lt Gen Yusuf Jamal (Arms Control Association: Pakistan Nuclear Security; CSIS Missile Threat: Shaheen 3).
- **Technical & Structural Architecture:** Organized into four core directorates—(1) Operations and Planning Directorate, (2) Computerized Command, Control, Communications, Information, Intelligence and Surveillance Directorate (C4I2SR, assisted by the Strategic Forces Communications Planning [SFCP] cell), (3) Strategic Weapons Development Directorate (auditing and coordinating PAEC, KRL, NESCOM, SUPARCO), and (4) Arms Control and Disarmament Affairs (ACDA) Directorate—plus the 25,000+ personnel SPD Security Division commanded by a two-star Major General operating the Personnel Reliability Program (PRP), Human Reliability Program (HRP), and three-tier perimeter defense and counter-intelligence around all nuclear sites (Arms Control Association: Pakistan Nuclear Security).
- **Causal Chain & Operational Chronology:** Created in 1999–2000 following the May 1998 Chagai tests and the 1999 Kargil crisis—and strengthened under the National Command Authority Act of 2010—the SPD centralized custody over KRL and PAEC, terminating autonomous scientific proliferation networks and establishing unified military-institutional oversight over an estimated ~170 nuclear warheads (FAS Pakistan Nuclear Weapons 2023).

175. Strategic Export Control Division (SECDIV)

- **Institutional Designation & Sovereign Mandate:** Multidisciplinary statutory division housed within Pakistan's Ministry of Foreign Affairs (MOFA) created in April 2007 to administer, license, enforce, and prosecute strategic export controls pursuant to the Export Control on Goods, Technologies, Material and Equipment related to Nuclear and Biological Weapons and their Delivery Systems Act, 2004 (MOFA Pakistan Strategic Exports Control Regime).
- **Geodetic Coordinates & Command Nodes:** Located at the Ministry of Foreign Affairs, Constitution Avenue, G-5/1, Islamabad (33°43'17"N 73°05'31"E) (MOFA Contact Directory).
- **Named Leadership & Key Personnel:** Governed by a high-level Oversight Board chaired by the Foreign Secretary of Pakistan (Amna Baloch) with representatives from SPD, PAEC, Pakistan Nuclear Regulatory Authority (PNRA), Ministry of Commerce, Ministry of Defence, and Federal Board of Revenue (Customs); operational leadership includes Director General (SECDIV) Brig. Haroon Rashid / Zafar Ali and Director (Enforcement) Asim Rehman (MOFA Contact Directory; MOFA Oversight Board Release).
- **Technical & Structural Architecture:** Enforces the Export Control Act 2004 (carrying penalties of up to 14 years imprisonment, Rs. 5 million fine, and total confiscation of offender property and assets worldwide), the Export Control (Licensing and Enforcement) Rules 2009 (revised 2022), and the National Control Lists (issued 2005 and revised in 2011, 2015, 2016, 2018, 2022, and on October 13, 2025 vide S.R.O. 1977(I)/2025 to harmonize with NSG, MTCR, and Australia Group standards), running an Internal Compliance Program (ICP) and Commodity Identification Training (CIT) for customs and border officers (MOFA Revised Control Lists Notification, Jan 30, 2026).
- **Causal Chain & Operational Chronology:** Directly triggered by the 2003–2004 exposure of the illicit A.Q. Khan centrifuge proliferation network and UN Security Council Resolution 1540 (2004), Pakistan enacted the Export Control Act in September 2004 and stood up SECDIV in 2007 under NCA/MOFA supervision to seal dual-use supply chains and enforce statutory end-user verification (Arms Control Association: Pakistan Nuclear Security).

176. NC3 Pakistan (National Command Authority — ECC, DCC, C4I2SR & PALS)

- **Institutional Designation & Sovereign Mandate:** Three-tiered national nuclear command, control, and communications (NC3) system of Pakistan, established on February 2, 2000, by the National Security Council and codified into statutory law by the National Command Authority Act, 2010 (Arms Control Association: Pakistan Nuclear Security; NTI PAEC Profile).

- **Geodetic Coordinates & Command Nodes:** Prime Minister's Secretariat (33°43'45"N 73°05'50"E), Islamabad, and Joint Staff Headquarters / SPD Secretariat, Chaklala, Rawalpindi (33°36'15"N 73°05'45"E), connected via redundant fiber-optic, microwave, and satellite C4I2SR links to Army Strategic Forces Command (ASFC), Air Force Strategic Force Command (AFSC), and Naval Strategic Force Command (NSFC) (Arms Control Association: Pakistan Nuclear Security).
- **Named Leadership & Key Personnel:** Chaired by the Prime Minister of Pakistan (Shehbaz Sharif; since the November 2009 presidential ordinance and 2010 NCA Act transferred the chairmanship from the President to the Prime Minister); includes the Minister of Foreign Affairs (Ishaq Dar), Minister of Defence (Khawaja Muhammad Asif), Minister of Finance, Minister of Interior, Chairman Joint Chiefs of Staff Committee (Gen Sahir Shamshad Mirza), Chief of Army Staff (Gen Asim Munir), Chief of Naval Staff (Adm Naveed Ashraf), Chief of Air Staff (ACM Zaheer Ahmad Baber Sidhu), and DG SPD (MOFA Arms Control and Disarmament; Arms Control Association: Pakistan Nuclear Security).
- **Technical & Structural Architecture:** Tier 1 comprises two committees: (a) the **Employment Control Committee (ECC)**, the apex politico-military body setting nuclear policy, Full Spectrum Deterrence doctrine, and wartime nuclear release authority by consensus/vote, and (b) the **Development Control Committee (DCC)**, the military-scientific body (comprising the PM, CJCSC, Service Chiefs, DG SPD, and heads of PAEC, KRL, NESCOM, SUPARCO) that translates ECC policy into weapon force goals and R&D; Tier 2 is the **SPD** with its **C4I2SR Directorate**; Tier 3 comprises the three **Service Strategic Forces Commands** (ASFC, AFSC, NSFC); warhead authentication enforces a two-man/three-man rule and indigenous Pakistani **Permissive Action System (PAS / PALS)** requiring a 12-digit alphanumeric authentication code generated by the NCA to mate and unlock warhead firing circuits (Arms Control Association: Pakistan Nuclear Security).
- **Causal Chain & Operational Chronology:** Institutionalized in February 2000 to establish formal command and custody over Pakistan's nuclear arsenal, the NCA convened during the February 2019 Balakot crisis to signal Full Spectrum Deterrence readiness while maintaining centralized SPD custody over warheads stored in underground cross-shaped bunkers at strategic missile garrisons (FAS Indian Nuclear Weapons 2024; FAS Pakistan Nuclear Weapons 2023).

177. Shaheen-I (Hatf-IV / Shaheen-IA)

- **Institutional Designation & Sovereign Mandate:** Single-stage, solid-fuel, road-mobile short-range ballistic missile (Hatf-IV) and its extended-range precision variant (Shaheen-IA) operated by Pakistan's Army Strategic Forces Command (CSIS Missile Threat: Shaheen 1 (Hatf 4)).
- **Geodetic Coordinates & Deployment Nodes:** Assembled by the National Development Complex (NDC) in the Kala Chitta Dahr mountain range near Fatehjang (33°37'30"N 72°32'00"E); flight-tested at Sonmiani Flight Test Range, Balochistan (25°11'38"N 66°44'47"E); deployed at ASFC garrisons such as Sargodha (32°02'30"N 72°40'00"E), Gujranwala, and Akro (25°31'40"N 68°08'10"E) (CSIS Missile Threat: Shaheen 1 (Hatf 4); FAS Pakistan Nuclear Weapons 2023).
- **Named Leadership & Key Personnel:** Developed by PAEC's National Development Complex (NDC) under Dr. Samar Mubarakmand and NESCOM, with operational command under ASFC and SPD (CSIS Missile Threat: Shaheen 1 (Hatf 4); SIPRI EU Non-Proliferation Paper No. 19).
- **Technical & Engineering Parameters:** Length 12.0 m; diameter 1.0 m; launch weight 9,500–10,000 kg; single-stage solid-propellant rocket motor derived from the Chinese DF-15 (M-9) architecture; payload 700–1,000 kg carrying a single 5–12 kt or 15–35 kt nuclear warhead in a separating conical reentry vehicle equipped with a Post-Separation Attitude Correction System (PSACS) using small vernier thrusters; range 650–750 km (Shaheen-I) and 900 km (Shaheen-IA); guidance via strapdown INS with terminal PSACS corrections yielding 200 m CEP (Shaheen-I) and <50 m CEP (Shaheen-IA); carried on a 4-axle (Shaheen-I) or 5-axle WS21200 (Shaheen-IA) road-mobile TEL (CSIS Missile Threat: Shaheen 1 (Hatf 4)).
- **Causal Chain & Operational Chronology:** Initiated by NDC in 1993 to give PAEC a solid-fuel ballistic missile superior in reaction time and survivability to KRL's liquid-fuel Ghauri, Shaheen-I was unveiled in March 1999, flight-tested on April 15, 1999, and inducted into ASFC in March 2003; the 900 km Shaheen-IA variant was first tested on April 25, 2012, and again on March 26, 2021, and featured in the 2022 Pakistan Day Parade (CSIS Missile Threat: Shaheen 1 (Hatf 4); FAS Pakistan Nuclear Weapons 2023).

178. Shaheen-II (Hatf-VI)

- **Institutional Designation & Sovereign Mandate:** Two-stage, solid-fuel, road-mobile medium-range ballistic missile serving as the primary long-range solid-propellant backbone of Pakistan's Army Strategic Forces Command (CSIS Missile Threat: Shaheen 2 (Hatf 6)).
- **Geodetic Coordinates & Deployment Nodes:** Manufactured at NDC Fatehjang (33°37'30"N 72°32'00"E) and Tarnawa (33°37'05"N 72°41'15"E); deployed at hardened ASFC missile garrisons including Khuzdar (27°46'55"N 66°38'15"E) and Sargodha (FAS Pakistan Nuclear Weapons 2023).
- **Named Leadership & Key Personnel:** Engineered by NDC/NESCOM under Dr. Samar Mubarakmand and supervised by SPD DG Lt Gen Khalid Kidwai and successors (CSIS Missile Threat: Shaheen 2 (Hatf 6)).
- **Technical & Engineering Parameters:** Length 17.2 m; diameter 1.4 m; launch weight 23,600–25,000 kg; two-stage solid-propellant rocket motors (utilizing a 1.4 m booster first stage and a modified 1.0 m Shaheen-I motor as the second stage and RV); payload 700–1,230 kg carrying a single 10–40 kt nuclear warhead; range 1,500–2,000 km (reported by ISPR as 1,500 km and assessed by US NASIC at 2,000 km); separating reentry vehicle equipped with four small liquid/gas attitude-control motors for terminal accuracy (~350 m to <100 m CEP) and BMD evasion; transported on a 6-axle road-mobile TEL (WS21200 / MAZ-547V configuration) (CSIS Missile Threat: Shaheen 2 (Hatf 6); Bulletin of the Atomic Scientists Pakistani Nuclear Weapons 2021).
- **Causal Chain & Operational Chronology:** First displayed in March 2000 and flight-tested on March 9, 2004, Shaheen-II completed operational training launches in November 2014 and May 2019, providing Pakistan with a rapid-reaction solid-fuel MRBM capable of striking major Indian strategic centers without the multi-hour liquid-fueling vulnerability of Ghauri (CSIS Missile Threat: Shaheen 2 (Hatf 6)).

179. Shaheen-III (MRBM)

- **Institutional Designation & Sovereign Mandate:** Two-stage, solid-fuel, road-mobile medium-range ballistic missile with a 2,750 km range, engineered as the longest-range operational ballistic missile in Pakistan's strategic arsenal (CSIS Missile Threat: Shaheen 3).

- **Geodetic Coordinates & Deployment Nodes:** Produced at NDC Fatehjang (33°37'30"N 72°32'00"E) and tested from Balochistan/Sonmiani (25°11'38"N 66°44'47"E) into the Arabian Sea (CSIS Missile Threat: Shaheen 3; FAS Pakistan Nuclear Weapons 2023).
- **Named Leadership & Key Personnel:** Strategic requirement defined by SPD Advisor Lt Gen Khalid Kidwai (who stated publicly at the March 2015 Carnegie International Nuclear Policy Conference that the 2,750 km range was specifically selected to reach India's Andaman and Nicobar Islands); engineered by NESCOM/NDC (CSIS Missile Threat: Shaheen 3).
- **Technical & Engineering Parameters:** Length ~19.0–20.0 m; diameter 1.4 m; launch weight ~26,000–27,000 kg; two-stage solid-propellant rocket motor; payload 700–1,000 kg carrying a single 10–40 kt nuclear warhead (and serving as the booster foundation for the MIRV-capable Ababeel missile tested in January 2017 and October 2023); maximum range 2,750 km; transported on a Chinese-supplied 8-axle (16x16 WS51200) road-mobile TEL (CSIS Missile Threat: Shaheen 3; FAS Pakistan Nuclear Weapons 2023).
- **Causal Chain & Operational Chronology:** First flight-tested on March 9, 2015, displayed at the March 23, 2016 Pakistan Day Parade, and subsequently flight-tested on December 11, 2015, January 20, 2021, and April 9, 2022, Shaheen-III was developed specifically to deny the Indian Strategic Forces Command any sanctuary or rear-area second-strike staging bases in eastern India or the Andaman and Nicobar archipelago in the Bay of Bengal (CSIS Missile Threat: Shaheen 3).

180. Ghauri (Hatf-V)

- **Institutional Designation & Sovereign Mandate:** Single-stage, liquid-fuel, road-mobile medium-range ballistic missile (Hatf-V) developed by Khan Research Laboratories (KRL) and operated by Pakistan's Army Strategic Forces Command (CSIS Missile Threat: Ghauri (Hatf 5)).
- **Geodetic Coordinates & Deployment Nodes:** Developed at KRL Kahuta (33°35'42"N 73°23'24"E); flight-tested near Jhelum/Tilla Jogian (32°56'30"N 73°43'30"E) and Balochistan; deployed at ASFC garrisons including Sargodha and Khuzdar (27°46'55"N 66°38'15"E) (CSIS Missile Threat: Ghauri (Hatf 5); FAS Pakistan Nuclear Weapons 2023).
- **Named Leadership & Key Personnel:** Spearheaded by Dr. Abdul Qadeer Khan at KRL during the 1990s in institutional competition with Dr. Samar Mubarakmand's PAEC/NDC solid-fuel program (CSIS Missile Threat: Ghauri (Hatf 5); SIPRI EU Non-Proliferation Paper No. 19).
- **Technical & Engineering Parameters:** Length 15.9 m; diameter 1.35 m; launch weight 15,850 kg; single-stage liquid-propellant engine burning TM-185 fuel (80% kerosene, 20% gasoline) and AK-27I inhibited red fuming nitric acid (IRFNA) oxidizer, structurally derived from the North Korean Nodong-1 (12–25 Nodong airframes/assemblies acquired by Pakistan in the 1990s); payload 700–1,200 kg separating warhead carrying a 12–35 kt HEU implosion nuclear weapon; range 1,250–1,500 km (with an aluminum-alloy Ghauri-II / Hatf-VA prototype tested to 1,800 km); inertial guidance with jet-vane thrust vectoring (~2,500 m baseline CEP, refined with upgraded digital INS in the 2018–2023 Ghauri Weapon System tests); carried on a 5-axle road-mobile TEL (CSIS Missile Threat: Ghauri (Hatf 5)).
- **Causal Chain & Operational Chronology:** Procured through the 1990s KRL–DPRK technology barter (in which KRL provided centrifuge equipment/expertise while North Korea supplied Nodong-1 missiles), Ghauri conducted its maiden test flight on April 6, 1998 (directly precipitating the May 1998 South Asian nuclear tests), entered ASFC service on January 8, 2003, and underwent ASFC training launches on April 15, 2015, October 8, 2018, and October 24, 2023 (CSIS Missile Threat: Ghauri (Hatf 5); FAS Pakistan Nuclear Weapons 2023).

181. Ghaznavi (Hatf-III)

- **Institutional Designation & Sovereign Mandate:** Single-stage, solid-fuel, road-mobile short-range ballistic missile (Hatf-III) delivering tactical and operational nuclear or conventional payloads up to 290–300 km (CSIS Missile Threat: Ghaznavi (Hatf 3)).
- **Geodetic Coordinates & Deployment Nodes:** Manufactured at NDC Fatehjang (33°37'30"N 72°32'00"E) and the M-11 assembly plant at Sargodha (32°02'30"N 72°40'00"E); deployed with ASFC units near the eastern border including Gujranwala (32°14'15"N 74°07'45"E) and Pano Aqil (27°50'20"N 69°08'30"E) (CSIS Missile Threat: Ghaznavi (Hatf 3); Bulletin of the Atomic Scientists Pakistani Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** Engineered by NDC/NESCOM and operated by the Army Strategic Forces Command under SPD oversight (CSIS Missile Threat: Ghaznavi (Hatf 3)).
- **Technical & Engineering Parameters:** Length 8.5 m; diameter 0.8 m; launch weight 4,650 kg; single-stage solid-propellant rocket motor directly derived from China's DF-11 (M-11, of which Pakistan acquired 34 missiles in 1991); payload 500–700 kg carrying a single 5–12 kt miniaturized nuclear warhead or conventional unitary/submunition warhead; range 290–300 km; strapdown inertial guidance with four aerodynamic control fins on the separating reentry vehicle achieving 250 m CEP (50 m with terminal optical/radar correlation); mounted on a 4-axle (8x8) road-mobile TEL (CSIS Missile Threat: Ghaznavi (Hatf 3)).
- **Causal Chain & Operational Chronology:** After terminating an early indigenous Hatf-3 concept in 1991 upon receiving Chinese M-11 missiles, Pakistan reverse-engineered and domestically produced the system as Ghaznavi, conducting its first acknowledged test flight on May 26, 2002, inducting it into ASFC in February 2004, and executing Pakistan's first night ballistic missile readiness launch on August 29, 2019, followed by training launches in January 2020 and February 2021 (CSIS Missile Threat: Ghaznavi (Hatf 3); Bulletin of the Atomic Scientists Pakistani Nuclear Weapons 2021).

182. Hatf-IX (Nasr)

- **Institutional Designation & Sovereign Mandate:** Single-stage, solid-fuel, multi-tube tactical nuclear short-range ballistic missile system engineered to execute Pakistan's "Full Spectrum Deterrence" doctrine by countering Indian Army "Cold Start" / Proactive Strategy armored thrusts at the battlefield level (CSIS Missile Threat: Nasr (Hatf 9)).
- **Geodetic Coordinates & Deployment Nodes:** Produced by NDC/NESCOM near Rawalpindi/Fatehjang (33°37'30"N 72°32'00"E) and deployed at forward ASFC garrisons within 60–70 km of the Indian border, including Gujranwala (32°14'15"N 74°07'45"E) and Pano Aqil (27°50'20"N 69°08'30"E) (FAS Pakistan Nuclear Weapons 2023).
- **Named Leadership & Key Personnel:** Conceived under SPD DG Lt Gen Khalid Kidwai and engineered by NESCOM/NDC using compact plutonium-239 pits derived from the Khushab heavy-water reactors and PINSTECH New Labs reprocessing plant (ISIS Report: Pakistan WGU and WGPu Inventory; CSIS Missile Threat: Nasr (Hatf 9)).
- **Technical & Engineering Parameters:** Length ~6.0 m; diameter 0.4 m (400 mm, derived from the Chinese WS-2 heavy multiple rocket launcher family); launch weight ~1,200 kg; single-stage solid-propellant rocket motor with in-flight maneuverability to defeat terminal air defenses; payload ~400 kg carrying a sub-kiloton to 5 kt linear-implosion or levitated-pit plutonium tactical nuclear warhead; range 60–70 km (extended from

60 km to 70 km in July 2017); launched in salvos from a 4-tube (quad-cell) or 2-tube (twin-box) container mounted on an 8x8 cross-country shoot-and-scoot TEL (CSIS Missile Threat: Nasr (Hatf 9); FAS Pakistan Nuclear Weapons 2023).

- **Causal Chain & Operational Chronology:** Unveiled in its maiden flight test on April 19, 2011, and tested in four-missile salvo mode on May 29, 2012, November 5, 2013, September 26, 2014, July 5, 2017, and January 24 & 31, 2019, Nasr operationalized Khushab-produced weapons-grade plutonium into a low-yield battlefield nuclear weapon while remaining under centralized NCA/SPD Permissive Action System control rather than delegated to field commanders (CSIS Missile Threat: Nasr (Hatf 9); FAS Pakistan Nuclear Weapons 2023).

183. Babur-1 (Hatf-VII / Babur-1A)

- **Institutional Designation & Sovereign Mandate:** Subsonic, low-observable, terrain-hugging ground-launched cruise missile (GLCM) designated Hatf-VII Babur-1 and upgraded as Babur-1A, providing precision nuclear and conventional land-attack capabilities (CSIS Missile Threat: Babur (Hatf 7)).
- **Geodetic Coordinates & Deployment Nodes:** Developed and manufactured by NESCOM's Air Weapons Complex (AWC) / National Development Complex at Wah/Fatehjang (33°46'20"N 72°45'10"E) and deployed at ASFC garrisons such as Akro, Sindh (25°31'40"N 68°08'10"E), where satellite imagery confirms 5-axle Babur TELs and cross-shaped underground storage bunkers (FAS Pakistan Nuclear Weapons 2023).
- **Named Leadership & Key Personnel:** Engineered by NESCOM under Dr. Samar Mubarakmand and operated by ASFC under SPD direction (CSIS Missile Threat: Babur (Hatf 7)).
- **Technical & Engineering Parameters:** Length 6.2 m; diameter 0.52 m; wingspan 2.67 m (pop-out mid-body wings); launch weight ~1,500 kg; propulsion via solid-propellant tandem booster and micro-turbofan/turbojet sustainer engine cruising at Mach 0.8 (880 km/h); payload 450–500 kg carrying a 5–12 kt miniaturized plutonium nuclear warhead or conventional warhead; operational range 350–450 km (Babur-1 / Babur-1A); guidance via INS, GPS/GLONASS, Terrain Contour Matching (TERCOM), and Digital Scene Matching and Area Co-relation (DSMAC) achieving single-digit-meter CEP; launched from a 4-tube (quad) box or 3-tube cylindrical canister launcher mounted on a 5-axle (10x10) road-mobile TEL (CSIS Missile Threat: Babur (Hatf 7); CSIS Hatf-7 Tag).
- **Causal Chain & Operational Chronology:** Initiated in the late 1990s (drawing on reverse-engineered US BGM-109 Tomahawk cruise missiles recovered in Balochistan/Afghanistan in 1998 and Chinese DH-10 technology) to counter India's BrahMos cruise missile and emerging ballistic missile defense shield, Babur-1 conducted an unannounced maiden flight test on August 11, 2005, entered service around 2010, and was flight-tested as the 450 km Babur-1A with upgraded avionics on February 11, 2021 (CSIS Missile Threat: Babur (Hatf 7); CSIS Hatf-7 Tag).

184. Babur-2 (Babur-1B Extended-Range GLCM)

- **Institutional Designation & Sovereign Mandate:** Extended-range ground-launched cruise missile (designated interchangeably by ISPR as Babur Weapon System-1B / Babur-2) doubling the strike radius of the baseline Babur-1 to 700 km (CSIS Missile Threat: Babur (Hatf 7); CSIS Hatf-7 Tag).
- **Geodetic Coordinates & Deployment Nodes:** Manufactured by NESCOM/NDC near Islamabad/Fatehjang (33°37'30"N 72°32'00"E) and tested at Sonmiani Flight Test Range (25°11'38"N 66°44'47"E) (CSIS Missile Threat: Pakistan Test Fires Hatf 7 Babur).
- **Named Leadership & Key Personnel:** Developed by NESCOM in coordination with SPD DG Lt Gen Mazhar Jamil and Lt Gen Nadeem Zaki Manj (CSIS Missile Threat: Pakistan Test Fires Hatf 7 Babur; FAS Pakistan Nuclear Weapons 2023).
- **Technical & Engineering Parameters:** Length ~6.5–7.0 m; diameter 0.52 m; refined low-observable airframe, extended fuel tankage, and higher-efficiency turbofan engine; payload 450–500 kg carrying a 5–12 kt nuclear warhead or conventional precision land/anti-ship warhead; range 700 km (reported up to 900 km in December 2021 extended-range test); TERCOM/DSMAC/INS guidance allowing precision terrain-hugging flight in the absence of satellite navigation; deployed in a 3-tube canisterized Mobile Launch Vehicle (MLV) on a 5-axle 10x10 TEL (CSIS Missile Threat: Babur (Hatf 7); FAS Pakistan Nuclear Weapons 2023).
- **Causal Chain & Operational Chronology:** First flight-tested on December 14, 2016, followed by an April 14, 2018 launch and a December 21, 2021 extended-range test, Babur-2/1B allows ASFC cruise missile batteries to hold low-altitude, radar-evading nuclear strike trajectories across deep Indian targets while also spawning the naval surface-launched Harbah ASCM/LACM tested from PNS Himmat in January 2018 (CSIS Hatf-7 Tag).

185. Babur-3 (SLCM)

- **Institutional Designation & Sovereign Mandate:** Submarine-launched nuclear-capable cruise missile (SLCM) developed for the Pakistan Navy's Naval Strategic Force Command (NSFC) to establish an assured sea-based second-strike nuclear capability (CSIS Missile Threat: Pakistan Conducts First Test of Babur-3).
- **Geodetic Coordinates & Deployment Nodes:** Commanded from Naval Strategic Force Command (NSFC) Headquarters, Naval Headquarters, Sector E-8, Islamabad (33°42'48"N 73°02'32"E), and deployed aboard Pakistan Navy Agosta-90B (Khalid-class: PNS Khalid, PNS Saad, PNS Hamzah) and incoming Hangor-class (Type 039B AIP) submarines operating from PNS Siddiq / Naval Base Ormara (25°12'35"N 64°39'50"E) and Karachi (24°48'20"N 66°58'10"E) (CSIS Missile Threat: Pakistan Conducts First Test of Babur-3; FAS Pakistan Nuclear Weapons 2023).
- **Named Leadership & Key Personnel:** Supervised by SPD and Naval Strategic Force Command (inaugurated May 19, 2012, under Admiral Muhammad Asif Sandila and Vice Admiral Tanveer Faiz) and endorsed by Prime Minister Nawaz Sharif and CJCSC Gen Zubair Mahmood Hayat during its maiden launch (CSIS Missile Threat: Pakistan Conducts First Test of Babur-3).
- **Technical & Engineering Parameters:** Length ~6.2 m; diameter 0.52 m (encapsulated in a buoyant torpedo-launched capsule compatible with 533 mm submarine torpedo tubes); underwater controlled solid booster ejection followed by surface broach, capsule separation, wing deployment, and turbofan ignition; payload 450 kg carrying a 5–12 kt miniaturized plutonium nuclear warhead; range 450 km; guidance via INS, TERCOM, DSMAC, and sea-skimming radar altimeter (CSIS Missile Threat: Pakistan Conducts First Test of Babur-3).
- **Causal Chain & Operational Chronology:** Following the May 2012 establishment of the Naval Strategic Force Command in direct response to India's INS Arihant SSBN program, Pakistan conducted the maiden underwater launch of Babur-3 from a submerged platform in the Indian Ocean on January 9, 2017, and a second successful submerged test on March 29, 2018, completing the technical foundation for Pakistan's sea-based nuclear triad leg (CSIS Missile Threat: Pakistan Conducts First Test of Babur-3).

186. Ra'ad (Hatf-VIII / Ra'ad-I & Ra'ad-II ALCM)

- **Institutional Designation & Sovereign Mandate:** Subsonic, stealth-contoured air-launched cruise missile (ALCM) existing in two operational blocks—Ra'ad-I (350 km) and Ra'ad-II (550–600 km)—giving the Pakistan Air Force Strategic Command (AFSC) a standoff airborne nuclear strike capability (CSIS Missile Threat: Ra'ad; FAS Pakistan Nuclear Weapons 2023).
- **Geodetic Coordinates & Deployment Nodes:** Developed by NESCOM's Air Weapons Complex (AWC) at Kamra/Wah (33°46'20"N 72°45'10"E) and deployed with PAF No. 15 Squadron ("Cobras") and No. 27 Squadron ("Zarrars") flying Mirage IIIEP/RP/5PA ROSE aircraft at PAF Base Rafiqui, Shorkot (30°45'29"N 72°16'57"E) and PAF Base Masroor, Karachi (24°53'37"N 66°56'20"E), as well as integration onto JF-17 Thunder Block II/III aircraft (FAS Pakistan Nuclear Weapons 2023).
- **Named Leadership & Key Personnel:** Engineered by NESCOM/AWC in coordination with the Pakistan Air Force Air Force Strategic Command (AFSC) and SPD (CSIS Missile Threat: Ra'ad).
- **Technical & Engineering Parameters:** Length 4.85 m (reported up to 7.36 m with test telemetry/booster); body cross-section faceted trapezoidal stealth fuselage (~0.5–0.76 m width); launch weight ~1,100 kg (Ra'ad-I/II); turbojet/turbofan engine cruising at Mach 0.8; Ra'ad-I utilizes a twin H-tail configuration with a 350 km range, while Ra'ad-II utilizes an X-tail configuration, lighter composite structures, and upgraded turbofan yielding a 550–600 km range; payload 450 kg carrying a 10–35 kt nuclear warhead; TERCOM/DSMAC/INS guidance with terminal imaging infrared (IIR) or active radar seeker (CSIS Missile Threat: Ra'ad; RECNA Pakistani Nuclear Weapons Capability).
- **Causal Chain & Operational Chronology:** First flight-tested from a PAF Mirage III on August 25, 2007 (completing six Ra'ad-I tests through January 19, 2016), Ra'ad evolved into the extended-range Ra'ad-II unveiled at the March 23, 2017 Pakistan Day Parade and flight-tested on February 18, 2020 (reported at 600 km range by ISPR), allowing aging PAF Mirage III/V strike jets and newer JF-17 fighters to launch nuclear strikes well outside the engagement envelope of Indian S-400 air defense batteries (FAS Pakistan Nuclear Weapons 2023).

187. Khan Research Laboratories (KRL)

- **Institutional Designation & Sovereign Mandate:** Classified uranium enrichment and strategic weapons engineering research institute (officially Dr. A.Q. Khan Research Laboratories, established in July 1976 as Engineering Research Laboratories [ERL] and renamed KRL on May 1, 1981), operating under NCA/SPD oversight (World Nuclear Association: Pakistan; ISIS Report: Pakistan WGU and WGPu Inventory).
- **Geodetic Coordinates & Facility Nodes:** Headquartered at Kahuta, Rawalpindi District, Punjab (33°35'42"N 73°23'24"E), ~30 km southeast of Islamabad (ISIS Report: Pakistan WGU and WGPu Inventory).
- **Named Leadership & Key Personnel:** Founded on July 31, 1976, by metallurgist Dr. Abdul Qadeer Khan (who directed KRL until March 2001) with construction executed by Brigadier (later Lt Gen) Zahid Ali Akbar Khan of the Pakistan Army Corps of Engineers under Prime Minister Zulfikar Ali Bhutto and President Gen Muhammad Zia-ul-Haq; succeeded in 2001 by Dr. Javed Iqbal Mirza, Dr. Muhammad Aslam, and subsequent SPD-appointed Chairmen (World Nuclear Association: Pakistan; SIPRI EU Non-Proliferation Paper No. 19).
- **Technical & Engineering Parameters:** Operates cascades of gas ultracentrifuges—progressing from aluminum-rotor P-1 centrifuges (adapted from Dutch URENCO SNOR/CNOR blueprints acquired by A.Q. Khan at FDO Amsterdam between 1972 and 1975) to maraging-steel-rotor P-2 centrifuges (~5 SWU/yr each, adapted from German G-2 designs, with 8,000–10,000 P-2 centrifuges documented in a single KRL cascade hall) and carbon-fiber/multi-bellows P-3 and P-4 centrifuges (12–25+ SWU/yr)—enriching uranium hexafluoride (UF6) to 90%+ weapons-grade uranium (WGU), alongside developing the Ghauri (Hatf-V) ballistic missile and the Baktar-Shikan/Anza tactical missiles (ISIS Report: Pakistan WGU and WGPu Inventory).
- **Causal Chain & Operational Chronology:**Separated from PAEC in July 1976 as Project-706 to accelerate uranium enrichment after India's 1974 nuclear test, KRL produced weapon-grade HEU by 1983–1984 (incorporating Chinese CHIC-4 implosion weapon design data), supplied the HEU fissile cores detonated by PAEC during the May 28, 1998 Chagai-I tests, and in 1999–2001 was subordinated to the NCA and SPD with its proliferation network dismantled and its remit focused strictly on unsafeguarded uranium enrichment and assigned defense R&D (SIPRI EU Non-Proliferation Paper No. 19; ISIS Report: Pakistan WGU and WGPu Inventory).

188. Kahuta (Kahuta Uranium Enrichment Facility)

- **Institutional Designation & Sovereign Mandate:** Physical industrial complex housing the gas centrifuge enrichment halls, UF6 feed/withdrawal plants, centrifuge manufacturing workshops, and vacuum metallurgy laboratories of Khan Research Laboratories in Kahuta, Punjab (ISIS Report: Pakistan WGU and WGPu Inventory).
- **Geodetic Coordinates & Facility Nodes:** Situated in a heavily fortified valley at Kahuta, Rawalpindi District (33°35'42"N 73°23'24"E), working in tandem with the Gadwal uranium enrichment plant near Wah (33°48'05"N 72°41'38"E) where LEU/MEU is topped off to 90%+ weapon-grade HEU (ISIS Report: Pakistan WGU and WGPu Inventory).
- **Named Leadership & Key Personnel:** Secured by the SPD Security Division (headed by a two-star Major General) with dedicated Pakistan Air Force and Army Air Defence Crotale/LY-80/HQ-9 surface-to-air missile batteries and a permanent no-fly zone (Arms Control Association: Pakistan Nuclear Security).
- **Technical & Engineering Parameters:** Receives uranium hexafluoride (UF6) converted by PAEC's Chemical Processing Plant (CPP) at Dera Ghazi Khan (30°03'20"N 70°38'10"E, processing yellowcake mined from the Bannu Basin and Suleman Range); operates tens of thousands of P-2, P-3, and P-4 supercritical gas centrifuges generating an estimated 45,000–100,000+ SWU/year, contributing to Pakistan's cumulative stockpile of ~5.3 ± 1.5 tonnes of weapons-grade HEU (NTI Nuclear Disarmament Pakistan; ISIS Report: Pakistan WGU and WGPu Inventory).
- **Causal Chain & Operational Chronology:** Selected by Brig Zahid Ali Akbar Khan and Dr. A.Q. Khan in 1976 for its natural terrain shielding and proximity to Rawalpindi, Kahuta survived early centrifuge cascade failures during the October 1979 earthquake, achieved sustained WGU production by 1984, expanded threefold in 1991, and continues to supply HEU for composite-core (U-235/Pu-239) and boosted-fission warheads across Pakistan's strategic missile force (ISIS Report: Pakistan WGU and WGPu Inventory; World Nuclear Association: Pakistan).

189. Pakistan Atomic Energy Commission (PAEC)

- **Institutional Designation & Sovereign Mandate:** Statutory federal scientific and nuclear engineering agency of Pakistan (established March 1956), responsible for uranium mining/milling/conversion, the Khushab heavy-water plutonium production complex, spent-fuel reprocessing at PINSTECH New Labs and Chashma, nuclear weapon design/testing (Chagai-I and Chagai-II), and civilian nuclear power generation (NTI PAEC Profile).
- **Geodetic Coordinates & Facility Nodes:** Headquartered at Constitution Avenue, Sector G-5/1, Islamabad (33°43'28"N 73°05'36"E); operates PINSTECH / New Laboratories at Nilore (33°39'10"N 73°15'40"E), Khushab Nuclear Complex (32°01'50"N 72°12'30"E), Chashma Nuclear Complex and Kundian Reprocessing Plant (32°23'25"N 71°27'45"E), Karachi Nuclear Power Complex (24°50'42"N 66°47'16"E), and Chagai Ras Koh (28°47'32"N 64°57'00"E) and Kharan (28°29'00"N 64°19'00"E) underground test sites (NTI PAEC Profile; SIPRI EU Non-Proliferation Paper No. 19).
- **Named Leadership & Key Personnel:** Founded under Dr. Nazir Ahmad (1956–1960) and Dr. Ishrat Hussain Usmani (1960–1972); weapon program built by Chairman Munir Ahmad Khan (1972–1991), Dr. Ishfaq Ahmad (1991–2001, who led the May 1998 Chagai tests alongside Member Technical Dr. Samar Mubarakmand), Parvez Butt (2001–2006), Anwar Ali (2006–2009), Dr. Ansar Parvez (2009–2015), Muhammad Naeem (2015–2022), and current Chairman Dr. Raja Ali Raza Anwar (NTI PAEC Profile; NTI PAEC Chief Report).
- **Technical & Engineering Parameters:** Manages the entire plutonium fuel cycle (from Dera Ghazi Khan uranium mining and Multan heavy-water production to Khushab irradiation and PINSTECH New Labs / Chashma PUREX reprocessing), fabricated the high-explosive implosion assemblies and diagnostic triggers for the five Chagai-I underground nuclear tests on May 28, 1998 (Ras Koh Hills, maximum yield ~40–45 kt) and the single Chagai-II plutonium/miniaturized test on May 30, 1998 (Kharan Desert, ~15–20 kt), and spawned the National Development Complex (NDC, established 1990) and National Engineering and Scientific Commission (NESCOM, established 2001) that manufacture the Shaheen, Ghaznavi, Nasr, Babur, and Ra'ad missile families (SIPRI EU Non-Proliferation Paper No. 19; NTI PAEC Profile).
- **Causal Chain & Operational Chronology:** Tasked by President Zulfikar Ali Bhutto at the January 20, 1972 Multan meeting to build a nuclear deterrent, PAEC developed the Directorate of Technical Development (DTD), Cold Test sites at Kirana Hills (31°57'50"N 72°41'30"E), the Ras Koh test tunnels (surveyed in 1976), the Khushab plutonium reactors, and the NDC solid-fuel missile complex, operating today under the NCA Development Control Committee and SPD (NTI PAEC Profile; SIPRI EU Non-Proliferation Paper No. 19).

190. Khushab Nuclear Complex (Khushab-I, II, III, IV)

- **Institutional Designation & Sovereign Mandate:** Unsafeguarded military nuclear reactor and heavy-water production complex operated by PAEC to produce weapons-grade plutonium-239 and tritium for Pakistan's miniaturized tactical, cruise, and MIRV nuclear warheads (World Nuclear Association: Pakistan; ISIS Report: Pakistan WGU and WGPu Inventory).
- **Geodetic Coordinates & Facility Nodes:** Located 30 km south of Jauharabad / Khushab in Punjab (32°01'50"N 72°12'30"E), ~200 km south of Islamabad; linked to the nearby heavy-water production plant (32°03'45"N 72°11'55"E) and to spent-fuel reprocessing plants at PINSTECH New Laboratories, Nilore (33°39'10"N 73°15'40"E) and Chashma/Kundian (32°23'25"N 71°27'45"E) (World Nuclear Association: Pakistan; NTI Nuclear Disarmament Pakistan).
- **Named Leadership & Key Personnel:** Operated by PAEC Reactor Directorate under NCA/SPD oversight and defended by the SPD Security Division and multi-layered air defense batteries (SIPRI EU Non-Proliferation Paper No. 19; Arms Control Association: Pakistan Nuclear Security).
- **Technical & Engineering Parameters:** Houses four unsafeguarded natural-uranium-fueled, heavy-water-moderated plutonium production reactors:
    1. **Khushab-I:** 40–50 MWt heavy-water reactor (construction started 1986, operational April 1998), located at 32°02'02"N 72°12'45"E;
    2. **Khushab-II:** ~40–50 MWt heavy-water reactor (construction started 2000–2002, operational late 2009/early 2010), located at 32°01'46"N 72°12'03"E;
    3. **Khushab-III:** ~40–50 MWt heavy-water reactor (construction started 2006, operational late 2013), adjacent to Khushab-II at 32°01'52"N 72°12'03"E;
    4. **Khushab-IV:** Larger ~90 MWt (assessed 50–90 MWt with expanded mechanical-draft cooling towers, construction started 2011, operational January 2015), located at 32°02'05"N 72°11'49"E; collectively generating ~35–50+ kg of weapons-grade plutonium-239 per year (total military plutonium stockpile estimated at ~580 kg, sufficient for 100+ compact plutonium warheads) (World Nuclear Association: Pakistan; NTI Nuclear Disarmament Pakistan).
- **Causal Chain & Operational Chronology:** Initiated by PAEC in 1986 to overcome the mass and diameter constraints of early HEU implosion warheads, Khushab-I went critical in April 1998—just before the Chagai tests—and the rapid construction of Khushab-II, III, and IV between 2000 and 2015 (paired with expanded reprocessing cells at Nilore New Labs and Chashma) directly enabled Pakistan to field compact, lightweight plutonium warheads for the 400 mm Hatf-IX Nasr tactical rocket, the 520 mm Babur-1/2/3 cruise missiles, the Ra'ad ALCM, and the Ababeel MIRV payload bus (ISIS Report: Pakistan WGU and WGPu Inventory; FAS Pakistan Nuclear Weapons 2023).

Section IX: State of Israel Strategic Deterrence (Samson Option), Dimona & Soreq Nuclear Research Centers, Sdot Micha (Wing 2), Jericho Ballistic Missiles, Dolphin/Dolphin-II Submarines, and Strategic Strike Aviation (Entities 191–208)

191. Samson Option (Strategic Deterrence Doctrine & Israeli NC3)

- **Institutional Designation & Sovereign Mandate:** Israel's strategic nuclear deterrence doctrine of last-resort massive retaliation against existential threats to the State of Israel, coupled with the peacetime/crisis policy of "nuclear opacity" (*amimut*) pledged since 1963 that "Israel will not be the first to introduce nuclear weapons to the Middle East" (NTI Nuclear Disarmament Israel; Wilson Center: Arnan Sini Azaryahu Archival Testimony).

- **Geodetic Coordinates & Command Nodes:** Directed from the Prime Minister's Office in Jerusalem (31°46'30"N 35°12'06"E), the Israel Defense Forces (IDF) / Ministry of Defense underground National Management Center ("The Pit" / *Bor*) at HaKirya, Tel Aviv (32°04'32"N 34°47'27"E), and the hardened national leadership bunker complex in the Jerusalem Hills (*Malmab* / Israel Atomic Energy Commission headquarters at Ramat Hachayal, Tel Aviv 32°06'35"N 34°50'10"E) (NTI Israel Atomic Energy Commission; Wilson Center: Arnan Sini Azaryahu Archival Testimony).
- **Named Leadership & Key Personnel:** Formulated by Prime Minister David Ben-Gurion, Shimon Peres, Ernst David Bergmann (first IAEC Chairman), Munya Mardor (RAFAEL Director), Yisrael Galili, and Yigal Allon; operational C2 requires a "double-key system" involving the Prime Minister of Israel (Benjamin Netanyahu), the Minister of Defense (Israel Katz), the Chief of the General Staff (Lt Gen Herzi Halevi / Eyal Zamir), the Director General of the Israel Atomic Energy Commission (Brig Gen [Ret.] Moshe Edri), and the Director of Defense Ministry Security (*Malmab*) (Wilson Center: Arnan Sini Azaryahu Archival Testimony; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Technical & Structural Architecture:** Governs an estimated stockpile of ~90 assembled nuclear warheads (backed by weapon-grade plutonium sufficient for 100–300 weapons) deployed across a strategic triad of Jericho-II/III ballistic missiles at Wing 2 Sdot Micha, nuclear-capable F-15I and F-16I/F-35I strike aircraft, and Dolphin/Dolphin-II submarines equipped with 650 mm tubes (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021; NTI Nuclear Disarmament Israel).
- **Causal Chain & Operational Chronology:** As documented by historian Avner Cohen and the declassified oral history of cabinet insider Arnan "Sini" Azaryahu, Israel's nuclear posture was tested during three historic crises: (1) late May 1967 prior to the Six-Day War, when RAFAEL Director Munya Mardor oversaw the emergency assembly of two rudimentary implosion devices for a potential Sinai demonstration blast ("Operation Shimshon"); (2) October 7, 1973 (the second day of the Yom Kippur War), when Defense Minister Moshe Dayan proposed preparing nuclear weapons for a demonstration option, which Prime Minister Golda Meir, Yisrael Galili, and Yigal Allon rejected while instructing Military Secretary Brig Gen Yisrael Lior to order IAEC Director General Shalheveth Freier to halt assembly under the double-key C2 system; and (3) January–February 1991 during the Gulf War Iraqi Scud attacks against Tel Aviv and Dimona (Wilson Center: Arnan Sini Azaryahu Archival Testimony; FAS Israel's Nuclear Weapons Guide).

192. Shimon Peres Negev Nuclear Research Center (Dimona)

- **Institutional Designation & Sovereign Mandate:** Israel's primary military fissile material production, plutonium reprocessing, tritium extraction, and warhead component fabrication complex (renamed the Shimon Peres Negev Nuclear Research Center in August 2018), operated by the Israel Atomic Energy Commission (IAEC) outside IAEA safeguards (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021; National Security Archive: US Discovery of Dimona).
- **Geodetic Coordinates & Facility Nodes:** Located in the Negev Desert 13 km southeast of the city of Dimona (31°00'05"N 35°08'40"E), protected by dedicated Arrow/Patriot/David's Sling/Yahalom air defense batteries and a strict lethal-exclusion no-fly zone (FAS Israel Archives; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** Procured under the secret October 3, 1957 Franco-Israeli agreement negotiated by Defense Ministry Director General Shimon Peres and Prime Minister David Ben-Gurion with French Prime Minister Maurice Bourgès-Maunoury and CEA engineer Pierre Guillaumat; first project director Col. Manes Pratt; operated under IAEC Director General Brig Gen (Ret.) Moshe Edri (National Security Archive: US Discovery of Dimona).
- **Technical & Engineering Parameters:** Centers on the **IRR-2 (Israel Research Reactor-2)**, a French-designed EL-102 heavy-water-moderated, natural-uranium-fueled reactor (modeled on the Marcoule G-1/EL-3 reactor, utilizing 20 tonnes of Norwegian heavy water procured via the June 1959 Norway-UK-Israel transfer), originally rated at 24 MWt when it achieved criticality in December 1963 and subsequently uprated via cooling system enhancements to 40–70 MWt (producing ~10–20 kg of weapons-grade Pu-239 and lithium-6/tritium annually); immediately adjacent sits **Machon 2 (Institute 2)**, an eight-level subterranean concrete building (disguised at surface level as a two-story laundry/service building with hidden elevators) Constructed by France's Saint-Gobain Techniques Nouvelles (SGN) housing hot cells for PUREX chemical reprocessing of irradiated fuel rods, plutonium metallurgy, and lithium-deuteride/tritium boosting component production; satellite imagery from 2021–2025 confirms a massive multi-story deep excavation (31°00'00"N 35°08'35"E) immediately southwest of the IRR-2 dome as part of a major modernization/life-extension campaign through 2040 (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021; National Security Archive: US Discovery of Dimona).
- **Causal Chain & Operational Chronology:** Excavated in secret starting in March 1958 (initially described to U.S. diplomats as a "textile plant" before U-2 reconnaissance flights and University of Michigan professor Henry Gomberg's intelligence reports exposed the reactor dome and underground reprocessing plant in late 1960), Dimona went critical in 1963, began extracting weapon-grade plutonium at Machon 2 in 1965–1966, and produced an estimated cumulative inventory of 800–900+ kg of weapon-grade plutonium powering Israel's strategic triad (National Security Archive: US Discovery of Dimona; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).

193. Soreq Nuclear Research Center (Yavne)

- **Institutional Designation & Sovereign Mandate:** Second national research center of the Israel Atomic Energy Commission (IAEC), operating the IAEA-safeguarded IRR-1 reactor alongside advanced proton/deuteron linear accelerators, high-energy laser/plasma physics, and—per declassified U.S. Department of Defense assessments—nuclear weapons physics design and theoretical modeling (NTI Soreq Nuclear Research Center; NTI Israel Atomic Energy Commission).
- **Geodetic Coordinates & Facility Nodes:** Located in the coastal sand dunes northwest of Yavne (31°54'12"N 34°42'22"E), immediately adjacent to Palmachim Airbase and missile test launch complex (31°53'52"N 34°41'26"E) (NTI Soreq Nuclear Research Center; FAS Israel Archives).
- **Named Leadership & Key Personnel:** Established under the July 12, 1955 US-Israel Atoms for Peace bilateral agreement under IAEC Chairman Prof. Ernst David Bergmann; overseen by the IAEC Director General (NTI Soreq Nuclear Research Center).
- **Technical & Engineering Parameters:** Houses the **IRR-1 (Israel Research Reactor-1)**, a U.S.-supplied AMF Atomics 5 MWt pool-type light-water-cooled and moderated research reactor fueled by 93% HEU (construction begun January 1958, criticality June 16, 1960), plus a 10 MeV proton cyclotron and the **Soreq Applied Research Accelerator Facility (SARAF)**, a two-phase German Accel Instruments / CEA superconducting radio-frequency (SRF) linear accelerator accelerating continuous-wave proton and deuteron beams up to 40 MeV at 5 mA beam currents for fast-neutron physics and material irradiation studies (NTI Soreq Nuclear Research Center).

- **Causal Chain & Operational Chronology:** Built in 1958–1960 to train Israel's first cadre of reactor physicists and radiochemists before they transferred to the classified Dimona project, Soreq interfaces with nearby Palmachim Airbase and Rafael's Yodefat warhead assembly facility (32°49'25"N 35°16'05"E) to provide high-energy density physics, neutron radiography, and non-nuclear hydrodynamics validation for Israel's strategic stockpile (NTI Soreq Nuclear Research Center; FAS Israel Archives).

194. Sdot Micha Airbase (Kanaf 2 / Area 209)

- **Institutional Designation & Sovereign Mandate:** Restricted Israeli Air Force strategic ballistic missile base and underground nuclear warhead storage complex (officially designated in Israeli Ministerial Committee on Privatization Resolutions M'Kh/24 of August 28, 2005, and M'Kh/2 of December 23, 2013, as "Area 209" / *Shetach Esh 209*, and in IAF hierarchy as *Kanaf 2* / Wing 2) (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021; GlobalMilitary Sdot Micha Airbase).
- **Geodetic Coordinates & Facility Nodes:** Located in the Elah Valley (*Emek HaElah*) near moshav Sdot Micha and Zekharia (Khirbet Zachariah) in the Judean foothills midway between Jerusalem and Ashdod (31°44'19"N 34°55'10"E), spanning 12,550 dunams (~16 sq km) in the Mateh Yehuda Regional Council (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** Founded in 1962 by Colonel (later Major General and IAF Commander) Benjamin "Benny" Peled; operates under the Commander of the Israeli Air Force (Maj Gen Tomer Bar / Maj Gen Omer Tischler) (MiGFlug Israeli Air Force; GlobalMilitary Sdot Micha Airbase).
- **Technical & Engineering Parameters:** Lacks a fixed-wing runway; features two limestone hill-ridge cavern clusters connected by ~1 km dispersal roads—a **Northern Cluster** (31°44'48"N 34°56'02"E) with 14 blast-protected limestone caves and a **Southern Cluster** (31°43'50"N 34°55'25"E) with 9 limestone caves (totaling **23 caves**, each with two blast-door portals housing two TELs per cave for a maximum capacity of **46 Jericho-II/III road-mobile launchers**)—plus high-bay drive-through missile checkout/warhead mating halls, a double-fenced underground **nuclear warhead storage complex** (31°45'22"N 34°54'52"E) with four tunnel portals 1.5 km northwest of the missile caves, solid-rocket motor static test stands, and co-located **Arrow-2** (deployed 2012) and **Arrow-3** (deployed 2017 in four hardened rectangular bunkers) exo-atmospheric anti-ballistic missile batteries (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Causal Chain & Operational Chronology:** Established in 1962 with missile cavern excavation at Hirbat Zachariah starting in late 1967 to house the newly developed French-Israeli MD-620 Jericho-1 missiles, Sdot Micha achieved operational status by 1969–1970, was placed on alert during the October 1973 Yom Kippur War, and underwent major tunnel portal, road, and warhead bunker modernizations between 2014 and 2021 to transition from Jericho-II MRBMs to three-stage Jericho-III ICBMs (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).

195. Wing 2 (Kanaf 2 — Israeli Air Force Ballistic Missile Wing)

- **Institutional Designation & Sovereign Mandate:** Dedicated surface-to-surface strategic ballistic missile wing (*Kanaf 2* / 2nd Wing) of the Israeli Air Force commanding all operational Jericho ballistic missile squadrons, nuclear warhead mating/maintenance detachments, and base defense units at Sdot Micha (GlobalMilitary Sdot Micha Airbase; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Geodetic Coordinates & Command Nodes:** Wing 2 Command Complex at Sdot Micha Airbase (31°44'19"N 34°55'10"E), reporting directly to IAF Headquarters at HaKirya, Tel Aviv (32°04'32"N 34°47'27"E) (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** Commanded by an Israeli Air Force Colonel (Wing Commander) reporting to the Commander of the Israeli Air Force and the IDF General Staff Operations Directorate; historically established under Col. Benny Peled (GlobalMilitary Sdot Micha Airbase).
- **Technical & Structural Architecture:** Exercises administrative, technical, and tactical launch control over three specialized ballistic missile squadrons—**150 Squadron**, **199 Squadron**, and **248 Squadron**—alongside the underground warhead custody detachment and the co-located Arrow-2/Arrow-3 air defense batteries protecting the strategic missile field (GlobalMilitary Sdot Micha Airbase).
- **Causal Chain & Operational Chronology:** Structured within the Israeli Air Force rather than the Ground Forces to centralize strategic deep-strike targeting and NC3 authentication under the IAF underground command center at HaKirya, Wing 2 executes dispersion drills in which Jericho TELs roll out from the 23 limestone caverns onto concealed firing pads or remote dispersal sites upon receiving authenticated double-key directives (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).

196. 150 Squadron (IAF Wing 2)

- **Institutional Designation & Sovereign Mandate:** First operational strategic ballistic missile squadron of the Israeli Air Force at Wing 2, operating road-mobile Jericho-II MRBMs and Jericho-III IRBMs/ICBMs (GlobalMilitary Sdot Micha Airbase; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Geodetic Coordinates & Deployment Nodes:** Dispersed within the limestone cavern complex at Sdot Micha Airbase (31°44'48"N 34°56'02"E / 31°43'50"N 34°55'25"E) (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** Commanded by an IAF Lieutenant Colonel under the Wing 2 Commander at Sdot Micha (GlobalMilitary Sdot Micha Airbase).
- **Technical & Engineering Parameters:** Operates half of the ~24–50 deployed Jericho-II/III road-mobile transporter-erector-launchers housed inside double-portal limestone caves with drive-through warhead mating bays (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021; NTI Nuclear Disarmament Israel).
- **Causal Chain & Operational Chronology:** Stood up at Sdot Micha at the end of the 1960s to field the initial batch of 24–30 Dassault MD-620 Jericho-1 missiles, 150 Squadron transitioned to the two-stage Jericho-2 in 1989–1990 and to the three-stage Jericho-3 beginning in 2011–2014 (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).

197. 199 Squadron (IAF Wing 2)

- **Institutional Designation & Sovereign Mandate:** Second operational strategic ballistic missile squadron of the Israeli Air Force at Wing 2, operating Jericho-II and Jericho-III nuclear-tipped ballistic missiles alongside 150 Squadron (GlobalMilitary Sdot Micha Airbase; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).

- **Geodetic Coordinates & Deployment Nodes:** Based at the Sdot Micha limestone cavern clusters (`31°43'50"N 34°55'25"E` / `31°44'48"N 34°56'02"E`), Mateh Yehuda District (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** Commanded by an IAF Lieutenant Colonel subordinate to Wing 2 (`Kanaf 2`) and IAF Headquarters (GlobalMilitary Sdot Micha Airbase).
- **Technical & Engineering Parameters:** Operates road-mobile Jericho-II/III TELs from blast-hardened horizontal limestone tunnels equipped with autonomous power, environmental control, and encrypted land-line/satellite fire-control links to HaKirya (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Causal Chain & Operational Chronology:** Activated to expand Israel's land-based missile force across the northern and southern cavern clusters at Sdot Micha, 199 Squadron ensures continuous rotational alert and dispersed launch readiness so that a portion of the Jericho force remains continuously ready for immediate exit and launch (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).

198. 248 Squadron (IAF Wing 2 & Palmachim)

- **Institutional Designation & Sovereign Mandate:** Strategic ballistic missile testing, propulsion evaluation, operational conversion, and space-launch support squadron of the Israeli Air Force assigned to Wing 2 (GlobalMilitary Sdot Micha Airbase; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Geodetic Coordinates & Deployment Nodes:** Operates across Sdot Micha Airbase (`31°44'19"N 34°55'10"E`, including the rocket motor test stands at Sdot Micha) and the coastal launch pad at Palmachim Airbase (`31°53'04"N 34°40'49"E`) (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** Operates jointly with engineers from Israel Aerospace Industries (IAI MLM Division at Beer Yaakov), Rafael Advanced Defense Systems, and Tomer (government-owned rocket propulsion company, formerly IMI Givon Division) (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Technical & Engineering Parameters:** Executes live-fire "rocket propulsion system" test launches of Jericho-III/IIIA and Shavit space launch vehicles westward over the Mediterranean Sea from Palmachim (conducted in January 2008, July 2013, May 2015, May 2017, December 2019, January 2020, and July 2024) as well as static solid-rocket motor firings at Sdot Micha (including the April 2021 static motor test) (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021; SIPRI Yearbook 2025 World Nuclear Forces).
- **Causal Chain & Operational Chronology:** Bridging IAI/Rafael/Tomer R&D with operational deployment at 150 and 199 Squadrons, 248 Squadron validates solid-fuel motor reliability, stage separation, reentry vehicle guidance, and Shavit Ofeq reconnaissance satellite launches from Palmachim (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).

199. Jericho-1 (YA-1 SRBM)

- **Institutional Designation & Sovereign Mandate:** Israel's first nuclear-capable short-range solid-fuel ballistic missile (designated YA-1 in Israel and MD-620 by Avions Marcel Dassault), deployed from 1970 until its retirement in the 1990s (CSIS Missile Threat: Jericho 1).
- **Geodetic Coordinates & Deployment Nodes:** Originally flight-tested at Île du Levant, France (`43°02'00"N 06°28'00"E`) from 1965 to 1968, and produced/deployed in Israel at Sdot Micha Airbase (`31°44'19"N 34°55'10"E`) (CSIS Missile Threat: Jericho 1; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** Contracted under the April 26, 1963 agreement signed by Shimon Peres and Marcel Dassault, with propulsion by Nord Aviation/SEPR and Israeli industrial takeover led by Israel Aircraft Industries (IAI) and RAFAEL after Charles de Gaulle's June 1967 arms embargo (CSIS Missile Threat: Jericho 1).
- **Technical & Engineering Parameters:** Length 13.4 m; diameter 0.8 m; launch weight 6,500–6,700 kg; two-stage solid-propellant rocket motor with four clipped-delta tail fins; payload 450–650 kg carrying a single 20 kt plutonium implosion nuclear warhead or HE warhead; range 500 km (sufficient to hold Cairo and Damascus at risk from Sdot Micha); inertial guidance yielding ~1,000 m CEP; deployed on road-mobile TELs (24 to 30 missiles produced by 1970, expanding to ~50–100 missiles) (CSIS Missile Threat: Jericho 1; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Causal Chain & Operational Chronology:** Initiated in 1962–1963 to guarantee an all-weather nuclear delivery vehicle immune to Arab air defenses, MD-620 / Jericho-1 was transferred to domestic Israeli production after the 1967 French embargo, reached operational capability at Sdot Micha in 1970 (confirmed by a declassified 1974 CIA memorandum), was alerted during the October 1973 war, and was phased out in the 1990s in favor of Jericho-2 (CSIS Missile Threat: Jericho 1; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).

200. Jericho-2 (YA-3 MRBM)

- **Institutional Designation & Sovereign Mandate:** Two-stage, solid-fuel, road-mobile medium-range ballistic missile (designated YA-3) serving as the primary land-based nuclear delivery vehicle of the IAF from 1989–1990 and the direct structural basis for the Shavit space launch vehicle (CSIS Missile Threat: Jericho 2).
- **Geodetic Coordinates & Deployment Nodes:** Manufactured by IAI MLM Division at Beer Yaakov (`31°56'25"N 34°50'15"E`) with solid rocket motors cast at Givon/Ramla (`31°55'10"N 34°54'30"E`); flight-tested from Palmachim (`31°53'04"N 34°40'49"E`) and Overberg Test Range, South Africa (`34°35'00"S 20°19'00"E`), and deployed at Sdot Micha (`31°44'19"N 34°55'10"E`) (CSIS Missile Threat: Jericho 2).
- **Named Leadership & Key Personnel:** Developed by Israel Aerospace Industries (IAI MLM Division), Israel Military Industries (IMI / Tomer), and RAFAEL (CSIS Missile Threat: Jericho 2).
- **Technical & Engineering Parameters:** Length 14.0–15.0 m; diameter 1.35 m; launch weight 22,000–26,000 kg; two-stage solid-propellant composite motors (identical to the first two stages of the three-stage Shavit SLV first launched on September 19, 1988); payload 1,000–1,500 kg separating reentry vehicle carrying a 20–200 kt boosted-fission or up to 1 MT two-stage thermonuclear warhead; range 1,500–3,500 km depending on payload weight; strapdown inertial guidance coupled with an active radar terminal seeker on the separating RV (modeled on Pershing II terminal guidance, achieving ~50–100 m CEP); deployed on road-mobile TELs inside Sdot Micha's limestone caverns (CSIS Missile Threat: Jericho 2).
- **Causal Chain & Operational Chronology:** Begun in 1977 (initially alongside Iran under Project Flower prior to the 1979 Iranian Revolution, and later supported through joint test flights with South Africa in 1989 as the RSA-3), Jericho-2 underwent Mediterranean flight tests from Palmachim between May 1987 and 1992 (flying 1,300+ km south of Crete), entered Wing 2 service in 1990 to hold the entire Middle East and western Iran at risk, and is undergoing progressive replacement by Jericho-3 through 2026 (CSIS Missile Threat: Jericho 2).

201. Jericho-3 (YA-4 IRBM/ICBM)

- **Institutional Designation & Sovereign Mandate:** Three-stage, solid-fuel intermediate-to-intercontinental-range ballistic missile (designated YA-4) providing Israel with strategic nuclear reach across the entire Middle East, Iran, Pakistan, and Europe/Russia west of the Urals (CSIS Missile Threat: Jericho 3; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Geodetic Coordinates & Deployment Nodes:** Produced at IAI MLM Beer Yaakov (31°56'25"N 34°50'15"E) and Tomer Ramla (31°55'10"N 34°54'30"E), tested from Palmachim Airbase (31°53'04"N 34°40'49"E), and deployed in upgraded caverns at Sdot Micha Airbase (31°44'19"N 34°55'10"E) (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** Engineered by IAI MLM Division, Rafael, and Tomer under the Israel Ministry of Defense Directorate of Defense Research & Development (MAFAT, headed by Brig Gen [Ret.] Dr. Daniel Gold) (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Technical & Engineering Parameters:** Length 15.5–16.0 m; diameter 1.56 m (sharing first- and second-stage motor lineage with the enlarged Shavit-2 SLV); launch weight 29,000 kg; three-stage solid-propellant rocket motors; payload 1,000–1,300 kg capable of carrying either a single 750 kg nuclear warhead (150–400 kt yield) or 2–3 independently targeted reentry vehicles (MIRVs) with penetration aids/decoys; operational range 4,800–6,500 km; inertial navigation with terminal radar/optical guidance; deployed on heavy road-mobile/cave-mobile TELs at Sdot Micha (CSIS Missile Threat: Jericho 3; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Causal Chain & Operational Chronology:** First test-launched from Palmachim on January 17, 2008, Jericho-3 achieved initial operational capability with Wing 2 in 2011; an upgraded motor variant (Jericho-3A) was flight-tested in July 2013, followed by propulsion tests in 2015, 2017, 2019, 2020, and 2024 and cave upgrades at Sdot Micha starting in 2014 to house the larger 1.56 m diameter canisters (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021; SIPRI Yearbook 2025 World Nuclear Forces).

202. Dolphin-Class (Type 800 Batch 1 Submarines)

- **Institutional Designation & Sovereign Mandate:** First batch of three German-built Type 800 diesel-electric attack/strategic cruise-missile submarines—**INS Dolphin**, **INS Leviathan**, and **INS Tekumah**—operated by the Israeli Navy's 7th Flotilla (*Shayetet 7*) as an invulnerable sea-based second-strike deterrent (NTI Israel Submarine Capabilities; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Geodetic Coordinates & Deployment Nodes:** Built by Howaldtswerke-Deutsche Werft (HDW) in Kiel (54°19'05"N 10°09'25"E) and Thyssen Nordseewerke in Emden, Germany; homeported at the hardened submarine pens of Haifa Naval Base (32°49'38"N 35°00'18"E), conducting patrols in the Mediterranean and transiting the Suez Canal to the Red Sea / Persian Gulf periphery (NTI Israel Submarine Capabilities; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** Approved and subsidized by German Chancellor Helmut Kohl following the 1991 Gulf War; former Chief of the Policy Planning Staff of the German Defense Ministry Hans Rühle publicly confirmed that German officials understood the vessels were designed to carry Israeli nuclear-capable cruise missiles; operated by Commander of the Israeli Navy (Vice Admiral David Saar Salama) and Commander of Shayetet 7 under Prime Minister/IDF General Staff C2 (NTI Israel Submarine Capabilities).
- **Technical & Engineering Parameters:** Comprises three hulls: **INS Dolphin** (commissioned May 29, 1999), **INS Leviathan** (commissioned June 29, 1999), and **INS Tekumah** (commissioned July 25, 2000); length 57.3 m; beam 6.8 m; draught 6.2 m; displacement 1,640 tonnes surfaced / 1,900 tonnes submerged; powered by three MTU 16V 396 SE 84 diesel engines (3.12 MW) and one Siemens Permasyn electric motor (2.85 MW) driving a single shaft to 20 knots submerged (4,500–8,000 nmi range at 8 knots surfaced, up to 30 days patrol endurance); uniquely armed in the bow with **ten torpedo tubes**: six standard **533 mm** tubes (for Atlas Elektronik DM2A4 Seehecht torpedoes and UGM-84 Harpoon missiles) and **four enlarged 650 mm (25.6-inch) hydraulic-ejection tubes** engineered specifically to launch nuclear-armed Popeye Turbo submarine-launched cruise missiles (total internal load of 16 torpedoes/missiles) with Atlas Elektronik ISUS 90-1 TCS fire control (NTI Israel Submarine Capabilities; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Causal Chain & Operational Chronology:** Donated (first two hulls) and co-funded (third hull) by Germany after revelations that German firms had assisted Iraq's chemical and Scud programs prior to 1991, the Dolphin Batch 1 class gave Israel an assured sea-based nuclear second-strike platform immune to a surprise ballistic missile attack on Sdot Micha (NTI Israel Submarine Capabilities).

203. Dolphin-II-Class (AIP Batch 2 Submarines)

- **Institutional Designation & Sovereign Mandate:** Second batch of three stretched, Air-Independent Propulsion (AIP)-equipped strategic submarines—**INS Tanin**, **INS Rahav**, and the lengthened VLS-equipped **INS Drakon**—operated by *Shayetet 7* to maintain continuous submerged strategic deterrence patrols (NTI Israel Submarine Capabilities; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Geodetic Coordinates & Deployment Nodes:** Built at ThyssenKrupp Marine Systems (TKMS / HDW) shipyard in Kiel, Germany (54°19'05"N 10°09'25"E) and based at the expanded AIP submarine pier and underground maintenance facility at Haifa Naval Base (32°49'38"N 35°00'18"E) (NTI Israel Submarine Capabilities).
- **Named Leadership & Key Personnel:** Contracted for the 4th and 5th boats in November 2005 under German Chancellor Gerhard Schröder ($1.17 billion, one-third German-subsidized) and for the 6th boat (INS Drakon) in March 2012 under Chancellor Angela Merkel and Prime Minister Benjamin Netanyahu (followed by a 2022 €3 billion contract for three next-generation **Dakar-class** submarines to replace Batch 1 in the 2030s) (NTI Israel Submarine Capabilities).
- **Technical & Engineering Parameters:** Length 68.6 m (11.3 m longer than Batch 1 on Tanin/Rahav, and ~72.6 m on INS Drakon); beam 6.8 m; displacement 2,050 tonnes surfaced / 2,400 tonnes submerged (higher on Drakon); integrates a Siemens polymer electrolyte membrane (PEM) hydrogen fuel-cell **Air-Independent Propulsion (AIP)** module enabling submerged endurance of **18+ days without snorkeling** (over four times longer than Batch 1) and top submerged speed of 25 knots; armed with **six 533 mm** and **four 650 mm** bow tubes for Popeye Turbo nuclear SLCMs; additionally, the third Dolphin-II boat, **INS Drakon** (launched at Kiel in August 2023), features an enlarged sail and a 2 m × 4 m × 11 m vertical hull plug housing 2 large or 4–8 smaller **Vertical Launch System (VLS)** missile silos (NTI Israel Submarine Capabilities; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Causal Chain & Operational Chronology:** Engineered to eliminate the acoustic and radar vulnerability of periscope-depth diesel snorkeling during long-endurance deterrence patrols off the Levant and Red Sea/Arabian Sea, the Dolphin-II fleet guarantees that at least one AIP-equipped nuclear-capable submarine remains continuously submerged on patrol at all times (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).

204. INS Tanin (Dolphin-II Lead AIP Submarine)

- **Institutional Designation & Sovereign Mandate:** Lead vessel of the Dolphin-II AIP submarine class (4th Dolphin-class overall), named after the biblical sea monster ("Crocodile/Leviathan") and two earlier Israeli submarines (NTI Israel Submarine Capabilities).
- **Geodetic Coordinates & Deployment Nodes:** Launched at HDW Kiel (54°19'05"N 10°09'25"E) in May 2012 and homeported at Haifa Naval Base (32°49'38"N 35°00'18"E) with Shayetet 7 (NTI Israel Submarine Capabilities).
- **Named Leadership & Key Personnel:** Delivered at Kiel on May 3, 2012, and formally welcomed into Haifa Naval Base on September 23, 2014, by Prime Minister Benjamin Netanyahu, Defense Minister Moshe Ya'alon, IDF Chief of Staff Lt Gen Benny Gantz, and Navy Commander Vice Admiral Ram Rothberg (NTI Israel Submarine Capabilities).
- **Technical & Engineering Parameters:** Length 68.6 m; beam 6.8 m; displacement 2,050 tonnes surfaced / 2,400 tonnes submerged; diesel-electric + Siemens PEM fuel-cell AIP propulsion (25 knots submerged, 4,500+ km range, 18+ days continuous submerged AIP endurance); armed with six 533 mm and four 650 mm bow tubes firing DM2A4 torpedoes, Harpoon ASCMs, and nuclear-armed Popeye Turbo SLCMs; crew of 35–45 officers and submariners (NTI Israel Submarine Capabilities; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Causal Chain & Operational Chronology:** Ordered on July 6, 2006, launched in May 2012, and commissioned in September 2014 after 18 months of North Sea/Baltic sea trials, INS Tanin inaugurated Israel's fuel-cell AIP operational capability, extending covert second-strike patrol endurance across the Eastern Mediterranean and Red Sea (NTI Israel Submarine Capabilities).

205. INS Rahav (Second Dolphin-II AIP Submarine)

- **Institutional Designation & Sovereign Mandate:** Second vessel of the Dolphin-II AIP submarine class (5th Dolphin-class overall), expanding Shayetet 7's AIP strategic fleet (NTI Israel Submarine Capabilities).
- **Geodetic Coordinates & Deployment Nodes:** Built at HDW/TKMS Kiel (54°19'05"N 10°09'25"E) and based at Haifa Naval Base (32°49'38"N 35°00'18"E) (NTI Israel Submarine Capabilities).
- **Named Leadership & Key Personnel:** Christened at Kiel on April 29, 2013, delivered in December 2015, and formally commissioned at Haifa on January 12, 2016, in the presence of President Reuven Rivlin, Prime Minister Benjamin Netanyahu, Defense Minister Moshe Ya'alon, and Navy Commander Vice Admiral Eli Sharvit (NTI Israel Submarine Capabilities).
- **Technical & Engineering Parameters:** Length 68.6 m; beam 6.8 m; displacement 2,050 tonnes surfaced / 2,400 tonnes submerged; MTU diesel-generators + Siemens PEM fuel-cell AIP; six 533 mm and four 650 mm torpedo/cruise-missile tubes supporting Popeye Turbo nuclear SLCMs; fitted with Israeli Elbit/Rafael electronic support measures (ESM), encrypted satellite/VLF strategic C2 comms, and ISUS-90-55 combat management suite (NTI Israel Submarine Capabilities; Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Causal Chain & Operational Chronology:** Arriving at Haifa on January 12, 2016, after a 3,000-nautical-mile voyage from Kiel, INS Rahav paired with INS Tanin to allow the Israeli Navy to maintain a dual-AIP submarine rotational cycle,including overt deterrence transits through the Suez Canal in 2020 and 2021 (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).

206. Popeye Turbo (Rafael SLCM / ALCM)

- **Institutional Designation & Sovereign Mandate:** Turbofan-powered strategic cruise missile developed by Rafael Advanced Defense Systems in air-launched (Popeye Turbo ALCM) and submarine-launched (Popeye Turbo SLCM) variants to arm Israel's Dolphin/Dolphin-II submarines with a 1,500 km nuclear second-strike weapon (CSIS Missile Threat: Popeye).
- **Geodetic Coordinates & Deployment Nodes:** Developed and assembled at Rafael's Haifa / Yodefat (32°49'25"N 35°16'05"E) and Kiryat Yam (32°51'30"N 35°04'30"E) complexes; stored and loaded at Haifa Naval Base (32°49'38"N 35°00'18"E) (FAS Israel Archives; CSIS Missile Threat: Popeye).
- **Named Leadership & Key Personnel:** Engineered by Rafael Advanced Defense Systems and integrated onto Dolphin submarines by Shayetet 7 and the Israel Atomic Energy Commission (CSIS Missile Threat: Popeye; NTI Israel Submarine Capabilities).
- **Technical & Engineering Parameters:** Derived from the solid-fuel AGM-142 Popeye (Have Nap, 4.82 m long, 0.533 m diameter, 1,360 kg), the lengthened submarine-launched **Popeye Turbo SLCM** measures ~6.25 m in length and **0.65 m (650 mm)** in diameter to fit the four 650 mm bow tubes of the Dolphin-class submarines; powered by a solid-propellant booster and a Williams/Bet-Shemesh micro-turbofan jet engine burning JP-10 high-density fuel; payload ~200 kg nuclear warhead (estimated 20–200 kt yield utilizing compact Dimona plutonium/tritium pits); operational range **1,500 km** (with the shorter air-launched Popeye Turbo ALCM ranging 200–350 km); guidance via INS, GPS, TERCOM, and terminal imaging infrared/CCD optical seeker (CSIS Missile Threat: Popeye; NTI Israel Submarine Capabilities).
- **Causal Chain & Operational Chronology:** After the U.S. Clinton administration declined Israel's 2000 request to purchase BGM-109 Tomahawk land-attack cruise missiles, Rafael adapted the turbofan Popeye 3 concept (originally pitched to the UK in 1994) into the 650 mm Popeye Turbo SLCM, which U.S. Navy and Pentagon officials observed during a 1,500 km submerged test launch from a Dolphin-class submarine in the Indian Ocean off Sri Lanka in May 2000, fielding the system in 2002 (CSIS Missile Threat: Popeye; NTI Israel Submarine Capabilities).

207. F-15I Ra'am (69 Squadron "Hammers")

- **Institutional Designation & Sovereign Mandate:** Twin-engine, long-range strategic strike fighter ("Thunder" / *Ra'am*, Israeli variant of the Boeing F-15E Strike Eagle) operated by the IAF's 69 Squadron ("The Hammers") as a primary long-range dual-capable nuclear and conventional deep-strike platform (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021; MiGFlug Israeli Air Force).
- **Geodetic Coordinates & Deployment Nodes:** Based at Hatzerim Airbase (Base 6) in the Negev Desert west of Beersheba (31°14'00"N 34°39'45"E), alongside the F-15A/B/C/D *Baz* 106 and 133 Squadrons at Tel Nof Airbase (31°50'22"N 34°49'18"E, which includes a high-security underground weapons storage area) (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** Procured under the January 27, 1994 decision of Prime Minister Yitzhak Rabin (21 aircraft ordered in May 1994 plus 4 in November 1995, delivered 1998–1999); operated by 69 Squadron under the Commander of Hatzerim Airbase and IAF Commander (MiGFlug Israeli Air Force).
- **Technical & Engineering Parameters:** Fleet of 25 aircraft (tail numbers 201–280); length 19.43 m; wingspan 13.05 m; max takeoff weight 36,700 kg; powered by two Pratt & Whitney F100-PW-229 afterburning turbofans (129.7 kN thrust each, Mach 2.5 top speed); unrefueled combat radius >1,800 km (range 4,450 km with conformal fuel tanks and drop tanks, extended further by IAF Boeing 707 *Re'em* and incoming KC-46A

Gideon tankers); fitted with Raytheon AN/APG-70I synthetic-aperture radar, Elbit central mission computers and Kaiser/Elbit holographic HUD, Elisra SPS-2110 integrated electronic warfare system, Rafael Litening/LANTIRN pods, GBU-28/GBU-31 bunker busters, Air LORA / Rampage / Rocks aeroballistic missiles, and nuclear gravity bombs (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021; MiGFlug Israeli Air Force).

- **Causal Chain & Operational Chronology:** Acquired after the 1991 Gulf War to give the IAF an unrefueled, all-weather night strike capability reaching Iran and Iraq, 69 Squadron's F-15I Ra'am jets executed Operation Outside the Box on September 6, 2007—destroying Syria's clandestine North Korean-built Al-Kibar plutonium reactor at Deir ez-Zor—and serve as the heavy airborne delivery arm for Israel's ~30 air-deliverable nuclear gravity bombs (to be augmented by 25 new F-15IA fighters contracted in December 2025) (MiGFlug Israeli Air Force; NTI Nuclear Disarmament Israel).

208. F-35I Adir (140, 116 & 117 Squadrons)

- **Institutional Designation & Sovereign Mandate:** Fifth-generation, single-engine stealth multirole strike fighter (*Adir* / "Mighty One", customized Israeli variant of the Lockheed Martin F-35A Lightning II) operating from Nevatim Airbase to replace legacy F-16C/D squadrons in strategic penetration and potential nuclear strike missions (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021; MiGFlug Israeli Air Force).
- **Geodetic Coordinates & Deployment Nodes:** Homebased at Nevatim Airbase (Base 28) in the northern Negev Desert southeast of Beersheba (31°12'30"N 35°00'44"E) across three operational squadrons in subterranean/hardened hangars: **140 Squadron** ("Golden Eagle"), **116 Squadron** ("Lions of the South"), and **117 Squadron** ("First Jet") (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).
- **Named Leadership & Key Personnel:** First two aircraft (AS-1 / 901 and AS-2 / 902) received at Nevatim on December 12, 2016, by Prime Minister Benjamin Netanyahu, Defense Minister Avigdor Lieberman, and IAF Commander Maj Gen Amir Eshel; first world combat employment announced by Maj Gen Amikam Norkin on May 22, 2018 (MiGFlug Israeli Air Force).
- **Technical & Engineering Parameters:** 50 aircraft contracted in the initial two-squadron program (all delivered by 2024–2025, with a third 25-jet contract signed in 2024 bringing the total order to 75, and a fourth squadron expansion planned); length 15.67 m; wingspan 10.7 m; powered by one Pratt & Whitney F135-PW-100 turbofan (191 kN afterburning thrust, Mach 1.6); fitted with AN/APG-81 AESA radar and unique Israeli open-architecture "plug-and-play" mission computer interfaces developed with IAI and Elbit, Elisra custom electronic warfare suite, Rafael Spice-1000 and internal/conformal fuel tanks extending unrefueled combat radius to reach Iran (~1,700+ km), and internal weapons bays compatible with the dual-capable F-35A Block 4 nuclear architecture (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021; MiGFlug Israeli Air Force).
- **Causal Chain & Operational Chronology:** Activated in 140 Squadron on December 6, 2017, 116 Squadron in January 2020, and 117 Squadron in July 2021, the F-35I Adir fleet at Nevatim (which FAS analysts identify alongside Tel Nof and Hatzerim as having potential nuclear tasking) gives the IAF the stealth capability to penetrate Russian-supplied S-300/S-400 air defense networks without external escort (Bulletin of the Atomic Scientists Israeli Nuclear Weapons 2021).

Section X: Republic of Korea (ROK) National Crisis Management Center (NCMC), Three-Axis System (Kill Chain, KAMD, KMPR), Hyunmoo-IV/V Bunker-Busters, and Hardened Wartime Command Bunkers (Entities 209–217)

209. National Crisis Management Center (NCMC)

- **Institutional Designation & Sovereign Mandate:** Supreme presidential underground national security situation room and C4I command bunker (*Gukga Wigi Gwalli Senteo* / 국가위기관리센터) operating under the Office of National Security (ONS) to convene the National Security Council (NSC) and transmit Commander-in-Chief military directives to the ROK Joint Chiefs of Staff (JCS) (The Korea Times: Presidential Underground Bunker Briefing; The Chosun Daily: President Lee Jae-myung Returns to Cheong Wa Dae).
- **Geodetic Coordinates & Command Nodes:** Dual-node presidential underground bunker architecture in Seoul: (1) the original hardened subterranean bunker beneath the **Blue House (Cheong Wa Dae)** at Jongno-gu, Seoul (37°35'12"N 126°58'29"E, first opened in 1976 and renovated for President Lee Jae-myung's return on December 29, 2025), and (2) the multi-basement hardened bunker beneath the **Yongsan Presidential Office** (formerly the Ministry of National Defense building) at Yongsan-gu, Seoul (37°32'05"N 126°58'39"E, utilized from May 10, 2022 to December 28, 2025) (Maeil Business Newspaper: Blue House NCMC Renovation, Dec 30, 2025; Korea JoongAng Daily: Yongsan Basement NCMC).
- **Named Leadership & Key Personnel:** Commanded by the President of the Republic of Korea (Lee Jae-myung, who assumed office in 2025 following the impeachment and removal of Yoon Suk-yeol) and operationally run by the Director of the Office of National Security (National Security Advisor), the Principal Deputy NSA, and the Head of the National Crisis Management Center (a seconded active-duty Brigadier General/Major General) (AP News: President Lee Returns to Blue House, Dec 2025; The Chosun Daily: Cheong Wa Dae Return).
- **Technical & Structural Architecture:** Directly linked via redundant encrypted military fiber-optic and satellite KJCCS (Korean Joint Command and Control System) data buses to the JCS Central Command Post in Yongsan, the Korea Strategic Command at Namtaeryeong (B-1 Bunker), the Air Force Master Control and Reporting Center (MCRC) at Osan, Gyeryongdae (B-2 Bunker), and ROK-US Combined Forces Command at CP TANGO, displaying real-time ballistic missile warning feeds from KAMD radars, 425 Project satellites, and US DSP/SBIRS early warning networks (The Korea Times: Presidential Underground Bunker Briefing).
- **Causal Chain & Operational Chronology:** Established in 1976 in the Blue House underground bunker, expanded into a comprehensive national crisis center in 2003 under President Roh Moo-hyun, relocated to the Yongsan basement floors on May 10, 2022, when President Yoon Suk-yeol took his midnight JCS command transfer briefing there, and refurbished and reactivated at Cheong Wa Dae (with Senior Secretaries consolidated in Yeomin 1 Building) on December 29, 2025, under President Lee Jae-myung, the NCMC is the apex political decision node that authorizes Kill Chain and KMPR strikes (The Chosun Daily: Cheong Wa Dae Return; Maeil Business Newspaper: Blue House NCMC Renovation, Dec 30, 2025).

210. Korea Massive Punishment and Retaliation (KMPR & Korea Strategic Command)

- **Institutional Designation & Sovereign Mandate:** Third axis (*Daeryang Eungjing Bobok* / 대량응징보복, also designated Overwhelming Response) of the ROK Three-Axis System and the core retaliatory mission of the **Republic of Korea Strategic Command (ROK STRATCOM)**, officially inaugurated on October 1, 2024, to execute massive conventional precision decapitation and bunker-busting retaliation against North Korean leadership and WMD infrastructure in the event of a DPRK nuclear or strategic attack (The Chosun Daily: S. Korea Launches Strategic Command, Oct 1, 2024; Hub and Spokes: Rise of South Korea's Strategic Command).
- **Geodetic Coordinates & Command Nodes:** Korea Strategic Command is headquartered at the Capital Defense Command / Namtaeryeong compound in Gwanak-gu / Gwacheon, southern Seoul (37°27'40"N 126°59'15"E), operating out of the hardened B-1 underground command bunker (The Chosun Daily: S. Korea Launches Strategic Command, Oct 1, 2024).
- **Named Leadership & Key Personnel:** Inaugurated on September 30 / October 1, 2024, under Defense Minister Kim Yong-hyun (and subsequently directed under Defense Minister Ahn Gyu-back and President Lee Jae-myung); inaugural Commander of Korea Strategic Command was ROK Air Force **Lieutenant General Jin Young-seung** (served Oct 1, 2024–Sept 2025, before his promotion to four-star General and appointment as Chairman of the Joint Chiefs of Staff on September 30, 2025) (The Chosun Daily: S. Korea Inaugurates Strategic Command; The Korea Herald: Hyunmoo-5 and Defense Minister Ahn Gyu-back).
- **Technical & Structural Architecture:** ROK Strategic Command unifies operational control over South Korea's strategic multi-domain deterrence assets: the Army Missile Strategic Command (operating Hyunmoo-II, Hyunmoo-III, Hyunmoo-IV, and the 8-ton-warhead Hyunmoo-V ballistic/cruise missiles), ROK Air Force F-35A stealth fighter wings and space operations units, ROK Navy Dosan Ahn Chang-ho-class (KSS-III) 3,000-ton SLBM-equipped submarines, cyber/electromagnetic warfare units, and the Joint Special Operations Brigade ("Spartan 3000" decapitation unit), while serving as the direct operational counterpart to **U.S. Strategic Command (USSTRATCOM)** for **Conventional-Nuclear Integration (CNI)** under the April 2023 Washington Declaration and the U.S.–ROK Nuclear Consultative Group (NCG) Guidelines (The Chosun Daily: S. Korea Launches Strategic Command, Oct 1, 2024; Hub and Spokes: Rise of South Korea's Strategic Command).
- **Causal Chain & Operational Chronology:** Formulated in September 2016 following North Korea's fifth nuclear test to hold every square meter of Pyongyang's leadership districts and deep underground command bunkers (such as Ryongsong Residence and Kangdong tunnels) at risk of non-nuclear obliteration, KMPR was elevated to a unified joint command on October 1, 2024, with the launch of ROK Strategic Command and the public unveiling of the Hyunmoo-5 monster missile (The Chosun Daily: S. Korea Launches Strategic Command, Oct 1, 2024; The War Zone: Hyunmoo-5 Fielded by South Korea).

211. Kill Chain (1st Axis & 425 Project ISR Constellation)

- **Institutional Designation & Sovereign Mandate:** First axis of the ROK Three-Axis System—an integrated sensor-to-shooter preemptive strike architecture designed to detect, track, target, and destroy North Korean nuclear and ballistic missile launchers within 30 minutes of detecting an imminent launch preparation (Arirang News: 425 Project and Kill Chain; The Korea Herald: Five-Satellite Network).
- **Geodetic Coordinates & Deployment Nodes:** Satellite ground control and Space Operations Center at Osan Air Base (37°05'26"N 127°01'46"E) and Daejeon Agency for Defense Development (ADD) / Defense Intelligence Agency nodes (36°23'10"N 127°18'15"E), feeding targeting coordinates to Korea Strategic Command at Namtaeryeong (37°27'40"N 126°59'15"E) (Korea JoongAng Daily: 425 Project Fourth Satellite).
- **Named Leadership & Key Personnel:** Managed by the Defense Acquisition Program Administration (DAPA), Agency for Defense Development (ADD), Korea Aerospace Industries (KAI), Hanwha Systems, and Thales Alenia Space; operationalized under Defense Minister Ahn Gyu-back and JCS Chairman Gen Jin Young-seung (United24 Media: South Korea Completes 425 Project; İlkha: 5th Satellite Launch).
- **Technical & Engineering Parameters:** Anchored in space by the **425 Project** (a 1.2 trillion won / $840 million military reconnaissance satellite constellation, phonetically named in Korean *Sa-I-O* after "SAR" and "EO"), comprising **five 800–1,000 kg low-Earth-orbit spy satellites** all launched aboard SpaceX Falcon 9 rockets from Vandenberg SFB and Cape Canaveral SFS:
  1. **Unit 1 (EO/IR):** Electro-Optical / Infrared satellite launched December 1, 2023 (30 cm resolution);
  2. **Unit 2 (SAR-1):** Synthetic Aperture Radar satellite launched April 7, 2024 (all-weather, day/night 30 cm radar imaging);
  3. **Unit 3 (SAR-2):** SAR satellite launched December 21, 2024;
  4. **Unit 4 (SAR-3):** SAR satellite launched April 22, 2025;
  5. **Unit 5 (SAR-4):** Final SAR satellite launched November 2, 2025; achieving a **2-hour revisit rate** across the entire DPRK regardless of cloud cover (which obscures the Korean Peninsula 70% of the year), supplemented by RQ-4B Global Hawk UAVs, E-737 Peace Eye AEW&C, and a follow-on constellation of 20 small satellites (2025–2028) and 40 microsatellites (2028–2030) to reduce revisit intervals to **30 minutes**, linked directly to F-35A/F-15K fighters (armed with Taurus KEPD 350K cruise missiles) and Hyunmoo ballistic/cruise missile brigades (United24 Media: South Korea Completes 425 Project; İlkha: 5th Satellite Launch).
- **Causal Chain & Operational Chronology:** Initiated after the 2010 Yeonpyeong Island shelling and North Korea's transition to solid-fuel mobile TELs, Kill Chain completed its full 5-satellite 425 Project space sensor layer on November 2, 2025, enabling independent ROK all-weather TEL detection and compressing the find-fix-track-target-engage-assess cycle to under 30 minutes (Arirang News: 425 Project and Kill Chain).

212. Korea Air and Missile Defense (KAMD)

- **Institutional Designation & Sovereign Mandate:** Second axis of the ROK Three-Axis System—a multi-layered, overlapping terminal-phase ballistic and cruise missile defense architecture protecting South Korean command centers, airbases, nuclear power plants, and population centers (CSIS Missile Threat: Missiles of South Korea).
- **Geodetic Coordinates & Deployment Nodes:** Commanded by the ROK Air Force Air Defense Missile Command through the **Korea Air and Missile Defense Operations Center (KAMDOC)** at Osan Air Base (37°05'26"N 127°01'46"E); Green Pine radar sites at Chungcheong/Gyeonggi; U.S. THAAD battery at Seongju, North Gyeongsang Province (36°04'38"N 128°12'50"E); L-SAM and M-SAM test range at Anheung/Taean (36°40'30"N 126°08'15"E) (CSIS Missile Threat: South Korea Test Fires L-SAM).
- **Named Leadership & Key Personnel:** Developed by the Agency for Defense Development (ADD), LIG Nex1, Hanwha Aerospace, and Hanwha Systems, operated by the ROKAF Air Defense Missile Command in real-time data link with the U.S. 35th Air Defense Artillery Brigade (CSIS Missile Threat: South Korea Receives First Cheongung-II Battery).
- **Technical & Engineering Parameters:** Integrates five sensor and interceptor layers:

1. **Early Warning Sensors:** Four Israeli ELTA **EL/M-2080 Green Pine** Block B/C L-band radars (>800–900 km detection range), U.S. AN/TPY-2 X-band radar at Seongju, and ROK Navy Sejong the Great-class / Jeongjo the Great-class (KDX-III Batch I/II) Aegis destroyers equipped with SPY-1D(V) and SM-3/SM-6 capability;
2. **Upper-Tier Exo/Endo-Atmospheric (40–150 km altitude):** U.S. Army **THAAD** battery at Seongju (intercept altitude 40–150 km, range 200 km) and indigenous **L-SAM Block II** (high-altitude glide-phase/upper-tier interceptor under development);
3. **Mid-Tier Upper Endo-Atmospheric (40–60+ km altitude):** Indigenous **L-SAM (Long-range Surface-to-Air Missile)**, tested at Taean on February 23, 2022, November 2022, May 2023, and completed development in November 2024 for deployment by 2026–2027, using a Divert and Attitude Control System (DACS) hit-to-kill kill vehicle to intercept ballistic missiles at **40–60 km altitude** and **150 km range**;
4. **Lower-Tier Terminal Phase (15–40 km altitude):** Eight batteries of **Patriot PAC-3 Configuration 3 / PAC-3 MSE** (20–40 km altitude) and seven+ batteries of indigenous **M-SAM II / Cheongung-II (KM-SAM Block II)** (developed by ADD/LIG Nex1 starting in 2012, first battery delivered November 26, 2020, intercepting missiles via hit-to-kill directional warhead at **15–20+ km altitude** and **40 km range**);
5. **Counter-Artillery / Low-Altitude Layer:** Low Altitude Missile Defense (LAMD, "Korean Iron Dome") and ADD's **Cheongwang** (Block-I fiber-laser anti-drone weapon fielded in late 2024) (CSIS Missile Threat: South Korea Test Fires L-SAM; CSIS Missile Threat: South Korea Receives First Cheongung-II Battery; Maeil Business Newspaper: Armed Forces Day 2026).

- **Causal Chain & Operational Chronology:** Initiated in 2006 as a sovereign Korean lower-tier shield and expanded into a multi-tiered hit-to-kill architecture following North Korea's deployment of KN-23 (Hwasong-11A) quasi-ballistic pull-up missiles, KAMD enables at least three sequential hit-to-kill engagement opportunities (THAAD/L-SAM at 50–150 km, followed by PAC-3 MSE and Cheongung-II at 15–40 km) against incoming DPRK salvos (CSIS Missile Threat: South Korea Test Fires L-SAM).

213. Hyunmoo-IV (Hyunmoo-4.1, 4.2 & 4.4)

- **Institutional Designation & Sovereign Mandate:** High-payload, solid-fuel precision bunker-buster ballistic missile family developed by the Agency for Defense Development (ADD) across three operational domain variants: **Hyunmoo-4.1** (land-based SRBM), **Hyunmoo-4.2** (surface-ship-launched ballistic missile), and **Hyunmoo-4.4** (submarine-launched ballistic missile, K-SLBM) (CSIS Missile Threat: South Korea Tests New Ballistic Missile; CSIS Hyunmoo-4 Tag).
- **Geodetic Coordinates & Deployment Nodes:** Tested at ADD Anheung Proving Ground, Taean, South Chungcheong Province (`36°40'30"N 126°08'15"E`); deployed with the ROK Army Missile Strategic Command (headquartered at Wonju, Gangwon Province, `37°20'30"N 127°55'15"E`), ROK Navy KDX-III Batch II (Jeongjo the Great-class) and KDDX destroyers, and ROK Navy KSS-III (Dosan Ahn Chang-ho-class) submarines at Jinhae (`35°08'15"N 128°39'00"E`) (CSIS Missile Threat: South Korea Tests New Ballistic Missile).
- **Named Leadership & Key Personnel:** Engineered by the Agency for Defense Development (ADD) and manufactured by Hanwha Aerospace under the oversight of the Ministry of National Defense and Korea Strategic Command (CSIS Hyunmoo-4 Tag).
- **Technical & Engineering Parameters:**
  1. **Hyunmoo-4.1 (Ground-based SRBM):** Length ~10–11 m; diameter 0.9–1.1 m; two-stage solid-propellant rocket motor with cold-launch canister and maneuvering reentry vehicle (MaRV); carries a **2,000 kg (2-ton) hardened kinetic earth-penetrating warhead** to an operational range of **800 km** (or lighter payloads well beyond 2,000 km following the May 2021 termination of the US-ROK Missile Guidelines); INS/GPS guidance with 1–5 m CEP;
  2. **Hyunmoo-4.2 (Ship-Launched Ballistic Missile):** Naval VLS-compatible high-payload ballistic missile installed in the KVLS-II cells of ROK Navy destroyers;
  3. **Hyunmoo-4.4 (K-SLBM):** Submarine-launched ballistic missile (diameter 0.8–0.9 m, range ~500 km, cold-gas ejected from the 6 VLS tubes of the 3,000-ton KSS-III Batch-I *Dosan Ahn Chang-ho* and *Ahn Mu* and 10 VLS tubes of the 3,600-ton KSS-III Batch-II *Jang Yeong-sil*), making South Korea the first non-nuclear-armed state in the world to field an operational SLBM (CSIS Missile Threat: South Korea Tests New Ballistic Missile; CSIS Hyunmoo-4 Tag).
- **Causal Chain & Operational Chronology:** Directly enabled by the November 2017 U.S.–ROK agreement led by Presidents Moon Jae-in and Donald Trump that lifted the 500 kg warhead weight cap on South Korean ballistic missiles (followed by the complete termination of all bilateral range/payload restrictions on May 21, 2021), ADD flight-tested Hyunmoo-4.1 at Taean in March 2020 and May 2020, and successfully executed the first submerged SLBM launch of Hyunmoo-4.4 from ROKS *Dosan Ahn Chang-ho* on September 15, 2021 (CSIS Hyunmoo-4 Tag).

214. Hyunmoo-V (Hyunmoo-5 "Monster Missile")

- **Institutional Designation & Sovereign Mandate:** Ultra-heavy, solid-fuel, cold-launched conventional ballistic missile ("Monster Missile" / *Goemul Misail*) carrying the world's heaviest conventional ballistic missile warhead (**8 to 9 tonnes**) to obliterate deeply buried North Korean leadership and nuclear storage bunkers under KMPR and Korea Strategic Command (IISS: South Korea's Hyunmoo-5 Breaks Cover, Oct 2024; The Korea Herald: S. Korea Fields Hyunmoo-5 Missile, Jan 2026).
- **Geodetic Coordinates & Deployment Nodes:** Publicly unveiled on October 1, 2024, at Seoul Air Base, Seongnam (`37°26'45"N 127°06'50"E`) during the 76th Armed Forces Day ceremony, flight-tested at ADD Anheung Proving Ground (`36°40'30"N 126°08'15"E`), and deployed beginning in late 2025–January 2026 to frontline units of the Army Missile Strategic Command under Korea Strategic Command (The Defense Post: South Korea Deploys Hyunmoo-5 to Frontline Units, Jan 2026; IISS: South Korea's Hyunmoo-5 Breaks Cover, Oct 2024).
- **Named Leadership & Key Personnel:** Developed by the Agency for Defense Development (ADD) and manufactured by Hanwha Aerospace; deployment and strategic doctrine ("achieving a balance of terror comparable in strength to tactical nuclear weapons") directed by Defense Minister Ahn Gyu-back and Korea Strategic Command (Sentry Magazine / Yonhap: Defense Minister Ahn Gyu-back on Hyunmoo-5; The Korea Herald: S. Korea Fields Hyunmoo-5 Missile, Jan 2026).
- **Technical & Engineering Parameters:** Total launch weight **~36,000 kg (36 tonnes)**; length ~15–16 m; diameter ~1.6 m; multi-stage high-energy solid-propellant rocket motor; carries an **8,000–9,000 kg (8 to 9-ton / >17,600 lb) dense heavy-metal kinetic penetration warhead** accelerating during terminal dive to near-hypersonic/Mach 10 velocities to generate seismic shockwaves and penetrate hardened granite/reinforced concrete underground facilities over 100 meters deep; operational range **300–600 km** with the full 8-ton warhead (and **3,000–5,500+ km** intermediate/intercontinental range if fitted with a standard 1-ton warhead); cold-launched via gas generator (ejecting the 36-ton missile ~30 m into the air before main engine ignition to protect the launcher) from a massive sealed cylindrical canister mounted on a **9-axle (18x18) Kia/Hanwha**

road-mobile Transporter-Erector-Launcher (TEL) equipped with steerable hydraulic axles for crab-walk diagonal movement (IISS: South Korea's Hyunmoo-5 Breaks Cover, Oct 2024; The War Zone: Hyunmoo-5 Fielded by South Korea).

- **Causal Chain & Operational Chronology:** Initiated around 2016 and fully unlocked by the May 2021 abolition of the US-ROK Missile Guidelines, Hyunmoo-5 was first disclosed in the 2022 National Assembly State Audit, completed R&D in late 2023, debuted publicly on two 9-axle TELs at the October 1, 2024 Armed Forces Day parade alongside the activation of Korea Strategic Command, and entered active frontline deployment in late 2025 / January 2026 (with up to 200 missiles planned for deployment through 2030) to enable simultaneous multi-missile seismic destruction of North Korean underground nuclear command nodes without violating the NPT (The Defense Post: South Korea Deploys Hyunmoo-5 to Frontline Units, Jan 2026; The Guardian: What is South Korea's Monster Missile).

215. Bunker B-1 (Namtaeryeong Wartime Command Bunker)

- **Institutional Designation & Sovereign Mandate:** Primary wartime underground command-and-control bunker of the Republic of Korea Ministry of National Defense (MoND), Joint Chiefs of Staff (JCS), Capital Defense Command (CDC), and Korea Strategic Command (*B-1 Beongkeo* / 문서고 B-1), excavated deep inside the granite core of Mount Gwanaksan (Hankyoreh: B1 Bunker Namtaeryeong; Korea JoongAng Daily: CP Tango and CDC B-1 Bunker).
- **Geodetic Coordinates & Facility Nodes:** Located inside the Capital Defense Command compound at Namtaeryeong Pass, bordering Gwanak-gu (Seoul) and Gwacheon (Gyeonggi Province) (37°27'40"N 126°59'15"E) (Hankyoreh: B1 Bunker Namtaeryeong).
- **Named Leadership & Key Personnel:** Managed by the Capital Defense Command (CDC) and utilized in wartime and during joint Ulchi Freedom Shield (UFS) exercises by the President of the ROK, Minister of National Defense, Chairman of the Joint Chiefs of Staff (Gen Jin Young-seung), and Commander of Korea Strategic Command (Korea JoongAng Daily: CP Tango and CDC B-1 Bunker; Hankyoreh: B1 Bunker Namtaeryeong).
- **Technical & Engineering Parameters:** Tunneled horizontally into solid granite beneath Mount Gwanaksan with only two heavily fortified blast-door tunnel portals; engineered to withstand direct nuclear, electromagnetic pulse (EMP), chemical, biological, and bunker-buster strikes; accommodates 500+ command staff for several months of sealed autonomous operation with independent air-filtration/overpressure systems, water/power generation, and full KJCCS/AKJCCS C4I war rooms (Hankyoreh: B1 Bunker Namtaeryeong).
- **Causal Chain & Operational Chronology:** Originally constructed between 1976 and 1978 by the ROK Army Corps of Engineers under President Park Chung-hee to provide a sovereign Korean wartime command post independent of USFK facilities, Bunker B-1 serves as the primary wartime relocation node for the ROK National Command Authority and the headquarters venue for the annual August Ulchi government-military wartime exercises (Korea JoongAng Daily: CP Tango and CDC B-1 Bunker).

216. Bunker B-2 (Gyeryongdae Wartime Command Bunker)

- **Institutional Designation & Sovereign Mandate:** Tri-service wartime underground command bunker (*B-2 Beongkeo*) located beneath the Gyeryongdae military headquarters complex, housing the integrated wartime command posts of the ROK Army, ROK Navy, and ROK Air Force (The Korea Times: President Visits Gyeryongdae Wartime Command Bunker, Aug 9, 2024).
- **Geodetic Coordinates & Facility Nodes:** Excavated into the southern base of Mount Gyeryongsan at Gyeryongdae, Sindoan-myeon, Gyeryong-si, South Chungcheong Province (36°18'25"N 127°14'10"E), 160 km south of Seoul (The Korea Times: President Visits Gyeryongdae Wartime Command Bunker, Aug 9, 2024; Maeil Business Newspaper: 78th Armed Forces Day at Gyeryongdae, Oct 1, 2026).
- **Named Leadership & Key Personnel:** Occupied by the Chief of Staff of the ROK Army, Chief of Naval Operations of the ROK Navy, and Chief of Staff of the ROK Air Force, and inspected by the President and Minister of National Defense during Ulchi Freedom Shield wartime command exercises (The Korea Times: President Visits Gyeryongdae Wartime Command Bunker, Aug 9, 2024).
- **Technical & Engineering Parameters:** Constructed under "Project 620" (initiated in 1983 under President Chun Doo-hwan and completed in 1989) to relocate the three service headquarters out of range of North Korean long-range artillery around Seoul; features a multi-story subterranean EMP-shielded granite bunker complex directly beneath the octagon-shaped tri-service headquarters building, providing strategic depth and redundant national C4I continuity should Seoul-area command nodes (NCMC and B-1 Namtaeryeong) suffer disruption (The Korea Times: President Visits Gyeryongdae Wartime Command Bunker, Aug 9, 2024).
- **Causal Chain & Operational Chronology:** Serving as the southern strategic anchor of South Korea's national NC3 hierarchy, Bunker B-2 integrates force-generation, logistics, and service-level missile/air/naval strike coordination with Bunker B-1 and CP TANGO during peacetime readiness inspections and wartime operations (The Korea Times: President Visits Gyeryongdae Wartime Command Bunker, Aug 9, 2024).

217. CP TANGO (Command Post Theater Air Naval Ground Operations)

- **Institutional Designation & Sovereign Mandate:** Hardened granite subterranean wartime command post of the **ROK–U.S. Combined Forces Command (CFC)**, United Nations Command (UNC), and United States Forces Korea (USFK), serving since 1978 as the combined operational brain commanding all U.S. and South Korean land, air, naval, and strategic forces in wartime (The Korea Times: Yoon Visits Wartime Bunker CP Tango, Aug 23, 2023; Korea JoongAng Daily: CP Tango Bunker).
- **Geodetic Coordinates & Facility Nodes:** Excavated deep into a granite mountain in Seongnam (Sujeong-gu / Bokjeong-dong area, immediately south of Seoul and adjacent to Seoul Air Base), Gyeonggi Province (37°27'15"N 127°07'30"E) (The Korea Times: Yoon Visits Wartime Bunker CP Tango, Aug 23, 2023).
- **Named Leadership & Key Personnel:** Commanded by the four-star U.S. Army General serving concurrently as Commander UNC/CFC/USFK (Gen Xavier T. Brunson, succeeding Gen Paul LaCamera and Gen Robert Abrams) alongside the four-star ROK Army General serving as Deputy Commander CFC (who is slated to assume the Commander role upon future conditions-based wartime OPCON transfer) (Korea JoongAng Daily: CP Tango Bunker; ROK Drop: CP Tango).
- **Technical & Engineering Parameters:** Built in the 1970s inside solid granite bedrock behind multiple sequential steel blast doors engineered to withstand a direct tactical nuclear detonation, chemical/biological attack, and EMP; equipped with independent water treatment, underground power generation, helicopter landing zones, SCIF intelligence fusion centers (directly tied to the U.S. National Reconnaissance Office, NSA, and USSTRATCOM), and the Theater Automated Command and Control Information Management System (TACCIMS) / Combined Enterprise Regional Information Exchange System-Korea (CENTRIXS-K), allowing two months of sealed subterranean operation (Korea JoongAng Daily: CP Tango Bunker; The Korea Times: Yoon Visits Wartime Bunker CP Tango, Aug 23, 2023).

- **Causal Chain & Operational Chronology:** Operationalized with the founding of the ROK–U.S. Combined Forces Command in November 1978, CP TANGO remains the primary combined wartime command post even after the peacetime relocation of USFK/CFC headquarters from Yongsan to Camp Humphreys in Pyeongtaek, synchronizing ROK Strategic Command's Three-Axis operations (Kill Chain, KAMD, KMPR) with U.S. extended nuclear deterrence assets under OPLAN 5015/5022 (The Korea Times: Yoon Visits Wartime Bunker CP Tango, Aug 23, 2023; Hub and Spokes: Rise of South Korea's Strategic Command).

Section XI: Consolidated Cross-Architecture Audit Matrices (All 217 Entities)

Subsystem Matrix XI.A: United States NC3, Early Warning, Cryptographic & War Plan Entities (Entities 1–29)

Entity #	Entity Name	Tier & Functional Class	Core Engineering, RF, Cryptographic & Hardening Parameters	Primary Operator / Geolocation	Deterministic Upstream → Downstream Causal Role
1	Nuclear Football	Tier-0 Mobile NCA Terminal	45 lb Zero Halliburton case in Kevlar-lined leather; 75-p. Black Book, PEF book, EAS folder, UHF/SHF link	WHMO Military Aides / WHCA (Mobile with POTUS/VP)	NMCC Attack Conference → POTUS Strike Option Selection & Auth
2	Biscuit	Tier-0 Physical Auth Token	Tamper-evident opaque snap-open SAS laminated card housing daily Gold Code columns	NSA / POTUS / VP / WHMO	NMCC Two-Letter Challenge → Verified Gold Code Response
3	Gold Codes	Tier-0 NCA Crypto Protocol	Daily-rotated NSA high-entropy alphanumeric challenge-response sequences	NSA (Fort Meade, MD) / POTUS / NMCC	Biscuit Lookup → NMCC Two-Man SAS Identity Validation
4	Emergency Action Message (EAM)	Tier-1 Execution Directive	30–150+ char encrypted order; HFGCS (4724/6739/8992/11175 kHz), VLF (14–60 kHz), AEHF (44/20 GHz)	CJCS (NMCC) / USSTRATCOM / NAOC	Validated NCA Order → Triad Launch Crew Execution
5	NMCC	Tier-1 Primary Fixed C2 Hub	27,000 sq ft Pentagon center; CCPDS-R, SACCS, DRSN, AEHF; DoD Dir S-5100.30 (Oct 1962)	Joint Staff J-3 (38°52'15"N 77°03'21"W)	Cheyenne Mt Warning → Decision Conference & EAM Broadcast
6	ANMCC	Tier-1 Alternate Fixed C2 Hub	140 psi greenstone bunker; dual 34-ton blast doors; DoD Dir S-5100.30 (Oct 1962)	Joint Staff J-3 / DISA (Site R, PA, 39°44'03"N 77°25'11"W)	Sensor Warning / NMCC Loss → Survivable EAM Broadcast
7	NAOC	Tier-1 Airborne NCA C2 Hub	14 kHz–44 GHz (67 antennas); 5-mi VLF wire; HEMP-shielded Boeing E-4B mission	1 ACCS / 95th Wing (Offutt AFB, NE)	Airborne NCA Conference → Direct MEECN EAM Broadcast
8	Looking Glass	Tier-1 Airborne USSTRATCOM C2	UHF ALCS (225–400 MHz) direct ICBM launch; continuous 24/7 airborne 1961–1990	USSTRATCOM / TF 124 / 625 STOS	NMCC/NAOC EAM or Ground C2 Loss → Direct Silo & Triad Launch
9	TACAMO	Tier-2 Airborne SSBN Relay	AN/ART-54 200 kW VLF (17–30 kHz); 28,000-ft LTWA / 4,000-ft STWA; 70% vertical orbit	STRATCOMWING 1 (VQ-3, VQ-4, VQ-7, Tinker AFB, OK)	EHF/HF EAM Receipt → VLF Seawater Broadcast to SSBNs
10	BMEWS	Tier-0 Polar Radar Network	420–450 MHz UHF; AN/FPS-50 & FPS-49; 3,000 nmi range; IBM 7090 → SSPARS/UEWR	Space Delta 4 (Thule, Clear, RAF Fylingdales)	Polar ICBM Detection → Cheyenne Mt & NMCC Warning

Entity #	Entity Name	Tier & Functional Class	Core Engineering, RF, Cryptographic & Hardening Parameters	Primary Operator / Geolocation	Deterministic Upstream → Downstream Causal Role
11	Permissive Action Link (PAL)	Tier-3 Warhead Crypto Lock	Cat A/B/C (4-digit), Cat D (6-digit), Cat F (12-digit + USG/ESD); NSAM-160 (6 Jun 1962)	Sandia National Labs / AFGSC / USAFE	EAM PAL Key Entry → Warhead Arming Circuit Closure
12	Sealed Authenticator System (SAS)	Tier-2 Message Auth System	Tamper-evident foil "cookies" stored in bright red dual-combination safes	NSA / NMCC / LCCs / SSBNs / Bombers	EAM SAS String → Local Foil Match & Launch Unlock
13	Two-Man Rule	Tier-1–3 Positive Control	DoDD 5210.41 / AFI 91-104; 12-ft LCC separation, 2-sec simultaneous key turn, 2-capsule vote	DoD / AFGSC / USN Submarine Force	Dual-Officer Verification → Physical Key Actuation
14	SIOP (SIOP-62 to OPLAN 8044)	Tier-1 Historical War Plan	Integrated Triad NTB (3,200+ warheads in SIOP-62; ~16,000 targets in 1985); MAO/SAO/LNO	JSTPS / SAC / USSTRATCOM (1960–2003)	Presidential Policy → Pre-Planned EAM Strike Options
15	OPLAN 8010-12	Tier-1 Active Strategic Plan	Whole-of-gov nuclear/conventional/cyber plan vs. RU/CN/KP/IR (promulgated 30 Jul 2012)	USSTRATCOM JFCC-GS / J5 (Offutt AFB, NE)	Adaptive Crisis Planning → Black Book Option Execution
16	NC3	Tier-0 Macro System-of-Systems	~160 systems across VLF/LF/HF/UHF/SHF/EHF, SACCS, AEHF, FAB-T, GASNT	CDRUSSTRATCOM (NC3 Enterprise Lead)	Sensor Detection → NCA Decision → Triad Force Execution
17	MEECN	Tier-2 Survivable Comm Net	CJCS VLF/LF Modes 9, 9 MMPM, 15, HIDAR; MILSTAR/AEHF 44/20 GHz; SLVR/FSBS	CJCS / DISA / USSTRATCOM J6	NMCC/NAOC EAM Output → Jam-Resistant Delivery to Triad
18	DEFCON (5 to 1)	Tier-1 Readiness Scale	5 tiers (JCS SM-833-59, 1959): FADE OUT (5), DOUBLE TAKE (4), ROUND HOUSE (3), FAST PACE (2), 1	CJCS / Combatant Commands	Strategic Warning → Synchronized Global Force Posture
19	COCKED PISTOL	Tier-1 Max Alert Codeword	Official JCS codeword for DEFCON 1 (White) / maximum strategic readiness	CJCS / NMCC / USSTRATCOM	Imminent Attack Confirmation → Immediate Launch Posture
20	E-4B Nightwatch	Tier-1 NCA Command Aircraft	Modified 747-200B (4 jets); 800,000 lb MTOW; 4× CF6-50E2; 111 crew; 72+ hr endurance	1 ACCS / 95th Wing (Offutt AFB, NE)	Ground C2 Threat → Airborne NCA Command & EAM Relay
21	E-6B Mercury	Tier-1/2 ABNCP & TACAMO Jet	707-320B derivative (16 jets); 342,000 lb MTOW; 4× CFM56-2A-2; ALCS + 200 kW VLF	STRATCOMWING 1 (Tinker AFB, OK)	AEHF EAM Receipt → VLF SSBN Relay & UHF ALCS Silo Launch
22	EC-135 Looking Glass	Tier-1 Cold War ABNCP Fleet	KC-135 derivative; ALCS Phase I–III (1967); PACCS AUXCP/ALCC/Radio Relay network	SAC 2 ACCS / 4 ACCS (1961–1998)	SAC HQ Destruction → Airborne PACCS Relay & Silo Launch

Entity #	Entity Name	Tier & Functional Class	Core Engineering, RF, Cryptographic & Hardening Parameters	Primary Operator / Geolocation	Deterministic Upstream → Downstream Causal Role
23	Raven Rock (Site R)	Tier-1 Hardened DoD Bunker	39°44'03"N 77°25'11"W; 140 psi granite; 5 caverns/buildings (A–E); dual 34-ton doors	WHS / 114th Sig Bn / Joint Staff (Adams County, PA)	Pentagon Evacuation → Underground DoD/ANMCC Command
24	Cheyenne Mountain Complex	Tier-1 Warning/Defense Bunker	38°44'40"N 104°50'48"W; 2,000 ft granite; 600 psi; dual 25-ton doors; 15 buildings on 1,380 springs	NORAD / USNORTHCOM / Space Base Delta 1 (CO)	Space/Radar Sensor Feeds → Attack Assessment to NMCC
25	Mount Weather EOC	Tier-1 Civilian COG Bunker	39°03'59"N 77°53'31"W; 434-acre Area A + 600,000 sq ft underground Area B; 34-ton door; FNARS	FEMA ONCP / DHS (Bluemont, VA)	National Emergency → Constitutional Succession & FNARS/EAS
26	USSTRATCOM	Tier-1 Unified Nuclear CCMD	41°07'06"N 95°54'45"W; 1M sq ft $1.3B LeMay C2F; TF 204/214/144/134/124	CDRUSSTRATCOM (Offutt AFB, NE)	POTUS Launch Order → Operational Triad Strike Execution
27	Defense Support Program (DSP)	Tier-0 GEO IR Warning Satellites	35,900 km GEO; 23 satellites (1970–2007); 6 rpm spin; 6,000-element PbS/HgCdTe (2.7/4.3 µm IR)	Space Delta 4 (2 SWS, Buckley SFB, CO)	Boost-Phase Plume Detection → Buckley MCS & Cheyenne Mt
28	SBIRS	Tier-0 GEO/HEO IR Constellation	6 GEO + 4 HEO payloads; dual scanning/staring HgCdTe SWIR/MWIR/See-to-Ground sensors	Space Delta 4 (2 SWS / 11 SWS, Buckley SFB, CO)	Pad-to-Midcourse IR Track → Radar Cueing & NMCC Warning
29	PAVE PAWS (AN/FPS-115 / UEWR)	Tier-0 UHF Phased Array Radars	420–450 MHz; 580 kW peak/face; 1,792 T/R modules/face; 240°/360° AESA; 3,000 nmi range	Space Delta 4 (6/7/12/13 SWS, RAF Fylingdales)	SLBM/ICBM Horizon Track → Dual-Phenomenology Confirmation

Subsystem Matrix XI.B: US Strategic Triad, Warheads & UK/French Deterrence Systems (Entities 30–55)

#	Entity / System	Sovereign State & Branch	Primary Contractor / Design Lab	Key Technical Spec / Yield / CEP	Command Authority & Operational Commander (2026)	Causal Interconnections & 2026 Operational Status
30	**ICBM Architecture**	USA (USAF / AFGSC)	Boeing, Northrop Grumman, Autonetics, Draper Lab	>5,500 km range; Mach 23 burnout; 90–200 m CEP; 400 active deployed missiles	POTUS → SECDEF → CDRUSSTRATCOM (Adm. Richard Correll) → AFGSC (Gen. Stephen Davis) → 20 AF (Maj. Gen. Colin Connor)	Prompt land-based counterforce sponge of the US triad; transitioning from LGM-30G Minuteman III to LGM-35A Sentinel (FAS US Nuclear Weapons 2025)
31	**Silo (LF & LCC)**	USA (USAF 20th AF)	Boeing, USACE CEBMCO, Northrop Grumman, Bechtel	80 ft deep × 12 ft diameter LF; 90–110 ton door; 1,000–2,000 psi hardening; 50 LFs/5 LCCs per squadron	20th Air Force (90th MW F.E. Warren, 91st MW Minot, 341st MW Malmstrom); 2-officer REACT crews + E-6B ALCS	450 active/warm Minuteman silos; USAF building 400 brand-new silos for LGM-35A Sentinel (prototype built March 2026) (AF.mil Sentinel Silo Prototype)

#	Entity / System	Sovereign State & Branch	Primary Contractor / Design Lab	Key Technical Spec / Yield / CEP	Command Authority & Operational Commander (2026)	Causal Interconnections & 2026 Operational Status
32	**Minuteman (I, II, III)**	USA (USAF 20th AF)	Boeing, Thiokol/ATK, Aerojet, Hercules, Autonetics	LGM-30G: 18.26 m, 36,030 kg, 13,000 km range, 1,150 kg throw-weight, 120–200 m CEP; W78 or W87-0	Gen. Stephen L. Davis (AFGSC) → Maj. Gen. Colin J. Connor (20 AF)	400 LGM-30G missiles deployed with 1 RV each (uploadable to 3 MIRVs post-New START expiry); serving until 2052 (AFGSC Minuteman III)
33	**Sentinel (LGM-35A)**	USA (USAF AFNWC / AFGSC)	Northrop Grumman, Aerojet Rocketdyne, Draper Lab, Lockheed Martin	3-stage composite solid ICBM; >9,656 km range; Mach 23; W87-0 & W87-1 in Mk 21/Mk 21A RV	DRPM White; CDRUSSTRATCOM (Adm. Richard Correll); AFGSC (Gen. Stephen Davis)	Stage 1/2/3 static tests completed (2023–2025); inert missile assembled at Vandenberg Sept 2026; flight test 2027; IOC early 2030s (AF.mil Inert Sentinel Assembly)
34	**SSBN Architecture**	USA, UK, France (USN, RN, MN)	GD Electric Boat, BAE Systems, Naval Group	Submerged nuclear-powered ballistic missile platforms; Blue/Gold dual crews; VLF/TACAMO NC3	USN: VADM Richard E. Seif (COMSUBFOR) & RADM Chris Cavanaugh (COMSUBPAC); PAE SSP: VADM Douglas L. Williams	Most survivable assured second-strike leg; carries >50% of US deployed warheads, 100% of UK warheads, and >80% of French warheads (SUBPAC SSBNs)
35	**Trident (C4 & D5/D5LE)**	USA & UK (USN & Royal Navy)	Lockheed Martin Space, Hercules/ATK, Draper Lab	D5LE: 13.42 m, 59,090 kg, 2,800 kg throw-weight, 12,000 km range, Mk 6 Mod 1 stellar-inertial **90 m CEP**	PAE SSP Director VADM Douglas L. Williams (relieved VADM Johnny Wolfe June 2026)	C4 retired 2005; D5LE arms 14 US *Ohio* and 4 UK *Vanguard* SSBNs (and future *Columbia*/*Dreadnought* via D5LE2 to 2084) (CSIS Trident D5)
36	**Ohio-class (SSBN-726)**	USA (USN SUBFOR / SUBPAC)	General Dynamics Electric Boat, GE (S8G PWR)	560 ft, 18,750 tons submerged, S8G PWR (60,000 shp), 20 active D5LE tubes + 4 torpedo tubes; 155 crew	COMSUBFOR (VADM Rick Seif, SUBGRU 10 Kings Bay) & COMSUBPAC (RADM Chris Cavanaugh, SUBGRU 9 Bangor)	14 active SSBNs (8 Pacific, 6 Atlantic) + 4 SSGNs; final ERO refueling completed Feb 2023; retiring late 2020s–2030s (FAS SSBN-726)
37	**Columbia-class (SSBN-826)**	USA (USN PEO SSBN)	GD Electric Boat, HII Newport News, Bechtel (S1B PWR)	560 ft, 20,810 tons submerged, 42-yr life-of-ship S1B core, electric drive pump-jet, 16 D5LE tubes (CMC)	PEO SSBN (RADM Todd Weeks); Naval Reactors (Adm. William Houston); COMSUBFOR (VADM Rick Seif)	12 boats planned (~$130B program); lead boat *USS District of Columbia* (SSBN-826) >50% built, first patrol ~2030–2031 (FAS US Nuclear Weapons 2025)
38	**B-52H Stratofortress**	USA (USAF 8th AF)	Boeing, Pratt & Whitney (TF33) / Rolls-Royce (F130)	185 ft span, 488,000 lb MTOW, 8,800 mi range; carries 20 × AGM-86B ALCM (W80-1) or AGM-181A LRSO (W80-4)	AFGSC (Gen. Stephen Davis) → 8th Air Force / J-GSOC (Maj. Gen. Ty W. Neuman) → 2nd BW & 5th BW	76 total airframes (46 nuclear-coded); upgrading to B-52J with AESA radar, F130 engines, and LRSO missile to fly past 2050 (AFNWC B-52H)
39	**B-2 Spirit (B-2A)**	USA (USAF 8th AF / 509th BW)	Northrop Grumman, Boeing, GE (4 × F118-GE-100)	172 ft flying wing, 336,500 lb MTOW, 60,000 lb payload; up to	8th Air Force (Maj. Gen. Ty W. Neuman) → 509th Bomb Wing	19–20 stealth bombers; certified for B61-12; won 2024 & 2025

#	Entity / System	Sovereign State & Branch	Primary Contractor / Design Lab	Key Technical Spec / Yield / CEP	Command Authority & Operational Commander (2026)	Causal Interconnections & 2026 Operational Status
				16 × B61-12 (30 m CEP via RATS) or B61-11	(Col. Joshua Wiitala) & 131st BW at Whiteman AFB	Omaha Trophies (presented May 2026) (USAF B-2 Spirit)
40	**B-21 Raider**	USA (USAF AFRCO / 8th AF)	Northrop Grumman, Pratt & Whitney, BAE Systems	6th-gen penetrating stealth bomber; 2-pilot crew; carries B61-12, B61-13, and AGM-181A LRSO	AFRCO (William Bailey); 8th Air Force (Maj. Gen. Ty W. Neuman, activated B-21 Detachment 1 Sept 2026)	First flight Nov 2023; 2nd test jet at Edwards Sept 2025; production expansion Oct 2026; beddown at Ellsworth, Whiteman, Dyess (USAF B-21 Raider)
41	**Peacekeeper (LGM-118A MX)**	USA (USAF SAC / 20th AF)	Martin Marietta, Thiokol, Aerojet, Hercules, Rocketdyne, Draper	21.8 m, 87,750 kg, 3,950 kg throw-weight, cold-gas steam launch, AIRS **90 m CEP**; 10 × W87-0/Mk 21 RVs	Deployed at 400th SMS, 90th MW, F.E. Warren AFB (1986–2005)	50 missiles retired by Sept 2005; W87/Mk 21 RVs transferred to Minuteman III; boosters reused as Minotaur IV (CSIS Peacekeeper)
42	**Titan II (LGM-25C)**	USA (USAF SAC)	Martin Marietta, Aerojet (LR87/LR91), AC Spark Plug / Delco	31.3 m, 149,700 kg, hypergolic Aerozine-50/N_2O_4, 15,000 km range, 900 m CEP; 1 × **9.0 Mt W53** in Mk 6 RV	308th SMW (Little Rock), 381st SMW (McConnell), 390th SMW (Davis-Monthan)	54 silos (1963–1987); retired after Searcy (1965) and Damascus (1980) accidents; Complex 571-7 preserved as museum (CSIS Titan II)
43	**W76 (W76-0 / -1 / -2)**	USA (DOE/NNSA & USN)	LANL (Charles Cremer) & Sandia; Lockheed Mk 4 / Mk 4A RV	Two-stage radiation implosion: **100 kt** (W76-0), **90 kt** (W76-1 with MC4700 super-fuze), **5–8 kt** (W76-2 primary-only)	NNSA (Brandon Williams); LANL (Dr. Thomas Mason); Sandia (Dr. James Peery); Pantex Plant	W76-1 LEP completed Dec 2018 (~1,536 active); W76-2 low-yield fielded 2019–2020 on Trident II D5LE (DOE W76-1 Fact Sheet)
44	**W78 (Mk 12A RV)**	USA (DOE/NNSA & USAF)	LANL & Sandia; General Electric Mk 12A RV	Two-stage radiation implosion (Pu-239 primary + HEU/LiD secondary, conventional HE); **335 kt yield**	LANL (Dr. Thomas Mason); Sandia (Dr. James Peery); 20th Air Force (Maj. Gen. Colin Connor)	1,083 built (1979–1982); ~200 deployed on Minuteman III + ~400 in upload hedge; to be replaced by W87-1 (Nuclear Weapon Archive W78)
45	**W87 (W87-0 & W87-1)**	USA (DOE/NNSA & USAF)	LLNL & Sandia; Avco Mk 21 & Lockheed Martin Mk 21A RV	Two-stage radiation implosion with TATB IHE (LX-17) and Fire-Resistant Pit; **300 kt** (upgradable to **475 kt**)	LLNL (Dr. Kim Budil); Sandia (Dr. James Peery); LANL PF-4 (new plutonium pit production)	~540 W87-0 built (200 deployed on MMIII); **W87-1** using newly manufactured Pu-239 pits in Phase 6.3 for LGM-35A Sentinel (Sandia Labs 2024)
46	**W88 (W88 Alt 370 / Mk 5)**	USA (DOE/NNSA & USN)	LANL & Sandia; Lockheed Mk 5 RV	Two-stage radiation implosion (prolate *Komodo* primary + spherical *Cursa* secondary); **475 kt yield**	LANL (Dr. Thomas Mason); Sandia (Dr. James Peery); PAE SSP (VADM Douglas Williams)	~400 built (1988–1989); **W88 Alt 370** FPU July 2021, **Last Production Unit completed Dec 2025** at Pantex (DOE W88 Alt 370 LPU Dec 2025)
47	**MIRV Bus & Physics**	USA, UK, France	Autonetics, Boeing, Lockheed, Draper Lab, ArianeGroup	Exo-atmospheric Post-Boost Vehicle (PBV) sequential stop-and-drop warhead	USSTRATCOM JFCC GS / J-GSOC (OPLAN 8010); FOST & RN Targeting Centres	First tested Aug 16, 1968 (MMIII & Poseidon); active on Trident II

#	Entity / System	Sovereign State & Branch	Primary Contractor / Design Lab	Key Technical Spec / Yield / CEP	Command Authority & Operational Commander (2026)	Causal Interconnections & 2026 Operational Status
				dispensing; high-β spin-stabilized cones		D5LE (up to 8 RVs) and French M51 (4–6 RVs) (FAS ICBM Intro)
48	**MARV Mechanics**	USA & France	Martin Marietta / Goodyear (Pershing II); Lockheed (Mk 500); McDonnell Douglas (AMaRV)	Endo-atmospheric aerodynamic lift flaps or cranked-cone roll control; Pershing II RADAG **30 m CEP**	US Army 56th FA Cmd (historical Pershing II); USN/USAF ABRES; French DGA (V-MAX / SYLEX)	Pershing II forced 1987 INF Treaty; AMaRV established aerodynamic foundation for modern C-HGB hypersonic glide vehicles (CSIS Pershing 2)
49	**Letters of Last Resort**	United Kingdom (10 Downing Street / RN)	Written personally by the UK Prime Minister on day one in office	4 handwritten directives in nested safes aboard the 4 *Vanguard* SSBNs; 4 options (retaliate, refrain from strike, CO judgment, allied command)	Prime Minister Sir Keir Starmer → CDS ACM Sir Richard Knighton → First Sea Lord Gen. Sir Gwyn Jenkins → SSBN CO	Destroyed unopened when PM leaves office; paired with absence of external PALs to guarantee second-strike retaliation (UK Defence Journal Starmer Letters)
50	**Vanguard-class SSBN**	United Kingdom (Royal Navy)	BAE Systems (VSEL Barrow), Rolls-Royce (PWR2 Core H), AWE	4 boats (S28–S31): 149.9 m, 15,900 t submerged, PWR2 reactor, 16 Trident II D5LE tubes, Holbrook/Mk 4A warheads (up to 100 kt)	First Sea Lord Gen. Sir Gwyn Jenkins; Chief of Defence Nuclear Madelaine McTernan; AWE CEO Nick Elliott	Sole UK nuclear platform at HMNB Clyde/RNAD Coulport since 1994; extended past 33 years until *Dreadnought* arrives (Royal Navy Vanguard Class)
51	**Dreadnought-class SSBN**	United Kingdom (Royal Navy / SDA)	BAE Systems Barrow, Rolls-Royce Derby (PWR3), AWE (Astraea)	4 boats (*Dreadnought*, *Valiant*, *Warspite*, *King George VI*): 153.6 m, 17,200 t, PWR3 reactor, 12 CMC tubes (3 Quad Packs), A21/Mk 7 Astraea warhead	Chief of Defence Nuclear Madelaine McTernan; SDA CEO Chris Gardner; First Sea Lord Gen. Sir Gwyn Jenkins	4th boat *King George VI* entered production Sept 22, 2025; £8.4B injected July 30, 2026; entering service early 2030s (Royal Navy Dreadnought Class)
52	**CASD (Operation Relentless)**	United Kingdom (Royal Navy)	Royal Navy Submarine Service, DNO, AWE, 43 Commando RM	Unbroken submerged SSBN deterrence patrols since April 1969 (57+ years); 260 warhead stockpile ceiling	PM Sir Keir Starmer → Defence Secretary John Healey → CDS ACM Sir Richard Knighton → COMSUBFLOT	Longest military operation in UK history; anchored at HMNB Clyde & RNAD Coulport (GOV.UK CASD 50)
53	**Force de frappe (Dissuasion)**	France (Armées / FOST & FAS / CEA DAM)	CEA/DAM, DGA, Naval Group, ArianeGroup, Dassault, MBDA	~290 warheads (FOST + FAS/FANu); **PC Jupiter** (Élysée); **RAMSES** mesh & **SYDEREC** aerostat NC3; **COFAS** Taverny & **CDAOA**	President Emmanuel Macron → CEMA Gén. Fabien Mandon → ALFOST (VAE Xavier Petit) & CFAS (GCA Stéphane Virem); CEA/DAM (Jérôme Demoment)	€7.38B nuclear budget in 2026; Civaux tritium program; Laser Mégajoule simulation; Luxeuil 4th nuclear air base by 2035 (FAS French Nuclear Weapons 2025)
54	**Triomphant-class & M51 SLBM**	France (Marine Nationale / FOST)	Naval Group (Cherbourg), TechnicAtome (K15), ArianeGroup (M51), CEA/DAM (TNO/TNO-2)	4 SNLEs (S616–S619): 138 m, 14,335 t submerged, 100 HLES steel, K15 150 MWt PWR, 16 **M51.2/M51.3** SLBMs (54 t, 9,000+ km, 4–6 TNO/TNO-2 MIRVs at ~100–150 kt)	President Emmanuel Macron → CEMA Gén. Fabien Mandon → ALFOST VAE Xavier Petit (Île Longue / Brest)	**M51.3 / TNO-2** entered service Oct 24, 2025; acceptance launch from *Le Vigilant* Oct 6, 2026; **M51.4** ordered Aug 2025 for SNLE 3G (French MoD M51.3 Launch Oct 2026)

#	Entity / System	Sovereign State & Branch	Primary Contractor / Design Lab	Key Technical Spec / Yield / CEP	Command Authority & Operational Commander (2026)	Causal Interconnections & 2026 Operational Status
55	**ASMP-A / ASMPA-R & TNA**	France (Armée de l'Air et de l'Espace / Marine Nationale)	MBDA France (missile), CEA/DAM (**TNA** warhead), Dassault (Rafale B/M)	5.38 m, 860 kg, liquid-kerosene integral rocket-ramjet, **Mach 3.0**, 500–600 km range; **TNA ~300 kt** IHE thermonuclear warhead	President Emmanuel Macron → CEMA Gén. Fabien Mandon → CFAS GCA Stéphane Virem (EC 1/4 & EC 2/4 at Saint-Dizier) & FANu (*Charles de Gaulle*)	**ASMPA-R** flight-tested May 22, 2024; hypersonic Mach 5+ **ASN4G** entered realization phase in March 2026 for Rafale F5 in 2035 (CSIS ASMP-A; Sénat Rapport N° 950)

Subsystem Matrix XI.C: USSR & Russian Federation NC3, SPRN & Strategic Triad Entities (Entities 56–97)

#	Entity	GRAU / Project Index	Lead Design Bureau & Chief Designer	Primary Coordinates	Role & Deterministic Causal Function in NC3 / SPRN / Triad	Verified Primary Source
56	**Cheget**	Subscriber Terminal of `Kazbek` (`15E601` assoc.)	NII Avtomatiki (`НИИА`) — V. I. Bessonov, V. S. Semenikhin	Kremlin (`55°45'07"N 37°37'03"E`), MoD (`55°43'38"N 37°35'16"E`), GS (`55°44'55"N 37°36'14"E`)	Portable executive terminal displaying SPRN alerts and transmitting 2-of-3 political launch sanctions via `Kavkaz-1/7`.	Wikipedia: Cheget
57	**Kazbek**	`АСБУ ВГК «Казбек»`	NII Avtomatiki (`НИИА`) — V. S. Semenikhin, V. I. Bessonov	General Staff Moscow (`55°44'55"N 37°36'14"E`), Chekhov (`55°09'27"N 37°22'48"E`)	Top-level automated NC3 system binding `Krokus` warnings, `Cheget` briefcases, and `Kavkaz` communications (`<180 s` cycle).	FAS Russian C3I Guide
58	**Perimeter**	`15Э601` (`15E601`) / `Периметр-РЦ`	KB Yuzhnoye (V. F. Utkin) & NPO Impuls (V. I. Melnik, T. N. Sokolov)	Vypolzovo (`57°52'00"N 33°40'00"E`), Kosvinsky Kamen (`59°31'00"N 59°03'43"E`)	Reserve fail-deadly retaliatory C2 complex firing `15A11`/`15Yu75` command rockets when nuclear strikes + GS decapitation occur.	Missilery: 15P011/15E601
59	**Dead Hand**	Western designation of `15Э601 Периметр`	4th TsNII MO & NPO Impuls — V. Ye. Yarynich, V. L. Kataev	Kosvinsky Kamen (`59°31'00"N 59°03'43"E`), Hoover/NSA Archives	Semi-automatic predelegation architecture transferring launch authority to underground sphere crew after verified decapitation.	NSA: Hoffman Excerpt
60	**Oko**	`73Д6` (`11Ф624`), `74Д6` (`11Ф649`), `71Х6` (`17Ф118`)	TsNII Kometa (A. I. Savin) & NPO Lavochkin (S. S. Kryukov, V. M. Kovtunenko)	Serpukhov-15 (`55°03'15"N 37°11'15"E`), Pivan-1 (`50°20'57"N 137°11'22"E`)	1st/2nd-gen HEO/GEO infrared early-warning satellite constellation (101 launched 1972–2012) detecting ICBM/SLBM plumes.	Wikipedia: Oko
61	**Tundra**	`14К032 ЕКС «Купол»` / Spacecraft `14Ф142`	JSC Corp. Kometa (V. P. Misnik) & RSC Energia (N. N. Sevastyanov)	Plesetsk Site 43 (`62°55'44"N 40°27'25"E`), Serpukhov-15 (`55°03'15"N 37°11'15"E`)	3rd-gen HEO IR early-warning constellation (6 launched 2015–2022) with dual-band staring IR + `БАОБУ` emergency C2 transponder.	RussianSpaceWeb: Tundra
62	**Kavkaz**	`АСУ СН «Кавказ-1..7»`	MNIIRS (A. P. Bilenko), NII Avtomatiki (V. I. Bessonov), TsNII Svyazi	Moscow Kremlin/GS (`55°45'07"N 37°37'03"E`), Chekhov (`55°09'27"N 37°22'48"E`)	7-subsystem hardened mobile, cable, tropospheric, airborne (`Kavkaz-4`), and satellite (`Kavkaz-6`) conference network (`Kavkaz-7`).	NIS: Russian NC3

#	Entity	GRAU / Project Index	Lead Design Bureau & Chief Designer	Primary Coordinates	Role & Deterministic Causal Function in NC3 / SPRN / Triad	Verified Primary Source
63	**Signal-A**	15Э557 (15E557) / 15Э1 (Сигнал)	OKB / NPO Impuls — T. N. Sokolov, V. I. Melnik	All 12 RVSN Divisions, Regimental CPs (15B52У/15B222), and ICBM silos	Packet-switched RVSN combat control system delivering launch orders, unblocking codes, and remote retargeting directly to silos.	RussianForces: 3rd-Gen C2
64	**Krokus**	Комплекс «Крокус»	TsNII Kometa & NII Avtomatiki — V. G. Repin, A. I. Savin	Solnechnogorsk (56°11'00"N 37°00'35"E) to GS (55°44'55"N 37°36'14"E)	Real-time SPRN missile-attack warning ingestion and executive display terminal subsystem feeding Kazbek.	Podvig: Russian SNF C2
65	**Agat**	БИУС «Омнибус» / «Округ» & П-310М «Агат»	JSC Concern Morinformsystem-Agat — G. V. Antsev	Moscow HQ (55°45'26"N 37°44'48"E), Severomorsk (69°04'10"N 33°25'00"E), Vilyuchinsk	Naval strategic C3 and SLBM pre-launch fire-control/unblocking combat management architecture aboard Russian SSBNs.	KMP: Concern Agat
66	**Monolit**	15Э556 (15E556) АСБУ «Монолит»	OKB / NPO Impuls — V. I. Melnik, T. N. Sokolov	Vlasikha (55°41'18"N 37°10'43"E), Balabanovo (55°11'19"N 36°36'37"E)	Strategic RVSN/GS central command-formation complex translating Kazbek sanctions into Signal-A and Vyuga strike packets.	SPbSTU: NPO Impuls
67	**Sirena**	15П011 (15A11) & 15П175 (15Ю75/15Ж75) + 15Б99	KB Yuzhnoye (V. F. Utkin), MITT (Yu. S. Solomonov), NPO Impuls (V. I. Melnik)	Vypolzovo (57°52'00"N 33°40'00"E), Baikonur Site 175 (45°59'20"N 63°33'40"E)	Silo (15A11) and road-mobile (15Yu75/15Zh75) command rockets lofting 1,412 kg 15B99 radio-command transmitter to 1,000 km.	Astronautix: 15A11
68	**15E601**	15Э601, 15П011, 15A11, 15Б99, 15П716, 15B75	NPO Impuls, KB Yuzhnoye, KBSM, MITT, NII Avtomatiki	Vypolzovo (57°52'00"N 33°40'00"E), Kosvinsky Kamen (59°31'00"N 59°03'43"E)	Complete GRAU sub-index hierarchy distinguishing 15E601 (Perimeter C2 complex) from Cheget/Kazbek and 15P011/15A11.	Ruzhany: 15P011/15E601
69	**Il-80**	Ил-80 (Ил-86ВКП «Аймак», Звено-С, Maxdome)	Ilyushin OKB (G. V. Novozhilov) & NPO Polet (4 built: RA-86146..86149)	Chkalovsky Air Base (55°52'42"N 38°03'42"E), Taganrog (47°11'54"N 38°50'55"E)	EMP-hardened windowless airborne General Staff command post with SATCOM canoe and 8 km VLF trailing-wire antenna.	Wikipedia: Ilyushin Il-80
70	**Il-82**	Ил-82 (Ил-76ВКП, Изделие 65С, Звено)	Ilyushin OKB (G. V. Novozhilov) & TAPOiCh (2 built: RA-76450, RA-76451)	Chkalovsky Air Base (55°52'42"N 38°03'42"E)	Airborne C3 relay and reserve command post on austere-field Il-76MD airframe augmenting Il-80.	Air Force Mag: Il-82
71	**Tu-142MR Orel**	Ту-142МР «Орёл» (ВПМР, БКСР-200, Фрегат, Bear-J)	Tupolev OKB (A. A. Tupolev, N. V. Kirsanov) & Taganrog TMZ	Kipelovo (59°11'18"N 39°07'24"E), Kamenny Ruchey (49°14'12"N 140°11'24"E)	Naval TACAMO relay aircraft deploying 8.6 km trailing wire with 100 kW Fregat VLF (14–30 kHz) transmitter to SSBNs.	Wikipedia: Tupolev Tu-142
72	**Kosvinsky Kamen**	Объект «Гранит» / ЗЦКП (в/ч 40274)	TsPI-20 / 26th TsNII MO, Glavspetsstroy, NPO Impuls	Mount Kosvinsky Kamen, Sverdlovsk Oblast (59°31'00"N 59°03'43"E, 1,519 m)	Deep-underground (300 m granite/dunite) shock-isolated General Staff/RVSN alternate C2 bunker and Perimeter nerve center.	Wikipedia: Kosvinsky Kamen

#	Entity	GRAU / Project Index	Lead Design Bureau & Chief Designer	Primary Coordinates	Role & Deterministic Causal Function in NC3 / SPRN / Triad	Verified Primary Source
73	**Mount Yamantau**	Белорецк-16 / ЗАТО Межгорье (УС-30, ГУСП)	Construction Directorate No. 30 (УС-30) — N. I. Makarov, A. L. Linets	Mount Yamantau, Bashkortostan (54°15'18"N 58°06'07"E, 1,640 m)	Deep-subterranean (1,000 m quartzite) Continuity-of-Government sanctuary for Presidential/Security Council survival.	Wikipedia: Mount Yamantau
74	**Vlasikha**	Одинцово-10 / Перхушково (ЦКП РВСН, в/ч 33925)	TsPI-20 & NPO Impuls — Cmdr. Col. Gen. S. V. Karakayev	Vlasikha, Moscow Oblast (55°41'14"N 37°11'02"E HQ; 55°41'18"N 37°10'43"E Bunker)	Main Headquarters and 4/12-tier underground Central Command Post of the RVSN hosting Monolit and Signal-A.	Wikipedia: Vlasikha
75	**12th GUMO**	12-е ГУ МО РФ (в/ч 31600, Объекты «С», РТБ)	MoD RF / VNIIEF / VNIITF / VNIIA — Col. Gen. I. A. Kolesnikov	Moscow HQ (55°44'51.8"N 37°36'06.4"E) + 12 National Arsenals	Exclusive custodian of all ~4,309–4,400 Russian nuclear warheads, PAL locks, rail transport, and bomber/tactical warhead mating.	Stimson: 12th GUMO
76	**SPRN**	СПРН / 820-й ГЦПРН (в/ч 44806, 15th VKS Army)	JSC RTI Mints (A. L. Mints, V. G. Repin, V. P. Sloka, S. F. Boev)	Solnechnogorsk-7 (56°11'00"N 37°00'35"E), Serpukhov-15 (55°03'15"N 37°11'15"E)	Integrated two-echelon (space IR + ground radar) Missile Attack Warning System feeding Krokus and Kazbek.	Wikipedia: GTsPRN
77	**Dnepr**	5Н86 «Днепр» (5N86, Hen House)	RTI Mints (Yu. V. Polyak) & DMZ Dnepropetrovsk	Olenegorsk (68°06'59.6"N 33°54'55.5"E), Balkhash (46°36'11.3"N 74°31'51.8"E), Sevastopol	1st-gen VHF (154–162 MHz) double sectoral horn radar (2 × 250 m wings, 120° azimuth, 4,000–6,000 km range).	Project Orlan: 5N86 Dnepr
78	**Dnestr**	5Н15 «Днестр» & 5Н15М «Днестр-М» (Hen House)	RTI Mints — Yu. V. Polyak, A. L. Mints	Sary-Shagan (46°00'04.7"N 73°38'52.1"E), Balkhash (46°37'53.2"N 74°30'44.5"E), Mishelevka	Foundational VHF (200/800 μs pulse) space-surveillance (SKKP, 1967) and early-warning (SPRN, 1971) radar cells.	Wikipedia: Dnestr Radar
79	**Daryal**	5Н79 «Дарьял» & 90Н6 «Дарьял-У» (Pechora)	RTI Mints — V. P. Sloka	Pechora (65°12'37.88"N 57°17'07.64"E), Gabala (40°52'11.08"N 47°48'06.11"E)	2nd-gen bistatic VHF (150–200 MHz) LPAR (30×40 m Tx, 80×80 m Rx, 50 MW, 6,000 km range) covering trans-polar axis.	Wikipedia: Daryal Radar
80	**Volga**	70М6 «Волга» (ОРТУ «Барановичи», в/ч 03522)	RTI Mints — V. P. Sloka	Hantsavichy / Ozerechye, Belarus (52°50'51.97"N 26°28'15.70"E)	Bistatic solid-state UHF (420–450 MHz) LPAR (120° azimuth, 4,800 km range) covering Western Europe & North Atlantic.	Wikipedia: Baranavichy Radar
81	**Voronezh**	77Я6-М / 77Я6-ВП (VHF) & 77Я6-ДМ (UHF)	RTI Mints (V. I. Karasev, S. F. Boev) & NPK NIIDAR (S. D. Saprykin)	Lekhtusi (60°16'31.7"N 30°32'45.7"E), Armavir (44°55'30.4"N 40°59'02.0"E), Kaliningrad + 7 more	3rd-gen modular high-factory-readiness (0.7 MW, 6,000–10,000 km) solid-state phased-array perimeter (10 stations).	Wikipedia: Voronezh Radar
82	**Project 955 Borei**	Проект 955 «Борей» & 955А «Борей-А» (Д-30)	TsKB MT Rubin (V. N. Zdornov, S. V. Sukhanov) & PO Sevmash	Gadzhiyevo (69°15'22"N 33°19'48"E), Vilyuchinsk (52°55'15"N 158°29'20"E)	4th-gen 24,000 t pump-jet SSBN (OK-650V reactor) carrying 16 × R-30 Bulava SLBMs (8+ boats active).	Wikipedia: Borei-Class

#	Entity	GRAU / Project Index	Lead Design Bureau & Chief Designer	Primary Coordinates	Role & Deterministic Causal Function in NC3 / SPRN / Triad	Verified Primary Source
83	**Project 667BDRM Delfin**	Проект 667БДРМ «Дельфин» (Д-9РМ/Д-29РМУ2)	TsKB MT Rubin (S. N. Kovalev) & PO Sevmash / Zvyozdochka	Gadzhiyevo, Murmansk Oblast (69°15'22"N 33°19'48"E)	3rd-gen 18,200 t twin-reactor (2 × VM-4SG) SSBN carrying 16 × R-29RMU2 Sineva / Layner SLBMs.	Wikipedia: Delta IV-Class
84	**Project 941 Akula**	Проект 941 «Акула» / 941УМ (Д-19, Typhoon)	TsKB MT Rubin (S. N. Kovalev) & PO Sevmash (6 built: TK-208..TK-20)	Zapadnaya Litsa (69°26'00"N 32°21'00"E), Severodvinsk (64°34'45"N 39°48'35"E)	48,000 t multi-hull titanium Arctic SSBN (20 × R-39 Rif); TK-208 Dmitry Donskoy served as Bulava testbed until 2023.	Wikipedia: Typhoon-Class
85	**R-30 Bulava**	3М30 (3М30) / РСМ-56 / Д-30 (SS-N-32)	JSC Corp. MITT (Yu. S. Solomonov) & Votkinsk Plant	Deployed on Project 955/955A at Gadzhiyevo (69°15'22"N 33°19'48"E) & Vilyuchinsk (52°55'15"N 158°29'20"E)	36.8 t 3-stage solid/liquid SLBM (8,300–9,300 km range, 6–10 × 100–150 kt MIRVs, dry cold launch).	CSIS: RSM-56 Bulava
86	**R-29RMU2 Sineva**	3М37 / РСМ-54 / Д-29РМУ2 (SS-N-23A Skiff)	JSC GRTs Makeev (V. G. Degtyar, V. P. Makeev) & Krasmash	Deployed on Project 667BDRM at Gadzhiyevo (69°15'22"N 33°19'48"E)	40.3 t 3-stage hypergolic liquid SLBM (8,300–11,547 km range, 2,800 kg throw-weight, 4 × 500 kt or 10 × 100 kt MIRVs).	CSIS: SS-N-23 Sineva
87	**R-29RMU2.1 Layner**	3М37У2.1 / РСМ-54 / Д-29РМУ2.1 (SS-N-23B)	JSC GRTs Makeev (V. G. Degtyar) & Krasmash	Deployed on Project 667BDRM at Gadzhiyevo (69°15'22"N 33°19'48"E)	Enhanced Sineva derivative (2014) with redesigned PBV carrying up to 12 × 100 kt MIRVs or mixed 500 kt + 100 kt payloads.	RussianForces: Liner SLBM
88	**R-36M2 Voevoda**	15А18М (15П018М, 15П718М, РС-20В, SS-18 Mod 5/6)	KB Yuzhnoye (V. F. Utkin) & Yuzhmash	Dombarovsky (51°05'38"N 59°50'33"E), Uzhur (55°19'48"N 89°48'32"E)	211.1 t 2-stage liquid heavy ICBM (8,800 kg throw-weight, 11,000–16,000 km, 10 × 800 kt MIRVs, 300–600 atm silo).	CSIS: R-36 / SS-18 Satan
89	**RS-28 Sarmat**	15А28 (15П278, 15П718М, SS-X-30 / SS-29)	JSC GRTs Makeev (V. G. Degtyar), NPO Mashinostroyeniya & Krasmash	Uzhur (55°19'48"N 89°48'32"E), Plesetsk Site 157 (62°53'11"N 41°45'56"E)	208.1 t 3-stage liquid heavy ICBM (18,000 km global/FOBS range, ~10 t throw-weight, 10–15 MIRVs or 3 × Avangard).	CSIS: RS-28 Sarmat
90	**RT-2PM2 Topol-M**	15Ж65 (15П165 Silo) & 15Ж55 (15П155 Mobile, SS-27 Mod 1)	JSC Corp. MITT (B. N. Lapygin, Yu. S. Solomonov) & Votkinsk Plant	Tatishchevo (51°40'00"N 45°34'00"E, 60 silos), Teykovo (56°51'00"N 40°32'00"E, 18 TELs)	47.2 t 3-stage fast-burn solid ICBM (11,000 km range, 1,200 kg throw-weight, 1 × 800 kt single MaRV warhead).	CSIS: RT-2PM2 Topol-M
91	**RS-24 Yars**	15Ж65М (Mobile 15П165М) & 15Ж67 (Silo 15П155М, SS-27 Mod 2)	JSC Corp. MITT (Yu. S. Solomonov) & Votkinsk Plant	Kozelsk (54°02'15"N 35°47'30"E), Teykovo (56°51'00"N 40°32'00"E) + 7 more divisions (~206 launchers)	49.0 t 3-stage solid ICBM (10,500–12,000 km range, 3–4 × 150–300 kt MIRVs) forming backbone of Russian mobile/silo force.	CSIS: RS-24 Yars
92	**BZhRK Molodets**	15П961 (15Ж61 Rail) & 15П060 (15Ж60 Silo, РТ-23УТТХ, SS-24)	KB Yuzhnoye (V. F. Utkin), KBSM (A. F. Utkin) & Pavlograd PMZ	Kostroma (57°45'00"N 40°55'00"E), Bershet (57°46'00"N 56°22'00"E), Krasnoyarsk (56°17'00"N 91°59'00"E)	Soviet 12-train rail-mobile ICBM system (3 × 104.5 t 15Zh61 per train, inflatable nosecone, 10 × 430–550 kt MIRVs; retired 2005).	FAS Guide: RT-23 / SS-24

#	Entity	GRAU / Project Index	Lead Design Bureau & Chief Designer	Primary Coordinates	Role & Deterministic Causal Function in NC3 / SPRN / Triad	Verified Primary Source
93	**BZhRK Barguzin**	`ОКР «Баргузин»` (`15П169` project, `15Ж65М`/`15Ж67`)	JSC Corp. MITT (Yu. S. Solomonov) & KBSM	Plesetsk Test Range (`62°55'32"N 40°34'40"E`, Nov 2016 ejection test; frozen Dec 2017)	Next-gen rail-mobile ICBM train carrying `6 × 49 t RS-24 Yars` (`24 MIRVs/train`) in standard 4-axle freight cars.	Wikipedia: BZhRK Barguzin
94	**15Yu71 Avangard**	`15Ю71` (`Изделие 4202`) in `15П771` (`15А35П`, SS-19 Mod 4)	JSC VPK NPO Mashinostroyeniya — G. A. Yefremov, A. G. Leonov	Dombarovsky (Yasny), Orenburg Oblast (`51°05'38"N 59°50'33"E`, 12 silos deployed)	`Mach 20–27` (`~2,000 kg`) carbon-composite hypersonic boost-glide vehicle (`1 × 800 kt–2 Mt` warhead) on `UR-100NUTTKh`.	CSIS: Avangard HGV
95	**Tu-160**	`Ty-160` / `Ty-160M` / `Ty-160M2` (`Изделие 70`, Blackjack)	PJSC Tupolev (V. I. Bliznyuk) & Kazan KA3 (`NK-32` / `NK-32-02`)	Engels-2 Air Base, Saratov Oblast (`51°28'54"N 46°12'36"E`), Kazan (`55°51'25"N 49°07'55"E`)	`275 t Mach 2.05` variable-sweep heavy strategic bomber carrying `12 × Kh-102` nuclear ALCMs on two internal `MKU-6-5U` drums.	Wikipedia: Tupolev Tu-160
96	**Tu-95MS**	`Ty-95MC` / `Ty-95MCM` (`Изделие ВП-021`, Bear-H6/H16)	PJSC Tupolev (N. V. Kirsanov), Aviakor Samara & Beriev Taganrog	Engels-2 (`51°28'54"N 46°12'36"E`), Ukrainka (`51°10'12"N 128°26'42"E`), Dyagilevo (`54°38'30"N 39°34'20"E`)	`187.7 t` four-turboprop (`NK-12MPM`) strategic bomber carrying `8 × Kh-102` stealth ALCMs on 4 external `AKU-5M` pylons (`+6` internal).	SIPRI: Russian Nuclear Forces
97	**Kh-102**	`X-102` (`Изделие 504Б`, AS-23B Kodiak; conv. `X-101` / `504A`)	JSC GosMKB Raduga (I. Seleznev, S. A. Bogatikov) & ODK-Saturn	Engels-2 RTB (`51°29'45"N 46°14'10"E`), Ukrainka RTB (`51°11'30"N 128°24'15"E`)	`2,400 kg` low-observable (`0.01 m²` RCS) turbofan (`TRDD-50A`) strategic ALCM (`2,800–4,500+ km` range, `250–450 kt` warhead).	CSIS: Kh-101 / Kh-102

Subsystem Matrix XI.D: People's Republic of China (PRC) & DPRK Strategic Entities (Entities 98–152)

ID	Entity Name	State	Category	Geolocation / Coordinates	Key Leadership / Institutional Control	Core Technical & Operational Metrics	Deterministic Causal Role in Strategic Architecture	Primary Verified Source
98	**PLARF**	PRC	Strategic Service Branch	Qinghe, Beijing (`40.0381°N, 116.3314°E`)	Gen. Wang Houbin (Cmdr), Gen. Xu Xisheng (PC), CMC	9 Bases (61–69), 41+ brigades, >120,000 personnel, 400+ ICBMs, 500 IRBMs	Executes land-based nuclear counterstrike & conventional theater precision strike	CASI PLARF Org
99	**CMC & JOCC**	PRC	Supreme C2 Authority	Bayi Bldg & Xishan Underground (`39.9850°N, 116.1400°E`)	Chairman Xi Jinping, Gen. Zhang Youxia, Gen. Liu Zhenli	Dual-command C2: direct CMC-to-brigade nuclear chain; JOCC-to-Theater conventional chain	Sole nuclear release authority; integrates 3–4 min EWCS launch-on-warning loop	CASI NC3 Report 2024
100	**CAEP (Mianyang)**	PRC	Warhead R&D & Production	Science City, Mianyang (`31.4775°N, 104.7697°E`)	Pres. Liu Cangli, SASTIND / CMC EDD; Yu Min, Deng Jiaxian	12 institutes, 6 Natl Key Labs, FBR-2 reactor, Dragon-I/II hydrotest, ~600–620 warheads	Designs, tests, and fabricates all PRC thermonuclear warheads (`#535`, `#5x5`)	CAEP Technical Profile
101	**Ninth Academy**	PRC	Nuclear Weapons Design Lineage	Haiyan (`36.9600°N, 100.9200°E`) to Mianyang (`31.4775°N, 104.7697°E`)	Li Jue, Deng Jiaxian, Zhou Guangzhao, Nie Rongzhen	9 nuclear physics/metallurgy/implosion divisions; 45 nuclear tests (1964–1996)	Engineered Project 596 (1964) & Test No. 6 H-bomb (1967); relocated to Third Line	Stokes NPEC Study

ID	Entity Name	State	Category	Geolocation / Coordinates	Key Leadership / Institutional Control	Core Technical & Operational Metrics	Deterministic Causal Role in Strategic Architecture	Primary Verified Source
102	**Base 22 (067 Base)**	PRC	Central Warhead Custodian	Baoji / Taibai, Shaanxi (34.3619°N, 107.2378°E)	Maj. Gen. Jia Qianrui, Hong Youdao; PLARF HQ	Unit 0674 (1958); moved from Qinghai to Shaanxi (1969); renamed Base 67 (2017)	Enforces physical separation of warheads from missile brigades in peacetime	CASI Warhead Mgmt 2026
103	**Taibai Depot**	PRC	Subterranean Warhead Vault	Hongchuan, Taibai County (34.0583°N, 107.3164°E)	Base 67 Technical Service Brigade & Unit 96038	>500–1,000 m granite overburden, tritium servicing, rail/road transfer, AD/ECM regt	Stores central warhead reserve; dispatches warheads to Base 61–66 inspection units	CASI Warhead Mgmt 2026
104	**Base 61**	PRC	Operational Missile Base	Huangshan, Anhui (29.7147°N, 118.3375°E)	MUCD 96601; Eastern Theater Command / PLARF HQ	7 brigades (611–617): DF-31AG, DF-26, DF-17, DF-16, DF-15B/C, DF-11A + UAV regt	Primary Taiwan Strait SRBM/MRBM/HGV strike force + mobile ICBM deterrent	CNS PLARF ORBAT 2023
105	**Base 62**	PRC	Operational Missile Base	Kunming, Yunnan (25.0440°N, 102.7580°E)	MUCD 96602; Southern Theater Command / PLARF HQ	7 brigades (621–627): DF-31AG, DF-31A, DF-26, DF-21D, DF-17, CJ-10A	Covers South China Sea, India, Indochina, and US/Guam via ICBM/IRBM/ASBM	FAS Nuclear Notebook 2024
106	**Base 63**	PRC	Operational Missile Base	Huaihua, Hunan (27.5694°N, 110.0017°E)	MUCD 96603; PLARF HQ / CMC JOCC	6 brigades (631–636): DF-5A/B/C silos (631, 633, 634), DF-31AG (632), CJ-10, DF-16	Central-southern heavy liquid ICBM silo & road-mobile ICBM strategic reserve	CASI PLARF Org
107	**Base 64**	PRC	Operational Missile Base	Lanzhou (36.0570°N, 103.8390°E) / Xining (36.6171°N, 101.7782°E)	MUCD 96604; Western Theater Command / PLARF HQ	7 brigades (641–647): DF-41 (644, 645), DF-31AG (641–643), DF-26 (646) + 320 silos	Deep inland ICBM sanctuary ranging CONUS/Europe/India via mobile TELs & silos	FAS Nuclear Notebook 2024
108	**Base 65**	PRC	Operational Missile Base	Shenyang, Liaoning (41.8350°N, 123.4310°E)	MUCD 96605; Northern Theater Command / PLARF HQ	6 brigades (651–656): DF-31AG/DF-41, DF-26, DF-21A/D, DF-17, DF-100	Polar CONUS ICBM vector + theater nuclear/conventional strike on Japan/ROK/Guam	CASI PLARF Org
109	**Base 66**	PRC	Operational Missile Base	Luoyang, Henan (34.6197°N, 112.4540°E)	MUCD 96606; Central Theater Command / PLARF HQ	6 brigades (661–666): DF-5A/B/C silos (661, 662), DF-31A/AG (663, 664), DF-26 (666)	Core central mountain DF-5B/C MIRV silo & road-mobile ICBM retaliatory force	FAS Nuclear Notebook 2024
110	**Base 67**	PRC	Stockpile Support Base	Baoji, Shaanxi (34.3619°N, 107.2378°E)	MUCD 96607; PLARF HQ / CMC EDD	3 brigades (2 storage/testing, 1 emergency) + Inspection Inst + 8 support regiments	Manages warhead inspection, transport, ECM/AD defense, and crisis mating delivery	CASI Warhead Mgmt 2026
111	**Base 68**	PRC	Engineering Support Base	Luoyang, Henan (34.6520°N, 112.4280°E)	MUCD 96608 (merged 308th Eng Cmd); PLARF HQ	6 Engineering Brigades (681–686) + Comms Eng Regt + 4 support regiments	Excavates "Underground Great Wall" tunnels and constructed 350+ new ICBM silos	CASI PLARF Org
112	**Base 69**	PRC	Test & Training Base	Yinchuan / Jilantai (39.7820°N, 105.6250°E) / Wuzhai	MUCD 96609; PLARF HQ	15–16 Jilantai training silos, OPFOR regt, 3 test/training districts	Certifies silo loading, OPFOR combat drills, and Dec 2024 multi-ICBM salvo launches	FAS Silo Construction
113	**Dongfeng (DF Family)**	PRC	Missile Industrial Architecture	Beijing (39.7950°N, 116.4020°E), Xi'an, Wuhan	CASC CALT (1st Acad), AASPT (4th Acad), CASIC 4th Acad	Liquid (UDMH/N2O4) & solid (NEPE carbon-epoxy) SRBM/MRBM/IRBM/ICBM/HGV series	Sovereign industrial-engineering backbone supplying all PLARF surface-to-surface missiles	CASI Missile Industry

ID	Entity Name	State	Category	Geolocation / Coordinates	Key Leadership / Institutional Control	Core Technical & Operational Metrics	Deterministic Causal Role in Strategic Architecture	Primary Verified Source
114	**DF-5B (CSS-4 Mod 3)**	PRC	Silo-Based MIRV ICBM	Hunan (`26.4308°N, 109.4617°E`) & Henan (`34.5169°N, 110.8942°E`)	CASC CALT; PLARF Brigades 631 & 661	2-stage liquid + PBV, 32.6 m x 3.35 m, 183,000 kg, 12,000–13,000 km, 5 MIRVs, 300 m CEP	Penetrates US GMD missile defense with 5 independent thermonuclear RVs per silo	CSIS DF-5 Profile
115	**DF-5C (CSS-4 Mod 4)**	PRC	Silo-Based Heavy/MIRV ICBM	Bases 63 & 66 (30 new silos in Hunan/Henan)	CASC CALT & CAEP; PLARF Bases 63 & 66	32.6 m x 3.35 m, 12,000–13,000+ km; tested w/ 10 MIRVs (2017) & multi-Mt warhead	Delivers 10-MIRV saturation or multi-megaton hard-target kill from expanded silos	FAS Nuclear Notebook 2026
116	**DF-31AG (CSS-10 Mod 2A)**	PRC	Road-Mobile & Silo ICBM	8+ brigades (`28.7607°N, 104.7914°E`) & >100 northern silos	CASC CALT / CASIC Wanshan; Bases 61–66	3-stage solid, 15 m x 2.0 m, 42,000 kg, 11,200+ km, 8-axle off-road TEL & DF-31B silo version	Backbone mobile ICBM (64+ TELs) & primary missile loaded in >100 northern silos	CSIS DF-31 Profile
117	**DF-41 (CSS-20)**	PRC	Road/Silo/Rail MIRV ICBM	Yumen (`40.2°N, 96.9°E`), Hami (`42.3°N, 92.8°E`), Ordos (`39.9°N, 108.3°E`)	CASC CALT; Brigades 644, 645, 652 & 320 northern silos	3-stage solid, 21 m x 2.25 m, ~80,000 kg, 12,000–15,000 km, up to 3 MIRVs (`#535`/`#5x5`)	Longest-range solid MIRV ICBM; 320 silos create massive counterforce sink	CSIS DF-41 Profile
118	**DF-26 (CSS-18)**	PRC	Dual-Capable IRBM / ASBM	7+ brigades (Korla, Qingzhou, Dalian, Jianshui, Xinyang)	CASC CALT; PLARF Bases 61, 62, 64, 65, 66	2-stage solid, 14 m x 1.4 m, 20,000 kg, 4,000 km, MaRV, 6-axle TEL, 250+ TELs / 500+ missiles	"Hot-swappable" nuclear (incl. `<10 kt` low-yield) & conventional precision strike / ASBM	CSIS DF-26 Profile
119	**DF-17 (CSS-22)**	PRC	Hypersonic Glide Vehicle MRBM	Yong'an (`25.9922°N, 117.3611°E`), Puning (`23.3725°N, 116.1650°E`)	CASC CALT & CAAA (11th Acad); Brigades 614, 627, 655	Solid booster + DF-ZF HGV, 11 m, 15,000 kg, 1,800–2,500 km, Mach 5–10 @ 60 km depressed apogee	Evades midcourse/terminal missile defenses via endoatmospheric hypersonic glide	CSIS DF-17 Profile
120	**Julang (JL Program)**	PRC	Sea-Based SLBM Program	Beijing (CALT/CASIC) & Dalian Test Base (`38.8180°N, 121.4950°E`)	Huang Weilu, Fang Xinran; PLAN Submarine Force	3 generations (JL-1, JL-2, JL-3); tested from Type 031/032 SSBs for Type 092/094/096 SSBNs	Provides sea-based assured second-strike leg of China's nuclear triad	CSIS JL-2 Profile
121	**JL-2 (CSS-N-14)**	PRC	Intercontinental SLBM	Yulin Naval Base, Hainan (`18.2108°N, 109.6844°E`)	CASC CALT; PLAN 2nd Submarine Base (Yulin)	3-stage solid (DF-31 derivative), 13 m x 2.0 m, 42,000 kg, 7,200+ km, 1 Mt or MIRV warhead	First operational intercontinental SLBM on Type 094 SSBN (12 tubes/boat)	FAS Nuclear Forces 2019
122	**JL-3 (CSS-N-20)**	PRC	MIRV Intercontinental SLBM	Yulin Naval Base, Hainan (`18.2108°N, 109.6844°E`)	CASC CALT & AASPT; PLAN SSBN Fleet / CMC JOCC	3-stage composite solid, aerospike nose, 10,000+ km range, MIRV (`#535`/`#5x5` warheads)	Enables Type 094/096 SSBNs to strike continental US from South China Sea/Bohai bastions	FAS Nuclear Notebook 2025
123	**Type 094 Jin-Class**	PRC	Operational SSBN Fleet	Yulin Naval Base Cavern, Sanya (`18.2044°N, 109.6922°E`)	CSSC 719th Inst (Zhang Jinlin); PLAN (6+ hulls in service)	137 m x 11.8 m, 11,000 t submerged, PWR, 12 SLBM tubes/boat (72 total tubes armed w/ JL-3)	Conducts continuous at-sea nuclear deterrence patrols from Hainan underground base	CSIS ChinaPower Nuclear
124	**Type 096 Tang-Class**	PRC	Next-Generation SSBN	Bohai Shipyard, Huludao (`40.7186°N, 120.9917°E`)	CSSC 719th Inst & BSHIC Huludao; PLAN	Wide-beam quieted hull, raft-mounting/pump-jet, 16–24	Overcomes Type 094 acoustic vulnerability for deep-ocean 40-year concurrent deterrence	FAS Nuclear Notebook 2026

ID	Entity Name	State	Category	Geolocation / Coordinates	Key Leadership / Institutional Control	Core Technical & Operational Metrics	Deterministic Causal Role in Strategic Architecture	Primary Verified Source
						SLBM tubes for JL-3 / follow-on SLBM		
125	**H-6N Bomber**	PRC	Nuclear Triad Bomber & ALBM	Neixiang Air Base, Henan (33.0950°N, 111.7980°E)	AVIC Xi'an; PLAAF 106th Bomber Bde / 8th Bomber Div	Air-refuelable twin-turbofan bomber w/ ventral cradle for CH-AS-X-13 / KD-21 ALBM (~3,000 km)	Restores airborne leg of PRC nuclear triad; delivers stand-off/low-yield nuclear ALBM	SIPRI Chinese Nuclear Forces
126	**PLA EW Network**	PRC	Space/Ground Warning & C2	Base 37 HQ, Lintong, Xi'an (34.3660°N, 109.2140°E)	PLAASF Base 37 (Unit 32035) & Base 26; CMC JOCC	TJS/Huoyan-1 GEO IR satellites (90s detection, 3–4 min alert) + 7+ LPARs	Enables dual-phenomenology Early-Warning Counterstrike (Launch-on-Warning) posture	CASI Base 37 Report 2024
127	**Huian LPAR**	PRC	Strategic Phased-Array Radar	Hui'an, Fujian (24.9750°N, 118.8850°E) & Lin'an (30.2865°N, 119.1286°E)	CETC 14th Inst; PLAASF Base 37 / Eastern Theater EW	Near-vertical UHF/P-band AESA, ~115°–130° ESE boresight (120° sector), 3,000–5,500 km range	Tracks Taiwan/Philippine Sea/Guam missiles & jams Taiwan's Leshan Pave Paws radar	CASI Ground Segment
128	**Shuangyashan LPAR**	PRC	Strategic Phased-Array Radar	Jiamusi / Shuangyashan border (46.527581°N, 130.756423°E)	CETC 14th Inst; PLAASF Base 37 (Unit 32035)	20°-sloped UHF/P-band AESA, ~35°–55° NE boresight (120° sector), 5,500 km range	Primary early-warning sensor for US ICBM/SLBM trajectories over North Pacific/Arctic	CASI Base 37 Report 2024
129	**Xinjiang LPAR**	PRC	Strategic Phased-Array Radar	Korla, Xinjiang (41.641194°N, 86.236749°E)	CETC 14th Inst; Korla Test Complex / Base 37	Sloped UHF/P-band AESA (built 2004–2009), ~210°–290° W/SW & test-range azimuth (120° sector)	Tracks Indian/western missiles, PLARF test RVs, and cues HQ-19/SC-19/DN-3 ABM intercepts	CASI Base 37 Report 2024
130	**Beidou-3 (BDS-3)**	PRC	GNSS PNT & Short-Message NC3	Beijing (40.0720°N, 116.2750°E) & GEO/IGSO/MEO Orbit	Yang Changfeng (CAST); PLAASF Base 35 & CMC JSD	35 satellites (24 MEO, 3 IGSO, 3+ GEO), Ka-band ISLs, 10 cm military PNT, 14,000-bit 2-way SMC	Provides jam-proof 10 cm missile guidance & two-way encrypted C2 to mobile TELs	PLA BeiDou Study
131	**ShenGuang (SG-I–IV)**	PRC	Laser ICF Weapons Physics	Shanghai SIOM (31.3850°N, 121.2460°E) & Mianyang (31.4820°N, 104.7760°E)	Wang Ganchang, He Xiantu, Zhang Jie; CAEP RCLF (8th Inst)	SG-I (1.6 kJ), SG-II/U (6–24 kJ + PW), SG-III (48-beam, 180 kJ UV), SG-IV (~1.8–2.0 MJ UV)	Simulates thermonuclear secondary ignition & certifies new MIRV warheads post-CTBT	Cambridge Petawatt Lasers
132	**Missile General Bureau**	DPRK	Supreme Missile Directorate	Pyongyang (39.0392°N, 125.7625°E)	Gen. Jang Chang Ha (Gen. Dir.), Col. Gen. Kim Jong Sik	Founded 2016.4.30 / unveiled Feb 2023; commands Red Flag ICBM companies & R&D institutes	Directs all DPRK strategic/tactical missile R&D, production, and operational combat launches	NK Leadership Watch MGB
133	**SAC & Haekbangashoe**	DPRK	National Nuclear Command	Changgwang-dong, Pyongyang (39.0145°N, 125.7442°E)	SAC Pres. Kim Jong Un, Pak Jong Chon, Jang Chang Ha	Sept 8, 2022 Nuclear Force Policy Law + "Haekbangashoe" (Nuclear Trigger) C2 system	Codifies preemptive nuclear doctrine & automatic retaliatory launch upon C2 decapitation	Taylor & Francis DPRK NC2
134	**Hwasong-14 (KN-20)**	DPRK	2-Stage Liquid ICBM	Tested at Panghyon (39.9275°N, 125.2194°E) & Mupyong-ri	Academy of National Defense Science / KPA Strategic Force	2-stage UDMH/N2O4 (single RD-250 + 4 verniers), 19.5 m x 1.7 m, 10,000–10,400 km, 8-axle TE	Proved DPRK intercontinental staging (July 4 & 28, 2017 tests) reaching North America	CSIS Hwasong-14 Profile

ID	Entity Name	State	Category	Geolocation / Coordinates	Key Leadership / Institutional Control	Core Technical & Operational Metrics	Deterministic Causal Role in Strategic Architecture	Primary Verified Source
135	**Hwasong-15 (KN-22)**	DPRK	2-Stage Heavy Liquid ICBM	Tested at Pyongsong & Sunan Airport (39.2241°N, 125.6700°E)	1st Red Flag Company / Missile General Bureau	2-stage liquid (twin-chamber gimbaled Paektusan), 22.5 m x 2.0–2.4 m, 72,000 kg, 13,000 km, 9-axle TEL	First DPRK ICBM able to reach entire continental US with 1,000 kg payload (2017, 2023)	CSIS Hwasong-15 Profile
136	**Hwasong-17 (KN-27/28)**	DPRK	Ultra-Heavy Liquid ICBM	Sunan Airport (39.2241°N, 125.6700°E) & Sanum-dong	Gen. Jang Chang Ha / MGB	2-stage liquid (4-nozzle 160-tf cluster), 24–26 m x 2.5 m, ~100,000 kg, 15,000+ km, 11-axle TEL	World's largest road-mobile liquid ICBM; sized for multi-warhead MIRV payloads	CSIS Hwasong-17 Profile
137	**Hwasong-18**	DPRK	3-Stage Solid-Fuel ICBM	Kangdong Launch Site, Pyongyang (39.0760°N, 125.9860°E)	2nd Red Flag Company / MGB (Jang Chang Ha, Kim Jong Sik)	3-stage solid cold-launched canister ICBM, ~25 m x 2.0 m, 15,000+ km, 9-axle wheeled TEL	Eliminates pre-launch liquid fueling vulnerability; tested 3x in 2023 (up to 6,648 km apogee)	CSIS Hwasong-18 Profile
138	**Hwasong-19**	DPRK	Heavy Solid-Fuel MIRV ICBM	Pyongyang / Kangdong (39.0760°N, 125.9860°E)	Kim Jong Un, Gen. Jang Chang Ha, Kim Jong Sik (MGB)	3-stage solid + PBV, 28+ m x 2.4–2.5 m, 15,000+ km, 11/12-axle TEL; 7,687.5 km apogee (Oct 31, 2024)	Combines solid-fuel rapid launch with heavy throw-weight for MIRVs and penetration aids	38 North Hwasong-19
139	**Pukguksong-1 (KN-11)**	DPRK	2-Stage Solid SLBM	Sinpo South Shipyard (40.0258°N, 128.1558°E)	Academy of National Defense Science / KPAN	2-stage solid, 9.3 m x 1.1 m, 1,000–1,200 km, cold-launch with 8 base grid fins	Validated underwater cold-gas launch from *8.24 Yongung* SSB (Aug 2016) & solid propulsion	Taylor & Francis SLBMs
140	**Pukguksong-2 (KN-15)**	DPRK	Tracked Solid-Fuel MRBM	Kusong (39.9275°N, 125.2194°E) & Lake Yonpung	KPA Strategic Force / Academy of National Defense Science	2-stage solid canisterized MRBM, ~10 m x 1.1–1.4 m, 1,200–2,000 km, tracked caterpillar TEL	Off-road solid-fuel regional nuclear strike platform replacing liquid-fueled Nodong	FAS DPRK Nuclear 2024
141	**Pukguksong-3 (KN-26)**	DPRK	Wide-Body Solid SLBM	Wonsan / Sinpo Bay (39.9800°N, 128.2000°E)	Academy of National Defense Science / KPAN	2-stage filament-wound composite solid SLBM, ~10.6 m x 1.4 m, 1,900–2,500 km range	Tested Oct 2, 2019 (910 km apogee); standardized 1.4 m diameter for new SSB fleet	Pukguksong-3 Profile
142	**Pukguksong-4 (4ㅅ)**	DPRK	Large-Diameter Solid SLBM	Pyongyang (39.0194°N, 125.7533°E) / Sinpo Shipyard	Missile General Bureau / KPAN	2-stage composite solid, ~1.5–1.8 m diameter, 3,500–5,400 km range (UN PoE S/2021/211)	Unveiled Oct 10, 2020; maximizes range and payload volume inside submarine launch tubes	Taylor & Francis SLBMs
143	**Pukguksong-5 (5ㅅ)**	DPRK	Elongated-Shroud Solid SLBM	Pyongyang (39.0194°N, 125.7533°E) / Sinpo Shipyard	Missile General Bureau / KPAN	2-stage composite solid, ~11 m x 1.5–1.8 m, pointed ogival shroud, 3,000–5,000+ km range	Unveiled Jan 14, 2021; enlarged nose shroud for MIRV / MaRV / penetration aid integration	TWZ Pukguksong-5 Analysis
144	**Pukguksong-6**	DPRK	Extended-Length Solid SLBM	Pyongyang (39.0194°N, 125.7533°E) / Sinpo Shipyard	Missile General Bureau / KPAN	Extended 2/3-stage solid SLBM (~12–13.5 m x 1.6–1.8 m), 4,000–6,000+ km estimated range	Unveiled April 25, 2022; sized for *Hero Kim Kun Ok* (No. 841) & future nuclear SSBN tubes	SIPRI DPRK Forces 2024
145	**Hwasal-1 (& Ra-3)**	DPRK	Nuclear Land-Attack Cruise Missile	South Hamgyong / West Coast launch corridors	Missile General Bureau (Jang Chang Ha / Kim Jong Sik)	Subsonic turbofan LACM, ~6 m x 0.52–0.6 m, 1,500 km range (7,580s flight), **Hwasan-31** warhead	Low-altitude terrain-hugging tactical nuclear strike & 600 m simulated airburst platform	SIPRI DPRK Forces 2024

ID	Entity Name	State	Category	Geolocation / Coordinates	Key Leadership / Institutional Control	Core Technical & Operational Metrics	Deterministic Causal Role in Strategic Architecture	Primary Verified Source
146	**Hwasal-2 (& Pulhwasal-3-31)**	DPRK	Strategic Nuclear Cruise Missile	Kimchaek (40.6689°N, 129.1897°E), Hamhung, Nampo	Missile General Bureau & KPAN	Subsonic turbofan LACM/SLCM, 0.6 m dia, 400 kg **Hwasan-31** warhead, 1,800–2,000 km (10,234s)	Extends low-altitude nuclear cruise strike from TELs, corvettes, destroyers & subs to Japan	Hwasal-2 Technical Profile
147	**Haeil (-1, -2, -5-23)**	DPRK	Nuclear Underwater Attack UUV	Sinpo (40.0258°N, 128.1558°E), Riwon, Hongwon, Kajin	Underwater Weapons System Inst / Kim Jong Un	Autonomous UUV (5.7–16 m x 0.9–1.5 m), 80–150 m depth, 4.6 kn, 600–1,000 km (71 hr), **Hwasan-31**	Clandestine underwater nuclear detonation against carrier strike groups & ROK/Japan ports	38 North Haeil Analysis
148	**Railway-Borne Regiment**	DPRK	Rail-Mobile SRBM Unit	Yangdok (39.2050°N, 126.6850°E) & Uiju (40.1850°N, 124.5300°E)	Marshal Pak Jong Chon, KPA Strategic Force / MGB	Diesel-locomotive boxcar TELs w/ 2x **KN-23 (Hwasong-11A)** (430–800 km); expanding to 18-missile bde	Uses mountain rail tunnels for survivable rapid-salvo nuclear/conventional KN-23 strikes	CSIS Rail-Mobile Analysis
149	**Punggye-ri Test Site**	DPRK	Underground Nuclear Test Site	Mt. Mantap, Kilju County (41.2794°N, 129.0869°E)	Nuclear Weapons Institute (Ri Hong Sop, Hong Sung Mu)	4 Tunnels (East, North, South, West); 6 tests (2006–2017) from 0.2–1 kt to 100–250 kt; Tunnel 3 ready	Certified plutonium, composite, and 2-stage thermonuclear warheads; Tunnel 3 restored (2022)	NTI Punggye-ri Profile
150	**Yongbyon Nuclear Center**	DPRK	Fissile Material Complex	Yongbyon County (39.7958°N, 125.7547°E)	General Dept of Atomic Energy & Nuclear Weapons Inst	5 MWe reactor (6 kg Pu/yr), IRT-2000, 100 MWt ELWR (2023), PUREX Lab, 2 Centrifuge Plants (2010 & 2025)	Produces Pu-239, HEU & tritium for ~50 assembled warheads (up to 90 warheads fissile stock)	38 North Yongbyon 2025
151	**Sohae Launching Ground**	DPRK	Space Launch & Engine Test Site	Cholsan County (39.6600°N, 124.7053°E)	NATA (Ryu Sang Hun) & MGB (Jang Chang Ha, Kim Jong Sik)	Original Pad, Yunsong VETS (39.6525°N, 124.7119°E), Solid Stand, Coastal Pad (39.638°N, 124.739°E)	Static-tested all ICBM engines; launched **Chollima-1 / Malligyong-1** into LEO (Nov 21, 2023)	CSIS Beyond Parallel Sohae
152	**Tongchang-ri Complex**	DPRK	Strategic Space/Missile Logistics Hub	Cholsan Peninsula (39.6600°N, 124.7053°E)	NATA, MGB & KPA Engineering Bureaus	Underground Rail Terminal (39.671°N, 124.708°E), port, assembly halls; bordering villages razed Mar 2026	Concealed rail/sea logistics & southward Yellow Sea launch corridor for SLVs and ICBM engines	38 North Sohae Razing 2026

Subsystem Matrix XI.E: India, Pakistan, Israel & Republic of Korea Strategic & NC3 Entities (Entities 153–217)

#	State	Entity Name	Class / Role	Primary Coordinates	Key Technical / Operational Parameter	Named Command / Institutional Authority	Primary Verified Source
153	India	National Command Authority (NCA)	Apex Nuclear C2 Authority	28°36'50"N 77°12'18"E	Established Jan 4, 2003; dual Political & Executive Councils	PM Narendra Modi (Est. by PM Atal Bihari Vajpayee)	MEA CCS Release
154	India	Strategic Forces Command (SFC)	Operational Tri-Service Nuclear Command	28°36'36"N 77°12'12"E	Manages ~172 warheads across Army, Navy, IAF legs	3-Star C-in-C (First: Air Marshal T.M. Asthana)	FAS India 2024
155	India	Political Council (NCA)	Sole Launch Authorization Body	28°36'50"N 77°12'18"E	Civilian nuclear release under NFU & massive retaliation	PM Narendra Modi + CCS Ministers	MEA CCS Release

#	State	Entity Name	Class / Role	Primary Coordinates	Key Technical / Operational Parameter	Named Command / Institutional Authority	Primary Verified Source
156	India	Executive Council (NCA)	Advisory & Directive Execution Body	28°37'21"N 77°12'45"E	Coordinates DAE/BARC, DRDO, SFC & Service Chiefs	Chaired by NSA Ajit Doval (First: Brajesh Mishra)	Nautilus NC3 India
157	India	Agni-I	Single-Stage Solid SRBM/MRBM	20°45'29"N 87°05'08"E	700–1,200 km range; 1,000 kg payload; 12,000 kg mass; 25 m CEP	DRDO ASL / SFC 334 Missile Group	CSIS Agni-I
158	India	Agni-II	Two-Stage Solid MRBM	20°45'29"N 87°05'08"E	2,000–3,500 km range; 1,000 kg MaRV; 16,000 kg; rail/road TEL	DRDO ASL / SFC 335 Missile Group	CSIS Agni-II
159	India	Agni-III	Two-Stage 2.0m Solid IRBM	20°45'29"N 87°05'08"E	3,000–3,500 km range; 1,500 kg payload; 48,300 kg; 40 m CEP	Avinash Chander / Dr. Tessy Thomas (DRDO)	CSIS Agni-III
160	India	Agni-IV	Two-Stage Composite Solid IRBM	20°45'29"N 87°05'08"E	3,500–4,000 km range; 1,000 kg payload; 17,000 kg; 8x8 TEL	Dr. Tessy Thomas (ASL Hyderabad)	CSIS Agni-IV
161	India	Agni-V	Three-Stage Canisterized MIRV ICBM	20°45'29"N 87°05'08"E	5,000–8,000 km range; 1,500 kg (3–4 MIRVs, Mission Divyastra)	Sheena Rani / Dr. Samir V. Kamat (DRDO)	PIB Mission Divyastra
162	India	Agni-Prime (Agni-P)	Two-Stage Canisterized Composite MRBM	20°45'29"N 87°05'08"E	1,000–2,000 km range; ~11,000 kg mass; sealed canister TEL	DRDO ASL / Indian Army SFC	CSIS Agni-P
163	India	Prithvi-II	Single-Stage Storable Liquid SRBM	21°26'45"N 87°01'05"E	250–350 km range; 500–1,000 kg payload; G-Fuel + RFNA; 8x8 TEL	Dr. A.P.J. Abdul Kalam / SFC 333 & 444 Groups	CSIS Prithvi
164	India	Dhanush	Ship-Launched Liquid SRBM (ShLBM)	14°46'54"N 74°07'32"E	250–400 km range; 500–1,000 kg payload on INS Subhadra/Suvarna	DRDO R&DE(E) / Indian Navy SFC	CSIS Dhanush
165	India	INS Arihant (S2)	Lead Arihant-Class SSBN (SSBN-80)	17°41'45"N 83°16'05"E	6,000 t; 83 MWt PWR; 4 VLS tubes (12 K-15 or 4 K-4 SLBMs)	ATV Programme / Eastern Naval Command	FAS India 2024
166	India	INS Arighat (S3)	Second Arihant-Class SSBN	17°41'45"N 83°16'05"E	Commissioned Aug 29, 2024; 6,000 t; 83 MWt PWR; 4 VLS tubes	Adm Dinesh K. Tripathi / DM Rajnath Singh	FAS India 2024
167	India	Project S4 (S4 & S4*)	Enlarged Arihant-Class SSBNs	17°28'25"N 82°56'28"E	~7,000 t; +16–18m hull plug; 8 VLS tubes (8 K-4 or 24 K-15)	ATV Programme / INS Varsha Rambilli	Bulletin India 2024
168	India	K-15 Sagarika (B-05)	Two-Stage Solid Short-Range SLBM	17°40'00"N 83°25'00"E	700–750 km range; 10.8m x 0.8m; 3 missiles per SSBN VLS tube	A.K. Chakrabarti (DRDO DRDL/ASL)	CSIS Sagarika
169	India	K-4	Two-Stage Solid IRBM-Class SLBM	17°40'00"N 83°25'00"E	3,500 km range; 12m x 1.3m; 17,000 kg mass; up to 2,000 kg payload	DRDO ASL / Indian Navy SSBN Force	FAS India 2024

#	State	Entity Name	Class / Role	Primary Coordinates	Key Technical / Operational Parameter	Named Command / Institutional Authority	Primary Verified Source
170	India	Mirage 2000H (2000I/TI)	Dual-Capable Nuclear Strike Fighter	26°17'36"N 78°13'42"E	51 upgraded jets; No. 1 & No. 7 Sqns IAF; 12–45 kt gravity bomb	IAF Maharajpur AFS Gwalior / SFC	FAS India 2024
171	India	Rafale DH (EH/DH)	4.5-Gen Dual-Capable Omnirole Fighter	30°22'05"N 76°49'00"E	36 jets (13 ISEs); No. 17 Sqn (Ambala) & No. 101 Sqn (Hasimara)	IAF Western & Eastern Air Commands / SFC	FAS India 2024
172	India	BARC (Trombay)	Fissile Material & Warhead Design Hub	19°00'28"N 72°55'07"E	100 MWt Dhruva reactor (~20–25 kg WGPu/yr); pit & PWR design	Director Vivek Bhasin / DAE Secy Dr. A.K. Mohanty	BARC Official
173	India	DRDO (ASL Hyderabad)	Missile, Implosion Lens & PAL Agency	17°19'55"N 78°30'35"E	Agni/Prithvi/K-series R&D, TBRL implosion lenses, PAL interlocks	Chairman Dr. Samir V. Kamat	DRDO Official
174	Pakistan	Strategic Plans Division (SPD)	Permanent Secretariat of NCA	33°36'15"N 73°05'45"E	4 Directorates (Ops, C4I2SR, SWD, ACDA) + 25,000-troop Security Div	Advisor Lt Gen Khalid Kidwai / DG SPD	Arms Control Assoc
175	Pakistan	SECDIV (MOFA)	Strategic Export Control Division	33°43'17"N 73°05'31"E	Enforces Export Control Act 2004 & Oct 2025 SRO 1977(I)/2025 Lists	Foreign Secy Oversight Board / DG Brig Haroon Rashid	MOFA SECDIV
176	Pakistan	NC3 Pakistan (NCA)	ECC, DCC, C4I2SR & PALS Architecture	33°43'45"N 73°05'50"E	3-tier NCA (2000 / Act 2010); 12-digit PAS/PAL unlock code	PM Shehbaz Sharif / CJCSC Gen Sahir Shamshad Mirza	Arms Control Assoc
177	Pakistan	Shaheen-I (Hatf-IV / IA)	Single-Stage Solid SRBM	33°37'30"N 72°32'00"E	750 km (Shaheen-I) / 900 km (Shaheen-IA); 9,500 kg; PSACS RV	NESCOM / NDC / Army Strategic Forces Command	CSIS Shaheen-1
178	Pakistan	Shaheen-II (Hatf-VI)	Two-Stage Solid Road-Mobile MRBM	27°46'55"N 66°38'15"E	1,500–2,000 km range; 23,600 kg; 17.2m x 1.4m; 6-axle TEL	NESCOM / NDC / ASFC	CSIS Shaheen-2
179	Pakistan	Shaheen-III	Two-Stage Solid Extended-Range MRBM	25°11'38"N 66°44'47"E	2,750 km range (reaches Andaman & Nicobar); 8-axle WS51200 TEL	Lt Gen Khalid Kidwai / NESCOM / ASFC	CSIS Shaheen-3
180	Pakistan	Ghauri (Hatf-V)	Single-Stage Liquid Road-Mobile MRBM	33°35'42"N 73°23'24"E	1,250–1,500 km range; 15,850 kg; TM-185/IRFNA; Nodong-1 heritage	Dr. A.Q. Khan (KRL) / ASFC	CSIS Ghauri
181	Pakistan	Ghaznavi (Hatf-III)	Single-Stage Solid Road-Mobile SRBM	32°02'30"N 72°40'00"E	290–300 km range; 4,650 kg; 8.5m x 0.8m (DF-11 derived); 4-axle TEL	NESCOM / NDC / ASFC	CSIS Ghaznavi
182	Pakistan	Hatf-IX (Nasr)	Multi-Tube Solid Tactical Nuclear SRBM	32°14'15"N 74°07'45"E	60–70 km range; 400mm diameter; 2- or 4-tube 8x8 TEL; Pu-239 pit	SPD / NESCOM / ASFC	CSIS Nasr
183	Pakistan	Babur-1 (Hatf-VII / 1A)	Ground-Launched Cruise Missile (GLCM)	25°31'40"N 68°08'10"E	350–450 km range; Mach 0.8 turbofan; TERCOM/DSMAC; 5-axle TEL	NESCOM / NDC / ASFC	CSIS Babur

#	State	Entity Name	Class / Role	Primary Coordinates	Key Technical / Operational Parameter	Named Command / Institutional Authority	Primary Verified Source
184	Pakistan	Babur-2 (Babur-1B)	Extended-Range GLCM	25°11'38"N 66°44'47"E	700–900 km range; upgraded avionics & aerodynamics; 3-tube 10x10 TEL	NESCOM / ASFC	CSIS Hatf-7 Tag
185	Pakistan	Babur-3	Submarine-Launched Cruise Missile (SLCM)	25°12'35"N 64°39'50"E	450 km range; 533mm torpedo-tube capsule; tested Jan 2017 & Mar 2018	Naval Strategic Force Command (NSFC)	CSIS Babur-3
186	Pakistan	Ra'ad (Hatf-VIII / I & II)	Air-Launched Cruise Missile (ALCM)	30°45'29"N 72°16'57"E	350 km (Ra'ad-I) / 550–600 km (Ra'ad-II); carried by Mirage III/V	NESCOM Air Weapons Complex / PAF AFSC	CSIS Ra'ad
187	Pakistan	Khan Research Labs (KRL)	Gas Centrifuge & Ghauri R&D Institute	33°35'42"N 73°23'24"E	Established 1976 (ERL); P-1/P-2/P-3/P-4 centrifuge enrichment	Founded by Dr. A.Q. Khan; overseen by SPD	ISIS Pakistan Report
188	Pakistan	Kahuta Enrichment Facility	Primary WGU Centrifuge Complex	33°35'42"N 73°23'24"E	Produces 90%+ HEU (~5.3 tonnes WGU national stock with Gadwal)	KRL / SPD Security Division	World Nuclear Assoc
189	Pakistan	PAEC	Atomic Energy & Plutonium/Test Agency	33°43'28"N 73°05'36"E	Conducted May 28 & 30, 1998 Chagai-I/II tests; runs PINSTECH & Khushab	Chairman Dr. Raja Ali Raza Anwar (Hist: Munir A. Khan)	NTI PAEC Profile
190	Pakistan	Khushab Nuclear Complex	Four Heavy-Water Plutonium Reactors	32°01'50"N 72°12'30"E	Khushab-I (1998), II (2010), III (2013), IV (2015); ~580 kg WGPu stock	PAEC / SPD Security Division	NTI Pakistan
191	Israel	Samson Option	Last-Resort Deterrence & Opacity C2	32°04'32"N 34°47'27"E	Double-key PM + DM authorization; ~90 warheads across nuclear triad	PM Benjamin Netanyahu / DM Israel Katz / IAEC DG	Wilson Center Archive
192	Israel	Dimona (NNRC)	IRR-2 Reactor & Machon 2 Reprocessing	31°00'05"N 35°08'40"E	24–70 MWt heavy-water reactor (1963) + 8-level underground Machon 2	IAEC DG Brig Gen (Ret.) Moshe Edri	NSA Discovery of Dimona
193	Israel	Soreq NRC (Yavne)	IRR-1 Reactor, SARAF & Weapon Physics	31°54'12"N 34°42'22"E	5 MWt IRR-1 (1960), 40 MeV SARAF linear accelerator, weapon design	Israel Atomic Energy Commission (IAEC)	NTI Soreq Profile
194	Israel	Sdot Micha Airbase	Ballistic Missile Base & Warhead Depot	31°44'19"N 34°55'10"E	23 limestone caves (14 North, 9 South; 46 TELs) + 4-tunnel warhead depot	Israeli Air Force (Area 209 / Kanaf 2)	Bulletin Israel 2021
195	Israel	Wing 2 (Kanaf 2)	IAF Strategic Ballistic Missile Wing	31°44'19"N 34°55'10"E	Commands 150, 199 & 248 Squadrons and nuclear warhead custody unit	Founded 1962 by Col. Benny Peled; IAF HQ	GlobalMilitary Sdot Micha
196	Israel	150 Squadron	First Operational Jericho Missile Sqn	31°44'48"N 34°56'02"E	Operates road-mobile Jericho-II/III TELs from limestone caves	IAF Wing 2 (Sdot Micha)	Bulletin Israel 2021

#	State	Entity Name	Class / Role	Primary Coordinates	Key Technical / Operational Parameter	Named Command / Institutional Authority	Primary Verified Source
197	Israel	199 Squadron	Second Operational Jericho Missile Sqn	31°43'50"N 34°55'25"E	Operates road-mobile Jericho-II/III TELs from limestone caves	IAF Wing 2 (Sdot Micha)	Bulletin Israel 2021
198	Israel	248 Squadron	Jericho Testing & Shavit Launch Sqn	31°53'04"N 34°40'49"E	Executes propulsion/flight tests at Palmachim & Sdot Micha	IAF Wing 2 / IAI MLM / Tomer	Bulletin Israel 2021
199	Israel	Jericho-1 (YA-1)	Two-Stage Solid SRBM (Dassault MD-620)	31°44'19"N 34°55'10"E	500 km range; 6,700 kg; 650 kg payload (20 kt); operational 1970–1990s	Marcel Dassault / IAI / IAF Wing 2	CSIS Jericho 1
200	Israel	Jericho-2 (YA-3)	Two-Stage Solid MRBM (Shavit Basis)	31°44'19"N 34°55'10"E	1,500–3,500 km range; 22,000–26,000 kg; 1,000 kg RV; active radar	IAI MLM Division / IAF Wing 2	CSIS Jericho 2
201	Israel	Jericho-3 (YA-4)	Three-Stage Solid IRBM/ICBM	31°44'19"N 34°55'10"E	4,800–6,500 km range; 29,000 kg; 1,000–1,300 kg payload (150–400 kt)	IAI MLM / Rafael / Tomer / IAF Wing 2	CSIS Jericho 3
202	Israel	Dolphin-Class (Batch 1)	Type 800 Diesel-Electric Submarines	32°49'38"N 35°00'18"E	3 boats (Dolphin, Leviathan, Tekumah); six 533mm + four 650mm tubes	HDW Kiel / Israeli Navy Shayetet 7 (Haifa)	NTI Israel Submarines
203	Israel	Dolphin-II-Class (Batch 2)	Stretched Fuel-Cell AIP Submarines	32°49'38"N 35°00'18"E	3 boats (Tanin, Rahav, Drakon w/ VLS); 2,400 t; 18+ days AIP submerged	TKMS/HDW Kiel / Shayetet 7	NTI Israel Submarines
204	Israel	INS Tanin	First Dolphin-II AIP Submarine	32°49'38"N 35°00'18"E	Commissioned Sept 2014; 68.6m; Siemens PEM AIP; four 650mm SLCM tubes	Israeli Navy Shayetet 7	NTI Israel Submarines
205	Israel	INS Rahav	Second Dolphin-II AIP Submarine	32°49'38"N 35°00'18"E	Commissioned Jan 2016; 68.6m; Siemens PEM AIP; four 650mm SLCM tubes	Israeli Navy Shayetet 7	NTI Israel Submarines
206	Israel	Popeye Turbo	650mm Turbofan Nuclear SLCM / ALCM	32°49'38"N 35°00'18"E	1,500 km SLCM range (tested May 2000 Indian Ocean); ~200 kg warhead	Rafael Advanced Defense Systems / Shayetet 7	CSIS Popeye
207	Israel	F-15I Ra'am	Long-Range Dual-Capable Strike Fighter	31°14'00"N 34°39'45"E	25 aircraft in 69 Squadron ("Hammers"); >1,800 km combat radius	IAF Hatzerim Airbase (Base 6)	MiGFlug IAF
208	Israel	F-35I Adir	5th-Gen Stealth Strike Fighter	31°12'30"N 35°00'44"E	50 delivered (75 ordered); 140, 116 & 117 Squadrons; custom IAF C4/EW	IAF Nevatim Airbase (Base 28)	Bulletin Israel 2021
209	ROK	National Crisis Mgmt Center	Presidential Underground C4I Bunker	37°35'12"N 126°58'29"E	Blue House (1976 / returned Dec 2025) & Yongsan basement bunkers	President Lee Jae-myung / ONS Director	Chosun Daily Cheong Wa Dae

#	State	Entity Name	Class / Role	Primary Coordinates	Key Technical / Operational Parameter	Named Command / Institutional Authority	Primary Verified Source
210	ROK	KMPR & Korea Strategic Cmd	3rd Axis Retaliation & Unified STRATCOM	37°27'40"N 126°59'15"E	Launched Oct 1, 2024; commands Hyunmoo, F-35A, KSS-III & US-ROK CNI	First Cdr Lt Gen (now JCS Chair Gen) Jin Young-seung	Chosun Daily STRATCOM
211	ROK	Kill Chain (425 Project)	1st Axis Preemptive Strike & 5 Spy Sats	37°05'26"N 127°01'46"E	1 EO/IR + 4 SAR satellites (completed Nov 2, 2025; 30cm res, 2hr revisit)	DAPA / ADD / ROK Strategic Command	United24 425 Project
212	ROK	KAMD	2nd Axis Multi-Layered Missile Defense	37°05'26"N 127°01'46"E	Green Pine, THAAD, L-SAM (40–60km), PAC-3 MSE, M-SAM II Cheongung-II	ROKAF Air Defense Missile Command (KAMDOC)	CSIS L-SAM
213	ROK	Hyunmoo-IV (4.1, 4.2, 4.4)	2-Ton Bunker-Buster SRBM & K-SLBM	36°40'30"N 126°08'15"E	4.1 (800 km, 2t warhead), 4.2 (Ship), 4.4 (KSS-III SLBM tested Sept 2021)	ADD / Hanwha / Army Missile Strategic Command	CSIS Hyunmoo-4
214	ROK	Hyunmoo-V ("Monster Missile")	36-Ton Ballistic Missile w/ 8-Ton Warhead	37°26'45"N 127°06'50"E	8–9 ton bunker-buster warhead; 9-axle TEL; unveiled Oct 2024, fielded 2025/26	Defense Minister Ahn Gyu-back / ROK STRATCOM	IISS Hyunmoo-5
215	ROK	Bunker B-1 (Namtaeryeong)	Capital Defense & Wartime JCS Bunker	37°27'40"N 126°59'15"E	Built 1978 inside Mt. Gwanaksan granite; EMP/nuclear hardened (500+ staff)	Capital Defense Command / JCS / MoND	Hankyoreh B1 Bunker
216	ROK	Bunker B-2 (Gyeryongdae)	Tri-Service HQ Wartime Bunker	36°18'25"N 127°14'10"E	Built 1989 under Project 620 beneath Mt. Gyeryongsan; Army/Navy/AF HQ	ROK Army, Navy & Air Force Chiefs of Staff	Korea Times Gyeryongdae
217	ROK	CP TANGO (Seongnam)	Combined Forces Command Wartime Bunker	37°27'15"N 127°07'30"E	Built 1970s inside Seongnam granite; nuclear blast doors; TACCIMS/CENTRIXS	Cdr UNC/CFC/USFK Gen Xavier Brunson & ROK Deputy	Korea Times CP Tango

Section XII: Unified Categorized Archival, Governmental, Institutional & Academic Source Registry

Registry XII.A: United States, United Kingdom & French Primary Defense, National Laboratory & Archival Sources

1. Official Defense, Military & Executive Publications (DoD, USAF, USN, UK MoD, French Ministry of the Armed Forces)

- **U.S. Air Force / Air Force Global Strike Command (AFGSC) / Air Force Nuclear Weapons Center (AFNWC):**
  - LGM-30G Minuteman III Fact Sheet – U.S. Air Force (AF.mil)
  - LGM-30G Minuteman III Fact Sheet – Air Force Global Strike Command
  - Minuteman III (LGM-30G) – Air Force Nuclear Weapons Center
  - Boeing LGM-30G Minuteman III – National Museum of the U.S. Air Force
  - LGM-35A Sentinel Fact Sheet – Air Force Global Strike Command
  - LGM-35A Sentinel Fact Sheet – Air Force Nuclear Weapons Center
  - Delivering Deterrence: Sentinel Restructure to Complete in 2026 – U.S. Air Force (February 17, 2026)
  - Sentinel Program Advances with Silo Prototype – U.S. Air Force (March 27, 2026)
  - USAF Assembles Inert Sentinel Missile, Moving with Pace Toward Flight Testing – U.S. Air Force (September 14, 2026)
  - Air Force, Northrop Grumman Advance Sentinel ICBM Modernization with Stage-Two Rocket Motor Test – U.S. Air Force (July 29, 2025)

- B-52H Stratofortress Fact Sheet – U.S. Air Force
- B-52H Stratofortress Fact Sheet – Air Force Nuclear Weapons Center
- B-2 Spirit Fact Sheet – U.S. Air Force
- B-21 Raider Fact Sheet – U.S. Air Force
- B-21 Summit Unites Military, Community Leaders to Prepare Dyess for Future Mission – AFGSC (October 19, 2026)
- Air Force Doctrine Publication (AFDP) 3-72: Nuclear Operations – U.S. Air Force
- Leadership Biographies (Gen. Stephen L. Davis, Lt. Gen. Jason R. Armagost, Maj. Gen. Ty W. Neuman, Maj. Gen. Colin J. Connor) – Air Force Global Strike Command (2026)
- USSTRATCOM Commander Adm. Rich Correll Tours Dyess AFB – AFGSC (June 30, 2026)
- 509th Bomb Wing Presented with Second Consecutive Omaha Trophy – AFGSC (June 2, 2026)

- **U.S. Navy / Strategic Systems Programs (SSP) / Submarine Forces / National Park Service:**
  - Ballistic Missile Submarines (SSBNs) – Commander, Submarine Force, U.S. Pacific Fleet (SUBPAC)
  - Vice Admiral Richard E. "Rick" Seif, Commander, Submarine Forces – U.S. Navy Flag Officer Biographies (February 24, 2026)
  - Submarine Force U.S. Pacific Fleet Changes Command (RADM Chris Cavanaugh) – SUBPAC (April 12, 2025)
  - Vice Adm. Douglas L. Williams Relieves Vice Adm. Johnny Wolfe Jr. at PAE Strategic Systems Programs Change of Command – U.S. Navy SSP (June 12, 2026)
  - The Delta-01 Launch Control Facility – Minuteman Missile National Historic Site (U.S. National Park Service)
  - The Delta-09 Missile Silo – Minuteman Missile National Historic Site (U.S. National Park Service)
- **UK Ministry of Defence & Royal Navy:**
  - Vanguard Class Nuclear Submarine – Royal Navy
  - Dreadnought Class Nuclear Submarine – Royal Navy
  - Types of UK Royal Navy Submarine – GOV.UK (October 27, 2025)
  - Continuous At Sea Deterrent 50: What You Need to Know (Operation Relentless) – GOV.UK
  - £8.4bn Injection into Dreadnoughts to Carry Nuclear Deterrent Programme into 2030s – Royal Navy (July 30, 2026)
  - Air Chief Marshal Sir Richard Knighton Takes Over as Chief of the Defence Staff – GOV.UK (September 2, 2025)
  - General Sir Gwyn Jenkins Appointed as First Sea Lord and Chief of the Naval Staff – GOV.UK (May 15, 2025)
  - Defence Nuclear Organisation (Chief of Defence Nuclear Madelaine McTernan CB) – GOV.UK
- **French Ministry of the Armed Forces (*Ministère des Armées*) & DGA:**
  - Le missile mer-sol balistique stratégique M51 – Direction générale de l'armement (DGA)
  - Adaptation au M51 des SNLE – Direction générale de l'armement (DGA)
  - Missile M51.3 : un tir d'acceptation réussi depuis le SNLE Le Vigilant – Ministère des Armées (October 6, 2026)
  - Communiqué : La DGA commande le développement du missile balistique stratégique M51.4 à ArianeGroup – Ministère des Armées (September 12, 2025)
  - Biographie du chef d'état-major des Armées (Général d'armée aérienne Fabien Mandon) – Ministère des Armées
  - Un nouvel amiral à la tête des forces sous-marines et de la Force océanique stratégique (VAE Xavier Petit, ALFOST) – Ministère des Armées (September 3, 2024)
  - Les Forces aériennes stratégiques, piliers de la dissuasion nationale (GCA Stéphane Virem, CFAS) – Ministère des Armées (October 17, 2024)
  - From Power Projection to Nuclear Deterrence: The Strategic Role of the FAS – Ministère des Armées

2. National Nuclear Laboratories & Warhead Design Authorities (DOE/NNSA, LANL, LLNL, Sandia, AWE, CEA/DAM)

- Warhead Modernization Fact Sheet – U.S. Department of Energy / NNSA (July 2024)
- W76-1 Life Extension Program Fact Sheet – U.S. Department of Energy / NNSA
- W88 Alteration 370 Program Fact Sheet – U.S. Department of Energy / NNSA
- NNSA Completes First Production Unit of W88 Alteration 370 – DOE/NNSA (July 13, 2021)
- NNSA Completes Last Production Unit of W88 Nuclear Warhead Upgrade – DOE/NNSA (December 16, 2025)
- Nuclear Deterrence Accomplishments (B61-12, W80-4, W87-1, Mk21 Fuze, W88 Alt 370) – Sandia National Laboratories (2024)
- Our Leadership (Dr. Kim Budil, Carolyn Zerkle, Brad Wallin) – Lawrence Livermore National Laboratory
- National Security Science: Fusion Issue (Dr. Thom Mason, Charlie Nakhleh, Adm. Rich Correll) – Los Alamos National Laboratory (2026)
- Atomic Weapons Establishment (AWE plc) Annual Report and Accounts – AWE Aldermaston (2025/2026)
- La Direction des Applications Militaires (CEA/DAM) au cœur de la dissuasion nucléaire française – CEA
- Organigramme du CEA (Jérôme Demoment, Directeur des Applications Militaires) – CEA (November 1, 2026)

3. Legislative, Parliamentary & Congressional Research Reports (CRS, Sénat, Assemblée Nationale)

- Navy Columbia Class (Ohio Replacement) Ballistic Missile Submarine (SSBN[X]) Program: Background and Issues for Congress – Congressional Research Service / NHHC

- Navy Ship Names: Background for Congress (CRS Report RS22478, USS District of Columbia SSBN-826) – Congressional Research Service
- Air Force B-21 Raider Long-Range Strike Bomber: Background and Issues for Congress – Congressional Research Service
- Rapport d'information N° 950 sur la dissuasion nucléaire – Sénat de la République française (September 16, 2026)
- Rapport d'information N° 1112 portant recueil d'auditions sur la dissuasion nucléaire – Assemblée nationale

4. Academic, Scientific & Technical Reference Archives (FAS Nuclear Notebook, Bulletin of the Atomic Scientists, CSIS Missile Threat, Nuclear Weapon Archive)

- FAS Nuclear Notebook: United States Nuclear Weapons, 2025 – Bulletin of the Atomic Scientists / Federation of American Scientists (January 13, 2025)
- FAS Nuclear Notebook: United Kingdom Nuclear Weapons, 2024 – Bulletin of the Atomic Scientists / Federation of American Scientists (November 12, 2024)
- FAS Nuclear Notebook: French Nuclear Weapons, 2025 – Bulletin of the Atomic Scientists / Taylor & Francis (July 15, 2025)
- Minuteman III | Missile Threat – Center for Strategic and International Studies (CSIS)
- LGM-118 Peacekeeper (MX) | Missile Threat – CSIS
- Titan II | Missile Threat – CSIS
- Trident D5 | Missile Threat – CSIS
- Pershing 2 | Missile Threat – CSIS
- M51 | Missile Threat – CSIS
- Air-Sol Moyenne Portée (ASMP/ ASMP-A) | Missile Threat – CSIS
- Intercontinental Ballistic Missiles Technical Introduction – Federation of American Scientists (FAS)
- LGM-30A/B Minuteman I, LGM-30F Minuteman II, LGM-30G Minuteman III, LGM-118A Peacekeeper, SM-68B Titan II – FAS Nuclear Forces Guide
- SSBN-726 Ohio-Class, Trident I C-4, Trident II D-5, Le Triomphant SNLE – FAS Nuclear Forces Guide
- The W76, W78, W87, W88 Warheads & Peacekeeper (MX) ICBM – The Nuclear Weapon Archive (Carey Sublette)
- The Future of the U.S. Intercontinental Ballistic Missile Force (MARV / AMaRV Analysis) – RAND / DTIC
- Ensuring Cyber Resilience in NATO's Command, Control and Communication Systems – Chatham House
- Le Président et la Bombe : Chapitre 19, Si l'ordre était donné (PC Jupiter, RAMSES, SYDEREC) – Bruno Tertrais & Jean Guisnel (Cairn.info)

Registry XII.B: USSR & Russian Federation NC3, SPRN, RVSN & Strategic Triad Primary & Academic Sources

I. Primary Archival, Treaty & Declassified Government Records

- **National Security Archive (George Washington University) / NARA:** David E. Hoffman, *The Dead Hand: The Untold Story of the Cold War Arms Race and Its Dangerous Legacy* — Archival Briefing Book Excerpt (2009/2015), https://nsarchive2.gwu.edu/NunnLugar/2015/59.%202009-00-00%20Excerpt%20from%20The%20Dead%20Hand%20by%20David%20E%20Hoffman.pdf
- **National Security Archive (GWU):** *Kazakhstan and Nunn-Lugar: A Non-Proliferation Success Story* (Electronic Briefing Book No. 528, 29 August 2015), https://nsarchive2.gwu.edu/NSAEBB/NSAEBB528-Kazakhstan-Nunn-Lugar-Non-Proliferation-Success/
- **Vitaly Leonidovich Kataev Archival Papers (CPSU Central Committee Defense Industry Department / RussianForces.org):** *A Note on Mobile Missile Systems («СПРАВКА по ракетным комплексам мобильного базирования»)*, Hoover Institution / Russian Strategic Nuclear Forces Project, https://russianforces.org/Kataev_K5.11_Mobile_missiles_e.pdf
- **U.S. Congress House Armed Services Committee Record (Federation of American Scientists):** *Russian Threats to United States Security in the Post-Cold War Era* (24 January 2000 — Testimony on Kosvinsky Kamen and Mount Yamantau), https://irp.fas.org/congress/2000_hr/hr_012400.htm
- **U.S. National Air and Space Intelligence Center (NASIC / CSIS Archive):** *Ballistic and Cruise Missile Threat* (NASIC-1031-0985-17, June 2017), http://missilethreat.csis.org/wp-content/uploads/2017/06/06262017_nasic.pdf
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II. Academic & Technical Research Institutions (FAS, Bulletin of the Atomic Scientists, CSIS, SIPRI, NTI, UNIDIR)

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